

References

- [1] AAAI. **35th AAAI Conference on Artificial Intelligence: Reproducibility Checklist**. <https://aaai.org/Conferences/AAAI-21/reproducibility-checklist/>, 2021. Last accessed: June 6th, 2021.
- [2] AAAI2020. *The Thirty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2020, The Thirty-Second Innovative Applications of Artificial Intelligence Conference, IAAI 2020, The Tenth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2020, New York, NY, USA, February 7-12, 2020*, 2020. AAAI Press. ISBN 978-1-57735-823-7.
- [3] E. H. L. Aarts and J. K. Lenstra, editors. *Local Search in Combinatorial Optimization*. John Wiley & Sons, Chichester, UK, 1997.
- [4] E. H. L. Aarts, J. H. M. Korst, and W. Michiels. **Simulated Annealing**. In E. K. Burke and G. Kendall, editors, *Search Methodologies*, pages 187–210. Springer, Boston, MA, 2005. doi:10.1007/0-387-28356-0_7.
- [5] H. A. Abbass. **The self-adaptive Pareto differential evolution algorithm**. In *IEEE CEC*, pages 831–836, Piscataway, NJ, 2002. IEEE Press.
- [6] H. A. Abbass, R. Sarker, and C. Newton. **PDE: a Pareto-frontier differential evolution approach for multi-objective optimization problems**. In *IEEE CEC*, pages 971–978, Piscataway, NJ, 2001. IEEE Press.
- [7] O. Abdelkhalik and A. Gad. **Dynamic-Size Multiple Populations Genetic Algorithm for Multigravity-Assist Trajectory Optimization**. *Journal of Guidance, Control, and Dynamics*, 35(2):520–529, 2012. doi:10.2514/1.54330.
- [8] A. Abraham and M. Paprzycki, editors. *Proceedings of the 5th International Conference on Intelligent Systems Design and Applications*, 2005.
- [9] A. Abraham, L. Jain, and R. Goldberg, editors. *Evolutionary Multiobjective Optimization*. Advanced Information and Knowledge Processing. Springer, London, UK, Jan. 2005.
- [10] A. Abraham, C. Grosan, and H. Ishibuchi, editors. *Hybrid Evolutionary Algorithms*, volume 75 of *Studies in Computational Intelligence*. Springer, 2007.
- [11] D. Abramson. **Constructing School Timetables Using Simulated Annealing: Sequential and Parallel Algorithms**. *Management Science*, 37(1):98–113, 1991.
- [12] D. Abramson, M. K. Amoorthy, and H. Dang. **Simulated annealing cooling schedules for the school timetabling problem**. *Asia-Pacific Journal of Operational Research*, 16(1):1–22, 1999.
- [13] A. Acan. **An external memory implementation in ant colony optimization**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 73–84. Springer, Heidelberg, Germany, 2004.
Keywords: memory-based ACO.
- [14] A. Acan. **An external partial permutations memory for ant colony optimization**. In G. R. Raidl and J. Gottlieb, editors, *Proceedings of EvoCOP 2005 – 5th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3448 of *LNCS*, pages 1–11. Springer, Heidelberg, Germany, 2005.
Keywords: memory-based ACO.
- [15] ACC2019. *2019 American Control Conference, ACC 2019, Philadelphia, PA, USA, July 10-12, 2019*, 2019. IEEE.

- [16] T. Achterberg. **SCIP: Solving constraint integer programs**. *Mathematical Programming Computation*, 1(1):1–41, July 2009.
- [17] T. Achterberg and T. Berthold. **Improving the feasibility pump**. *Discrete Optimization*, 4(1):77–86, 2007.
- [18] ACM. **Artifact Review and Badging Version 1.1**. <https://www.acm.org/publications/policies/artifact-review-and-badging-current>, Aug. 2020.
- [19] H.-G. Acosta-Mesa, F. Rechy-Ramírez, E. Mezura-Montes, N. Cruz-Ramírez, and R. Hernández Jiménez. **Application of time series discretization using evolutionary programming for classification of precancerous cervical lesions**. *Journal of Biomedical Informatics*, 49:73–83, 2014. doi:10.1016/j.jbi.2014.03.004.
Keywords: irace.
- [20] B. Addis, M. Locatelli, and F. Schoen. **Disk Packing in a Square: A New Global Optimization Approach**. *INFORMS Journal on Computing*, 20(4):516–524, 2008. doi:10.1287/ijoc.1080.0263.
- [21] B. Adenso-Díaz. **Restricted Neighborhood in the Tabu Search for the Flowshop Problem**. *Eur. J. Oper. Res.*, 62(1):27–37, 1992.
- [22] B. Adenso-Díaz and M. Laguna. **Fine-Tuning of Algorithms Using Fractional Experimental Design and Local Search**. *Operations Research*, 54(1):99–114, 2006. doi:10.1287/opre.1050.0243.
Keywords: Calibra, heuristic search, parameter setting, Taguchi design of experiments.
- [23] S. Adriaensen, A. Biedenkapp, G. Shala, N. Awad, T. Eimer, M. T. Lindauer, and F. Hutter. **Automated dynamic algorithm configuration**. *J. Artif. Intell. Res.*, 75:1633–1699, 2022. doi:10.1613/jair.1.13922.
- [24] M. Affenzeller, S. M. Winkler, A. V. Kononova, H. Trautmann, T. Tušar, P. Machado, and T. Bäck, editors. *Parallel Problem Solving from Nature - PPSN XVIII, 18th International Conference, PPSN 2024, Hagenberg, Austria, September 14-18, 2024, Proceedings, Part II*, volume 15149 of *LNCS*. Springer, Cham, Switzerland, 2024.
- [25] B. Afsar, K. Miettinen, and F. Ruiz. **Assessing the Performance of Interactive Multiobjective Optimization Methods: A Survey**. *ACM Comput. Surv.*, 54(4), 2021. doi:10.1145/3448301.
Keywords: decision-makers, Interactive methods, performance assessment, preference information, multiobjective optimization problems.
- [26] B. Afsar, J. Silvennoinen, G. Misitano, F. Ruiz, A. B. Ruiz, and K. Miettinen. **Designing empirical experiments to compare interactive multiobjective optimization methods**. *J. Oper. Res. Soc.*, 74(11):2327–2338, Nov. 2022. doi:10.1080/01605682.2022.2141145.
- [27] P. J. Agrell. **On redundancy in multi criteria decision making**. *Eur. J. Oper. Res.*, 98(3): 571–586, 1997. doi:10.1016/0377-2217(95)00340-1.
Keywords: Multi criteria decision making, Redundancy, objective reduction, Vector optimisation.
- [28] H. E. Aguirre. **Advances on Many-objective Evolutionary Optimization**. In C. Blum and E. Alba, editors, *GECCO Companion*, pages 641–666. ACM Press, New York, NY, 2013.
Keywords: many-objective evolutionary optimization.
- [29] H. E. Aguirre and K. Takadama, editors. *Genetic and Evolutionary Computation Conference, GECCO 2018, Kyoto, Japan, July 15-19, 2018*. ACM Press, New York, NY, 2018. doi:10.1145/3205455.

- [30] H. E. Aguirre and K. Tanaka. **Working principles, behavior, and performance of MOEAs on MNK-landscapes**. *Eur. J. Oper. Res.*, 181(3):1670–1690, 2007. doi:[10.1016/j.ejor.2006.08.004](https://doi.org/10.1016/j.ejor.2006.08.004).
- [31] H. E. Aguirre and K. Tanaka. **Many-Objective Optimization by Space Partitioning and Adaptive ϵ -Ranking on MNK-Landscapes**. In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCs*, pages 407–422. Springer, Heidelberg, Germany, 2009.
- [32] H. E. Aguirre, S. Zapotecas, A. Liefooghe, S. Verel, and K. Tanaka. **Approaches for Many-Objective Optimization: Analysis and Comparison on MNK-Landscapes**. In S. Bonnevey et al., editors, *Artificial Evolution: 12th International Conference, Evolution Artificielle, EA, 2015*, volume 9554 of *LNCs*, pages 14–28. Springer, Cham, Switzerland, 2016. doi:[10.1007/978-3-319-31471-6_2](https://doi.org/10.1007/978-3-319-31471-6_2).
- [33] S. Ahmadi and I. H. Osman. **Density Based Problem Space Search for the Capacitated Clustering p -Median Problem**. *Annals of Operations Research*, 131:21–43, 2004.
- [34] A. AhmadiTeshnizi, W. Gao, and M. Udell. **OptiMUS: Scalable Optimization Modeling with (MI)LP Solvers and Large Language Models**. In *Proceedings of the 41st International Conference on Machine Learning, ICML 2024*, Proceedings of Machine Learning Research, pages 577–596. PMLR, 2024. doi:[10.48550/arXiv.2402.10172](https://doi.org/10.48550/arXiv.2402.10172).
- [35] A. Aho, J. Hopcroft, and J. Ullman. *Data structures and algorithms*. Addison-Wesley, Reading, MA, 1983.
- [36] A. Ahrari, S. Elsayed, R. Sarker, D. Essam, and C. A. Coello Coello. **Weighted pointwise prediction method for dynamic multiobjective optimization**. *Inf. Sci.*, 546:349–367, 2021.
- [37] R. K. Ahuja, T. Magnanti, and J. B. Orlin. *Network Flows: Theory, Algorithms and Applications*. Prentice-Hall, 1993.
- [38] R. K. Ahuja, O. Ergun, and A. P. Punnen. **A Survey of Very Large-scale Neighborhood Search Techniques**. *Discrete Applied Mathematics*, 123(1–3):75–102, 2002.
- [39] U. Aickelin, E. K. Burke, and J. Li. **Improved Squeaky Wheel Optimisation for Driver Scheduling**. In T. P. Runarsson, H.-G. Beyer, E. K. Burke, J.-J. Merelo, D. Whitley, and X. Yao, editors, *Parallel Problem Solving from Nature – PPSN IX*, volume 4193 of *LNCs*, pages 182–191. Springer, Heidelberg, Germany, 2006.
- [40] R. M. Aiex, M. G. C. Resende, and C. C. Ribeiro. **TTT plots: a perl program to create time-to-target plots**. *Optimization Letters*, 1(4):355–366, Oct. 2006. doi:[10.1007/s11590-006-0031-4](https://doi.org/10.1007/s11590-006-0031-4).
Keywords: ECDF, runtime distribution.
- [41] S. Aine, R. Kumar, and P. P. Chakrabarti. **Adaptive parameter control of evolutionary algorithms to improve quality-time trade-off**. *Applied Soft Computing*, 9(2):527–540, 2009. doi:[10.1016/j.asoc.2008.07.001](https://doi.org/10.1016/j.asoc.2008.07.001).
Keywords: anytime.
- [42] H. Aissi and B. Roy. **Robustness in Multi-criteria Decision Aiding**. In M. Ehrgott, J. R. Figueira, and S. Greco, editors, *Trends in Multiple Criteria Decision Analysis*, volume 142 of *International Series in Operations Research & Management Science*, chapter 4, pages 87–121. Springer, US, 2010.

- [43] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. **Optuna: A Next-generation Hyperparameter Optimization Framework**. In Teredesai et al., editors, *25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631. ACM Press, New York, NY, July 2019. doi:[10.1145/3292500.3330701](https://doi.org/10.1145/3292500.3330701).
- [44] S. M. Aktürk, A. Atamtürk, and S. Gürel. **A Strong Conic Quadratic Reformulation for Machine-Job Assignment with Controllable Processing Times**. Research Report BCOL.07.01, University of California-Berkeley, 2007.
- [45] N. Al Moubayed, A. Petrovski, and J. McCall. **D^2 MOPSO: MOPSO based on decomposition and dominance with archiving using crowding distance in objective and solution spaces**. *Evol. Comput.*, 22(1):47–77, 2014.
- [46] I. Alaya, C. Solnon, and K. Ghédira. **Ant algorithm for the multi-dimensional knapsack problem**. In B. Filipič and J. Šilc, editors, *International Conference on Bioinspired Optimization Methods and their Applications (BIOMA 2004)*, pages 63–72, 2004. URL <https://books.google.be/books?id=0ZLsAAAACAAJ>.
- [47] I. Alaya, C. Solnon, and K. Ghédira. **Ant Colony Optimization for Multi-Objective Optimization Problems**. In *19th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2007)*, volume 1, pages 450–457. IEEE Computer Society Press, Los Alamitos, CA, 2007.
- [48] E. Alba and F. Chicano. **ACOhg: dealing with huge graphs**. In D. Thierens et al., editors, *GECCO*, pages 10–17. ACM Press, New York, NY, 2007. doi:[10.1145/1276958.1276961](https://doi.org/10.1145/1276958.1276961).
- [49] E. Alba, F. Chicano, and G. J. Luque, editors. *Smart Cities: First International Conference, Smart-CT 2016, Málaga, Spain, June 15-17, 2016, Proceedings*. LNCS. Springer, Cham, Switzerland, 2016.
- [50] S. Albers. **Online Algorithms: A Survey**. *Mathematical Programming*, 97(1):3–26, 2003.
- [51] A. Albrecht and K. Steinhöfel, editors. *Second International Symposium, SAGA 2003, Hatfield, UK, September 22-23, 2003, Proceedings*, volume 2827 of LNCS, 2003. Springer Verlag. doi:[10.1007/b13596](https://doi.org/10.1007/b13596).
- [52] A. A. Albrecht, P. C. R. Lane, and K. Steinhöfel. **Analysis of Local Search Landscapes for k-SAT Instances**. *Mathematics in Computer Science*, 3(4):465–488, 2010. doi:[10.1007/s11786-010-0040-7](https://doi.org/10.1007/s11786-010-0040-7).
- [53] A. Aleti and I. Moser. **A systematic literature review of adaptive parameter control methods for evolutionary algorithms**. *ACM Comput. Surv.*, 49(3, Article 56):35, Oct. 2016. doi:[10.1145/2996355](https://doi.org/10.1145/2996355).
- [54] A. Alexandrescu. **Fast Deterministic Selection**. In C. S. Iliopoulos, S. P. Pissis, S. J. Puglisi, and R. Raman, editors, *16th International Symposium on Experimental Algorithms (SEA 2017)*, volume 75 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 24:1–24:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2017. doi:[10.4230/LIPIcs.SEA.2017.24](https://doi.org/10.4230/LIPIcs.SEA.2017.24). *Keywords*: Quickselect, Median of Medians, Median of Ninthers.
- [55] V. Alexandrov, M. Lees, V. Krzhizhanovskaya, J. Dongarra, and P. M. A. Sloot, editors. *2013 International Conference on Computational Science*, volume 18 of *Procedia Computer Science*, 2013. Elsevier.
- [56] P. Alfaro-Fernández, R. Ruiz, F. Pagnozzi, and T. Stützle. **Automatic Algorithm Design for Hybrid Flowshop Scheduling Problems**. *Eur. J. Oper. Res.*, 282(3):835–845, 2020. doi:[10.1016/j.ejor.2019.10.004](https://doi.org/10.1016/j.ejor.2019.10.004). *Keywords*: Scheduling, Hybrid flowshop, Automatic algorithm configuration, Automatic Algorithm Design.

- [57] A. Ali and M. Meilă. **Experiments with Kemeny ranking: What Works When?** *Mathematical Social Science*, 64(1):28–40, July 2012. doi:10.1016/j.mathsocsci.2011.08.008.
Keywords: Borda ranking, Kemeny ranking.
Annotation: Computational Foundations of Social Choice.
- [58] M. Alissa, K. Sim, and E. Hart. **Algorithm Selection Using Deep Learning without Feature Extraction.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 198–206. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321845.
- [59] A. Allahverdi and H. Aydilek. **Algorithms for no-wait flowshops with total completion time subject to makespan.** *International Journal of Advanced Manufacturing Technology*, pages 1–15, 2013.
- [60] J. Allen, A. Moussa, and X. Liu. **Human-in-the-Loop Learning of Qualitative Preference Models.** In R. Barták and K. W. Brawner, editors, *Proceedings of the Thirty-Second International Florida Artificial Intelligence Research Society Conference*, pages 108–111. AAAI Press, 2019. doi:10.48550/arXiv.1909.09064.
- [61] S. Allen, E. K. Burke, M. R. Hyde, and G. Kendall. **Evolving reusable 3d packing heuristics with genetic programming.** In F. Rothlauf, editor, *GECCO*, pages 931–938. ACM Press, New York, NY, 2009. doi:10.1145/1569901.1570029.
Keywords: hyper-heuristic.
- [62] R. Allmendinger. *Tuning Evolutionary Search for Closed-Loop Optimization.* PhD thesis, The University of Manchester, UK, Jan. 2012. URL <https://www.escholar.manchester.ac.uk/uk-ac-man-scw:156551>.
Annotation: Supervised by Joshua D. Knowles.
- [63] R. Allmendinger and J. D. Knowles. **Evolutionary Optimization on Problems Subject to Changes of Variables.** In R. Schaefer, C. Cotta, J. Kołodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 151–160. Springer, Heidelberg, Germany, 2010.
- [64] R. Allmendinger and J. D. Knowles. **Evolutionary Search in Lethal Environments.** In *International Conference on Evolutionary Computation Theory and Applications*, pages 63–72. SciTePress, 2011. doi:10.5220/0003673000630072.
- [65] R. Allmendinger and J. D. Knowles. **Policy Learning in Resource-Constrained Optimization.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 1971–1979. ACM Press, New York, NY, 2011. ISBN 978-1-4503-0557-0. doi:10.1145/2001576.2001841.
- [66] R. Allmendinger and J. D. Knowles. **On Handling Ephemeral Resource Constraints in Evolutionary Search.** *Evol. Comput.*, 21(3):497–531, Sept. 2013. ISSN 1063-6560, 1530-9304. doi:10.1162/EVCQ_a_00097.
- [67] R. Allmendinger, A. Jaskiewicz, A. Liefoghe, and C. Tammer. **What if we increase the number of objectives? Theoretical and empirical implications for many-objective combinatorial optimization.** *Comput. Oper. Res.*, 145:105857, 2022. doi:10.1016/j.cor.2022.105857.
- [68] C. Almeder. **A hybrid optimization approach for multi-level capacitated lot-sizing problems.** *Eur. J. Oper. Res.*, 200(2):599–606, 2010. doi:10.1016/j.ejor.2009.01.019.
Keywords: Ant colony optimization, Manufacturing, Material requirements planning, Mixed-integer programming.
- [69] F. Almeida et al., editors. *Proceedings of HM 2006 – 3rd International Workshop on Hybrid Metaheuristics*, volume 4030 of *LNCS*. Springer, Heidelberg, Germany, 2006.

- [70] A. Alsheddy and E. Tsang. **Guided Pareto local search and its application to the 0/1 multi-objective knapsack problems.** In M. Caserta and S. Voß, editors, *MIC 2009*, Hamburg, Germany, 2010. University of Hamburg.
- [71] S. Alupoeai and S. Katkoori. **Ant Colony System Application to Marcocell Overlap Removal.** *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 12(10): 1118–1122, 2004.
- [72] S. Amani, M. Alizadeh, and C. Thrampoulidis. **Linear Stochastic Bandits Under Safety Constraints.** In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NeurIPS 32)*, pages 9256–9266, 2019.
- [73] A. R. S. Amaral. **The Corridor Allocation Problem.** *Comput. Oper. Res.*, 39(12):3325–3330, 2012. doi:10.1016/j.cor.2012.04.016.
Keywords: Facility layout, Double row layout, Integer programming.
- [74] C. Amir, A. Badr, and I. Farag. **A Fuzzy Logic Controller for Ant Algorithms.** *Computing and Information Systems*, 11(2):26–34, 2007.
- [75] J. H. Andersen and R. S. Powell. **The Use of Continuous Decision Variables in an Optimising Fixed Speed Pump Scheduling Algorithm.** In R. S. Powell and K. S. Hindi, editors, *Computing and Control for the Water Industry*, pages 119–128. Research Studies Press Ltd., 1999.
- [76] K. Andersen, R. V. V. Vidal, and V. B. Iversen. **Design of a Teleprocessing Communication Network Using Simulated Annealing.** In R. V. V. Vidal, editor, *Applied Simulated Annealing*, pages 201–215. Springer, 1993.
- [77] K. A. Andersen, K. Jörnsten, and M. Lind. **On bicriterion minimal spanning trees: An approximation.** *Comput. Oper. Res.*, 23(12):1171–1182, 1996.
- [78] H. Andersson, K. Fagerholt, and K. Hobbesland. **Integrated maritime fleet deployment and speed optimization: Case study from RoRo shipping.** *Comput. Oper. Res.*, 55:233–240, Mar. 2015. doi:10.1016/j.cor.2014.03.017.
- [79] C. Andrieu, N. de Freitas, A. Doucet, and M. I. Jordan. **An Introduction to MCMC for Machine Learning.** *Mach. Learn.*, 50(1-2):5–43, 2003.
- [80] Y. P. Aneja and K. P. K. Nair. **Bicriteria Transportation Problem.** *Management Science*, 25(1):73–78, 1979.
- [81] E. Angel, E. Bampis, and L. Gourvés. **Approximating the Pareto curve with local search for the bicriteria TSP(1,2) problem.** *Theor. Comput. Sci.*, 310(1-3):135–146, 2004. doi:10.1016/S0304-3975(03)00376-1.
Keywords: Archiving, Local search, Multicriteria TSP, Approximation algorithms.
- [82] P. Angelov et al., editors. *Evolving and Autonomous Learning Systems (EALS), 2014 IEEE Symposium on*, 2014. IEEE.
- [83] D. Anghinolfi, A. Boccalatte, M. Paolucci, and C. Vecchiola. **Performance Evaluation of an Adaptive Ant Colony Optimization Applied to Single Machine Scheduling.** In X. Li et al., editors, *Simulated Evolution and Learning, 7th International Conference, SEAL 2008*, volume 5361 of *LNCS*, pages 411–420. Springer, Heidelberg, Germany, 2008.
- [84] D. Angus. **Population-Based Ant Colony Optimisation for Multi-objective Function Optimisation.** In M. Randall, H. A. Abbass, and J. Wiles, editors, *Progress in Artificial Life (ACAL)*, volume 4828 of *LNCS*, pages 232–244. Springer, Heidelberg, Germany, 2007. doi:10.1007/978-3-540-76931-6_21.

- [85] D. Angus and C. Woodward. **Multiple Objective Ant Colony Optimisation**. *Swarm Intelligence*, 3(1):69–85, 2009. doi:[10.1007/s11721-008-0022-4](https://doi.org/10.1007/s11721-008-0022-4).
- [86] M. F. Anjos and M. V. C. Vieira. **Mathematical optimization approaches for facility layout problems: The state-of-the-art and future research directions**. *Eur. J. Oper. Res.*, 261(1):1–16, 2017.
- [87] J. Ansel, S. Kamil, K. Veeramachaneni, J. Ragan-Kelley, J. Bosboom, U.-M. O’Reilly, and S. Amarasinghe. **OpenTuner: An extensible framework for program autotuning**. In *Proceedings of the 23rd International Conference on Parallel Architectures and Compilation*, pages 303–315, New York, NY, 2014. ACM Press. doi:[10.1145/2628071.2628092](https://doi.org/10.1145/2628071.2628092).
- [88] C. Ansótegui, M. Sellmann, and K. Tierney. **A Gender-Based Genetic Algorithm for the Automatic Configuration of Algorithms**. In I. P. Gent, editor, *Principles and Practice of Constraint Programming, CP 2009*, volume 5732 of *LNCS*, pages 142–157. Springer, Heidelberg, Germany, 2009. doi:[10.1007/978-3-642-04244-7_14](https://doi.org/10.1007/978-3-642-04244-7_14).
Keywords: GGA.
- [89] C. Ansótegui, Y. Malitsky, and M. Sellmann. **MaxSAT by Improved Instance-Specific Algorithm Configuration**. In D. Stracuzzi et al., editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2594–2600. AAAI Press, 2014.
- [90] C. Ansótegui, Y. Malitsky, H. Samulowitz, M. Sellmann, and K. Tierney. **Model-Based Genetic Algorithms for Algorithm Configuration**. In Q. Yang and M. Wooldridge, editors, *Proc. of IJCAI-15*, pages 733–739. IJCAI/AAAI Press, Menlo Park, CA, 2015.
Keywords: GGA++.
- [91] C. Ansótegui, J. Gabàs, Y. Malitsky, and M. Sellmann. **MaxSAT by Improved Instance-Specific Algorithm Configuration**. *Artif. Intell.*, 235:26–39, June 2016. doi:[10.1016/j.artint.2015.12.006](https://doi.org/10.1016/j.artint.2015.12.006).
Keywords: ISAC++.
Annotation: Proposed ISAC++.
- [92] C. Ansótegui, M. Sellmann, and K. Tierney. **GGA: Gender-based Genetic Algorithm Configurator**. https://bitbucket.org/gga_ac/, 2017. Version visited last on July 2017.
- [93] K. Anstreicher, N. Brixius, J.-P. Goux, and J. Linderoth. **Solving large quadratic assignment problems on computational grids**. *Mathematical Programming Series B*, 91(3):563–588, Feb. 2002. doi:[10.1007/s101070100255](https://doi.org/10.1007/s101070100255).
- [94] Apache Software Foundation. **Hadoop**, 2008. URL <https://hadoop.apache.org>.
- [95] Apache Software Foundation. **Spark**, 2012. URL <https://spark.apache.org>.
- [96] J. S. Appleby, D. V. Blake, and E. A. Newman. **Techniques for producing school timetables on a computer and their application to other scheduling problems**. *The Computer Journal*, 3(4):237–245, 1961. doi:[10.1093/comjnl/3.4.237](https://doi.org/10.1093/comjnl/3.4.237).
- [97] D. Applegate and W. J. Cook. **A Computational Study of the Job-Shop Scheduling Problem**. *ORSA Journal on Computing*, 3(2):149–156, 1991.
- [98] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. **Finding Cuts in the TSP**. Technical Report 95–05, DIMACS Center, Rutgers University, Piscataway, NJ, USA, Mar. 1995.
- [99] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. **On the Solution of Traveling Salesman Problems**. *Documenta Mathematica*, Extra Volume ICM III:645–656, 1998.

- [100] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. **Finding Tours in the TSP**. Technical Report 99885, Forschungsinstitut für Diskrete Mathematik, University of Bonn, Germany, 1999.
- [101] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. **Implementing the Dantzig-Fulkerson-Johnson Algorithm for Large Traveling Salesman Problems**. *Mathematical Programming Series B*, 97(1–2):91–153, 2003. doi:10.1007/s10107-003-0440-4.
- [102] D. Applegate, W. J. Cook, and A. Rohe. **Chained Lin-Kernighan for Large Traveling Salesman Problems**. *INFORMS Journal on Computing*, 15(1):82–92, 2003. doi:10.1287/ijoc.15.1.82.15157.
- [103] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, Princeton, NJ, 2006.
- [104] D. Applegate, R. E. Bixby, V. Chvátal, W. J. Cook, D. Espinoza, M. Goycoolea, and K. Helsgaun. **Certification of an Optimal TSP Tour Through 85,900 Cities**. *Operations Research Letters*, 37(1):11–15, 2009.
- [105] D. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. **Concorde TSP Solver**. <http://www.math.uwaterloo.ca/tsp/concorde.html>, 2014. Version visited last on 15 April 2014.
- [106] J. April, F. Glover, J. P. Kelly, and M. Laguna. **Simulation-based optimization: Practical introduction to simulation optimization**. In S. E. Chick, P. J. Sanchez, D. M. Ferrin, and D. J. Morrice, editors, *Proceedings of the 35th Winter Simulation Conference: Driving Innovation*, volume 1, pages 71–78, New York, NY, Dec. 2003. ACM Press. doi:10.1109/WSC.2003.1261410.
- [107] H. R. Arabnia, editor. *Proceedings of the International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA'98)*. CSREA Press, 1998.
- [108] H. R. Arabnia and R. Joshua, editors. *Proceedings of the 2005 International Conference on Artificial Intelligence, ICAI 2005*. CSREA Press, 2005. ISBN 1-932415-66-1.
- [109] C. R. Aragon and R. G. Seidel. **Randomized search trees**. In *30th Annual Symposium on Foundations of Computer Science*, pages 540–545, 1989. doi:10.1109/SFCS.1989.63531. *Keywords*: Binary search trees, Treap, Dynamic programming.
- [110] C. Aranha, C. L. Camacho-Villalón, F. Campelo, M. Dorigo, R. Ruiz, M. Sevaux, K. Sörensen, and T. Stützle. **Metaphor-based Metaheuristics, a Call for Action: the Elephant in the Room**. *Swarm Intelligence*, 16(1):1–6, 2022. doi:10.1007/s11721-021-00202-9.
- [111] I. Araya and M.-C. Riff. **A beam search approach to the container loading problem**. *Comput. Oper. Res.*, 43:100–107, Mar. 2014. doi:10.1016/j.cor.2013.09.003.
- [112] I. Araya, K. Guerrero, and E. Nuñez. **VCS: A new heuristic function for selecting boxes in the single container loading problem**. *Comput. Oper. Res.*, 82:27–35, June 2017. doi:10.1016/j.cor.2017.01.002.
- [113] C. Archetti, M. Savelsbergh, and M. G. Speranza. **The Vehicle Routing Problem with Occasional Drivers**. *Eur. J. Oper. Res.*, 254(2):472–480, 2016. doi:10.1016/j.ejor.2016.03.049.
- [114] D. Ariely, G. Loewenstein, and D. Prelec. **Coherent Arbitrariness: Stable Demand Curves Without Stable Preferences**. *The Quarterly Journal of Economics*, 118(1):73–106, 2003. doi:10.1162/00335530360535153. *Keywords*: anchoring.
- [115] F. Arnold and K. Sörensen. **Knowledge-guided local search for the vehicle routing problem**. *Comput. Oper. Res.*, 105:32–46, 2019. doi:10.1016/j.cor.2019.01.002.

- [116] F. Arnold and K. Sörensen. **What makes a VRP solution good? The generation of problem-specific knowledge for heuristics.** *Comput. Oper. Res.*, 106:280–288, 2019. doi:[10.1016/j.cor.2018.02.007](https://doi.org/10.1016/j.cor.2018.02.007).
- [117] F. Arnold, I. Santana, K. Sörensen, and T. Vidal. **PILS: Exploring high-order neighborhoods by pattern mining and injection.** *Arxiv preprint arXiv:1912.11462 [cs.AI]*, 2019. doi:[10.48550/arXiv.1912.11462](https://doi.org/10.48550/arXiv.1912.11462).
- [118] S. Arora and B. Barak. *Computational complexity: a modern approach*. Cambridge University Press, 2009.
- [119] M. A. Arostegui Jr, S. N. Kadipasaoglu, and B. M. Khumawala. **An empirical comparison of tabu search, simulated annealing, and genetic algorithms for facilities location problems.** *International Journal of Production Economics*, 103(2):742–754, 2006.
- [120] A. B. Arrieta, N. Díaz-Rodríguez, J. D. Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera. **Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI.** *Information Fusion*, 58:82–115, June 2020. doi:[10.1016/j.inffus.2019.12.012](https://doi.org/10.1016/j.inffus.2019.12.012).
- [121] J. E. C. Arroyo and V. A. Armentano. **A partial enumeration heuristic for multi-objective flowshop scheduling problems.** *J. Oper. Res. Soc.*, 55(9):1000–1007, 2004.
- [122] J. E. C. Arroyo and V. A. Armentano. **Genetic local search for multi-objective flowshop scheduling problems.** *Eur. J. Oper. Res.*, 167(3):717–738, 2005.
Keywords: Multicriteria Scheduling.
- [123] J. E. C. Arroyo and J. Y.-T. Leung. **An Effective Iterated Greedy Algorithm for Scheduling Unrelated Parallel Batch Machines with Non-identical Capacities and Unequal Ready Times.** *Computers & Industrial Engineering*, 105:84–100, 2017.
- [124] E. Arza, J. Ceberio, A. Pérez, and E. Irurozki. **Approaching the quadratic assignment problem with kernels of mallows models under the hamming distance.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:[10.1145/3319619.3321976](https://doi.org/10.1145/3319619.3321976).
Keywords: QAP, EDA, Mallows.
- [125] E. Arza, J. Ceberio, E. Irurozki, and A. Pérez. **Comparing Two Samples Through Stochastic Dominance: A Graphical Approach.** *Journal of Computational and Graphical Statistics*, pages 1–38, June 2022. doi:[10.1080/10618600.2022.2084405](https://doi.org/10.1080/10618600.2022.2084405).
- [126] A. Asadpour and H. Nazerzadeh. **Maximizing Stochastic Monotone Submodular Functions.** *Management Science*, 62(8):2374–2391, 2016. doi:[10.1287/mnsc.2015.2254](https://doi.org/10.1287/mnsc.2015.2254).
- [127] Y. Asahiro, K. Iwama, and E. Miyano. **Random Generation of Test Instances with Controlled Attributes.** In D. S. Johnson and M. A. Trick, editors, *Cliques, Coloring, and Satisfiability: Second DIMACS Implementation Challenge*, volume 26 of *DIMACS Series on Discrete Mathematics and Theoretical Computer Science*, pages 377–393. American Mathematical Society, Providence, RI, 1996.
- [128] N. Ascheuer. *Hamiltonian Path Problems in the On-line Optimization of Flexible Manufacturing Systems*. PhD thesis, Technische Universität Berlin, Berlin, Germany, 1995.
- [129] N. Ascheuer, M. Fischetti, and M. Grötschel. **Solving asymmetric travelling salesman problem with time windows by branch-and-cut.** *Mathematical Programming*, 90:475–506, 2001.

- [130] J. M. Assael, Z. Wang, and N. de Freitas. **Heteroscedastic Treed Bayesian Optimisation**. *Arxiv preprint arXiv:1410.7172*, 2014. doi:10.48550/arXiv.1410.7172.
Keywords: Treed-GP.
- [131] A. Atamtürk. **On the facets of the mixed-integer knapsack polyhedron**. *Mathematical Programming*, 98(1):145–175, 2003. doi:10.1007/s10107-003-0400-z.
- [132] R. Atkinson, J. E. van Zyl, G. A. Walters, and D. A. Savic. **Genetic algorithm optimisation of level-controlled pumping station operation**. In *Water network modelling for optimal design and management*, pages 79–90. Centre for Water Systems, Exeter, UK, 2000.
- [133] C. Audet and D. Orban. **Finding Optimal Algorithmic Parameters Using Derivative-Free Optimization**. *SIAM J. Optim.*, 17(3):642–664, 2006. doi:10.1137/040620886.
Keywords: mesh adaptive direct search; pattern search.
- [134] C. Audet, C.-K. Dang, and D. Orban. **Algorithmic Parameter Optimization of the DFO Method with the OPAL Framework**. In K. Naono, K. Teranishi, J. Cavazos, and R. Suda, editors, *Software Automatic Tuning: From Concepts to State-of-the-Art Results*, pages 255–274. Springer, 2010.
- [135] C. Audet, C.-K. Dang, and D. Orban. **Optimization of Algorithms with OPAL**. *Mathematical Programming Computation*, 6(3):233–254, 2014.
- [136] C. Audet, J. Bigeon, D. Cartier, S. Le Digabel, and L. Salomon. **Performance indicators in multiobjective optimization**. *Eur. J. Oper. Res.*, 292(2):397–422, 2021. doi:10.1016/j.ejor.2020.11.016.
- [137] P. Audze and V. Eglājs. **New approach to the design of multifactor experiments**. *Problems of Dynamics and Strengths*, 35:104–107, 1977. (in Russian).
- [138] P. Auer. **Using Confidence Bounds for Exploitation-Exploration Trade-offs**. *J. Mach. Learn. Res.*, 3:397–422, Nov. 2002.
- [139] P. Auer, N. Cesa-Bianchi, and P. Fischer. **Finite-time analysis of the multiarmed bandit problem**. *Mach. Learn.*, 47(2-3):235–256, 2002.
- [140] A. Auger and B. Doerr, editors. *Theory of Randomized Search Heuristics: Foundations and Recent Developments*, volume 1 of *Series on Theoretical Computer Science*. World Scientific Publishing Co., Singapore, 2011.
- [141] A. Auger and N. Hansen. **A restart CMA evolution strategy with increasing population size**. In *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, pages 1769–1776, Piscataway, NJ, Sept. 2005. IEEE Press. doi:10.1109/CEC.2005.1554902.
Keywords: IPOP-CMA-ES.
- [142] A. Auger and N. Hansen. **Performance evaluation of an advanced local search evolutionary algorithm**. In *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, pages 1777–1784, Piscataway, NJ, Sept. 2005. IEEE Press.
Keywords: LR-CMAES.
- [143] A. Auger, J. Bader, D. Brockhoff, and E. Zitzler. **Articulating User Preferences in Many-Objective Problems by Sampling the Weighted Hypervolume**. In F. Rothlauf, editor, *GECCO*, pages 555–562. ACM Press, New York, NY, 2009. doi:10.1145/1569901.1569979.

- [144] A. Auger, J. Bader, D. Brockhoff, and E. Zitzler. **Investigating and Exploiting the Bias of the Weighted Hypervolume to Articulate User Preferences**. In F. Rothlauf, editor, *GECCO*, pages 563–570. ACM Press, New York, NY, 2009.
- [145] A. Auger, J. Bader, D. Brockhoff, and E. Zitzler. **Theory of the hypervolume indicator: optimal μ -distributions and the choice of the reference point**. In F. Rothlauf, editor, *GECCO*, pages 87–102. ACM Press, New York, NY, 2009.
- [146] A. Auger, J. Bader, D. Brockhoff, and E. Zitzler. **Hypervolume-based multiobjective optimization: Theoretical foundations and practical implications**. *Theor. Comput. Sci.*, 425:75–103, 2012. doi:[10.1016/j.tcs.2011.03.012](https://doi.org/10.1016/j.tcs.2011.03.012).
- [147] A. Auger, D. Brockhoff, M. López-Ibáñez, K. Miettinen, B. Naujoks, and G. Rudolph. **Which questions should be asked to find the most appropriate method for decision making and problem solving? (Working Group “Algorithm Design Methods”)**. In S. Greco, J. D. Knowles, K. Miettinen, and E. Zitzler, editors, *Learning in Multiobjective Optimization (Dagstuhl Seminar 12041)*, volume 2(1) of *Dagstuhl Reports*, pages 92–93. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2012. doi:[10.4230/DagRep.2.1.50](https://doi.org/10.4230/DagRep.2.1.50).
- [148] A. Auger, D. Brockhoff, N. Hansen, D. Tutar, T. Tušar, and T. Wagner. **GECCO Workshop on Real-Parameter Black-Box Optimization Benchmarking (BBOB 2016): Focus on multi-objective problems**. <https://numbbo.github.io/workshops/BBOB-2016/>, 2016.
- [149] A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors. *Parallel Problem Solving from Nature - PPSN XV 15th International Conference, Coimbra, Portugal, September 8-12, 2018, Proceedings, Part I*, volume 11101 of *LNCS*. Springer, Cham, Switzerland, 2018.
- [150] A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors. *Parallel Problem Solving from Nature - PPSN XV 15th International Conference, Coimbra, Portugal, September 8-12, 2018, Proceedings, Part II*, volume 11102 of *LNCS*. Springer, Cham, Switzerland, 2018.
- [151] M. Avci and S. Topaloglu. **A Multi-start Iterated Local Search Algorithm for the Generalized Quadratic Multiple Knapsack Problem**. *Comput. Oper. Res.*, 83:54–65, 2017.
- [152] A. Avramescu, R. Allmendinger, and M. López-Ibáñez. **Managing Manufacturing and Delivery of Personalised Medicine: Current and Future Models**. *Arxiv preprint arXiv:2105.12699 [econ.GN]*, 2021. URL <https://arxiv.org/abs/2105.12699>.
- [153] A. Avramescu, R. Allmendinger, and M. López-Ibáñez. **A Multi-Objective Multi-Type Facility Location Problem for the Delivery of Personalised Medicine**. In P. Castillo and J. L. Jiménez Laredo, editors, *Applications of Evolutionary Computation*, volume 12694 of *LNCS*, pages 388–403. Springer, Cham, Switzerland, 2021. doi:[10.1007/978-3-030-72699-7_25](https://doi.org/10.1007/978-3-030-72699-7_25). Supplementary material: <https://doi.org/10.5281/zenodo.4495162>.
Keywords: Personalised medicine, Biopharmaceuticals Supply chain, Facility location-allocation, Evolutionary multi-objective optimisation.
- [154] D. Aydın. **Composite artificial bee colony algorithms: From component-based analysis to high-performing algorithms**. *Applied Soft Computing*, 32:266–285, 2015. doi:[10.1016/j.asoc.2015.03.051](https://doi.org/10.1016/j.asoc.2015.03.051).
Keywords: irace.
- [155] D. Aydın, G. Yavuz, S. Özyön, C. Yasar, and T. Stützle. **Artificial Bee Colony Framework to Non-convex Economic Dispatch Problem with Valve Point Effects: A Case Study**. In P. A. N. Bosman, editor, *GECCO Companion*, pages 1311–1318. ACM Press, New York, NY, 2017.

- [156] D. Aydın, G. Yavuz, and T. Stützle. **ABC-X: A Generalized, Automatically Configurable Artificial Bee Colony Framework**. *Swarm Intelligence*, 11(1):1–38, 2017.
- [157] M. Ayodele. **Penalty Weights in QUBO Formulations: Permutation Problems**. In L. Pérez Cáceres and S. Verel, editors, *Proceedings of EvoCOP 2022 – 22nd European Conference on Evolutionary Computation in Combinatorial Optimization*, LNCS, pages 159–174. Springer, Cham, Switzerland, 2022.
- [158] M. Ayodele, R. Allmendinger, M. López-Ibáñez, and M. Parizy. **Multi-Objective QUBO Solver: Bi-Objective Quadratic Assignment Problem**. In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022*, pages 467–475. ACM Press, New York, NY, 2022. doi:10.1145/3512290.3528698.
Keywords: digital annealer, multi-objective, bi-objective QAP, QUBO.
- [159] M. Ayodele, R. Allmendinger, M. López-Ibáñez, and M. Parizy. **A Study of Scalarisation Techniques for Multi-objective QUBO Solving**. In O. Grothe, S. Nickel, S. Rebennack, and O. Stein, editors, *Operations Research Proceedings 2022, OR 2022*, Lecture Notes in Operations Research, pages 393–399. Springer, Cham, Switzerland, 2022. doi:10.1007/978-3-031-24907-5_47.
- [160] M. Ayodele, R. Allmendinger, M. López-Ibáñez, and M. Parizy. **A Study of Scalarisation Techniques for Multi-Objective QUBO Solving**. *Arxiv preprint arXiv:2210.11321*, 2022. doi:10.48550/arXiv.2210.11321.
- [161] M. Ayodele, R. Allmendinger, M. López-Ibáñez, M. Parizy, and A. Liefoghe. **Applying Ising Machines to Multi-Objective QUBOs**. In S. Silva and L. Paquete, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2023*, pages 2166–2174. ACM Press, New York, NY, 2023. doi:10.1145/3583133.3596312.
Keywords: digital annealer, multi-objective, bi-objective QAP, QUBO.
Annotation: ISBN: 979-8-4007-0120-7.
- [162] M. Aziz and M.-H. Tayarani-N. **An adaptive memetic Particle Swarm Optimization algorithm for finding large-scale Latin hypercube designs**. *Engineering Applications of Artificial Intelligence*, 36:222–237, 2014. doi:10.1016/j.engappai.2014.07.021.
Keywords: F-race.
- [163] A. Aziz-Alaoui, C. Doerr, and J. Dreö. **Towards Large Scale Automated Algorithm Design by Integrating Modular Benchmarking Frameworks**. In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1365–1374. ACM Press, New York, NY, 2021. doi:10.1145/3449726.3463155.
- [164] D. Babić. **Spear theorem prover**. <https://www.domagoj-babic.com/index.php/ResearchProjects/Spear>, 2008.
- [165] D. Babić and A. J. Hu. **Structural Abstraction of Software Verification Conditions**. In *Computer Aided Verification: 19th International Conference, CAV 2007*, pages 366–378, 2007.
Annotation: Spear-svw instances, http://www.cs.ubc.ca/labs/beta/Projects/ParamILS/benchmark_instances/SpearSWV/SWV-scrambled-first302.tar.gz, http://www.cs.ubc.ca/labs/beta/Projects/ParamILS/benchmark_instances/SpearSWV/SWV-scrambled-last302.tar.gz.
- [166] D. Babić and F. Hutter. **Spear Theorem Prover**. In *SAT’08: Proceedings of the SAT 2008 Race*, 2008. Supplementary material: <https://www.domagoj-babic.com/index.php/ResearchProjects/Spear>.
Annotation: Unreviewed paper.
- [167] F. Bacchus and T. Walsh, editors. *International Conference on Theory and Applications of Satisfiability Testing*, volume 3569, 2005.

- [168] F. Bach and D. Blei, editors. *Proceedings of the 32nd International Conference on Machine Learning, ICML 2015, Lille, France, 7-9 July 2015*, volume 37, 2015. PMLR.
- [169] A. Bachem, B. Steckemetz, and M. Wottawa. **An efficient parallel cluster-heuristic for large Traveling Salesman Problems**. Technical Report 94-150, University of Koln, Germany, 1994.
Keywords: Genetic Edge Recombination (ERX).
- [170] F. Bachoc, C. Helbert, and V. Picheny. **Gaussian process optimization with failures: Classification and convergence proof**. *J. Glob. Optim.*, 2020. doi:10.1007/s10898-020-00920-0.
Keywords: crashed simulation; latent gaussian process; automotive fan design; industrial application; GP classification; Expected Feasible Improvement with Gaussian Process Classification with signs; EFI GPC sign.
- [171] T. Bäck. *Evolutionary algorithms in theory and practice: evolution strategies, evolutionary programming, genetic algorithms*. Oxford University Press, 1996.
- [172] T. Bäck, editor. *Proceedings of the 7th International Conference on Genetic Algorithms (ICGA), East Lansing, MI, USA, July 19-23, 1997*. Morgan Kaufmann Publishers, San Francisco, CA, 1997.
- [173] T. Bäck, T. Fukuda, and Z. Michalewicz, editors. *Proceedings of the 1996 IEEE International Conference on Evolutionary Computation (ICEC'96)*. IEEE Press, Piscataway, NJ, 1996.
- [174] T. Bäck, D. B. Fogel, and Z. Michalewicz. *Handbook of evolutionary computation*. IOP Publishing, 1997.
- [175] T. Bäck, Z. Michalewicz, and X. Yao, editors. *Proceedings of the 1997 IEEE International Conference on Evolutionary Computation (ICEC'97)*. IEEE Press, Piscataway, NJ, 1997.
- [176] T. Bäck, M. Preuss, A. Deutz, H. Wang, C. Doerr, M. T. M. Emmerich, and H. Trautmann, editors. *Parallel Problem Solving from Nature - PPSN XVI 16th International Conference, Leiden, The Netherlands, September 5-9, 2020, Proceedings*, volume 12269 of LNCS. Springer, Cham, Switzerland, 2020.
- [177] J. Bader and E. Zitzler. **HypE: An Algorithm for Fast Hypervolume-Based Many-Objective Optimization**. *Evol. Comput.*, 19(1):45–76, 2011. doi:10.1162/EVCO_a_00009.
- [178] H. Baharmand, T. Comes, and M. Lauras. **Bi-objective multi-layer location-allocation model for the immediate aftermath of sudden-onset disasters**. *Transportation Research Part E: Logistics and Transportation Review*, 127:86–110, 2019. doi:10.1016/j.tre.2019.05.002.
- [179] E. K. Baker. **An Exact Algorithm for the Time-Constrained Traveling Salesman Problem**. *Operations Research*, 31(5):938–945, 1983. doi:10.1287/opre.31.5.938.
Keywords: makespan optimization.
- [180] M. Baker. **Is there a reproducibility crisis?** *Nature*, 533:452–454, 2016.
- [181] P. Balaprakash, M. Birattari, T. Stützle, and M. Dorigo. **Incremental local search in ant colony optimization: Why it fails for the quadratic assignment problem**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of LNCS, pages 156–166. Springer, Heidelberg, Germany, 2006.

- [182] P. Balaprakash, M. Birattari, and T. Stützle. **Improvement Strategies for the F-Race Algorithm: Sampling Design and Iterative Refinement.** In T. Bartz-Beielstein, M. J. Blesa, C. Blum, B. Naujoks, A. Roli, G. Rudolph, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 4771 of *LNCS*, pages 108–122. Springer, Heidelberg, Germany, 2007. doi:10.1007/978-3-540-75514-2_9.
Keywords: Iterated Race.
- [183] P. Balaprakash, M. Birattari, T. Stützle, and M. Dorigo. **Adaptive Sampling Size and Importance Sampling in Estimation-based Local Search for the Probabilistic Traveling Salesman Problem.** *Eur. J. Oper. Res.*, 199(1):98–110, 2009.
- [184] P. Balaprakash, M. Birattari, T. Stützle, Z. Yuan, and M. Dorigo. **Estimation-based Ant Colony Optimization Algorithms for the Probabilistic Travelling Salesman Problem.** *Swarm Intelligence*, 3(3):223–242, 2009.
- [185] P. Balaprakash, M. Birattari, T. Stützle, and M. Dorigo. **Estimation-based Metaheuristics for the Probabilistic Travelling Salesman Problem.** *Comput. Oper. Res.*, 37(11):1939–1951, 2010. doi:10.1016/j.cor.2009.12.005.
- [186] P. Balaprakash, M. Birattari, T. Stützle, and M. Dorigo. **Estimation-based Metaheuristics for the Single Vehicle Routing Problem with Stochastic Demands and Customers.** *Computational Optimization and Applications*, 61(2):463–487, 2015. doi:10.1007/s10589-014-9719-z.
- [187] E. Balas and M. C. Carrera. **A Dynamic Subgradient-based Branch and Bound Procedure for Set Covering.** *Operations Research*, 44(6):875–890, 1996.
- [188] E. Balas and A. Ho. **Set Covering Algorithms Using Cutting Planes, Heuristics, and Subgradient Optimization: A Computational Study.** In M. W. Padberg, editor, *Combinatorial optimization*, volume 12 of *Mathematical Programming Studies*, pages 37–60. Springer, Berlin/Heidelberg, 1980. doi:10.1007/BFb0120886.
- [189] E. Balas and C. Martin. **Pivot and Complement—A Heuristic for 0–1 Programming.** *Management Science*, 26(1):86–96, 1980.
- [190] E. Balas and M. W. Padberg. **Set Partitioning: A Survey.** *SIAM Review*, 18:710–760, 1976.
- [191] E. Balas and N. Simonetti. **Linear Time Dynamic-Programming Algorithms for New Classes of Restricted TSPs: A Computational Study.** *INFORMS Journal on Computing*, 13(1):56–75, 2001. doi:10.1287/ijoc.13.1.56.9748.
Keywords: tsptw.
- [192] E. Balas and A. Vazacopoulos. **Guided Local Search with Shifting Bottleneck for Job Shop Scheduling.** *Management Science*, 44(2):262–275, 1998.
- [193] B. Balcik and B. M. Beamon. **Facility location in humanitarian relief.** *International Journal of Logistics*, 11(2):101–121, 2008.
- [194] S. C. Bankes. **Tools and techniques for developing policies for complex and uncertain systems.** *Proceedings of the National Academy of Sciences*, 99(suppl 3):7263–7266, 2002. doi:10.1073/pnas.092081399.
- [195] W. Banzhaf, J. M. Daida, A. E. Eiben, M. H. Garzon, V. Honavar, M. J. Jakiela, and R. E. Smith, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 1999, 13-17 July 1999, Orlando, Florida, USA.* Morgan Kaufmann Publishers, San Francisco, CA, 1999.

- [196] P. Baptiste and L. K. Hguny. **A branch and bound algorithm for the F/no_idle/C_{max}**. In *Proceedings of the international conference on industrial engineering and production management, IEPM'97*, pages 429–438, Lyon, 1997.
- [197] B. Barán and M. Schaerer. **A multiobjective ant colony system for vehicle routing problem with time windows**. In *Proceedings of the Twenty-first IASTED International Conference on Applied Informatics*, pages 97–102, Innsbruck, Austria, 2003.
- [198] C. Barba-González, V. Ojalehto, J. García-Nieto, A. J. Nebro, K. Miettinen, and J. F. Aldana-Montes. **Artificial Decision Maker Driven by PSO: An Approach for Testing Reference Point Based Interactive Methods**. In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11101 of *LNCS*, pages 274–285. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-319-99253-2_22.
Keywords: machine decision-maker.
- [199] E. Bareinboim and J. Pearl. **Transportability of causal effects: Completeness results**. In J. Hoffmann and B. Selman, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 698,704. AAAI Press, 2012.
- [200] E. Bareinboim and J. Pearl. **Causal inference and the data-fusion problem**. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016. doi:10.1073/pnas.1510507113.
- [201] C. Barnhart, E. L. Johnson, G. L. Nemhauser, M. W. P. Savelsbergh, and P. H. Vance. **Branch-and-price: Column generation for solving huge integer programs**. *Operations Research*, 46(3):316–329, 1998.
- [202] R. S. Barr, B. L. Golden, J. P. Kelly, M. G. C. Resende, and J. William R. Stewart. **Designing and Reporting on Computational Experiments with Heuristic Methods**. *J. Heuristics*, 1(1):9–32, 1995. doi:10.1007/BF02430363.
- [203] R. Barták and K. W. Brawner, editors. *Proceedings of the Thirty-Second International Florida Artificial Intelligence Research Society Conference, Sarasota, Florida, USA, May 19-22 2019*, 2019. AAAI Press.
- [204] E. Bartholomew and J. H. Kwakkel. **On considering robustness in the search phase of Robust Decision Making: A comparison of Many-Objective Robust Decision Making, multi-scenario Many-Objective Robust Decision Making, and Many Objective Robust Optimization**. *Environmental Modelling & Software*, 127:104699, 2020. doi:10.1016/j.envsoft.2020.104699.
- [205] P. L. Bartlett, F. C. N. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, editors. *Advances in Neural Information Processing Systems 25: 26th Annual Conference on Neural Information Processing Systems 2012*. Curran Associates, Red Hook, NY, 2012.
- [206] T. Bartz-Beielstein. *Experimental Research in Evolutionary Computation: The New Experimentalism*. Springer, Berlin, Germany, 2006.
Keywords: SPO.
- [207] T. Bartz-Beielstein. **How to Create Generalizable Results**. In J. Kacprzyk and W. Pedrycz, editors, *Springer Handbook of Computational Intelligence*, pages 1127–1142. Springer, Berlin/Heidelberg, 2015.
Keywords: Mixed-effects models, random-effects model, problem instance generation.
- [208] T. Bartz-Beielstein and S. Markon. **Tuning search algorithms for real-world applications: A regression tree based approach**. In *Proceedings of the 2004 Congress on Evolutionary Computation (CEC 2004)*, pages 1111–1118, Piscataway, NJ, Sept. 2004. IEEE Press.

- [209] T. Bartz-Beielstein and M. Preuss. **Considerations of budget allocation for sequential parameter optimization (SPO)**. In L. Paquete, M. Chiarandini, and D. Basso, editors, *Empirical Methods for the Analysis of Algorithms, Workshop EMAA 2006, Proceedings*, pages 35–40, Reykjavik, Iceland, 2006.
- [210] T. Bartz-Beielstein and M. Preuss. **Experimental Analysis of Optimization Algorithms: Tuning and Beyond**. In Y. Borenstein and A. Moraglio, editors, *Theory and Principled Methods for the Design of Metaheuristics*, Natural Computing Series, pages 205–245. Springer, Berlin/Heidelberg, 2014. doi:[10.1007/978-3-642-33206-7_10](https://doi.org/10.1007/978-3-642-33206-7_10).
- [211] T. Bartz-Beielstein and M. Zaefferer. **Model-based methods for continuous and discrete global optimization**. *Applied Soft Computing*, 55:154–167, June 2017. doi:[10.1016/j.asoc.2017.01.039](https://doi.org/10.1016/j.asoc.2017.01.039).
- [212] T. Bartz-Beielstein, C. Lasarczyk, and M. Preuss. **Sequential Parameter Optimization**. In *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, pages 773–780, Piscataway, NJ, Sept. 2005. IEEE Press.
- [213] T. Bartz-Beielstein, M. J. Blesa, C. Blum, B. Naujoks, A. Roli, G. Rudolph, and M. Sampels, editors. *Hybrid Metaheuristics HM 2007, 4th International Workshop*, volume 4771 of *LNCS*. Springer, Heidelberg, Germany, 2007.
- [214] T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors. *Experimental Methods for the Analysis of Optimization Algorithms*. Springer, Berlin/Heidelberg, 2010.
- [215] T. Bartz-Beielstein, O. Flasch, P. Koch, and W. Konen. **SPOT: A Toolbox for Interactive and Automatic Tuning in the R Environment**. In *Proceedings 20. Workshop Computational Intelligence*, pages 264–273, Karlsruhe, 2010. KIT Scientific Publishing.
- [216] T. Bartz-Beielstein, C. Lasarczyk, and M. Preuss. **The Sequential Parameter Optimization Toolbox**. In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 337–360. Springer, Berlin/Heidelberg, 2010. doi:[10.1007/978-3-642-02538-9_14](https://doi.org/10.1007/978-3-642-02538-9_14).
Keywords: SPOT.
- [217] T. Bartz-Beielstein, J. Ziegenhirt, W. Konen, O. Flasch, P. Koch, and M. Zaefferer. *SPOT: Sequential Parameter Optimization*, 2011. URL <http://cran.r-project.org/package=SPOT>. R package.
- [218] T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors. *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*. Springer, Heidelberg, Germany, 2014.
- [219] T. Bartz-Beielstein, C. Doerr, D. van den Berg, J. Bossek, S. Chandrasekaran, T. Eftimov, A. Fischbach, P. Kerschke, W. La Cava, M. López-Ibáñez, K. M. Malan, J. H. Moore, B. Naujoks, P. Orzechowski, V. Volz, M. Wagner, and T. Weise. **Benchmarking in Optimization: Best Practice and Open Issues**. *Arxiv preprint arXiv:2007.03488 [cs.NE]*, 2020. URL <https://arxiv.org/abs/2007.03488>.
- [220] T. Bartz-Beielstein, B. Filipič, P. Korošec, and E.-G. Talbi, editors. *High-Performance Simulation-Based Optimization*. Springer International Publishing, Cham, Switzerland, 2020.
- [221] M. Basseur, A. Goëffon, A. Liefooghe, and S. Verel. **On Set-based Local Search for Multiobjective Combinatorial Optimization**. In C. Blum and E. Alba, editors, *GECCO*, pages 471–478. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8. doi:[10.1145/2463372.2463430](https://doi.org/10.1145/2463372.2463430).

- [222] V. Basto-Fernandes, I. Yevseyeva, A. Deutz, and M. T. M. Emmerich. **A survey of diversity oriented optimization: Problems, indicators, and algorithms.** In *EVOLVE - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computation VII*, volume 662 of *Studies in Computational Intelligence*, pages 3–23. Springer, Cham, Switzerland, 2017. doi:10.1007/978-3-319-49325-1_1.
- [223] A. Basu and L. N. Frazer. **Rapid Determination of the Critical Temperature in Simulated Annealing Inversion.** *Science*, 249(4975):1409–1412, 1990.
- [224] K. Basu and A. B. Owen. **Transformations and Hardy-Krause Variation.** *SIAM Journal on Numerical Analysis*, 54(3):1946–1966, 2016. doi:10.1137/15M1052184.
Keywords: hypercube, hypersphere, simplex, quasi-Monte Carlo.
- [225] E. Batteux, E. Ferguson, and R. J. Tunney. **Do our risk preferences change when we make decisions for others? A meta-analysis of self-other differences in decisions involving risk.** *PLoS One*, 14(5):e0216566, 2019. doi:10.1371/journal.pone.0216566.
- [226] M. Battistotti, M. López-Ibáñez, K. Smith-Miles, J. Handl, and M. A. Muñoz. **Constructing Streams of Optimization Instances for Benchmarking Algorithm Selection and Configuration Approaches in Streaming Scenarios.** In T. Hu and L. Trujillo, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026*. ACM Press, New York, NY, 2026. doi:10.1145/3795095.3805073.
- [227] M. Battistutta, A. Schaerf, and T. Urli. **Feature-based tuning of single-stage simulated annealing for examination timetabling.** In E. Özcan, E. K. Burke, and B. McCollum, editors, *PATAT 2014: Proceedings of the 10th International Conference of the Practice and Theory of Automated Timetabling*, pages 53–61. PATAT, 2014.
Keywords: F-race.
- [228] M. Battistutta, A. Schaerf, and T. Urli. **Feature-based Tuning of Single-stage Simulated Annealing for Examination Timetabling.** *Annals of Operations Research*, 252(2):239–254, 2017.
- [229] R. Battiti and P. Campigotto. **Reactive search optimization: Learning while optimizing. An experiment in interactive multi-objective optimization.** In M. Caserta and S. Voß, editors, *MIC 2009*, Hamburg, Germany, 2010. University of Hamburg.
- [230] R. Battiti and A. Passerini. **Brain-Computer Evolutionary Multiobjective Optimization: A Genetic Algorithm Adapting to the Decision Maker.** *IEEE Trans. Evol. Comput.*, 14(5):671–687, 2010. doi:10.1109/TEVC.2010.2058118.
Keywords: BC-EMOA.
Annotation: Errata: DTLZ6 and DTLZ7 in the paper are actually DTLZ7 and DTLZ8 in [822].
- [231] R. Battiti and M. Protasi. **Reactive Search, A History-Based Heuristic for MAX-SAT.** *ACM Journal of Experimental Algorithmics*, 2, 1997.
- [232] R. Battiti and G. Tecchiolli. **Simulated annealing and Tabu search in the long run: A comparison on QAP tasks.** *Computer and Mathematics with Applications*, 28(6):1–8, 1994. doi:10.1016/0898-1221(94)00147-2.
- [233] R. Battiti and G. Tecchiolli. **The Reactive Tabu Search.** *ORSA Journal on Computing*, 6(2): 126–140, 1994.
- [234] R. Battiti and G. Tecchiolli. **The continuous reactive tabu search: blending combinatorial optimization and stochastic search for global optimization.** *Annals of Operations Research*, 63(2):151–188, 1996. doi:10.1007/BF02125453.

- [235] R. Battiti, M. Brunato, and F. Mascia. *Reactive Search and Intelligent Optimization*, volume 45 of *Operations Research/Computer Science Interfaces*. Springer, New York, NY, 2008. doi:10.1007/978-0-387-09624-7.
- [236] R. Battiti, D. E. Kvasov, and Y. D. Sergeyev, editors. *11th International Conference, LION 11, Nizhny Novgorod, Russia, June 19-21, 2017, Revised Selected Papers*, volume 10556 of *LNCS*. Springer, Cham, Switzerland, 2017.
- [237] R. Battiti, M. Brunato, I. Kotsireas, and P. M. Pardalos, editors. *12th International Conference, LION 12, Kalamata, Greece, June 10-15, 2018*, volume 11353 of *LNCS*. Springer, Cham, Switzerland, 2018.
- [238] E. B. Baum. **Iterated Descent: A Better Algorithm for Local Search in Combinatorial Optimization Problems**. Manuscript, 1986.
- [239] E. B. Baum. **Towards Practical “Neural” Computation for Combinatorial Optimization Problems**. In *Neural Networks for Computing, AIP Conference Proceedings*, pages 53–64, 1986.
- [240] J. Baumgartner and M. Sheeran, editors. *FMCAD’07: Proceedings of the 7th International Conference Formal Methods in Computer Aided Design*, Austin, Texas, USA, 2007. IEEE Computer Society, Washington, DC, USA.
- [241] W. J. Baumol. **Management models and industrial applications of linear programming**. *Naval Research Logistics Quarterly*, 9(1):63–64, 1962. doi:10.1002/nav.3800090109.
- [242] J. Bautista and J. Pereira. **Ant algorithms for a time and space constrained assembly line balancing problem**. *Eur. J. Oper. Res.*, 177(3):2016–2032, 2007. doi:10.1016/j.ejor.2005.12.017.
- [243] J. Baxter. **Local Optima Avoidance in Depot Location**. *J. Oper. Res. Soc.*, 32(9):815–819, 1981.
- [244] A. Baykasoglu, T. Dereli, and I. Sabuncu. **A multiple objective ant colony optimization approach to assembly line balancing problems**. In *35th International Conference on Computers and Industrial Engineering (CIE35)*, pages 263–268, Istanbul, Turkey, 2005.
- [245] B. Beachkofski and R. Grandhi. **Improved Distributed Hypercube Sampling**. In *Proceedings of the 43rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*. AIAA paper 2002-1274, American Institute of Aeronautics and Astronautics, 2002.
- [246] J. Bealt, D. Shaw, C. M. Smith, and M. López-Ibáñez. **Peer Reviews for Making Cities Resilient: A Systematic Literature Review**. *International Journal of Emergency Management*, 15(4):334–359, 2019. doi:10.1504/IJEM.2019.104201.
Keywords: city resilience, city peer review, disaster risk governance.
- [247] J. E. Beasley. **OR-Library: distributing test problems by electronic mail**. *J. Oper. Res. Soc.*, pages 1069–1072, 1990. Currently available from <http://people.brunel.ac.uk/~mastjjb/jeb/info.html>.
- [248] J. E. Beasley. **Heuristic algorithms for the unconstrained binary quadratic programming problem**. Technical report, The Management School, Imperial College, London, England, 1998.
- [249] J. E. Beasley and P. C. Chu. **A Genetic Algorithm for the Set Covering Problem**. *Eur. J. Oper. Res.*, 94(2):392–404, 1996.
- [250] J. E. Beasley and P. C. Chu. **A Genetic Algorithm for the Multidimensional Knapsack Problem**. *J. Heuristics*, 4(1):63–86, 1998.

- [251] S. Becker, J. Gottlieb, and T. Stützle. **Applications of Racing Algorithms: An Industrial Perspective.** In E.-G. Talbi, P. Liardet, P. Collet, E. Lutton, and M. Schoenauer, editors, *Artificial Evolution*, volume 3871 of *LNCS*, pages 271–283. Springer, Heidelberg, Germany, 2005.
- [252] D. D. Bedworth and J. E. Bailey. *Integrated Production Control Systems: Management, Analysis, Design*, volume 2. John Wiley & Sons, New York, NY, 1982.
- [253] A. Beham, M. Affenzeller, and S. Wagner. **Instance-based algorithm selection on quadratic assignment problem landscapes.** In P. A. N. Bosman, editor, *GECCO Companion*, pages 1471–1478. ACM Press, New York, NY, 2017.
- [254] J. Behnamian and S. M. T. Fatemi Ghomi. **Hybrid Flowshop Scheduling with Machine and Resource-dependent Processing Times.** *Applied Mathematical Modelling*, 35(3):1107–1123, 2011.
- [255] M. Behrisch, L. Bieker, J. Erdmann, and D. Krajzewicz. **SUMO - Simulation of Urban Mobility: An Overview.** In *SIMUL 2011, The Third International Conference on Advances in System Simulation*, pages 63–68, Barcelona, Spain, 2011. ThinkMind.
- [256] N. Beldiceanu, N. Jussien, and E. Pinson, editors. *Integration of AI and OR Techniques in Constraint Programming for Combinatorial Optimization Problems, 9th International Conference, CPAIOR 2012, Nantes, France, May 28 – June 1, 2012. Proceedings*, volume 7298 of *LNCS*. Springer, Heidelberg, Germany, 2012. ISBN 978-3-642-29827-1.
- [257] R. K. Belew and M. D. Vose, editors. *Proceedings of the 4th Workshop on Foundations of Genetic Algorithms, San Diego, CA, USA, August 5 1996*. Morgan Kaufmann Publishers, 1996.
- [258] N. Belkhir, J. Dreo, P. Savéant, and M. Schoenauer. **Feature Based Algorithm Configuration: A Case Study with Differential Evolution.** In J. Handl, E. Hart, P. R. Lewis, M. López-Ibáñez, G. Ochoa, and B. Paechter, editors, *PPSN*, volume 9921 of *LNCS*, pages 156–166. Springer, Heidelberg, Germany, 2016. ISBN 978-3-319-45822-9. doi:10.1007/978-3-319-45823-6.
- [259] N. Belkhir, J. Dreo, P. Savéant, and M. Schoenauer. **Per Instance Algorithm Configuration of CMA-ES with Limited Budget.** In P. A. N. Bosman, editor, *GECCO*, pages 681–688. ACM Press, New York, NY, 2017.
- [260] R. Bellio, S. Ceschia, L. Di Gaspero, A. Schaerf, and T. Urli. **Feature-based tuning of simulated annealing applied to the curriculum-based course timetabling problem.** *Comput. Oper. Res.*, 65:83–92, 2016.
- [261] R. Bellman. **The theory of dynamic programming.** *Bulletin of the American Mathematical Society*, 60:503–515, 1954.
- [262] A. Belov, D. Diepold, M. Heule, and M. Jarvisalo, editors. *Proceedings of SAT Competition 2014: Solver and Benchmark Descriptions*, volume B-2014-2 of *Science Series of Publications B*, 2014. University of Helsinki.
- [263] V. Belton and T. J. Stewart. **Goal and Reference Point Methods.** In *Multiple Criteria Decision Analysis: An Integrated Approach*, chapter 7, pages 209–232. Springer, 2002. ISBN 978-1-4615-1495-4. doi:10.1007/978-1-4615-1495-4_7.
- [264] V. Belton and T. J. Stewart. *Multiple Criteria Decision Analysis: An Integrated Approach*. Springer, 2002. ISBN 978-1-4615-1495-4.

- [265] V. Belton, J. Branke, P. Eskelinen, S. Greco, J. Molina, F. Ruiz, and R. Słowiński. **Interactive Multiobjective Optimization from a Learning Perspective**. In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*, pages 405–433. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-88908-3_15.
- [266] F. Ben Abdelaziz, S. Krichen, and J. Chaouachi. **A hybrid heuristic for multiobjective knapsack problems**. In M. G. C. Resende and J. Pinho de Souza, editors, *Proc. of MIC 1997, the 2nd Metaheuristics International Conference*, pages 205–212, 1997. doi:10.1007/978-1-4615-5775-3_14.
- [267] N. Benabbou, C. Leroy, and T. Lust. **An Interactive Regret-Based Genetic Algorithm for Solving Multi-Objective Combinatorial Optimization Problems**. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2335–2342. AAAI Press, 2020. ISBN 978-1-57735-823-7. doi:10.1609/aaai.v34i03.5612.
Keywords: interactive, multi-objective, decision-makers.
- [268] A. J. Benavides and M. Ritt. **Iterated Local Search Heuristics for Minimizing Total Completion Time in Permutation and Non-permutation Flow Shops**. In R. I. Brafman, C. Domshlak, P. Haslum, and S. Zilberstein, editors, *Proceedings of the Twenty-Fifth International Conference on Automated Planning and Scheduling, ICAPS 2015*, pages 34–41. AAAI Press, Menlo Park, CA, 2015.
Keywords: irace.
- [269] A. J. Benavides and M. Ritt. **Two Simple and Effective Heuristics for Minimizing the Makespan in Non-permutation Flow Shops**. *Comput. Oper. Res.*, 66:160–169, 2016. doi:10.1016/j.cor.2015.08.001.
- [270] A. Benavoli, G. Corani, F. Mangili, and M. Zaffalon. **A Bayesian nonparametric procedure for comparing algorithms**. In F. Bach and D. Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning, ICML 2015*, volume 37, pages 1264–1272. PMLR, 2015.
Keywords: racing.
- [271] C. J. Bendell, S. Liu, T. Aumentado-Armstrong, B. Istrate, P. T. Cernek, S. Khan, S. Picioreanu, M. Zhao, and R. A. Murgita. **Transient protein-protein interface prediction: datasets, features, algorithms, and the RAD-T predictor**. *BMC Bioinformatics*, 15:82, 2014.
- [272] J. F. Benders. **Partitioning Procedures for Solving Mixed-variables Programming Problems**. *Numerische Mathematik*, 4(3):238–252, 1962.
- [273] S. Benedettini, A. Roli, and L. Di Gaspero. **Two-level ACO for Haplotype Inference under Pure Parsimony**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 6th International Conference, ANTS 2008*, volume 5217 of *LNCS*, pages 179–190. Springer, Heidelberg, Germany, 2008.
- [274] S. Benedettini, A. Roli, and C. Blum. **A Randomized Iterated Greedy Algorithm for the Founder Sequence Reconstruction Problem**. In C. Blum and R. Battiti, editors, *LION 4*, volume 6073 of *LNCS*, pages 37–51. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-13800-3.
- [275] Y. Bengio and Y. LeCun, editors. *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015.
- [276] Y. Bengio, A. Lodi, and A. Prouvost. **Machine learning for combinatorial optimization: A methodological tour d’horizon**. *Eur. J. Oper. Res.*, 290(2):405–421, 2021. doi:10.1016/j.ejor.2020.07.063.

Keywords: Combinatorial optimization, Machine learning, Branch and bound, Mixed-integer programming solvers.

- [277] U. Benlic and J.-K. Hao. **Breakout Local Search for the Quadratic Assignment Problem**. *Applied Mathematics and Computation*, 219(9):4800–4815, 2013.
- [278] J. L. Bentley. **Multidimensional Divide-and-conquer**. *Commun. ACM*, 23(4):214–229, 1980. doi:10.1145/358841.358850.
- [279] J. L. Bentley. **Experiments on Traveling Salesman Heuristics**. In D. S. Johnson, editor, *Proceedings of the First Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 91–99. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 1990.
- [280] J. L. Bentley. **Fast Algorithms for Geometric Traveling Salesman Problems**. *ORSA Journal on Computing*, 4(4):387–411, 1992.
- [281] F. Bergadano and L. De Raedt, editors. *European Conference on Machine Learning, Catania, Italy, April 6-8, 1994. Proceedings*, volume 784 of *LNCS*. Springer, Berlin/Heidelberg, 1994.
- [282] L. Berger, J. Emmerling, and M. Tavoni. **Managing catastrophic climate risks under model uncertainty aversion**. *Management Science*, 63(3):749–765, 2017.
- [283] D. Bergman, A. A. Cire, W.-J. van Hoes, and J. Hooker. *Decision Diagrams for Optimization*. Springer, Cham, Switzerland, 2016. ISBN 978-3-319-42849-9. doi:10.1007/978-3-319-42849-9.
- [284] J. S. Bergstra and Y. Bengio. **Random Search for Hyper-Parameter Optimization**. *J. Mach. Learn. Res.*, 13:281–305, 2012.
- [285] J. S. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl. **Algorithms for Hyper-Parameter Optimization**. In J. Shawe-Taylor, R. S. Zemel, P. L. Bartlett, F. Pereira, and K. Q. Weinberger, editors, *Advances in Neural Information Processing Systems (NIPS 24)*, pages 2546–2554. Curran Associates, Red Hook, NY, 2011. URL <http://papers.nips.cc/paper/4443-algorithms-for-hyper-parameter-optimization.pdf>.
- [286] J. S. Bergstra, D. Yasmin, and D. Cox. **Making a science of model search: Hyperparameter optimization in hundreds of dimensions for vision architectures**. In S. Dasgupta and D. McAllester, editors, *Proceedings of the 30th International Conference on Machine Learning, ICML 2013*, volume 28, pages 115–123, 2013.
- [287] F. Berkenkamp, A. Krause, and A. P. Schoellig. **Bayesian Optimization with Safety Constraints: Safe and Automatic Parameter Tuning in Robotics**. *Arxiv preprint arXiv:1602.04450*, 2016. URL <http://arxiv.org/abs/1602.04450>.
Keywords: Safe Optimization, SafeOpt.
- [288] F. Berkenkamp, A. P. Schoellig, and A. Krause. **Safe controller optimization for quadrotors with Gaussian processes**. In *2016 IEEE International Conference on Robotics and Automation (ICRA)*, pages 491–496. IEEE, 2016. doi:10.1109/ICRA.2016.7487170.
Keywords: Safe Optimization, SafeOpt.
- [289] F. Berkenkamp, A. Krause, and A. P. Schoellig. **Bayesian optimization with safety constraints: safe and automatic parameter tuning in robotics**. *Mach. Learn.*, June 2021. doi:10.1007/s10994-021-06019-1.
Annotation: Preprint: <http://arxiv.org/abs/1602.04450>.
- [290] J. O. Berkey and P. Y. Wang. **Two-dimensional finite bin-packing algorithms**. *J. Oper. Res. Soc.*, 38(5):423–429, 1987. doi:10.2307/2582731.

- [291] H. Bersini, M. Dorigo, S. Langerman, G. Seront, and L. M. Gambardella. **Results of the First International Contest on Evolutionary Optimisation**. In T. Bäck, T. Fukuda, and Z. Michalewicz, editors, *Proceedings of the 1996 IEEE International Conference on Evolutionary Computation (ICEC'96)*, pages 611–615. IEEE Press, Piscataway, NJ, 1996.
- [292] L. Bertacco, M. Fischetti, and A. Lodi. **A feasibility pump heuristic for general mixed-integer problems**. *Discrete Optimization*, 4(1):63–76, 2007.
- [293] T. Berthold, A. M. Gleixner, S. Heinz, and T. Koch, editors. *Integration of AI and OR Techniques in Constraint Programming for Combinatorial Optimization Problems, 8th International Conference, CPAIOR 2011, Berlin, Germany, May 23 – 27, 2011. Proceedings*. LNCS. Springer, Heidelberg, Germany, 2011.
- [294] F. Berto, Z. Cao, C. Hua, H. Kim, J. Park, G. Song, J. Wang, and H. Ye. **ReEvo: Large Language Models as Hyper-Heuristics with Reflective Evolution**. In A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, pages 43571–43608, 2024. doi:10.52202/079017-1381.
- [295] F. Berto, C. Hua, J. Park, L. Luttmann, Y. Ma, F. Bu, J. Wang, H. Ye, M. Kim, S. Choi, N. G. Zepeda, A. Hottung, J. Zhou, J. Bi, Y. Hu, F. Liu, H. Kim, J. Son, H. Kim, D. Angioni, W. Kool, Z. Cao, Q. Zhang, J. Kim, J. Zhang, K. Shin, C. Wu, S. Ahn, G. Song, C. Kwon, K. Tierney, L. Xie, and J. Park. **RL4CO: An Extensive Reinforcement Learning for Combinatorial Optimization Benchmark**. In Y. Sun et al., editors, *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 5278–5289. ACM Press, New York, NY, Aug. 2025. doi:10.1145/3711896.3737433.
- [296] D. Bertsekas. *Dynamic Programming and Optimal Control*. Athena Scientific, Belmont, MA, 1995.
- [297] D. Bertsekas. *Network Optimization: Continuous and Discrete Models*. Athena Scientific, Belmont, MA, 1998.
- [298] D. P. Bertsekas, J. N. Tsitsiklis, and C. Wu. **Rollout Algorithms for Combinatorial Optimization**. *J. Heuristics*, 3(3):245–262, 1997.
- [299] D. Bertsimas and N. Kallus. **From predictive to prescriptive analytics**. *Management Science*, 66(3):1025–1044, 2020.
- [300] N. Beume and G. Rudolph. **Faster S-Metric Calculation by Considering Dominated Hypervolume as Klee’s Measure Problem**. In B. Kovalerchuk, editor, *Proceedings of the Second IASTED Conference on Computational Intelligence*, pages 231–236. ACTA Press, Anaheim, 2006.
- [301] N. Beume, C. M. Fonseca, M. López-Ibáñez, L. Paquete, and J. Vahrenhold. **On the Complexity of Computing the Hypervolume Indicator**. Technical Report CI-235/07, University of Dortmund, Dec. 2007. Published in *IEEE Trans. Evol. Comput.* [303].
- [302] N. Beume, B. Naujoks, and M. T. M. Emmerich. **SMS-EMOA: Multiobjective selection based on dominated hypervolume**. *Eur. J. Oper. Res.*, 181(3):1653–1669, 2007. doi:10.1016/j.ejor.2006.08.008.
- [303] N. Beume, C. M. Fonseca, M. López-Ibáñez, L. Paquete, and J. Vahrenhold. **On the complexity of computing the hypervolume indicator**. *IEEE Trans. Evol. Comput.*, 13(5):1075–1082, 2009. doi:10.1109/TEVC.2009.2015575.
- [304] H.-G. Beyer and U.-M. O’Reilly, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2005*. ACM Press, New York, NY, 2005.

- [305] H.-G. Beyer and H.-P. Schwefel. **Evolution Strategies: A Comprehensive Introduction.** *Natural Computing*, 1:3–52, 2002.
- [306] H.-G. Beyer, H.-P. Schwefel, and I. Wegener. **How to analyse evolutionary algorithms.** *Theor. Comput. Sci.*, 287(1):101–130, 2002.
- [307] L. C. T. Bezerra. *A component-wise approach to multi-objective evolutionary algorithms: from flexible frameworks to automatic design.* PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2016.
Annotation: Supervised by Thomas Stützle and Manuel López-Ibáñez.
- [308] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Generation of Multi-Objective ACO Algorithms for the Biobjective Knapsack.** In M. Dorigo et al., editors, *ANTS 2012*, volume 7461 of *LNCS*, pages 37–48. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32650-9_4. Supplementary material: <https://iridia.ulb.ac.be/supp/IridiaSupp2012-008/>.
- [309] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Generation of MOACO Algorithms for the Biobjective Bidimensional Knapsack Problem: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2012-008/>, 2012.
- [310] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **An Analysis of Local Search for the Bi-objective Bidimensional Knapsack: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2012-016/>, 2013.
- [311] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Deconstructing Multi-Objective Evolutionary Algorithms: An Iterative Analysis on the Permutation Flowshop: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2013-010/>, 2013.
- [312] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **An Analysis of Local Search for the Bi-objective Bidimensional Knapsack Problem.** In M. Middendorf and C. Blum, editors, *EvoCOP*, volume 7832 of *LNCS*, pages 85–96. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-37198-1_8.
- [313] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Component-Wise Design of Multi-Objective Evolutionary Algorithms.** Technical Report TR/IRIDIA/2014-012, IRIDIA, Université Libre de Bruxelles, Belgium, Aug. 2014.
- [314] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Deconstructing Multi-Objective Evolutionary Algorithms: An Iterative Analysis on the Permutation Flowshop.** In P. M. Pardalos, M. G. C. Resende, C. Vogiatzis, and J. L. Walteros, editors, *LION 8*, volume 8426 of *LNCS*, pages 57–172. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-09584-4_16. Supplementary material: <https://iridia.ulb.ac.be/supp/IridiaSupp2013-010/>.
- [315] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Design of Evolutionary Algorithms for Multi-Objective Combinatorial Optimization.** In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 508–517. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-10762-2_50.
- [316] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Design of Evolutionary Algorithms for Multi-Objective Combinatorial Optimization.** <https://iridia.ulb.ac.be/supp/IridiaSupp2014-007/>, 2014.
- [317] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Component-Wise Design of Multi-Objective Evolutionary Algorithms.** <https://github.com/iridia-ulb/automoea-tevc-2016>, 2015.

- [318] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **To DE or Not to DE? Multi-objective Differential Evolution Revisited from a Component-Wise Perspective: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2015-001/>, 2015.
- [319] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **To DE or Not to DE? Multi-objective Differential Evolution Revisited from a Component-Wise Perspective.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 48–63. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-15934-8_4.
- [320] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Comparing Decomposition-Based and Automatically Component-Wise Designed Multi-Objective Evolutionary Algorithms.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 396–410. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-15934-8_27.
- [321] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Component-Wise Design of Multi-Objective Evolutionary Algorithms.** *IEEE Trans. Evol. Comput.*, 20(3):403–417, 2016. doi:10.1109/TEVC.2015.2474158. Supplementary material: <https://github.com/iridia-ulb/automoea-tevc-2016>.
- [322] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **A Large-Scale Experimental Evaluation of High-Performing Multi- and Many-Objective Evolutionary Algorithms.** <https://iridia.ulb.ac.be/supp/IridiaSupp2015-007/>, 2017.
- [323] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **An Empirical Assessment of the Properties of Inverted Generational Distance Indicators on Multi- and Many-objective Optimization.** In H. Trautmann, G. Rudolph, K. Klamroth, O. Schütze, M. M. Wiecek, Y. Jin, and C. Grimme, editors, *EMO*, volume 10173 of *LNCS*, pages 31–45. Springer International Publishing, Cham, Switzerland, 2017. doi:10.1007/978-3-319-54157-0_3.
- [324] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **A Large-Scale Experimental Evaluation of High-Performing Multi- and Many-Objective Evolutionary Algorithms.** Technical Report TR/IRIDIA/2017-005, IRIDIA, Université Libre de Bruxelles, Belgium, 2017.
- [325] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Configuration of Multi-objective Optimizers and Multi-objective Configuration.** Technical Report TR/IRIDIA/2017-011, IRIDIA, Université Libre de Bruxelles, Belgium, Nov. 2017. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2017-011.pdf>. Published as a book chapter [332].
- [326] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **A Large-Scale Experimental Evaluation of High-Performing Multi- and Many-Objective Evolutionary Algorithms.** *Evol. Comput.*, 26(4):621–656, 2018. doi:10.1162/evco_a_00217. Supplementary material: <https://iridia.ulb.ac.be/supp/IridiaSupp2015-007/>.
- [327] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatically Designing State-of-the-Art Multi- and Many-Objective Evolutionary Algorithms.** Technical Report TR/IRIDIA/2018-001, IRIDIA, Université Libre de Bruxelles, Belgium, Jan. 2018. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2018-001.pdf>. Published in *Evolutionary Computation* journal [331].
- [328] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatically Designing State-of-the-Art Multi- and Many-Objective Evolutionary Algorithms: Supplementary material.** <https://github.com/iridia-ulb/automoea-ecj-2020>, 2019.

- [329] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Archiver Effects on the Performance of State-of-the-art Multi- and Many-objective Evolutionary Algorithms.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321789. Supplementary material: <https://iridia.ulb.ac.be/supp/IridiaSupp2019-004/>.
- [330] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Archiver Effects on the Performance of State-of-the-art Multi- and Many-objective Evolutionary Algorithms: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2019-004/>, 2019.
- [331] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatically Designing State-of-the-Art Multi- and Many-Objective Evolutionary Algorithms.** *Evol. Comput.*, 28(2):195–226, 2020. doi:10.1162/evco_a_00263. Supplementary material: <https://github.com/iridia-ulb/automoea-ecj-2020>.
- [332] L. C. T. Bezerra, M. López-Ibáñez, and T. Stützle. **Automatic Configuration of Multi-objective Optimizers and Multi-objective Configuration.** In T. Bartz-Beielstein, B. Filipič, P. Korošec, and E.-G. Talbi, editors, *High-Performance Simulation-Based Optimization*, pages 69–92. Springer International Publishing, Cham, Switzerland, 2020. doi:10.1007/978-3-030-18764-4_4.
- [333] L. Bianchi, L. M. Gambardella, and M. Dorigo. **An Ant Colony Optimization Approach to the Probabilistic Traveling Salesman Problem.** In J.-J. Merelo et al., editors, *Parallel Problem Solving from Nature – PPSN VII*, volume 2439 of *LNCS*, pages 883–892. Springer, Heidelberg, Germany, 2002. Annotation: IC.34.
- [334] L. Bianchi, M. Birattari, M. Manfrin, M. Mastrolilli, L. Paquete, O. Rossi-Doria, and T. Schiavinotto. **Hybrid Metaheuristics for the Vehicle Routing Problem with Stochastic Demands.** *Journal of Mathematical Modelling and Algorithms*, 5(1):91–110, 2006.
- [335] L. Bianchi, M. Dorigo, L. M. Gambardella, and W. J. Gutjahr. **A survey on metaheuristics for stochastic combinatorial optimization.** *Natural Computing*, 8(2):239–287, 2009.
- [336] A. Biedenkapp, M. T. Lindauer, K. Eggensperger, F. Hutter, C. Fawcett, and H. H. Hoos. **Efficient Parameter Importance Analysis via Ablation with Surrogates.** In S. P. Singh and S. Markovitch, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press, Feb. 2017. doi:10.1609/aaai.v31i1.10657.
- [337] A. Biedenkapp, J. Marben, M. T. Lindauer, and F. Hutter. **CAVE: Configuration assessment, visualization and evaluation.** In R. Battiti, M. Brunato, I. Kotsireas, and P. M. Pardalos, editors, *LION 12*, volume 11353 of *LNCS*, pages 115–130. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-030-05348-2_10.
- [338] A. Biedenkapp, H. F. Bozkurt, T. Eimer, F. Hutter, and M. T. Lindauer. **Dynamic Algorithm Configuration: Foundation of a New Meta-Algorithmic Framework.** In G. D. Giacomo, A. Catala, B. Dilkina, M. Milano, S. Barro, A. Bugarín, and J. Lang, editors, *Proceedings of the 24th European Conference on Artificial Intelligence (ECAI)*, volume 325 of *Frontiers in Artificial Intelligence and Applications*, pages 427–434. IOS Press, 2020.
- [339] A. Biere. **Yet another Local Search Solver and Lingeling and Friends Entering the SAT Competition 2014.** In A. Belov, D. Diepold, M. Heule, and M. Järvisalo, editors, *Proc. of SAT Competition 2014: Solver and Benchmark Descriptions*, volume B-2014-2 of *Science Series of Publications B*, pages 39–40. University of Helsinki, 2014.
- [340] G. Bilchev and I. C. Parmee. **The Ant Colony Metaphor for Searching Continuous Design Spaces.** In T. C. Fogarty, editor, *Evolutionary Computing, AISB Workshop*, volume 993 of *LNCS*, pages 25–39. Springer, Berlin, Germany, 1995. doi:10.1007/3-540-60469-3_22.

- [341] M. Binois, D. Ginsbourger, and O. Roustant. **Quantifying uncertainty on Pareto fronts with Gaussian process conditional simulations.** *Eur. J. Oper. Res.*, 243(2):386–394, 2015. [doi:10.1016/j.ejor.2014.07.032](https://doi.org/10.1016/j.ejor.2014.07.032).
Keywords: Attainment function, Expected Hypervolume Improvement, Kriging, Multi-objective optimization, Vorob’ev expectation.
- [342] M. Birattari. **The race Package for R: Racing Methods for the Selection of the Best.** Technical Report TR/IRIDIA/2003-037, IRIDIA, Université Libre de Bruxelles, Belgium, 2003.
- [343] M. Birattari. *The Problem of Tuning Metaheuristics as Seen from a Machine Learning Perspective.* PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2004.
Annotation: Supervised by Marco Dorigo.
- [344] M. Birattari. **On the Estimation of the Expected Performance of a Metaheuristic on a Class of Instances. How Many Instances, How Many Runs?** Technical Report TR/IRIDIA/2004-001, IRIDIA, Université Libre de Bruxelles, Belgium, 2004.
- [345] M. Birattari. *Tuning Metaheuristics: A Machine Learning Perspective*, volume 197 of *Studies in Computational Intelligence*. Springer, Berlin/Heidelberg, 2009. [doi:10.1007/978-3-642-00483-4](https://doi.org/10.1007/978-3-642-00483-4).
Annotation: Based on the PhD thesis [343].
- [346] M. Birattari, G. A. Di Caro, and M. Dorigo. **Toward the formal foundation of Ant Programming.** In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCIS*, pages 188–201. Springer, Heidelberg, Germany, 2002.
- [347] M. Birattari, T. Stützle, L. Paquete, and K. Varrentrapp. **A Racing Algorithm for Configuring Metaheuristics.** In W. B. Langdon et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2002*, pages 11–18. Morgan Kaufmann Publishers, San Francisco, CA, 2002.
Keywords: F-race.
- [348] M. Birattari, P. Balaprakash, and M. Dorigo. **The ACO/F-RACE algorithm for combinatorial optimization under uncertainty.** In K. F. Doerner, M. Gendreau, P. Greistorfer, W. J. Gutjahr, R. F. Hartl, and M. Reimann, editors, *Metaheuristics – Progress in Complex Systems Optimization*, volume 39 of *Operations Research/Computer Science Interfaces Series*, pages 189–203. Springer, New York, NY, 2006.
- [349] M. Birattari, M. Zlochin, and M. Dorigo. **Towards a theory of practice in metaheuristics design: A machine learning perspective.** *Theoretical Informatics and Applications*, 40(2): 353–369, 2006.
- [350] M. Birattari, P. Pellegrini, and M. Dorigo. **On the invariance of ant colony optimization.** *IEEE Trans. Evol. Comput.*, 11(6):732–742, 2007. [doi:10.1109/TEVC.2007.892762](https://doi.org/10.1109/TEVC.2007.892762).
- [351] M. Birattari, P. Balaprakash, T. Stützle, and M. Dorigo. **Estimation Based Local Search for Stochastic Combinatorial Optimization.** *INFORMS Journal on Computing*, 20(4):644–658, 2008.
- [352] M. Birattari, Z. Yuan, P. Balaprakash, and T. Stützle. **F-Race and Iterated F-Race: An Overview.** In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 311–336. Springer, Berlin/Heidelberg, 2010. [doi:10.1007/978-3-642-02538-9_13](https://doi.org/10.1007/978-3-642-02538-9_13).
Keywords: F-race, iterated F-race, irace, tuning.
- [353] M. Birattari, Z. Yuan, P. Balaprakash, and T. Stützle. **Parameter Adaptation in Ant Colony Optimization.** In M. Caserta and S. Voß, editors, *MIC 2009*, Hamburg, Germany, 2010. University of Hamburg.

- [354] M. Birattari, M. Chiarandini, M. Saerens, and T. Stützle. **Learning Graphical Models for Algorithm Configuration**. In T. Berthold, A. M. Gleixner, S. Heinz, and T. Koch, editors, *CPAIOR 2011*, LNCS. Springer, Heidelberg, Germany, 2011.
- [355] M. Birattari, M. Chiarandini, M. Saerens, and T. Stützle. **Learning graphical models for parameter tuning**. Technical Report TR/IRIDIA/2011-002, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-002.pdf>.
- [356] S. Bird, E. Klein, and E. Loper. *Natural language processing with Python: analyzing text with the natural language toolkit*. O'Reilly Media, Inc., 2009.
- [357] F. Biscani, D. Izzo, and C. H. Yam. **A Global Optimisation Toolbox for Massively Parallel Engineering Optimisation**. In *ICATT*, 2010. URL <http://arxiv.org/abs/1004.3824>.
Keywords: PaGMO.
- [358] F. Biscani, D. Izzo, and C. H. Yam. **A Global Optimisation Toolbox for Massively Parallel Engineering Optimisation**. *Arxiv preprint arXiv:1004.3824*, 2010. URL <http://arxiv.org/abs/1004.3824>.
Keywords: PaGMO.
- [359] B. Bischl, O. Mersmann, H. Trautmann, and M. Preuss. **Algorithm Selection Based on Exploratory Landscape Analysis and Cost-sensitive Learning**. In T. Soule and J. H. Moore, editors, *GECCO*, pages 313–320. ACM Press, New York, NY, 2012.
Keywords: continuous optimization, landscape analysis, algorithm selection.
- [360] B. Bischl, M. Lang, J. Bossek, L. Judt, J. Richter, T. Kuehn, and E. Studerus. *mlr: Machine Learning in R*, 2013. URL <http://cran.r-project.org/package=mlr>. R package.
- [361] B. Bischl, P. Kerschke, L. Kotthoff, M. T. Lindauer, Y. Malitsky, A. Fréchette, H. H. Hoos, F. Hutter, K. Leyton-Brown, K. Tierney, and J. Vanschoren. **ASlib: A Benchmark Library for Algorithm Selection**. *Artif. Intell.*, 237:41–58, 2016.
- [362] B. Bischl, M. Lang, L. Kotthoff, J. Schiffner, J. Richter, E. Studerus, G. Casalicchio, and Z. M. Jones. **mlr: Machine Learning in R**. *J. Mach. Learn. Res.*, 17(170):1–5, 2016.
- [363] B. Bischl, M. Lang, J. Bossek, D. Horn, K. Schork, J. Richter, and P. Kerschke. *ParamHelpers: Helpers for Parameters in Black-Box Optimization, Tuning and Machine Learning*, 2017. URL <https://cran.r-project.org/package=ParamHelpers>. R package version 1.10.
- [364] B. Bischl, J. Richter, J. Bossek, D. Horn, J. Thomas, and M. Lang. **mlrMBO: A Modular Framework for Model-Based Optimization of Expensive Black-Box Functions**. *Arxiv preprint arXiv:1703.03373 [stat.ML]*, 2017. URL <http://arxiv.org/abs/1703.03373>.
- [365] B. Bischl, M. Binder, M. Lang, T. Pielok, J. Richter, S. Coors, J. Thomas, T. Ullmann, M. Becker, A.-L. Boulesteix, D. Deng, and M. T. Lindauer. **Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges**. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 13(2):e1484, 2023.
- [366] E. E. Bischoff and M. S. W. Ratcliff. **Issues in the development of approaches to container loading**. *Omega*, 23(4):377–390, Aug. 1995. doi:10.1016/0305-0483(95)00015-g.
Keywords: single container loading, 3DBPP.
- [367] C. M. Bishop. *Pattern recognition and machine learning*. Springer, 2006.
- [368] E. Bıyık, J. Margoliash, S. R. Alimo, and D. Sadigh. **Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models**. In *2019 American Control Conference (ACC)*, pages 1792–1799. IEEE, 2019. doi:10.23919/ACC.2019.8815276.

- [369] C. Blackmore, O. Ray, and K. Eder. **Automatically Tuning the GCC Compiler to Optimize the Performance of Applications Running on Embedded Systems.** *Arxiv preprint arXiv:1703.08228*, 2017. URL <https://arxiv.org/abs/1703.08228>.
- [370] J. Blank and K. Deb. **pymoo: Multi-Objective Optimization in Python.** *IEEE Access*, 8: 89497–89509, 2020. doi:10.1109/ACCESS.2020.2990567.
- [371] J. Blank, K. Deb, Y. Dhebar, S. Bandaru, and H. Seada. **Generating Well-Spaced Points on a Unit Simplex for Evolutionary Many-Objective Optimization.** *IEEE Trans. Evol. Comput.*, 25(1):48–60, 2021. doi:10.1109/TEVC.2020.2992387.
Keywords: diversity preservation, reference points, Riesz s-energy.
- [372] X. Blasco, J. M. Herrero, J. Sanchis, and M. Martínez. **A new graphical visualization of n-dimensional Pareto front for decision-making in multiobjective optimization.** *Inf. Sci.*, 178(20):3908–3924, 2008.
- [373] M. J. Blesa and C. Blum. **Ant Colony Optimization for the Maximum Edge-Disjoint Paths Problem.** In G. R. Raidl et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2004*, volume 3005 of *LNCS*, pages 160–169. Springer, Heidelberg, Germany, 2004.
- [374] M. J. Blesa and C. Blum. **Finding edge-disjoint paths in networks by means of artificial ant colonies.** *Journal of Mathematical Modelling and Algorithms*, 6(3):361–391, 2007.
- [375] M. J. Blesa, C. Blum, C. Cotta, A. J. Fernández, J. E. Gallardo, A. Roli, and M. Sampels, editors. *Hybrid Metaheuristics HM 2008, 5th International Workshop*, volume 5296 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [376] M. J. Blesa, C. Blum, L. Di Gaspero, A. Roli, M. Sampels, and A. Schaerf, editors. *Hybrid Metaheuristics, 6th International Workshop, HM 2009, Udine, Italy, October 16-17, 2009. Proceedings*, volume 5818 of *LNCS*. Springer, Heidelberg, Germany, 2009.
- [377] M. J. Blesa, C. Blum, P. Festa, A. Roli, and M. Sampels, editors. *Hybrid Metaheuristics, 8th International Workshop, HM 2013, Ischia, Italy, May 23-25, 2013. Proceedings*, volume 7919 of *LNCS*. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-38515-5.
- [378] M. J. Blesa, C. Blum, and S. Voß, editors. *Hybrid Metaheuristics, 9th International Workshop, HM 2014, Hamburg, Germany, June 11-13, 2014. Proceedings*, volume 8457 of *LNCS*. Springer, Heidelberg, Germany, 2014. ISBN 978-3-319-07643-0.
- [379] S. Bleuler, M. Laumanns, L. Thiele, and E. Zitzler. **PISA – A Platform and Programming Language Independent Interface for Search Algorithms.** In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 494–508. Springer, Heidelberg, Germany, 2003.
- [380] L. Bliet, P. da Costa, R. Refaei Afshar, R. Reijnen, Y. Zhang, T. Catshoek, D. Vos, S. Verwer, F. Schmitt-Ulms, A. Hottung, T. Shah, M. Sellmann, K. Tierney, C. Perreault-Lafleur, C. Leboeuf, F. Bobbio, J. Pepin, W. A. Silva, R. Gama, H. L. Fernandes, M. Zaefferer, M. López-Ibáñez, and E. Irurozki. **The First AI4TSP Competition: Learning to Solve Stochastic Routing Problems.** *Artif. Intell.*, 319:103918, 2023. ISSN 0004-3702. doi:10.1016/j.artint.2023.103918.
Keywords: AI for TSP competition, Travelling salesman problem, Routing problem, Stochastic combinatorial optimization, Surrogate-based optimization, Deep reinforcement learning.
- [381] J. Blitzer, R. McDonald, and F. Pereira. **Domain adaptation with structural correspondence learning.** In D. Jurafsky and E. Gaussier, editors, *Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing, EMNLP2006*, Empirical Methods in Natural Language Processing, pages 120–128, 2006.

- [382] A. Blot, H. H. Hoos, L. Jourdan, M.-E. Kessaci-Marmion, and H. Trautmann. **MO-ParamILS: A Multi-objective Automatic Algorithm Configuration Framework**. In P. Festa, M. Sellmann, and J. Vanschoren, editors, *LION 10*, volume 10079 of *LNCS*, pages 32–47. Springer, Cham, Switzerland, 2016. doi:[10.1007/978-3-319-50349-3_3](https://doi.org/10.1007/978-3-319-50349-3_3).
- [383] A. Blot, L. Jourdan, and M.-E. Kessaci-Marmion. **Automatic design of multi-objective local search algorithms: case study on a bi-objective permutation flowshop scheduling problem**. In P. A. N. Bosman, editor, *GECCO*, pages 227–234. ACM Press, New York, NY, 2017. doi:[10.1145/3071178.3071323](https://doi.org/10.1145/3071178.3071323).
- [384] A. Blot, A. Pernet, L. Jourdan, M.-E. Kessaci-Marmion, and H. H. Hoos. **Automatically Configuring Multi-objective Local Search Using Multi-objective Optimisation**. In H. Trautmann, G. Rudolph, K. Klamroth, O. Schütze, M. M. Wiecek, Y. Jin, and C. Grimme, editors, *EMO*, volume 10173 of *LNCS*, pages 61–76. Springer International Publishing, Cham, Switzerland, 2017.
- [385] A. Blot, M. López-Ibáñez, M.-E. Kessaci-Marmion, and L. Jourdan. **New Initialisation Techniques for Multi-Objective Local Search: Application to the Bi-objective Permutation Flowshop**. In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11101 of *LNCS*, pages 323–334. Springer, Cham, Switzerland, 2018. doi:[10.1007/978-3-319-99253-2_26](https://doi.org/10.1007/978-3-319-99253-2_26).
- [386] A. Blum, editor. *41st Annual Symposium on Foundations of Computer Science, FOCS 2000, 12-14 November 2000, Redondo Beach, California, USA*, 2000. IEEE Computer Society Press.
- [387] C. Blum. **Beam-ACO—Hybridizing Ant Colony Optimization with Beam Search: An Application to Open Shop Scheduling**. *Comput. Oper. Res.*, 32(6):1565–1591, 2005.
- [388] C. Blum. **Beam-ACO for simple assembly line balancing**. *INFORMS Journal on Computing*, 20(4):618–627, 2008. doi:[10.1287/ijoc.1080.0271](https://doi.org/10.1287/ijoc.1080.0271).
- [389] C. Blum and E. Alba, editors. *Genetic and Evolutionary Computation Conference, GECCO 2013, Proceedings, Amsterdam, The Netherlands, July 6-10, 2013*. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8.
- [390] C. Blum and E. Alba, editors. *Genetic and Evolutionary Computation Conference, GECCO 2013, Companion Material Proceedings, Amsterdam, The Netherlands, July 6-10, 2013*. ACM Press, New York, NY, 2013.
- [391] C. Blum and R. Battiti, editors. *4th International Conference, LION 4, Venice, Italy, January 18-22, 2010. Selected Papers*, volume 6073 of *LNCS*. Springer, Heidelberg, Germany, 2010. doi:[10.1007/978-3-642-13800-3](https://doi.org/10.1007/978-3-642-13800-3).
- [392] C. Blum and M. Dorigo. **The hyper-cube framework for ant colony optimization**. *IEEE Transactions on Systems, Man, and Cybernetics – Part B*, 34(2):1161–1172, 2004.
- [393] C. Blum and M. Dorigo. **Search Bias in Ant Colony Optimization: On the Role of Competition-Balanced Systems**. *IEEE Trans. Evol. Comput.*, 9(2):159–174, 2005.
- [394] C. Blum and M. López-Ibáñez. **Ant Colony Optimization**. In *The Industrial Electronics Handbook: Intelligent Systems*. CRC Press, 2nd edition, 2011. ISBN 9781439802830. URL <http://www.crcpress.com/product/isbn/9781439802830>.
- [395] C. Blum and M. Mastrolilli. **Using Branch & Bound Concepts in Construction-Based Metaheuristics: Exploiting the Dual Problem Knowledge**. In T. Bartz-Beielstein, M. J. Blesa, C. Blum, B. Naujoks, A. Roli, G. Rudolph, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 4771 of *LNCS*, pages 123–139. Springer, Heidelberg, Germany, 2007.

- [396] C. Blum and D. Merkle, editors. *Swarm Intelligence—Introduction and Applications*. Natural Computing Series. Springer Verlag, Berlin, Germany, 2008.
- [397] C. Blum and G. Ochoa, editors. *Evolutionary Computation in Combinatorial Optimization – 14th European Conference, EvoCOP 2014, Granada, Spain, April 24-25, 2014, Proceedings*, volume 8600 of *LNCS*. Springer, Heidelberg, Germany, 2014.
- [398] C. Blum and G. Ochoa. **A comparative analysis of two matheuristics by means of merged local optima networks**. *Eur. J. Oper. Res.*, 290(1):36–56, 2021.
- [399] C. Blum and G. R. Raidl. *Hybrid Metaheuristics—Powerful Tools for Optimization*. Artificial Intelligence: Foundations, Theory, and Algorithms. Springer, Berlin, Germany, 2016.
- [400] C. Blum and A. Roli. **Metaheuristics in Combinatorial Optimization: Overview and Conceptual Comparison**. *ACM Comput. Surv.*, 35(3):268–308, 2003.
- [401] C. Blum and A. Roli. **Hybrid metaheuristics: an introduction**. In C. Blum, M. J. Blesa, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics: An emergent approach for optimization*, volume 114 of *Studies in Computational Intelligence*, pages 1–30. Springer, Berlin, Germany, 2008.
- [402] C. Blum and M. Sampels. **An Ant Colony Optimization Algorithm for Shop Scheduling Problems**. *Journal of Mathematical Modelling and Algorithms*, 3(3):285–308, 2004. doi:10.1023/B:JMAA.0000038614.39977.6f.
- [403] C. Blum and M. Yábar Vallès. **Multi-level Ant Colony Optimization for DNA Sequencing by Hybridization**. In F. Almeida et al., editors, *Hybrid Metaheuristics*, volume 4030 of *LNCS*, pages 94–109. Springer, Heidelberg, Germany, 2006. doi:10.1007/11890584_8.
- [404] C. Blum, J. Bautista, and J. Pereira. **Beam-ACO applied to assembly line balancing**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*, pages 96–107. Springer, Heidelberg, Germany, 2006. doi:10.1007/11839088_9.
- [405] C. Blum, C. Cotta, A. J. Fernández, and J. E. Gallardo. **A probabilistic beam search algorithm for the shortest common supersequence problem**. In C. Cotta et al., editors, *Proc. of EvoCOP 2007 – Seventh European Conference on Evolutionary Computation in Combinatorial Optimisation*, volume 4446 of *LNCS*, pages 36–47. Springer, Berlin, Germany, 2007.
- [406] C. Blum, M. J. Blesa, and M. López-Ibáñez. **Beam Search for the Longest Common Subsequence Problem**. Technical Report LSI-08-29, Department LSI, Universitat Politècnica de Catalunya, 2008. Published in *Comput. Oper. Res.* [409].
- [407] C. Blum, M. J. Blesa, A. Roli, and M. Sampels, editors. *Hybrid Metaheuristics: An emergent approach for optimization*, volume 114 of *Studies in Computational Intelligence*. Springer, Berlin, Germany, 2008.
- [408] C. Blum, M. Yábar Vallès, and M. J. Blesa. **An ant colony optimization algorithm for DNA sequencing by hybridization**. *Comput. Oper. Res.*, 35(11):3620–3635, 2008.
- [409] C. Blum, M. J. Blesa, and M. López-Ibáñez. **Beam search for the longest common subsequence problem**. *Comput. Oper. Res.*, 36(12):3178–3186, 2009. doi:10.1016/j.cor.2009.02.005.
- [410] C. Blum, J. Puchinger, G. R. Raidl, and A. Roli. **Hybrid Metaheuristics in Combinatorial Optimization: A Survey**. *Applied Soft Computing*, 11(6):4135–4151, 2011.

- [411] C. Blum, B. Calvo, and M. J. Blesa. **FrogCOL and FrogMIS: New Decentralized Algorithms for Finding Large Independent Sets in Graphs.** *Swarm Intelligence*, 9(2-3): 205–227, 2015. doi:10.1007/s11721-015-0110-1.
Keywords: irace.
- [412] C. Blum, P. Pinacho, M. López-Ibáñez, and J. A. Lozano. **Construct, Merge, Solve & Adapt: A New General Algorithm for Combinatorial Optimization.** *Comput. Oper. Res.*, 68: 75–88, 2016. doi:10.1016/j.cor.2015.10.014.
Keywords: irace, CMSA.
- [413] A. F. Bocchese, C. Fawcett, M. Vallati, A. E. Gerevini, and H. H. Hoos. **Performance robustness of AI planners in the 2014 International Planning Competition.** *AI Communications*, 31(6):445–463, Dec. 2018. doi:10.3233/AIC-170537.
- [414] E. J. W. Boers et al., editors. *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2001*, volume 2037 of *LNCS*. Springer, Heidelberg, Germany, 2001.
- [415] K. D. Boese. *Models for Iterative Global Optimization*. PhD thesis, University of California, Computer Science Department, Los Angeles, CA, 1996.
- [416] K. D. Boese, A. B. Kahng, and S. Muddu. **A New Adaptive Multi-Start Technique for Combinatorial Global Optimization.** *Operations Research Letters*, 16(2):101–113, 1994.
Keywords: big-valley hypothesis, TSP, landscape analysis.
- [417] I. O. Bohachevsky, M. E. Johnson, and M. L. Stein. **Generalized Simulated Annealing for Function Optimization.** *Technometrics*, 28(3):209–217, 1986.
- [418] M. Bohanec. **Decision making: a computer-science and information-technology viewpoint.** *Interdisciplinary Description of Complex Systems*, 7(2):22–37, 2009.
- [419] B. Bollobás. *Random Graphs*. Cambridge University Press, New York, NY, 2nd edition, 2001.
- [420] B. Bonet and S. Koenig, editors. *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, AAAI 2015, Austin, Texas, USA, January 25-30, 2015*, 2015. AAAI Press.
- [421] S. Bonnevey et al., editors. *Artificial Evolution: 12th International Conference, Evolution Artificielle, EA 2015, Lyon, France, October 26-28, 2015. Revised Selected Papers*, volume 9554 of *LNCS*. Springer, Cham, Switzerland, 2016.
- [422] G. Booch, J. E. Rumbaugh, and I. Jacobson. *The Unified Modeling Language User Guide*. Addison-Wesley, 2nd edition, 2005.
- [423] M. Borenstein, L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. *Introduction to Meta-Analysis*. Wiley, 2009.
- [424] Y. Borenstein and A. Moraglio, editors. *Theory and Principled Methods for the Design of Metaheuristics*. Natural Computing Series. Springer, Berlin/Heidelberg, 2014.
- [425] P. C. Borges. **CHESS - Changing Horizon Efficient Set Search: A simple principle for multiobjective optimization.** *J. Heuristics*, 6(3):405–418, 2000.
- [426] P. C. Borges and M. P. Hansen. **A basis for future successes in multiobjective combinatorial optimization.** Technical Report IMM-REP-1998-8, Institute of Mathematical Modelling, Technical University of Denmark, Lyngby, Denmark, 1998.
- [427] A. Borodin and R. El-Yaniv. *Online computation and competitive analysis*. Cambridge University Press, New York, NY, 1998. ISBN 0-521-56392-5.

- [428] E. Boros, P. L. Hammer, and G. Tavares. **Local search heuristics for Quadratic Unconstrained Binary Optimization (QUBO)**. *J. Heuristics*, 13(2):99–132, 2007. doi:[10.1007/s10732-007-9009-3](https://doi.org/10.1007/s10732-007-9009-3).
- [429] A. Bortfeldt and H. Gehring. **A hybrid genetic algorithm for the container loading problem**. *Eur. J. Oper. Res.*, 131(1):143–161, 2001. doi:[10.1016/S0377-2217\(00\)00055-2](https://doi.org/10.1016/S0377-2217(00)00055-2).
- [430] A. Bortfeldt and G. Wäscher. **Constraints in container loading - A state-of-the-art review**. *Eur. J. Oper. Res.*, 229(1):1–20, 2013. doi:[10.1016/j.ejor.2012.12.006](https://doi.org/10.1016/j.ejor.2012.12.006).
- [431] B. E. Boser, I. Guyon, and V. Vapnik. **A Training Algorithm for Optimal Margin Classifiers**. In D. Haussler, editor, *COLT'92*, pages 144–152. ACM Press, 1992. doi:[10.1145/130385.130401](https://doi.org/10.1145/130385.130401).
Annotation: Proposed SVM.
- [432] P. A. N. Bosman, editor. *Genetic and Evolutionary Computation Conference, GECCO 2017, Berlin, Germany, July 15-19, 2017*. ACM Press, New York, NY, 2017.
- [433] P. A. N. Bosman, editor. *Genetic and Evolutionary Computation Conference, GECCO 2017, Berlin, Germany, July 15-19, 2017*. ACM Press, New York, NY, 2017.
- [434] P. A. N. Bosman and D. Thierens. **The balance between proximity and diversity in multiobjective evolutionary algorithms**. *IEEE Trans. Evol. Comput.*, 7(2):174–188, Apr. 2003. doi:[10.1109/TEVC.2003.810761](https://doi.org/10.1109/TEVC.2003.810761).
Keywords: IGD.
Annotation: Proposed IGD but with a different name. The name IGD was given by [676].
- [435] J. Bossek. *smoof: Single and Multi-Objective Optimization Test Functions*, 2016. URL <http://CRAN.R-project.org/package=smoof>. R package version 1.2.
- [436] J. Bossek, P. Kerschke, A. Neumann, M. Wagner, F. Neumann, and H. Trautmann. **Evolving Diverse TSP Instances by Means of Novel and Creative Mutation Operators**. In T. Friedrich, C. Doerr, and D. V. Arnold, editors, *Proceedings of the 15th ACM/SIGEVO Conference on Foundations of Genetic Algorithms*, pages 58–71. ACM, 2019.
- [437] H. M. Botee and E. Bonabeau. **Evolving Ant Colony Optimization**. *Advances in Complex Systems*, 1:149–159, 1998.
- [438] M. Botte and A. Schöbel. **Dominance for multi-objective robust optimization concepts**. *Eur. J. Oper. Res.*, 273(2):430–440, 2019.
- [439] S. Bouamama, C. Blum, and A. Boukerram. **A Population-based Iterated Greedy Algorithm for the Minimum Weight Vertex Cover Problem**. *Applied Soft Computing*, 12(6):1632–1639, 2012.
- [440] K. Bouleimen and H. Lecocq. **A new efficient simulated annealing algorithm for the resource-constrained project scheduling problem and its multiple mode version**. *Eur. J. Oper. Res.*, 149(2):268–281, 2003. doi:[10.1016/S0377-2217\(02\)00761-0](https://doi.org/10.1016/S0377-2217(02)00761-0).
Keywords: multi-mode resource-constrained project scheduling, project scheduling, simulated annealing.
- [441] P. F. Boulos, C. H. Orr, W. de Schaetzen, J. G. Chatila, M. Moore, P. Hsiung, and D. Thomas. **Optimal pump operation of water distribution systems using genetic algorithms**. In *AWWA Distribution System Symp.*, Denver, USA, 2001. American Water Works Association.
- [442] G. Bous, P. Fortemps, F. Glineur, and M. Pirlot. **ACUTA: A novel method for eliciting additive value functions on the basis of holistic preference statements**. *Eur. J. Oper. Res.*, 206(2):435–444, 2010.

- [443] C. Boutilier, editor. *IJCAI 2009, Proceedings of the 21st International Joint Conference on Artificial Intelligence, Pasadena, California, USA, July 11-17, 2009*, 2009. AAAI Press, Menlo Park, CA.
- [444] V. Bowman and J. Joseph. **On the Relationship of the Tchebycheff Norm and the Efficient Frontier of Multiple-Criteria Objectives.** In H. Thiriez and S. Zionts, editors, *Multiple Criteria Decision Making*, volume 130 of *LNEMS*, pages 76–86. Springer, Berlin/Heidelberg, 1976. doi:10.1007/978-3-642-87563-2_5.
- [445] G. E. P. Box and N. R. Draper. *Response surfaces, mixtures, and ridge analyses*. John Wiley & Sons, 2007.
- [446] G. E. P. Box, W. G. Hunter, and J. S. Hunter. *Statistics for experimenters: an introduction to design, data analysis, and model building*. John Wiley & Sons, New York, NY, 1978.
- [447] B. Bozkurt, J. W. Fowler, E. S. Gel, B. Kim, M. Köksalan, and J. Wallenius. **Quantitative comparison of approximate solution sets for multicriteria optimization problems with weighted Tchebycheff preference function.** *Operations Research*, 58(3):650–659, 2010. Annotation: Proposed IPF indicator.
- [448] L. Bradstreet, L. Barone, L. While, S. Huband, and P. Hingston. **Use of the WFG Toolkit and PISA for Comparison of MOEAs.** In *IEEE Symposium on Computational Intelligence in Multicriteria Decision-Making, IEEE MCDM*, pages 382–389, 2007.
- [449] R. I. Brafman, F. Roberts, and A. Tsoukiàs, editors. *Algorithmic Decision Theory, Third International Conference, ADT 2011, Piscataway, New Jersey, USA, October 26-28, 2011*, volume 6992 of *LNAI*. Springer, Heidelberg, Germany, 2011.
- [450] R. I. Brafman, C. Domshlak, P. Haslum, and S. Zilberstein, editors. *Proceedings of the Twenty-Fifth International Conference on Automated Planning and Scheduling, ICAPS 2015, Jerusalem, Israel, June 7-11, 2015*. AAAI Press, Menlo Park, CA, 2015.
- [451] S. C. Brailsford, W. J. Gutjahr, M. S. Rauner, and W. Zeppelzauer. **Combined Discrete-event Simulation and Ant Colony Optimisation Approach for Selecting Optimal Screening Policies for Diabetic Retinopathy.** *Computational Management Science*, 4(1):59–83, 2006.
- [452] A. Brandt. **Multilevel Computations: Review and Recent Developments.** In S. F. McCormick, editor, *Multigrid Methods: Theory, Applications, and Supercomputing, Proceedings of the 3rd Copper Mountain Conference on Multigrid Methods*, volume 110 of *Lecture Notes in Pure and Applied Mathematics*, pages 35–62. Marcel Dekker, New York, NY, 1988.
- [453] J. Branke and K. Deb. **Integrating User Preferences into Evolutionary Multi-Objective Optimization.** In Y. Jin, editor, *Knowledge Incorporation in Evolutionary Computation*, pages 461–477. Springer, Berlin/Heidelberg, 2005. doi:10.1007/978-3-540-44511-1_21.
- [454] J. Branke and J. Elomari. **Simultaneous tuning of metaheuristic parameters for various computing budgets.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 263–264. ACM Press, New York, NY, 2011. doi:10.1145/2001858.2002006. Keywords: meta-optimization, offline parameter optimization.
- [455] J. Branke and J. Elomari. **Racing with a Fixed Budget and a Self-Adaptive Significance Level.** In P. M. Pardalos and G. Nicosia, editors, *LION 7*, volume 7997 of *LNCS*. Springer, Heidelberg, Germany, 2013.
- [456] J. Branke and C. Schmidt. **Faster Convergence by Means of Fitness Estimation.** *Soft Computing*, 9(1):13–20, Jan. 2005. ISSN 1432-7643, 1433-7479. doi:10.1007/s00500-003-0329-4.

- [457] J. Branke, T. Kaussler, and H. Schmeck. **Guidance in evolutionary multi-objective optimization.** *Adv. Eng. Softw.*, 32:499–507, 2001.
- [458] J. Branke, C. Schmidt, and H. Schmeck. **Efficient fitness estimation in noisy environments.** In E. D. Goodman, editor, *Proceedings of the 3rd Annual Conference on Genetic and Evolutionary Computation, GECCO 2001*, pages 243–250. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [459] J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors. *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [460] J. Branke, S. Greco, R. Słowiński, and P. Zielniewicz. **Interactive evolutionary multiobjective optimization driven by robust ordinal regression.** *Bulletin of the Polish Academy of Sciences: Technical Sciences*, 58(3):347–358, 2010. doi:10.2478/v10175-010-0033-3.
- [461] J. Branke, S. Corrente, S. Greco, R. Słowiński, and P. Zielniewicz. **Using Choquet integral as preference model in interactive evolutionary multiobjective optimization.** Technical report, WBS, University of Warwick, 2014.
- [462] J. Branke, S. Corrente, S. Greco, M. Kadziński, M. López-Ibáñez, V. Mousseau, M. Munerato, and R. Słowiński. **Behavior-Realistic Artificial Decision-Makers to Test Preference-Based Multi-objective Optimization Method (Working Group “Machine Decision-Making”).** In S. Greco, K. Klamroth, J. D. Knowles, and G. Rudolph, editors, *Understanding Complexity in Multiobjective Optimization (Dagstuhl Seminar 15031)*, volume 5(1) of *Dagstuhl Reports*, pages 110–116. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2015. doi:10.4230/DagRep.5.1.96.
Keywords: multiple criteria decision making, evolutionary multiobjective optimization.
- [463] J. Branke, S. Greco, R. Słowiński, and P. Zielniewicz. **Learning Value Functions in Interactive Evolutionary Multiobjective Optimization.** *IEEE Trans. Evol. Comput.*, 19(1):88–102, 2015.
- [464] J. Branke, S. Corrente, S. Greco, R. Słowiński, and P. Zielniewicz. **Using Choquet integral as preference model in interactive evolutionary multiobjective optimization.** *Eur. J. Oper. Res.*, 250(3):884–901, 2016. doi:10.1016/j.ejor.2015.10.027.
- [465] J. Branke, S. S. Farid, and N. Shah. **Industry 4.0: a vision for personalized medicine supply chains?** *Cell and Gene Therapy Insights*, 2(2):263–270, 2016. doi:10.18609/cgti.2016.027.
- [466] J. Branke, S. Nguyen, C. W. Pickardt, and M. Zhang. **Automated Design of Production Scheduling Heuristics: A Review.** *IEEE Trans. Evol. Comput.*, 20(1):110–124, 2016. doi:10.1109/TEVC.2015.2429314.
- [467] J.-P. Brans and B. Mareschal. *PROMETHEE-GAIA. Une méthode d’aide à la décision en présence de critères multiples.* Editions Ellipses, Paris, France, 2002. ISBN 2-7298-1253-9.
- [468] J.-P. Brans and B. Mareschal. **PROMETHEE Methods.** In J. R. Figueira, S. Greco, and M. Ehrgott, editors, *Multiple Criteria Decision Analysis, State of the Art Surveys*, chapter 5, pages 163–195. Springer, 2005.
- [469] R. Braune and G. Zäpfel. **Shifting Bottleneck Scheduling for Total Weighted Tardiness Minimization—A Computational Evaluation of Subproblem and Re-optimization Heuristics.** *Comput. Oper. Res.*, 66:130–140, 2016.

- [470] Y. Bravo, J. Ferrer, G. J. Luque, and E. Alba. **Smart Mobility by Optimizing the Traffic Lights: A New Tool for Traffic Control Centers.** In E. Alba, F. Chicano, and G. J. Luque, editors, *Smart Cities (Smart-CT 2016)*, LNCS, pages 147–156. Springer, Cham, Switzerland, 2016. doi:10.1007/978-3-319-39595-1_15.
Keywords: Multi-objective optimization, Smart mobility, Traffic lights planning.
- [471] P. Brazdil, J. Gama, and B. Henery. **Characterizing the applicability of classification algorithms using meta-level learning.** In F. Bergadano and L. De Raedt, editors, *Machine Learning: ECML-94*, volume 784 of LNCS, pages 83–102. Springer, Berlin/Heidelberg, 1994. doi:10.1007/3-540-57868-4_52.
- [472] L. Breiman. **Random Forests.** *Mach. Learn.*, 45(1):5–32, 2001. doi:10.1023/A:1010933404324.
- [473] L. Breiman, J. Friedman, C. J. Stone, and R. A. Olshen. *Classification and regression trees*. CRC Press, 1984.
- [474] M. Brendel and M. Schoenauer. **Instance-based Parameter Tuning for Evolutionary AI Planning.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO Companion*, pages 591–598. ACM Press, New York, NY, 2011. doi:10.1145/2001858.2002053.
- [475] M. Brendel and M. Schoenauer. **Learn-and-Optimize: A Parameter Tuning Framework for Evolutionary AI Planning.** In J.-K. Hao, P. Legrand, P. Collet, N. Monmarché, E. Lutton, and M. Schoenauer, editors, *Artificial Evolution: 10th International Conference, Evolution Artificielle, EA, 2011*, volume 7401 of LNCS, pages 145–155. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-35533-2_13.
- [476] G. Brewka, S. Coradeschi, A. Perini, and P. Traverso, editors. *Proceedings of the 17th European Conference on Artificial Intelligence, ECAI 2006, Riva del Garda, Italy, August 29 - September 1, 2006*. IOS Press, 2006.
- [477] K. Bringmann and T. Friedrich. **Approximating the Least Hypervolume Contributor: NP-Hard in General, But Fast in Practice.** In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of LNCS, pages 6–20. Springer, Heidelberg, Germany, 2009.
Annotation: Extended version published in [485].
- [478] K. Bringmann and T. Friedrich. **Don’t be greedy when calculating hypervolume contributions.** In I. I. Garibay, T. Jansen, R. P. Wiegand, and A. S. Wu, editors, *Proceedings of the Tenth ACM SIGEVO Workshop on Foundations of Genetic Algorithms (FOGA)*, pages 103–112. ACM, 2009. ISBN 978-1-60558-414-0.
Annotation: Extended version published in [480].
- [479] K. Bringmann and T. Friedrich. **Approximating the volume of unions and intersections of high-dimensional geometric objects.** *Computational Geometry*, 43(6–7):601–610, 2010. doi:10.1016/j.comgeo.2010.03.004.
Annotation: Proved that the hypervolume indicator is #P-hard, proposed FPRAS for the hypervolume.
- [480] K. Bringmann and T. Friedrich. **An efficient algorithm for computing hypervolume contributions.** *Evol. Comput.*, 18(3):383–402, 2010.
- [481] K. Bringmann and T. Friedrich. **The Maximum Hypervolume Set Yields Near-optimal Approximation.** In M. Pelikan and J. Branke, editors, *GECCO*, pages 511–518. ACM Press, New York, NY, 2010.
Annotation: Proved that hypervolume approximates the additive ϵ -indicator, converging quickly as N increases, that is, sets that maximize hypervolume are near optimal on additive ϵ too, with the gap diminishing as quickly as $O(1/N)$.

- [482] K. Bringmann and T. Friedrich. **Tight bounds for the approximation ratio of the hypervolume indicator.** In R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 607–616. Springer, Heidelberg, Germany, 2010.
- [483] K. Bringmann and T. Friedrich. **Convergence of Hypervolume-Based Archiving Algorithms I: Effectiveness.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 745–752. ACM Press, New York, NY, 2011. doi:10.1145/2001576.2001678.
Annotation: Extended version published as [486].
- [484] K. Bringmann and T. Friedrich. **Convergence of Hypervolume-Based Archiving Algorithms II: Competitiveness.** In T. Soule and J. H. Moore, editors, *GECCO*, pages 457–464. ACM Press, New York, NY, 2012. doi:10.1145/2330163.2330229.
Annotation: Extended version published as [486].
- [485] K. Bringmann and T. Friedrich. **Approximating the Least Hypervolume Contributor: NP-Hard in General, But Fast in Practice.** *Theor. Comput. Sci.*, 425:104–116, 2012. doi:10.1016/j.tcs.2010.09.026.
- [486] K. Bringmann and T. Friedrich. **Convergence of hypervolume-based archiving algorithms.** *IEEE Trans. Evol. Comput.*, 18(5):643–657, 2014. doi:10.1109/TEVC.2014.2341711.
Keywords: competitive ratio.
Annotation: Proof that all nondecreasing $(\mu + \lambda)$ archiving algorithms with $\lambda < \mu$ are ineffective.
- [487] K. Bringmann, T. Friedrich, F. Neumann, and M. Wagner. **Approximation-guided Evolutionary Multi-objective Optimization.** In T. Walsh, editor, *Proc. of IJCAI-11*, pages 1198–1203. IJCAI/AAAI Press, Menlo Park, CA, 2011.
- [488] K. Bringmann, T. Friedrich, and P. Klitzke. **Generic postprocessing via subset selection for hypervolume and epsilon-indicator.** In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 518–527. Springer, Heidelberg, Germany, 2014.
- [489] K. Bringmann, T. Friedrich, and P. Klitzke. **Two-dimensional subset selection for hypervolume and epsilon-indicator.** In C. Igel and D. V. Arnold, editors, *GECCO*. ACM Press, New York, NY, 2014. doi:10.1145/2576768.2598276.
- [490] K. Bringmann, S. Cabello, and M. T. M. Emmerich. **Maximum volume subset selection for anchored boxes.** *Arxiv preprint arXiv:1803.00849*, 2018. doi:10.48550/arXiv.1803.00849.
Keywords: hypervolume subset selection.
- [491] A. Britto and A. Pozo. **Using archiving methods to control convergence and diversity for many-objective problems in particle swarm optimization.** In *IEEE CEC*, pages 1–8, Piscataway, NJ, 2012. IEEE Press. doi:10.1109/CEC.2012.6256149.
- [492] D. R. Broad, G. C. Dandy, and H. R. Maier. **A Metamodeling Approach to Water Distribution System Optimization.** In *6th Annual Symposium on Water Distribution Systems Analysis*. ASCE, June 2004.
- [493] E. Brochu, V. Cora, and N. de Freitas. **A Tutorial on Bayesian Optimization of Expensive Cost Functions, with Application to Active User Modeling and Hierarchical Reinforcement Learning.** *Arxiv preprint arXiv:1012.2599*, Dec. 2010. URL <https://arxiv.org/abs/1012.2599>.
- [494] D. Brockhoff. **A Bug in the Multiobjective Optimizer IBEA: Salutory Lessons for Code Release and a Performance Re-Assessment.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 187–201. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-15934-8_13.

- [495] D. Brockhoff and T. Tušar. **Benchmarking algorithms from the platypus framework on the biobjective bbob-biobj testbed.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1905–1911. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.
Keywords: unbounded archive.
- [496] D. Brockhoff and E. Zitzler. **Are All Objectives Necessary? On Dimensionality Reduction in Evolutionary Multiobjective Optimization.** In T. P. Runarsson, H.-G. Beyer, E. K. Burke, J.-J. Merelo, D. Whitley, and X. Yao, editors, *Parallel Problem Solving from Nature – PPSN IX*, volume 4193 of *LNCS*, pages 533–542. Springer, Heidelberg, Germany, 2006.
- [497] D. Brockhoff and E. Zitzler. **Dimensionality Reduction in Multiobjective Optimization: The Minimum Objective Subset Problem.** In K.-H. Waldmann and U. M. Stocker, editors, *Operations Research Proceedings 2006*, pages 423–429. Springer, Berlin/Heidelberg, 2007. doi:10.1007/978-3-540-69995-8_68.
Keywords: objective reduction.
- [498] D. Brockhoff and E. Zitzler. **Improving hypervolume-based multiobjective evolutionary algorithms by using objective reduction methods.** In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 2086–2093, Piscataway, NJ, 2007. IEEE Press. doi:10.1109/CEC.2007.4424730.
Keywords: objective reduction.
- [499] D. Brockhoff and E. Zitzler. **Objective Reduction in Evolutionary Multiobjective Optimization: Theory and Applications.** *Evol. Comput.*, 17(2):135–166, 2009. doi:10.1162/evco.2009.17.2.135.
- [500] D. Brockhoff, T. Friedrich, N. Hebbinghaus, C. Klein, F. Neumann, and E. Zitzler. **Do Additional Objectives Make a Problem Harder?** In D. Thierens et al., editors, *GECCO*, pages 765–772. ACM Press, New York, NY, 2007. doi:10.1145/1276958.1277114.
- [501] D. Brockhoff, D. K. Saxena, K. Deb, and E. Zitzler. **On Handling a Large Number of Objectives A Posteriori and During Optimization.** In J. D. Knowles, D. Corne, K. Deb, and D. R. Chair, editors, *Multiobjective Problem Solving from Nature*, Natural Computing Series, pages 377–403. Springer, Berlin/Heidelberg, 2008. doi:10.1007/978-3-540-72964-8_18.
- [502] D. Brockhoff, M. López-Ibáñez, B. Naujoks, and G. Rudolph. **Runtime Analysis of Simple Interactive Evolutionary Biobjective Optimization Algorithms.** In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part I*, volume 7491 of *LNCS*, pages 123–132. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32937-1_13.
- [503] D. Brockhoff, T. Wagner, and H. Trautmann. **On the properties of the R2 indicator.** In T. Soule and J. H. Moore, editors, *GECCO*, pages 465–472. ACM Press, New York, NY, 2012.
Annotation: Proof that R2 is weakly Pareto compliant.
- [504] D. Brockhoff, J. Bader, L. Thiele, and E. Zitzler. **Directed Multiobjective Optimization Based on the Weighted Hypervolume Indicator.** *Journal of Multi-Criteria Decision Analysis*, 20(5-6):291–317, 2013. doi:10.1002/mcda.1502.
Keywords: hypervolume, preference-based search, multi objective optimization, evolutionary algorithm.
- [505] D. Brockhoff, T. Wagner, and H. Trautmann. **R2 indicator-based multiobjective search.** *Evol. Comput.*, 23(3):369–395, 2015.
- [506] D. Brockhoff, T. Tušar, D. Tušar, T. Wagner, N. Hansen, and A. Auger. **Biobjective performance assessment with the COCO platform.** *Arxiv preprint arXiv:1605.01746*, 2016. doi:10.48550/arXiv.1605.01746.

- [507] D. Brockhoff, R. Calandra, M. López-Ibáñez, F. Neumann, and S. Ulaganathan. **Meta-modeling for (interactive) multi-objective optimization (WG5)**. In K. Klamroth, J. D. Knowles, G. Rudolph, and M. M. Wiecek, editors, *Personalized Multiobjective Optimization: An Analytics Perspective (Dagstuhl Seminar 18031)*, volume 8(1) of *Dagstuhl Reports*, pages 85–94. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2018. doi:[10.4230/DagRep.8.1.33](https://doi.org/10.4230/DagRep.8.1.33).
Keywords: multiple criteria decision making, evolutionary multiobjective optimization.
- [508] C. E. Brodley, editor. *Machine Learning, Proceedings of the Twenty-first International Conference, ICML 2004, Banff, Alberta, Canada, July 4-8, 2004*, New York, NY, 2004. ACM Press.
- [509] C. G. Broyden. **The Convergence of a Class of Double-rank Minimization Algorithms: 2. The New Algorithm**. *IMA Journal of Applied Mathematics*, 6(3):222–231, Sept. 1970. doi:[10.1093/imamat/6.3.222](https://doi.org/10.1093/imamat/6.3.222).
Keywords: BFGS.
Annotation: One of the four papers that proposed BFGS.
- [510] C. G. Broyden. **The Convergence of a Class of Double-rank Minimization Algorithms 1. General Considerations**. *IMA Journal of Applied Mathematics*, 6(1):76–90, Mar. 1970. doi:[10.1093/imamat/6.1.76](https://doi.org/10.1093/imamat/6.1.76).
Keywords: Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm.
- [511] P. Brucker, J. Hurink, and F. Werner. **Improving Local Search Heuristics for some Scheduling Problems — Part I**. *Discrete Applied Mathematics*, 65(1–3):97–122, 1996.
- [512] P. Brucker, J. Hurink, and F. Werner. **Improving Local Search Heuristics for some Scheduling Problems — Part II**. *Discrete Applied Mathematics*, 72(1–2):47–69, 1997.
- [513] A. Brum and M. Ritt. **Automatic Design of Heuristics for Minimizing the Makespan in Permutation Flow Shops**. In *Proceedings of the 2018 Congress on Evolutionary Computation (CEC 2018)*, pages 1–8, Piscataway, NJ, 2018. IEEE Press. doi:[10.1109/CEC.2018.8477787](https://doi.org/10.1109/CEC.2018.8477787).
- [514] A. Brum and M. Ritt. **Automatic Algorithm Configuration for the Permutation Flow Shop Scheduling Problem Minimizing Total Completion Time**. In A. Liefoghe and M. López-Ibáñez, editors, *Proceedings of EvoCOP 2018 – 18th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 10782 of *LNCS*, pages 85–100. Springer, Heidelberg, Germany, 2018. doi:[10.1007/978-3-319-77449-7_6](https://doi.org/10.1007/978-3-319-77449-7_6).
- [515] M. J. Brusco, L. W. Jacobs, and G. M. Thompson. **A Morphing Procedure to Supplement a Simulated Annealing Heuristic for Cost- and Coverage-correlated Set Covering Problems**. *Annals of Operations Research*, 86:611–627, 1999.
- [516] J. T. Buchanan. **An experimental evaluation of interactive MCDM methods and the decision making process**. *J. Oper. Res. Soc.*, 45(9):1050–1059, 1994.
- [517] J. T. Buchanan. **A naive approach for solving MCDM problems: the GUESS method**. *J. Oper. Res. Soc.*, 48:202–206, 1997.
- [518] J. T. Buchanan and J. L. Corner. **The effects of anchoring in interactive MCDM solution methods**. *Comput. Oper. Res.*, 24(10):907–918, Oct. 1997. doi:[10.1016/S0305-0548\(97\)00014-2](https://doi.org/10.1016/S0305-0548(97)00014-2).
- [519] A. L. Buchsbaum and M. T. Goodrich. **Three-Dimensional Layers of Maxima**. *Algorithmica*, 39:275–289, 2004. doi:[10.1007/s00453-004-1082-5](https://doi.org/10.1007/s00453-004-1082-5).
Annotation: Proposed the best-known $O(n \log n)$ algorithm for 3D.

- [520] T. N. Bui and J. R. Rizzo, Jr. **Finding Maximum Cliques with Distributed Ants.** In K. Deb et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2004, Part I*, volume 3102 of *LNCS*, pages 24–35. Springer, Heidelberg, Germany, 2004.
- [521] B. Bullnheimer, R. F. Hartl, and C. Strauss. **An Improved Ant System Algorithm for the Vehicle Routing Problem.** *Annals of Operations Research*, 89:319–328, 1999.
- [522] B. Bullnheimer, R. F. Hartl, and C. Strauss. **A new rank-based version of the Ant System: A computational study.** *Central European Journal for Operations Research and Economics*, 7(1):25–38, 1999.
- [523] W. Burgard and D. Roth, editors. *Proceedings of the Twenty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2011, San Francisco, California, USA, August 07-11, 2011*, 2011. AAAI Press.
- [524] L. Buriol, P. M. França, and P. Moscato. **A New Memetic Algorithm for the Asymmetric Traveling Salesman Problem.** *J. Heuristics*, 10(5):483–506, 2004.
- [525] R. E. Burkard and U. Fincke. **The asymptotic probabilistic behaviour of quadratic sum assignment problems.** *Zeitschrift für Operations Research*, 27(1):73–81, 1983.
- [526] R. E. Burkard and F. Rendl. **A Thermodynamically Motivated Simulation Procedure for Combinatorial Optimization Problems.** *Eur. J. Oper. Res.*, 17(2):169–174, 1984. doi:10.1016/0377-2217(84)90231-5.
Keywords: 2-exchange delta evaluation for QAP.
- [527] R. E. Burkard, S. E. Karisch, and F. Rendl. **QAPLIB—a Quadratic Assignment Problem Library.** *J. Glob. Optim.*, 10(4):391–403, 1997.
- [528] R. E. Burkard, E. Çela, P. M. Pardalos, and L. S. Pitsoulis. **The quadratic assignment problem.** In P. M. Pardalos and D.-Z. Du, editors, *Handbook of Combinatorial Optimization*, volume 2, pages 241–338. Kluwer Academic Publishers, 1998.
- [529] E. K. Burke and Y. Bykov. **The Late Acceptance Hill-Climbing Heuristic.** Technical Report CSM-192, University of Stirling, 2012.
- [530] E. K. Burke and Y. Bykov. **The Late Acceptance Hill-Climbing Heuristic.** *Eur. J. Oper. Res.*, 258(1):70–78, 2017.
- [531] E. K. Burke and W. Erben, editors. *Practice and Theory of Automated Timetabling III, Third International Conference, PATAT 2000, Konstanz, Germany, August 16-18, 2000, Selected Papers*, volume 2079 of *LNCS*. Springer, 2000.
- [532] E. K. Burke and G. Kendall, editors. *Search Methodologies: Introductory Tutorials in Optimization and Decision Support Techniques*. Springer, Boston, MA, 2005. doi:10.1007/0-387-28356-0.
- [533] E. K. Burke, M. R. Hyde, G. Kendall, and J. R. Woodward. **Automatic Heuristic Generation with Genetic Programming: Evolving a Jack-of-all-trades or a Master of One.** In D. Thierens et al., editors, *GECCO*, pages 1559–1565. ACM Press, New York, NY, 2007. doi:10.1145/1276958.1277273.
- [534] E. K. Burke, M. R. Hyde, G. Kendall, and J. R. Woodward. **A Genetic Programming Hyper-Heuristic Approach for Evolving 2-D Strip Packing Heuristics.** *IEEE Trans. Evol. Comput.*, 14(6):942–958, 2010. doi:10.1109/TEVC.2010.2041061.
- [535] E. K. Burke, M. R. Hyde, and G. Kendall. **Grammatical Evolution of Local Search Heuristics.** *IEEE Trans. Evol. Comput.*, 16(7):406–417, 2012. doi:10.1109/TEVC.2011.2160401.

- [536] E. K. Burke, M. R. Hyde, G. Kendall, and J. R. Woodward. **Automating the Packing Heuristic Design Process with Genetic Programming.** *Evol. Comput.*, 20(1):63–89, 2012. doi:10.1162/evco_a_00044.
Keywords: one-, two-, or three-dimensional knapsack and bin packing problems.
- [537] E. K. Burke, M. Gendreau, M. R. Hyde, G. Kendall, G. Ochoa, E. Özcan, and R. Qu. **Hyper-heuristics: A Survey of the State of the Art.** *J. Oper. Res. Soc.*, 64(12):1695–1724, 2013. doi:10.1057/jors.2013.71.
- [538] E. K. Burke, M. R. Hyde, G. Kendall, G. Ochoa, E. Özcan, and J. R. Woodward. **A Classification of Hyper-Heuristic Approaches: Revisited.** In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 272 of *International Series in Operations Research & Management Science*, chapter 14, pages 453–477. Springer, 2019. doi:10.1007/978-3-319-91086-4_14.
- [539] E. Buson, R. Roberti, and P. Toth. **A Reduced-Cost Iterated Local Search Heuristic for the Fixed-Charge Transportation Problem.** *Operations Research*, 62(5):1095–1106, 2014.
- [540] M. Buzdalov. **Towards better estimation of statistical significance when comparing evolutionary algorithms.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1782–1788. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.3326899.
- [541] R. Caballero, F. Ruiz, and R. E. Steuer, editors. *Advances in Multiple Objective and Goal Programming*, volume 455 of *LNEMS*. Springer, Heidelberg, Germany, 1997.
- [542] R. Caballero, M. Luque, J. Molina, and F. Ruiz. **PROMOIN: An Interactive System for Multiobjective Programming.** *Information Technologies and Decision Making*, 1:635–656, 2002.
Keywords: preferences, multi interactive methods framework.
- [543] S. Cagnoni et al., editors. *Real-World Applications of Evolutionary Computing, EvoWorkshops 2000: EvoIASP, EvoSCONDI, EvoTel, EvoSTIM, EvoROB, and EvoFlight, Edinburgh, Scotland, UK, April 17, 2000, Proceedings*, volume 1803 of *LNCS*. Springer, Heidelberg, Germany, 2000.
- [544] S. Cagnoni et al., editors. *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2002*, volume 2279 of *LNCS*. Springer, Heidelberg, Germany, 2002.
- [545] S. Cagnoni et al., editors. *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2003*, volume 2611 of *LNCS*. Springer, Heidelberg, Germany, 2003.
- [546] S. Cahon, N. Melab, and E.-G. Talbi. **ParadisEO: A Framework for the Reusable Design of Parallel and Distributed Metaheuristics.** *J. Heuristics*, 10(3):357–380, 2004. doi:10.1023/B:HEUR.0000026900.92269.ec.
- [547] X. Cai, Y. Li, Z. Fan, and Q. Zhang. **An external archive guided multiobjective evolutionary algorithm based on decomposition for combinatorial optimization.** *IEEE Trans. Evol. Comput.*, 19(4):508–523, 2015.
- [548] X. Cai, Y. Xiao, M. Li, H. Hu, H. Ishibuchi, and X. Li. **A grid-based inverted generational distance for multi/many-objective optimization.** *IEEE Trans. Evol. Comput.*, 25(1):21–34, 2021.
Annotation: weakly Pareto-compliant indicator.
- [549] X. Cai, Y. Xiao, Z. Li, Q. Sun, H. Xu, M. Li, and H. Ishibuchi. **A kernel-based indicator for multi/many-objective optimization.** *IEEE Trans. Evol. Comput.*, 2021.

- [550] Z. Cai, H. Huang, Y. Qin, and X. Ma. **Ant Colony Optimization Based on Adaptive Volatility Rate of Pheromone Trail**. *International Journal of Communications, Network and System Sciences*, 2(8):792–796, 2009.
- [551] L. Caires, G. F. Italiano, L. Monteiro, C. Palamidessi, and M. Yung, editors. *Proceedings of the 32nd International Colloquium on Automata, Languages and Programming, ICALP 2005*, volume 3580 of *LNCS*. Springer, Heidelberg, Germany, 2005.
- [552] P. Calabar and T. C. Son, editors. *12th International Conference, LPNMR 2013, Corunna, Spain, September 15-19, 2013. Proceedings*, volume 8148 of *LNAI*. Springer, Heidelberg, Germany, 2013.
- [553] B. Calvo, O. M. Shir, J. Ceberio, C. Doerr, H. Wang, T. Bäck, and J. A. Lozano. **Bayesian Performance Analysis for Black-box Optimization Benchmarking**. In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1789–1797. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.
Keywords: bayesian inference, benchmarking, black-box optimization, evolutionary algorithms, performance measures, plackett-luce model.
- [554] C. L. Camacho-Villalón, M. Dorigo, and T. Stützle. **Why the Intelligent Water Drops Cannot Be Considered as a Novel Algorithm**. In M. Dorigo, M. Birattari, A. L. Christensen, A. Reina, and V. Trianni, editors, *ANTS 2018*, volume 11172 of *LNCS*, pages 302–314. Springer, Heidelberg, Germany, 2018.
- [555] C. L. Camacho-Villalón, M. Dorigo, and T. Stützle. **The intelligent water drops algorithm: why it cannot be considered a novel algorithm**. *Swarm Intelligence*, 13:173–192, 2019.
- [556] C. L. Camacho-Villalón, T. Stützle, and M. Dorigo. **Grey Wolf, Firefly and Bat Algorithms: Three Widespread Algorithms that Do Not Contain Any Novelty**. In M. Dorigo, T. Stützle, M. J. Blesa, C. Blum, H. Hamann, and M. K. Heinrich, editors, *Swarm Intelligence, 12th International Conference, ANTS 2020*, volume 12421 of *LNCS*, pages 121–133. Springer, Heidelberg, Germany, 2020.
- [557] C. L. Camacho-Villalón, T. Stützle, and M. Dorigo. **PSO-X: A Component-Based Framework for the Automatic Design of Particle Swarm Optimization Algorithms**. *IEEE Trans. Evol. Comput.*, 26(3):402–416, 2021. doi:10.1109/TEVC.2021.3102863.
- [558] C. L. Camacho-Villalón, T. Stützle, and M. Dorigo. **PSO-X: A Component-Based Framework for the Automatic Design of Particle Swarm Optimization Algorithms**. Technical Report TR/IRIDIA/2021-002, IRIDIA, Université Libre de Bruxelles, Belgium, 2021. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2021-002.pdf>.
Annotation: Published as [557].
- [559] C. L. Camacho-Villalón, T. Stützle, and M. Dorigo. **Cuckoo Search $\equiv (\mu + \lambda)$ -Evolution Strategy – A Rigorous Analysis of an Algorithm That Has Been Misleading the Research Community for More Than 10 Years and Nobody Seems to Have Noticed**. Technical Report TR/IRIDIA/2021-006, IRIDIA, Université Libre de Bruxelles, Belgium, 2021. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2021-006.pdf>.
- [560] C. L. Camacho-Villalón, M. Dorigo, and T. Stützle. **An analysis of why cuckoo search does not bring any novel ideas to optimization**. *Comput. Oper. Res.*, page 105747, 2022.
- [561] C. L. Camacho-Villalón, M. Dorigo, and T. Stützle. **Exposing the grey wolf, moth-flame, whale, firefly, bat, and antlion algorithms: six misleading optimization techniques inspired by bestial metaphors**. *International Transactions in Operational Research*, 2022. doi:10.1111/itor.13176.
- [562] E. Cambria, B. Schuller, Y. Xia, and C. Havasi. **New avenues in opinion mining and sentiment analysis**. *IEEE Intelligent Systems*, 28(2):15–21, 2013. doi:10.1109/MIS.2013.30.

- [563] A. M. Campbell and P. C. Jones. **Prepositioning supplies in preparation for disasters.** *Eur. J. Oper. Res.*, 209(2):156–165, 2011.
- [564] F. Campelo and E. F. Wanner. **Sample size calculations for the experimental comparison of multiple algorithms on multiple problem instances.** *J. Heuristics*, 26(6):851–883, 2020. doi:[10.1007/s10732-020-09454-w](https://doi.org/10.1007/s10732-020-09454-w).
- [565] F. Campelo, A. R. Trindade, and M. López-Ibáñez. **Pseudoreplication in Racing Methods for Tuning Metaheuristics.** In preparation, 2017.
- [566] F. Campelo, L. S. Batista, and C. Aranha. **The MOEADr Package: A Component-Based Framework for Multiobjective Evolutionary Algorithms Based on Decomposition.** *Journal of Statistical Software*, 92, 2020. doi:[10.18637/jss.v092.i06](https://doi.org/10.18637/jss.v092.i06).
- [567] P. Campigotto and A. Passerini. **Adapting to a realistic decision maker: experiments towards a reactive multi-objective optimizer.** In C. Blum and R. Battiti, editors, *LION 4*, volume 6073 of *LNCS*, pages 338–341. Springer, Heidelberg, Germany, 2010. doi:[10.1007/978-3-642-13800-3](https://doi.org/10.1007/978-3-642-13800-3).
- [568] J. L. Cano Rodríguez et al. **poliastro: Astrodynamics in Python.** Zenodo, 2015.
- [569] E. Cantú-Paz. *Efficient and Accurate Parallel Genetic Algorithms.* Kluwer Academic Publishers, Boston, MA, 2000.
- [570] E. Cantú-Paz et al., editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2003, Part I*, volume 2723 of *LNCS*. Springer, Heidelberg, Germany, 2003.
- [571] L. Cao, W. Kusters, and J. Lijffijt, editors. *Proceedings of the 32nd Benelux Conference on Artificial Intelligence, BNAIC 2020, Leiden, The Netherlands, 19-20 November 2020*, 2020. URL <https://bnaic.liacs.leidenuniv.nl/wordpress/wp-content/uploads/bnaic2020proceedings.pdf>.
- [572] Z. Cao, S. Jiang, J. Zhang, and H. Guo. **A unified framework for vehicle rerouting and traffic light control to reduce traffic congestion.** *IEEE Transactions on Intelligent Transportation Systems*, 18(7):1958–1973, 2017.
- [573] G. Caporossi. **Variable Neighborhood Search for Extremal Vertices : The AutoGraphX-III System.** *Comput. Oper. Res.*, 78:431–438, 2017.
- [574] F. Caraffini, A. V. Kononova, and D. Corne. **Infeasibility and structural bias in differential evolution.** *Inf. Sci.*, 496:161–179, 2019. doi:[10.1016/j.ins.2019.05.019](https://doi.org/10.1016/j.ins.2019.05.019).
- [575] I. Caragiannis, A. D. Procaccia, and N. Shah. **When Do Noisy Votes Reveal the Truth?** In M. J. Kearns, R. P. McAfee, and É. Tardos, editors, *Proceedings of the Fourteenth ACM Conference on Electronic Commerce*, pages 143–160. ACM Press, New York, NY, 2013. doi:[10.1145/2482540.2482570](https://doi.org/10.1145/2482540.2482570).
Keywords: computer social choice, mallows model, sample complexity.
- [576] P. Cardoso, M. Jesus, and A. Marquez. **MONACO: multi-objective network optimisation based on an ACO.** In *Proc. X Encuentros de Geometría Computacional*, Seville, Spain, 2003.
- [577] J. Carlier. **The One-machine Sequencing Problem.** *Eur. J. Oper. Res.*, 11(1):42–47, 1982.
- [578] W. B. Carlton and J. W. Barnes. **Solving the traveling-salesman problem with time windows using tabu search.** *IIE Transactions*, 28:617–629, 1996.
- [579] R. Carnell. *lhs: Latin Hypercube Samples*, 2016. URL <http://r-forge.r-project.org/projects/lhs/>. R package version 0.14.

- [580] Y. Caseau and F. Laburthe. **Heuristics for large constrained vehicle routing problems.** *J. Heuristics*, 5(3):281–303, 1999.
- [581] Y. Caseau, G. Silverstein, and F. Laburthe. **Learning Hybrid Algorithms for Vehicle Routing Problems.** *Theory and Practice of Logic Programming*, 1(6):779–806, 2001.
- [582] M. Caserta and S. Voß, editors. *Proceedings of MIC 2009, the 8th Metaheuristics International Conference*, Hamburg, Germany, 2010. University of Hamburg.
- [583] P. Castillo and J. L. Jiménez Laredo, editors. *Applications of Evolutionary Computation – 24th International Conference, EvoApplications 2021, Held as Part of EvoStar 2021, Virtual Event, April 7-9, 2021, Proceedings*, volume 12694 of LNCS. Springer, Cham, Switzerland, 2021.
- [584] D. Cattaruzza, N. Absi, D. Feillet, and D. Vigo. **An Iterated Local Search for the Multi-commodity Multi-trip Vehicle Routing Problem with Time Windows.** *Comput. Oper. Res.*, 51:257–267, 2014.
- [585] M. Cattolico et al., editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2006*. ACM Press, New York, NY, 2006.
- [586] A. M. Caunhye, X. Nie, and S. Pokharel. **Optimization models in emergency logistics: A literature review.** *Socio-Economic Planning Sciences*, 46(1):4–13, 2012.
- [587] CCIE. *Proceedings of the 2010 International Conference on Computing, Control and Industrial Engineering*, Los Alamitos, CA, 2010. IEEE Computer Society Press.
- [588] J. Ceberio, E. Irurozki, A. Mendiburu, and J. A. Lozano. **A distance-based ranking model estimation of distribution algorithm for the flowshop scheduling problem.** *IEEE Trans. Evol. Comput.*, 18(2):286–300, 2014. doi:10.1109/TEVC.2013.2260548.
Keywords: Estimation of distribution algorithms, Generalized Mallows model, Permutation flowshop scheduling problem, Permutations-based optimization problems.
- [589] J. Ceberio, A. Mendiburu, and J. A. Lozano. **Kernels of Mallows Models for Solving Permutation-based Problems.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 505–512. ACM Press, New York, NY, 2015.
- [590] E. Çela. *The Quadratic Assignment Problem: Theory and Algorithms*. Kluwer Academic Publishers, Dordrecht, The Netherlands, 1998.
- [591] M. Ceriotti and M. Vasile. **Automated Multigravity Assist Trajectory Planning with a Modified Ant Colony Algorithm.** *Journal of Aerospace Computing, Information, and Communication*, 7(9):261–293, 2010. doi:10.2514/1.48448.
- [592] S. Ceschia and A. Schaerf. **Modeling and solving the dynamic patient admission scheduling problem under uncertainty.** *Artificial Intelligence in Medicine*, 56(3):199–205, 2012. doi:10.1016/j.artmed.2012.09.001.
Keywords: F-race.
- [593] S. Ceschia, L. Di Gaspero, and A. Schaerf. **Design, Engineering, and Experimental Analysis of a Simulated Annealing Approach to the Post-Enrolment Course Timetabling Problem.** *Comput. Oper. Res.*, 39(7):1615–1624, 2012.
- [594] S. Ceschia, A. Schaerf, and T. Stützle. **Local Search Techniques for a Routing-packing Problem.** *Computers & Industrial Engineering*, 66(4):1138–1149, 2013.
- [595] A. Cesta, A. Oddi, and S. F. Smith. **Iterative Flattening: A Scalable Method for Solving Multi-Capacity Scheduling Problems.** In H. A. Kautz and B. W. Porter, editors, *Proc. of AAAI 2000 – Seventeenth National Conference on Artificial Intelligence*, pages 742–747. AAAI Press/MIT Press, Menlo Park, CA, 2000.

- [596] U. K. Chakraborty, editor. *Advances in differential evolution*. Springer, Heidelberg, Germany, 2008.
- [597] S. Chand and M. Wagner. **Evolutionary many-objective optimization: A quick-start guide**. *Surveys in Operations Research and Management Science*, 20(2):35–42, 2015. doi:10.1016/j.sorms.2015.08.001.
- [598] S. T. H. Chang. **Optimizing the Real Time Operation of a Pumping Station at a Water Filtration Plant using Genetic Algorithms**. Honors thesis, Department of Civil and Environmental Engineering, The University of Adelaide, 1999.
- [599] T.-J. Chang, N. Meade, J. E. Beasley, and Y. M. Sharaiha. **Heuristics for cardinality constrained portfolio optimisation**. *Comput. Oper. Res.*, 27(13):1271–1302, 2000.
Keywords: Portfolio optimisation, CCMVPOP, Efficient frontier.
- [600] D. V. Chase and L. E. Ormsbee. **Optimal pump operation of water distribution systems with multiple storage tanks**. In *Proceedings of American Water Works Association Computer Specialty Conference*, pages 205–214, Denver, USA, 1989. AWWA.
- [601] D. V. Chase and L. E. Ormsbee. **An alternate formulation of time as a decision variable to facilitate real-time operation of water supply systems**. In *Proceedings of the 18th Annual Conference of Water Resources Planning and Management*, pages 923–927, New York, NY, 1991. ASCE.
- [602] D. V. Chase and L. E. Ormsbee. **Computer-generated pumping schedules for satisfying operation objectives**. *J. Am. Water Works Assoc.*, 85(7):54–61, 1993.
- [603] S. Chaudhuri and K. Deb. **An interactive evolutionary multi-objective optimization and decision making procedure**. *Applied Soft Computing*, 10(2):496–511, 2010.
- [604] C. H. Che, W. Huang, A. Lim, and W. Zhu. **The Multiple Container Loading Cost Minimization Problem**. *Eur. J. Oper. Res.*, 214(3):501–511, 2011. doi:10.1016/j.ejor.2011.04.017.
Keywords: S-GRASP.
- [605] P. C. Cheeseman, B. Kanefsky, and W. M. Taylor. **Where the Really Hard Problems Are**. In J. Mylopoulos and R. Reiter, editors, *Proc. of IJCAI-91*, pages 331–340. Morgan Kaufmann Publishers, 1995.
- [606] R. Chelouah and P. Siarry. **Tabu search applied to global optimization**. *Eur. J. Oper. Res.*, 123(2):256–270, 2000.
- [607] D. Chen, M. Buzdalov, C. Doerr, and N. Dang. **Using Automated Algorithm Configuration for Parameter Control**. In F. Chicano, T. Friedrich, T. Kötzing, and F. Rothlauf, editors, *Proceedings of the 17th ACM/SIGEVO Conference on Foundations of Genetic Algorithms*, pages 38–49. ACM, 2023. doi:10.1145/3594805.3607127.
- [608] F. Chen, Y. Gao, Z.-q. Chen, and S.-f. Chen. **SCGA: Controlling genetic algorithms with Sarsa(0)**. In *Computational Intelligence for Modelling, Control and Automation, 2005 and International Conference on Intelligent Agents, Web Technologies and Internet Commerce, International Conference on*, volume 1, pages 1177–1183. IEEE, 2005. doi:10.1109/CIMCA.2005.1631422.
- [609] H. Chen, R. H. L. Chiang, and V. C. Storey. **Business Intelligence and Analytics: From Big Data to Big Impact**. *MIS Quarterly*, 36(4):1165–1188, 2012.

- [610] L. Chen, X. H. Xu, and Y. X. Chen. **An adaptive ant colony clustering algorithm.** In I. Cloete, K.-P. Wong, and M. Berthold, editors, *Proceedings of the International Conference on Machine Learning and Cybernetics*, pages 1387–1392. IEEE Press, 2004.
- [611] L. Chen, B. Xin, J. Chen, and J. Li. **A virtual-decision-maker library considering personalities and dynamically changing preference structures for interactive multiobjective optimization.** In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 636–641, Piscataway, NJ, 2017. IEEE Press. doi:10.1109/CEC.2017.7969370.
Keywords: machine DM, interactive EMOA.
- [612] N. Chen, W.-N. Chen, Y.-J. Gong, Z.-H. Zhan, J. Zhang, Y. Li, and Y.-S. Tan. **An evolutionary algorithm with double-level archives for multiobjective optimization.** *IEEE Trans. Cybern.*, 45(9):1851–1863, 2014.
- [613] R. Chen, K. Li, and X. Yao. **Dynamic Multiobjectives Optimization With a Changing Number of Objectives.** *IEEE Trans. Evol. Comput.*, 22(1):157–171, 2017. doi:10.1109/TEVC.2017.2669638.
Annotation: two co-evolving populations (two archive).
- [614] R.-M. Chen and F.-R. Hsieh. **An exchange local search heuristic based scheme for permutation flow shop problems.** *Applied Mathematics & Information Sciences*, 8(1): 209–215, 2014.
- [615] T. Chen, M. Li, and X. Yao. **Standing on the shoulders of giants: Seeding search-based multi-objective optimization with prior knowledge for software service composition.** *Information and Software Technology*, 114:155–175, 2019.
Annotation: Example of deterioration in archiving.
- [616] W. Chen, H. Ishibuchi, and K. Shang. **Modified Distance-based Subset Selection for Evolutionary Multi-objective Optimization Algorithms.** In *Proceedings of the 2020 Congress on Evolutionary Computation (CEC 2020)*, pages 1–8, Piscataway, NJ, 2020. IEEE Press.
Keywords: IGD+.
- [617] W. Chen, H. Ishibuchi, and K. Shang. **Clustering-Based Subset Selection in Evolutionary Multiobjective Optimization.** In *2021 IEEE International Conference on Systems, Man, and Cybernetics*, pages 468–475. IEEE, 2021.
- [618] X. Chen and A. Stafylopatis, editors. *Computational Intelligence (SSCI), 2016 IEEE Symposium Series on*, 2016.
- [619] Y. Chen, J.-K. Hao, and F. Glover. **A hybrid metaheuristic approach for the capacitated arc routing problem.** *Eur. J. Oper. Res.*, 553(1):25–39, 2016. doi:10.1016/j.ejor.2016.02.015.
Keywords: irace.
- [620] C.-B. Cheng and C.-P. Mao. **A modified ant colony system for solving the travelling salesman problem with time windows.** *Mathematical and Computer Modelling*, 46: 1225–1235, 2007. doi:10.1016/j.mcm.2006.11.035.
- [621] F. Y. Cheng and X. S. Li. **Generalized center method for multiobjective engineering optimization.** *Engineering Optimization*, 31(5):641–661, 1999. doi:10.1080/03052159908941390.
- [622] R. Cheng, Y. Jin, M. Olhofer, and B. Sendhoff. **A Reference Vector Guided Evolutionary Algorithm for Many-Objective Optimization.** *IEEE Trans. Evol. Comput.*, 20(5):773–791, 2016. doi:10.1109/TEVC.2016.2519378.

- [623] W. Cheng, J. Hühn, and E. Hüllermeier. **Decision Tree and Instance-Based Learning for Label Ranking**. In A. P. Danyluk, L. Bottou, and M. L. Littman, editors, *Proceedings of the 26th International Conference on Machine Learning, ICML 2009*, pages 161–168, New York, NY, 2009. ACM Press. doi:10.1145/1553374.1553395.
- [624] C. Chevalier, D. Ginsbourger, J. Bect, and I. Molchanov. **Estimating and Quantifying Uncertainties on Level Sets Using the Vorob’ev Expectation and Deviation with Gaussian Process Models**. In D. Uciniski, A. C. Atkinson, and M. Patan, editors, *mODa 10—Advances in Model-Oriented Design and Analysis*, pages 35–43. Springer International Publishing, Heidelberg, Germany, 2013. doi:10.1007/978-3-319-00218-7_5.
- [625] T.-C. Chiang. **nsga3cpp: A C++ implementation of NSGA-III**. <http://web.ntnu.edu.tw/~tcchiang/publications/nsga3cpp/nsga3cpp.htm>, 2014.
- [626] M. Chiarandini. *Stochastic Local Search Methods for Highly Constrained Combinatorial Optimisation Problems*. PhD thesis, FB Informatik, TU Darmstadt, Germany, 2005.
- [627] M. Chiarandini and Y. Goegebeur. **Mixed Models for the Analysis of Optimization Algorithms**. In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 225–264. Springer, Berlin/Heidelberg, 2010. doi:10.1007/978-3-642-02538-9.
Annotation: Preliminary version available as *Tech. Rep.* MF-2009-07-001 at the The Danish Mathematical Society.
- [628] M. Chiarandini, M. Birattari, K. Socha, and O. Rossi-Doria. **An Effective Hybrid Algorithm for University Course Timetabling**. *Journal of Scheduling*, 9(5):403–432, Oct. 2006. doi:10.1007/s10951-006-8495-8.
Keywords: 2003 international timetabling competition, F-race.
- [629] M. Chica, O. Cordon, S. Damas, and J. Bautista. **A New Diversity Induction Mechanism for a Multi-objective Ant Colony Algorithm to Solve a Real-world time and Space Assembly Line Balancing Problem**. *Memetic Computing*, 3(1):15–24, 2011. ISSN 1865-9284.
- [630] F. Chicano and K. Krawiec, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2021, Lille, France, July 10-14, 2021*. ACM Press, New York, NY, 2021.
- [631] F. Chicano and K. Krawiec, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO 2021, Lille, France, July 10-14, 2021*. ACM Press, New York, NY, 2021.
- [632] F. Chicano, D. Whitley, and E. Alba. **A Methodology to Find the Elementary Landscape Decomposition of Combinatorial Optimization Problems**. *Evol. Comput.*, 19(4):597–637, 2011.
- [633] F. Chicano, G. J. Luque, and E. Alba. **Autocorrelation Measures for the Quadratic Assignment Problem**. *Applied Mathematics Letters*, 25:698–705, 2012. doi:10.1016/j.aml.2011.09.053.
- [634] F. Chicano, B. Derbel, and S. Verel. **Fourier Transform-based Surrogates for Permutation Problems**. In S. Silva and L. Paquete, editors, *GECCO*, pages 275–283. ACM Press, New York, NY, 2023. doi:10.1145/3583131.3590425.
Keywords: Asteroid Routing Problem, Single Machine Total Weighted Tardiness, Walsh-Hadamard transform.
Annotation: ISBN: 9798400701191.
- [635] F. Chicano, T. Friedrich, T. Kötzing, and F. Rothlauf, editors. *Foundations of Genetic Algorithms, 17th ACM/SIGEVO International Workshop, FOGA 2023, Potsdam, Germany*. ACM, 2023.

- [636] S. E. Chick, P. J. Sanchez, D. M. Ferrin, and D. J. Morrice, editors. *Proceedings of the 35th Winter Simulation Conference: Driving Innovation*, volume 1, New York, NY, Dec. 2003. ACM Press.
- [637] C. D. Chio, S. Cagnoni, C. Cotta, M. Ebner, A. Ekárt, A. I. Esparcia-Alcázar, C. K. Goh, J.-J. Merelo, F. Neri, M. Preuss, J. Togelius, and G. N. Yannakakis, editors. *Applications of Evolutionary Computation, EvoApplications 2010: EvoCOMPLEX, EvoGAMES, EvoIASP, EvoINTELLIGENCE, EvoNUM, and EvoSTOC, Istanbul, Turkey, April 7-9, 2010, Proceedings, Part I*, volume 6024 of *LNCSE*. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-12239-2.
- [638] D. S. Chivilikhin, V. I. Ulyantsev, and A. A. Shalyto. **Modified ant colony algorithm for constructing finite state machines from execution scenarios and temporal formulas.** *Automation and Remote Control*, 77(3):473–484, 2016. doi:10.1134/S0005117916030097.
Keywords: irace.
- [639] F. Chollet et al. **Keras.** <https://keras.io>, 2015.
- [640] M. Christen, O. Schenk, and H. Burkhart. **PATUS: A Code Generation and Autotuning Framework for Parallel Iterative Stencil Computations on Modern Microarchitectures.** In F. Mueller, editor, *Proceedings of the 2011 IEEE International Parallel & Distributed Processing Symposium, IPDPS '11*, pages 676–687. IEEE Computer Society, 2011. doi:10.1109/IPDPS.2011.70.
- [641] J. Christiaens and G. V. Berghe. **Slack Induction by String Removals for Vehicle Routing Problems.** Technical Report 7-05-2018, Department of Computing Science, KU Leuven, Gent, Belgium, 2018.
- [642] N. Christofides. **Worst-case analysis of a new heuristic for the travelling salesman problem.** Technical Report 388, Graduate School of Industrial Administration, Carnegie-Mellon University, Pittsburgh, PA, 1976.
- [643] N. Christofides, A. Mingozzi, and P. Toth. **State-space relaxation procedures for the computation of bounds to routing problems.** *Networks*, 11(2):145–164, 1981. doi:10.1002/net.3230110207.
Keywords: makespan optimization.
- [644] T. Chugh. *Handling expensive multiobjective optimization problems with evolutionary algorithms.* PhD thesis, University of Jyväskylä, 2017.
- [645] T. Chugh. **Scalarizing Functions in Bayesian Multiobjective Optimization.** In *Proceedings of the 2020 Congress on Evolutionary Computation (CEC 2020)*, pages 1–8, Piscataway, NJ, 2020. IEEE Press. doi:10.1109/CEC48606.2020.9185706.
- [646] T. Chugh and M. López-Ibáñez. **Maximising Hypervolume and Minimising ϵ -Indicators using Bayesian Optimisation over Sets.** In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1326–1334. ACM Press, New York, NY, 2021. doi:10.1145/3449726.3463178. Supplementary material: <https://doi.org/10.5281/zenodo.4675569>.
Keywords: multi-objective, surrogate models, epsilon, hypervolume.
- [647] T. Chugh, Y. Jin, K. Miettinen, J. Hakanen, and K. Sindhya. **A Surrogate-Assisted Reference Vector Guided Evolutionary Algorithm for Computationally Expensive Many-Objective Optimization.** *IEEE Trans. Evol. Comput.*, 22(1):129–142, Feb. 2018.
- [648] T. Chugh, K. Sindhya, J. Hakanen, and K. Miettinen. **A survey on handling computationally expensive multiobjective optimization problems with evolutionary algorithms.** *Soft Computing*, 23(9):3137–3166, 2019. doi:10.1007/s00500-017-2965-0.

- [649] S. Chusanapiputt, D. Nualhong, S. Jantarang, and S. Phoomvuthisarn. **Selective self-adaptive approach to ant system for solving unit commitment problem.** In M. Cattolico et al., editors, *GECCO*, pages 1729–1736. ACM Press, New York, NY, 2006.
- [650] A. D. Cianciolo, R. Powell, and M. K. Lockwood. **Mars Science Laboratory Launch-Arrival Space Study: A Pork Chop Plot Analysis.** In *2006 IEEE Aerospace Conference*, pages 1–11. IEEE, 2006. doi:[10.1109/AERO.2006.1655797](https://doi.org/10.1109/AERO.2006.1655797).
- [651] C. Cintrano, J. Ferrer, M. López-Ibáñez, and E. Alba. **Hybridization of Racing Methods with Evolutionary Operators for Simulation Optimization of Traffic Lights Programs.** In C. Zarges and S. Verel, editors, *Proceedings of EvoCOP 2021 – 21th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 12692 of *LNCS*, pages 17–33. Springer, Cham, Switzerland, 2021. doi:[10.1007/978-3-030-72904-2_2](https://doi.org/10.1007/978-3-030-72904-2_2).
Keywords: Hybrid algorithms, Evolutionary algorithms, Simulation optimization, Uncertainty, Traffic light planning.
Annotation: Extended version published in Evolutionary Computation journal [652].
- [652] C. Cintrano, J. Ferrer, M. López-Ibáñez, and E. Alba. **Hybridization of Evolutionary Operators with Elitist Iterated Racing for the Simulation Optimization of Traffic Lights Programs.** *Evol. Comput.*, 31(1):31–51, 2023. doi:[10.1162/evco_a_00314](https://doi.org/10.1162/evco_a_00314).
Keywords: irace, Simulation optimization, Uncertainty, Traffic light planning.
- [653] J. Cirasella, D. S. Johnson, L. A. McGeoch, and W. Zhang. **The Asymmetric Traveling Salesman Problem: Algorithms, Instance Generators, and Tests.** In A. L. Buchsbaum and J. Snoeyink, editors, *Algorithm Engineering and Experimentation, Third International Workshop, ALENEX 2001, Washington, DC, USA, January 5-6, 2001, Revised Papers*, volume 2153 of *LNCS*, pages 32–59, Berlin, Germany, 2001. Springer. doi:[10.1007/3-540-44808-X_3](https://doi.org/10.1007/3-540-44808-X_3).
- [654] A. A. Cire and W.-J. van Hoeve. **Multivalued Decision Diagrams for Sequencing Problems.** *Operations Research*, 61(6):1259–1462, 2013. doi:[10.1287/opre.2013.1221](https://doi.org/10.1287/opre.2013.1221).
- [655] J. Claerbout and M. Karrenbach. **Electronic documents give reproducible research a new meaning.** In *SEG Technical Program Expanded Abstracts 1992*, pages 601–604. Society of Exploration Geophysicists, 1992. doi:[10.1190/1.1822162](https://doi.org/10.1190/1.1822162).
Annotation: Proposed a reproducibility taxonomy, defined reproducibility and taxonomy.
- [656] R. M. Clark, L. A. Rossman, and L. J. Wymer. **Modeling distribution system water quality: regulatory implications.** *Journal of Water Resources Planning and Management, ASCE*, 121(6):423–428, 1995.
- [657] M. Clerc. **Standard Particle Swarm Optimisation.** hal-00764996, Sept. 2012. URL <https://hal.archives-ouvertes.fr/hal-00764996>.
- [658] M. Clerc and J. Kennedy. **The particle swarm - explosion, stability, and convergence in a multidimensional complex space.** *IEEE Trans. Evol. Comput.*, 6(1):58–73, Feb. 2002. doi:[10.1109/4235.985692](https://doi.org/10.1109/4235.985692).
- [659] M. Clerc and J. Kennedy. **Standard PSO 2011.** Particle Swarm Central, 2011. URL <http://www.particleswarm.info/>.
- [660] D. Cliff, P. Husbands, J.-A. Meyer, and S. Wilson, editors. *Proceedings of the third international conference on Simulation of adaptive behavior: From Animals to Animats 3.* MIT Press, Cambridge, MA, 1994.
- [661] J. Climaco, editor. *Proceedings of the 13th International Conference on Multiple Criteria Decision Making (MCDM'97).* Springer Verlag, 1997.

- [662] I. Cloete, K.-P. Wong, and M. Berthold, editors. *Proceedings of the 3rd International Conference on Machine Learning and Cybernetics*, 2004. IEEE Press.
- [663] J. J. Cochran, editor. *Wiley Encyclopedia of Operations Research and Management Science*. John Wiley & Sons, 2011. doi:10.1002/9780470400531.
- [664] A. Cockburn, P. Dragicevic, L. Besançon, and C. Gutwin. **Threats of a Replication Crisis in Empirical Computer Science**. *Commun. ACM*, 63(8):70–79, July 2020. doi:10.1145/3360311.
- [665] B. Codenotti, G. Manzini, L. Margara, and G. Resta. **Perturbation: An Efficient Technique for the Solution of Very Large Instances of the Euclidean TSP**. *INFORMS Journal on Computing*, 8(2):125–133, 1996.
- [666] H. Coelho, R. Studer, and M. Wooldridge, editors. *Proceedings of the 19th European Conference on Artificial Intelligence*. IOS Press, 2010.
- [667] C. A. Coello Coello. **Handling Preferences in Evolutionary Multiobjective Optimization: A Survey**. In *IEEE CEC*, pages 30–37, Piscataway, NJ, July 2000. IEEE Press.
- [668] C. A. Coello Coello. **Theoretical and numerical constraint-handling techniques used with evolutionary algorithms: a survey of the state of the art**. *Computer Methods in Applied Mechanics and Engineering*, 191(11-12):1245–1287, 2002. doi:10.1016/S0045-7825(01)00323-1.
- [669] C. A. Coello Coello. **Special Issue on Evolutionary Multiobjective Optimization**. *IEEE Trans. Evol. Comput.*, 7(2), 2003.
- [670] C. A. Coello Coello. **Evolutionary multi-objective optimization: a historical view of the field**. *IEEE Computational Intelligence Magazine*, 1(1):28–36, 2006.
- [671] C. A. Coello Coello, editor. *5th International Conference, LION 5, Rome, Italy, January 17-21, 2011. Selected Papers*, volume 6683 of *LNCS*. Springer, Heidelberg, Germany, 2011.
- [672] C. A. Coello Coello. **Multi-objective Evolutionary Algorithms in Real-World Applications: Some Recent Results and Current Challenges**. In *Advances in Evolutionary and Deterministic Methods for Design, Optimization and Control in Engineering and Sciences*, pages 3–18. Springer, 2015. doi:10.1007/978-3-319-11541-2_1.
- [673] C. A. Coello Coello. **Recent Results and Open Problems in Evolutionary Multiobjective Optimization**. In C. Martín-Vide, R. Neruda, and M. A. Vega-Rodríguez, editors, *Theory and Practice of Natural Computing - 6th International Conference, TPNC 2017*, volume 10687 of *LNCS*, pages 3–21. Springer International Publishing, Cham, Switzerland, 2017.
- [674] C. A. Coello Coello, editor. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2020, Cancún, Mexico, July 8-12, 2020*. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.
- [675] C. A. Coello Coello, editor. *2020 IEEE Symposium Series on Computational Intelligence, SSCI 2020, Canberra, Australia, December 1-4, 2020*, 2020. IEEE Press.
- [676] C. A. Coello Coello and M. Reyes-Sierra. **A Study of the Parallelization of a Coevolutionary Multi-objective Evolutionary Algorithm**. In R. Monroy, G. Arroyo-Figueroa, L. E. Sucar, and H. Sossa, editors, *Proc. of MICAI*, volume 2972 of *LNAI*, pages 688–697. Springer, Heidelberg, Germany, 2004.
Keywords: IGD.
Annotation: Proposed the name Inverted Generational Distance (IGD), but the metric was already proposed by [434].

- [677] C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors. *Evolutionary Multi-Criterion Optimization, Third International Conference, EMO 2005, Guanajuato, Mexico, March 9-11, 2005. Proceedings*, volume 3410 of *LNCS*. Springer, Berlin/Heidelberg, 2005.
- [678] C. A. Coello Coello, G. B. Lamont, and D. A. Van Veldhuizen. *Evolutionary Algorithms for Solving Multi-Objective Problems*. Springer, New York, NY, 2nd edition, 2007. doi:10.1007/978-0-387-36797-2.
- [679] C. A. Coello Coello, C. Dhaenens, and L. Jourdan, editors. *Advances in Multi-Objective Nature Inspired Computing*, volume 272 of *Studies in Computational Intelligence*. Springer, 2010.
- [680] C. A. Coello Coello et al., editors. *Parallel Problem Solving from Nature, PPSN XII, 12th International Conference, Taormina, Italy, September 1-5, 2012, Proceedings, Part I*, volume 7491 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [681] C. A. Coello Coello et al., editors. *Parallel Problem Solving from Nature - PPSN XII - 12th International Conference, Taormina, Italy, September 1-5, 2012, Proceedings, Part II*, volume 7492 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [682] G. Cohen. **Optimal Control of Water Supply Networks**. In S. G. Tzafestas, editor, *Optimization and Control of Dynamic Operational Research Models*, volume 4, chapter 8, pages 251–276. North-Holland Publishing Company, Amsterdam, 1982.
- [683] P. R. Cohen. *Empirical Methods for Artificial Intelligence*. MIT Press, Cambridge, MA, 1995.
- [684] W. W. Cohen and H. Hirsh, editors. *Proceedings of the 11th International Conference on Machine Learning, ICML 1994, New Brunswick, NJ, USA, San Francisco, CA, 1994*. Morgan Kaufmann Publishers.
- [685] W. W. Cohen, A. McCallum, and S. T. Roweis, editors. *Proceedings of the 25th International Conference on Machine Learning, ICML 2008, Helsinki, Finland, July 05-09, 2008*, New York, NY, 2008. ACM Press.
- [686] A. Cohn, editor. *Proceedings, The Twenty-First National Conference on Artificial Intelligence and the Eighteenth Innovative Applications of Artificial Intelligence Conference, July 16-20, 2006, Boston, Massachusetts, USA*, volume 6, 2006. AAAI Press/MIT Press, Menlo Park, CA.
- [687] H. Cohn and M. J. Fielding. **Simulated Annealing: Searching for an Optimal Temperature**. *SIAM J. Optim.*, 9(3):779–802, 1999.
- [688] S. Colas, N. Monmarché, P. Gaucher, and M. Slimane. **Artificial Ants for the Optimization of Virtual Keyboard Arrangement for Disabled People**. In N. Monmarché, E.-G. Talbi, P. Collet, M. Schoenauer, and E. Lutton, editors, *Artificial Evolution*, volume 4926 of *LNCS*, pages 87–99. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-79305-2.
- [689] P. Collet, N. Monmarché, P. Legrand, M. Schoenauer, and E. Lutton, editors. *Artificial Evolution: 9th International Conference, Evolution Artificielle, EA, 2009, Strasbourg, France, October 26-28, 2009. Revised Selected Papers*, volume 5975 of *LNCS*. Springer, Heidelberg, Germany, 2010.
- [690] J. M. Colmenar, M. Laguna, and R. Martín-Santamaría. **Changeover minimization in the production of metal parts for car seats**. *Computers & Industrial Engineering*, 198:110634, Dec. 2024. doi:10.1016/j.cie.2024.110634.
Keywords: VNS, Hexaly, Gurobi.
- [691] A. F. Colombo and B. W. Karney. **Impacts of Leaks on Energy Consumption in Pumped Systems with Storage**. *Journal of Water Resources Planning and Management, ASCE*, 131(2):146–155, Mar. 2005.

- [692] A. Colorni, M. Dorigo, and V. Maniezzo. **Distributed Optimization by Ant Colonies**. In F. J. Varela and P. Bourguine, editors, *Proceedings of the First European Conference on Artificial Life*, pages 134–142. MIT Press, Cambridge, MA, 1992.
- [693] A. Colorni, M. Dorigo, V. Maniezzo, and M. Trubian. **Ant System for Job-shop Scheduling**. *JORBEL — Belgian Journal of Operations Research, Statistics and Computer Science*, 34(1): 39–53, 1994.
- [694] R. K. Congram, C. N. Potts, and S. van de Velde. **An Iterated Dynasearch Algorithm for the Single-Machine Total Weighted Tardiness Scheduling Problem**. *INFORMS Journal on Computing*, 14(1):52–67, 2002.
- [695] A. R. Conn, K. Scheinberg, and L. N. Vicente. *Introduction to Derivative-Free Optimization*. MPS–SIAM Series on Optimization. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 2009.
- [696] D. T. Connolly. **An Improved Annealing Scheme for the QAP**. *Eur. J. Oper. Res.*, 46(1): 93–100, 1990.
- [697] W. J. Conover. *Practical Nonparametric Statistics*. John Wiley & Sons, New York, NY, 3rd edition, 1999.
- [698] R. J. Cook and V. T. Farewell. **Multiplicity Considerations in the Design and Analysis of Clinical Trials**. *Journal of the Royal Statistical Society: Series A*, 159:93–110, 1996.
Annotation: multiplicity; multiple endpoints; multiple treatments; p-value adjustment; type I error; argues that if results are intended to be interpreted marginally, there may be no need for controlling experimentwise error rate.
- [699] S. A. Cook. **The Complexity of Theorem-proving Procedures**. In *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, STOC '71, pages 151–158. ACM, 1971.
[doi:10.1145/800157.805047](https://doi.org/10.1145/800157.805047).
- [700] W. J. Cook. **The Traveling Salesman Problem**. <http://www.math.uwaterloo.ca/tsp>, 2010.
Version visited last on 15 April 2014.
- [701] W. J. Cook. *In Pursuit of the Traveling Salesman*. Princeton University Press, Princeton, NJ, 2012.
- [702] W. J. Cook. **Computing in Combinatorial Optimization**. In B. Steffen and G. Woeginger, editors, *Computing and Software Science: State of the Art and Perspectives*, volume 10000 of *LNCS*, pages 27–47. Springer, Cham, Switzerland, 2019. [doi:10.1007/978-3-319-91908-9_3](https://doi.org/10.1007/978-3-319-91908-9_3).
- [703] V. Coppé, X. Gillard, and P. Schaus. **Decision Diagram-Based Branch-and-Bound with Caching for Dominance and Suboptimality Detection**. *INFORMS Journal on Computing*, 36(6):1522–1542, 2024. [doi:10.1287/ijoc.2022.0340](https://doi.org/10.1287/ijoc.2022.0340).
- [704] D. Coppersmith, L. K. Fleischer, and A. Rurda. **Ordering by Weighted Number of Wins Gives a Good Ranking for Weighted Tournaments**. *ACM Transactions on Algorithms*, 6(3):1–13, July 2010. [doi:10.1145/1798596.1798608](https://doi.org/10.1145/1798596.1798608).
Keywords: Approximation algorithms, Borda's method, feedback arc set problem, rank aggregation, tournaments.
- [705] J. Cordeau and M. Maischberger. **A Parallel Iterated Tabu Search Heuristic for Vehicle Routing Problems**. *Comput. Oper. Res.*, 39(9):2033–2050, 2012.
- [706] J. Cordeau, G. Laporte, and A. Mercier. **A unified tabu search heuristic for vehicle routing problems with time windows**. *J. Oper. Res. Soc.*, 52(8):928–936, 2001.

- [707] O. Cordón and S. Damas. **Image Registration with Iterated Local Search**. *J. Heuristics*, 12(1–2):73–94, 2006.
- [708] O. Cordón, I. F. de Viana, F. Herrera, and L. Moreno. **A New ACO Model Integrating Evolutionary Computation Concepts: The Best-Worst Ant System**. In M. Dorigo et al., editors, *Abstract proceedings of ANTS 2000 – From Ant Colonies to Artificial Ants: Second International Workshop on Ant Algorithms*, pages 22–29. IRIDIA, Université Libre de Bruxelles, Belgium, Sept. 2000.
- [709] O. Cordón, F. Herrera, and T. Stützle. **Special Issue on Ant Colony Optimization: Models and Applications**. *Mathware & Soft Computing*, 9(3):137–268, 2002.
- [710] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein. *Introduction to algorithms*. MIT Press, Cambridge, MA, 2009.
- [711] D. Corne and J. D. Knowles. **Some Multiobjective Optimizers are Better than Others**. In *IEEE CEC*, pages 2506–2512, Piscataway, NJ, Dec. 2003. IEEE Press.
- [712] D. Corne and J. D. Knowles. **No free lunch and free leftovers theorems for multiobjective optimisation problems**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 327–341. Springer, Heidelberg, Germany, 2003. doi:10.1007/3-540-36970-8_23.
- [713] D. Corne and A. Reynolds. **Evaluating optimization algorithms: bounds on the performance of optimizers on unseen problems**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 707–710. ACM Press, New York, NY, 2011. doi:10.1145/2001858.2002073. Supplementary material: <http://is.gd/evalopt>.
- [714] D. Corne, M. Dorigo, and F. Glover, editors. *New Ideas in Optimization*. McGraw Hill, London, UK, 1999.
- [715] D. Corne, J. D. Knowles, and M. J. Oates. **The Pareto Envelope-Based Selection Algorithm for Multiobjective Optimization**. In M. Schoenauer et al., editors, *PPSN*, volume 1917 of *LNCS*, pages 839–848. Springer, Heidelberg, Germany, 2000. Annotation: IC.29.
- [716] D. Corne, N. R. Jerram, J. D. Knowles, and M. J. Oates. **PESA-II: Region-Based Selection in Evolutionary Multiobjective Optimization**. In E. D. Goodman, editor, *Proceedings of the 3rd Annual Conference on Genetic and Evolutionary Computation, GECCO 2001*, pages 283–290. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [717] J. L. Corner and J. T. Buchanan. **Capturing decision maker preference: Experimental comparison of decision analysis and MCDM techniques**. *Eur. J. Oper. Res.*, 98(1):85–97, 1997.
- [718] J. a. Correia, S. Smith, and R. Qaddoura, editors. *Applications of Evolutionary Computation – 26th European Conference, EvoApplications 2023, Held as Part of EvoStar 2023, Brno, Czech Republic, April 12-14, 2023, Proceedings*, volume 13989 of *LNCS*. Springer Nature, Switzerland, 2023.
- [719] P. Corry and E. Kozan. **Ant Colony Optimisation for Machine Layout Problems**. *Computational Optimization and Applications*, 28(3):287–310, 2004.
- [720] J. Corstjens, N. Dang, B. Depaire, A. Caris, and P. De Causmaecker. **A combined approach for analysing heuristic algorithms**. *J. Heuristics*, 25(4):591–628, 2019. doi:10.1007/s10732-018-9388-7.

- [721] J. Corstjens, B. Depaire, A. Caris, and K. Sörensen. **A multilevel evaluation method for heuristics with an application to the VRPTW**. *International Transactions in Operational Research*, 27(1):168–196, 2020. doi:10.1111/itor.12631.
- [722] A. D. Corte and K. Sörensen. **Optimisation of gravity-fed water distribution network design: A critical review**. *Eur. J. Oper. Res.*, 228(1):1–10, 2013. doi:10.1016/j.ejor.2012.11.046.
- [723] A. D. Corte and K. Sörensen. **An Iterated Local Search Algorithm for Water Distribution Network Design Optimization**. *Networks*, 67(3):187–198, 2016.
- [724] A. D. Corte and K. Sörensen. **An Iterated Local Search Algorithm for multi-period water distribution network design optimization**. *Water*, 8(8):359, 2016. doi:10.3390/w8080359.
- [725] C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, editors. *Advances in Neural Information Processing Systems 28: Annual Conference on Neural Information Processing Systems 2015, December 7-12, 2015, Montreal, Quebec, Canada*, 2015. URL <http://papers.nips.cc/book/advances-in-neural-information-processing-systems-28-2015>.
- [726] COSEAL. **CO**nfiguration and **SE**lection of **AL**gorithms. <http://www.coseal.net>, 2017.
- [727] D. Costa and A. Hertz. **Ants can color graphs**. *J. Oper. Res. Soc.*, 48:295–305, 1997.
- [728] W. E. Costa, M. C. Goldbarg, and E. F. G. Goldbarg. **Hybridizing VNS and path-relinking on a particle swarm framework to minimize total flowtime**. *Expert Systems with Applications*, 39(18):13118–13126, 2012.
- [729] C. Cotta and P. Cowling, editors. *EvoCOP*, volume 5482 of *LNCS*. Springer, Heidelberg, Germany, 2009.
- [730] C. Cotta et al., editors. *Proceedings of EvoCOP 2007 – Seventh European Conference on Evolutionary Computation in Combinatorial Optimisation*, volume 4446 of *LNCS*. Springer, Berlin, Germany, 2007.
- [731] J. D. Cowan, G. Tesauro, and J. Alspector, editors. *Advances in Neural Information Processing Systems*, volume 6. Morgan Kaufmann Publishers, San Francisco, CA, 1994.
- [732] P. I. Cowling, G. Kendall, and E. Soubeiga. **A Hyperheuristic Approach to Scheduling a Sales Summit**. In E. K. Burke and W. Erben, editors, *PATAT 2000: Proceedings of the 3rd International Conference of the Practice and Theory of Automated Timetabling*, volume 2079 of *LNCS*, pages 176–190. Springer, 2000. doi:10.1007/3-540-44629-X_11.
Annotation: First mention of the term hyper-heuristic.
- [733] S. P. Coy, B. L. Golden, G. C. Runger, and E. A. Wasil. **Using Experimental Design to Find Effective Parameter Settings for Heuristics**. *J. Heuristics*, 7(1):77–97, 2001.
- [734] I. B. Crabtree. **Resource Scheduling: Comparing Simulated Annealing with Constraint Programming**. *BT Technology Journal*, 13(1):121–127, 1995.
- [735] M. J. Crawley. *The R Book*. Wiley, 2nd edition, 2012.
- [736] D. E. Critchlow, M. A. Fligner, and J. S. Verducci. **Probability Models on Rankings**. *Journal of Mathematical Psychology*, 35:294–318, 1991.
- [737] G. A. Croes. **A Method for Solving Traveling Salesman Problems**. *Operations Research*, 6:791–812, 1958.

- [738] H. P. Crowder, R. S. Dembo, and J. M. Mulvey. **Reporting computational experiments in mathematical programming.** *Mathematical Programming*, 15(1):316–329, 1978. doi:10.1007/BF01609036.
Keywords: reproducibility.
- [739] W. B. Crowston, F. Glover, G. L. Thompson, and J. D. Trawick. **Probabilistic and Parametric Learning Combinations of Local Job Shop Scheduling Rules.** ONR Research Memorandum No. 117, GSIA, Carnegie-Mellon University, Pittsburgh, PA, USA, 1963.
- [740] C. Cruz, J. R. González, and D. A. Pelta. **Optimization in Dynamic Environments: A Survey on Problems, Methods and Measures.** *Soft Computing*, 15(7):1427–1448, 2011.
- [741] F. Cruz, A. Subramanian, B. P. Bruck, and M. Iori. **A Heuristic Algorithm for a Single Vehicle Static Bike Sharing Rebalancing Problem.** *Comput. Oper. Res.*, 79:19–33, 2017.
- [742] J. C. Culberson. **Iterated Greedy Graph Coloring and the Difficulty Landscape.** Technical Report 92-07, Department of Computing Science, The University of Alberta, Edmonton, Alberta, Canada, 1992.
- [743] J. C. Culberson. **On the Futility of Blind Search: An Algorithmic View of “No Free Lunch”.** *Evol. Comput.*, 6(2):109–127, 1998. doi:10.1162/evco.1998.6.2.109.
Keywords: NFL.
- [744] J. C. Culberson and F. Luo. **Exploring the k -colorable Landscape with Iterated Greedy.** In D. S. Johnson and M. A. Trick, editors, *Cliques, Coloring, and Satisfiability: Second DIMACS Implementation Challenge*, volume 26 of *DIMACS Series on Discrete Mathematics and Theoretical Computer Science*, pages 245–284. American Mathematical Society, Providence, RI, 1996.
- [745] J. C. Culberson, A. Beacham, and D. Papp. **Hiding our Colors.** In *Proceedings of the CP’95 Workshop on Studying and Solving Really Hard Problems*, pages 31–42, Cassis, France, Sept. 1995.
- [746] J. Cumming. *Understanding the New Statistics – Effect Sizes, Confidence Intervals, and Meta-analysis.* Taylor & Francis, 2012.
- [747] P. Czyżżak and A. Jaskiewicz. **Pareto simulated annealing – a metaheuristic technique for multiple-objective combinatorial optimization.** *Journal of Multi-Criteria Decision Analysis*, 7(1):34–47, 1998.
- [748] R. D’Agostino and E. S. Pearson. **Tests for Departure from Normality. Empirical Results for the Distributions of b_2 and $\sqrt{b_1}$.** *Biometrika*, 60(3):613–622, Dec. 1973. doi:10.2307/2335012.
- [749] M. Damas, M. Salmerón, J. Ortega, G. Olivares, and H. Pomares. **Parallel Dynamic Water Supply Scheduling in a Cluster of Computers.** *Concurrency and Computation: Practice and Experience*, 13(15):1281–1302, Dec. 2001. ISSN 1532-0626 (print), 1532-0634 (electronic).
- [750] S. B. Damelin, F. J. Hickernell, D. L. Ragozin, and X. Zeng. **On Energy, Discrepancy and Group Invariant Measures on Measurable Subsets of Euclidean Space.** *Journal of Fourier Analysis and Applications*, 16(6):813–839, 2010.
Keywords: Capacity; Cubature; Discrepancy; Distribution; Group invariant kernel; Group invariant measure; Energy minimizer; Equilibrium measure; Numerical integration; Positive definite; Potential field; Riesz kernel; Reproducing Hilbert space; Signed measure.
- [751] G. C. Dandy and M. S. Gibbs. **Optimizing System Operations and Water Quality.** In P. Bizier and P. DeBarry, editors, *Proceedings of World Water and Environmental Resources Congress*. ASCE, Philadelphia, USA, 2003. doi:10.1061/40685(2003)127. on CD-ROM.

- [752] N. Dang and P. De Causmaecker. **Analysis of Algorithm Components and Parameters: Some Case Studies**. In N. F. Matsatsinis, Y. Marinakis, and P. M. Pardalos, editors, *LION 13*, volume 11968 of *LNCS*, pages 288–303. Springer, Cham, Switzerland, 2019. doi:10.1007/978-3-030-05348-2_25.
- [753] N. Dang and C. Doerr. **Hyper-parameter tuning for the $(1 + (\lambda, \lambda))$ GA**. In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 889–897. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321725.
Keywords: irace; theory.
- [754] N. Dang Thi Thanh. *Data analytics for algorithm design*. PhD thesis, KU Leuven, Belgium, 2018.
Annotation: Supervised by Patrick De Causmaecker.
- [755] N. Dang Thi Thanh and P. De Causmaecker. **Motivations for the Development of a Multi-objective Algorithm Configurator**. In B. Vitoriano, E. Pinson, and F. Valente, editors, *ICORES 2014 - Proceedings of the 3rd International Conference on Operations Research and Enterprise Systems*, pages 328–333. SciTePress, 2014.
- [756] N. Dang Thi Thanh and P. De Causmaecker. **Characterization of Neighborhood Behaviours in a Multi-neighborhood Local Search Algorithm**. In P. Festa, M. Sellmann, and J. Vanschoren, editors, *LION 10*, volume 10079 of *LNCS*, pages 234–239. Springer, Cham, Switzerland, 2016.
- [757] N. Dang Thi Thanh, L. Pérez Cáceres, P. De Causmaecker, and T. Stützle. **Configuring irace Using Surrogate Configuration Benchmarks**. In P. A. N. Bosman, editor, *GECCO*, pages 243–250. ACM Press, New York, NY, 2017. doi:10.1145/3071178.3071238.
Keywords: irace.
- [758] E. Danna, E. Rothberg, and C. Le Pape. **Exploring relaxation induced neighborhoods to improve MIP solutions**. *Mathematical Programming*, 102(1):71–90, 2005.
- [759] A. Dantas and A. Pozo. **On the use of fitness landscape features in meta-learning based algorithm selection for the quadratic assignment problem**. *Theor. Comput. Sci.*, 805: 62–75, 2020. doi:10.1016/j.tcs.2019.10.033.
- [760] A. L. Dantas and A. T. R. Pozo. **A Meta-Learning Algorithm Selection Approach for the Quadratic Assignment Problem**. In *Proceedings of the 2018 Congress on Evolutionary Computation (CEC 2018)*, pages 1–8, Piscataway, NJ, 2018. IEEE Press.
- [761] G. B. Dantzig and P. Wolfe. **Decomposition Principle for Linear Programs**. *Operations Research*, 8(1):101–111, 1960.
- [762] A. P. Danyluk, L. Bottou, and M. L. Littman, editors. *Proceedings of the 26th Annual International Conference on Machine Learning, ICML 2009, Montreal, Quebec, Canada, June 14-18, 2009*, New York, NY, 2009. ACM Press.
- [763] F. Daolio, S. Verel, G. Ochoa, and M. Tomassini. **Local Optima Networks and the Performance of Iterated Local Search**. In T. Soule and J. H. Moore, editors, *GECCO*, pages 369–376. ACM Press, New York, NY, 2012.
- [764] F. Daolio, A. Liefooghe, S. Verel, H. E. Aguirre, and K. Tanaka. **Problem Features versus Algorithm Performance on Rugged Multiobjective Combinatorial Fitness Landscapes**. *Evol. Comput.*, 25(4):555–585, 2017. doi:10.1162/evco_a_00193.
- [765] I. Das and J. E. Dennis. **A closer look at drawbacks of minimizing weighted sums of objectives for Pareto set generation in multicriteria optimization problems**. *Structural Optimization*, 14(1):63–69, 1997. doi:10.1007/BF01197559.

- [766] I. Das and J. E. Dennis. **Normal-boundary intersection: A new method for generating the Pareto surface in nonlinear multicriteria optimization problems.** *SIAM J. Optim.*, 8(3):631–657, 1998. doi:[10.1137/S1052623496307510](https://doi.org/10.1137/S1052623496307510).
Keywords: simplex lattice design.
- [767] S. Das and P. N. Suganthan. **Differential Evolution: A Survey of the State-of-the-art.** *IEEE Trans. Evol. Comput.*, 15(1), Feb. 2011.
- [768] S. Das, S. S. Mullick, and P. N. Suganthan. **Recent advances in differential evolution—An updated survey.** *Swarm and Evolutionary Computation*, 27:1–30, 2016.
- [769] S. Dasgupta and D. McAllester, editors. *Proceedings of the 30th International Conference on Machine Learning, ICML 2013, Atlanta, GA, USA, 16-21 June 2013*, volume 28, 2013.
- [770] S. Dash. **Exponential Lower Bounds on the Lengths of Some Classes of Branch-and-Cut Proofs.** *Mathematics of Operations Research*, 30(3):678–700, 2005.
- [771] C. Daskalakis, I. Diakonikolas, and M. Yannakakis. **How good is the Chord algorithm?** *SIAM J. Comput.*, 45(3):811–858, 2016.
- [772] S. Daulton, M. Balandat, and E. Bakshy. **Differentiable Expected Hypervolume Improvement for Parallel Multi-Objective Bayesian Optimization.** In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems (NeurIPS 33)*, pages 9851–9864, 2020.
- [773] J. Daunizeau, H. E. M. den Ouden, M. Pessiglione, S. J. Kiebel, K. J. Friston, and K. E. Stephan. **Observing the observer (II): deciding when to decide.** *PLoS One*, 5(12):e15555, 2010. doi:[10.1371/journal.pone.0015555](https://doi.org/10.1371/journal.pone.0015555).
- [774] J. Daunizeau, H. E. M. den Ouden, M. Pessiglione, K. E. Stephan, S. J. Kiebel, and K. J. Friston. **Observing the observer (I): meta-Bayesian models of learning and decision-making.** *PLoS One*, 5(12):e15554, 2010. doi:[10.1371/journal.pone.0015554](https://doi.org/10.1371/journal.pone.0015554).
- [775] A. P. Davies and E. E. Bischoff. **Weight distribution considerations in container loading.** *Eur. J. Oper. Res.*, 114(3):509–527, May 1999. doi:[10.1016/s0377-2217\(98\)00139-8](https://doi.org/10.1016/s0377-2217(98)00139-8).
Keywords: Single container loading, 3DBPP.
- [776] J.-C. de Borda. **Mémoire sur les Élections au Scrutin.** *Histoire de l’Académie Royal des Sciences*, 1781.
Keywords: ranking.
- [777] A. R. R. de Freitas, P. J. Fleming, and F. G. Guimarães. **Aggregation trees for visualization and dimension reduction in many-objective optimization.** *Inf. Sci.*, 298:288–314, 2015.
- [778] N. de Freitas and K. Murphy, editors. *Proceedings of the Twenty-Eighth Conference on Uncertainty in Artificial Intelligence (UAI’12), Catalina Island, CA August 14-18 2012*, 2013. AUAI Press.
- [779] K. A. De Jong. *Evolutionary computation: a unified approach*. MIT Press, Cambridge, MA, 2006.
- [780] K. A. De Jong and W. M. Spears. **A formal analysis of the role of multi-point crossover in genetic algorithms.** *Annals of Mathematics and Artificial Intelligence*, 5(1):1–26, 1992.
- [781] K. A. De Jong, R. Poli, and J. E. Rowe, editors. *Foundations of Genetic Algorithms, 7th International Workshop, FOGA 2002, Torremolinos, Spain, September 2-4, 2002, Proceedings*. Morgan Kaufmann Publishers, 2002.

- [782] R. H. R. de Lima and A. T. R. Pozo. **A study on auto-configuration of Multi-Objective Particle Swarm Optimization Algorithm.** In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 718–725, Piscataway, NJ, 2017. IEEE Press. doi:10.1109/CEC.2017.7969381.
- [783] E. B. de Moraes Barbosa, E. L. F. Senne, and M. B. Silva. **Improving the Performance of Metaheuristics: An Approach Combining Response Surface Methodology and Racing Algorithms.** *International Journal of Engineering Mathematics*, 2015:1–9, 2015. doi:10.1155/2015/167031.
Keywords: F-race.
Annotation: Article ID 167031.
- [784] J. de Nobel, D. Vermetten, H. Wang, C. Doerr, and T. Bäck. **Tuning as a means of assessing the benefits of new ideas in interplay with existing algorithmic modules.** In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1375–1384. ACM Press, New York, NY, 2021. doi:10.1145/3449726.3463167. Supplementary material: <https://doi.org/10.5281/zenodo.4524959>.
- [785] J. de Nobel, D. Vermetten, H. Wang, C. Doerr, and T. Bäck. **Data and Code from Tuning as a means of assessing the benefits of new ideas in interplay with existing algorithmic modules.** Zenodo, Feb. 2021.
- [786] J. de Nobel, F. Ye, D. Vermetten, H. Wang, C. Doerr, and T. Bäck. **IOHexperimenter: Benchmarking Platform for Iterative Optimization Heuristics.** *Arxiv preprint arXiv:2111.04077*, 2021. doi:10.48550/arXiv.2111.04077.
Annotation: Published in ECJ [787].
- [787] J. de Nobel, F. Ye, D. Vermetten, H. Wang, C. Doerr, and T. Bäck. **IOHexperimenter: Benchmarking Platform for Iterative Optimization Heuristics.** *Evol. Comput.*, pages 1–6, 2024. doi:10.1162/evco_a_00342.
- [788] A. de Perthuis de Laillevault, B. Doerr, and C. Doerr. **Money for Nothing: Speeding Up Evolutionary Algorithms Through Better Initialization.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 815–822. ACM Press, New York, NY, 2015.
- [789] A. G. C. de Sá, W. J. G. S. Pinto, L. O. V. B. Oliveira, and G. Pappa. **RECIPE: A Grammar-Based Framework for Automatically Evolving Classification Pipelines.** In J. McDermott, M. Castelli, L. Sekanina, E. Haasdijk, and P. García-Sánchez, editors, *Proceedings of the 20th European Conference on Genetic Programming, EuroGP 2017*, volume 10196 of *LNCS*, pages 246–261. Springer, Heidelberg, Germany, 2017. ISBN 978-3-319-55695-6. doi:10.1007/978-3-319-55696-3_16.
- [790] W. de Schaetzen, D. A. Savic, and G. A. Walters. **A genetic algorithm approach to pump scheduling in water supply.** In V. Babovic and L. C. Larsen, editors, *Hydroinformatics '98*, pages 897–899, Rotterdam, Balkema, 1998.
- [791] M. De Souza and M. Ritt. **An Automatically Designed Recombination Heuristic for the Test-Assignment Problem.** In *Proceedings of the 2018 Congress on Evolutionary Computation (CEC 2018)*, pages 1–8, Piscataway, NJ, 2018. IEEE Press. doi:10.1109/CEC.2018.8477801.
- [792] M. De Souza and M. Ritt. **Automatic Grammar-Based Design of Heuristic Algorithms for Unconstrained Binary Quadratic Programming.** In A. Liefvooghe and M. López-Ibáñez, editors, *Proceedings of EvoCOP 2018 – 18th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 10782 of *LNCS*, pages 67–84. Springer, Heidelberg, Germany, 2018. doi:10.1007/978-3-319-77449-7_5.

- [793] M. De Souza and M. Ritt. **Hybrid Heuristic for Unconstrained Binary Quadratic Programming – Source Code of HHBQP**. <https://github.com/souzamarcelo/hhbqp>, 2018.
- [794] M. De Souza, M. Ritt, and M. López-Ibáñez. **CAPOPT: Capping Methods for the Automatic Configuration of Optimization Algorithms**. <https://github.com/souzamarcelo/capopt>, 2020.
- [795] M. De Souza, M. Ritt, M. López-Ibáñez, and L. Pérez Cáceres. **ACVIZ: A Tool for the Visual Analysis of the Configuration of Algorithms with irace – Source Code**. <https://github.com/souzamarcelo/acviz>, 2020.
- [796] M. De Souza, M. Ritt, M. López-Ibáñez, and L. Pérez Cáceres. **ACVIZ: Algorithm Configuration Visualizations for irace: Supplementary material**. Zenodo, Sept. 2020.
- [797] M. De Souza, M. Ritt, and M. López-Ibáñez. **Capping Methods for the Automatic Configuration of Optimization Algorithms – Supplementary Material**. <https://github.com/souzamarcelo/supp-cor-capopt>, 2021.
- [798] M. De Souza, M. Ritt, M. López-Ibáñez, and L. Pérez Cáceres. **ACVIZ: A Tool for the Visual Analysis of the Configuration of Algorithms with irace**. *Oper. Res. Perspect.*, 8:100186, 2021. doi:10.1016/j.orp.2021.100186. Supplementary material: <https://zenodo.org/record/4714582>.
- [799] M. De Souza, M. Ritt, and M. López-Ibáñez. **Capping Methods for the Automatic Configuration of Optimization Algorithms**. *Comput. Oper. Res.*, 139:105615, 2022. doi:10.1016/j.cor.2021.105615. Supplementary material: <https://github.com/souzamarcelo/supp-cor-capopt>.
- [800] S. De Vries and R. V. Vohra. **Combinatorial Auctions: A Survey**. *INFORMS Journal on Computing*, 15(3):284–309, 2003.
- [801] A. Dean and D. Voss. *Design and Analysis of Experiments*. Springer, London, UK, 1999. doi:10.1007/b97673.
- [802] T. Dean and M. S. Boddy. **An Analysis of Time-Dependent Planning**. In H. E. Shrobe, T. M. Mitchell, and R. G. Smith, editors, *Proceedings of the 7th National Conference on Artificial Intelligence, AAAI-88*, pages 49–54. AAAI Press/MIT Press, Menlo Park, CA, 1988. URL <http://www.aaai.org/Conferences/AAAI/aaai88.php>.
Keywords: anytime, performance profiles.
- [803] K. Deb. **Multi-objective genetic algorithms: problem difficulties and construction of test problems**. *Evol. Comput.*, 7(3):205–230, 1999.
Annotation: Naive definition of PLO-set.
- [804] K. Deb. **An efficient constraint handling method for genetic algorithms**. *Computer Methods in Applied Mechanics and Engineering*, 186(2/4):311–338, 2000. doi:10.1016/S0045-7825(99)00389-8.
- [805] K. Deb. *Multi-Objective Optimization Using Evolutionary Algorithms*. Wiley, Chichester, UK, 2001.
- [806] K. Deb. **Multi-objective optimization**. In E. K. Burke and G. Kendall, editors, *Search Methodologies*, pages 273–316. Springer, Boston, MA, 2005. doi:10.1007/0-387-28356-0_10.
- [807] K. Deb. **Introduction to evolutionary multiobjective optimization**. In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*, pages 59–96. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-88908-3_3.

- [808] K. Deb and R. B. Agrawal. **Simulated binary crossover for continuous search spaces.** *Complex Systems*, 9(2):115–148, 1995.
Keywords: SBX.
- [809] K. Deb and S. Agrawal. **A Niche-Penalty Approach for Constraint Handling in Genetic Algorithms.** In A. Dobnikar, N. C. Steele, D. W. Pearson, and R. F. Albrecht, editors, *Artificial Neural Nets and Genetic Algorithms (ICANNGA-99)*, pages 235–243. Springer Verlag, 1999.
[doi:10.1007/978-3-7091-6384-9](#).
Keywords: polynomial mutation.
- [810] K. Deb and D. Deb. **Analysing mutation schemes for real-parameter genetic algorithms.** *Intern. J. Artif. Intell. Soft. Comput.*, 4(1):1–28, 2014.
Annotation: Proposed Gaussian mutation.
- [811] K. Deb and H. Jain. **An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point-Based Nondominated Sorting Approach, Part I: Solving Problems With Box Constraints.** *IEEE Trans. Evol. Comput.*, 18(4):577–601, 2014. [doi:10.1109/TEVC.2013.2281535](#).
Annotation: Proposed NSGA-III.
- [812] K. Deb and S. Jain. **Multi-Speed Gearbox Design Using Multi-Objective Evolutionary Algorithms.** Technical Report 2002001, KanGAL, Feb. 2002.
- [813] K. Deb and M. Köksalan. **Guest Editorial: Special Issue on Preference-based Multiobjective Evolutionary Algorithms.** *IEEE Trans. Evol. Comput.*, 14(5):669–670, Oct. 2010. [doi:10.1109/TEVC.2010.2070371](#).
- [814] K. Deb and C. Myburgh. **Breaking the billion-variable barrier in real-world optimization using a customized evolutionary algorithm.** In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO*, pages 653–660. ACM Press, New York, NY, 2016.
- [815] K. Deb and A. Sinha. **Solving Bilevel Multi-Objective Optimization Problems Using Evolutionary Algorithms.** In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCS*, pages 110–124. Springer, Heidelberg, Germany, 2009.
- [816] K. Deb and J. Sundar. **Reference point based multi-objective optimization using evolutionary algorithms.** In M. Cattolico et al., editors, *GECCO*, pages 635–642. ACM Press, New York, NY, 2006. [doi:10.1145/1143997.1144112](#).
Annotation: Proposed R-NSGA-II.
- [817] K. Deb and S. Tiwari. **Omni-optimizer: A generic evolutionary algorithm for single and multi-objective optimization.** *Eur. J. Oper. Res.*, 185(3):1062–1087, 2008. [doi:10.1016/j.ejor.2006.06.042](#).
Keywords: epsilon-dominance, archiving.
Annotation: Archiving method with epsilon dominance and density in the decision and objective spaces.
- [818] K. Deb, S. Agarwal, A. Pratap, and T. Meyarivan. **A fast elitist non-dominated sorting genetic algorithm for multi-objective optimization: NSGA-II.** In M. Schoenauer et al., editors, *PPSN*, volume 1917 of *LNCS*, pages 849–858. Springer, Heidelberg, Germany, 2000.
[doi:10.1007/3-540-45356-3_83](#).
Annotation: IC.29.
- [819] K. Deb, L. Thiele, M. Laumanns, and E. Zitzler. **Scalable Test Problems for Evolutionary Multi-Objective Optimization.** Technical Report 112, Computer Engineering and Networks Laboratory (TIK), Swiss Federal Institute of Technology (ETH), Zürich, Switzerland, 2001.
Keywords: DTLZ benchmark, Do not cite this TR! It is incorrect and it is superseded by [822].

- [820] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan. **A fast and elitist multi-objective genetic algorithm: NSGA-II**. *IEEE Trans. Evol. Comput.*, 6(2):182–197, 2002. doi:[10.1109/4235.996017](https://doi.org/10.1109/4235.996017).
- [821] K. Deb, M. Mohan, and S. Mishra. **Evaluating the ϵ -domination based multi-objective evolutionary algorithm for a quick computation of Pareto-optimal solutions**. *Evol. Comput.*, 13(4):501–525, Dec. 2005. doi:[10.1162/106365605774666895](https://doi.org/10.1162/106365605774666895).
Keywords: ϵ -dominance, ϵ -MOEA.
- [822] K. Deb, L. Thiele, M. Laumanns, and E. Zitzler. **Scalable Test Problems for Evolutionary Multiobjective Optimization**. In A. Abraham, L. Jain, and R. Goldberg, editors, *Evolutionary Multiobjective Optimization*, Advanced Information and Knowledge Processing, pages 105–145. Springer, London, UK, Jan. 2005. doi:[10.1007/1-84628-137-7_6](https://doi.org/10.1007/1-84628-137-7_6).
Keywords: DTLZ benchmark.
- [823] K. Deb, R. Tewari, M. Dixit, and J. Dutta. **Finding trade-off solutions close to KKT points using evolutionary multi-objective optimization**. In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 2109–2116, Piscataway, NJ, 2007. IEEE Press.
- [824] K. Deb, S. Gupta, D. Daum, J. Branke, A. Mall, and D. Padmanabhan. **Reliability-based optimization using evolutionary algorithms**. *IEEE Trans. Evol. Comput.*, 13(5):1054–1074, Oct. 2009. doi:[10.1109/TEVC.2009.2014361](https://doi.org/10.1109/TEVC.2009.2014361).
- [825] K. Deb, L. Zhu, and S. Kulkarni. **Handling Multiple Scenarios in Evolutionary Multi-Objective Numerical Optimization**. *IEEE Trans. Evol. Comput.*, 22(6):920–933, 2018. doi:[10.1109/TEVC.2017.2776921](https://doi.org/10.1109/TEVC.2017.2776921).
Keywords: scenario-based.
- [826] K. Deb, E. D. Goodman, C. A. Coello Coello, K. Klamroth, K. Miettinen, S. Mostaghim, and P. M. Reed, editors. *Evolutionary Multi-Criterion Optimization – 10th International Conference, EMO 2019, East Lansing, MI, USA, March 10-13, 2019, Proceedings*, volume 11411 of LNCS. Springer International Publishing, Cham, Switzerland, 2019. ISBN 978-3-030-12597-4. doi:[10.1007/978-3-030-12598-1](https://doi.org/10.1007/978-3-030-12598-1).
- [827] K. Deb et al., editors. *Genetic and Evolutionary Computation Conference, GECCO 2004, Seattle, WA, USA, June 26-30, 2004, Proceedings, Part I*, volume 3102 of LNCS. Springer, Heidelberg, Germany, 2004.
- [828] K. Deb et al., editors. *Genetic and Evolutionary Computation Conference, GECCO 2004, Seattle, WA, USA, June 26-30, 2004, Proceedings, Part II*, volume 3103 of LNCS. Springer, Heidelberg, Germany, 2004.
- [829] R. Dechter, editor. *Principles and Practice of Constraint Programming, CP 2000, 6th International Conference, Singapore, September 18-21, 2000, Proceedings*, volume 1894 of LNCS. Springer, Heidelberg, Germany, 2000.
- [830] W. A. Dees, Jr. and P. G. Karger. **Automated Rip-up and Reroute Techniques**. In *DAC’82, Proceedings of the 19th Design Automation Workshop*, pages 432–439. IEEE Press, 1982.
- [831] V. Dekhtyarenko. **Verification of weight coefficients in multicriteria optimization problems**. *Computer-Aided Design*, 13(6):339–344, 1981.
- [832] R. F. Dell and M. H. Karwan. **An interactive MCDM weight space reduction method utilizing a Tchebycheff utility function**. *Naval Research Logistics*, 37(2):263–277, 1990.
- [833] F. Della Croce, T. Garaix, and A. Grosso. **Iterated Local Search and Very Large Neighborhoods for the Parallel-machines Total Tardiness Problem**. *Comput. Oper. Res.*, 39(6):1213–1217, 2012.

- [834] M. Dell’Amico and M. Trubian. **Applying Tabu Search to the Job Shop Scheduling Problem.** *Annals of Operations Research*, 41:231–252, 1993.
- [835] M. Dell’Amico, M. Iori, S. Martello, and M. Monaci. **Heuristic and Exact Algorithms for the Identical Parallel Machine Scheduling Problem.** *INFORMS Journal on Computing*, 20(3):333–344, 2016.
- [836] M. Dell’Amico, M. Iori, S. Novellani, and T. Stützle. **A destroy and repair algorithm for the Bike sharing Rebalancing Problem.** *Comput. Oper. Res.*, 71:146–162, 2016. doi:10.1016/j.cor.2016.01.011.
Keywords: irace.
- [837] M. Delorme, M. Iori, and S. Martello. **Bin packing and cutting stock problems: Mathematical models and exact algorithms.** *Eur. J. Oper. Res.*, 255(1):1–20, 2016. doi:10.1016/j.ejor.2016.04.030.
- [838] M. Delorme, M. Iori, and S. Martello. **BPPLIB: a library for bin packing and cutting stock problems.** *Optimization Letters*, 12(2):235–250, 2018. doi:10.1007/s11590-017-1192-z.
- [839] X. Delorme, X. Gandibleux, and F. Degoutin. **Evolutionary, constructive and hybrid procedures for the bi-objective set packing problem.** *Eur. J. Oper. Res.*, 204(2):206–217, 2010.
Annotation: This paper cannot be found on internet!! Does it exist?
- [840] R. A. DeMillo, editor. *Proceedings of the sixteenth annual ACM Symposium on Theory of Computing*, 1984. ACM Press.
- [841] S. Dempe, G. Eichfelder, and J. Fliege. **On the effects of combining objectives in multi-objective optimization.** *Mathematical Methods of Operations Research*, 82(1):1–18, 2015.
- [842] M. L. den Besten. *Simple Metaheuristics for Scheduling.* PhD thesis, FB Informatik, TU Darmstadt, Germany, 2004. URL <http://tuprints.ulb.tu-darmstadt.de/516/>.
- [843] M. L. den Besten, T. Stützle, and M. Dorigo. **Ant Colony Optimization for the Total Weighted Tardiness Problem.** In M. Schoenauer et al., editors, *PPSN*, volume 1917 of *LNCS*, pages 611–620. Springer, Heidelberg, Germany, 2000.
Annotation: IC.29.
- [844] M. L. den Besten, T. Stützle, and M. Dorigo. **Design of Iterated Local Search Algorithms: An Example Application to the Single Machine Total Weighted Tardiness Problem.** In E. J. W. Boers et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2001*, volume 2037 of *LNCS*, pages 441–452. Springer, Heidelberg, Germany, 2001.
- [845] J.-L. Deneubourg, S. Aron, S. Goss, and J.-M. Pasteels. **The Self-Organizing Exploratory Pattern of the Argentine Ant.** *Journal of Insect Behavior*, 3(2):159–168, 1990. doi:10.1007/BF01417909.
- [846] J. Deng and Q. Zhang. **Approximating Hypervolume and Hypervolume Contributions Using Polar Coordinate.** *IEEE Trans. Evol. Comput.*, 23(5):913–918, Oct. 2019. doi:10.1109/tevc.2019.2895108.
Annotation: Proposed approximating the hypervolume using scalarizations.
- [847] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. **Imagenet: A large-scale hierarchical image database.** In *Computer Vision and Pattern Recognition, 2009. CVPR 2009. IEEE Conference on*, pages 248–255. IEEE, 2009.

- [848] R. Denysiuk, L. Costa, and I. Espírito Santo. **Many-objective optimization using differential evolution with variable-wise mutation restriction**. In C. Blum and E. Alba, editors, *GECCO*, pages 591–598. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8.
- [849] U. Derigs and U. Vogel. **Experience with a Framework for Developing Heuristics for Solving Rich Vehicle Routing Problems**. *J. Heuristics*, 20(1):75–106, 2014.
- [850] J. Derrac, S. García, D. Molina, and F. Herrera. **A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms**. *Swarm and Evolutionary Computation*, 1(1):3–18, 2011.
- [851] A. Deshwal, S. Belakaria, J. R. Doppa, and D. H. Kim. **Bayesian Optimization over Permutation Spaces**. *Arxiv preprint arXiv:2112.01049*, 2021. doi:[10.48550/arXiv.2112.01049](https://doi.org/10.48550/arXiv.2112.01049).
Keywords: BOPS, CEGO.
- [852] P. Detti, F. Papalini, and G. Z. Manrique de Lara. **A multi-depot dial-a-ride problem with heterogeneous vehicles and compatibility constraints in healthcare**. *Omega*, 70:1–14, 2017. doi:[10.1016/j.omega.2016.08.008](https://doi.org/10.1016/j.omega.2016.08.008).
- [853] S. Devaraj, M. Fan, and R. Kohli. **Examination of online channel preference: using the structure-conduct-outcome framework**. *Decision Support Systems*, 42(2):1089–1103, 2006. doi:[10.1016/j.dss.2005.09.004](https://doi.org/10.1016/j.dss.2005.09.004).
- [854] S. Dewez. *On the toll setting problem*. PhD thesis, Faculté de Sciences, Université Libre de Bruxelles, 2014.
Annotation: Supervised by Dr. Martine Labbé.
- [855] C. Dhaenens, L. Jourdan, and M.-E. Marmion, editors. *9th International Conference, LION 9, Lille, France, January 12-15, 2015. Revised Selected Papers*, volume 8994 of *LNCS*. Springer, Heidelberg, Germany, 2015.
- [856] I. S. Dhillon, Y. Koren, R. Ghani, T. E. Senator, P. Bradley, R. Parekh, J. He, R. L. Grossman, and R. Uthurusamy, editors. *The 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD 2013*. ACM Press, New York, NY, 2013.
- [857] G. A. Di Caro and M. Dorigo. **AntNet: Distributed Stigmergetic Control for Communications Networks**. *J. Artif. Intell. Res.*, 9:317–365, 1998.
- [858] G. A. Di Caro, F. Ducatelle, and L. M. Gambardella. **AntHocNet: An adaptive nature-inspired algorithm for routing in mobile ad hoc networks**. *European Transactions on Telecommunications*, 16(5):443–455, 2005.
- [859] C. Di Chio et al., editors. *EvoApplications 2012: EvoCOMNET, EvoCOMPLEX, EvoFIN, EvoGAMES, EvoHOT, EvoIASP, EvoNUM, EvoPAR, EvoRISK, EvoSTIM, and EvoSTOC, Málaga, Spain, April 11-13, 2012, Proceedings*, volume 7248 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [860] L. Di Gaspero and A. Schaerf. **EASYLOCAL++: An object-oriented framework for flexible design of local search algorithms**. *Software — Practice & Experience*, 33(8):733–765, July 2003.
Keywords: software engineering, local search, easylocal.
- [861] L. Di Gaspero and A. Schaerf. **Easysyn++: A tool for automatic synthesis of stochastic local search algorithms**. In T. Stützle, M. Birattari, and H. H. Hoos, editors, *Engineering Stochastic Local Search Algorithms. Designing, Implementing and Analyzing Effective Heuristics. SLS 2007*, volume 4638 of *LNCS*, pages 177–181. Springer, Heidelberg, Germany, 2007.

- [862] L. Di Gaspero, M. Chiarandini, and A. Schaerf. **A Study on the Short-Term Prohibition Mechanisms in Tabu Search**. In G. Brewka, S. Coradeschi, A. Perini, and P. Traverso, editors, *Proceedings of the 17th European Conference on Artificial Intelligence, ECAI 2006, Riva del Garda, Italy, August 29 - September 1, 2006*, pages 83–87. IOS Press, 2006.
- [863] L. Di Gaspero, A. Schaerf, and T. Stützle, editors. *Proceedings of MIC 2011, the 9th Metaheuristics International Conference*, 2011.
- [864] L. Di Gaspero, A. Rendl, and T. Urli. **Constraint-Based Approaches for Balancing Bike Sharing Systems**. In C. Schulte, editor, *Principles and Practice of Constraint Programming*, volume 8124 of *LNCS*, pages 758–773. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-40627-0_56.
Keywords: F-race.
- [865] L. Di Gaspero, A. Rendl, and T. Urli. **A Hybrid ACO+CP for Balancing Bicycle Sharing Systems**. In M. J. Blesa, C. Blum, P. Festa, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 7919 of *LNCS*, pages 198–212. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-38515-5. doi:10.1007/978-3-642-38516-2_16.
Keywords: F-race.
- [866] F. di Pierro, S.-T. Khu, and D. A. Savic. **An investigation on preference order ranking scheme for multiobjective evolutionary optimization**. *IEEE Trans. Evol. Comput.*, 11(1): 17–45, 2007.
- [867] I. Diakonikolas and M. Yannakakis. **Succinct approximate convex Pareto curves**. In *Proceedings of the nineteenth annual ACM-SIAM symposium on Discrete algorithms*, pages 74–83. Society for Industrial and Applied Mathematics, 2008.
- [868] I. Diakonikolas and M. Yannakakis. **Small approximate Pareto sets for biobjective shortest paths and other problems**. *SIAM J. Comput.*, 39(4):1340–1371, 2009.
- [869] L. C. Dias, V. Mousseau, J. R. Figueira, and J. N. Clímaco. **An aggregation/disaggregation approach to obtain robust conclusions with ELECTRE TRI**. *Eur. J. Oper. Res.*, 138(2): 332–348, Apr. 2002.
- [870] D. Díaz, P. Villedor, P. Areces, J. Rodil, and M. Suárez. **An ACO Algorithm to Solve an Extended Cutting Stock Problem for Scrap Minimization in a Bar Mill**. In M. Dorigo et al., editors, *ANTS 2014*, volume 8667 of *LNCS*, pages 13–24. Springer, Heidelberg, Germany, 2014.
- [871] J. E. Diaz and M. López-Ibáñez. **Incorporating Decision-Maker’s Preferences into the Automatic Configuration of Bi-Objective Optimisation Algorithms**. *Eur. J. Oper. Res.*, 289(3):1209–1222, 2021. doi:10.1016/j.ejor.2020.07.059. Supplementary material: <https://doi.org/10.5281/zenodo.3749288>.
- [872] J. E. Diaz, J. Handl, and D.-L. Xu. **Evolutionary robust optimization in production planning: interactions between number of objectives, sample size and choice of robustness measure**. *Comput. Oper. Res.*, 79:266–278, 2017. doi:10.1016/j.cor.2016.06.020.
Keywords: Evolutionary multi-objective optimization, Production planning, Robust optimization, Simulation-based optimization, Uncertainty modelling.
- [873] J. E. Diaz, J. Handl, and D.-L. Xu. **Integrating meta-heuristics, simulation and exact techniques for production planning of a failure-prone manufacturing system**. *Eur. J. Oper. Res.*, 266(3):976–989, 2018. ISSN 0377-2217. doi:10.1016/j.ejor.2017.10.062.
Keywords: Genetic algorithms, Combinatorial optimization, Production planning, Simulation-based optimization, Uncertainty modelling.

- [874] B. Dilkina, E. B. Khalil, and G. L. Nemhauser. **Comments on: On learning and branching: a survey.** *TOP*, 25:242–246, 2017.
Annotation: Comments on [1993].
- [875] J.-Y. Ding, S. Song, J. N. D. Gupta, R. Zhang, R. Chiong, and C. Wu. **An Improved Iterated Greedy Algorithm with a Tabu-based Reconstruction Strategy for the No-wait Flowshop Scheduling Problem.** *Applied Soft Computing*, 30:604–613, 2015.
- [876] R. Ding, H. Dong, J. He, and T. Li. **A novel two-archive strategy for evolutionary many-objective optimization algorithm based on reference points.** *Applied Soft Computing*, 78:447–464, 2019. doi:10.1016/j.asoc.2019.02.040.
- [877] C. L. do Val Lopes, F. V. C. Martins, E. F. Wanner, and K. Deb. **Analyzing dominance move (MIP-DoM) indicator for multi-and many-objective optimization.** *IEEE Trans. Evol. Comput.*, 2021.
- [878] D. Doblas, A. J. Nebro, M. López-Ibáñez, J. García-Nieto, and C. A. Coello Coello. **Automatic Design of Multi-objective Particle Swarm Optimizers.** In M. Dorigo, H. Hamann, M. López-Ibáñez, J. García-Nieto, A. Engelbrecht, C. Pinciroli, V. Strobel, and C. L. Camacho-Villalón, editors, *Swarm Intelligence, 13th International Conference, ANTS 2022*, volume 13491 of *LNCS*, pages 28–40. Springer, Cham, Switzerland, 2022. doi:10.1007/978-3-031-20176-9_3.
- [879] A. Dobnikar, N. C. Steele, D. W. Pearson, and R. F. Albrecht, editors. *Artificial Neural Nets and Genetic Algorithms (ICANNGA-99), Proceedings of the International Conference in Portorož, Slovenia, 1999.* Springer Verlag, 1999. doi:10.1007/978-3-7091-6384-9.
- [880] K. F. Doerner, R. F. Hartl, and M. Reimann. **Are COMPETants more competent for problem solving? The case of a multiple objective transportation problem.** *Central European Journal for Operations Research and Economics*, 11(2):115–141, 2003.
- [881] K. F. Doerner, W. J. Gutjahr, R. F. Hartl, C. Strauss, and C. Stummer. **Pareto Ant Colony Optimization: A Metaheuristic Approach to Multiobjective Portfolio Selection.** *Annals of Operations Research*, 131:79–99, 2004.
- [882] K. F. Doerner, M. Gendreau, P. Greistorfer, W. J. Gutjahr, R. F. Hartl, and M. Reimann, editors. *6th Metaheuristics International Conference (MIC 2005)*, Vienna, Austria, 2005.
- [883] K. F. Doerner, G. Fuellerer, M. Gronalt, R. F. Hartl, and M. Iori. **Metaheuristics for the Vehicle Routing Problem with Loading Constraints.** *Networks*, 49(4):294–307, 2006.
- [884] K. F. Doerner, M. Gendreau, P. Greistorfer, W. J. Gutjahr, R. F. Hartl, and M. Reimann, editors. *Metaheuristics – Progress in Complex Systems Optimization*, volume 39 of *Operations Research/Computer Science Interfaces Series*. Springer, New York, NY, 2006.
- [885] K. F. Doerner, W. J. Gutjahr, R. F. Hartl, C. Strauss, and C. Stummer. **Pareto ant colony optimization with ILP preprocessing in multiobjective project portfolio selection.** *Eur. J. Oper. Res.*, 171:830–841, 2006.
- [886] K. F. Doerner, W. J. Gutjahr, R. F. Hartl, C. Strauss, and C. Stummer. **Nature-Inspired Metaheuristics in Multiobjective Activity Crashing.** *Omega*, 36(6):1019–1037, 2008.
- [887] K. F. Doerner, D. Merkle, and T. Stützle. **Special issue on Ant Colony Optimization.** *Swarm Intelligence*, 3(1), 2009.
- [888] B. Doerr and F. Neumann, editors. *Theory of Evolutionary Computation.* Springer International Publishing, 2020. doi:10.1007/978-3-030-29414-4.

- [889] B. Doerr, F. Neumann, D. Sudholt, and C. Witt. **Runtime analysis of the 1-ANT ant colony optimizer**. *Theor. Comput. Sci.*, 412(1):1629–1644, 2011.
- [890] B. Doerr, D. Johannsen, and C. Winzen. **Multiplicative drift analysis**. *Algorithmica*, 64(4): 673–697, 2012.
- [891] B. Doerr, T. Kötzing, J. Lengler, and C. Winzen. **Black-box complexities of combinatorial problems**. *Theor. Comput. Sci.*, 471:84–106, 2013.
- [892] B. Doerr, C. Doerr, and F. Ebel. **From black-box complexity to designing new genetic algorithms**. *Theor. Comput. Sci.*, 567:87–104, 2015. doi:10.1016/j.tcs.2014.11.028.
- [893] B. Doerr, C. Gießen, C. Witt, and J. Yang. **The $(1+\lambda)$ evolutionary algorithm with self-adjusting mutation rate**. *Algorithmica*, 81(2):593–631, 2019.
- [894] B. Doerr, C. Doerr, and J. Yang. **Optimal parameter choices via precise black-box analysis**. *Theor. Comput. Sci.*, 801:1–34, 2020. doi:10.1016/j.tcs.2019.06.014.
- [895] C. Doerr, H. Wang, F. Ye, S. van Rijn, and T. Bäck. **IOHprofiler: A Benchmarking and Profiling Tool for Iterative Optimization Heuristics**. *Arxiv preprint arXiv:1806.07555*, Oct. 2018. doi:10.48550/arXiv.1810.05281.
Keywords: Benchmarking; Heuristics.
- [896] J.-P. Doignon, A. Pekeč, and M. Regenwetter. **The repeated insertion model for rankings: Missing link between two subset choice models**. *Psychometrika*, 69(1):33–54, Mar. 2004. doi:10.1007/bf02295838.
- [897] E. D. Dolan and J. J. Moré. **Benchmarking optimization software with performance profiles**. *Mathematical Programming*, 91(2):201–213, 2002.
Keywords: performance profiles; convergence.
Annotation: This methodology has been criticized in <https://doi.org/10.1145/2950048>.
- [898] P. Domingos and G. Hulten. **Mining high-speed data streams**. In R. Ramakrishnan, S. J. Stolfo, R. J. Bayardo, and I. Parsa, editors, *The 6th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD 2000*, pages 71–80. ACM Press, New York, NY, 2000.
- [899] A. V. Donati, R. Montemanni, N. Casagrande, A. E. Rizzoli, and L. M. Gambardella. **Time dependent vehicle routing problem with a multi ant colony system**. *Eur. J. Oper. Res.*, 185(3):1174–1191, 2008.
- [900] X. Dong, H. Huang, and P. Chen. **An Iterated Local Search Algorithm for the Permutation Flowshop Problem with Total Flowtime Criterion**. *Comput. Oper. Res.*, 36(5):1664–1669, 2009.
- [901] X. Dong, Ping, H. Huang, and M. Nowak. **A Multi-restart Iterated Local Search Algorithm for the Permutation Flow Shop Problem Minimizing Total Flow Time**. *Comput. Oper. Res.*, 40(2):627–632, 2013.
- [902] M. Dorigo. *Optimization, Learning and Natural Algorithms*. PhD thesis, Dipartimento di Elettronica, Politecnico di Milano, Italy, 1992. In Italian.
- [903] M. Dorigo. **Ant Colony Optimization**. *Scholarpedia*, 2(3):1461, 2007. doi:10.4249/scholarpedia.1461.
- [904] M. Dorigo. **Swarm intelligence: A few things you need to know if you want to publish in this journal**. *Swarm Intelligence*, Nov. 2016. URL https://static.springer.com/sgw/documents/1593723/application/pdf/Additional_submission_instructions.pdf.

- [905] M. Dorigo and C. Blum. **Ant colony optimization theory: A survey.** *Theor. Comput. Sci.*, 344(2-3):243–278, 2005.
- [906] M. Dorigo and G. A. Di Caro. **The Ant Colony Optimization Meta-Heuristic.** In D. Corne, M. Dorigo, and F. Glover, editors, *New Ideas in Optimization*, pages 11–32. McGraw Hill, London, UK, 1999.
Annotation: Also available as Technical Report IRIDIA/99-1, Université Libre de Bruxelles, Belgium.
- [907] M. Dorigo and L. M. Gambardella. **Ant Colony System.** Technical Report IRIDIA/96-05, IRIDIA, Université Libre de Bruxelles, Belgium, 1996.
- [908] M. Dorigo and L. M. Gambardella. **Ant Colonies for the Traveling Salesman Problem.** *BioSystems*, 43(2):73–81, 1997. doi:10.1016/S0303-2647(97)01708-5.
- [909] M. Dorigo and L. M. Gambardella. **Ant Colony System: A Cooperative Learning Approach to the Traveling Salesman Problem.** *IEEE Trans. Evol. Comput.*, 1(1):53–66, 1997.
Keywords: Ant Colony System.
- [910] M. Dorigo and T. Stützle. **The Ant Colony Optimization Metaheuristic: Algorithms, Applications and Advances.** In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 251–285. Kluwer Academic Publishers, Norwell, MA, 2002.
- [911] M. Dorigo and T. Stützle. *Ant Colony Optimization*. MIT Press, Cambridge, MA, 2004.
Annotation: 305 p.
- [912] M. Dorigo, V. Maniezzo, and A. Coloni. **The Ant System: An autocatalytic optimizing process.** Technical Report 91-016 Revised, Dipartimento di Elettronica, Politecnico di Milano, Italy, 1991.
- [913] M. Dorigo, V. Maniezzo, and A. Coloni. **Positive Feedback as a Search Strategy.** Technical Report 91-016, Dipartimento di Elettronica, Politecnico di Milano, Italy, 1991.
- [914] M. Dorigo, V. Maniezzo, and A. Coloni. **Ant System: Optimization by a Colony of Cooperating Agents.** *IEEE Transactions on Systems, Man, and Cybernetics – Part B*, 26(1): 29–41, 1996.
- [915] M. Dorigo, G. A. Di Caro, and L. M. Gambardella. **Ant Algorithms for Discrete Optimization.** *Artificial Life*, 5(2):137–172, 1999.
- [916] M. Dorigo, T. Stützle, and G. A. Di Caro. **Special Issue on “Ant Algorithms”.** *Future Generation Computer Systems*, 16(8), 2000.
Keywords: swarm intelligence, ant colony optimization.
- [917] M. Dorigo, L. M. Gambardella, M. Middendorf, and T. Stützle. **Guest Editorial: Special Section on Ant Colony Optimization.** *IEEE Trans. Evol. Comput.*, 6(4):317–320, 2002. doi:10.1109/TEVC.2002.802446.
Keywords: ant colony optimization, swarm intelligence.
- [918] M. Dorigo, M. Birattari, and T. Stützle. **Ant Colony Optimization: Artificial Ants as a Computational Intelligence Technique.** *IEEE Computational Intelligence Magazine*, 1(4): 28–39, 2006.
- [919] M. Dorigo, M. A. Montes de Oca, S. Oliveira, and T. Stützle. **Ant Colony Optimization.** In J. J. Cochran, editor, *Wiley Encyclopedia of Operations Research and Management Science*, volume 1, pages 114–125. John Wiley & Sons, 2011. doi:10.1002/9780470400531.

- [920] M. Dorigo, M. Birattari, X. Li, M. López-Ibáñez, K. Ohkura, C. Pinciroli, and T. Stützle, editors. *Swarm Intelligence, 10th International Conference, ANTS 2016, Brussels, Belgium, September 7-9, 2016, Proceedings*, volume 9882 of *LNCS*. Springer, Heidelberg, Germany, 2016. doi:10.1007/978-3-319-44427-7.
- [921] M. Dorigo, M. Birattari, X. Li, M. López-Ibáñez, K. Ohkura, C. Pinciroli, and T. Stützle. **ANTS 2016 Special Issue: Editorial**. *Swarm Intelligence*, Nov. 2017. doi:10.1007/s11721-017-0146-5.
- [922] M. Dorigo, M. Birattari, A. L. Christensen, A. Reina, and V. Trianni, editors. *Swarm Intelligence, 11th International Conference, ANTS 2018, Rome, Italy, October 29-31, 2018, Proceedings*, volume 11172 of *LNCS*. Springer, Heidelberg, Germany, 2018.
- [923] M. Dorigo, T. Stützle, M. J. Blesa, C. Blum, H. Hamann, and M. K. Heinrich, editors. *Swarm Intelligence, 12th International Conference, ANTS 2020, Barcelona, Spain, October 26-28, 2020, Proceedings*, volume 12421 of *LNCS*. Springer, Heidelberg, Germany, 2020.
- [924] M. Dorigo, H. Hamann, M. López-Ibáñez, J. García-Nieto, A. Engelbrecht, C. Pinciroli, V. Strobel, and C. L. Camacho-Villalón, editors. *Swarm Intelligence, 13th International Conference, ANTS 2022, Málaga, Spain, November 2-4, 2022, Proceedings*, volume 13491 of *LNCS*. Springer, Cham, Switzerland, 2022. doi:10.1007/978-3-031-20176-9.
- [925] M. Dorigo et al., editors. *Abstract proceedings of ANTS 2000 – From Ant Colonies to Artificial Ants: Second International Workshop on Ant Algorithms, Brussels, Belgium, September 7-9, 2000*, Sept. 2000. IRIDIA, Université Libre de Bruxelles, Belgium.
- [926] M. Dorigo et al., editors. *Ant Algorithms, Third International Workshop, ANTS 2002, Brussels, Belgium, September 12-14, 2002, Proceedings*, volume 2463 of *LNCS*. Springer, Heidelberg, Germany, 2002.
- [927] M. Dorigo et al., editors. *Ant Colony Optimization and Swarm Intelligence, 4th International Workshop, ANTS 2004*, volume 3172 of *LNCS*. Springer, Heidelberg, Germany, 2004.
- [928] M. Dorigo et al., editors. *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*. Springer, Heidelberg, Germany, 2006.
- [929] M. Dorigo et al., editors. *Ant Colony Optimization and Swarm Intelligence, 6th International Conference, ANTS 2008*, volume 5217 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [930] M. Dorigo et al., editors. *Ant Colony Optimization and Swarm Intelligence, 7th International Conference, ANTS 2010*, volume 6234 of *LNCS*. Springer, Heidelberg, Germany, 2010.
- [931] M. Dorigo et al., editors. *Swarm Intelligence, 8th International Conference, ANTS 2012*, volume 7461 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [932] M. Dorigo et al., editors. *Swarm Intelligence, 9th International Conference, ANTS 2014*, volume 8667 of *LNCS*. Springer, Heidelberg, Germany, 2014.
- [933] M. Doumpos and C. Zopounidis. **Preference disaggregation and statistical learning for multicriteria decision support: A review**. *Eur. J. Oper. Res.*, 209(3):203–214, 2011.
- [934] E. Dovgan, T. Tušar, and B. Filipič. **Parameter tuning in an evolutionary algorithm for commodity transportation optimization**. *Evol. Comput.*, pages 1–8, 2010.
- [935] J. Dreö. *Adaptation de la métaheuristique des colonies de fourmis pour l’optimisation difficile en variables continues: Application en génie biologique et médical*. PhD thesis, LERISS - Laboratoire d’étude et de recherche en instrumentation, signaux et systèmes, Université Paris XII Val de Marne, Dec. 2003. URL <https://tel.archives-ouvertes.fr/tel-00093143>.

Keywords: metaheuristic ; continuous optimization ; global optimization ; imagery ; registration ; ant colony algorithm ; estimation of distribution algorithm ; evolutionary computation ; métaheuristique ; optimisation continue ; optimisation globale ; imagerie ; biomédical ; recalage ; algorithme de colonie de fourmis ; algorithme à estimation de distribution ; algorithme évolutionnaire.

- [936] J. Dreó. **Using performance fronts for parameter setting of stochastic metaheuristics.** In F. Rothlauf, editor, *GECCO Companion*, pages 2197–2200. ACM Press, New York, NY, 2009. doi:10.1145/1570256.1570301.
- [937] J. Dreó and P. Siarry. **A New Ant Colony Algorithm Using the Heterarchical Concept Aimed at Optimization of Multim minima Continuous Functions.** In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 216–221. Springer, Heidelberg, Germany, 2002.
- [938] J. Dreó and P. Siarry. **Continuous interacting ant colony algorithm based on dense heterarchy.** *Future Generation Computer Systems*, 20(5):841–856, 2004.
- [939] J. Dreó, C. Doerr, and Y. Semet. **Coupling the design of benchmark with algorithm in landscape-aware solver design.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1419–1420. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.
- [940] J. Dreó, A. Liefoghe, S. Verel, M. Schoenauer, J.-J. Merelo, A. Quemy, B. Bouvier, and J. Gmys. **Paradiseo: from a modular framework for evolutionary computation to the automated design of metaheuristics: 22 years of Paradiseo.** In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1522–1530. ACM Press, New York, NY, 2021. doi:10.1145/3449726.3463276.
Keywords: metaheuristics, evolutionary computation, software framework, automated algorithm design.
- [941] S. Droste, T. Jansen, and I. Wegener. **A new framework for the valuation of algorithms for black-box-optimization.** In K. A. De Jong, R. Poli, and J. E. Rowe, editors, *Proceedings of the Seventh Workshop on Foundations of Genetic Algorithms (FOGA)*, pages 253–270. Morgan Kaufmann Publishers, 2002.
- [942] S. Droste, T. Jansen, and I. Wegener. **Upper and lower bounds for randomized search heuristics in black-box optimization.** *ACM Trans. Comput. Syst.*, 39(4):525–544, 2006.
- [943] M. M. Drugan and D. Thierens. **Path-Guided Mutation for Stochastic Pareto Local Search Algorithms.** In R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 485–495. Springer, Heidelberg, Germany, 2010.
- [944] M. M. Drugan and D. Thierens. **Stochastic Pareto local search: Pareto neighbourhood exploration and perturbation strategies.** *J. Heuristics*, 18(5):727–766, 2012.
- [945] C. Drummond. **Replicability is not Reproducibility: Nor is it Good Science.** In *Proceedings of the Evaluation Methods for Machine Learning Workshop at the 26th ICML*, Montreal, Canada, 2009. URL <http://www.site.uottawa.ca/~cdrummon/pubs/ICMLws09.pdf>.
- [946] J. Du and J. Y.-T. Leung. **Minimizing Total Tardiness on One Machine is NP-Hard.** *Mathematics of Operations Research*, 15(3):483–495, 1990.
- [947] A. Duarte, J. Sánchez-Oro, N. Mladenović, and R. Todosijević. **Variable Neighborhood Descent.** In R. Martí, P. M. Pardalos, and M. G. C. Resende, editors, *Handbook of Heuristics*, pages 341–367. Springer International Publishing, 2018. ISBN 978-3-319-07125-1. doi:10.1007/978-3-319-07124-4_9.

- [948] J. Dubois-Lacoste. **Weight Setting Strategies for Two-Phase Local Search: A Study on Biobjective Permutation Flowshop Scheduling**. Technical Report TR/IRIDIA/2009-024, IRIDIA, Université Libre de Bruxelles, Belgium, 2009.
- [949] J. Dubois-Lacoste. **A study of Pareto and Two-Phase Local Search Algorithms for Biobjective Permutation Flowshop Scheduling**. Master’s thesis, IRIDIA, Université Libre de Bruxelles, Belgium, 2009.
- [950] J. Dubois-Lacoste. **Effective Stochastic Local Search Algorithms For Bi-Objective Permutation Flowshop Scheduling**. Rapport d’avancement de recherches présenté pour la formation doctorale en sciences de l’ingénieur, IRIDIA, Université Libre de Bruxelles, Belgium, 2010.
- [951] J. Dubois-Lacoste. *Anytime Local Search for Multi-Objective Combinatorial Optimization: Design, Analysis and Automatic Configuration*. PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2014.
Annotation: Supervised by Thomas Stützle and Manuel López-Ibáñez.
- [952] J. Dubois-Lacoste and T. Stützle. **Tuning of a Stigmergy-based Traffic Light Controller as a Dynamic Optimization Problem**. In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 1–8, Piscataway, NJ, 2017. IEEE Press.
- [953] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Effective Hybrid Stochastic Local Search Algorithms for Biobjective Permutation Flowshop Scheduling**. In M. J. Blesa, C. Blum, L. Di Gaspero, A. Roli, M. Sampels, and A. Schaerf, editors, *Hybrid Metaheuristics*, volume 5818 of *LNCS*, pages 100–114. Springer, Heidelberg, Germany, 2009. doi:10.1007/978-3-642-04918-7_8.
- [954] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Effective Hybrid Stochastic Local Search Algorithms for Biobjective Permutation Flowshop Scheduling**. Technical Report TR/IRIDIA/2009-020, IRIDIA, Université Libre de Bruxelles, Belgium, June 2009. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2009-020.pdf>. Published in the proceedings of Hybrid Metaheuristics 2009 [953].
- [955] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Supplementary material: Improving the Anytime Behavior of Two-Phase Local Search**. <https://iridia.ulb.ac.be/supp/IridiaSupp2010-012>, 2010.
- [956] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Supplementary material: A Hybrid TP+PLS Algorithm for Bi-objective Flow-shop Scheduling Problems**. <https://iridia.ulb.ac.be/supp/IridiaSupp2010-001>, 2010.
- [957] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Adaptive “Anytime” Two-Phase Local Search**. In C. Blum and R. Battiti, editors, *LION 4*, volume 6073 of *LNCS*, pages 52–67. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-13800-3_5.
- [958] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Adaptive “Anytime” Two-Phase Local Search**. Technical Report TR/IRIDIA/2009-026, IRIDIA, Université Libre de Bruxelles, Belgium, 2010. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2009-026.pdf>. Published in the proceedings of LION 4 [957].
- [959] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **A Hybrid TP+PLS Algorithm for Bi-objective Flow-Shop Scheduling Problems**. Technical Report TR/IRIDIA/2010-019, IRIDIA, Université Libre de Bruxelles, Belgium, 2010. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2010-019.pdf>. Published in Computers & Operations Research [962].

- [960] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Improving the Anytime Behavior of Two-Phase Local Search**. Technical Report TR/IRIDIA/2010-022, IRIDIA, Université Libre de Bruxelles, Belgium, 2010. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2010-022.pdf>. Published in *Annals of Mathematics and Artificial Intelligence* [961].
- [961] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Improving the Anytime Behavior of Two-Phase Local Search**. *Annals of Mathematics and Artificial Intelligence*, 61(2):125–154, 2011. doi:10.1007/s10472-011-9235-0.
- [962] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **A Hybrid TP+PLS Algorithm for Bi-objective Flow-Shop Scheduling Problems**. *Comput. Oper. Res.*, 38(8):1219–1236, 2011. doi:10.1016/j.cor.2010.10.008. Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2010-001/>.
- [963] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Automatic Configuration of State-of-the-art Multi-Objective Optimizers Using the TP+PLS Framework**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 2019–2026. ACM Press, New York, NY, 2011. doi:10.1145/2001576.2001847.
- [964] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Pareto Local Search Algorithms for Anytime Bi-objective Optimization**. In J.-K. Hao and M. Middendorf, editors, *EvoCOP*, volume 7245 of *LNCS*, pages 206–217. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-29124-1_18.
- [965] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Combining Two Search Paradigms for Multi-objective Optimization: Two-Phase and Pareto Local Search**. In E.-G. Talbi, editor, *Hybrid Metaheuristics*, volume 434 of *Studies in Computational Intelligence*, pages 97–117. Springer Verlag, 2013. doi:10.1007/978-3-642-30671-6_3. URL <http://www.springer.com/engineering/computational+intelligence+and+complexity/book/978-3-642-30670-9>.
- [966] J. Dubois-Lacoste, H. H. Hoos, and T. Stützle. **On the Empirical Scaling Behaviour of State-of-the-art Local Search Algorithms for the Euclidean TSP**. In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 377–384. ACM Press, New York, NY, 2015. doi:10.1145/2739480.2754747.
- [967] J. Dubois-Lacoste, M. López-Ibáñez, and T. Stützle. **Anytime Pareto Local Search**. *Eur. J. Oper. Res.*, 243(2):369–385, 2015. doi:10.1016/j.ejor.2014.10.062. *Keywords*: Pareto local search.
- [968] J. Dubois-Lacoste, F. Pagnozzi, and T. Stützle. **Supplementary material: An iterated greedy algorithm with optimization of partial solutions for the permutation flowshop problem**. <https://iridia.ulb.ac.be/supp/IridiaSupp2013-006>, 2017.
- [969] J. Dubois-Lacoste, F. Pagnozzi, and T. Stützle. **An Iterated Greedy Algorithm with Optimization of Partial Solutions for the Permutation Flowshop Problem**. *Comput. Oper. Res.*, 81:160–166, 2017. doi:10.1016/j.cor.2016.12.021. Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2013-006>.
- [970] F. Duddeck. **Multidisciplinary optimization of car bodies**. *Struct. Multidisc. Optim.*, 35(4):375–389, 2008. doi:10.1007/s00158-007-0130-6. *Annotation*: Evolutionary optimization of car bodies at General Motors.
- [971] G. Dueck. **New Optimization Heuristics: the Great Deluge Algorithm and the Record-To-Record Travel**. *Journal of Computational Physics*, 104(1):86–92, 1993.
- [972] G. Dueck and T. Scheuer. **Threshold Accepting: A General Purpose Optimization Algorithm Appearing Superior to Simulated Annealing**. *Journal of Computational Physics*, 90(1):161–175, 1990.

- [973] G. Dueck, M. Maehler, J. Schneider, G. Schrimpf, and H. Stamm-Wilbrandt. **Optimization with Ruin Recreate**. US Patent 6,418,398 B1, July 2002. Filed on October 1, 1999 and granted on July 9, 2002; Assignee is IBM Corporation, Armonk, NY (US).
- [974] C. Duin and S. Voß. **The Pilot Method: A Strategy for Heuristic Repetition with Application to the Steiner Problem in Graphs**. *Networks*, 34(3):181–191, 1999.
- [975] R. R. P. R. Duivenvoorden, F. Berkenkamp, N. Carion, A. Krause, and A. P. Schoellig. **Constrained Bayesian Optimization with Particle Swarms for Safe Adaptive Controller Tuning**. *IFAC-PapersOnLine*, 50(1):11800–11807, 2017. doi:10.1016/j.ifacol.2017.08.1991.
Keywords: Adaptive Control, Constrained Bayesian Optimization, Safety, Gaussian Process, Particle Swarm Optimization, Policy Search, Reinforcement learning.
Annotation: 20th IFAC World Congress.
- [976] Y. Dumas, J. Desrosiers, E. Gelinas, and M. M. Solomon. **An Optimal Algorithm for the Traveling Salesman Problem with Time Windows**. *Operations Research*, 43(2):367–371, 1995. doi:10.1287/opre.43.2.367.
- [977] I. Dumitrescu and T. Stützle. **Combinations of Local Search and Exact Algorithms**. In G. R. Raidl and J. Gottlieb, editors, *Proceedings of EvoCOP 2003 – 3rd European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 2611 of *LNCS*, pages 211–223. Springer, Heidelberg, Germany, 2003. doi:10.1007/3-540-36605-9_20.
- [978] I. Dumitrescu and T. Stützle. **Usage of Exact Algorithms to Enhance Stochastic Local Search Algorithms**. In V. Maniezzo, T. Stützle, and S. Voß, editors, *Matheuristics—Hybridizing Metaheuristics and Mathematical Programming*, volume 10 of *Annals of Information Systems*, pages 103–134. Springer, New York, NY, 2009.
- [979] O. J. Dunn. **Multiple Comparisons Among Means**. *Journal of the American Statistical Association*, 56(293):52–64, 1961.
- [980] O. J. Dunn. **Multiple Comparisons Using Rank Sums**. *Technometrics*, 6(3):241–252, 1964.
- [981] J. J. Durillo and A. J. Nebro. **jMetal: A Java framework for multi-objective optimization**. *Adv. Eng. Softw.*, 42(10):760–771, 2011. ISSN 0965-9978. doi:10.1016/j.advengsoft.2011.05.014.
- [982] J. J. Durillo, J. García-Nieto, A. J. Nebro, C. A. Coello Coello, F. Luna, and E. Alba. **Multi-Objective Particle Swarm Optimizers: An Experimental Comparison**. In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCS*, pages 495–509. Springer, Heidelberg, Germany, 2009. ISBN 978-3-642-01020-0.
- [983] J. J. Durillo, A. J. Nebro, F. Luna, and E. Alba. **On the Effect of the Steady-State Selection Scheme in Multi-Objective Genetic Algorithms**. In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCS*, pages 183–197. Springer, Heidelberg, Germany, 2009.
- [984] C. Dwork, R. Kumar, M. Naor, and D. Sivakumar. **Rank aggregation methods for the Web**. In V. Y. Shen, N. Saito, M. R. Lyu, and M. E. Zurko, editors, *Proceedings of the Tenth International World Wide Web Conference, WWW 10*, pages 613–622. ACM Press, New York, NY, 2001. ISBN 1-58113-348-0. doi:10.1145/371920.372165.
Keywords: Kemeny ranking,multi-word queries,rank aggregation,ranking functions,spam.
- [985] J. G. Dy and A. Krause, editors. *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10-15, 2018*, volume 80 of *Proceedings of Machine Learning Research*, 2018. PMLR.

- [986] J. S. Dyer. **Interactive Goal Programming**. *Management Science*, 19(1):62–70, 1972. doi:10.1287/mnsc.19.1.62.
- [987] H. W. E., S. J. E., and K. N., editors. *Recent Advances in Memetic Algorithms*, volume 166 of *Studies in Fuzziness and Soft Computing*. Springer, Berlin/Heidelberg, 2005.
- [988] R. C. Eberhart and J. Kennedy. **A New Optimizer Using Particle Swarm Theory**. In *Proceedings of the Sixth International Symposium on Micro Machine and Human Science*, pages 39–43, 1995.
- [989] ECML PKDD. *Machine Learning and Knowledge Discovery in Databases - European Conference, ECML PKDD 2015, Porto, Portugal, September 7-11, 2015, Proceedings, Part III*, volume 9286 of *LNCS*, 2015. Springer.
- [990] S. Edelkamp and R. Barták, editors. *Proceedings of the Seventh Annual Symposium on Combinatorial Search, SOCS 2014, Prague, Czech Republic, 15-17 August 2014*, 2014. AAAI Press.
- [991] W. Edwards. **How to Use Multiattribute Utility Measurement for Social Decisionmaking**. *IEEE Transactions on Systems, Man, and Cybernetics*, 7(5):326–340, May 1977. doi:10.1109/TSMC.1977.4309720.
Annotation: Proposed SMART method.
- [992] K. Eggensperger, F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Efficient Benchmarking of Hyperparameter Optimizers via Surrogates**. In B. Bonet and S. Koenig, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1114–1120. AAAI Press, 2015. doi:10.1609/aaai.v29i1.9375.
- [993] K. Eggensperger, M. T. Lindauer, and F. Hutter. **Pitfalls and best practices in algorithm configuration**. *J. Artif. Intell. Res.*, 64:861–893, 2019.
- [994] R. W. Eglese. **Simulated Annealing: a Tool for Operational Research**. *Eur. J. Oper. Res.*, 46(3):271–281, 1990.
- [995] W. Ehm. **Reproducibility from the perspective of meta-analysis**. In H. Atmanspacher and S. Maasen, editors, *Reproducibility – Principles, problems, practices and prospects*, pages 141–168. Wiley, 2016.
- [996] J. F. Ehmke, A. M. Campbell, and B. W. Thomas. **Vehicle routing to minimize time-dependent emissions in urban areas**. *Eur. J. Oper. Res.*, 251(2):478–494, June 2016. doi:10.1016/j.ejor.2015.11.034.
- [997] M. Ehrgott. *Multicriteria Optimization*, volume 491 of *LNEMS*. Springer, Berlin, Germany, 2000.
- [998] M. Ehrgott. *Multicriteria Optimization*. Springer, Berlin, Germany, 2nd edition, 2005. doi:10.1007/3-540-27659-9.
- [999] M. Ehrgott. **A discussion of scalarization techniques for multiple objective integer programming**. *Annals of Operations Research*, 147(1):343–360, 2006.
- [1000] M. Ehrgott and X. Gandibleux. **Approximative Solution Methods for Combinatorial Multicriteria Optimization**. *TOP*, 12(1):1–88, 2004.
- [1001] M. Ehrgott and X. Gandibleux. **Hybrid Metaheuristics for Multi-objective Combinatorial Optimization**. In C. Blum, M. J. Blesa, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics: An emergent approach for optimization*, volume 114 of *Studies in Computational Intelligence*, pages 221–259. Springer, Berlin, Germany, 2008. doi:10.1007/978-3-540-78295-7_8.

- [1002] M. Ehrgott and K. Klamroth. **Connectedness of Efficient Solutions in Multiple Criteria Combinatorial Optimization**. *Eur. J. Oper. Res.*, 97(1):159–166, 1997. doi:10.1016/S0377-2217(96)00116-6.
- [1003] M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors. *Evolutionary Multi-Criterion Optimization. 5th International Conference, EMO 2009*, volume 5467 of *LNCS*. Springer, Heidelberg, Germany, 2009.
- [1004] M. Ehrgott, J. R. Figueira, and S. Greco, editors. *Trends in Multiple Criteria Decision Analysis*, volume 142 of *International Series in Operations Research & Management Science*. Springer, US, 2010.
- [1005] A. E. Eiben and M. Jelasity. **A critical note on experimental research methodology in EC**. In *IEEE CEC*, pages 582–587, Piscataway, NJ, 2002. IEEE Press. doi:10.1109/cec.2002.1006991.
Annotation: Discusses reproducibility, generalizability and separation between training (for tuning and experimentation) and testing instances (for comparisons).
- [1006] A. E. Eiben and G. Rudolph. **Theory of evolutionary algorithms: A bird’s eye view**. *Theor. Comput. Sci.*, 229(1-2):3–9, 1999.
- [1007] A. E. Eiben and S. K. Smit. **Parameter Tuning for Configuring and Analyzing Evolutionary Algorithms**. *Swarm and Evolutionary Computation*, 1(1):19–31, 2011. doi:10.1016/j.swevo.2011.02.001.
- [1008] A. E. Eiben and J. E. Smith. *Introduction to Evolutionary Computing*. Springer, 2003. ISBN 3540401849.
- [1009] A. E. Eiben and J. E. Smith. *Introduction to Evolutionary Computing*. Natural Computing Series. Springer, 2nd edition, 2007.
- [1010] A. E. Eiben, T. Bäck, M. Schoenauer, and H.-P. Schwefel, editors. *Parallel Problem Solving from Nature – PPSN V, 5th International Conference Amsterdam, The Netherlands September 27-30, 1998 Proceedings*, volume 1498 of *LNCS*. Springer, Heidelberg, Germany, 1998.
- [1011] A. E. Eiben, R. Hinterding, and Z. Michalewicz. **Parameter Control in Evolutionary Algorithms**. *IEEE Trans. Evol. Comput.*, 3(2):124–141, 1999.
- [1012] A. E. Eiben, M. Horvath, W. Kowalczyk, and M. C. Schut. **Reinforcement learning for online control of evolutionary algorithms**. In *International Workshop on Engineering Self-Organising Applications*, pages 151–160. Springer, 2006.
- [1013] A. E. Eiben, Z. Michalewicz, M. Schoenauer, and J. E. Smith. **Parameter Control in Evolutionary Algorithms**. In F. Lobo, C. F. Lima, and Z. Michalewicz, editors, *Parameter Setting in Evolutionary Algorithms*, pages 19–46. Springer, Berlin, Germany, 2007.
- [1014] S. Eker and J. H. Kwakkel. **Including robustness considerations in the search phase of Many-Objective Robust Decision Making**. *Environmental Modelling & Software*, 105: 201–216, 2018.
Keywords: scenario-based.
- [1015] M. El-Abd. **Opposition-based Artificial Bee Colony Algorithm**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 109–116. ACM Press, New York, NY, 2011.
- [1016] M. El Yafrani, M. Scoczynski, I. Sung, M. Wagner, C. Doerr, and P. Nielsen. **MATE: A Model-Based Algorithm Tuning Engine**. In C. Zarges and S. Verel, editors, *Proceedings of EvoCOP 2021 – 21th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 12692 of *LNCS*, pages 51–67. Springer, Cham, Switzerland, 2021. doi:10.

1007/978-3-030-72904-2_4.

Keywords: Genetic programming, Model-based tuning, Parameter tuning.

- [1017] E. Elkind, editor. *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI 2023, 19th-25th August 2023, Macao, SAR, China, 2023*. IJCAI/AAAI Press, Menlo Park, CA.
- [1018] J. L. Elman. **Distributed representations, simple recurrent networks, and grammatical structure**. *Mach. Learn.*, 7(2-3):195–225, 1991.
- [1019] N. Elsokkary, F. S. Khan, D. La Torre, T. S. Humble, and J. Gottlieb. **Financial Portfolio Management using D-Wave’s Quantum Optimizer: The Case of Abu Dhabi Securities Exchange**. Technical report, Oak Ridge National Lab, Oak Ridge, TN, USA, 2017. URL <https://www.osti.gov/biblio/1423041>.
- [1020] V. A. Emelichev and V. A. Perepelitsa. **Complexity of Vector Optimization Problems on Graphs**. *Optimization*, 22(6):906–918, 1991. doi:10.1080/02331939108843732.
- [1021] V. A. Emelichev and V. A. Perepelitsa. **On the Cardinality of the Set of Alternatives in Discrete Many-criterion Problems**. *Discrete Mathematics and Applications*, 2(5):461–471, 1992.
- [1022] M. T. M. Emmerich and A. H. Deutz. **A tutorial on multiobjective optimization: Fundamentals and evolutionary methods**. *Natural Computing*, 17(3):585–609, 2018.
- [1023] M. T. M. Emmerich and C. M. Fonseca. **Computing Hypervolume Contributions in Low Dimensions: Asymptotically Optimal Algorithm and Complexity Results**. In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 121–135. Springer, Berlin/Heidelberg, 2011. doi:10.1007/978-3-642-19893-9_9.
- [1024] M. T. M. Emmerich, K. C. Giannakoglou, and B. Naujoks. **Single- and multiobjective evolutionary optimization assisted by Gaussian random field metamodels**. *IEEE Trans. Evol. Comput.*, 10(4):421–439, 2006. doi:10.1109/TEVC.2005.859463.
- [1025] M. T. M. Emmerich, A. H. Deutz, and J. W. Klinkenberg. **Hypervolume-based expected improvement: Monotonicity properties and exact computation**. In *Proceedings of the 2011 Congress on Evolutionary Computation (CEC 2011)*, pages 2147–2154, Piscataway, NJ, 2011. IEEE Press. doi:10.1109/CEC.2011.5949880.
Annotation: Proposed Expected Hypervolume Improvement (EHVI).
- [1026] M. T. M. Emmerich, A. Deutz, O. Schütze, P. Legrand, E. Tantar, and A.-A. Tantar. *EVOLVE - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computation VII*, volume 662 of *Studies in Computational Intelligence*. Springer, Cham, Switzerland, 2017.
- [1027] M. T. M. Emmerich et al., editors. *Evolutionary Multi-Criterion Optimization – 12th International Conference, EMO 2023, Leiden, The Netherlands, March 20-24, 2023, Proceedings*, volume 13970 of *LNCS*. Springer International Publishing, Cham, Switzerland, 2023.
- [1028] A. Engau and M. M. Wiecek. **2D decision-making for multicriteria design optimization**. *Struct. Multidisc. Optim.*, 34:301–315, 2007. doi:10.1007/s00158-006-0078-y.
- [1029] A. Engau and M. M. Wiecek. **Interactive coordination of objective decompositions in multiobjective programming**. *Management Science*, 54(7):1350–1363, 2008.
- [1030] S. Eppe, Y. De Smet, and T. Stützle. **A bi-objective optimization model to eliciting decision maker’s preferences for the PROMETHEE II method**. In R. I. Brafman, F. Roberts, and A. Tsoukiàs, editors, *Algorithmic Decision Theory, Third International Conference, ADT 2011*, volume 6992 of *LNAI*, pages 56–66. Springer, Heidelberg, Germany, 2011.

- [1031] S. Eppe, M. López-Ibáñez, T. Stützle, and Y. De Smet. **An Experimental Study of Preference Model Integration into Multi-Objective Optimization Heuristics.** In *Proceedings of the 2011 Congress on Evolutionary Computation (CEC 2011)*, pages 2751–2758, Piscataway, NJ, 2011. IEEE Press. doi:[10.1109/CEC.2011.5949963](https://doi.org/10.1109/CEC.2011.5949963).
- [1032] D. Eriksson, M. Pearce, J. Gardner, R. D. Turner, and M. Poloczek. **Scalable Global Optimization via Local Bayesian Optimization.** In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NeurIPS 32)*, pages 5496–5507, 2019. Annotation: Arxiv preprint arXiv: <https://arxiv.org/abs/1910.01739>.
- [1033] E. Ertin, A. N. Dean, M. L. Moore, and K. L. Priddy. **Dynamic Optimization for Optimal Control of Water Distribution Systems.** In K. L. Priddy, P. E. Keller, and P. J. Angeline, editors, *Applications and Science of Computational Intelligence IV, Proceedings of SPIE*, volume 4390, pages 142–149, Mar. 2001.
- [1034] ESANN. *Proceedings of 22th European Symposium on Artificial Neural Networks, ESANN 2014, Bruges, Belgium, April 23-25, 2014*, 2014.
- [1035] ESANN. *Proceedings of 23rd European Symposium on Artificial Neural Networks, ESANN 2015, Bruges, Belgium, April 22-24, 2015*, 2015.
- [1036] V. Esat and M. Hall. **Water resources system optimization using genetic algorithms.** In A. Verwey, A. Minns, V. Babovic, and C. Maksimović, editors, *Hydroinformatics’94*, pages 225–231, Balkema, Rotterdam, The Netherlands, 1994.
- [1037] L. J. Eshelman, editor. *Proceedings of the 6th International Conference on Genetic Algorithms (ICGA), Pittsburgh, PA, USA, July 15-19, 1995*. Morgan Kaufmann Publishers, San Francisco, CA, Pittsburgh, PA, 1995.
- [1038] L. J. Eshelman and J. D. Schaffer. **Real-Coded Genetic Algorithms and Interval-Schemata.** In D. Whitley, editor, *Foundations of Genetic Algorithms (FOGA)*, pages 187–202. Morgan Kaufmann Publishers, 1993. ISBN 1-55860-263-1.
- [1039] L. J. Eshelman, A. Caruana, and J. D. Schaffer. **Biases in the Crossover Landscape.** In J. D. Schaffer, editor, *Proceedings of the Third International Conference on Genetic Algorithms (ICGA’89)*, pages 86–91. Morgan Kaufmann Publishers, San Mateo, CA, 1989.
- [1040] A. I. Esparcia-Alcázar and A. M. Mora, editors. *17th European Conference, EvoApplications 2014, Granada, Spain, April 23-25, 2014, Revised Selected Papers*, volume 8602 of *LNCs*. Springer, Heidelberg, Germany, 2014.
- [1041] I. Essafi, Y. Mati, and S. Dauzère-Pérés. **A Genetic Local Search Algorithm for Minimizing Total Weighted Tardiness in the Job-shop Scheduling Problem.** *Comput. Oper. Res.*, 35(8):2599–2616, 2008.
- [1042] R. M. Everson, J. E. Fieldsend, and S. Singh. **Full Elite Sets for Multi-objective Optimisation.** In *Adaptive Computing in Design and Manufacture V*, pages 343–354. Springer, London, UK, 2002. doi:[10.1007/978-0-85729-345-9_29](https://doi.org/10.1007/978-0-85729-345-9_29). Annotation: Extended version published as [1096].
- [1043] C. J. Eyckelhof and M. Snoek. **Ant Systems for a Dynamic TSP: Ants Caught in a Traffic Jam.** In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCs*, pages 88–99. Springer, Heidelberg, Germany, 2002.
- [1044] J. G. Falcón-Cardona, S. Zapotecas-Martínez, and A. García-Nájera. **Pareto compliance from a practical point of view.** In F. Chicano and K. Krawiec, editors, *GECCO*, pages 395–402. ACM Press, New York, NY, 2021. doi:[10.1145/3449639.3459276](https://doi.org/10.1145/3449639.3459276).

- [1045] S. Falkner, M. T. Lindauer, and F. Hutter. **SpySMAC: Automated configuration and performance analysis of SAT solvers**. In M. Heule and S. Weaver, editors, *Theory and Applications of Satisfiability Testing – SAT 2015*, volume 9340 of *LNCS*, pages 215–222. Springer, Cham, Switzerland, 2015. doi:[10.1007/978-3-319-24318-4_16](https://doi.org/10.1007/978-3-319-24318-4_16).
- [1046] W. Fan and A. Bifet. **Mining big data: current status, and forecast to the future**. *ACM SIGKDD Explorations Newsletter*, 14(2):1–5, 2013.
- [1047] G. Fandel and T. Gal, editors. *Multiple Criteria Decision Making Theory and Application, Proceedings of the Third Conference Hagen/Königswinter, West Germany, August 20-24, 1979*. Number 177 in *LNEMS*. Springer, Heidelberg, Germany, 1980.
- [1048] D. Fanelli. **Negative results are disappearing from most disciplines and countries**. *Scientometrics*, 90(3):891–904, 2012. doi:[10.1007/s11192-011-0494-7](https://doi.org/10.1007/s11192-011-0494-7).
- [1049] K. T. Fang and Y. Wang. *Number-Theoretic Methods in Statistics*. Chapman & Hall/CRC, London, UK, 1994.
Keywords: hypersphere, Monte-Carlo sampling, low-discrepancy sequences.
- [1050] H. Faria, Jr, S. Binato, M. G. C. Resende, and D. J. Falcão. **Power transmission network design by a greedy randomized adaptive path relinking approach**. *IEEE Transactions on Power Systems*, 20(1):43–49, 2005.
- [1051] M. Farina and P. Amato. **On the Optimal Solution Definition for Many-criteria Optimization Problems**. In *Proceedings of the NAFIPS-FLINT International Conference’2002*, pages 233–238, Piscataway, New Jersey, June 2002. IEEE Service Center. doi:[10.1109/nafigps.2002.1018061](https://doi.org/10.1109/nafigps.2002.1018061).
Annotation: First to mention exponential number of nondominated solutions with respect to many objectives?
- [1052] R. Farmani, G. A. Walters, and D. A. Savic. **Evolutionary multi-objective optimization of the design and operation of water distribution network: total cost vs. reliability vs. water quality**. *Journal of Hydroinformatics*, 8(3):165–179, 2006.
- [1053] V. E. Farrugia, H. P. Martínez, and G. N. Yannakakis. **The Preference Learning Toolbox**. *Arxiv preprint arXiv:1506.01709*, 2015. doi:[10.48550/arXiv.1506.01709](https://doi.org/10.48550/arXiv.1506.01709).
- [1054] D. Favaretto, E. Moretti, and P. Pellegrini. **Ant colony system approach for variants of the traveling salesman problem with time windows**. *Journal of Information and Optimization Sciences*, 27(1):35–54, 2006.
- [1055] D. Favaretto, E. Moretti, and P. Pellegrini. **Ant Colony System for a VRP with Multiple Time Windows and Multiple Visits**. *Journal of Interdisciplinary Mathematics*, 10(2):263–284, 2007.
- [1056] D. Favaretto, E. Moretti, and P. Pellegrini. **On the explorative behavior of MAX-MIN Ant System**. In T. Stützle, M. Birattari, and H. H. Hoos, editors, *SLS 2009*, volume 5752 of *LNCS*, pages 115–119. Springer, Heidelberg, Germany, 2009.
- [1057] C. Fawcett and H. H. Hoos. **Analysing Differences between Algorithm Configurations through Ablation**. In *Proc. of MIC 2013, the 10th Metaheuristics International Conference*, pages 123–132, 2013.
- [1058] C. Fawcett and H. H. Hoos. **Analysing Differences Between Algorithm Configurations through Ablation**. *J. Heuristics*, 22(4):431–458, 2016. doi:[10.1007/s10732-014-9275-9](https://doi.org/10.1007/s10732-014-9275-9).
Keywords: Ablation analysis, Automated algorithm configuration, Empirical analysis, Parameter importance.

- [1059] C. Fawcett, M. Helmert, H. H. Hoos, E. Karpas, G. Röger, and J. Seipp. **FD-Autotune: Domain-Specific Configuration using Fast-Downward**. In E. Karpas, S. Jiménez Celorrio, and S. Kambhampati, editors, *Proc. of ICAPS-PAL11*, 2011.
- [1060] U. Feige. **A Threshold of $\ln n$ for Approximating Set Cover**. *J. ACM*, 45(4):634–652, 1998. doi:[10.1145/285055.285059](https://doi.org/10.1145/285055.285059).
Keywords: approximation ratio, set cover.
- [1061] P. M. Fenwick. **A new data structure for cumulative frequency tables**. *Software — Practice & Experience*, 24(3):327–336, 1994. doi:[10.1002/spe.4380240306](https://doi.org/10.1002/spe.4380240306).
Keywords: Binary indexed tree, Arithmetic coding, Cumulative frequencies.
Annotation: Independently proposed Fenwick trees after [2709].
- [1062] T. A. Feo and M. G. C. Resende. **A Probabilistic Heuristic for a Computationally Difficult Set Covering Problem**. *Operations Research Letters*, 8(2):67–71, 1989.
Annotation: Proposed GRASP.
- [1063] T. A. Feo and M. G. C. Resende. **Greedy Randomized Adaptive Search Procedures**. *J. Glob. Optim.*, 6(2):109–113, 1995.
- [1064] T. A. Feo, M. G. C. Resende, and S. H. Smith. **A Greedy Randomized Adaptive Search Procedure for Maximum Independent Set**. *Operations Research*, 42:860–878, Oct. 1994.
Keywords: GRASP.
- [1065] E. Fernandez, J. Navarro, and S. Bernal. **Multicriteria Sorting Using a Valued Indifference Relation Under a Preference Disaggregation Paradigm**. *Eur. J. Oper. Res.*, 198(2): 602–609, 2009.
- [1066] S. Fernández, S. Álvarez, D. Díaz, M. Iglesias, and B. Ena. **Scheduling a Galvanizing Line by Ant Colony Optimization**. In M. Dorigo et al., editors, *ANTS 2014*, volume 8667 of *LNC3*, pages 146–157. Springer, Heidelberg, Germany, 2014. doi:[10.1007/978-3-319-09952-1_13](https://doi.org/10.1007/978-3-319-09952-1_13).
- [1067] S. Fernández, S. Álvarez, E. Malatsetxebarria, P. Valledor, and D. Díaz. **Performance Comparison of Ant Colony Algorithms for the Scheduling of Steel Production Lines**. In J. L. Jiménez Laredo, S. Silva, and A. I. Esparcia-Alcázar, editors, *GECCO Companion*. ACM Press, New York, NY, 2015. doi:[10.1145/2739482.2764658](https://doi.org/10.1145/2739482.2764658).
Keywords: irace.
- [1068] S. Fernández, P. Valledor, D. Díaz, E. Malatsetxebarria, and M. Iglesias. **Criticality of Response Time in the usage of Metaheuristics in Industry**. In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO Companion*, pages 937–940. ACM Press, New York, NY, 2016.
- [1069] V. Fernandez-Viagas and J. M. Framiñán. **On Insertion Tie-breaking Rules in Heuristics for the Permutation Flowshop Scheduling Problem**. *Comput. Oper. Res.*, 45:60–67, 2014.
- [1070] V. Fernandez-Viagas and J. M. Framiñán. **A Beam-search-based Constructive Heuristic for the PFSP to Minimise Total Flowtime**. *Comput. Oper. Res.*, 81:167–177, 2017.
- [1071] V. Fernandez-Viagas and J. M. Framiñán. **Iterated-greedy-based algorithms with beam search initialization for the permutation flowshop to minimise total tardiness**. *Expert Systems with Applications*, 94:58–69, 2018.
- [1072] V. Fernandez-Viagas, R. Ruiz, and J. M. Framiñán. **A New Vision of Approximate Methods for the Permutation Flowshop to Minimise Makespan: State-of-the-art and Computational Evaluation**. *Eur. J. Oper. Res.*, 257(3):707–721, 2017.

- [1073] V. Fernandez-Viagas, J. M. S. Valente, and J. M. Framiñán. **Iterated-greedy-based algorithms with Beam Search Initialization for the Permutation Flowshop to Minimise Total Tardiness.** *Expert Systems with Applications*, 94:58–69, 2018.
- [1074] J. C. Ferreira, C. M. Fonseca, and A. Gaspar-Cunha. **Methodology to select solutions from the Pareto-optimal set: a comparative study.** In D. Thierens et al., editors, *GECCO*, pages 789–796. ACM Press, New York, NY, 2007.
- [1075] R. Ferreira da Silva and S. Urrutia. **A General VNS Heuristic for the Traveling Salesman Problem with Time Windows.** *Discrete Optimization*, 7(4):203–211, 2010. doi:10.1016/j.disopt.2010.04.002.
Keywords: TSPTW, GVNS.
- [1076] A. Ferrer, D. Guimarans, H. Ramalhinho Lourenço, and A. A. Juan. **A BRILS Metaheuristic for Non-smooth Flow-shop Problems with Failure-risk Costs.** *Expert Systems with Applications*, 44:177–186, 2016.
- [1077] J. Ferrer, J. García-Nieto, E. Alba, and F. Chicano. **Intelligent Testing of Traffic Light Programs: Validation in Smart Mobility Scenarios.** *Mathematical Problems in Engineering*, 2016:1–19, 2016. doi:10.1155/2016/3871046.
- [1078] J. Ferrer, M. López-Ibáñez, and E. Alba. **Reliable Simulation-Optimization of Traffic Lights in a Real-World City.** *Applied Soft Computing*, 78:697–711, 2019. doi:10.1016/j.asoc.2019.03.016. Supplementary material: <https://github.com/MLopez-Ibanez/irace-sumo>.
- [1079] F. J. Ferri, P. Pudil, M. Hatef, and J. Kittler. **Comparative study of techniques for large-scale feature selection.** In E. S. Gelsema and L. S. Kanal, editors, *Pattern Recognition in Practice IV*, volume 16 of *Machine Intelligence and Pattern Recognition*, pages 403–413. North-Holland, 1994. doi:10.1016/b978-0-444-81892-8.50040-7.
Annotation: Describes sequential forward / backward selection.
- [1080] P. Festa, M. Sellmann, and J. Vanschoren, editors. *10th International Conference, LION 10, Ischia, Italy, May 29 - June 1, 2016. Revised Selected Papers*, volume 10079 of *LNCS*. Springer, Cham, Switzerland, 2016.
- [1081] M. Feurer and F. Hutter. **Hyperparameter Optimization.** In F. Hutter, L. Kotthoff, and J. Vanschoren, editors, *Automated Machine Learning*, pages 3–33. Springer, 2019. doi:10.1007/978-3-030-05318-5.
- [1082] M. Feurer, A. Klein, K. Eggenberger, J. Springenberg, M. Blum, and F. Hutter. **Efficient and robust automated machine learning.** In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NIPS 28)*, pages 2962–2970, 2015. URL <http://papers.nips.cc/book/advances-in-neural-information-processing-systems-28-2015>.
- [1083] M. Feurer, A. Klein, K. Eggenberger, J. Springenberg, M. Blum, and F. Hutter. **Auto-sklearn: Efficient and Robust Automated Machine Learning.** In F. Hutter, L. Kotthoff, and J. Vanschoren, editors, *Automated Machine Learning*, pages 113–134. Springer, 2019. doi:10.1007/978-3-030-05318-5.
- [1084] Á. Fialho. *Adaptive operator selection for optimization.* PhD thesis, Université Paris Sud-Paris XI, 2010.
- [1085] Á. Fialho, L. Da Costa, M. Schoenauer, and M. Sebag. **Analyzing Bandit-based Adaptive Operator Selection Mechanisms.** *Annals of Mathematics and Artificial Intelligence*, 60(1–2): 25–64, 2010.

- [1086] Á. Fialho, R. Ros, M. Schoenauer, and M. Sebag. **Comparison-based adaptive strategy selection with bandits in differential evolution.** In R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 194–203. Springer, Heidelberg, Germany, 2010.
- [1087] Á. Fialho, M. Schoenauer, and M. Sebag. **Fitness-AUC bandit adaptive strategy selection vs. the probability matching one within differential evolution: an empirical comparison on the BBOB-2010 noiseless testbed.** In M. Pelikan and J. Branke, editors, *GECCO Companion*, pages 1535–1542. ACM Press, New York, NY, 2010.
- [1088] Á. Fialho, M. Schoenauer, and M. Sebag. **Toward comparison-based adaptive operator selection.** In M. Pelikan and J. Branke, editors, *GECCO*, pages 767–774. ACM Press, New York, NY, 2010.
Annotation: Proposed F-AUC and F-SR.
- [1089] M. J. Fielding. **Simulated Annealing with an Optimal Fixed Temperature.** *SIAM J. Optim.*, 11(2):289–307, 2000.
- [1090] J. E. Fieldsend. **University staff teaching allocation: formulating and optimising a many-objective problem.** In P. A. N. Bosman, editor, *GECCO*, pages 1097–1104. ACM Press, New York, NY, 2017. doi:10.1145/3071178.3071230.
Annotation: Example of NSGA-III deteriorating.
- [1091] J. E. Fieldsend. **Data structures for non-dominated sets: implementations and empirical assessment of two decades of advances.** In C. A. Coello Coello, editor, *GECCO*, pages 489–497. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.3390150.
Keywords: Unbounded archive.
- [1092] J. E. Fieldsend. **On the Active Use of an ND-Tree-Based Archive for Multi-Objective Optimisation.** In P. Legrand, A. Liefooghe, E. Keedwell, J. Lepagnot, L. Idoumghar, N. Monmarché, and E. Lutton, editors, *EA 2023: Artificial Evolution*, LNCS, pages 1–14. Springer, 2023. doi:10.1007/978-3-031-42616-2_1.
- [1093] J. E. Fieldsend and R. M. Everson. **Visualising high-dimensional Pareto relationships in two-dimensional scatterplots.** In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 558–572. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4. doi:10.1007/978-3-642-37140-0_42.
- [1094] J. E. Fieldsend and M. Wagner, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022, Boston, Massachusetts, July 9-13, 2022.* ACM Press, New York, NY, 2022. doi:10.1145/3512290.
- [1095] J. E. Fieldsend and M. Wagner, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO 2022, Boston, Massachusetts, July 9-13, 2022.* ACM Press, New York, NY, 2022. ISBN 9781450392686. doi:10.1145/3520304.
- [1096] J. E. Fieldsend, R. M. Everson, and S. Singh. **Using unconstrained elite archives for multiobjective optimization.** *IEEE Trans. Evol. Comput.*, 7(3):305–323, 2003. doi:10.1109/TEVC.2003.810733.
- [1097] J. R. Figueira, S. Greco, and M. Ehrgott, editors. *Multiple Criteria Decision Analysis, State of the Art Surveys.* Springer, 2005.
- [1098] J. R. Figueira, C. M. Fonseca, P. Halffmann, K. Klamroth, L. Paquete, S. Ruzika, B. Schulze, M. Stiglmayr, and D. Willems. **Easy to say they are Hard, but Hard to see they are Easy-Towards a Categorization of Tractable Multiobjective Combinatorial**

- Optimization Problems.** *Journal of Multi-Criteria Decision Analysis*, 24(1-2):82–98, 2017. doi:[10.1002/mcda.1574](https://doi.org/10.1002/mcda.1574).
- [1099] R. Fikes and W. G. Lehnert, editors. *Proceedings of the 11th National Conference on Artificial Intelligence*, 1993. AAAI Press/MIT Press, Menlo Park, CA.
- [1100] J. Filipe and J. Kacprzyk, editors. *Proceedings of the International Joint Conference on Computational Intelligence (IJCCI-2010)*, 2010. SciTePress.
- [1101] B. Filipič and J. Šilc, editors. *Bioinspired optimization methods and their applications: Proceedings of the International Conference on Bioinspired Optimization Methods and their Applications - BIOMA 2004, 11-12 October 2004, Ljubljana, Slovenia*, 2004. URL <https://books.google.be/books?id=OZLsAAAACAAJ>.
- [1102] A. Fink and S. Voß. **HotFrame: A Heuristic Optimization Framework.** In S. Voß and D. L. Woodruff, editors, *Optimization Software Class Libraries*, pages 81–154. Kluwer Academic Publishers, Boston, MA, 2002.
- [1103] A. Fischbach and T. Bartz-Beielstein. **Improving the reliability of test functions generators.** *Applied Soft Computing*, 92:106315, 2020.
- [1104] M. Fischetti and A. Lodi. **Local Branching.** *Mathematical Programming Series B*, 98:23–47, 2003.
- [1105] M. Fischetti and M. Monaci. **Proximity search for 0-1 mixed-integer convex programming.** *J. Heuristics*, 20(6):709–731, 2014.
- [1106] M. Fischetti and M. Monaci. **Exploiting Erraticism in Search.** *Operations Research*, 62(1):114–122, 2014. doi:[10.1287/opre.2013.1231](https://doi.org/10.1287/opre.2013.1231).
Annotation: <http://mat.tepper.cmu.edu/blog/?p=1695>.
- [1107] M. Fischetti and D. Salvagnin. **Feasibility pump 2.0.** *Mathematical Programming Computation*, 1(2–3):201–222, 2009.
- [1108] M. Fischetti, F. Glover, and A. Lodi. **The feasibility pump.** *Mathematical Programming*, 104(1):91–104, 2005.
- [1109] M. Fischetti, M. Monaci, and D. Salvagnin. **Three Ideas for the Quadratic Assignment Problem.** *Operations Research*, 60(4):954–964, 2012.
- [1110] B. Fisset, C. Dhaenens, and L. Jourdan. **MO-Mine_{clust}: A Framework for Multi-Objective Clustering.** In C. Dhaenens, L. Jourdan, and M.-E. Marmion, editors, *LION 9*, volume 8994 of *LNCS*, pages 293–305. Springer, Heidelberg, Germany, 2015.
Keywords: irace.
- [1111] T. Fitzgerald, Y. Malitsky, B. O’Sullivan, and K. Tierney. **ReACT: Real-Time Algorithm Configuration through Tournaments.** In S. Edelkamp and R. Barták, editors, *Proceedings of the Seventh Annual Symposium on Combinatorial Search, SOCS 2014*, pages 62–70. AAAI Press, 2014. doi:[10.1609/SOCS.V5I1.18314](https://doi.org/10.1609/SOCS.V5I1.18314).
- [1112] T. Fitzgerald, Y. Malitsky, and B. O’Sullivan. **ReACTR: Realtime algorithm configuration through tournament rankings.** In Q. Yang and M. Wooldridge, editors, *Proc. of IJCAI-15*, pages 304–310. IJCAI/AAAI Press, Menlo Park, CA, 2015.
- [1113] W. Fitzgibbon, Y. A. Kuznetsov, P. Neittaanmäki, and O. Pironneau, editors. *Modeling, Simulation and Optimization for Science and Technology*, volume 34 of *Computational Methods in Applied Sciences*. Springer, 2014.

- [1114] M. Fleischer. **The Measure of Pareto Optima. Applications to Multi-objective Metaheuristics.** In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 519–533. Springer, Heidelberg, Germany, 2003.
- [1115] P. J. Fleming, R. C. Purshouse, and R. J. Lygoe. **Many-objective optimization: An engineering design perspective.** In C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors, *EMO*, volume 3410 of *LNCS*, pages 14–32. Springer, Berlin/Heidelberg, 2005.
- [1116] R. Fletcher. **A new approach to variable metric algorithms.** *The Computer Journal*, 13(3):317–322, Sept. 1970. doi:10.1093/comjnl/13.3.317.
Keywords: BFGS.
Annotation: One of the four papers that proposed BFGS.
- [1117] R. Fletcher. *Practical methods of optimization.* John Wiley & Sons, New York, NY, 1987.
Annotation: BFGS.
- [1118] C. Fleurent and F. Glover. **Improved constructive multistart strategies for the quadratic assignment problem using adaptive memory.** *INFORMS Journal on Computing*, 11(2):198–204, 1999.
- [1119] J. Fliege. **The effects of adding objectives to an optimisation problem on the solution set.** *Operations Research Letters*, 35(6):782–790, 2007.
- [1120] M. A. Fligner and J. S. Verducci. **Distance Based Ranking Models.** *Journal of the Royal Statistical Society: Series B (Methodological)*, 48(3):359–369, 1986. doi:10.1111/j.2517-6161.1986.tb01420.x.
Keywords: Mallows model, ranking, probabilistic models.
- [1121] M. M. Flood. **The Travelling Salesman Problem.** *Operations Research*, 4:61–75, 1956.
- [1122] D. Floreano and L. Keller. **Evolution of Adaptive Behaviour in Robots by Means of Darwinian Selection.** *PLoS Biol.*, 8(1):e1000292, 2010. doi:10.1371/journal.pbio.1000292.
- [1123] D. Floreano and F. Mondada. **Automatic creation of an autonomous agent: Genetic evolution of a neural network driven robot.** In D. Cliff, P. Husbands, J.-A. Meyer, and S. Wilson, editors, *Proceedings of the third international conference on Simulation of adaptive behavior: From Animals to Animats 3*, pages 421–430. MIT Press, Cambridge, MA, 1994.
Annotation: LIS-CONF-1994-003.
- [1124] D. Floreano and J. Urzelai. **Evolutionary robots with on-line self-organization and behavioral fitness.** *Neural Networks*, 13(4-5):431–443, 2000.
- [1125] B. E. Flores. **A pragmatic view of accuracy measurement in forecasting.** *Omega*, 14(2):93–98, 1986.
Annotation: Proposed symmetric mean absolute percentage error (SMAPE).
- [1126] F. Focacci, F. Laburthe, and A. Lodi. **Local Search and Constraint Programming.** In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 369–403. Kluwer Academic Publishers, Norwell, MA, 2002.
- [1127] F. Focacci, A. Lodi, and M. Milano. **A Hybrid Exact Algorithm for the TSPTW.** *INFORMS Journal on Computing*, 14:403–417, 2002.
- [1128] T. C. Fogarty, editor. *Evolutionary Computing, AISB Workshop, Sheffield, UK, April 3-4, 1995, Selected Papers*, volume 993 of *LNCS*. Springer, Berlin, Germany, 1995.
- [1129] D. B. Fogel. *Evolutionary Computation. Toward a New Philosophy of Machine Intelligence.* IEEE Press, 1995.

- [1130] D. B. Fogel, A. J. Owens, and M. J. Walsh. *Artificial Intelligence Through Simulated Evolution*. John Wiley & Sons, 1966.
- [1131] D. B. Fogel et al., editors. *WCCI*, Piscataway, NJ, 2002. IEEE Press.
- [1132] J. P. Folch, S. Zhang, R. Lee, B. Shafei, D. Walz, C. Tsay, M. van der Wilk, and R. Misener. **SnAKe: Bayesian Optimization with Pathwise Exploration**. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 35226–35239, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/e4bab1843c8d5a69f5abfd0824593493-Abstract-Conference.html.
- [1133] J. P. Folch, J. A. C. Odgers, S. Zhang, R. M. Lee, B. Shafei, D. Walz, C. Tsay, M. van der Wilk, and R. Misener. **Practical Path-based Bayesian Optimization**. In *NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World*, 2023. URL <https://openreview.net/forum?id=E8zSTm2bGu>.
- [1134] C. M. Fonseca and P. J. Fleming. **Genetic Algorithms for Multiobjective Optimization: Formulation, Discussion and Generalization**. In S. Forrest, editor, *Proceedings of the Fifth International Conference on Genetic Algorithms (ICGA'93)*, pages 416–423. Morgan Kaufmann Publishers, 1993. ISBN 1-55860-299-2.
Annotation: Proposes MOGA and P-MOGA.
- [1135] C. M. Fonseca and P. J. Fleming. **An overview of evolutionary algorithms in multiobjective optimization**. *Evol. Comput.*, 3(1):1–16, 1995.
Annotation: Proposed FON benchmark problem.
- [1136] C. M. Fonseca and P. J. Fleming. **On the Performance Assessment and Comparison of Stochastic Multiobjective Optimizers**. In H.-M. Voigt et al., editors, *Parallel Problem Solving from Nature – PPSN IV*, volume 1141 of *LNCS*, pages 584–593. Springer, Heidelberg, Germany, 1996.
- [1137] C. M. Fonseca and P. J. Fleming. **Multiobjective Optimization and Multiple Constraint Handling with Evolutionary Algorithms (II): Application Example**. *IEEE Transactions on Systems, Man, and Cybernetics – Part A*, 28(1):38–44, Jan. 1998. doi:10.1109/3468.650320.
- [1138] C. M. Fonseca and P. J. Fleming. **Multiobjective Optimization and Multiple Constraint Handling with Evolutionary Algorithms (I): A Unified Formulation**. *IEEE Transactions on Systems, Man, and Cybernetics – Part A*, 28(1):26–37, Jan. 1998. doi:10.1109/3468.650319.
- [1139] C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors. *Evolutionary Multi-Criterion Optimization, Second International Conference, EMO 2003, Faro, Portugal, April 2003: proceedings*, volume 2632 of *LNCS*. Springer, Heidelberg, Germany, 2003.
- [1140] C. M. Fonseca, V. Grunert da Fonseca, and L. Paquete. **Exploring the Performance of Stochastic Multiobjective Optimisers with the Second-Order Attainment Function**. In C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors, *EMO*, volume 3410 of *LNCS*, pages 250–264. Springer, Berlin/Heidelberg, 2005. doi:10.1007/978-3-540-31880-4_18.
- [1141] C. M. Fonseca, L. Paquete, and M. López-Ibáñez. **An improved dimension-sweep algorithm for the hypervolume indicator**. In *IEEE CEC*, pages 1157–1163, Piscataway, NJ, July 2006. IEEE Press. doi:10.1109/CEC.2006.1688440.
- [1142] C. M. Fonseca, A. P. Guerreiro, M. López-Ibáñez, and L. Paquete. **On the Computation of the Empirical Attainment Function**. In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 106–120. Springer, Berlin/Heidelberg, 2011. doi:10.1007/978-3-642-19893-9_8.

- [1143] J. R. Fonseca Cacho and K. Taghva. **The State of Reproducible Research in Computer Science.** In S. Latifi, editor, *17th International Conference on Information Technology-New Generations (ITNG 2020)*, Advances in Intelligent Systems and Computing, pages 519–524. Springer International Publishing, 2020. doi:10.1007/978-3-030-43020-7_68.
Keywords: Docker, Improving transparency, OCR, Open science, Replicability, Reproducibility.
- [1144] S. Forrest, editor. *Proceedings of the 5th International Conference on Genetic Algorithms (ICGA), Urbana-Champaign, IL, USA, June 1993*, 1993. Morgan Kaufmann Publishers. ISBN 1-55860-299-2.
- [1145] A. I. J. Forrester and A. J. Keane. **Recent advances in surrogate-based optimization.** *Progress in Aerospace Sciences*, 45(1-3):50–79, 2009. doi:10.1016/j.paerosci.2008.11.001.
Keywords: Kriging; Gaussian Process; EGO; Design of Experiments.
- [1146] M. Förster, B. Bickel, B. Hardung, and G. Kókai. **Self-adaptive ant colony optimisation applied to function allocation in vehicle networks.** In D. Thierens et al., editors, *GECCO*, pages 1991–1998. ACM Press, New York, NY, 2007.
- [1147] F.-A. Fortin, S. Grenier, and M. Parizeau. **Generalizing the improved run-time complexity algorithm for non-dominated sorting.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 615–622. ACM Press, New York, NY, 2015. doi:10.1145/2463372.2463454.
Annotation: Proposed $O(n \log^{m-1} n)$ algorithm.
- [1148] M. Foster, M. Hughes, G. O’Brien, P. S. Oliveto, J. Pyle, D. Sudholt, and J. Williams. **Do sophisticated evolutionary algorithms perform better than simple ones?** In C. A. Coello Coello, editor, *GECCO*, pages 184–192. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.
- [1149] R. Fourer, D. M. Gay, and B. W. Kernighan. *AMPL: A Modeling Language for Mathematical Programming*. Duxbury, 2nd edition, 2002.
- [1150] J. W. Fowler, E. S. Gel, M. Köksalan, P. Korhonen, J. L. Marquis, and J. Wallenius. **Interactive evolutionary multi-objective optimization for quasi-concave preference functions.** *Eur. J. Oper. Res.*, 206(2):417–425, 2010. doi:10.1016/j.ejor.2010.02.027.
Keywords: Interactive optimization, Multi-objective optimization, Evolutionary optimization, Knapsack problem.
- [1151] B. L. Fox. **Uniting probabilistic methods for optimization.** In *Proceedings of the 24th conference on Winter simulation*, pages 500–505. ACM, 1992.
- [1152] B. L. Fox. **Integrating and accelerating tabu search, simulated annealing, and genetic algorithms.** *Annals of Operations Research*, 41(2):47–67, 1993.
- [1153] B. L. Fox. **Simulated annealing: folklore, facts, and directions.** In *Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing*, pages 17–48. Springer, 1995.
- [1154] M. Fox and D. Poole, editors. *Proceedings of the Twenty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2010, Atlanta, Georgia, USA, July 11-15, 2010*, 2010. AAAI Press.
- [1155] J. M. Framiñán, J. N. D. Gupta, and R. Leisten. **A Review and Classification of Heuristics for Permutation Flow-shop Scheduling with Makespan Objective.** *J. Oper. Res. Soc.*, 55(12):1243–1255, 2004.
- [1156] J. M. Framiñán, R. Leisten, and R. Ruiz. *Manufacturing Scheduling Systems: An Integrated View on Models, Methods, and Tools*. Springer, New York, NY, 2014.

- [1157] G. Francesca, M. Brambilla, A. Brutschy, V. Trianni, and M. Birattari. **AutoMoDe: A Novel Approach to the Automatic Design of Control Software for Robot Swarms.** *Swarm Intelligence*, 8(2):89–112, 2014. doi:[10.1007/s11721-014-0092-4](https://doi.org/10.1007/s11721-014-0092-4).
- [1158] G. Francesca, M. Brambilla, A. Brutschy, L. Garattoni, R. Miletitch, G. Podevijn, A. Reina, T. Soleymani, M. Salvaro, C. Pincirolì, F. Mascia, V. Trianni, and M. Birattari. **AutoMoDe-Chocolate: Automatic Design of Control Software for Robot Swarms.** *Swarm Intelligence*, 2015. doi:[10.1007/s11721-015-0107-9](https://doi.org/10.1007/s11721-015-0107-9).
Keywords: Swarm robotics; Automatic design; AutoMoDe.
- [1159] L. A. Franco and R. P. Hämmäläinen. **Engaging with Behavioral Operational Research: On Methods, Actors and Praxis.** In M. Kunc, J. Malpass, and L. White, editors, *Behavioral Operational Research*, pages 3–25. Palgrave Macmillan, London, UK, 2016. doi:[10.1057/978-1-137-53551-1_1](https://doi.org/10.1057/978-1-137-53551-1_1).
- [1160] A. Franzin. *Empirical Analysis of Stochastic Local Search Behaviour: Connecting Structure, Components and Landscape.* PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2021.
- [1161] A. Franzin. **Empirical Analysis of Stochastic Local Search Behaviour: Connecting Structure, Components and Landscape.** *4OR: A Quarterly Journal of Operations Research*, 2022. doi:[10.1007/s10288-022-00511-7](https://doi.org/10.1007/s10288-022-00511-7).
- [1162] A. Franzin and T. Stützle. **Exploration of Metaheuristics through Automatic Algorithm Configuration Techniques and Algorithmic Frameworks.** In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO Companion*, pages 1341–1347. ACM Press, New York, NY, 2016.
- [1163] A. Franzin and T. Stützle. **Comparison of Acceptance Criteria in Randomized Local Searches.** In E. Lutton, P. Legrand, P. Parrend, N. Monmarché, and M. Schoenauer, editors, *EA 2017: Artificial Evolution*, volume 10764 of *LNCS*, pages 16–29. Springer, Heidelberg, Germany, 2017.
- [1164] A. Franzin and T. Stützle. **Revisiting Simulated Annealing: a Component-Based Analysis: Supplementaty Material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2018-001>, 2018.
- [1165] A. Franzin and T. Stützle. **Revisiting Simulated Annealing: a Component-Based Analysis.** Technical Report TR/IRIDIA/2018-010, IRIDIA, Université Libre de Bruxelles, Belgium, 2018. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2018-010.pdf>.
- [1166] A. Franzin and T. Stützle. **Revisiting Simulated Annealing: A Component-Based Analysis.** *Comput. Oper. Res.*, 104:191–206, 2019. doi:[10.1016/j.cor.2018.12.015](https://doi.org/10.1016/j.cor.2018.12.015).
- [1167] A. Franzin and T. Stützle. **Towards transferring algorithm configurations across problems.** In M. Vlastelica, J. Song, A. Ferber, B. Amos, G. Martius, B. Dilkina, and Y. Yue, editors, *Learning Meets Combinatorial Algorithms Workshop at NeurIPS 2020*, pages 1–6, 2020.
- [1168] A. Franzin and T. Stützle. **A causal framework for understanding optimisation algorithms.** In F. Heintz, M. Milano, and B. O’Sullivan, editors, *Trustworthy AI – Integrating Learning, Optimization and Reasoning. TAILOR 2020*, volume 12641 of *LNCS*, pages 140–145. Springer, Cham, Switzerland, 2021.
- [1169] A. Franzin and T. Stützle. **A Landscape-based Analysis of Fixed Temperature and Simulated Annealing: Supplementaty Material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2021-002>, 2021.

- [1170] A. Franzin and T. Stützle. **A Landscape-based Analysis of Fixed Temperature and Simulated Annealing**. Technical Report TR/IRIDIA/2021-005, IRIDIA, Université Libre de Bruxelles, Belgium, 2021. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2021-005.pdf>.
- [1171] A. Franzin and T. Stützle. **A Landscape-based Analysis of Fixed Temperature and Simulated Annealing**. *Eur. J. Oper. Res.*, 304(2):395–410, 2023. doi:10.1016/j.ejor.2022.04.014.
- [1172] A. Franzin, F. Sambo, and B. Di Camillo. **bnstruct: an R package for Bayesian Network structure learning in the presence of missing data**. *Bioinformatics*, 33(8):1250–1252, 2016.
- [1173] A. Franzin, L. Pérez Cáceres, and T. Stützle. **Effect of Transformations of Numerical Parameters in Automatic Algorithm Configuration**. Technical Report TR/IRIDIA/2017-006, IRIDIA, Université Libre de Bruxelles, Belgium, Mar. 2017. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2017-006.pdf>.
- [1174] A. Franzin, L. Pérez Cáceres, and T. Stützle. **Effect of Transformations of Numerical Parameters in Automatic Algorithm Configuration**. *Optimization Letters*, 12(8):1741–1753, 2018. doi:10.1007/s11590-018-1240-3.
- [1175] A. Franzin, R. Gyory, J.-C. Nadé, G. Aubert, G. Klenkle, and H. Bersini. **Philéas: Anomaly Detection for IoT Monitoring**. In L. Cao, W. Kusters, and J. Lijffijt, editors, *Proceedings of the 32nd Benelux Conference on Artificial Intelligence, BNAIC 2020, Leiden, The Netherlands, 19-20 November 2020*, pages 56–70, 2020. URL <https://bnaic.liacs.leidenuniv.nl/wordpress/wp-content/uploads/bnaic2020proceedings.pdf>.
- [1176] C. B. Fraser. *Subsequences and Supersequences of Strings*. PhD thesis, University of Glasgow, 1995.
- [1177] P. I. Frazier. **A Tutorial on Bayesian Optimization**. *Arxiv preprint arXiv:1807.02811*, 2018. doi:10.48550/arXiv.1807.02811.
- [1178] B. Freisleben and P. Merz. **A Genetic Local Search Algorithm for Solving Symmetric and Asymmetric Traveling Salesman Problems**. In T. Bäck, T. Fukuda, and Z. Michalewicz, editors, *Proceedings of the 1996 IEEE International Conference on Evolutionary Computation (ICEC’96)*, pages 616–621. IEEE Press, Piscataway, NJ, 1996.
- [1179] A. R. R. Freitas, P. J. Fleming, and F. G. Guimarães. **A Non-parametric Harmony-Based Objective Reduction Method for Many-Objective Optimization**. In *2013 IEEE International Conference on Systems, Man, and Cybernetics*, pages 651–656. IEEE Press, 2013. doi:10.1109/SMC.2013.116.
- [1180] B. J. Frey and D. Dueck. **Clustering by Passing Messages Between Data Points**. *Science*, 315(5814):972–976, Feb. 2007. doi:10.1126/science.1136800.
Keywords: clustering; affinity propagation.
- [1181] M. Friedman. **The use of ranks to avoid the assumption of normality implicit in the analysis of variance**. *Journal of the American Statistical Association*, 32(200):675–701, 1937. doi:10.1080/01621459.1937.10503522.
- [1182] T. Friedrich, T. Kötzing, M. S. Krejca, and A. M. Sutton. **Robustness of Ant Colony Optimization to Noise**. In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 17–24. ACM Press, New York, NY, 2015. doi:10.1145/2739480.2754723.
Keywords: ant colony optimization, noisy fitness, run time analysis, theory.

- [1183] T. Friedrich, F. Neumann, and A. M. Sutton, editors. *Genetic and Evolutionary Computation Conference, GECCO 2016, Proceedings, Denver, CO, USA, July 20-24, 2016*. ACM Press, New York, NY, 2016.
- [1184] T. Friedrich, F. Neumann, and A. M. Sutton, editors. *Genetic and Evolutionary Computation Conference, GECCO 2016, Denver, CO, USA, July 20-24, 2016, Companion Material Proceedings*. ACM Press, New York, NY, 2016.
- [1185] T. Friedrich, T. Kötzing, and M. Wagner. **A Generic Bet-and-Run Strategy for Speeding Up Stochastic Local Search**. In S. P. Singh and S. Markovitch, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 801–807. AAAI Press, Feb. 2017.
- [1186] T. Friedrich, A. Göbel, F. Quinzan, and M. Wagner. **Heavy-Tailed Mutation Operators in Single-Objective Combinatorial Optimization**. In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11101 of *LNCS*, pages 134–145. Springer, Cham, Switzerland, 2018.
- [1187] T. Friedrich, F. Quinzan, and M. Wagner. **Escaping Large Deceptive Basins of Attraction with Heavy-tailed Mutation Operators**. In H. E. Aguirre and K. Takadama, editors, *GECCO*, pages 293–300. ACM Press, New York, NY, 2018. doi:10.1145/3205455.3205515.
Keywords: combinatorial optimization, heavy-tailed mutation, single-objective optimization, experiments-motivated theory, irace.
- [1188] T. Friedrich, C. Doerr, and D. V. Arnold, editors. *Foundations of Genetic Algorithms, 15th ACM/SIGEVO International Workshop, FOGA 2019, Potsdam, Germany*. ACM, 2019.
- [1189] M. Friendly. **Statistical graphics for multivariate data**. In *SAS Conference Proceedings: SAS Users Group International 16 (SUGI 16)*, 1991.
Annotation: February 17-20, 1991, New Orleans, Louisiana, 297 papers.
- [1190] M. Frigo and S. G. Johnson. **The Design and Implementation of FFTW3**. *Proceedings of the IEEE*, 93(2):216–231, 2005. Special issue on “Program Generation, Optimization, and Platform Adaptation”.
- [1191] H. Frikha, H. Chabchoub, and J.-M. Martel. **Inferring criteria’s relative importance coefficients in PROMETHEE II**. *International Journal of Operational Research*, 7(2): 257–275, 2010.
- [1192] Z. Fu, R. Eglese, and L. Y. O. Li. **A unified tabu search algorithm for vehicle routing problems with soft time windows**. *J. Oper. Res. Soc.*, 59(5):663–673, 2008.
- [1193] D. Fudenberg and J. Tirole. *Game Theory*. MIT Press, Cambridge, MA, 1983.
- [1194] G. Fuellerer, K. F. Doerner, R. F. Hartl, and M. Iori. **Metaheuristics for vehicle routing problems with three-dimensional loading constraints**. *Eur. J. Oper. Res.*, 201(3):751–759, 2009. doi:10.1016/j.ejor.2009.03.046.
- [1195] G. Fuellerer, K. F. Doerner, R. F. Hartl, and M. Iori. **Ant colony optimization for the two-dimensional loading vehicle routing problem**. *Comput. Oper. Res.*, 36(3):655–673, 2009.
- [1196] N. Fujimoto and K. Nanai. **Solving QUBO with GPU parallel MOPSO**. In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1788–1794. ACM Press, New York, NY, 2021.
- [1197] A. S. Fukunaga. **Evolving Local Search Heuristics for SAT Using Genetic Programming**. In K. Deb et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2004, Part II*, volume 3103 of *LNCS*, pages 483–494. Springer, Heidelberg, Germany, 2004.

- [1198] A. S. Fukunaga. **Automated Discovery of Local Search Heuristics for Satisfiability Testing.** *Evol. Comput.*, 16(1):31–61, Mar. 2008. doi:10.1162/evco.2008.16.1.31.
- [1199] N. E. Furlong, E. A. Lovelace, and K. L. Lovelace. *Research Methods and Statistics: An Integrated Approach*. Harcourt College Publishers, 2000.
- [1200] A. Furnham and H. C. Boo. **A literature review of the anchoring effect.** *Journal of Socio-Economics*, 40(1):35–42, 2011. doi:10.1016/j.socec.2010.10.008.
- [1201] J. Fürnkranz and E. Hüllermeier, editors. *Preference Learning*. Springer, Heidelberg, Germany, 2011. ISBN 978-3-642-14125-6.
- [1202] J. Fürnkranz and T. Joachims, editors. *Proceedings of the 27th international conference on machine learning, ICML 2010*, New York, NY, 2010. ACM Press.
- [1203] J. Fürnkranz, T. Scheffer, and M. Spiliopoulou, editors. *17th European Conference on Machine Learning, Berlin, Germany, September 18-22, 2006 Proceedings*, volume 4212 of *LNCS*, 2006. ISBN 978-3-540-46056-5.
- [1204] G. Fursin, Y. Kashnikov, A. W. Memon, Z. Chamski, O. Temam, M. Namolaru, E. Yom-Tov, B. Mendelson, A. Zaks, E. Courtois, F. Bodin, P. Barnard, E. Ashton, E. Bonilla, J. Thomson, C. K. I. Williams, and M. O’Boyle. **Milepost GCC: Machine Learning Enabled Self-tuning Compiler.** *International Journal of Parallel Programming*, 39(3):296–327, 2011. doi:10.1007/s10766-010-0161-2.
- [1205] D. Gaertner and K. Clark. **On Optimal Parameters for Ant Colony Optimization Algorithms.** In H. R. Arabnia and R. Joshua, editors, *Proceedings of the 2005 International Conference on Artificial Intelligence, ICAI 2005*, pages 83–89. CSREA Press, 2005. ISBN 1-932415-66-1.
- [1206] M. Gagliolo and C. Legrand. **Algorithm Survival Analysis.** In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 161–184. Springer, Berlin/Heidelberg, 2010. doi:10.1007/978-3-642-02538-9_7.
- [1207] M. Gagliolo and J. Schmidhuber. **Learning dynamic algorithm portfolios.** *Annals of Mathematics and Artificial Intelligence*, 47(3-4):295–328, 2007. doi:10.1007/s10472-006-9036-z.
Annotation: fully dynamic and online algorithm selection technique, with no separate training phase: all candidate algorithms are run in parallel, while a model incrementally learns their runtime distributions.
- [1208] C. Gagné, W. L. Price, and M. Gravel. **Comparing an ACO algorithm with other heuristics for the single machine scheduling problem with sequence-dependent setup times.** *J. Oper. Res. Soc.*, 53:895–906, 2002.
- [1209] T. Gal and H. Leberling. **Redundant objective functions in linear vector maximum problems and their determination.** *Eur. J. Oper. Res.*, 1(3):176–184, 1977. doi:10.1016/0377-2217(77)90025-X.
- [1210] P. Galinier and J.-K. Hao. **Hybrid evolutionary algorithms for graph coloring.** *Journal of Combinatorial Optimization*, 3(4):379–397, 1999. doi:10.1023/A:1009823419804.
- [1211] L. M. Gambardella and M. Dorigo. **Ant-Q: A Reinforcement Learning Approach to the Traveling Salesman Problem.** In A. Prieditis and S. Russell, editors, *Proceedings of the Twelfth International Conference on Machine Learning (ML-95)*, pages 252–260. Morgan Kaufmann Publishers, Palo Alto, CA, 1995.

- [1212] L. M. Gambardella and M. Dorigo. **Solving Symmetric and Asymmetric TSPs by Ant Colonies**. In T. Bäck, T. Fukuda, and Z. Michalewicz, editors, *Proceedings of the 1996 IEEE International Conference on Evolutionary Computation (ICEC'96)*, pages 622–627. IEEE Press, Piscataway, NJ, 1996.
Annotation: IC.18.
- [1213] L. M. Gambardella and M. Dorigo. **Ant Colony System Hybridized with a New Local Search for the Sequential Ordering Problem**. *INFORMS Journal on Computing*, 12(3): 237–255, 2000.
- [1214] L. M. Gambardella, É. D. Taillard, and G. Agazzi. **MACS-VRPTW: A Multiple Ant Colony System for Vehicle Routing Problems with Time Windows**. In D. Corne, M. Dorigo, and F. Glover, editors, *New Ideas in Optimization*, pages 63–76. McGraw Hill, London, UK, 1999.
- [1215] L. M. Gambardella, R. Montemanni, and D. Weyland. **Coupling Ant Colony Systems with Strong Local Searches**. *Eur. J. Oper. Res.*, 220(3):831–843, 2012. doi:10.1016/j.ejor.2012.02.038.
- [1216] X. Gandibleux, N. Mezdaoui, and A. Fréville. **A tabu search procedure to solve multiobjective combinatorial optimization problem**. In R. Caballero, F. Ruiz, and R. E. Steuer, editors, *Advances in Multiple Objective and Goal Programming*, volume 455 of *LNEMS*, pages 291–300. Springer, Heidelberg, Germany, 1997.
- [1217] X. Gandibleux, A. Jaskiewicz, A. Fréville, and R. Słowiński. **Special Issue on Multiple Objective Metaheuristics**. *J. Heuristics*, 6(3), 2000.
- [1218] X. Gandibleux, H. Morita, and N. Katoh. **Use of a genetic heritage for solving the assignment problem with two objectives**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 43–57. Springer, Heidelberg, Germany, 2003.
- [1219] X. Gandibleux, X. Delorme, and V. T'Kindt. **An Ant Colony Optimisation Algorithm for the Set Packing Problem**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 49–60. Springer, Heidelberg, Germany, 2004.
- [1220] X. Gandibleux, M. Sevaux, K. Sörensen, and V. T'Kindt, editors. *Metaheuristics for Multiobjective Optimisation*, volume 535 of *LNEMS*. Springer, Berlin/Heidelberg, 2004.
- [1221] H. Gao, H. Nie, and K. Li. **Visualisation of Pareto Front Approximation: A Short Survey and Empirical Comparisons**. In *Proceedings of the 2019 Congress on Evolutionary Computation (CEC 2019)*, pages 1750–1757, Piscataway, NJ, June 2019. IEEE Press. doi:10.1109/CEC.2019.8790298.
- [1222] H. Gao, H. Nie, and K. Li. **Visualisation of Pareto Front Approximation: A Short Survey and Empirical Comparisons**. *Arxiv preprint arXiv:1903.01768*, 2019. URL <http://arxiv.org/abs/1903.01768>.
- [1223] K. Gao, Y. Zhang, A. Sadollah, and R. Su. **Optimizing urban traffic light scheduling problem using harmony search with ensemble of local search**. *Applied Soft Computing*, 48:359–372, Nov. 2016. doi:10.1016/j.asoc.2016.07.029.
Keywords: harmony search algorithm, traffic light scheduling.
- [1224] J. García and F. Fernández. **A comprehensive survey on safe reinforcement learning**. *J. Mach. Learn. Res.*, 16(1):1437–1480, 2015.
- [1225] S. García, D. Molina, M. Lozano, and F. Herrera. **A study on the use of non-parametric tests for analyzing the evolutionary algorithms' behaviour: a case study on the CEC'2005 Special Session on Real Parameter Optimization**. *J. Heuristics*, 15(617):617–644, 2009. doi:10.1007/s10732-008-9080-4.

- [1226] S. García, A. Fernández, J. Luengo, and F. Herrera. **Advanced nonparametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power.** *Inf. Sci.*, 180(10):2044–2064, 2010.
- [1227] C. García-Martínez, O. Cordón, and F. Herrera. **A taxonomy and an empirical analysis of multiple objective ant colony optimization algorithms for the bi-criteria TSP.** *Eur. J. Oper. Res.*, 180(1):116–148, 2007.
- [1228] C. García-Martínez, F. J. Rodríguez, and M. Lozano. **Arbitrary function optimisation with metaheuristics: No free lunch and real-world problems.** *Soft Computing*, 16(12):2115–2133, 2012. doi:[10.1007/s00500-012-0881-x](https://doi.org/10.1007/s00500-012-0881-x).
- [1229] C. García-Martínez, F. Glover, F. J. Rodríguez, M. Lozano, and R. Martí. **Strategic Oscillation for the Quadratic Multiple Knapsack Problem.** *Computational Optimization and Applications*, 58(1):161–185, 2014.
- [1230] C. García-Martínez, F. J. Rodríguez, and M. Lozano. **Tabu-enhanced Iterated Greedy Algorithm: A Case Study in the Quadratic Multiple Knapsack Problem.** *Eur. J. Oper. Res.*, 232(3):454–463, 2014.
- [1231] J. García-Nieto, E. Alba, and A. C. Olivera. **Swarm intelligence for traffic light scheduling: Application to real urban areas.** *Engineering Applications of Artificial Intelligence*, 25(2):274–283, Mar. 2012.
Keywords: Cycle program optimization, Particle swarm optimization, Realistic traffic instances, SUMO microscopic simulator of urban mobility, Traffic light scheduling.
- [1232] J. García-Nieto, A. C. Olivera, and E. Alba. **Optimal Cycle Program of Traffic Lights With Particle Swarm Optimization.** *IEEE Trans. Evol. Comput.*, 17(6):823–839, Dec. 2013. doi:[10.1109/TEVC.2013.2260755](https://doi.org/10.1109/TEVC.2013.2260755).
- [1233] J. García-Nieto, E. López-Camacho, M. J. Godoy García, A. J. Nebro, J. J. Durillo, and J. F. Aldana-Montes. **A study of archiving strategies in multi-objective PSO for molecular docking.** In M. Dorigo, M. Birattari, X. Li, M. López-Ibáñez, K. Ohkura, C. Pinciroli, and T. Stützle, editors, *ANTS 2016*, volume 9882 of *LNCS*, pages 40–52. Springer, Heidelberg, Germany, 2016. doi:[10.1007/978-3-319-44427-7_4](https://doi.org/10.1007/978-3-319-44427-7_4).
- [1234] M. R. Garey and D. S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. Freeman & Co, San Francisco, CA, 1979.
- [1235] M. R. Garey, D. S. Johnson, and R. Sethi. **The Complexity of Flowshop and Jobshop Scheduling.** *Mathematics of Operations Research*, 1:117–129, 1976.
- [1236] I. I. Garibay, T. Jansen, R. P. Wiegand, and A. S. Wu, editors. *Foundations of Genetic Algorithms, 10th ACM SIGEVO International Workshop, FOGA 2009, Orlando, Florida, USA, January 9-11, 2009, Proceedings*. ACM, 2009. ISBN 978-1-60558-414-0.
- [1237] J. Garnier and L. Kallel. **Efficiency of Local Search with Multiple Local Optima.** *SIAM Journal Discrete Mathematics*, 15(1):122–141, 2001. doi:[10.1137/S0895480199355225](https://doi.org/10.1137/S0895480199355225).
- [1238] D. Garrett and D. Dasgupta. **Multiobjective landscape analysis and the generalized assignment problem.** In V. Maniezzo, R. Battiti, and J.-P. Watson, editors, *LION 2*, volume 5313 of *LNCS*, pages 110–124. Springer, Heidelberg, Germany, 2008.
- [1239] B. A. Garro, H. Sossa, and R. A. Vazquez. **Evolving ant colony system for optimizing path planning in mobile robots.** In *Electronics, Robotics and Automotive Mechanics Conference*, pages 444–449, Los Alamitos, CA, 2007. IEEE Computer Society. doi:[10.1109/CERMA.2007.60](https://doi.org/10.1109/CERMA.2007.60).

- [1240] A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors. *Evolutionary Multi-Criterion Optimization – 8th International Conference, EMO 2015, Guimarães, Portugal, March 29 – April 1, 2015. Proceedings, Part I*, volume 9018 of *LNCS*. Springer, Heidelberg, Germany, 2015.
- [1241] A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors. *Evolutionary Multi-Criterion Optimization – 8th International Conference, EMO 2015, Guimarães, Portugal, March 29 – April 1, 2015. Proceedings, Part II*, volume 9019 of *LNCS*. Springer, Heidelberg, Germany, 2015.
- [1242] W. Gaul, A. Bachem, W. Habenicht, W. Runge, and W. W. Stahl, editors. *Operations Research Proceedings 1991, Papers of the 20th Annual Meeting*. Springer, Berlin/Heidelberg, 1992.
- [1243] M. Gebser, R. Kaminski, B. Kaufmann, T. Schaub, M. T. Schneider, and S. Ziller. **A portfolio solver for answer set programming: Preliminary report**. In P. Calabar and T. C. Son, editors, *Logic Programming and Nonmonotonic Reasoning*, volume 8148 of *LNAI*, pages 352–357. Springer, Heidelberg, Germany, 2013.
- [1244] P. Geibel. **Reinforcement Learning for MDPs with Constraints**. In J. Fürnkranz, T. Scheffer, and M. Spiliopoulou, editors, *Machine Learning: ECML 2006*, volume 4212 of *LNCS*, pages 646–653, 2006. ISBN 978-3-540-46056-5. doi:10.1007/11871842_63.
Keywords: Safe RL.
- [1245] M. J. Geiger. **Decision Support for Multi-objective Flow Shop Scheduling by the Pareto Iterated Local Search Methodology**. *Computers & Industrial Engineering*, 61(3): 805–812, 2011.
- [1246] M. J. Geiger. **A Multi-threaded Local Search Algorithm and Computer Implementation for the Multi-mode, Resource-constrained Multi-project Scheduling Problem**. *Eur. J. Oper. Res.*, 256:729–741, 2017.
- [1247] S. Geman and D. Geman. **Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images**. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6(6):721–741, 1984.
- [1248] M. Gen and L. Lin. **Multiobjective evolutionary algorithm for manufacturing scheduling problems: state-of-the-art survey**. *Journal of Intelligent Manufacturing*, 25(5):849–866, 2014.
- [1249] M. Gendreau and J.-Y. Potvin. **Tabu Search**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, pages 41–59. Springer, New York, NY, 2nd edition, 2010.
- [1250] M. Gendreau and J.-Y. Potvin, editors. *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*. Springer, New York, NY, 2nd edition, 2010.
- [1251] M. Gendreau and J.-Y. Potvin, editors. *Handbook of Metaheuristics*, volume 272 of *International Series in Operations Research & Management Science*. Springer, 2019.
- [1252] M. Gendreau, A. Hertz, G. Laporte, and M. Stan. **A Generalized Insertion Heuristic for the Traveling Salesman Problem with Time Windows**. *Operations Research*, 46:330–335, 1998.
- [1253] M. Gendreau, F. Guertin, J.-Y. Potvin, and É. D. Taillard. **Parallel tabu search for real-time vehicle routing and dispatching**. *Transportation Science*, 33(4):381–390, 1999.
- [1254] M. Gendreau, F. Guertin, J.-Y. Potvin, and R. Séguin. **Neighborhood search heuristics for a dynamic vehicle dispatching problem with pick-ups and deliveries**. *Transportation Research Part C: Emerging Technologies*, 14(3):157–174, 2006.

- [1255] M. Gendreau, G. Ghiani, and E. Guerriero. **Time-dependent routing problems: A review.** *Comput. Oper. Res.*, 64:189–197, Dec. 2015. doi:[10.1016/j.cor.2015.06.001](https://doi.org/10.1016/j.cor.2015.06.001).
- [1256] I. P. Gent, editor. *Principles and Practice of Constraint Programming – CP 2009, 15th International Conference, CP 2009, Lisbon, Portugal, September 20-24, 2009, Proceedings*, volume 5732 of *LNCS*. Springer, Heidelberg, Germany, 2009. doi:[10.1007/978-3-642-04244-7](https://doi.org/10.1007/978-3-642-04244-7).
- [1257] I. P. Gent, S. A. Grant, E. MacIntyre, P. Prosser, P. Shaw, B. M. Smith, and T. Walsh. **How Not To Do It.** Technical Report 97.27, School of Computer Studies, University of Leeds, May 1997.
- [1258] I. P. Gent, H. H. Hoos, P. Prosser, and T. Walsh. **Morphing: Combining Structure and Randomness.** In *Proceedings of the Sixteenth National Conference on Artificial Intelligence*, pages 654–660, 1999.
- [1259] J. E. Gentle. *Random Number Generation and Monte Carlo Methods*. Springer, New York, NY, 2006.
- [1260] R. Genuer, J.-M. Poggi, and C. Tuleau-Malot. **Variable selection using random forests.** *Pattern Recognition Letters*, 31(14):2225–2236, 2010.
- [1261] A. Genz and F. Bretz. **Methods That Approximate the Integral.** In *Computation of Multivariate Normal and t Probabilities*, volume 195 of *Lecture Notes in Statistics*, pages 29–54. Springer, Berlin/Heidelberg, 2009. doi:[10.1007/978-3-642-01689-9_4](https://doi.org/10.1007/978-3-642-01689-9_4).
- [1262] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner. **Mathematical exploration and discovery at scale.** *Arxiv preprint arXiv:2511.02864 [cs.NE]*, 2025. doi:[10.48550/arXiv.2511.02864](https://doi.org/10.48550/arXiv.2511.02864).
- [1263] S. J. Gershman, E. J. Horvitz, and J. B. Tenenbaum. **Computational rationality: A converging paradigm for intelligence in brains, minds, and machines.** *Science*, 349(6245):273–278, 2015. doi:[10.1126/science.aac6076](https://doi.org/10.1126/science.aac6076).
- [1264] D. Geschwender, F. Hutter, L. Kotthoff, Y. Malitsky, H. H. Hoos, and K. Leyton-Brown. **Algorithm Configuration in the Cloud: A Feasibility Study.** In P. M. Pardalos, M. G. C. Resende, C. Vogiatzis, and J. L. Walteros, editors, *LION 8*, volume 8426 of *LNCS*, pages 41–46. Springer, Heidelberg, Germany, 2014. doi:[10.1007/978-3-319-09584-4_5](https://doi.org/10.1007/978-3-319-09584-4_5).
- [1265] P. Geurts, D. Ernst, and L. Wehenkel. **Extremely randomized trees.** *Mach. Learn.*, 63(1):3–42, Mar. 2006. doi:[10.1007/s10994-006-6226-1](https://doi.org/10.1007/s10994-006-6226-1).
Annotation: Proposed ExtraTrees.
- [1266] S. Ghemawat, H. Gobioff, and S.-T. Leung. **The Google File System.** *SIGOPS Oper. Syst. Rev.*, 37(5):29–43, Dec. 2003.
- [1267] K. Ghoseiri and B. Nadjari. **An ant colony optimization algorithm for the bi-objective shortest path problem.** *Applied Soft Computing*, 10(4):1237–1246, 2010.
- [1268] G. D. Giacomo, A. Catala, B. Dilkina, M. Milano, S. Barro, A. Bugarín, and J. Lang, editors. *Proceedings of the 24th European Conference on Artificial Intelligence (ECAI)*, volume 325 of *Frontiers in Artificial Intelligence and Applications*. IOS Press, 2020.
- [1269] K. C. Giannakoglou, D. T. Tsahalis, J. Periaux, K. D. Papaliliou, and T. Fogarty, editors. *Evolutionary Methods for Design, Optimisation and Control with Application to Industrial Problems. Proceedings of the EUROGEN 2001 Conference*. CIMNE, Barcelona, Spain, 2002. ISBN 84-89925-97-6.

- [1270] M. S. Gibbs, G. C. Dandy, H. R. Maier, and J. B. Nixon. **Calibrating genetic algorithms for water distribution system optimisation**. In *7th Annual Symposium on Water Distribution Systems Analysis*. ASCE, May 2005.
- [1271] X. Gillard. *Discrete Optimization with Decision Diagrams: Design of a Generic Solver, Improved Bounding Techniques, and Discovery of Good Feasible Solutions with Large Neighborhood Search*. PhD thesis, Université Catholique de Louvain, 2022.
- [1272] X. Gillard, V. Coppé, P. Schaus, and A. A. Cire. **Improving the Filtering of Branch-And-Bound MDD Solver**. In P. J. Stuckey, editor, *CPAIOR 2021*, volume 12735 of *LNCS*, pages 231–247. Springer, Cham, Switzerland, 2021. doi:10.1007/978-3-030-78230-6_15.
- [1273] N. Girerd, M. Rabilloud, P. Pibart, P. Mathieu, and P. Roy. **Quantification of Treatment Effect Modification on Both an Additive and Multiplicative Scale**. *PLoS One*, 11(4): 1–14, Apr. 2016. doi:10.1371/journal.pone.0153010.
- [1274] T. Glasmachers. **A Fast Incremental BSP Tree Archive for Non-dominated Points**. In H. Trautmann, G. Rudolph, K. Klamroth, O. Schütze, M. M. Wiecek, Y. Jin, and C. Grimme, editors, *EMO*, volume 10173 of *LNCS*, pages 252–266. Springer International Publishing, Cham, Switzerland, 2017. doi:10.1007/978-3-319-54157-0_18.
Keywords: archiving.
- [1275] A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, editors. *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10-15, 2024*, 2024.
- [1276] X. Glorot and Y. Bengio. **Understanding the difficulty of training deep feedforward neural networks**. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pages 249–256, 2010.
- [1277] F. Glover. **Heuristics for Integer Programming Using Surrogate Constraints**. *Decision Sciences*, 8:156–166, 1977.
- [1278] F. Glover. **Future Paths for Integer Programming and Links to Artificial Intelligence**. *Comput. Oper. Res.*, 13(5):533–549, 1986.
- [1279] F. Glover. **Tabu Search – Part I**. *INFORMS Journal on Computing*, 1(3):190–206, 1989. doi:10.1287/ijoc.1.3.190.
- [1280] F. Glover. **Tabu Search – Part II**. *INFORMS Journal on Computing*, 2(1):4–32, 1990.
- [1281] F. Glover. **A Template for Scatter Search and Path Relinking**. In J.-K. Hao, E. Lutton, E. M. A. Ronald, M. Schoenauer, and D. Snyers, editors, *Artificial Evolution*, volume 1363 of *LNCS*, pages 1–51. Springer, Heidelberg, Germany, 1998. doi:10.1007/BFb0026589.
- [1282] F. Glover and J.-K. Hao. **The case for Strategic Oscillation**. *Annals of Operations Research*, 183(1):163–173, 2011.
- [1283] F. Glover and G. A. Kochenberger. **Critical Even Tabu Search for Multidimensional Knapsack Problems**. In I. H. Osman and J. P. Kelly, editors, *Metaheuristics: Theory & Applications*, pages 407–427. Kluwer Academic Publishers, Norwell, MA, 1996.
- [1284] F. Glover and G. A. Kochenberger, editors. *Handbook of Metaheuristics*. Kluwer Academic Publishers, Norwell, MA, 2002.
- [1285] F. Glover and G. A. Kochenberger, editors. *Handbook of Metaheuristics*. Springer, Boston, MA, 2003. doi:10.1007/b101874.

- [1286] F. Glover and M. Laguna. *Tabu Search*. Kluwer Academic Publishers, Boston, MA, 1997.
- [1287] F. Glover, G. A. Kochenberger, and B. Alidaee. **Adaptive Memory Tabu Search for Binary Quadratic Programs**. *Management Science*, 44(3):336–345, 1998.
- [1288] F. Glover, M. Laguna, and R. Martí. **Scatter Search and Path Relinking: Advances and Applications**. In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 1–35. Kluwer Academic Publishers, Norwell, MA, 2002.
- [1289] F. Glover, Z. Lü, and J.-K. Hao. **Diversification-driven tabu search for unconstrained binary quadratic problems**. *4OR: A Quarterly Journal of Operations Research*, 8(3):239–253, 2010. doi:10.1007/s10288-009-0115-y.
- [1290] GNU Project, Free Software Foundation. **GCC, the GNU Compiler Collection**. <https://gcc.gnu.org>, 1987.
- [1291] M. Goerigk and A. Schöbel. **Recovery-to-optimality: A new two-stage approach to robustness with an application to aperiodic timetabling**. *Comput. Oper. Res.*, 52:1–15, 2014.
- [1292] E. F. G. Goldberg, G. R. Souza, and M. C. Goldberg. **Particle Swarm for the Traveling Salesman Problem**. In J. Gottlieb and G. R. Raidl, editors, *Proceedings of EvoCOP 2006 – 6th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3906 of *LNCS*, pages 99–110. Springer, Heidelberg, Germany, 2006.
- [1293] D. E. Goldberg. *Genetic Algorithms in Search, Optimization and Machine Learning*. Addison-Wesley, Boston, MA, 1989.
- [1294] D. E. Goldberg. **Probability matching, the magnitude of reinforcement, and classifier system bidding**. *Mach. Learn.*, 5(4):407–425, 1990.
- [1295] D. Goldfarb. **A Family of Variable-Metric Methods Derived by Variational Means**. *Mathematics of Computation*, 24(109):23–26, 1970.
Keywords: BFGS.
Annotation: One of the four papers that proposed BFGS.
- [1296] F. E. Goldman and L. W. Mays. **The Application of Simulated Annealing to the Optimal Operation of Water Systems**. In *Proceedings of 26th Annual Water Resources Planning and Management Conference*, Tempe, USA, June 2000. ASCE.
- [1297] M. H. Goldwasser, D. S. Johnson, and C. C. McGeoch, editors. *Data Structures, Near Neighbor Searches, and Methodology: Fifth and Sixth DIMACS Implementation Challenges, Proceedings of a DIMACS Workshop, USA, 1999*, volume 59 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*. American Mathematical Society, Providence, RI, 2002.
- [1298] D. Golovin, B. Solnik, S. Moitra, G. Kochanski, J. Karro, and D. Sculley. **Google Vizier: A Service for Black-Box Optimization**. In S. Matwin, S. Yu, and F. Farooq, editors, *23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1487–1495. ACM Press, 2017. doi:10.1145/3097983.3098043.
- [1299] M. C. Golumbic et al., editors. *Fifth International Symposium on Artificial Intelligence and Mathematics, AIM 1998, Fort Lauderdale, Florida, USA, January 4-6, 1998*, 1998.
- [1300] C. Gomes and M. Sellmann, editors. *Integration of AI and OR Techniques in Constraint Programming for Combinatorial Optimization Problems, 10th International Conference, CPAIOR 2013, Yorktown Heights, NY, USA, May 18-22, 2013. Proceedings*, volume 7874 of *LNCS*. Springer, Heidelberg, Germany, 2013.

- [1301] R. E. Gomory. **An algorithm for integer solutions to linear programs**. In R. L. Graves and P. Wolfe, editors, *Recent Advances in Mathematical Programming*, pages 260–302. McGraw Hill, New York, NY, 1963.
- [1302] W. Gong, Á. Fialho, and Z. Cai. **Adaptive strategy selection in differential evolution**. In M. Pelikan and J. Branke, editors, *GECCO*, pages 409–416. ACM Press, New York, NY, 2010. doi:10.1145/1830483.1830559.
- [1303] Z. Gong, N. Zhang, and F. Chiclana. **The optimization ordering model for intuitionistic fuzzy preference relations with utility functions**. *Knowledge-Based Systems*, 162:174–184, 2018. ISSN 0950-7051. doi:10.1016/j.knosys.2018.07.012.
Keywords: Intuitionistic fuzzy preference relation, Utility function, Ranking, Multiplicative consistency, Non-archimedean infinitesimal.
Annotation: Special Issue on intelligent decision-making and consensus under uncertainty in inconsistent and dynamic environments.
- [1304] E. D. Goodman, editor. *Proceedings of the 3rd Annual Conference on Genetic and Evolutionary Computation, GECCO 2001*. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [1305] Google. **TensorFlow**. <https://www.tensorflow.org>, 2017.
- [1306] M. Gorges-Schleuter. **Asparagos96 and the Travelling Salesman Problem**. In T. Bäck, Z. Michalewicz, and X. Yao, editors, *Proceedings of the 1997 IEEE International Conference on Evolutionary Computation (ICEC’97)*, pages 171–174. IEEE Press, Piscataway, NJ, 1997.
- [1307] J. Gorski, K. Klamroth, and S. Ruzika. **Connectedness of Efficient Solutions in Multiple Objective Combinatorial Optimization**. *Journal of Optimization Theory and Applications*, 150(3):475–497, 2011. doi:10.1007/s10957-011-9849-8.
- [1308] A. Gosavi. **Reinforcement Learning: A Tutorial Survey and Recent Advances**. *INFORMS Journal on Computing*, 21(2):178–192, 2009. doi:10.1287/ijoc.1080.0305.
- [1309] J. Gottlieb and G. R. Raidl, editors. *Proceedings of EvoCOP 2004 – 4th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3004 of *LNCS*. Springer, Heidelberg, Germany, 2004.
- [1310] J. Gottlieb and G. R. Raidl, editors. *Proceedings of EvoCOP 2006 – 6th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3906 of *LNCS*. Springer, Heidelberg, Germany, 2006.
- [1311] J. Gottlieb, M. Puchta, and C. Solnon. **A Study of Greedy, Local Search, and Ant Colony Optimization Approaches for Car Sequencing Problems**. In S. Cagnoni et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2003*, volume 2611 of *LNCS*, pages 246–257. Springer, Heidelberg, Germany, 2003.
- [1312] G. Gottlob and T. Walsh, editors. *IJCAI-03, Proceedings of the Eighteenth International Joint Conference on Artificial Intelligence, Acapulco, Mexico, August 9-15, 2003*, 2003. Morgan Kaufmann Publishers.
- [1313] N. Gould and J. Scott. **A Note on Performance Profiles for Benchmarking Software**. *ACM Trans. Math. Software*, 42, 2016. doi:10.1145/2950048.
- [1314] N. I. M. Gould, D. Orban, and P. L. Toint. **CUTEr and SifDec: A constrained and unconstrained testing environment, revisited**. *ACM Trans. Math. Software*, 29:373–394, 2003.

- [1315] R. B. Gramacy and H. K. H. Lee. **Bayesian Treed Gaussian Process Models With an Application to Computer Modeling.** *Journal of the American Statistical Association*, 103: 1119–1130, 2008. doi:[10.1198/016214508000000689](https://doi.org/10.1198/016214508000000689).
Keywords: Treed-GP.
- [1316] A. Grasas, A. A. Juan, and H. Ramalhinho Lourenço. **SimILS: A Simulation-based Extension of the Iterated Local Search Metaheuristic for Stochastic Combinatorial Optimization.** *Journal of Simulation*, 10(1):69–77, 2016.
- [1317] J. Gratch and S. A. Chien. **Adaptive Problem-solving for Large-scale Scheduling Problems: A Case Study.** *J. Artif. Intell. Res.*, 4:365–396, 1996.
Annotation: Earliest hyper-heuristic?
- [1318] J. Gratch and G. DeJong. **COMPOSER: A probabilistic solution to the utility problem in speed-up learning.** In W. R. Swartout, editor, *Proceedings of the 10th National Conference on Artificial Intelligence*, pages 235–240. AAAI Press/MIT Press, Menlo Park, CA, 1992.
Annotation: Earliest hyper-heuristic?
- [1319] M. Gravel, W. L. Price, and C. Gagné. **Scheduling continuous casting of aluminum using a multiple objective ant colony optimization metaheuristic.** *Eur. J. Oper. Res.*, 143(1): 218–229, 2002. doi:[10.1016/S0377-2217\(01\)00329-0](https://doi.org/10.1016/S0377-2217(01)00329-0).
- [1320] A. Graves, A.-r. Mohamed, and G. Hinton. **Speech recognition with deep recurrent neural networks.** In *Acoustics, speech and signal processing (icassp), 2013 ieee international conference on*, pages 6645–6649. IEEE, 2013.
- [1321] R. L. Graves and P. Wolfe, editors. *Recent Advances in Mathematical Programming*. McGraw Hill, New York, NY, 1963.
- [1322] S. Greco, B. Matarazzo, and R. Słowiński. **Interactive evolutionary multiobjective optimization using dominance-based rough set approach.** In H. Ishibuchi et al., editors, *Proceedings of the 2010 Congress on Evolutionary Computation (CEC 2010)*, pages 1–8, Piscataway, NJ, 2010. IEEE Press.
- [1323] S. Greco, M. Kadziński, V. Mousseau, and R. Słowiński. **ELECTRE^{GKMS}: Robust ordinal regression for outranking methods.** *Eur. J. Oper. Res.*, 214(1):118–135, 2011.
- [1324] S. Greco, J. D. Knowles, K. Miettinen, and E. Zitzler, editors. *Learning in Multiobjective Optimization (Dagstuhl Seminar 12041)*, volume 2(1) of *Dagstuhl Reports*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2012. doi:[10.4230/DagRep.2.1.50](https://doi.org/10.4230/DagRep.2.1.50).
- [1325] S. Greco, V. Mousseau, and R. Słowiński. **Robust ordinal regression for value functions handling interacting criteria.** *Eur. J. Oper. Res.*, 239(3):711–730, 2014. doi:[10.1016/j.ejor.2014.05.022](https://doi.org/10.1016/j.ejor.2014.05.022).
- [1326] S. Greco, K. Klamroth, J. D. Knowles, and G. Rudolph, editors. *Understanding Complexity in Multiobjective Optimization (Dagstuhl Seminar 15031)*, volume 5(1) of *Dagstuhl Reports*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2015. doi:[10.4230/DagRep.5.1.96](https://doi.org/10.4230/DagRep.5.1.96).
Keywords: multiple criteria decision making, evolutionary multiobjective optimization.
- [1327] G. W. Greenwood, X. Hu, and J. G. D’Ambrosio. **Fitness functions for multiple objective optimization problems: Combining preferences with Pareto rankings.** In R. K. Belew and M. D. Vose, editors, *Foundations of Genetic Algorithms (FOGA)*, pages 437–455. Morgan Kaufmann Publishers, 1996.
- [1328] J. J. Grefenstette, editor. *Proceedings of the First International Conference on Genetic Algorithms and Their Applications, July 24-26, 1985, Carnegie-Mellon University, Pittsburgh, PA, 1985*. Lawrence Erlbaum Associates. ISBN 0805804269.
Annotation: Download a scanned copy from: <http://gpbib.cs.ucl.ac.uk/icga/>.

- [1329] J. J. Grefenstette. **Optimization of Control Parameters for Genetic Algorithms.** *IEEE Transactions on Systems, Man, and Cybernetics*, 16(1):122–128, 1986. doi:10.1109/TSMC.1986.289288.
Keywords: parameter tuning.
- [1330] J. J. Grefenstette, editor. *Proceedings of the Second International Conference on Genetic Algorithms, July 28-31, 1987, Massachusetts Institute of Technology, Cambridge, MA, 1987.* Lawrence Erlbaum Associates. ISBN 9780805801583.
Annotation: Download a scanned copy from: <http://gpbib.cs.ucl.ac.uk/icga/>.
- [1331] A. Gretton and C. C. Robert, editors. *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics, AISTATS 2016, Cadiz, Spain, May 9-11, 2016*, volume 51 of *Proceedings of Machine Learning Research*, 2016. PMLR.
- [1332] D. R. Grimes, C. T. Bauch, and J. P. A. Ioannidis. **Modelling science trustworthiness under publish or perish pressure.** *Royal Society Open Science*, 5:171511, 2018.
- [1333] A. Grosso, F. Della Croce, and R. Tadei. **An Enhanced Dynasearch Neighborhood for the Single-Machine Total Weighted Tardiness Scheduling Problem.** *Operations Research Letters*, 32(1):68–72, 2004.
- [1334] A. Grosso, A. R. M. J. U. Jamali, and M. Locatelli. **Finding Maximin Latin Hypercube Designs by Iterated Local Search Heuristics.** *Eur. J. Oper. Res.*, 197(2):541–547, 2009.
- [1335] O. Grothe, S. Nickel, S. Rebennack, and O. Stein, editors. *Operations Research 2022, Selected Papers of the Annual International Conference of the German Operations Research Society (GOR), Karlsruhe, Germany, September 6-9, 2022.* Lecture Notes in Operations Research. Springer, Cham, Switzerland, 2022.
- [1336] P. Groves, B. Kayyali, D. Knott, and S. Van Kuiken. **The "big data" revolution in healthcare.** *McKinsey Quarterly*, 2, 2013.
- [1337] B. Groz and S. Maniu. **Hypervolume subset selection with small subsets.** *Evol. Comput.*, 27(4):611–637, 2019.
- [1338] V. Grunert da Fonseca and C. M. Fonseca. **A link between the multivariate cumulative distribution function and the hitting function for random closed sets.** *Statistics & Probability Letters*, 57(2):179–182, 2002. doi:10.1016/S0167-7152(02)00046-9.
- [1339] V. Grunert da Fonseca and C. M. Fonseca. **A characterization of the outcomes of stochastic multiobjective optimizers through a reduction of the hitting function test sets.** Technical report, CSI, Universidade do Algarve, 2004.
Keywords: high-order EAF.
- [1340] V. Grunert da Fonseca and C. M. Fonseca. **The Attainment-Function Approach to Stochastic Multiobjective Optimizer Assessment and Comparison.** In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 103–130. Springer, Berlin/Heidelberg, 2010. doi:10.1007/978-3-642-02538-9_5.
- [1341] V. Grunert da Fonseca and C. M. Fonseca. **The Relationship between the Covered Fraction, Completeness and Hypervolume Indicators.** In J.-K. Hao, P. Legrand, P. Collet, N. Monmarché, E. Lutton, and M. Schoenauer, editors, *Artificial Evolution: 10th International Conference, Evolution Artificielle, EA, 2011*, volume 7401 of *LNCS*, pages 25–36. Springer, Heidelberg, Germany, 2012.

- [1342] V. Grunert da Fonseca and C. M. Fonseca. **The Relationship between the Covered Fraction, Completeness and Hypervolume Indicators**. In J.-K. Hao, P. Legrand, P. Collet, N. Monmarché, E. Lutton, and M. Schoenauer, editors, *Artificial Evolution: 10th International Conference, Evolution Artificielle, EA, 2011*, volume 7401 of *LNCS*. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-35533-2_3.
Keywords: hypervolume, empiricial attainment function.
- [1343] V. Grunert da Fonseca, C. M. Fonseca, and A. O. Hall. **Inferential Performance Assessment of Stochastic Optimisers and the Attainment Function**. In E. Zitzler, K. Deb, L. Thiele, C. A. Coello Coello, and D. Corne, editors, *EMO*, volume 1993 of *LNCS*, pages 213–225. Springer, Berlin/Heidelberg, 2001. doi:10.1007/3-540-44719-9_15.
Keywords: EAF.
Annotation: Proposed looking at anytime behavior as a multi-objective problem.
- [1344] C. Guéret, N. Monmarché, and M. Slimane. **Ants Can Play Music**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 310–317. Springer, Heidelberg, Germany, 2004.
- [1345] A. P. Guerreiro and C. M. Fonseca. **Computing and Updating Hypervolume Contributions in Up to Four Dimensions**. *IEEE Trans. Evol. Comput.*, 22(3):449–463, June 2018. doi:10.1109/tevc.2017.2729550.
Annotation: Proposed HV3D⁺ with $O(n \log n)$ complexity and HV4D⁺ with $O(n^2)$ complexity.
- [1346] A. P. Guerreiro, C. M. Fonseca, and L. Paquete. **The Hypervolume Indicator: Computational Problems and Algorithms**. *ACM Comput. Surv.*, 54(6):1–42, 2021. doi:10.1145/3453474.
- [1347] A. P. Guerreiro, V. Manquinho, and J. R. Figueira. **Exact hypervolume subset selection through incremental computations**. *Comput. Oper. Res.*, 136:105–471, Dec. 2021. doi:10.1016/j.cor.2021.105471.
- [1348] G. Guillén-Gosálbez. **A novel MILP-based objective reduction method for multi-objective optimization: Application to environmental problems**. *Computers & Chemical Engineering*, 35(8):1469–1477, 2011. ISSN 0098-1354. doi:10.1016/j.compchemeng.2011.02.001.
Keywords: Environmental engineering, Life cycle assessment, Multi-objective optimization, Objective reduction.
- [1349] A. Gunawan, K. M. Ng, and K. L. Poh. **A Hybridized Lagrangian Relaxation and Simulated Annealing Method for the Course Timetabling Problem**. *Comput. Oper. Res.*, 39(12):3074–3088, 2012.
- [1350] O. E. Gundersen, Y. Gil, and D. W. Aha. **On Reproducible AI: Towards Reproducible Research, Open Science, and Digital Scholarship in AI Publications**. *AI Magazine*, 39(3):56–68, Sept. 2018. doi:10.1609/aimag.v39i3.2816.
Annotation: The reproducibility guidelines can be found here: https://folk.idi.ntnu.no/odderik/reproducibility_guidelines.pdf and a short how-to can be found here: https://folk.idi.ntnu.no/odderik/reproducibility_guidelines_how_to.html.
- [1351] M. Guntsch and J. Branke. **New Ideas for Applying Ant Colony Optimization to the Probabilistic TSP**. In S. Cagnoni et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2003*, volume 2611 of *LNCS*, pages 165–175. Springer, Heidelberg, Germany, 2003.
- [1352] M. Guntsch and M. Middendorf. **Pheromone Modification Strategies for Ant Algorithms Applied to Dynamic TSP**. In E. J. W. Boers et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2001*, volume 2037 of *LNCS*, pages 213–222. Springer,

Heidelberg, Germany, 2001.

Annotation: Also available as Tech. Rep. AIDA-00-07, Intellectics Group, Darmstadt University of Technology, Germany.

- [1353] M. Guntsch and M. Middendorf. **A Population Based Approach for ACO**. In S. Cagnoni et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2002*, volume 2279 of *LNCS*, pages 71–80. Springer, Heidelberg, Germany, 2002.
- [1354] M. Guntsch and M. Middendorf. **Applying Population Based ACO to Dynamic Optimization Problems**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 111–122. Springer, Heidelberg, Germany, 2002.
- [1355] M. Guntsch and M. Middendorf. **Solving Multi-Objective Permutation Problems with Population Based ACO**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 464–478. Springer, Heidelberg, Germany, 2003.
- [1356] Y. Guo and F. Farooq, editors. *KDD’18: The 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, London United Kingdom, August 19-23, 2018*. ACM Press, New York, NY, July 2018.
- [1357] J. N. D. Gupta. **Flowshop schedules with sequence dependent setup times**. *Journal of Operations Research Society of Japan*, 29:206–219, 1986.
- [1358] Gurobi. **Gurobi Optimizer**. <http://www.gurobi.com/products/gurobi-optimizer>, 2017.
- [1359] D. Gusfield. **Algorithms on Strings, Trees, and Sequences**. In *Computer Science and Computational Biology*. Cambridge University Press, 1997.
- [1360] P. Gustavsson and A. Syberfeldt. **A New Algorithm Using the Non-Dominated Tree to Improve Non-Dominated Sorting**. *Evol. Comput.*, 26(1):89–116, 2018. [doi:10.1162/evco_a_00204](https://doi.org/10.1162/evco_a_00204).
- [1361] G. Gutin and A. Punnen, editors. *The Traveling Salesman Problem and its Variations*. Kluwer Academic Publishers, Dordrecht, The Netherlands, 2002.
- [1362] G. Gutin, A. Yeo, and A. Zverovich. **Traveling salesman should not be greedy: domination analysis of greedy-type heuristics for the TSP**. *Discrete Applied Mathematics*, 117(1–3), 2002.
- [1363] W. J. Gutjahr. **A Graph-based Ant System and its Convergence**. *Future Generation Computer Systems*, 16(8):873–888, 2000.
- [1364] W. J. Gutjahr. **ACO Algorithms with Guaranteed Convergence to the Optimal Solution**. *Inf. Process. Lett.*, 82(3):145–153, 2002.
- [1365] W. J. Gutjahr. **A converging ACO algorithm for stochastic combinatorial optimization**. In A. Albrecht and K. Steinhöfel, editors, *Stochastic Algorithms: Foundations and Applications*, volume 2827 of *LNCS*, pages 10–25. Springer Verlag, 2003. [doi:10.1007/b13596](https://doi.org/10.1007/b13596).
- [1366] W. J. Gutjahr. **S-ACO: An Ant-Based Approach to Combinatorial Optimization Under Uncertainty**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 238–249. Springer, Heidelberg, Germany, 2004.
- [1367] W. J. Gutjahr. **On the finite-time dynamics of ant colony optimization**. *Methodology and Computing in Applied Probability*, 8(1):105–133, 2006.
- [1368] W. J. Gutjahr. **Mathematical runtime analysis of ACO algorithms: survey on an emerging issue**. *Swarm Intelligence*, 1(1):59–79, 2007.

- [1369] W. J. Gutjahr. **First steps to the runtime complexity analysis of ant colony optimization.** *Comput. Oper. Res.*, 35(9):2711–2727, 2008.
- [1370] W. J. Gutjahr and M. S. Rauner. **An ACO algorithm for a dynamic regional nurse-scheduling problem in Austria.** *Comput. Oper. Res.*, 34(3):642–666, 2007. doi:10.1016/j.cor.2005.03.018.
Annotation: Logistics of Health Care Management Part Special Issue: Logistics of Health Care Management.
- [1371] W. J. Gutjahr and G. Sebastiani. **Runtime analysis of ant colony optimization with best-so-far reinforcement.** *Methodology and Computing in Applied Probability*, 10(3):409–433, 2008.
- [1372] I. Guyon, J. Weston, S. Barnhill, and V. Vapnik. **Gene selection for cancer classification using support vector machines.** *Mach. Learn.*, 46(1):389–422, 2002.
Keywords: recursive feature elimination.
- [1373] I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, and R. Garnett, editors. *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, 2017.
- [1374] H. Haario, E. Saksman, and J. Tamminen. **An adaptive Metropolis algorithm.** *Bernoulli*, 7(2):223–242, 2001.
- [1375] E. Haasdijk, A. Atta-ul Qayyum, and A. E. Eiben. **Racing to improve on-line, on-board evolutionary robotics.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 187–194. ACM Press, New York, NY, 2011.
- [1376] S. Häckel, M. Fischer, D. Zechel, and T. Teich. **A multi-objective ant colony approach for Pareto-optimization using dynamic programming.** In C. Ryan, editor, *GECCO*, pages 33–40. ACM Press, New York, NY, 2008.
- [1377] J. Hadar and W. R. Russell. **Rules for ordering uncertain prospects.** *The American Economic Review*, 59(1):25–34, 1969.
Keywords: stochastic dominance.
- [1378] D. Hadka and P. M. Reed. **Diagnostic Assessment of Search Controls and Failure Modes in Many-Objective Evolutionary Optimization.** *Evol. Comput.*, 20(3):423–452, 2012.
- [1379] D. Hadka and P. M. Reed. **Borg: An Auto-Adaptive Many-Objective Evolutionary Computing Framework.** *Evol. Comput.*, 21(2):231–259, 2013.
- [1380] D. Hadka, P. M. Reed, and T. W. Simpson. **Diagnostic assessment of the Borg MOEA for many-objective product family design problems.** In *IEEE CEC*, pages 1–10, Piscataway, NJ, 2012. IEEE Press.
- [1381] Y. Haimes, L. Lasdon, and D. Da Wismer. **On a bicriterion formation of the problems of integrated system identification and system optimization.** *IEEE Transactions on Systems, Man, and Cybernetics*, 1(3):296–297, 1971. doi:10.1109/TSMC.1971.4308298.
Keywords: epsilon-constraint method.
- [1382] B. Hajek. **Cooling Schedules for Optimal Annealing.** *Mathematics of Operations Research*, 13(2):311–329, 1988.
- [1383] B. Hajek and G. Sasaki. **Simulated annealing—to cool or not.** *System & Control Letters*, 12(5):443–447, 1989.

- [1384] P. Hajela and C.-Y. Lin. **Genetic search strategies in multicriterion optimal design.** *Structural Optimization*, 4(2):99–107, 1992.
- [1385] G. T. Hall, P. S. Oliveto, and D. Sudholt. **On the impact of the cutoff time on the performance of algorithm configurators.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 907–915. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321879.
Keywords: theory, automatic configuration, capping.
- [1386] G. T. Hall, P. S. Oliveto, and D. Sudholt. **Fast Perturbative Algorithm Configurators.** In T. Bäck, M. Preuss, A. Deutz, H. Wang, C. Doerr, M. T. M. Emmerich, and H. Trautmann, editors, *Parallel Problem Solving from Nature – PPSN XVI*, volume 12269 of *LNCS*, pages 19–32. Springer, Cham, Switzerland, 2020. doi:10.1007/978-3-030-58112-1_2.
- [1387] G. T. Hall, P. S. Oliveto, and D. Sudholt. **Analysis of the performance of algorithm configurators for search heuristics with global mutation operators.** In C. A. Coello Coello, editor, *GECCO*, pages 823–831. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.3390218.
- [1388] G. T. Hall, P. S. Oliveto, and D. Sudholt. **On the impact of the performance metric on efficient algorithm configuration.** *Artif. Intell.*, 303:103629, Feb. 2022. doi:10.1016/j.artint.2021.103629.
Keywords: irace.
- [1389] M. E. Halstead, M. López-Ibáñez, G. Farmer, and P. A. Warren. **Multiobjective Optimisation for Others: How Anchoring Effects Change Based on Who Guides the Interaction.** *Journal of Multi-Criteria Decision Analysis*, 33(2), June 2026. doi:10.1002/mcda.70036.
- [1390] M. E. Halstead, M. López-Ibáñez, G. Farmer, and P. A. Warren. **Anchoring in interactive multi-criteria decision-making and optimization: preferences and initial satisfaction.** *J. Oper. Res. Soc.*, 2026. doi:10.1080/01605682.2026.2730523.
- [1391] J. H. Halton. **Algorithm 247: Radical-inverse quasi-random point sequence.** *Commun. ACM*, 7(12):701–702, Dec. 1964. doi:10.1145/355588.365104.
Keywords: quasi-random, low-discrepancy.
- [1392] H. W. Hamacher and G. Ruhe. **On spanning tree problems with multiple objectives.** *Annals of Operations Research*, 52(4):209–230, 1994.
- [1393] Y. Hamadi and M. Schoenauer, editors. *6th International Conference, LION 6, Paris, France, January 16-20, 2012. Selected Papers*, volume 7219 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [1394] Y. Hamadi, E. Monfroy, and F. Saubion, editors. *Autonomous Search*. Springer, Berlin, Germany, 2012.
- [1395] R. P. Hämmäläinen and T. J. Lahtinen. **Path dependence in Operational Research—How the modeling process can influence the results.** *Oper. Res. Perspect.*, 3:14–20, Jan. 2016. doi:10.1016/j.orp.2016.03.001.
Keywords: Behavioral Biases, Behavioral Operational Research, Ethics in modelling, OR practice, Path dependence, Systems perspective.
- [1396] R. P. Hämmäläinen, J. Luoma, and E. Saarinen. **On the importance of behavioral operational research: The case of understanding and communicating about dynamic systems.** *Eur. J. Oper. Res.*, 228(3):623–634, Aug. 2013. doi:10.1016/j.ejor.2013.02.001.
- [1397] G. Hamerly and C. Elkan. **Learning the k in k-means.** In S. Thrun, L. Saul, and B. Schölkopf, editors, *Advances in Neural Information Processing Systems (NIPS 16)*. MIT Press, 2003.

- [1398] H. Hammami and T. Stützle. **A Computational Study of Neighborhood Operators for Job-Shop Scheduling Problems with Regular Objectives**. In B. Hu and M. López-Ibáñez, editors, *EvoCOP*, volume 10197 of *LNCS*, pages 1–17. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-55453-2.
- [1399] J. S. Hammond, R. L. Keeney, and H. Raiffa. **Even swaps: a rational method for making trade-offs**. *Harvard Business Review*, 76(2), Apr. 1998. ISSN 0017-8012.
- [1400] J. Handl and J. D. Knowles. **Modes of Problem Solving with Multiple Objectives: Implications for Interpreting the Pareto Set and for Decision Making**. In J. D. Knowles, D. Corne, K. Deb, and D. R. Chair, editors, *Multiobjective Problem Solving from Nature*, Natural Computing Series, pages 131–151. Springer, Berlin/Heidelberg, 2008. doi:10.1007/978-3-540-72964-8_7.
- [1401] J. Handl and X. Li, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2024, Melbourne, Australia, July 14-18, 2024*. ACM Press, New York, NY, 2024.
- [1402] J. Handl and X. Li, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO 2024, Melbourne, Australia, July 14-18, 2024*. ACM Press, New York, NY, 2024. Annotation: ISBN: 979-8-4007-0495-6.
- [1403] J. Handl, E. Hart, P. R. Lewis, M. López-Ibáñez, G. Ochoa, and B. Paechter, editors. *Parallel Problem Solving from Nature - PPSN XIV 14th International Conference, Edinburgh, UK, September 17-21, 2016, Proceedings*, volume 9921 of *LNCS*. Springer, Heidelberg, Germany, 2016. ISBN 978-3-319-45822-9. doi:10.1007/978-3-319-45823-6.
- [1404] T. Hanne. **On the convergence of multiobjective evolutionary algorithms**. *Eur. J. Oper. Res.*, 117(3):553–564, 1999. doi:10.1016/S0377-2217(98)00262-8. Keywords: archiving, efficiency presserving.
- [1405] T. Hanne. **Global Multiobjective Optimization with Evolutionary Algorithms: Selection Mechanisms and Mutation Control**. In E. Zitzler, K. Deb, L. Thiele, C. A. Coello Coello, and D. Corne, editors, *EMO*, volume 1993 of *LNCS*, pages 197–212. Springer, Berlin/Heidelberg, 2001.
- [1406] T. Hanne. **A multiobjective evolutionary algorithm for approximating the efficient set**. *Eur. J. Oper. Res.*, 176(3):1723–1734, 2007.
- [1407] M. P. Hansen. **Tabu search for multiobjective optimization: MOTS**. In J. Climaco, editor, *Proceedings of the 13th International Conference on Multiple Criteria Decision Making (MCDM'97)*, pages 574–586. Springer Verlag, 1997.
- [1408] M. P. Hansen. *Metaheuristics for multiple objective combinatorial optimization*. PhD thesis, Institute of Mathematical Modelling, Technical University of Denmark, Mar. 1998.
- [1409] M. P. Hansen and A. Jaszkiewicz. **Evaluating the quality of approximations to the non-dominated set**. Technical Report IMM-REP-1998-7, Institute of Mathematical Modelling, Technical University of Denmark, Lyngby, Denmark, 1998. Annotation: Proposed R2 indicator.
- [1410] N. Hansen. **The CMA evolution strategy: a comparing review**. In *Towards a new evolutionary computation*, pages 75–102. Springer, 2006.
- [1411] N. Hansen. **Benchmarking a BI-population CMA-ES on the BBOB-2009 function testbed**. In F. Rothlauf, editor, *GECCO Companion*, pages 2389–2396. ACM Press, New York, NY, 2009. Keywords: bipop-cma-es.

- [1412] N. Hansen and A. Ostermeier. **Adapting Arbitrary Normal Mutation Distributions in Evolution Strategies: The Covariance Matrix Adaptation**. In T. Bäck, T. Fukuda, and Z. Michalewicz, editors, *Proceedings of the 1996 IEEE International Conference on Evolutionary Computation (ICEC'96)*, pages 312–317. IEEE Press, Piscataway, NJ, 1996. doi:[10.1109/ICEC.1996.542381](https://doi.org/10.1109/ICEC.1996.542381).
Keywords: Evolution strategies, Evolutionary algorithms, self-adaptation, stochastic processes, Covariance matrix, matrix algebra, derandomised adaptation, mutation distribution, rotation invariance, electronic switching systems.
Annotation: Proposed CMA-ES.
- [1413] N. Hansen and A. Ostermeier. **Completely derandomized self-adaptation in evolution strategies**. *Evol. Comput.*, 9(2):159–195, 2001. doi:[10.1162/106365601750190398](https://doi.org/10.1162/106365601750190398).
Keywords: CMA-ES.
- [1414] N. Hansen, A. Auger, S. Finck, and R. Ros. **Real-Parameter Black-Box Optimization Benchmarking 2009: Experimental setup**. Technical Report RR-6828, INRIA, France, 2009.
- [1415] N. Hansen, S. Finck, R. Ros, and A. Auger. **Real-Parameter Black-Box Optimization Benchmarking 2009: Noiseless Functions Definitions**. Technical Report RR-6829, INRIA, France, 2009. Updated February 2010.
Annotation: <http://coco.gforge.inria.fr/bbob2012-downloads>.
- [1416] N. Hansen, A. Auger, R. Ros, S. Finck, and P. Pošík. **Comparing Results of 31 Algorithms from the Black-Box Optimization Benchmarking BBOB-2009**. In M. Pelikan and J. Branke, editors, *GECCO Companion*, pages 1689–1696. ACM Press, New York, NY, 2010. doi:[10.1145/1830761.1830790](https://doi.org/10.1145/1830761.1830790).
Keywords: benchmarking, black-box optimization.
- [1417] N. Hansen, R. Ros, N. Mauny, M. Schoenauer, and A. Auger. **Impacts of invariance in search: When CMA-ES and PSO face ill-conditioned and non-separable problems**. *Applied Soft Computing*, 11(8):5755–5769, 2011.
- [1418] N. Hansen, A. Auger, O. Mersmann, T. Tušar, and D. Brockhoff. **COCO: A platform for comparing continuous optimizers in a black-box setting**. *Arxiv preprint arXiv:1603.08785*, 2016. Published as [1420].
- [1419] N. Hansen, Y. Akimoto, and P. Baudis. **CMA-ES/pycma on Github**. Zenodo, Feb. 2019.
- [1420] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. **COCO: A platform for comparing continuous optimizers in a black-box setting**. *Optimization Methods and Software*, 36(1):1–31, 2020. doi:[10.1080/10556788.2020.1808977](https://doi.org/10.1080/10556788.2020.1808977).
- [1421] N. Hansen, A. Auger, D. Brockhoff, and T. Tušar. **Anytime Performance Assessment in Blackbox Optimization Benchmarking**. *IEEE Trans. Evol. Comput.*, 26(6):1293–1305, Dec. 2022. doi:[10.1109/TEVC.2022.3210897](https://doi.org/10.1109/TEVC.2022.3210897).
- [1422] P. Hansen and B. Jaumard. **Algorithms for the Maximum Satisfiability Problem**. *Computing*, 44:279–303, 1990.
- [1423] P. Hansen and N. Mladenović. **Variable neighborhood search: Principles and applications**. *Eur. J. Oper. Res.*, 130(3):449–467, 2001.
- [1424] P. Hansen and N. Mladenović. **Variable Neighborhood Search**. In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 145–184. Kluwer Academic Publishers, Norwell, MA, 2002.

- [1425] P. Hansen, N. Mladenović, J. Brimberg, and J. A. M. Pérez. **Variable Neighborhood Search**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, pages 61–86. Springer, New York, NY, 2nd edition, 2010.
- [1426] J.-K. Hao and M. Middendorf, editors. *Evolutionary Computation in Combinatorial Optimization – 12th European Conference, EvoCOP 2012, Málaga, Spain, April 11-13, 2012, Proceedings*, volume 7245 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [1427] J.-K. Hao and J. Pannier. **Simulated Annealing and Tabu Search for Constraint Solving**. In M. C. Golumbic et al., editors, *Fifth International Symposium on Artificial Intelligence and Mathematics, AIM 1998, Fort Lauderdale, Florida, USA, January 4-6, 1998*, pages 1–15, 1998.
- [1428] J.-K. Hao, E. Lutton, E. M. A. Ronald, M. Schoenauer, and D. Snyers, editors. *Artificial Evolution, Third European Conference, AE’97, Nîmes, France, 22-24 October 1997, Selected Papers*, volume 1363 of *LNCS*. Springer, Heidelberg, Germany, 1998.
- [1429] J.-K. Hao, P. Legrand, P. Collet, N. Monmarché, E. Lutton, and M. Schoenauer, editors. *Artificial Evolution: 10th International Conference, Evolution Artificielle, EA, 2011, Angers, France, October 24-26, 2011. Revised Selected Papers*, volume 7401 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [1430] Z. Hao, R. Cai, and H. Huang. **An Adaptive Parameter Control Strategy for ACO**. In *Proceedings of the International Conference on Machine Learning and Cybernetics*, pages 203–206. IEEE Press, 2006.
- [1431] Z. Hao, H. Huang, Y. Qin, and R. Cai. **An ACO Algorithm with Adaptive Volatility Rate of Pheromone Trail**. In Y. Shi, G. D. van Albada, J. Dongarra, and P. M. A. Sloot, editors, *Computational Science – ICCS 2007, 7th International Conference, Proceedings, Part IV*, volume 4490 of *LNCS*, pages 1167–1170. Springer, Heidelberg, Germany, 2007.
- [1432] K. Haraguchi. **Iterated Local Search with Trellis-Neighborhood for the Partial Latin Square Extension Problem**. *J. Heuristics*, 22(5):727–757, 2016.
- [1433] D. P. Hardin and E. B. Saff. **Discretizing Manifolds via Minimum Energy Points**. *Notices of the American Mathematical Society*, 51(10):1186–1194, 2004.
- [1434] E. Hart and K. Sim. **A Hyper-Heuristic Ensemble Method for Static Job-Shop Scheduling**. *Evol. Comput.*, 24(4):609–635, 2016. doi:10.1162/EVCO_a_00183.
- [1435] E. Hart, I. Miguel, C. Stone, and Q. Renau. **Towards optimisers that ‘Keep Learning’**. In S. Silva and L. Paquete, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2023*, pages 1636–1638. ACM Press, New York, NY, 2023. doi:10.1145/3583133.3596344.
Annotation: ISBN: 979-8-4007-0120-7.
- [1436] J. P. Hart and A. W. Shogan. **Semi-greedy heuristics: An empirical study**. *Operations Research Letters*, 6(3):107–114, 1987.
- [1437] W. D. Harvey and M. L. Ginsberg. **Limited Discrepancy Search**. In C. S. Mellish, editor, *Proc. of IJCAI-95*, pages 607–615. Morgan Kaufmann Publishers, 1995.
- [1438] H. Hashimoto, M. Yagiura, and T. Ibaraki. **An Iterated Local Search Algorithm for the Time-dependent Vehicle Routing Problem with Time Windows**. *Discrete Optimization*, 5(2):434–456, 2008.
- [1439] S. Hasija and C. Rajendran. **Scheduling in flowshops to minimize total tardiness of jobs**. *Int. J. Prod. Res.*, 42(11):2289–2301, 2004. doi:10.1080/00207540310001657595.

- [1440] D. Haussler, editor. *Proceedings of the Fifth Annual ACM Conference on Computational Learning Theory, COLT 1992, Pittsburgh, PA, USA, July 27-29, 1992*, 1992. ACM Press.
- [1441] S. Haykin. **A comprehensive foundation**. *Neural Networks*, 2:41, 2004.
- [1442] Ö. Hazir, Y. Günalay, and E. Erel. **Customer order scheduling problem: a comparative metaheuristics study**. *International Journal of Advanced Manufacturing Technology*, 37(5): 589–598, May 2008. doi:10.1007/s00170-007-0998-8.
Keywords: ACO, Customer order scheduling, Genetic algorithms, Meta-heuristics, Simulated annealing, Tabu search.
- [1443] X. He, K. Zhao, and X. Chu. **AutoML: A survey of the state-of-the-art**. *Knowledge-Based Systems*, 212:106622, 2021. ISSN 0950-7051. doi:10.1016/j.knosys.2020.106622.
- [1444] Z. He and G. G. Yen. **Many-Objective Evolutionary Algorithm: Objective Space Reduction and Diversity Improvement**. *IEEE Trans. Evol. Comput.*, 20(1):145–160, 2016.
- [1445] V. Heidrich-Meisner and C. Igel. **Hoeffding and Bernstein races for selecting policies in evolutionary direct policy search**. In A. P. Danyluk, L. Bottou, and M. L. Littman, editors, *Proceedings of the 26th International Conference on Machine Learning, ICML 2009*, pages 401–408, New York, NY, 2009. ACM Press. doi:10.1145/1553374.1553426.
Keywords: automated algorithm configuration, CMA-ES, racing.
- [1446] F. Heintz, M. Milano, and B. O’Sullivan, editors. *Trustworthy AI - Integrating Learning, Optimization and Reasoning First International Workshop, TAILOR 2020, Virtual Event, September 4-5, 2020, Revised Selected Papers*, volume 12641 of *LNCS*. Springer, Cham, Switzerland, 2021.
- [1447] S. R. Hejazi and S. Saghafian. **Flowshop-scheduling Problems with Makespan Criterion: A Review**. *Int. J. Prod. Res.*, 43(14):2895–2929, 2005.
- [1448] M. Held and R. M. Karp. **The Traveling-Salesman Problem and Minimum Spanning Trees**. *Operations Research*, 18(6):1138–1162, 1970.
- [1449] C. Helmberg and F. Rendl. **Solving quadratic (0,1)-problems by semidefinite programs and cutting planes**. *Mathematical Programming*, 82(3):291–315, 1998.
- [1450] M. Helmert and G. Röger, editors. *Proceedings of the Sixth Annual Symposium on Combinatorial Search, SOCS 2013, Leavenworth, Washington, USA, July 11-13, 2013*, 2013. AAAI Press.
- [1451] K. Helsgaun. **An Effective Implementation of the Lin-Kernighan Traveling Salesman Heuristic**. *Eur. J. Oper. Res.*, 126:106–130, 2000.
- [1452] K. Helsgaun. **General k -opt Submoves for the Lin-Kernighan TSP Heuristic**. *Mathematical Programming Computation*, 1(2-3):119–163, 2009.
- [1453] K. Helsgaun. **Source Code of the Lin-Kernighan-Helsgaun Traveling Salesman Heuristic**. <http://webhotel4.ruc.dk/~keld/research/LKH/>, 2018.
- [1454] K. Helsgaun. **Efficient Recombination in the Lin-Kernighan-Helsgaun Traveling Salesman Heuristic**. In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11101 of *LNCS*, pages 95–107. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-319-99253-2_8.
- [1455] S. Helwig, J. Branke, and S. Mostaghim. **Experimental Analysis of Bound Handling Techniques in Particle Swarm Optimization**. *IEEE Trans. Evol. Comput.*, 17(2):259–271, Apr. 2013. doi:10.1109/tevc.2012.2189404.

- [1456] D. Henderson, S. H. Jacobson, and A. W. Johnson. **The Theory and Practice of Simulated Annealing**. In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 287–319. Springer, Boston, MA, 2003. doi:10.1007/b101874.
- [1457] D. Hennes and D. Izzo. **Interplanetary trajectory planning with Monte Carlo tree search**. In Q. Yang and M. Wooldridge, editors, *Proc. of IJCAI-15*, pages 769–775. IJCAI/AAAI Press, Menlo Park, CA, 2015.
- [1458] D. Hennes, D. Izzo, and D. Landau. **Fast approximators for optimal low-thrust hops between main belt asteroids**. In X. Chen and A. Stafylopatis, editors, *Computational Intelligence (SSCI), 2016 IEEE Symposium Series on*, pages 1–7, 2016. doi:10.1109/SSCI.2016.7850107.
- [1459] J. L. Henning. **SPEC CPU2000: measuring CPU performance in the New Millennium**. *Computer*, 33(7):28–35, 2000. doi:10.1109/2.869367.
- [1460] R. Herbrich, T. Graepel, and K. Obermayer. **Support vector learning for ordinal regression**. In *ICANN'99: Proceedings of the 9th International Conference on Artificial Neural Networks*, pages 97–102, 1999. doi:10.1049/cp:19991091.
Keywords: support vector machine;metric regression;support vector learning;ordinal regression;information retrieval;risk functional;machine learning;pattern classification;.
Annotation: Proposed the pairwise transform for learning-to-rank.
- [1461] C. Hernández and O. Schütze. **A bounded archive based for bi-objective problems based on distance and e-dominance to avoid cyclic behavior**. In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022*, pages 583–591. ACM Press, New York, NY, 2022. doi:10.1145/3512290.3528840.
- [1462] H. Hernández and C. Blum. **Ant colony optimization for multicasting in static wireless ad-hoc networks**. *Swarm Intelligence*, 3(2):125–148, 2009.
- [1463] C. I. Hernández Castellanos and O. Schütze. **A Bounded Archiver for Hausdorff Approximations of the Pareto Front for Multi-Objective Evolutionary Algorithms**. *Mathematical and Computational Applications*, 27(3):48, 2022. doi:10.3390/mca27030048.
- [1464] C. I. Hernández Castellanos, O. Schütze, J. Q. Sun, and S. Ober-Blöbaum. **Non-epsilon dominated evolutionary algorithm for the set of approximate solutions**. *Mathematical and Computational Applications*, 25(1):3, 2020.
Keywords: archiving, multimodal.
- [1465] M. A. Heroux. **Editorial: ACM TOMS Replicated Computational Results Initiative**. *ACM Trans. Math. Software*, 41(3):1–5, June 2015. doi:10.1145/2743015.
- [1466] A. Herrán, J. M. Colmenar, and A. Duarte. **An efficient Variable Neighborhood Search for the Space-Free Multi-Row Facility Layout problem**. *Eur. J. Oper. Res.*, 2021. doi:10.1016/j.ejor.2021.03.027.
- [1467] F. Herrera, M. Lozano, and J. L. Verdegay. **Tackling Real-Coded Genetic Algorithms: Operators and Tools for Behavioural Analysis**. *Artificial Intelligence Review*, 12:265–319, 1998.
Keywords: genetic algorithms, real coding, continuous search spaces, mutation, recombination.
- [1468] F. Herrera, M. Lozano, and A. M. Sánchez. **A taxonomy for the crossover operator for real-coded genetic algorithms: An experimental study**. *International Journal of Intelligent Systems*, 18(3):309–338, 2003. doi:10.1002/int.10091.

- [1469] F. Herrera, M. Lozano, and D. Molina. **Test suite for the special issue of Soft Computing on scalability of evolutionary algorithms and other metaheuristics for large scale continuous optimization problems.** <http://sci2s.ugr.es/eamhco/>, 2010.
Keywords: SOCO benchmark.
- [1470] A. Hertz and D. de Werra. **Using Tabu Search Techniques for Graph Coloring.** *Computing*, 39(4):345–351, 1987.
- [1471] M. Heule and S. Weaver, editors. *Theory and Applications of Satisfiability Testing – SAT 2015*, volume 9340 of *LNCS*. Springer, Cham, Switzerland, 2015.
- [1472] R. Heumüller, S. Nielebock, J. Krüger, and F. Ortmeier. **Publish or perish, but do not forget your software artifacts.** *Empirical Software Engineering*, 25(6):4585–4616, 2020. doi:10.1007/s10664-020-09851-6.
- [1473] D. P. Heyman and M. J. Sobel. *Stochastic models in operations research: stochastic optimization*, volume 2. Courier Corporation, 2003.
- [1474] C. Hicks. **A Genetic Algorithm tool for optimising cellular or functional layouts in the capital goods industry.** *International Journal of Production Economics*, 104(2):598–614, 2006. doi:10.1016/j.ijpe.2005.03.010.
- [1475] R. M. Hierons, M. Li, X. Liu, J. A. Parejo, S. Segura, and X. Yao. **Many-objective test suite generation for software product lines.** *ACM Transactions on Software Engineering and Methodology*, 29(1):1–46, 2020.
- [1476] G. E. Hinton and R. R. Salakhutdinov. **Reducing the dimensionality of data with neural networks.** *Science*, 313(5786):504–507, 2006.
- [1477] D. S. Hochbaum, editor. *Approximation Algorithms For NP-hard Problems*. PWS Publishing Co., 1996.
- [1478] W. Hoeffding. **Probability inequalities for sums of bounded random variables.** *Journal of the American Statistical Association*, 58(301):13–30, 1963.
- [1479] S. Hoeffler and D. Ariely. **Constructing stable preferences: A look into dimensions of experience and their impact on preference stability.** *Journal of Consumer Psychology*, 8(2):113–139, 1999. doi:10.1207/s15327663jcp0802_01.
- [1480] S. Hoeffler, D. Ariely, and P. West. **Path dependent preferences: The role of early experience and biased search in preference development.** *Organizational Behavior and Human Decision Processes*, 101(2):215–229, 2006. doi:10.1016/j.obhdp.2006.04.002.
- [1481] J. Hoffmann and B. Selman, editors. *Proceedings of the Twenty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2012, Toronto, Ontario, Canada, July 22-26, 2012*, 2012. AAAI Press.
- [1482] J. Holland. *Adaptation in Natural and Artificial Systems*. University of Michigan Press, 1975.
- [1483] M. Hollander and D. A. Wolfe. *Nonparametric statistical inference*. John Wiley & Sons, New York, NY, 1973. Second edition (1999).
- [1484] R. C. Holte and A. Howe, editors. *Proceedings of the Twenty-Second AAAI Conference on Artificial Intelligence, July 22-26, 2007, Vancouver, British Columbia, Canada*, 2007. AAAI Press/MIT Press, Menlo Park, CA.
- [1485] I. Hong, A. B. Kahng, and B. R. Moon. **Improved large-step Markov chain variants for the symmetric TSP.** *J. Heuristics*, 3(1):63–81, 1997.

- [1486] G. Hooker. **Discovering Additive Structure in Black Box Functions**. In W. Kim, R. Kohavi, J. Gehrke, and W. DuMouchel, editors, *Proceedings of the tenth ACM SIGKDD international conference on Knowledge discovery and data mining, KDD'04*, pages 575–580. ACM Press, New York, NY, 2004. doi:10.1145/1014052.1014122.
Keywords: diagnostics, functional ANOVA, feature selection, interpretation, visualization, additive models, draphical models.
- [1487] G. Hooker. **Generalized functional ANOVA diagnostics for high-dimensional functions of dependent variables**. *Journal of Computational and Graphical Statistics*, 16(3):709–732, 2012. doi:10.1198/106186007X237892.
- [1488] J. N. Hooker. **Needed: An Empirical Science of Algorithms**. *Operations Research*, 42(2): 201–212, 1994.
- [1489] J. N. Hooker. **Testing Heuristics: We Have It All Wrong**. *J. Heuristics*, 1(1):33–42, 1996. doi:10.1007/BF02430364.
- [1490] H. H. Hoos. **Programming by Optimisation: Towards a new Paradigm for Developing High-Performance Software**. In *MIC 2011, the 9th Metaheuristics International Conference*, 2011. URL <http://mic2011.diegm.uniud.it/uploads/plenaries/Hoos-MIC2011.pdf>. Plenary talk.
- [1491] H. H. Hoos. **Automated Algorithm Configuration and Parameter Tuning**. In Y. Hamadi, E. Monfroy, and F. Saubion, editors, *Autonomous Search*, pages 37–71. Springer, Berlin, Germany, 2012. doi:10.1007/978-3-642-21434-9_3.
- [1492] H. H. Hoos. **Programming by optimization**. *Commun. ACM*, 55(2):70–80, Feb. 2012. doi:10.1145/2076450.2076469.
- [1493] H. H. Hoos and T. Stützle. **Evaluating Las Vegas Algorithms — Pitfalls and Remedies**. In G. F. Cooper and S. Moral, editors, *Proceedings of the Fourteenth Conference on Uncertainty in Artificial Intelligence*, pages 238–245. Morgan Kaufmann Publishers, San Francisco, CA, 1998.
- [1494] H. H. Hoos and T. Stützle. *Stochastic Local Search: Foundations and Applications*. Elsevier, Amsterdam, The Netherlands, 2004.
Annotation: superseded by [1495].
- [1495] H. H. Hoos and T. Stützle. *Stochastic Local Search—Foundations and Applications*. Morgan Kaufmann Publishers, San Francisco, CA, 2005.
- [1496] H. H. Hoos and T. Stützle. **On the Empirical Scaling of Run-time for Finding Optimal Solutions to the Traveling Salesman Problem**. *Eur. J. Oper. Res.*, 238(1):87–94, 2014.
- [1497] H. H. Hoos and T. Stützle. **On the Empirical Time Complexity of Finding Optimal Solutions vs. Proving Optimality for Euclidean TSP Instances**. *Optimization Letters*, 9(6):1247–1254, 2015.
- [1498] H. H. Hoos, M. T. Lindauer, and T. Schaub. **Clasppfolio 2: Advances in Algorithm Selection for Answer Set Programming**. *Theory and Practice of Logic Programming*, 14(4-5):560–585, 2014.
- [1499] H. H. Hoos, F. Hutter, and K. Leyton-Brown. **Automated Configuration and Selection of SAT Solvers**. In *Handbook of Satisfiability*, pages 481–507. IOS Press, Feb. 2021. doi:10.3233/faia200995.
- [1500] J. Horn, N. Nafpliotis, and D. E. Goldberg. **A niched Pareto genetic algorithm for multiobjective optimization**. In *WCCI*, pages 82–87, Piscataway, NJ, June 1994. IEEE Press. doi:10.1109/ICEC.1994.350037.

- [1501] C. Horoba and F. Neumann. **Benefits and drawbacks for the use of epsilon-dominance in evolutionary multi-objective optimization.** In C. Ryan, editor, *GECCO*, pages 641–648. ACM Press, New York, NY, 2008.
Annotation: Proposed ϵ -box.
- [1502] K. Hoste and L. Eeckhout. **Cole: Compiler Optimization Level Exploration.** In M. L. Soffa and E. Duesterwald, editors, *Proceedings of the 6th Annual IEEE/ACM International Symposium on Code Generation and Optimization*, CGO '08, pages 165–174, New York, NY, 2008. ACM Press. doi:10.1145/1356058.1356080.
- [1503] A. Hottung and K. Tierney. **Neural large neighborhood search for routing problems.** *Artif. Intell.*, 313:103786, Dec. 2022. doi:10.1016/j.artint.2022.103786.
- [1504] A. Hottung, S. Tanaka, and K. Tierney. **Deep learning assisted heuristic tree search for the container pre-marshalling problem.** *Comput. Oper. Res.*, 113:104781, 2020. doi:10.1016/j.cor.2019.104781.
- [1505] A. Hottung, F. Berto, C. Hua, N. G. Zepeda, D. Wetzels, M. Römer, H. Ye, D. Zago, M. Poli, S. Massaroli, J. Park, and K. Tierney. **VRPagent: LLM-Driven Discovery of Heuristic Operators for Vehicle Routing Problems.** *Arxiv preprint arXiv:2510.07073 [cs.AI]*, 2025. doi:10.48550/arXiv.2510.07073.
- [1506] S. P. Hozo, B. Djulbegovic, and I. Hozo. **Estimating the mean and variance from the median, range, and the size of a sample.** *BMC Medical Research Methodology*, 5(1):13, 2005.
- [1507] B. Hu and M. López-Ibáñez, editors. *Evolutionary Computation in Combinatorial Optimization – 17th European Conference, EvoCOP 2017, Amsterdam, The Netherlands, April 19-21, 2017, Proceedings*, volume 10197 of *LNCS*. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-55453-2.
- [1508] T. Hu and L. Trujillo, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026, 13-17, 2026, San José, Costa Rica*. ACM Press, New York, NY, 2026.
- [1509] T. C. Hu, A. B. Kahng, and C.-W. A. Tsao. **Old Bachelor Acceptance: A New Class of Non-Monotone Threshold Accepting Methods.** *ORSA Journal on Computing*, 7(4): 417–425, 1995.
- [1510] W. Hu, L. Yan, H. Wang, B. Du, and D. Tao. **Real-time traffic jams prediction inspired by Biham, Middleton and Levine (BML) model.** *Inf. Sci.*, 2017.
Keywords: BML model, Prediction, Real-time, Traffic jam, Urban traffic network.
- [1511] W. Hu, H. Wang, Z. Qiu, C. Nie, and L. Yan. **A quantum particle swarm optimization driven urban traffic light scheduling model.** *Neural Computing & Applications*, 2018. doi:10.1007/s00521-016-2508-0.
Keywords: BML, Optimization, Simulation, Traffic congestion, Updating rules.
- [1512] L. Hua and Y. Wang. *Applications of Number Theory to Numerical Analysis*. Springer, Berlin, Germany, 1981. doi:10.1007/978-3-642-67829-5.
- [1513] C. Huang, Y. Li, and X. Yao. **A Survey of Automatic Parameter Tuning Methods for Metaheuristics.** *IEEE Trans. Evol. Comput.*, 24(2):201–216, 2020. doi:10.1109/TEVC.2019.2921598.
- [1514] D. Huang, T. T. Allen, W. I. Notz, and N. Zeng. **Global Optimization of Stochastic Black-Box Systems via Sequential Kriging Meta-Models.** *J. Glob. Optim.*, 34(3):441–466, 2006. doi:10.1007/s10898-005-2454-3.

- [1515] D.-S. Huang, K. Li, and G. W. Irwin, editors. *International Conference on Computational Science (3)*, volume 4115 of *LNCS*. Springer, Heidelberg, Germany, 2006.
- [1516] H. Huang, X. Yang, Z. Hao, and R. Cai. **A Novel ACO Algorithm with Adaptive Parameter**. In D.-S. Huang, K. Li, and G. W. Irwin, editors, *International Conference on Computational Science (3)*, volume 4115 of *LNCS*, pages 12–21. Springer, Heidelberg, Germany, 2006.
- [1517] K.-S. Huang, C.-B. Yang, and K. tsung Tseng. **Fast algorithms for finding the common subsequences of multiple sequences**. In *Proceedings of the International Computer Symposium*, pages 1006–1011. IEEE Press, 2004.
- [1518] S. Huband, P. Hingston, L. Barone, and L. While. **A Review of Multiobjective Test Problems and a Scalable Test Problem Toolkit**. *IEEE Trans. Evol. Comput.*, 10(5):477–506, 2006. doi:10.1109/TEVC.2005.861417.
Annotation: Proposed the WFG benchmark suite.
- [1519] B. Huberman, R. Lukose, and T. Hogg. **An Economic Approach to Hard Computational Problems**. *Science*, 275:51–54, 1997.
- [1520] D. L. Huerta-Muñoz, R. Z. Ríos-Mercado, and R. Ruiz. **An Iterated Greedy Heuristic for a Market Segmentation Problem with Multiple Attributes**. *Eur. J. Oper. Res.*, 261(1): 75–87, 2017.
- [1521] E. J. Hughes. **Multiple single objective Pareto sampling**. In *IEEE CEC*, pages 2678–2684, Piscataway, NJ, Dec. 2003. IEEE Press.
- [1522] E. J. Hughes. **MSOPS-II: A general-purpose many-objective optimiser**. In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 3944–3951, Piscataway, NJ, 2007. IEEE Press.
- [1523] E. J. Hughes. **Many-objective directed evolutionary line search**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 761–768. ACM Press, New York, NY, 2011.
- [1524] E. Hüllermeier, R. Kruse, and F. Hoffmann, editors. *13th International Conference on Information Processing and Management of Uncertainty, IPMU 2010, Germany, June 28-July 2, 2010. Proceedings*, volume 6178 of *LNAI*. Springer, Heidelberg, Germany, 2010.
- [1525] J. Humeau, A. Liefoghe, E.-G. Talbi, and S. Verel. **ParadisEO-MO: From Fitness Landscape Analysis to Efficient Local Search Algorithms**. Rapport de recherche RR-7871, INRIA, France, 2012.
- [1526] J. Humeau, A. Liefoghe, E.-G. Talbi, and S. Verel. **ParadisEO-MO: From Fitness Landscape Analysis to Efficient Local Search Algorithms**. *J. Heuristics*, 19(6):881–915, June 2013. doi:10.1007/s10732-013-9228-8.
- [1527] Y. Hung, V. R. Joseph, and S. N. Melkote. **Design and Analysis of Computer Experiments With Branching and Nested Factors**. *Technometrics*, 51(4):354–365, 2009. doi:10.1198/TECH.2009.07097.
- [1528] M. Hunt and M. López-Ibáñez. **Modeling a Decision-Maker in Goal Programming by means of Computational Rationality**. In I. Palomares, editor, *International Alan Turing Conference on Decision Support and Recommender systems*, pages 17–20, London, UK, Nov. 2019. Alan Turing Institute. ISBN 978-1-5262-0820-0.
- [1529] S. H. Hurlbert. **Pseudoreplication and the Design of Ecological Field Experiments**. *Ecological Monographs*, 54(2):187–211, 1984.

- [1530] M. Hurtgen and J.-C. Maun. **Optimal PMU placement using Iterated Local Search.** *International Journal of Electrical Power & Energy Systems*, 32(8):857–860, 2010.
- [1531] M. S. Hussin and T. Stützle. **Hierarchical Iterated Local Search for the Quadratic Assignment Problem.** In M. J. Blesa, C. Blum, L. Di Gaspero, A. Roli, M. Sampels, and A. Schaerf, editors, *Hybrid Metaheuristics*, volume 5818 of *LNCS*, pages 115–129. Springer, Heidelberg, Germany, 2009. doi:10.1007/978-3-642-04918-7_9.
- [1532] M. S. Hussin and T. Stützle. **Tabu Search vs. Simulated Annealing for Solving Large Quadratic Assignment Instances.** Technical Report TR/IRIDIA/2010-020, IRIDIA, Université Libre de Bruxelles, Belgium, 2010.
- [1533] M. S. Hussin and T. Stützle. **Tabu Search vs. Simulated Annealing for Solving Large Quadratic Assignment Instances.** *Comput. Oper. Res.*, 43:286–291, 2014.
- [1534] F. Hutter. **SAT benchmarks used in automated algorithm configuration.** <http://www.cs.ubc.ca/labs/beta/Projects/AAC/SAT-benchmarks.html>, 2007.
- [1535] F. Hutter. *Automated Configuration of Algorithms for Solving Hard Computational Problems.* PhD thesis, University of British Columbia, Department of Computer Science, Vancouver, Canada, Oct. 2009.
- [1536] F. Hutter and S. Ramage. *Manual for SMAC.* University of British Columbia, 2015. URL <http://www.cs.ubc.ca/labs/beta/Projects/SMAC/v2.10.03/manual.pdf>. SMAC version 2.10.03.
- [1537] F. Hutter, D. Babić, H. H. Hoos, and A. J. Hu. **Boosting Verification by Automatic Tuning of Decision Procedures.** In J. Baumgartner and M. Sheeran, editors, *FMCAD’07: Proceedings of the 7th International Conference Formal Methods in Computer Aided Design*, pages 27–34, Austin, Texas, USA, 2007. IEEE Computer Society, Washington, DC, USA.
- [1538] F. Hutter, H. H. Hoos, and T. Stützle. **Automatic Algorithm Configuration Based on Local Search.** In R. C. Holte and A. Howe, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1152–1157. AAAI Press/MIT Press, Menlo Park, CA, 2007.
- [1539] F. Hutter, H. H. Hoos, K. Leyton-Brown, and K. P. Murphy. **An experimental investigation of model-based parameter optimisation: SPO and beyond.** In F. Rothlauf, editor, *GECCO*, pages 271–278. ACM Press, New York, NY, 2009. doi:10.1145/1569901.1569940.
- [1540] F. Hutter, H. H. Hoos, K. Leyton-Brown, and T. Stützle. **ParamILS: An Automatic Algorithm Configuration Framework.** *J. Artif. Intell. Res.*, 36:267–306, Oct. 2009. doi:10.1613/jair.2861.
- [1541] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Automated Configuration of Mixed Integer Programming Solvers.** In A. Lodi, M. Milano, and P. Toth, editors, *CPAIOR 2010*, volume 6140 of *LNCS*, pages 186–202. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-13520-0_23.
Keywords: MIP, ParamILS.
- [1542] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Tradeoffs in the Empirical Evaluation of Competing Algorithm Designs.** *Annals of Mathematics and Artificial Intelligence*, 60(1–2): 65–89, 2010.
- [1543] F. Hutter, H. H. Hoos, K. Leyton-Brown, and K. Murphy. **Time-Bounded Sequential Parameter Optimization.** In C. Blum and R. Battiti, editors, *LION 4*, volume 6073 of *LNCS*, pages 281–298. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-13800-3_30.

- [1544] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Sequential Model-Based Optimization for General Algorithm Configuration**. In C. A. Coello Coello, editor, *LION 5*, volume 6683 of *LNCS*, pages 507–523. Springer, Heidelberg, Germany, 2011. doi:[10.1007/978-3-642-25566-3_40](https://doi.org/10.1007/978-3-642-25566-3_40).
Keywords: SMAC, ROAR.
- [1545] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Parallel Algorithm Configuration**. In Y. Hamadi and M. Schoenauer, editors, *LION 6*, volume 7219 of *LNCS*, pages 55–70. Springer, Heidelberg, Germany, 2012.
- [1546] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Bayesian Optimization With Censored Response Data**. *Arxiv preprint arXiv:1310.1947*, 2013. URL <http://arxiv.org/abs/1310.1947>.
- [1547] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Identifying Key Algorithm Parameters and Instance Features using Forward Selection**. In P. M. Pardalos and G. Nicosia, editors, *LION 7*, volume 7997 of *LNCS*, pages 364–381. Springer, Heidelberg, Germany, 2013. doi:[10.1007/978-3-642-44973-4_40](https://doi.org/10.1007/978-3-642-44973-4_40).
Keywords: parameter importance.
- [1548] F. Hutter, H. H. Hoos, and K. Leyton-Brown. **An Efficient Approach for Assessing Hyperparameter Importance**. In E. P. Xing and T. Jebara, editors, *Proceedings of the 31st International Conference on Machine Learning, ICML 2014*, volume 32, pages 754–762. PMLR, 2014. URL <https://proceedings.mlr.press/v32/hutter14.html>.
Keywords: fANOVA, parameter importance.
- [1549] F. Hutter, M. López-Ibáñez, C. Fawcett, M. T. Lindauer, H. H. Hoos, K. Leyton-Brown, and T. Stützle. **AClib: A Benchmark Library for Algorithm Configuration**. In P. M. Pardalos, M. G. C. Resende, C. Vogiatzis, and J. L. Walteros, editors, *LION 8*, volume 8426 of *LNCS*, pages 36–40. Springer, Heidelberg, Germany, 2014. doi:[10.1007/978-3-319-09584-4_4](https://doi.org/10.1007/978-3-319-09584-4_4).
- [1550] F. Hutter, L. Xu, H. H. Hoos, and K. Leyton-Brown. **Algorithm Runtime Prediction: Methods & Evaluation**. *Artif. Intell.*, 206:79–111, 2014. doi:[10.1016/j.artint.2013.10.003](https://doi.org/10.1016/j.artint.2013.10.003).
Keywords: Empirical performance models; Mixed integer programming; SAT.
- [1551] F. Hutter, H. H. Hoos, K. Leyton-Brown, and T. Stützle. **ParamILS**. <http://www.cs.ubc.ca/labs/beta/Projects/ParamILS/>, 2017. Version visited last on July 2017.
- [1552] F. Hutter, M. T. Lindauer, A. Balint, S. Bayless, H. H. Hoos, and K. Leyton-Brown. **The Configurable SAT Solver Challenge (CSSC)**. *Artif. Intell.*, 243:1–25, 2017. doi:[10.1016/j.artint.2016.09.006](https://doi.org/10.1016/j.artint.2016.09.006).
- [1553] F. Hutter, L. Kotthoff, and J. Vanschoren, editors. *Automated Machine Learning: Methods, Systems, Challenges*. Springer, 2019. doi:[10.1007/978-3-030-05318-5](https://doi.org/10.1007/978-3-030-05318-5).
- [1554] C. Iacopino and P. Palmer. **The Dynamics of Ant Colony Optimization Algorithms Applied to Binary Chains**. *Swarm Intelligence*, 6(4):343–377, 2012.
- [1555] C. Iacopino, P. Palmer, N. Policella, A. Donati, and A. Brewer. **How Ants Can Manage Your Satellites**. *Acta Futura*, 9:59–72, 2014. doi:[10.2420/AF09.2014.59](https://doi.org/10.2420/AF09.2014.59).
Keywords: ACO, Space.
- [1556] T. Ibaraki. **A Personal Perspective on Problem Solving by General Purpose Solvers**. *International Transactions in Operational Research*, 17(3):303–315, 2010.

- [1557] T. Ibaraki, S. Imahori, K. Nonobe, K. Sobue, T. Uno, and M. Yagiura. **An Iterated Local Search Algorithm for the Vehicle Routing Problem with Convex Time Penalty Functions**. *Discrete Applied Mathematics*, 156(11):2050–2069, 2008.
- [1558] IBM. **ILOG CPLEX Optimizer**. <http://www.ibm.com/software/integration/optimization/cplex-optimizer/>, 2017.
- [1559] ICANN. *Proceedings of the 9th International Conference on Artificial Neural Networks: ICANN '99, Location: Edinburgh, UK, 7-10 Sept. 1999*, 1999.
- [1560] ICLR. *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*, 2017.
- [1561] ICLR. *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*, 2024.
- [1562] ICML. *Proceedings of the 41st International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, Proceedings of Machine Learning Research, 2024. PMLR.
- [1563] ICMLC. *Proceedings of the International Conference on Machine Learning and Cybernetics*, 2006. IEEE Press.
- [1564] J. Ide and A. Schöbel. **Robustness for uncertain multi-objective optimization: a survey and analysis of different concepts**. *OR Spectrum*, 38(1):235–271, 2016. doi:10.1007/s00291-015-0418-7.
- [1565] IEEE CEC. *Proceedings of the 1999 Congress on Evolutionary Computation (CEC 1999)*, Piscataway, NJ, 1999. IEEE Press.
- [1566] IEEE CEC. *Proceedings of the 2000 Congress on Evolutionary Computation (CEC 2000)*, Piscataway, NJ, July 2000. IEEE Press.
- [1567] IEEE CEC. *Proceedings of the 2001 Congress on Evolutionary Computation (CEC 2001)*, Piscataway, NJ, 2001. IEEE Press.
- [1568] IEEE CEC. *IEEE CEC*, Piscataway, NJ, 2002. IEEE Press.
- [1569] IEEE CEC. *Proceedings of the 2003 Congress on Evolutionary Computation (CEC 2003), Canberra, ACT, Australia*, Piscataway, NJ, Dec. 2003. IEEE Press.
- [1570] IEEE CEC. *Proceedings of the 2004 Congress on Evolutionary Computation (CEC 2004)*, Piscataway, NJ, Sept. 2004. IEEE Press.
- [1571] IEEE CEC. *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, Piscataway, NJ, Sept. 2005. IEEE Press.
- [1572] IEEE CEC. *Proceedings of the 2006 Congress on Evolutionary Computation (CEC 2006)*, Piscataway, NJ, July 2006. IEEE Press.
- [1573] IEEE CEC. *Proceedings of the IEEE Congress on Evolutionary Computation, CEC 2007, 25-28 September 2007, Singapore*, Piscataway, NJ, 2007. IEEE Press.
- [1574] IEEE CEC. *Proceedings of the IEEE Congress on Evolutionary Computation, CEC 2008, June 1-6, 2008, Hong Kong, China*, Piscataway, NJ, 2008. IEEE Press.
- [1575] IEEE CEC. *IEEE CEC*, Piscataway, NJ, 2009. IEEE Press.
- [1576] IEEE CEC. *Proceedings of the 2011 Congress on Evolutionary Computation (CEC 2011), New Orleans, LA, USA*, Piscataway, NJ, 2011. IEEE Press.

- [1577] IEEE CEC. *Proceedings of the 2012 Congress on Evolutionary Computation (CEC 2012)*, Piscataway, NJ, 2012. IEEE Press.
- [1578] IEEE CEC. *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, Piscataway, NJ, 2013. IEEE Press.
- [1579] IEEE CEC. *Proceedings of the 2014 Congress on Evolutionary Computation (CEC 2014)*, Piscataway, NJ, 2014. IEEE Press.
- [1580] IEEE CEC. *IEEE Congress on Evolutionary Computation, CEC 2015, Sendai, Japan, May 25-28, 2015 Computation (CEC 2015)*, Piscataway, NJ, 2015. IEEE Press.
- [1581] IEEE CEC. *IEEE Congress on Evolutionary Computation, CEC 2016, Vancouver, BC, Canada, July 24-29, 2016*, Piscataway, NJ, 2016. IEEE Press. ISBN 978-1-5090-0623-6.
- [1582] IEEE CEC. *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, Piscataway, NJ, 2017. IEEE Press.
- [1583] IEEE CEC. *Proceedings of the 2018 Congress on Evolutionary Computation (CEC 2018)*, Piscataway, NJ, 2018. IEEE Press.
- [1584] IEEE CEC. *Proceedings of the 2019 Congress on Evolutionary Computation (CEC 2019)*, Piscataway, NJ, June 2019. IEEE Press.
- [1585] IEEE CEC. *Proceedings of the 2020 Congress on Evolutionary Computation (CEC 2020)*, Piscataway, NJ, 2020. IEEE Press.
- [1586] IEEE CEC. *Proceedings of the 2021 Congress on Evolutionary Computation (CEC 2021)*, Piscataway, NJ, 2021. IEEE Press.
- [1587] C. Igel. **Multi-objective Model Selection for Support Vector Machines**. In C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors, *EMO*, volume 3410 of *LNCIS*, pages 534–546. Springer, Berlin/Heidelberg, 2005. doi:10.1007/978-3-540-31880-4_37.
Annotation: Early work on multi-objective hyper-parameter optimization (AutoML).
- [1588] C. Igel and D. V. Arnold, editors. *Genetic and Evolutionary Computation Conference, GECCO 2014, Proceedings, Vancouver, BC, Canada, July 12-16, 2014*. ACM Press, New York, NY, 2014.
- [1589] C. Igel, N. Hansen, and S. Roth. **Covariance Matrix Adaptation for Multi-objective Optimization**. *Evol. Comput.*, 15(1):1–28, 2007.
- [1590] C. Igel, V. Heidrich-Meisner, and T. Glasmachers. **Shark**. *J. Mach. Learn. Res.*, 9:993–996, June 2008. URL <http://www.jmlr.org/papers/volume9/igel08a/igel08a.pdf>.
- [1591] IJCNN. *Proceedings of the International Joint Conference on Neural Networks, IJCNN 2006, part of the IEEE World Congress on Computational Intelligence, WCCI 2006, Vancouver, BC, Canada, 16-21 July 2006*, 2006. IEEE. doi:10.1109/IJCNN11286.2006.
- [1592] K. Ikeda, H. Kita, and S. Kobayashi. **Failure of Pareto-based MOEAs: Does non-dominated really mean near to optimal?** In *IEEE CEC*, pages 957–962, Piscataway, NJ, 2001. IEEE Press.
Keywords: dominance resistance.
- [1593] N. Ilich and S. P. Simonovic. **Evolutionary Algorithm for minimization of pumping cost**. *Journal of Computing in Civil Engineering, ASCE*, 12(4):232–240, Oct. 1998.
- [1594] J. Illian, A. Penttinen, H. Stoyan, and D. Stoyan. *Statistical Analysis and Modelling of Spatial Point Patterns*. Wiley, 2008.

- [1595] T. Imamichi, M. Yagiura, and H. Nagamochi. **An Iterated Local Search Algorithm Based on Nonlinear Programming for the Irregular Strip Packing Problem.** *Discrete Optimization*, 6(4):345–361, 2009.
- [1596] Innovation 24. **LocalSolver.** <http://www.localsolver.com>, 2016. Last visited, August 15, 2016.
- [1597] A. Inselberg. **The Plane with Parallel Coordinates.** *The Visual Computer*, 1(2):69–91, 1985.
- [1598] Intel. **Intel Software Autotuning Tool.** <https://software.intel.com/en-us/articles/intel-software-autotuning-tool/>, 2010.
- [1599] J. P. A. Ioannidis. **Why Most Published Research Findings Are False.** *PLoS Medicine*, 2(8):e124, 2005. doi:10.1371/journal.pmed.0020124.
- [1600] S. Iredi, D. Merkle, and M. Middendorf. **Bi-Criterion Optimization with Multi Colony Ant Algorithms.** In E. Zitzler, K. Deb, L. Thiele, C. A. Coello Coello, and D. Corne, editors, *EMO*, volume 1993 of *LNCS*, pages 359–372. Springer, Berlin/Heidelberg, 2001. *Keywords:* BicriterionAnt.
- [1601] S. Irnich. **A Unified Modeling and Solution Framework for Vehicle Routing and Local Search-Based Metaheuristics.** *INFORMS Journal on Computing*, 20(2):270–287, 2008.
- [1602] E. Irurozki and M. López-Ibáñez. **Unbalanced Mallows Models for Optimizing Expensive Black-Box Permutation Problems.** In F. Chicano and K. Krawiec, editors, *GECCO*, pages 225–233. ACM Press, New York, NY, 2021. doi:10.1145/3449639.3459366. Supplementary material: <https://doi.org/10.5281/zenodo.4500974>. *Keywords:* UMM, Permutation, Expensive, Black-box.
- [1603] E. Irurozki, B. Calvo, and J. A. Lozano. **Sampling and Learning Mallows and Generalized Mallows Models Under the Cayley Distance.** *Methodology and Computing in Applied Probability*, 20(1):1–35, June 2016. doi:10.1007/s11009-016-9506-7.
- [1604] E. Irurozki, B. Calvo, and J. A. Lozano. **PerMallows: An R Package for Mallows and Generalized Mallows Models.** *Journal of Statistical Software*, 71, 2019. ISSN 15487660. doi:10.18637/jss.v071.i12. *Keywords:* Cayley, Generalized Mallows, Hamming, Kendall’s τ , Learning, Mallows, Permutation, R, Ranking, Sampling, Ulam.
- [1605] E. Irurozki, J. Lobo, A. Perez, and J. Del Ser. **Rank aggregation for non-stationary data streams.** *Arxiv preprint arXiv:1910.08795 [stat.ML]*, 2020. URL <https://arxiv.org/abs/1910.08795>. *Keywords:* uborda.
- [1606] H. Ishibuchi and T. Murata. **A multi-objective genetic local search algorithm and its application to flowshop scheduling.** *IEEE Transactions on Systems, Man, and Cybernetics – Part C*, 28(3):392–403, 1998.
- [1607] H. Ishibuchi, S. Misaki, and H. Tanaka. **Modified simulated annealing algorithms for the flow shop sequencing problem.** *Eur. J. Oper. Res.*, 81(2):388–398, 1995.
- [1608] H. Ishibuchi, N. Tsukamoto, and Y. Nojima. **Evolutionary many-objective optimization: A short review.** In *Proceedings of the 2008 Congress on Evolutionary Computation (CEC 2008)*, pages 2419–2426, Piscataway, NJ, 2008. IEEE Press. doi:10.1109/CEC.2008.4631121.
- [1609] H. Ishibuchi, N. Tsukamoto, Y. Sakane, and Y. Nojima. **Hypervolume approximation using achievement scalarizing functions for evolutionary many-objective optimization.** In *IEEE CEC*, pages 530–537, Piscataway, NJ, 2009. IEEE Press. doi:10.1109/CEC.2009.4982991.

- [1610] H. Ishibuchi, N. Akedo, and Y. Nojima. **Behavior of Multiobjective Evolutionary Algorithms on Many-Objective Knapsack Problems.** *IEEE Trans. Evol. Comput.*, 19(2):264–283, 2015. doi:10.1109/TEVC.2014.2315442.
- [1611] H. Ishibuchi, H. Masuda, and Y. Nojima. **A Study on Performance Evaluation Ability of a Modified Inverted Generational Distance Indicator.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 695–702. ACM Press, New York, NY, 2015.
- [1612] H. Ishibuchi, H. Masuda, Y. Tanigaki, and Y. Nojima. **Modified Distance Calculation in Generational Distance and Inverted Generational Distance.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 110–125. Springer, Heidelberg, Germany, 2015.
Keywords: Performance metrics, multi-objective, IGD, IGD+.
Annotation: Proposed IGD+.
- [1613] H. Ishibuchi, Y. Setoguchi, H. Masuda, and Y. Nojima. **Performance of decomposition-based many-objective algorithms strongly depends on Pareto front shapes.** *IEEE Trans. Evol. Comput.*, 21(2):169–190, 2017.
- [1614] H. Ishibuchi, R. Imada, Y. Setoguchi, and Y. Nojima. **How to specify a reference point in hypervolume calculation for fair performance comparison.** *Evol. Comput.*, 26(3):411–440, 2018.
- [1615] H. Ishibuchi, L. He, and K. Shang. **Regular Pareto Front Shape is not Realistic.** In *Proceedings of the 2019 Congress on Evolutionary Computation (CEC 2019)*, pages 2034–2041, Piscataway, NJ, June 2019. IEEE Press. doi:10.1109/cec.2019.8790342.
Keywords: inverted simplex, inverted triangular.
- [1616] H. Ishibuchi, L. M. Pang, and K. Shang. **A new framework of evolutionary multi-objective algorithms with an unbounded external archive.** In G. D. Giacomo, A. Catala, B. Dilkina, M. Milano, S. Barro, A. Bugarín, and J. Lang, editors, *Proceedings of the 24th European Conference on Artificial Intelligence (ECAI)*, volume 325 of *Frontiers in Artificial Intelligence and Applications*. IOS Press, 2020.
- [1617] H. Ishibuchi et al., editors. *Proceedings of the 2010 Congress on Evolutionary Computation (CEC 2010)*, Piscataway, NJ, 2010. IEEE Press.
- [1618] K. Ito, F. Harashima, and K. Tanie, editors. *1999 IEEE International Conference on Systems, Man, and Cybernetics, October 12–15, 1999, Tokyo, Japan*, 1999. IEEE Press.
- [1619] P. Ivie and D. Thain. **Reproducibility in Scientific Computing.** *ACM Comput. Surv.*, 51(3):1–36, 2019. doi:10.1145/3186266.
- [1620] S. K. Iyer and B. Saxena. **Improved genetic algorithm for the permutation flowshop scheduling problem.** *Comput. Oper. Res.*, 31(4):593–606, 2004. doi:10.1016/S0305-0548(03)00016-9.
- [1621] D. Izzo. **Revisiting Lambert’s Problem.** *Celestial Mechanics and Dynamical Astronomy*, 121:1–15, 2015.
- [1622] D. Izzo, V. M. Becerra, D. R. Myatt, S. J. Nasuto, and J. M. Bishop. **Search space pruning and global optimisation of multiple gravity assist spacecraft trajectories.** *J. Glob. Optim.*, 38:283–296, 2007. doi:10.1007/s10898-006-9106-0.
- [1623] D. Izzo, L. F. Simões, M. Mörtens, G. C. de Croon, A. Heritier, and C. H. Yam. **Search for a Grand Tour of the Jupiter Galilean Moons.** In C. Blum and E. Alba, editors, *GECCO*, pages 1301–1308. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8. doi:10.1145/2463372.2463524.

- [1624] D. Izzo, I. Getzner, D. Hennes, and L. F. Simões. **Evolving solutions to TSP variants for active space debris removal.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 1207–1214. ACM Press, New York, NY, 2015.
- [1625] C. H. Jackson. **Multi-State Models for Panel Data: The msm Package for R.** *Journal of Statistical Software*, 38(8):1–29, 2011. URL <http://www.jstatsoft.org/v38/i08/>.
- [1626] R. H. F. Jackson, P. T. Boggs, S. G. Nash, and S. Powell. **Guidelines for Reporting Results of Computational Experiments. Report of the Ad Hoc Committee.** *Mathematical Programming*, 49(3):413–425, 1991.
- [1627] W. G. Jackson, E. Özcan, and R. I. John. **Move acceptance in local search metaheuristics for cross-domain search.** *Expert Systems with Applications*, 109:131–151, 2018.
- [1628] L. W. Jacobs and M. J. Brusco. **A Local Search Heuristic for Large Set-Covering Problems.** *Naval Research Logistics*, 42(7):1129–1140, 1995.
- [1629] K. E. Jacowitz and D. Kahneman. **Measures of Anchoring in Estimation Tasks.** *Personality and Social Psychology Bulletin*, 21(11):1161–1166, Nov. 1995. ISSN 0146-1672. doi:10.1177/01461672952111004.
- [1630] S. Jacquin, L. Jourdan, and E.-G. Talbi. **Dynamic Programming Based Metaheuristic for Energy Planning Problems.** In A. I. Esparcia-Alcázar and A. M. Mora, editors, *Applications of Evolutionary Computation*, volume 8602 of *LNCS*, pages 165–176. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-662-45523-4_14.
Keywords: irace.
- [1631] D. M. Jaeggi, G. T. Parks, T. Kipouros, and P. J. Clarkson. **The development of a multi-objective Tabu Search algorithm for continuous optimisation problems.** *Eur. J. Oper. Res.*, 185(3):1192–1212, 2008.
- [1632] S. Jain, R. Munos, F. Stephan, and T. Zeugmann, editors. *Algorithmic Learning Theory - 24th International Conference, ALT 2013, Singapore, October 6-9, 2013. Proceedings*, volume 8139 of *LNCS*. Springer, Berlin, Germany, 2013. doi:10.1007/978-3-642-40935-6.
- [1633] S. Jajodia, I. Minis, G. Harhalakis, and J.-M. Proth. **CLASS: computerized layout solutions using simulated annealing.** *Int. J. Prod. Res.*, 30(1):95–108, 1992.
- [1634] K. G. Jamieson and A. Talwalkar. **Non-stochastic Best Arm Identification and Hyperparameter Optimization.** In A. Gretton and C. C. Robert, editors, *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, volume 51 of *Proceedings of Machine Learning Research*, pages 240–248. PMLR, 2016. URL <http://proceedings.mlr.press/v51/jamieson16.html>.
Annotation: Proposed successive halving.
- [1635] A. Jaskiewicz. **Genetic local search for multi-objective combinatorial optimization.** *Eur. J. Oper. Res.*, 137(1):50–71, 2002.
- [1636] A. Jaskiewicz. **On the performance of multiple-objective genetic local search on the 0/1 knapsack problem – A comparative experiment.** *IEEE Trans. Evol. Comput.*, 6(4):402–412, 2002.
- [1637] A. Jaskiewicz. **Many-Objective Pareto Local Search.** *Eur. J. Oper. Res.*, 271(3):1001–1013, 2018. doi:10.1016/j.ejor.2018.06.009.
Annotation: Shows how to efficiently find the best solution according to Chebyshev scalarisation using an ND-tree archive.

- [1638] A. Jaszekiewicz and J. Branke. **Interactive Multiobjective Evolutionary Algorithms.** In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*, pages 179–193. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-88908-3_7.
- [1639] A. Jaszekiewicz and T. Lust. **ND-Tree-Based Update: A Fast Algorithm for the Dynamic Nondominance Problem.** *IEEE Trans. Evol. Comput.*, 22(5):778–791, 2018. doi:10.1109/TEVC.2018.2799684.
- [1640] A. Jaszekiewicz and P. Zielniewicz. **Improving the Efficiency of the Distance-Based Hypervolume Estimation Using ND-Tree.** *IEEE Trans. Evol. Comput.*, 29(3):726–733, 2025. doi:10.1109/TEVC.2024.3391857.
Keywords: Data structures, Chebycheff functions.
- [1641] A. Jaszekiewicz, H. Ishibuchi, and Q. Zhang. **Multiobjective memetic algorithms.** In F. Neri, C. Cotta, and P. Moscato, editors, *Handbook of Memetic Algorithms*, volume 379 of *Studies in Computational Intelligence*, pages 201–217. Springer, 2011.
- [1642] M. T. Jensen. **Reducing the run-time complexity of multiobjective EAs: The NSGA-II and other algorithms.** *IEEE Trans. Evol. Comput.*, 7(5):503–515, 2003. doi:10.1109/TEVC.2003.817234.
- [1643] M. T. Jensen. **Helper-Objectives: Using Multi-Objective Evolutionary Algorithms for Single-Objective Optimisation.** *Journal of Mathematical Modelling and Algorithms*, 3(4): 323–347, 2004.
Keywords: multi-objectivization.
- [1644] M. Jerrum. **Large cliques elude the Metropolis process.** *Random Structures & Algorithms*, 3(4):347–359, 1992.
- [1645] M. Jerrum and A. Sinclair. **The Markov chain Monte Carlo method: an approach to approximate counting and integration.** In D. S. Hochbaum, editor, *Approximation Algorithms For NP-hard Problems*, pages 482–520. PWS Publishing Co., 1996.
- [1646] M. Jerrum and G. Sorkin. **The Metropolis algorithm for graph bisection.** *Discrete Applied Mathematics*, 82(1):155–175, 1998.
- [1647] A. D. Jesus, A. Liefvooghe, B. Derbel, and L. Paquete. **Algorithm Selection of Anytime Algorithms.** In C. A. Coello Coello, editor, *GECCO*, pages 850–858. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.
- [1648] A. Jhingran et al., editors. *ACM Conference on Electronic Commerce (EC-00)*. ACM Press, New York, NY, 2000.
- [1649] S. Jiang, Y. S. Ong, J. Zhang, and L. Feng. **Consistencies and Contradictions of Performance Metrics in Multiobjective Optimization.** *IEEE Trans. Cybern.*, 44(12): 2391–2404, 2014.
- [1650] S. Jiang, J. Zou, S. Yang, and X. Yao. **Evolutionary Dynamic Multi-Objective Optimisation: A Survey.** *ACM Comput. Surv.*, 55(4), Nov. 2022. doi:10.1145/3524495.
Keywords: evolutionary algorithm, evolutionary dynamic multi-objective optimisation, dynamic environment, Multi-objective optimisation.
- [1651] J. L. Jiménez Laredo, S. Silva, and A. I. Esparcia-Alcázar, editors. *Genetic and Evolutionary Computation Conference, GECCO 2015, Madrid, Spain, July 11-15, 2015, Companion Material Proceedings*. ACM Press, New York, NY, 2015.

- [1652] J. L. Jiménez Laredo et al., editors. *Applications of Evolutionary Computation – 25th European Conference, EvoApplications 2022, Held as Part of EvoStar 2022, Madrid, Spain, April 20-22, 2022, Proceedings*, volume 13224 of *LNCS*. Springer Nature, Switzerland, 2022.
- [1653] H. Jin and M.-L. Wong. **Adaptive, convergent, and diversified archiving strategy for multiobjective evolutionary algorithms.** *Expert Systems with Applications*, 37(12): 8462–8470, 2010.
- [1654] Y. Jin. **A Comprehensive Survey of Fitness Approximation in Evolutionary Computation.** *Soft Computing*, 9(1):3–12, 2005.
- [1655] Y. Jin. **Surrogate-Assisted Evolutionary Computation: Recent Advances and Future Challenges.** *Swarm and Evolutionary Computation*, 1(2):61–70, June 2011. ISSN 2210-6502. doi:10.1016/j.swevo.2011.05.001.
Keywords: Evolutionary computation, Expensive optimization problems, Machine learning, Meta-models, Model management, Surrogates.
- [1656] Y. Jin and J. Branke. **Evolutionary Optimization in Uncertain Environments—A Survey.** *IEEE Trans. Evol. Comput.*, 9(5):303–317, 2005.
- [1657] Y. Jin, H. Wang, T. Chugh, D. Guo, and K. Miettinen. **Data-Driven Evolutionary Optimization: An Overview and Case Studies.** *IEEE Trans. Evol. Comput.*, 23(3):442–458, June 2019. doi:10.1109/tevc.2018.2869001.
- [1658] A. W. Johnson and S. H. Jacobson. **On the Convergence of Generalized Hill Climbing Algorithms.** *Discrete Applied Mathematics*, 119(1):37–57, 2002.
- [1659] D. S. Johnson. **Optimal Two- and Three-stage Production Scheduling with Setup Times Included.** *Naval Research Logistics Quarterly*, 1:61–68, 1954.
- [1660] D. S. Johnson. **Local Optimization and the Traveling Salesman Problem.** In M. Paterson, editor, *Automata, Languages and Programming, 17th International Colloquium*, volume 443 of *LNCS*, pages 446–461, Heidelberg, Germany, 1990. Springer.
- [1661] D. S. Johnson. **A Theoretician’s Guide to the Experimental Analysis of Algorithms.** In M. H. Goldwasser, D. S. Johnson, and C. C. McGeoch, editors, *Data Structures, Near Neighbor Searches, and Methodology: Fifth and Sixth DIMACS Implementation Challenges*, volume 59 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 215–250. American Mathematical Society, Providence, RI, 2002. doi:10.1090/dimacs/059/11.
- [1662] D. S. Johnson and L. A. McGeoch. **The Traveling Salesman Problem: A Case Study in Local Optimization.** In E. H. L. Aarts and J. K. Lenstra, editors, *Local Search in Combinatorial Optimization*, pages 215–310. John Wiley & Sons, Chichester, UK, 1997.
- [1663] D. S. Johnson and L. A. McGeoch. **Experimental Analysis of Heuristics for the STSP.** In G. Gutin and A. Punnen, editors, *The Traveling Salesman Problem and its Variations*, pages 369–443. Kluwer Academic Publishers, Dordrecht, The Netherlands, 2002.
- [1664] D. S. Johnson and M. A. Trick, editors. *Cliques, Coloring, and Satisfiability: Second DIMACS Implementation Challenge*, volume 26 of *DIMACS Series on Discrete Mathematics and Theoretical Computer Science*. American Mathematical Society, Providence, RI, 1996.
- [1665] D. S. Johnson, C. H. Papadimitriou, and M. Yannakakis. **How Easy is Local Search?** *J. Comput. Syst. Sci.*, 37(1):79–100, 1988.
- [1666] D. S. Johnson, C. R. Aragon, L. A. McGeoch, and C. Schevon. **Optimization by Simulated Annealing: An Experimental Evaluation: Part I, Graph Partitioning.** *Operations Research*, 37(6):865–892, 1989. doi:10.1287/opre.39.3.378.

- [1667] D. S. Johnson, C. R. Aragon, L. A. McGeoch, and C. Schevon. **Optimization by Simulated Annealing: An Experimental Evaluation: Part II, Graph Coloring and Number Partitioning.** *Operations Research*, 39(3):378–406, 1991.
- [1668] D. S. Johnson, L. A. McGeoch, C. Rego, and F. Glover. **8th DIMACS Implementation Challenge: The Traveling Salesman Problem.** <http://dimacs.rutgers.edu/archive/Challenges/TSP>, 2001.
Keywords: TSP Challenge, RUE, RCE, generators.
- [1669] D. S. Johnson, G. Gutin, L. A. McGeoch, A. Yeo, W. Zhang, and A. Zverovitch. **Experimental Analysis of Heuristics for the ATSP.** In G. Gutin and A. Punnen, editors, *The Traveling Salesman Problem and its Variations*, pages 445–487. Kluwer Academic Publishers, Dordrecht, The Netherlands, 2002.
- [1670] M. E. Johnson, L. M. Moore, and D. Ylvisaker. **Minimax and maximin distance designs.** *Journal of Statistical Planning and Inference*, 26(2):131–148, 1990.
Keywords: Bayesian design.
- [1671] C. Joncour, J. Kritter, S. Michel, and X. Schepler. **Generalized Relax-and-Fix Heuristic.** *Comput. Oper. Res.*, 149:106038, 2023.
- [1672] D. R. Jones, M. Schonlau, and W. J. Welch. **Efficient Global Optimization of Expensive Black-Box Functions.** *J. Glob. Optim.*, 13(4):455–492, 1998. doi:10.1023/A:1008306431147.
Keywords: EGO, bayesian optimization.
Annotation: Proposed EGO algorithm.
- [1673] N. C. Jones and P. A. Pevzner. *An introduction to bioinformatics algorithms.* MIT Press, Cambridge, MA, 2004.
- [1674] T. Jones and S. Forrest. **Fitness Distance Correlation as a Measure of Problem Difficulty for Genetic Algorithms.** In L. J. Eshelman, editor, *Proceedings of the Sixth International Conference on Genetic Algorithms (ICGA’95)*, pages 184–192. Morgan Kaufmann Publishers, San Francisco, CA, Pittsburgh, PA, 1995.
- [1675] J. Jookan, P. Leyman, and P. De Causmaecker. **A new class of hard problem instances for the 0–1 knapsack problem.** *Eur. J. Oper. Res.*, 301(3):841–854, 2022.
- [1676] J. Jookan, P. Leyman, T. Wauters, and P. De Causmaecker. **Exploring search space trees using an adapted version of Monte Carlo tree search for combinatorial optimization problems.** *Comput. Oper. Res.*, 150:106070, 2023. doi:10.1016/j.cor.2022.106070.
- [1677] D. E. Joslin and D. P. Clements. **Squeaky Wheel Optimization.** *J. Artif. Intell. Res.*, 10: 353–373, 1999.
- [1678] Journal of Heuristics. **Journal of Heuristics. Policies on Heuristic Search Research.** <http://www.springer.com/journal/10732>, 2015. Version visited last on June 10, 2015.
- [1679] P. W. Jowitt and G. Germanopoulos. **Optimal pump scheduling in water supply networks.** *Journal of Water Resources Planning and Management, ASCE*, 118(4):406–422, 1992.
- [1680] A. A. Juan, H. R. Lourenço, M. Mateo, R. Luo, and Q. Castellà. **Using Iterated Local Search for Solving the Flow-shop Problem: Parallelization, Parametrization, and Randomization Issues.** *International Transactions in Operational Research*, 21(1):103–126, 2014.

- [1681] A. A. Juan, J. Faulin, S. E. Grasman, M. Rabe, and G. Figueira. **A review of simheuristics: Extending metaheuristics to deal with stochastic combinatorial optimization problems.** *Oper. Res. Perspect.*, 2:62–72, 2015. doi:[10.1016/j.orp.2015.03.001](https://doi.org/10.1016/j.orp.2015.03.001).
Keywords: Metaheuristics; Simulation; Combinatorial optimization; Stochastic problems.
- [1682] H. Juillé and J. B. Pollack. **A Sampling-Based Heuristic for Tree Search Applied to Grammar Induction.** In J. Mostow and C. Rich, editors, *Proc. of AAAI 1998 – Fifteenth National Conference on Artificial Intelligence*, pages 776–783. AAAI Press/MIT Press, Menlo Park, CA, 1998.
- [1683] B. A. Julstrom. **What Have You Done for Me Lately? Adapting Operator Probabilities in a Steady-State Genetic Algorithm.** In L. J. Eshelman, editor, *Proceedings of the Sixth International Conference on Genetic Algorithms (ICGA’95)*, pages 81–87. Morgan Kaufmann Publishers, San Francisco, CA, Pittsburgh, PA, 1995.
- [1684] M. Jünger, G. Reinelt, and S. Thienel. **Provably Good Solutions for the Traveling Salesman Problem.** *Zeitschrift für Operations Research*, 40(2):183–217, 1994.
- [1685] D. Jurafsky and E. Gaussier, editors. *Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing, EMNLP2006*, Empirical Methods in Natural Language Processing, 2006.
- [1686] E. A. Kabova, J. C. Cole, O. Korb, M. López-Ibáñez, A. C. Williams, and K. Shankland. **Improved performance of crystal structure solution from powder diffraction data through parameter tuning of a simulated annealing algorithm.** *J. Appl. Cryst.*, 50(5):1411–1420, Oct. 2017. doi:[10.1107/S1600576717012602](https://doi.org/10.1107/S1600576717012602).
Keywords: crystal structure determination, powder diffraction, simulated annealing, parameter tuning, irace.
- [1687] J. Kacprzyk and W. Pedrycz, editors. *Springer Handbook of Computational Intelligence*. Springer, Berlin/Heidelberg, 2015.
- [1688] S. Kadioglu, Y. Malitsky, M. Sellmann, and K. Tierney. **ISAC: Instance-Specific Algorithm Configuration.** In H. Coelho, R. Studer, and M. Wooldridge, editors, *Proceedings of the 19th European Conference on Artificial Intelligence*, pages 751–756. IOS Press, 2010.
- [1689] D. Kahneman. **Maps of bounded rationality: Psychology for behavioral economics.** *The American Economic Review*, 93(5):1449–1475, 2003.
- [1690] D. Kahneman and A. Tversky. **Prospect theory: An analysis of decision under risk.** *Econometrica*, 47(2):263–291, 1979. doi:[10.2307/1914185](https://doi.org/10.2307/1914185).
- [1691] H. Kaji, K. Ikeda, and H. Kita. **Avoidance of constraint violation for experiment-based evolutionary multi-objective optimization.** In *IEEE CEC*, pages 2756–2763, Piscataway, NJ, 2009. IEEE Press. doi:[10.1109/CEC.2009.4983288](https://doi.org/10.1109/CEC.2009.4983288).
Keywords: Safe Optimization, evolutionary computation, constraint violation, experiment-based evolutionary multiobjective optimization, evolutionary algorithm, risky-constraint violation, Constraint optimization, Diesel engines, Calibration, Evolutionary computation, Electric breakdown, Optimization methods, Uncertainty, Computational fluid dynamics.
- [1692] J. Kallestad, R. Hasibi, A. Hemmati, and K. Sörensen. **A general deep reinforcement learning hyperheuristic framework for solving combinatorial optimization problems.** *Eur. J. Oper. Res.*, 309(1):446–468, Aug. 2023. doi:[10.1016/j.ejor.2023.01.017](https://doi.org/10.1016/j.ejor.2023.01.017).
Keywords: Deep RL, hyper-heuristic, ALNS.
- [1693] J. Kallrath, editor. *Modeling Languages in Mathematical Optimization*, volume 88 of *Applied Optimization*. Kluwer Academic Publishers, 2004.

- [1694] Q. Kang, H. He, and J. Wei. **An Effective Iterated Greedy Algorithm for Reliability-oriented Task Allocation in Distributed Computing Systems.** *Journal of Parallel and Distributed Computing*, 73(8):1106–1115, 2013.
- [1695] D. Karaboga and B. Akay. **A Survey: Algorithms Simulating Bee Swarm Intelligence.** *Artificial Intelligence Review*, 31(1–4):61–85, 2009.
- [1696] K. Karabulut. **A hybrid iterated greedy algorithm for total tardiness minimization in permutation flowshops.** *Computers & Industrial Engineering*, 98(Supplement C):300 – 307, 2016.
- [1697] K. Karabulut and F. M. Tasgetiren. **A Variable Iterated Greedy Algorithm for the Traveling Salesman Problem with Time Windows.** *Inf. Sci.*, 279:383–395, 2014.
- [1698] G. Karafotias, S. K. Smit, and A. E. Eiben. **A generic approach to parameter control.** In C. Di Chio et al., editors, *Applications of Evolutionary Computation*, volume 7248 of *LNCS*, pages 366–375. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-29178-4_37.
- [1699] G. Karafotias, A. E. Eiben, and M. Hoogendoorn. **Generic parameter control with reinforcement learning.** In C. Igel and D. V. Arnold, editors, *GECCO*, pages 1319–1326. ACM Press, New York, NY, 2014.
- [1700] G. Karafotias, M. Hoogendoorn, and A. E. Eiben. **Parameter Control in Evolutionary Algorithms: Trends and Challenges.** *IEEE Trans. Evol. Comput.*, 19(2):167–187, 2015.
- [1701] G. Karafotias, M. Hoogendoorn, and A. E. Eiben. **Evaluating reward definitions for parameter control.** In A. M. Mora and G. Squillero, editors, *Applications of Evolutionary Computation*, volume 9028 of *LNCS*, pages 667–680. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-16549-3_54.
- [1702] İ. Karahan and M. Köksalan. **A territory defining multiobjective evolutionary algorithms and preference incorporation.** *IEEE Trans. Evol. Comput.*, 14(4):636–664, 2010. doi:10.1109/TEVC.2009.2033586.
Keywords: TDEA.
- [1703] D. Karapetyan, A. J. Parkes, and T. Stützle. **Algorithm Configuration: Learning policies for the quick termination of poor performers.** In R. Battiti, M. Brunato, I. Kotsireas, and P. M. Pardalos, editors, *LION 12*, volume 11353 of *LNCS*, pages 220–224. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-030-05348-2_20.
- [1704] M. Karimi-Mamaghan, M. Mohammadi, P. Meyer, A. M. Karimi-Mamaghan, and E.-G. Talbi. **Machine learning at the service of meta-heuristics for solving combinatorial optimization problems: A state-of-the-art.** *Eur. J. Oper. Res.*, 296(2):393–422, 2022. doi:10.1016/j.ejor.2021.04.032.
Keywords: Meta-heuristics, Machine learning, Combinatorial optimization problems, State-of-the-art.
- [1705] N. Karmarkar. **A new polynomial-time algorithm for linear programming.** In R. A. DeMillo, editor, *Proceedings of the sixteenth annual ACM Symposium on Theory of Computing*, pages 302–311. ACM Press, 1984.
- [1706] Z. Karnin, T. Koren, and O. Somekh. **Almost optimal exploration in multi-armed bandits.** In S. Dasgupta and D. McAllester, editors, *Proceedings of the 30th International Conference on Machine Learning, ICML 2013*, volume 28, pages 1238–1246, 2013.
Annotation: Sequential Halving, Successive Halving.
- [1707] R. M. Karp. **Reducibility among combinatorial problems.** In R. E. Miller and W. Thatcher, James, editors, *Proc. of a symposium on the Complexity of Computer Computations, held March 20-22, 1972, at the IBM Thomas J. Watson Research Center, Yorktown Heights, New York, USA*, The IBM Research Symposia Series, pages 85–103. Springer, 1972.

- [1708] E. Karpas, S. Jiménez Celorrio, and S. Kambhampati, editors. *Proceedings of the 3rd Workshop on Learning and Planning, collocated with the 21st International Conference on Automated Planning and Scheduling (ICAPS-PAL'11)*, 2011.
- [1709] O. Karpenko, J. Shi, and Y. Dai. **Prediction of MHC class II binders using the ant colony search strategy**. *Artificial Intelligence in Medicine*, 35(1):147–156, 2005.
- [1710] J. R. Kasprzyk, P. M. Reed, G. W. Characklis, and B. R. Kirsch. **Many-objective de Novo water supply portfolio planning under deep uncertainty**. *Environmental Modelling & Software*, 34:87–104, 2012.
Keywords: scenario-based.
- [1711] J. R. Kasprzyk, S. Nataraj, P. M. Reed, and R. J. Lempert. **Many objective robust decision making for complex environmental systems undergoing change**. *Environmental Modelling & Software*, 42:55–71, 2013.
Keywords: scenario-based.
- [1712] K. Katayama and H. Narihisa. **Iterated Local Search Approach using Genetic Transformation to the Traveling Salesman Problem**. In W. Banzhaf, J. M. Daida, A. E. Eiben, M. H. Garzon, V. Honavar, M. J. Jakiela, and R. E. Smith, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 1999*, volume 1, pages 321–328. Morgan Kaufmann Publishers, San Francisco, CA, 1999.
- [1713] S. A. Kauffman. *The Origins of Order*. Oxford University Press, 1993.
- [1714] H. A. Kautz and B. W. Porter, editors. *Proceedings of the Seventeenth National Conference on Artificial Intelligence and Twelfth Conference on Innovative Applications of Artificial Intelligence, July 30 – August 3, 2000, Austin, Texas, USA*, 2000. AAAI Press/MIT Press, Menlo Park, CA.
- [1715] M. D. Kazantzis, A. R. Simpson, D. Kwong, and S. M. Tan. **A new methodology for optimizing the daily operations of a pumping plant**. In *Proceedings of 2002 Conference on Water Resources Planning*, Roanoke, USA, May 2002. ASCE.
- [1716] A. Kaznatcheev, D. A. Cohen, and P. Jeavons. **Representing Fitness Landscapes by Valued Constraints to Understand the Complexity of Local Search**. *J. Artif. Intell. Res.*, 69: 1077–1102, 2020. doi:[10.1613/jair.1.12156](https://doi.org/10.1613/jair.1.12156).
- [1717] L. Ke, C. Archetti, and Z. Feng. **Ants can solve the team orienteering problem**. *Computers & Industrial Engineering*, 54(3):648–665, 2008. doi:[10.1016/j.cie.2007.10.001](https://doi.org/10.1016/j.cie.2007.10.001).
Keywords: Ant colony optimization, Ant system, Heuristics, Team orienteering problem.
- [1718] L. Ke, Z. Feng, Z. Xu, K. Shang, and Y. Wang. **A multiobjective ACO algorithm for rough feature selection**. In *Circuits, Communications and System (PACCS), 2010 Second Pacific-Asia Conference on*, volume 1, pages 207–210, 2010.
- [1719] M. J. Kearns, R. P. McAfee, and É. Tardos, editors. *Proceedings of the fourteenth ACM Conference on Electronic Commerce, EC 2013, Philadelphia, PA, USA, June 16-20, 2013*. ACM Press, New York, NY, 2013. doi:[10.1145/2492002](https://doi.org/10.1145/2492002).
- [1720] E. Kee, S. Airey, and W. Cyre. **An adaptive genetic algorithm**. In E. D. Goodman, editor, *Proceedings of the 3rd Annual Conference on Genetic and Evolutionary Computation, GECCO 2001*, pages 391–397. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [1721] R. L. Keeney. **Analysis of preference dependencies among objectives**. *Operations Research*, 29:1105–1120, 1981.

- [1722] R. E. Keller and R. Poli. **Linear genetic programming of parsimonious metaheuristics.** In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 4508–4515, Piscataway, NJ, 2007. IEEE Press. doi:10.1109/CEC.2007.4425062.
- [1723] R. E. Keller and R. Poli. **Cost-Benefit Investigation of a Genetic-Programming Hyperheuristic.** In E. Lutton, P. Legrand, P. Parrend, N. Monmarché, and M. Schoenauer, editors, *EA 2017: Artificial Evolution*, volume 10764 of *LNCS*, pages 13–24. Springer, Heidelberg, Germany, 2017.
- [1724] H. Kellerer, U. Pferschy, and D. Pisinger. *Knapsack problems*. Springer, 2004.
- [1725] G. Kendall, G. Vanden Berghe, and B. McCollum, editors. *Multidisciplinary International Conference on Scheduling: Theory and Applications (MISTA 2013)*, Gent, Belgium, 2013.
- [1726] G. Kendall, R. Bai, J. Błazewicz, P. De Causmaecker, M. Gendreau, R. John, J. Li, B. McCollum, E. Pesch, R. Qu, N. Sabar, G. Vanden Berghe, and A. Yee. **Good Laboratory Practice for Optimization Research.** *J. Oper. Res. Soc.*, 67(4):676–689, 2016. doi:10.1057/jors.2015.77.
- [1727] M. G. Kendall. *Rank correlation methods*. Griffin, London, 1948.
- [1728] J. Kennedy and R. C. Eberhart. **Particle Swarm Optimization.** In *Proceedings of International Conference on Neural Networks (ICNN’95)*, pages 1942–1948, Piscataway, NJ, 1995. IEEE Press. doi:10.1109/ICNN.1995.488968.
Annotation: Proposed PSO.
- [1729] J. Kennedy and R. C. Eberhart. **A discrete binary version of the particle swarm algorithm.** In *Proceedings of the 1997 IEEE International Conference on Systems, Man, and Cybernetics*, pages 4104–4108, Piscataway, NJ, 1997. IEEE Press.
- [1730] J. Kennedy, R. C. Eberhart, and Y. Shi. *Swarm Intelligence*. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [1731] B. W. Kernighan and S. Lin. **An Efficient Heuristic Procedure for Partitioning Graphs.** *Bell Systems Technology Journal*, 49(2):213–219, 1970.
- [1732] N. L. Kerr. **HARKing: Hypothesizing After the Results are Known.** *Personality and Social Psychology Review*, 2(3):196–217, Aug. 1998. doi:10.1207/s15327957pspr0203_4.
- [1733] M. Kerrisk. **pthread - POSIX Threads.** In *Linux Programmer’s Manual*, Section 7. <https://man7.org/linux/man-pages/man7/pthreads.7.html>, 2021. (Last accessed Feb 22 2023).
- [1734] P. Kerschke and H. Trautmann. **The R-package FLACCO for exploratory landscape analysis with applications to multi-objective optimization problems.** In *Proceedings of the 2016 Congress on Evolutionary Computation (CEC 2016)*, pages 5262–5269, Piscataway, NJ, 2016. IEEE Press. ISBN 978-1-5090-0623-6. doi:10.1109/CEC.2016.7748359.
- [1735] P. Kerschke and H. Trautmann. **Automated Algorithm Selection on Continuous Black-Box Problems by Combining Exploratory Landscape Analysis and Machine Learning.** *Evol. Comput.*, 27(1):99–127, 2019. doi:10.1162/evco_a_00236.
- [1736] P. Kerschke, H. Wang, M. Preuss, C. Grimme, A. H. Deutz, H. Trautmann, and M. T. M. Emmerich. **Towards Analyzing Multimodality of Continuous Multiobjective Landscapes.** In J. Handl, E. Hart, P. R. Lewis, M. López-Ibáñez, G. Ochoa, and B. Paechter, editors, *PPSN*, volume 9921 of *LNCS*, pages 962–972. Springer, Heidelberg, Germany, 2016. ISBN 978-3-319-45822-9. doi:10.1007/978-3-319-45823-6_90.
- [1737] P. Kerschke, H. H. Hoos, F. Neumann, and H. Trautmann. **Automated Algorithm Selection: Survey and Perspectives.** *Evol. Comput.*, 27(1):3–45, Mar. 2019. doi:10.1162/evco_a_00242.

- [1738] P. Kerschke, H. Wang, M. Preuss, C. Grimme, A. H. Deutz, H. Trautmann, and M. T. M. Emmerich. **Search Dynamics on Multimodal Multiobjective Problems**. *Evol. Comput.*, 27(4):577–609, 2019. doi:[10.1162/evco_a_00234](https://doi.org/10.1162/evco_a_00234).
- [1739] V. Khare, X. Yao, and K. Deb. **Performance Scaling of Multi-objective Evolutionary Algorithms**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 376–390. Springer, Heidelberg, Germany, 2003.
- [1740] M. Khichane, P. Albert, and C. Solnon. **Integration of ACO in a Constraint Programming Language**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 6th International Conference, ANTS 2008*, volume 5217 of *LNCS*, pages 84–95. Springer, Heidelberg, Germany, 2008.
- [1741] M. Khichane, P. Albert, and C. Solnon. **An ACO-Based Reactive Framework for Ant Colony Optimization: First Experiments on Constraint Satisfaction Problems**. In T. Stützle, editor, *LION 3*, volume 5851 of *LNCS*, pages 119–133. Springer, Heidelberg, Germany, 2009. doi:[10.1007/978-3-642-11169-3_9](https://doi.org/10.1007/978-3-642-11169-3_9).
- [1742] A. R. KhudaBukhsh, L. Xu, H. H. Hoos, and K. Leyton-Brown. **SATenstein: Automatically Building Local Search SAT Solvers from Components**. In C. Boutilier, editor, *Proc. of IJCAI-09*, pages 517–524. AAAI Press, Menlo Park, CA, 2009.
- [1743] A. R. KhudaBukhsh, L. Xu, H. H. Hoos, and K. Leyton-Brown. **SATenstein: Automatically Building Local Search SAT Solvers from Components**. *Artif. Intell.*, 232:20–42, 2016. doi:[10.1016/j.artint.2015.11.002](https://doi.org/10.1016/j.artint.2015.11.002).
- [1744] P. Kilby and T. Urli. **Fleet design optimisation from historical data using constraint programming and large neighbourhood search**. *Constraints*, pages 1–20, 2015. doi:[10.1007/s10601-015-9203-0](https://doi.org/10.1007/s10601-015-9203-0).
Keywords: F-race.
- [1745] J. Kim, M. McCourt, T. You, S. Kim, and S. Choi. **Bayesian Optimization with Approximate Set Kernels**. *Mach. Learn.*, 2021. doi:[10.1007/s10994-021-05949-0](https://doi.org/10.1007/s10994-021-05949-0).
- [1746] J.-S. Kim, J.-H. Park, and D.-H. Lee. **Iterated Greedy Algorithms to Minimize the Total Family Flow Time for Job-shop Scheduling with Job Families and Sequence-dependent Set-ups**. *Engineering Optimization*, 49(10):1719–1732, 2017.
- [1747] M. Kim, J. Park, and J. Kim. **Learning Collaborative Policies to Solve NP-hard Routing Problems**. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. W. Vaughan, editors, *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
Keywords: Deep RL, TSP, prize collecting, PCTSP, CVRP, routing, attention model.
- [1748] W. Kim, R. Kohavi, J. Gehrke, and W. DuMouchel, editors. *KDD04: ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Seattle WA USA, August 22-25, 2004*. ACM Press, New York, NY, 2004.
- [1749] Y. Kim, R. Allmendinger, and M. López-Ibáñez. **Safe Learning and Optimization Techniques: Towards a Survey of the State of the Art**. *Arxiv preprint arXiv:2101.09505 [cs.LG]*, 2020. URL <https://arxiv.org/abs/2101.09505>.
- [1750] Y. Kim, R. Allmendinger, and M. López-Ibáñez. **Safe Learning and Optimization Techniques: Towards a Survey of the State of the Art**. In F. Heintz, M. Milano, and B. O’Sullivan, editors, *Trustworthy AI – Integrating Learning, Optimization and Reasoning. TAILOR 2020*, volume 12641 of *LNCS*, pages 123–139. Springer, Cham, Switzerland, 2021. doi:[10.1007/978-3-030-73959-1_12](https://doi.org/10.1007/978-3-030-73959-1_12).

- [1751] Y. Kim, R. Allmendinger, and M. López-Ibáñez. **Are Evolutionary Algorithms Safe Optimizers?** In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022*, pages 814–822. ACM Press, New York, NY, 2022. doi:[10.1145/3512290.3528818](https://doi.org/10.1145/3512290.3528818).
Keywords: Bayesian optimization, constrained optimization, benchmarking, safety constraints, safe optimization.
- [1752] Y.-D. Kim. **Heuristics for Flowshop Scheduling Problems Minimizing Mean Tardiness.** *J. Oper. Res. Soc.*, 44(1):19–28, 1993. doi:[10.1057/jors.1993.3](https://doi.org/10.1057/jors.1993.3).
- [1753] D. P. Kingma and J. Ba. **Adam: A method for stochastic optimization.** *Arxiv preprint arXiv:1412.6980 [cs.LG]*, 2014. URL <https://arxiv.org/abs/1412.6980>.
Annotation: Published as a conference paper at the 3rd International Conference for Learning Representations, San Diego, 2015 [1754].
- [1754] D. P. Kingma and J. Ba. **Adam: A Method for Stochastic Optimization.** In Y. Bengio and Y. LeCun, editors, *3rd International Conference on Learning Representations, ICLR 2015, Conference Track Proceedings*, 2015.
- [1755] E. Kipouridis, A. Kosmatopoulos, A. N. Papadopoulos, and K. Tsihlias. **Dynamic layers of maxima with applications to dominating queries.** *Computational Geometry*, 93:101699, Feb. 2021. doi:[10.1016/j.comgeo.2020.101699](https://doi.org/10.1016/j.comgeo.2020.101699).
- [1756] S. Kirkpatrick. **Optimization by Simulated Annealing: Quantitative Studies.** *Journal of Statistical Physics*, 34(5-6):975–986, 1984.
- [1757] S. Kirkpatrick and G. Toulouse. **Configuration Space Analysis of Travelling Salesman Problems.** *Journal de Physique*, 46(8):1277–1292, 1985.
- [1758] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. **Optimization by Simulated Annealing.** *Science*, 220(4598):671–680, 1983. doi:[10.1126/science.220.4598.671](https://doi.org/10.1126/science.220.4598.671).
Annotation: Proposed Simulated Annealing.
- [1759] J. M. Kivikangas, E. Vilkkumaa, J. Blank, V. Harjunen, P. Malo, K. Deb, N. J. Ravaja, and J. Wallenius. **Effects of many conflicting objectives on decision-makers’ cognitive burden and decision consistency.** *Eur. J. Oper. Res.*, 322(1):182–197, 2025. ISSN 0377-2217. doi:[10.1016/j.ejor.2024.10.039](https://doi.org/10.1016/j.ejor.2024.10.039).
Keywords: Multiple criteria analysis, Multiple Criteria Decision Making, Evolutionary Multi-Objective Optimization, Cognitive burden, Psychophysiological measurements.
- [1760] K. Klamroth, S. Mostaghim, B. Naujoks, S. Poles, R. C. Purshouse, G. Rudolph, S. Ruzika, S. Sayın, M. M. Wiecek, and X. Yao. **Multiobjective optimization for interwoven systems.** *Journal of Multi-Criteria Decision Analysis*, 24(1-2):71–81, 2017. doi:[10.1002/mcda.1598](https://doi.org/10.1002/mcda.1598).
- [1761] K. Klamroth, J. D. Knowles, G. Rudolph, and M. M. Wiecek, editors. *Personalized Multiobjective Optimization: An Analytics Perspective (Dagstuhl Seminar 18031)*, volume 8(1) of *Dagstuhl Reports*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2018. doi:[10.4230/DagRep.8.1.33](https://doi.org/10.4230/DagRep.8.1.33).
Keywords: multiple criteria decision making, evolutionary multiobjective optimization.
- [1762] A. J. Kleywegt, A. Shapiro, and T. Homem-de-Mello. **The Sample Average Approximation Method for Stochastic Discrete Optimization.** *SIAM J. Optim.*, 12(2):479–502, 2002.
- [1763] M. J. Knol, T. J. VanderWeele, R. H. H. Groenwold, O. H. Klungel, M. M. Rovers, and D. E. Grobbee. **Estimating measures of interaction on an additive scale for preventive exposures.** *European Journal of Epidemiology*, 26(6):433–438, 2011.

- [1764] J. D. Knowles. *Local-Search and Hybrid Evolutionary Algorithms for Pareto Optimization*. PhD thesis, University of Reading, UK, 2002.
Annotation: (Examiners: Prof. K. Deb and Prof. K. Warwick).
- [1765] J. D. Knowles. **A summary-attainment-surface plotting method for visualizing the performance of stochastic multiobjective optimizers**. In A. Abraham and M. Paprzycki, editors, *Proceedings of the 5th International Conference on Intelligent Systems Design and Applications*, pages 552–557, 2005. doi:10.1109/ISDA.2005.15. Supplementary material: https://www.cs.bham.ac.uk/~jdk/plot_attainments/.
- [1766] J. D. Knowles. **ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems**. *IEEE Trans. Evol. Comput.*, 10(1): 50–66, 2006. doi:10.1109/TEVC.2005.851274.
Keywords: ParEGO, online, metamodel.
- [1767] J. D. Knowles. **Closed-loop evolutionary multiobjective optimization**. *IEEE Computational Intelligence Magazine*, 4:77–91, 2009. doi:10.1109/MCI.2009.933095.
Keywords: resource-constraints.
- [1768] J. D. Knowles and D. Corne. **The Pareto Archived Evolution Strategy: A New Baseline Algorithm for Multiobjective Optimisation**. In *Proceedings of the 1999 Congress on Evolutionary Computation (CEC 1999)*, pages 98–105, Piscataway, NJ, 1999. IEEE Press.
Annotation: first mention of Adaptive Grid Archiving.
- [1769] J. D. Knowles and D. Corne. **Approximating the Nondominated Front Using the Pareto Archived Evolution Strategy**. *Evol. Comput.*, 8(2):149–172, 2000. doi:10.1162/106365600568167.
Annotation: Proposed PAES.
- [1770] J. D. Knowles and D. Corne. **M-PAES: A memetic algorithm for multiobjective optimization**. In *IEEE CEC*, pages 325–332, Piscataway, NJ, July 2000. IEEE Press.
- [1771] J. D. Knowles and D. Corne. **On Metrics for Comparing Non-Dominated Sets**. In *IEEE CEC*, pages 711–716, Piscataway, NJ, 2002. IEEE Press.
- [1772] J. D. Knowles and D. Corne. **Instance Generators and Test Suites for the Multiobjective Quadratic Assignment Problem**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 295–310. Springer, Heidelberg, Germany, 2003.
- [1773] J. D. Knowles and D. Corne. **Properties of an Adaptive Archiving Algorithm for Storing Nondominated Vectors**. *IEEE Trans. Evol. Comput.*, 7(2):100–116, Apr. 2003.
Keywords: S-metric, hypervolume.
Annotation: Proposed to use S-metric (hypervolume metric) for environmental selection.
- [1774] J. D. Knowles and D. Corne. **Bounded Pareto Archiving: Theory and Practice**. In X. Gandibleux, M. Sevaux, K. Sörensen, and V. T’Kindt, editors, *Metaheuristics for Multiobjective Optimisation*, volume 535 of *LNEMS*, pages 39–64. Springer, Berlin/Heidelberg, 2004. doi:10.1007/978-3-642-17144-4_2.
- [1775] J. D. Knowles and D. Corne. **Memetic algorithms for multiobjective optimization: issues, methods and prospects**. In H. W. E., S. J. E., and K. N., editors, *Recent Advances in Memetic Algorithms*, volume 166 of *Studies in Fuzziness and Soft Computing*, pages 313–352. Springer, Berlin/Heidelberg, 2005. doi:10.1007/3-540-32363-5_14.
- [1776] J. D. Knowles and D. Corne. **Quantifying the Effects of Objective Space Dimension in Evolutionary Multiobjective Optimization**. In S. Obayashi et al., editors, *EMO*, volume 4403 of *LNCS*, pages 757–771. Springer, Heidelberg, Germany, 2007.

- [1777] J. D. Knowles, R. A. Watson, and D. Corne. **Reducing Local Optima in Single-Objective Problems by Multi-objectivization**. In E. Zitzler, K. Deb, L. Thiele, C. A. Coello Coello, and D. Corne, editors, *EMO*, volume 1993 of *LNCS*, pages 269–283. Springer, Berlin/Heidelberg, 2001. doi:10.1007/3-540-44719-9_19.
Annotation: Proposed multi-objectivization.
- [1778] J. D. Knowles, D. Corne, and M. Fleischer. **Bounded archiving using the Lebesgue measure**. In *IEEE CEC*, pages 2490–2497, Piscataway, NJ, Dec. 2003. IEEE Press.
- [1779] J. D. Knowles, L. Thiele, and E. Zitzler. **A tutorial on the performance assessment of stochastic multiobjective optimizers**. TIK-Report 214, Computer Engineering and Networks Laboratory (TIK), Swiss Federal Institute of Technology (ETH), Zürich, Switzerland, Feb. 2006. Revised version.
- [1780] J. D. Knowles, D. Corne, and K. Deb. **Introduction: Problem solving, EC and EMO**. In J. D. Knowles, D. Corne, K. Deb, and D. R. Chair, editors, *Multiobjective Problem Solving from Nature*, Natural Computing Series, pages 1–28. Springer, Berlin/Heidelberg, 2008. doi:10.1007/978-3-540-72964-8_1.
- [1781] J. D. Knowles, D. Corne, K. Deb, and D. R. Chair, editors. *Multiobjective Problem Solving from Nature*. Natural Computing Series. Springer, Berlin/Heidelberg, 2008.
- [1782] J. D. Knowles, D. Corne, and A. P. Reynolds. **Noisy Multiobjective Optimization on a Budget of 250 Evaluations**. In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCS*, pages 36–50. Springer, Heidelberg, Germany, 2009.
- [1783] G. A. Kochenberger, F. Glover, B. Alidaee, and C. Rego. **A unified modeling and solution framework for combinatorial optimization problems**. *OR Spektrum*, 26(2):237–250, 2004.
- [1784] G. A. Kochenberger, J.-K. Hao, F. Glover, M. Lewis, Z. Lü, H. Wang, and Y. Wang. **The unconstrained binary quadratic programming problem: a survey**. *Journal of Combinatorial Optimization*, 28(1):58–81, 2014. doi:10.1007/s10878-014-9734-0.
- [1785] M. Köksalan. **Multiobjective Combinatorial Optimization: Some Approaches**. *Journal of Multi-Criteria Decision Analysis*, 15:69–78, 2009. doi:10.1002/mcda.425.
- [1786] M. Köksalan and İ. Karahan. **An Interactive Territory Defining Evolutionary Algorithm: iTDEA**. *IEEE Trans. Evol. Comput.*, 14(5):702–722, Oct. 2010. doi:10.1109/TEVC.2010.2070070.
- [1787] A. Kolen and E. Pesch. **Genetic Local Search in Combinatorial Optimization**. *Discrete Applied Mathematics*, 48(3):273–284, 1994.
- [1788] R. Kolisch and S. Hartmann. **Experimental investigation of heuristics for resource-constrained project scheduling: An update**. *Eur. J. Oper. Res.*, 174(1):23–37, Oct. 2006. doi:10.1016/j.ejor.2005.01.065.
Keywords: Computational evaluation, Heuristics, Project scheduling, Resource constraints.
- [1789] J. B. Kollat and P. M. Reed. **A framework for visually interactive decision-making and design using evolutionary multi-objective optimization (VIDEO)**. *Environmental Modelling & Software*, 22(12):1691–1704, 2007.
Keywords: glyph plot.
- [1790] D. Koller and N. Friedman. *Probabilistic graphical models: principles and techniques*. MIT Press, 2009.

- [1791] V. Koltun and C. H. Papadimitriou. **Approximately dominating representatives**. *Theor. Comput. Sci.*, 371(3):148–154, 2007.
- [1792] A. V. Kononova, T. Bäck, J. E. Fieldsend, and F. Chicano, editors. *FOGA’25, Proceedings of the 18th ACM/SIGEVO Conference on Foundations of Genetic Algorithms, Leiden, Netherlands August 27–29, 2025*. ACM, 2025.
- [1793] T. C. Koopmans and M. J. Beckmann. **Assignment Problems and the Location of Economic Activities**. *Econometrica*, 25:53–76, 1957.
Annotation: Introduced the Quadratic Assignment Problem (QAP).
- [1794] M. Koppen and K. Yoshida. **Visualization of Pareto-sets in evolutionary multi-objective optimization**. In *7th International Conference on Hybrid Intelligent Systems (HIS 2007)*, pages 156–161. IEEE, 2007.
- [1795] O. Korb, T. Stützle, and T. E. Exner. **PLANTS: Application of ant colony optimization to structure-based drug design**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*, pages 247–258. Springer, Heidelberg, Germany, 2006. doi:10.1007/11839088_22.
- [1796] O. Korb, T. Stützle, and T. E. Exner. **An Ant Colony Optimization Approach to Flexible Protein–Ligand Docking**. *Swarm Intelligence*, 1(2):115–134, 2007.
- [1797] O. Korb, T. Stützle, and T. E. Exner. **Empirical Scoring Functions for Advanced Protein-Ligand Docking with PLANTS**. *Journal of Chemical Information and Modeling*, 49(2):84–96, 2009.
- [1798] O. Korb, P. Monecke, G. Hessler, T. Stützle, and T. E. Exner. **pharmACOpore: Multiple Flexible Ligand Alignment Based on Ant Colony Optimization**. *Journal of Chemical Information and Modeling*, 50(9):1669–1681, 2010.
- [1799] P. Korhonen and J. Wallenius. **A pareto race**. *Naval Research Logistics*, 35(6):615–623, 1988. doi:10.1002/1520-6750(198812)35:6<615::AID-NAV3220350608>3.0.CO;2-K.
- [1800] P. Korhonen and J. Wallenius. **Behavioral Issues in MCDM: Neglected Research Questions**. In J. Climaco, editor, *Multicriteria Analysis*, pages 412–422. Springer, Berlin/Heidelberg, 1997. ISBN 978-3-642-60667-0. doi:10.1007/978-3-642-60667-0_39.
Keywords: Aspiration Level, Decision Tool, Nondominated Solution, Prefer Solution, Prospect Theory.
- [1801] P. Korhonen, H. Moskowitz, and J. Wallenius. **Choice Behavior in Interactive Multiple-Criteria Decision Making**. *Annals of Operations Research*, 23(1):161–179, Dec. 1990. doi:10.1007/BF02204844.
- [1802] P. J. Korhonen, K. Silvennoinen, J. Wallenius, and A. Öörni. **Can a linear value function explain choices? An experimental study**. *Eur. J. Oper. Res.*, 219(2):360–367, June 2012. doi:10.1016/j.ejor.2011.12.040.
Keywords: Binary choices, Inconsistency, Linear value function, Multiple criteria, Weights.
- [1803] P. J. Korhonen, J. Wallenius, T. Genc, and P. Xu. **On rational behavior in multi-attribute riskless choice**. *Eur. J. Oper. Res.*, 288(1):331–342, Jan. 2021. doi:10.1016/j.ejor.2020.05.056.
Keywords: Choice behavior, Decreasing marginal values, Loss aversion, Multiple criteria decision making, Riskless choice, Tradeoff, Win-win.
- [1804] F. Korn, B.-U. Pagel, and C. Faloutsos. **On the “dimensionality curse” and the “self-similarity blessing”**. *IEEE Transactions on Knowledge and Data Engineering*, 13(1):96–111, 2001. doi:10.1109/69.908983.

- [1805] J. Kornbluth. **Sequential multi-criterion decision making**. *Omega*, 13(6):569–574, 1985.
[doi:10.1016/0305-0483\(85\)90045-3](https://doi.org/10.1016/0305-0483(85)90045-3).
Keywords: machine decision making.
- [1806] P. Korošec, J. Šilc, and B. Robič. **Mesh-Partitioning with the Multiple Ant-Colony Algorithm**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 430–431. Springer, Heidelberg, Germany, 2004.
- [1807] P. Korošec, J. Šilc, and B. Robič. **Solving the mesh-partitioning problem with an ant-colony algorithm**. *Parallel Computing*, 30:785–801, 2004.
- [1808] P. Korošec, J. Šilc, K. Oblak, and F. Kosel. **The differential ant-stigmergy algorithm: an experimental evaluation and a real-world application**. In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 157–164, Piscataway, NJ, 2007. IEEE Press.
- [1809] A. Kostovska, D. Vermetten, C. Doerr, S. Džeroski, P. Panov, and T. Eftimov. **OPTION: optimization algorithm benchmarking ontology**. In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 239–240. ACM Press, New York, NY, 2021.
- [1810] A. Kostovska, D. Vermetten, S. Džeroski, P. Panov, T. Eftimov, and C. Doerr. **Using Knowledge Graphs for Performance Prediction of Modular Optimization Algorithms**. In J. a. Correia, S. Smith, and R. Qaddoura, editors, *EvoApplications 2023: Applications of Evolutionary Computation*, volume 13989 of *LNCS*, pages 253–268. Springer Nature, Switzerland, 2023.
- [1811] L. Kotthoff. **Algorithm Selection for Combinatorial Search Problems: A Survey**. *AI Magazine*, 35(3):48–60, 2014.
- [1812] L. Kotthoff, C. Thornton, H. H. Hoos, F. Hutter, and K. Leyton-Brown. **Auto-WEKA 2.0: Automatic model selection and hyperparameter optimization in WEKA**. *J. Mach. Learn. Res.*, 17:1–5, 2016.
- [1813] T. Kötzing, F. Neumann, H. Röglin, and C. Witt. **Theoretical Analysis of Two ACO Approaches for the Traveling Salesman Problem**. *Swarm Intelligence*, 6(1):1–21, 2012.
[doi:10.1007/s11721-011-0059-7](https://doi.org/10.1007/s11721-011-0059-7).
- [1814] P. Kouvelis and G. Yu. *Robust discrete optimization and its applications*. Nonconvex optimization and its applications. Kluwer Academic Publishers, Dordrecht, The Netherlands, 1997.
- [1815] O. Kovářik and M. Skrbek. **Ant Colony Optimization with Castes**. In V. Kurkova-Pohlova and J. Koutník, editors, *ICANN’08: Proceedings of the 18th International Conference on Artificial Neural Networks, Part I*, volume 5163 of *LNCS*, pages 435–442. Springer, Heidelberg, Germany, 2008.
- [1816] K. Kowalski, S. Stagl, R. Madlener, and I. Omann. **Sustainable energy futures: Methodological challenges in combining scenarios and participatory multi-criteria analysis**. *Eur. J. Oper. Res.*, 197(3):1063–1074, 2009.
- [1817] S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors. *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.
- [1818] J. Koza. *Genetic Programming: On the Programming of Computers By the Means of Natural Selection*. MIT Press, Cambridge, MA, 1992.
- [1819] J. R. Koza, editor. *Genetic Programming 1998: Proceedings of the Third Annual Conference, Late Breaking Papers*, Stanford University, California, July 1998. Stanford University Bookstore.

- [1820] S. Koziel and X.-S. Yang, editors. *Computational Optimization, Methods and Algorithms*, volume 356 of *Studies in Computational Intelligence*. Springer, Berlin/Heidelberg, 2011. doi:10.1007/978-3-642-20859-1.
- [1821] S. Koziel, D. E. Ciaurri, and L. Leifsson. **Surrogate-Based Methods**. In S. Koziel and X.-S. Yang, editors, *Computational Optimization, Methods and Algorithms*, volume 356 of *Studies in Computational Intelligence*, pages 33–59. Springer, Berlin/Heidelberg, 2011. doi:10.1007/978-3-642-20859-1.
- [1822] D. Kraft. **A software package for sequential quadratic programming**. Technical Report DFVLR-FB 88-28, DLR German Aerospace Center, Institute for Flight Mechanics, Koln, Germany, 1988.
- [1823] D. Krajzewicz, J. Erdmann, M. Behrisch, and L. Bieker. **Recent development and applications of SUMO - Simulation of Urban MObility**. *International Journal On Advances in Systems and Measurements*, 5(3-4):128–138, 2012.
- [1824] D. Krajzewicz, M. Heinrich, M. Milano, P. Bellavista, T. Stützle, J. Härri, T. Spyropoulos, R. Blokpoel, S. Hausberger, and M. Fellendorf. **COLOMBO: Investigating the Potential of V2X for Traffic Management Purposes assuming low penetration Rates**. In *Proceedings of ITS Europe 2013*, Dublin, Ireland, 2013.
- [1825] D. Krajzewicz, A. Leich, R. Blokpoel, M. Milano, and T. Stützle. **COLOMBO: Exploiting Vehicular Communications at Low Equipment Rates for Traffic Management Purposes**. In T. Schulze, B. Müller, and G. Meyer, editors, *Advanced Microsystems for Automotive Applications 2015: Smart Systems for Green and Automated Driving*, pages 117–130. Springer International Publishing, Cham, Switzerland, 2016.
- [1826] O. Kramer. **Iterated Local Search with Powell’s Method: A Memetic Algorithm for Continuous Global Optimization**. *Memetic Computing*, 2(1):69–83, 2010. doi:10.1007/s12293-010-0032-9.
- [1827] O. Kramer, B. Gloger, and A. Goebels. **An Experimental Analysis of Evolution Strategies and Particle Swarm Optimisers Using Design of Experiments**. In D. Thierens et al., editors, *GECCO*, pages 674–681. ACM Press, New York, NY, 2007.
- [1828] J. Krarup and P. M. Pruzan. **Computer-aided Layout Design**. In M. L. Balinski and C. Lemarechal, editors, *Mathematical Programming in Use*, volume 9 of *Mathematical Programming Studies*, pages 75–94. Springer, Berlin/Heidelberg, 1978.
- [1829] N. Krasnogor and P. L. Lanzi, editors. *Genetic and Evolutionary Computation Conference, GECCO 2011, Proceedings, Dublin, Ireland, July 12-16, 2011*. ACM Press, New York, NY, 2011.
- [1830] N. Krasnogor and P. L. Lanzi, editors. *13th Annual Genetic and Evolutionary Computation Conference, GECCO 2011, Companion Material Proceedings, Dublin, Ireland, July 12-16, 2011*. ACM Press, New York, NY, 2011.
- [1831] N. Krasnogor, B. Melián-Batista, J. A. Moreno-Pérez, J. M. Moreno-Vega, and D. A. Pelta, editors. *Nature Inspired Cooperative Strategies for Optimization (NICSO 2008)*, volume 236 of *Studies in Computational Intelligence*. Springer, Berlin, Germany, 2009. doi:10.1007/978-3-642-03211-0.
- [1832] O. Krause, T. Glasmachers, N. Hansen, and C. Igel. **Unbounded population MO-CMA-ES for the bi-objective BBOB test suite**. In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO Companion*, pages 1177–1184. ACM Press, New York, NY, 2016.
Keywords: archiving.

- [1833] S. Kreipl. **A Large Step Random Walk for Minimizing Total Weighted Tardiness in a Job Shop.** *Journal of Scheduling*, 3(3):125–138, 2000.
- [1834] J. Krettek, J. Braun, F. Hoffmann, and T. Bertram. **Interactive Incorporation of User Preferences in Multiobjective Evolutionary Algorithms.** In J. Mehnen, M. Köppen, A. Saad, and A. Tiwari, editors, *Applications of Soft Computing*, volume 58 of *Advances in Intelligent and Soft Computing*, pages 379–388. Springer, Berlin/Heidelberg, 2009.
- [1835] J. Krettek, J. Braun, F. Hoffmann, and T. Bertram. **Preference Modeling and Model Management for Interactive Multi-objective Evolutionary Optimization.** In E. Hüllermeier, R. Kruse, and F. Hoffmann, editors, *Information Processing and Management of Uncertainty, 13th International Conference, IPMU2010*, volume 6178 of *LNAI*, pages 574–583. Springer, Heidelberg, Germany, 2010.
- [1836] S. Kritzinger, F. Tricoire, K. F. Doerner, R. F. Hartl, and T. Stützle. **A Unified Framework for Routing Problems with a Fixed Fleet Size.** *International Journal of Metaheuristics*, 6(3):160–209, 2017.
- [1837] J. B. Kruskal. **On the shortest spanning subtree of a graph and the traveling salesman problem.** *Proceedings of the American Mathematical Society*, 7(1):48–50, 1956.
- [1838] W. H. Kruskal and J. M. Tanur. *Linear Hypotheses*, volume 1. Free Press, 1978.
- [1839] H. W. Kuhn. **The hungarian method for the assignment problem.** *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
- [1840] M. Kuhn. **Building Predictive Models in R Using the caret Package.** *Journal of Statistical Software*, 28(5):1–26, 2008.
- [1841] T. Kuhn, C. M. Fonseca, L. Paquete, S. Ruzika, M. M. Duarte, and J. R. Figueira. **Hypervolume subset selection in two dimensions: Formulations and algorithms.** *Evol. Comput.*, 24(3):411–425, 2016.
- [1842] J. Kuhpfahl and C. Bierwirth. **A Study on Local Search Neighborhoods for the Job Shop Scheduling Problem with Total Weighted Tardiness Objective.** *Comput. Oper. Res.*, 66:44–57, 2016.
- [1843] B. Kuipers and B. L. Webber, editors. *Proceedings of the Fourteenth National Conference on Artificial Intelligence and Ninth Innovative Applications of Artificial Intelligence Conference, AAAI 97, IAAI 97, July 27-31, 1997, Providence, Rhode Island, 1997.* AAAI Press/MIT Press, Menlo Park, CA.
- [1844] S. Kukkonen and J. Lampinen. **GDE3: the third evolution step of generalized differential evolution.** In *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, pages 443–450, Piscataway, NJ, Sept. 2005. IEEE Press.
- [1845] R. Kumar and P. K. Singh. **Pareto Evolutionary Algorithm Hybridized with Local Search for Biobjective TSP.** In A. Abraham, C. Grosan, and H. Ishibuchi, editors, *Hybrid Evolutionary Algorithms*, volume 75 of *Studies in Computational Intelligence*, pages 361–398. Springer, 2007. doi:10.1007/978-3-540-73297-6_14.
- [1846] R. Kumar and S. Vassilvitskii. **Generalized Distances between Rankings.** In M. Rappa, P. Jones, J. Freire, and S. Chakrabarti, editors, *Proceedings of the 19th International Conference on World Wide Web, WWW 2010.* ACM Press, New York, NY, 2010.
- [1847] M. Kunc, J. Malpass, and L. White, editors. *Behavioral Operational Research Theory, Methodology and Practice.* Palgrave Macmillan, London, UK, 2016.

- [1848] H. T. Kung, F. Luccio, and F. P. Preparata. **On Finding the Maxima of a Set of Vectors.** *J. ACM*, 22(4):469–476, 1975. doi:10.1145/321906.321910.
- [1849] V. Kurkova-Pohlova and J. Koutnik, editors. *ICANN’08: Proceedings of the 18th International Conference on Artificial Neural Networks, Part I*, volume 5163 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [1850] V. Kurkova-Pohlova and J. Koutnik, editors. *ICANN’08: Proceedings of the 18th International Conference on Artificial Neural Networks, Part II*, volume 5164 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [1851] D. V. Kurpel, C. T. Scarpin, J. E. Pécora Junior, C. M. Schenekemberg, and L. C. Coelho. **The Exact Solutions of Several Types of Container Loading Problems.** *Eur. J. Oper. Res.*, 284(1):87–107, 2020. doi:10.1016/j.ejor.2019.12.012.
Keywords: Single Stock-Size Cutting Stock Problem, 3D container loading.
- [1852] F. Kursawe. **A variant of evolution strategies for vector optimization.** In H.-P. Schwefel and R. Männer, editors, *Parallel Problem Solving from Nature – PPSN I*, pages 193–197. Springer, Berlin/Heidelberg, 1991. doi:10.1007/BFb0029752.
Annotation: Proposed KUR benchmark.
- [1853] I. Kurtulus and E. W. Davis. **Multi-Project Scheduling: Categorization of Heuristic Rules Performance.** *Management Science*, 28(2):161–172, 1982. doi:10.1287/mnsc.28.2.161.
Keywords: project management, research and development.
- [1854] H. J. Kushner. **A New Method of Locating the Maximum Point of an Arbitrary Multippeak Curve in the Presence of Noise.** *Journal of Basic Engineering*, 86(1):97–106, Mar. 1964. ISSN 0021-9223. doi:10.1115/1.3653121.
- [1855] J. H. Kwakkel. **The Exploratory Modeling Workbench: An open source toolkit for exploratory modeling, scenario discovery, and (multi-objective) robust decision making.** *Environmental Modelling & Software*, 96:239–250, 2017.
- [1856] M. Labbé and A. Violin. **Bilevel programming and price setting problems.** *4OR: A Quarterly Journal of Operations Research*, 11(1):1–30, 2013. doi:10.1007/s10288-012-0213-0.
- [1857] M. Labbé, P. Marcotte, and G. Savard. **A Bilevel Model of Taxation and Its Application to Optimal Highway Pricing.** *Management Science*, 44(12):1608–1622, 1998. doi:10.1287/mnsc.44.12.1608.
- [1858] R. Lacour, K. Klamroth, and C. M. Fonseca. **A box decomposition algorithm to compute the hypervolume indicator.** *Comput. Oper. Res.*, 79:347–360, 2017. doi:10.1016/j.cor.2016.06.021.
- [1859] B. Lacroix, D. Molina, and F. Herrera. **Dynamically updated region based memetic algorithm for the 2013 CEC Special Session and Competition on Real Parameter Single Objective Optimization.** In *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, pages 1945–1951, Piscataway, NJ, 2013. IEEE Press.
- [1860] B. Lacroix, D. Molina, and F. Herrera. **Region based memetic algorithm for real-parameter optimisation.** *Inf. Sci.*, 262:15–31, 2014. doi:10.1016/j.ins.2013.11.032.
Keywords: irace.
- [1861] S. R. Ladd. **ACOVEA (Analysis of Compiler Options via Evolutionary Algorithm).** <https://github.com/Acovea/libacovea>, 2000.
- [1862] M. Laguna. **Editor’s Note on the MIC 2013 Special Issue of the Journal of Heuristics (Volume 22, Issue 4, August 2016).** *J. Heuristics*, 22(5):665–666, 2016.

- [1863] T. J. Lahtinen and R. P. Hämmäläinen. **Path dependence and biases in the even swaps decision analysis method.** *Eur. J. Oper. Res.*, 249(3):890–898, 2016. doi:[10.1016/j.ejor.2015.09.056](https://doi.org/10.1016/j.ejor.2015.09.056).
- [1864] X. Lai and J.-K. Hao. **Iterated Maxima Search for the Maximally Diverse Grouping Problem.** *Eur. J. Oper. Res.*, 254(3):780–800, 2016.
- [1865] A. H. Land and A. G. Doig. **An Automatic Method of Solving Discrete Programming Problems.** *Econometrica*, 28(3):497–520, 1960.
- [1866] B. Lang. **Space-Partitioned ND-Trees for the Dynamic Nondominance Problem.** *IEEE Trans. Evol. Comput.*, 26(5):1004–1014, 2022. doi:[10.1109/TEVC.2022.3145631](https://doi.org/10.1109/TEVC.2022.3145631).
- [1867] M. Lang, H. Kotthaus, P. Marwedel, C. Weihs, J. Rahnenführer, and B. Bischl. **Automatic Model Selection for High-Dimensional Survival Analysis.** *Journal of Statistical Computation and Simulation*, 85(1):62–76, 2014. doi:[10.1080/00949655.2014.929131](https://doi.org/10.1080/00949655.2014.929131).
- [1868] W. B. Langdon and M. Harman. **Optimising Software with Genetic Programming.** *IEEE Trans. Evol. Comput.*, 19(1):118–135, 2015.
- [1869] W. B. Langdon et al., editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2002.* Morgan Kaufmann Publishers, San Francisco, CA, 2002.
- [1870] A. Langevin, F. Soumis, and J. Desrosiers. **Classification of travelling salesman problem formulations.** *Operations Research Letters*, 9(2):127–132, 1990.
- [1871] A. Langevin, M. Desrochers, J. Desrosiers, S. Gélinas, and F. Soumis. **A Two-Commodity Flow Formulation for the Traveling Salesman and Makespan Problems with Time Windows.** *Networks*, 23(7):631–640, 1993.
- [1872] J. Langford and J. Pineau, editors. *Proceedings of the 29th International Conference on Machine Learning, ICML 2012, Edinburgh, Scotland, UK, June 26 - July 1, 2012*, 2012. Omnipress.
- [1873] K. E. Lansey and K. Awumah. **Optimal Pump Operations Considering Pump Switches.** *Journal of Water Resources Planning and Management, ASCE*, 120(1):17–35, Jan. / Feb. 1994.
- [1874] G. Laporte. **Fifty Years of Vehicle Routing.** *Transportation Science*, 43(4):408–416, 2009.
- [1875] R. M. Lark and D. J. Lapworth. **A new statistic to express the uncertainty of kriging predictions for purposes of survey planning.** In *EGU General Assembly Conference Abstracts*, May 2014. URL <https://ui.adsabs.harvard.edu/abs/2014EGUGA..16.2183L>.
- [1876] C. Larman. *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development.* Prentice Hall, Englewood Cliffs, NJ, 3rd edition, 2004.
- [1877] H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors. *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, Virtual*, 2020.
- [1878] P. Larrañaga and J. A. Lozano. *Estimation of Distribution Algorithms: A New Tool for Evolutionary Computation.* Kluwer Academic Publishers, Boston, MA, 2002.
- [1879] A. LaTorre, S. Muelas, and J.-M. Peña. **A MOS-based dynamic memetic differential evolution algorithm for continuous optimization: a scalability test.** *Soft Computing*, 15(11):2187–2199, 2011.
- [1880] M. Laumanns. **Stochastic convergence of random search to fixed size Pareto set approximations.** *Arxiv preprint arXiv:0711.2949*, 2007. URL <https://arxiv.org/abs/0711.2949>.

- [1881] M. Laumanns and R. Zenklusen. **Stochastic convergence of random search methods to fixed size Pareto front approximations.** (submitted), Nov. 2010.
- [1882] M. Laumanns and R. Zenklusen. **Stochastic convergence of random search methods to fixed size Pareto front approximations.** *Eur. J. Oper. Res.*, 213(2):414–421, 2011. doi:10.1016/j.ejor.2011.03.039.
- [1883] M. Laumanns, E. Zitzler, and L. Thiele. **A unified model for multi-objective evolutionary algorithms with elitism.** In *IEEE CEC*, pages 46–53, Piscataway, NJ, July 2000. IEEE Press.
- [1884] M. Laumanns, L. Thiele, K. Deb, and E. Zitzler. **Combining Convergence and Diversity in Evolutionary Multiobjective Optimization.** *Evol. Comput.*, 10(3):263–282, 2002. doi:10.1162/106365602760234108.
Keywords: archiving, ϵ -dominance, ϵ -approximation, ϵ -Pareto.
Annotation: Proposed ϵ -approx and ϵ -Pareto archivers.
- [1885] M. Laumanns, L. Thiele, E. Zitzler, and K. Deb. **Archiving with guaranteed convergence and diversity in multi-objective optimization.** In W. B. Langdon et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2002*, pages 439–447. Morgan Kaufmann Publishers, San Francisco, CA, 2002.
- [1886] M. Laumanns, L. Thiele, and E. Zitzler. **Running time analysis of multiobjective evolutionary algorithms on pseudo-boolean functions.** *IEEE Trans. Evol. Comput.*, 8(2):170–182, 2004.
- [1887] M. Laumanns, L. Thiele, and E. Zitzler. **Running time analysis of evolutionary algorithms on a simplified multiobjective knapsack problem.** *Natural Computing*, 3(1):37–51, 2004.
- [1888] B. Laurent and J.-K. Hao. **Iterated Local Search for the Multiple Depot Vehicle Scheduling Problem.** *Computers & Industrial Engineering*, 57(1):277–286, 2009.
- [1889] E. L. Lawler and D. E. Wood. **Branch-and-Bound Methods: A Survey.** *Operations Research*, 14(4):699–719, 1966. doi:10.1287/opre.14.4.699.
- [1890] E. L. Lawler, J. K. Lenstra, A. H. G. Rinnooy Kan, and D. B. Shmoys. *The Traveling Salesman Problem*. John Wiley & Sons, Chichester, UK, 1985.
- [1891] S. E. Lazic. **The problem of pseudoreplication in neuroscientific studies: is it affecting your analysis?** *BMC Neuroscience*, 11(5):397–407, 2004. doi:10.1186/1471-2202-11-5.
- [1892] V. Leal do Forte, F. M. Tavares Montenegro, J. A. de Moura Brito, and N. Maculan. **Iterated Local Search Algorithms for the Euclidean Steiner Tree Problem in n Dimensions.** *International Transactions in Operational Research*, 23(6):1185–1199, 2016.
- [1893] Y. LeCun, Y. Bengio, et al. **Convolutional networks for images, speech, and time series.** *The handbook of brain theory and neural networks*, 3361(10):255–258, 1995.
- [1894] Y. LeCun, Y. Bengio, and G. Hinton. **Deep learning.** *Nature*, 521(7553):436–444, 2015.
- [1895] D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, and R. Garnett, editors. *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, 2016.
- [1896] T. Lee, M. Leok, and N. H. McClamroch. **A combinatorial optimal control problem for spacecraft formation reconfiguration.** In *2007 46th IEEE Conference on Decision and Control*. IEEE, 2007. doi:10.1109/cdc.2007.4434143.
Keywords: bilevel.

- [1897] P. Legrand, A. Liefoghe, E. Keedwell, J. Lepagnot, L. Idoumghar, N. Monmarché, and E. Lutton, editors. *Artificial Evolution: International Conference, Évolution Artificielle, EA 2023*. LNCS. Springer, 2023.
- [1898] P. Legrand et al., editors. *Artificial Evolution: 11th International Conference, Evolution Artificielle, EA 2013, Bordeaux, France, October 21-23, 2013. Revised Selected Papers*, volume 8752 of LNCS. Springer, Heidelberg, Germany, 2013.
- [1899] G. Leguizamón and E. Alba. **Ant Colony Based Algorithms for Dynamic Optimization Problems**. In E. Alba, A. Nakib, and P. Siarry, editors, *Metaheuristics for Dynamic Optimization*, volume 433 of *Studies in Computational Intelligence*, pages 189–210. Springer, Berlin/Heidelberg, 2013. doi:10.1007/978-3-642-30665-5_9.
- [1900] G. Leguizamón and Z. Michalewicz. **A New Version of Ant System for Subset Problems**. In *Proceedings of the 1999 Congress on Evolutionary Computation (CEC 1999)*, pages 1459–1464, Piscataway, NJ, 1999. IEEE Press.
- [1901] P. K. Lehre and C. Witt. **Black-box search by unbiased variation**. *Algorithmica*, 64(4): 623–642, 2012.
- [1902] F. T. Leighton. **A Graph Coloring Algorithm for Large Scheduling Problems**. *Journal of Research of the National Bureau of Standards*, 84(6):489–506, 1979.
- [1903] R. J. Lempert, S. Popper, and S. C. Banks. *Shaping the Next One Hundred Years: New Methods for Quantitative, Long Term Policy Analysis*. RAND, 2003.
- [1904] R. J. Lempert, D. G. Groves, S. W. Popper, and S. C. Banks. **A general analytic method for generating robust strategies and narrative scenarios**. *Management Science*, 52(4): 514–528, 2006.
- [1905] C. Leon, S. Martin, J. M. Elena, and J. Luque. **EXPLORE: Hybrid expert system for water networks management**. *Journal of Water Resources Planning and Management, ASCE*, 126(2):65–74, 2000.
- [1906] L. Lessing, I. Dumitrescu, and T. Stützle. **A Comparison Between ACO Algorithms for the Set Covering Problem**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of LNCS, pages 1–12. Springer, Heidelberg, Germany, 2004.
- [1907] S. C. H. Leung and D. Zhang. **A fast layer-based heuristic for non-guillotine strip packing**. *Expert Systems with Applications*, 38(10):13032–13042, Sept. 2011. doi:10.1016/j.eswa.2011.04.105.
Keywords: Container loading problem.
- [1908] L. Levin. **Universal’nyie perebornyie zadachi**. *Problemy Peredachi Informatsii*, 9:265–266, 1973.
- [1909] D. Lewandowski, D. Kurowicka, and H. Joe. **Generating Random Correlation Matrices Based on Vines and Extended Onion Method**. *Journal of Multivariate Analysis*, 100(9): 1989–2001, 2009. doi:10.1016/j.jmva.2009.04.008.
Keywords: Correlation matrix; Dependence vines; Onion method; Partial correlation; LKJ.
- [1910] R. M. R. Lewis. *A Guide to Graph Colouring: Algorithms and Applications*. Springer, Cham, Switzerland, 2016. doi:10.1007/978-3-319-25730-3.
Annotation: Supplementary material available at [1911].
- [1911] R. M. R. Lewis. **Suite of Graph Colouring Algorithms – Supplementary Material to the Book “A Guide to Graph Colouring: Algorithms and Applications”**. <http://rhydLewis.eu/resources/gCol.zip>, 2016.

- [1912] K. Leyton-Brown, M. Pearson, and Y. Shoham. **Towards a Universal Test Suite for Combinatorial Auction Algorithms**. In A. Jhingran et al., editors, *ACM Conference on Electronic Commerce (EC-00)*, pages 66–76. ACM Press, New York, NY, 2000. doi:10.1145/352871.352879.
Annotation: CPLEX-regions200 benchmark set, <http://www.cs.ubc.ca/labs/beta/Projects/ParamILS/results.html>.
- [1913] K. Leyton-Brown, E. Nudelman, and Y. Shoham. **Learning the Empirical Hardness of Optimization Problems: The Case of Combinatorial Auctions**. In P. van Hentenryck, editor, *Principles and Practice of Constraint Programming, CP 2002*, LNCS, pages 556–572. Springer, Heidelberg, Germany, 2002.
- [1914] K. Leyton-Brown, E. Nudelman, G. Andrew, J. McFadden, and Y. Shoham. **A Portfolio Approach to Algorithm Selection**. In G. Gottlob and T. Walsh, editors, *Proc. of IJCAI-03*, pages 1542–1543. Morgan Kaufmann Publishers, 2003.
Annotation: First example of modern algorithm selection for optimisation?
- [1915] B. Li, J. Li, K. Tang, and X. Yao. **An Improved Two Archive Algorithm for Many-Objective Optimization**. In *Proceedings of the 2014 Congress on Evolutionary Computation (CEC 2014)*, pages 2869–2876, Piscataway, NJ, 2014. IEEE Press.
- [1916] B. Li, J. Li, K. Tang, and X. Yao. **Many-Objective Evolutionary Algorithms: A Survey**. *ACM Comput. Surv.*, 48(1):1–35, 2015. doi:10.1145/2792984.
- [1917] B. Li, K. Tang, J. Li, and X. Yao. **Stochastic ranking algorithm for many-objective optimization based on multiple indicators**. *IEEE Trans. Evol. Comput.*, 20(6):924–938, 2016.
- [1918] H. Li and Q. Zhang. **Multiobjective Optimization Problems with Complicated Pareto sets, MOEA/D and NSGA-II**. *IEEE Trans. Evol. Comput.*, 13(2):284–302, 2009.
- [1919] J. D. Li. **A two-step rejection procedure for testing multiple hypotheses**. *Journal of Statistical Planning and Inference*, 138(6):1521–1527, 2008.
- [1920] K. Li, R. Chen, G. Fu, and X. Yao. **Two-archive evolutionary algorithm for constrained multiobjective optimization**. *IEEE Trans. Evol. Comput.*, 23(2):303–315, 2018.
- [1921] L. Li, I. Yevseyeva, V. Basto-Fernandes, H. Trautmann, N. Jing, and M. T. M. Emmerich. **Building and using an ontology of preference-based multiobjective evolutionary algorithms**. In H. Trautmann, G. Rudolph, K. Klamroth, O. Schütze, M. M. Wiecek, Y. Jin, and C. Grimme, editors, *EMO*, volume 10173 of *LNCS*, pages 406–421. Springer International Publishing, Cham, Switzerland, 2017.
- [1922] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar. **Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization**. *J. Mach. Learn. Res.*, 18(185):1–52, 2018.
Keywords: racing.
- [1923] M. Li. **Is Our Archiving Reliable? Multiobjective Archiving Methods on “Simple” Artificial Input Sequences**. *ACM Trans. Evol. Learn. Optim.*, 1(3):1–19, 2021. doi:10.1145/3465335.
- [1924] M. Li and X. Yao. **Dominance Move: A Measure of Comparing Solution Sets in Multiobjective Optimization**. *arXiv preprint arXiv:1702.00477*, 2017.

- [1925] M. Li and X. Yao. **An Empirical Investigation of the Optimality and Monotonicity Properties of Multiobjective Archiving Methods.** In K. Deb, E. D. Goodman, C. A. Coello Coello, K. Klamroth, K. Miettinen, S. Mostaghim, and P. M. Reed, editors, *Evolutionary Multi-criterion Optimization, EMO 2019*, volume 11411 of *LNCS*, pages 15–26. Springer International Publishing, Cham, Switzerland, 2019. ISBN 978-3-030-12597-4. doi:10.1007/978-3-030-12598-1_2.
- [1926] M. Li and X. Yao. **Quality Evaluation of Solution Sets in Multiobjective Optimisation: A Survey.** *ACM Comput. Surv.*, 52(2):1–38, 2019. doi:10.1145/3300148.
- [1927] M. Li and X. Yao. **What weights work for you? Adapting weights for any Pareto front shape in decomposition-based evolutionary multiobjective optimisation.** *Evol. Comput.*, 28(2):227–253, 2020.
- [1928] M. Li, S. Yang, X. Liu, and R. Shen. **A Comparative Study on Evolutionary Algorithms for Many-Objective Optimization.** In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 261–275. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4.
- [1929] M. Li, S. Yang, and X. Liu. **Shift-based density estimation for Pareto-based algorithms in many-objective optimization.** *IEEE Trans. Evol. Comput.*, 18(3):348–365, 2014.
Annotation: Proposed SDE indicator algorithm.
- [1930] M. Li, S. Yang, and X. Liu. **A performance comparison indicator for Pareto front approximations in many-objective optimization.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 703–710. ACM Press, New York, NY, 2015.
Annotation: Proposed PCI indicator.
- [1931] M. Li, S. Yang, and X. Liu. **Pareto or non-Pareto: Bi-criterion evolution in multiobjective optimization.** *IEEE Trans. Evol. Comput.*, 20(5):645–665, 2016.
- [1932] M. Li, C. Grosan, S. Yang, X. Liu, and X. Yao. **Multi-line distance minimization: A visualized many-objective test problem suite.** *IEEE Trans. Evol. Comput.*, 22(1):61–78, 2018.
Annotation: highly degenerate Pareto fronts.
- [1933] M. Li, T. Chen, and X. Yao. **How to evaluate solutions in Pareto-based search-based software engineering? A critical review and methodological guidance.** *IEEE Trans. Software Eng.*, 48(5):1771–1799, 2020. doi:10.1109/TSE.2020.3036108.
- [1934] M. Li, M. López-Ibáñez, and X. Yao. **Multi-Objective Archiving.** *IEEE Trans. Evol. Comput.*, 28(3):696–717, June 2024. doi:10.1109/TEVC.2023.3314152.
- [1935] X. Li, L. Chen, H. Xu, and J. N. D. Gupta. **Trajectory Scheduling Methods for Minimizing Total Tardiness in a Flowshop.** *Oper. Res. Perspect.*, 2:13–23, 2015. ISSN 2214–7160. doi:10.1016/j.orp.2014.12.001.
- [1936] X. Li et al., editors. *Simulated Evolution and Learning, 7th International Conference, SEAL 2008*, volume 5361 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [1937] Y. Li and W. Li. **Adaptive Ant Colony Optimization Algorithm Based on Information Entropy: Foundation and Application.** *Fundamenta Informaticae*, 77(3):229–242, 2007.
- [1938] Z. Li, Y. Wang, J. Yu, Y. Zhang, and X. Li. **A Novel Cloud-Based Fuzzy Self-Adaptive Ant Colony System.** In *ICNC’08: Proceedings of the 2008 Fourth International Conference on Natural Computation*, volume 7, pages 460–465, Washington, DC, 2008. IEEE Computer Society.

- [1939] Z. Li, M. Shahidehpour, S. Bahramirad, and A. Khodaei. **Optimizing Traffic Signal Settings in Smart Cities.** *IEEE Transactions on Smart Grid*, 3053(4):1–1, 2016. doi:[10.1109/TSG.2016.2526032](https://doi.org/10.1109/TSG.2016.2526032).
- [1940] Z. Li, J. Zou, S. Yang, and J. Zheng. **A two-archive algorithm with decomposition and fitness allocation for multi-modal multi-objective optimization.** *Inf. Sci.*, 574:413–430, 2021.
- [1941] C.-J. Liao, C.-T. Tseng, and P. Luarn. **A Discrete Version of Particle Swarm Optimization for Flowshop Scheduling Problems.** *Comput. Oper. Res.*, 34(10):3099–3111, 2007.
- [1942] T. Liao. *Population-based Heuristic Algorithms for Continuous and Mixed Discrete-Continuous Optimization Problem.* PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2013.
- [1943] T. Liao and T. Stützle. **Benchmark results for a simple hybrid algorithm on the CEC 2013 benchmark set for real-parameter optimization.** In *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, pages 1938–1944, Piscataway, NJ, 2013. IEEE Press.
- [1944] T. Liao, D. Molina, M. A. Montes de Oca, and T. Stützle. **A Note on the Effects of Enforcing Bound Constraints on Algorithm Comparisons using the IEEE CEC’05 Benchmark Function Suite.** Technical Report TR/IRIDIA/2011-010, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-010.pdf>. Published in *Evol. Comput.* [1951].
- [1945] T. Liao, D. Molina, M. A. Montes de Oca, and T. Stützle. **Computational Results for an Automatically Tuned IPOP-CMA-ES on the CEC’05 Benchmark Set.** Technical Report TR/IRIDIA/2011-022, IRIDIA, Université Libre de Bruxelles, Belgium, 2011.
- [1946] T. Liao, M. A. Montes de Oca, D. Aydın, T. Stützle, and M. Dorigo. **An Incremental Ant Colony Algorithm with Local Search for Continuous Optimization.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 125–132. ACM Press, New York, NY, 2011.
- [1947] T. Liao, M. A. Montes de Oca, and T. Stützle. **Tuning Parameters across Mixed Dimensional Instances: A Performance Scalability Study of Sep-G-CMA-ES.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO Companion*, pages 703–706. ACM Press, New York, NY, 2011.
Annotation: Workshop on Scaling Behaviours of Landscapes, Parameters and Algorithms.
- [1948] T. Liao, D. Aydın, and T. Stützle. **Artificial Bee Colonies for Continuous Optimization: Experimental Analysis and Improvements.** *Swarm Intelligence*, 7(4):327–356, 2013.
- [1949] T. Liao, M. A. Montes de Oca, and T. Stützle. **Computational results for an automatically tuned CMA-ES with increasing population size on the CEC’05 benchmark set.** *Soft Computing*, 17(6):1031–1046, 2013. doi:[10.1007/s00500-012-0946-x](https://doi.org/10.1007/s00500-012-0946-x).
- [1950] T. Liao, T. Stützle, M. A. Montes de Oca, and M. Dorigo. **A Unified Ant Colony Optimization Algorithm for Continuous Optimization.** Technical Report TR/IRIDIA/2013-002, IRIDIA, Université Libre de Bruxelles, Belgium, 2013.
- [1951] T. Liao, D. Molina, M. A. Montes de Oca, and T. Stützle. **A Note on the Effects of Enforcing Bound Constraints on Algorithm Comparisons using the IEEE CEC’05 Benchmark Function Suite.** *Evol. Comput.*, 22(2):351–359, 2014.
- [1952] T. Liao, K. Socha, M. A. Montes de Oca, T. Stützle, and M. Dorigo. **Ant Colony Optimization for Mixed-Variable Optimization Problems.** *IEEE Trans. Evol. Comput.*, 18(4):503–518, 2014.
Keywords: ACOR.

- [1953] T. Liao, T. Stützle, M. A. Montes de Oca, and M. Dorigo. **A Unified Ant Colony Optimization Algorithm for Continuous Optimization.** *Eur. J. Oper. Res.*, 234(3): 597–609, 2014.
- [1954] T. Liao, D. Molina, and T. Stützle. **Performance Evaluation of Automatically Tuned Continuous Optimizers on Different Benchmark Sets.** *Applied Soft Computing*, 27: 490–503, 2015.
- [1955] A. Liefooghe and M. López-Ibáñez, editors. *Evolutionary Computation in Combinatorial Optimization – 18th European Conference, EvoCOP 2018, Parma, Italy, April 4-6, 2018, Proceedings*, volume 10782 of *LNCS*. Springer, Heidelberg, Germany, 2018. doi:10.1007/978-3-319-77449-7.
- [1956] A. Liefooghe and M. López-Ibáñez. **Many-objective (Combinatorial) Optimization is Easy.** In S. Silva and L. Paquete, editors, *GECCO*, pages 704–712. ACM Press, New York, NY, 2023. doi:10.1145/3583131.3590475.
Annotation: ISBN: 9798400701191.
- [1957] A. Liefooghe, S. Mesmoudi, J. Humeau, L. Jourdan, and E.-G. Talbi. **A Study on Dominance-based Local Search Approaches for Multiobjective Combinatorial Optimization.** In T. Stützle, M. Birattari, and H. H. Hoos, editors, *SLS 2009*, volume 5752 of *LNCS*, pages 120–124. Springer, Heidelberg, Germany, 2009.
- [1958] A. Liefooghe, L. Jourdan, and E.-G. Talbi. **A Software Framework Based on a Conceptual Unified Model for Evolutionary Multiobjective Optimization: ParadisEO-MOEO.** *Eur. J. Oper. Res.*, 209(2):104–112, 2011.
- [1959] A. Liefooghe, L. Paquete, M. Simões, and J. R. Figueira. **Connectedness and Local Search for Bicriteria Knapsack Problems.** In P. Merz and J.-K. Hao, editors, *EvoCOP*, volume 6622 of *LNCS*, pages 48–59. Springer, Heidelberg, Germany, 2011. doi:10.1007/978-3-642-20364-0_5.
- [1960] A. Liefooghe, J. Humeau, S. Mesmoudi, L. Jourdan, and E.-G. Talbi. **On dominance-based multiobjective local search: design, implementation and experimental analysis on scheduling and traveling salesman problems.** *J. Heuristics*, 18(2):317–352, 2012. doi:10.1007/s10732-011-9181-3.
- [1961] A. Liefooghe, B. Derbel, S. Verel, H. E. Aguirre, and K. Tanaka. **What Makes an Instance Difficult for Black-box 0–1 Evolutionary Multiobjective Optimizers?** In P. Legrand et al., editors, *Artificial Evolution: 11th International Conference, Evolution Artificielle, EA, 2013*, volume 8752 of *LNCS*, pages 3–15. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-319-11683-9_1.
- [1962] A. Liefooghe, S. Verel, and J.-K. Hao. **A hybrid metaheuristic for multiobjective unconstrained binary quadratic programming.** *Applied Soft Computing*, 16:10–19, 2014.
- [1963] A. Liefooghe, S. Verel, L. Paquete, and J.-K. Hao. **Experiments on Local Search for Bi-objective Unconstrained Binary Quadratic Programming.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 171–186. Springer, Heidelberg, Germany, 2015.
- [1964] A. Liefooghe, B. Derbel, S. Verel, H. E. Aguirre, and K. Tanaka. **Towards Landscape-Aware Automatic Algorithm Configuration: Preliminary Experiments on Neutral and Rugged Landscapes.** In B. Hu and M. López-Ibáñez, editors, *EvoCOP*, volume 10197 of *LNCS*, pages 215–232. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-55453-2_15.

- [1965] A. Liefvooghe, B. Derbel, S. Verel, M. López-Ibáñez, H. E. Aguirre, and K. Tanaka. **On Pareto Local Optimal Solutions Networks**. In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11102 of *LNCS*, pages 232–244. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-319-99259-4_19.
- [1966] A. Liefvooghe, M. López-Ibáñez, L. Paquete, and S. Verel. **Dominance, Epsilon, and Hypervolume Local Optimal Sets in Multi-objective Optimization, and How to Tell the Difference**. In H. E. Aguirre and K. Takadama, editors, *GECCO*, pages 324–331. ACM Press, New York, NY, 2018. doi:10.1145/3205455.3205572.
- [1967] A. Liefvooghe, F. Daolio, B. Derbel, S. Verel, H. E. Aguirre, and K. Tanaka. **Landscape-Aware Performance Prediction for Evolutionary Multi-objective Optimization**. *IEEE Trans. Evol. Comput.*, 24(6):1063–1077, 2020.
- [1968] A. Liefvooghe, S. Verel, B. Lacroix, A. Zavoianu, and J. McCall. **Landscape features and automated algorithm selection for multi-objective interpolated continuous optimisation problems**. In F. Chicano and K. Krawiec, editors, *GECCO*, pages 421–429. ACM Press, New York, NY, 2021. doi:10.1145/3449639.3459353.
- [1969] B. Likar and J. Kocijan. **Predictive control of a gas–liquid separation plant based on a Gaussian process model**. *Computers & Chemical Engineering*, 31(3):142–152, 2007. doi:10.1016/j.compchemeng.2006.05.011.
- [1970] D. J. Lilja. *Measuring Computer Performance: A Practitioner’s Guide*. Cambridge University Press, 2000. doi:10.1017/CB09780511612398.
- [1971] A. Lim, B. Rodrigues, and Y. Wang. **A multi-faced buildup algorithm for three-dimensional packing problems**. *Omega*, 31(6):471–481, Dec. 2003. doi:10.1016/j.omega.2003.08.004.
Keywords: Container loading problem, maximal space representation, cover representation.
- [1972] J. Lin. **Divergence Measures based on the Shannon Entropy**. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991. doi:10.1109/18.61115.
Annotation: Proposed Jensen-Shannon distance.
- [1973] S. Lin and B. W. Kernighan. **An Effective Heuristic Algorithm for the Traveling Salesman Problem**. *Operations Research*, 21(2):498–516, 1973.
- [1974] M. T. Lindauer, H. H. Hoos, F. Hutter, and T. Schaub. **AutoFolio: Algorithm Configuration for Algorithm Selection**. In B. Bonet and S. Koenig, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press, 2015.
- [1975] M. T. Lindauer, H. H. Hoos, F. Hutter, and T. Schaub. **AutoFolio: An Automatically Configured Algorithm Selector**. *J. Artif. Intell. Res.*, 53:745–778, 2015.
- [1976] M. T. Lindauer, H. H. Hoos, K. Leyton-Brown, and T. Schaub. **Automatic Construction of Parallel Portfolios via Algorithm Configuration**. *Artif. Intell.*, 244:272–290, 2017. doi:10.1016/j.artint.2016.05.004.
Keywords: parHydra, Algorithm configuration, Algorithm portfolios, Automated parallelization, Parallel SAT solving, Programming by optimization.
- [1977] M. T. Lindauer, J. N. van Rijn, and L. Kotthoff. **The algorithm selection competitions 2015 and 2017**. *Artif. Intell.*, 272:86–100, 2019.
- [1978] M. T. Lindauer, K. Eggenberger, M. Feurer, A. Biedenkapp, D. Deng, C. Benjamins, T. Ruhkopf, R. Sass, and F. Hutter. **SMAC3: A Versatile Bayesian Optimization Package for Hyperparameter Optimization**. *J. Mach. Learn. Res.*, 23:1–9, 2022.

- [1979] W. Ling and H. Luo. **An Adaptive Parameter Control Strategy for Ant Colony Optimization.** In *CIS'07: Proceedings of the 2007 International Conference on Computational Intelligence and Security*, pages 142–146, Washington, DC, 2007. IEEE Computer Society.
- [1980] A. Lissovoi and C. Witt. **Runtime Analysis of Ant Colony Optimization on Dynamic Shortest Path Problems.** *Theor. Comput. Sci.*, 561(Part A):73–85, 2015. doi:10.1016/j.tcs.2014.06.035.
- [1981] J. D. C. Little, K. G. Murty, D. W. Sweeney, and C. Karel. **An Algorithm for the Traveling Salesman Problem.** *Operations Research*, 11:972–989, 1963.
- [1982] C. Liu, K. Smith-Miles, T. Wauters, and A. M. Costa. **A block-building constraint programming model for the container loading problem.** *Comput. Oper. Res.*, 182:107111, Oct. 2025. doi:10.1016/j.cor.2025.107111.
- [1983] D. Liu et al., editors. *Proceedings of the International Joint Conference on Neural Networks (IJCNN 2008), Hong Kong, China, June 1-6, 2008*, 2008. IEEE Press.
- [1984] F. Liu, X. Tong, M. Yuan, X. Lin, F. Luo, Z. Wang, Z. Lu, and Q. Zhang. **Evolution of Heuristics: Towards Efficient Automatic Algorithm Design Using Large Language Model.** In *Proceedings of the 41st International Conference on Machine Learning, ICML 2024*, Proceedings of Machine Learning Research, pages 32201–32223. PMLR, 2024. doi:10.48550/arXiv.2401.02051.
- [1985] J. Liu and C. R. Reeves. **Constructive and Composite Heuristic Solutions to the P// Σ Ci Scheduling Problem.** *Eur. J. Oper. Res.*, 132(2):439–452, 2001. doi:10.1016/S0377-2217(00)00137-5.
- [1986] P. Liu, H. Wang, and W. Qiyu. **Bayesian Optimization with Switching Cost: Regret Analysis and Lookahead Variants.** In E. Elkind, editor, *Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI-2023)*, pages 4011–4018. IJCAI/AAAI Press, Menlo Park, CA, 2023. doi:10.24963/ijcai.2023/446.
- [1987] S. Liu, D. Maljovec, B. Wang, P.-T. Bremer, and V. Pascucci. **Visualizing High-Dimensional Data: Advances in the Past Decade.** *IEEE Transactions on Visualization and Computer Graphics*, 23(3), 2017. doi:10.1109/TVCG.2016.2640960.
- [1988] Y. Liu, G. G. Yen, and D. Gong. **A multimodal multiobjective evolutionary algorithm using two-archive and recombination strategies.** *IEEE Trans. Evol. Comput.*, 23(4): 660–674, 2018.
- [1989] Z. Liu and J. Tang. **IJCAI 2021 Reproducibility Guidelines, 35th International Joint Conference on Artificial Intelligence.** <https://ijcai-21.org/wp-content/uploads/2020/12/20201226-IJCAI-Reproducibility.pdf>, 2021.
- [1990] F. Lobo, C. F. Lima, and Z. Michalewicz, editors. *Parameter Setting in Evolutionary Algorithms*. Springer, Berlin, Germany, 2007.
- [1991] M. Locatelli and F. Schoen. **Random Linkage: a family of acceptance/rejection algorithms for global optimisation.** *Mathematical Programming*, 85(2), 1999. *Keywords:* Multi-Level Single-Linkage (MLSL).
- [1992] A. Lodi and A. Tramontani. **Performance Variability in Mixed-Integer Programming.** In H. Topaloglu, editor, *Theory Driven by Influential Applications*, pages 1–12. INFORMS, 2013.
- [1993] A. Lodi and G. Zarpellon. **On Learning and Branching: A Survey.** *TOP*, 25:207–236, 2017.

- [1994] A. Lodi, S. Martello, and D. Vigo. **Heuristic and metaheuristic approaches for a class of two-dimensional bin packing problems.** *INFORMS Journal on Computing*, 11(4):345–357, 1999. doi:10.1287/ijoc.11.4.345.
- [1995] A. Lodi, S. Martello, and M. Monaci. **Two-dimensional packing problems: A survey.** *Eur. J. Oper. Res.*, 141(2):241–252, 2002. doi:10.1016/S0377-2217(02)00123-6.
- [1996] A. Lodi, S. Martello, and D. Vigo. **TSPack: a unified tabu search code for multi-dimensional bin packing problems.** *Annals of Operations Research*, 131(1-4):203–213, 2004. doi:10.1023/B:ANOR.0000039519.03572.08.
- [1997] A. Lodi, S. Martello, and D. Vigo. **Two- and Three-Dimensional Bin Packing – Source Code of TSPack.** <https://site.unibo.it/operations-research/en/research/library-of-codes-and-instances-1/tspack-tar.gz/@download/file/TSPack.tar.gz>, 2004.
- [1998] A. Lodi, M. Milano, and P. Toth, editors. *Integration of AI and OR Techniques in Constraint Programming for Combinatorial Optimization Problems, 7th International Conference, CPAIOR 2010*, volume 6140 of *LNCS*. Springer, Heidelberg, Germany, 2010.
- [1999] P.-L. Loh and S. Nowozin. **Faster Hoeffding Racing: Bernstein Races via Jackknife Estimates.** In S. Jain, R. Munos, F. Stephan, and T. Zeugmann, editors, *Proc. of Algorithmic Learning Theory*, volume 8139 of *LNCS*, pages 203–217. Springer, Berlin, Germany, 2013. doi:10.1007/978-3-642-40935-6.
- [2000] J. D. Lohn, G. S. Hornby, and D. S. Linden. **Human-competitive Evolved Antennas.** *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*, 22(3):235–247, 2008. doi:10.1017/s0890060408000164.
Annotation: Evolutionary optimization of antennas for NASA.
- [2001] M. López-Ibáñez. *Multi-objective Ant Colony Optimization*. Diploma thesis, Intellectics Group, Computer Science Department, Technische Universität Darmstadt, Germany, 2004.
- [2002] M. López-Ibáñez. **High Performance Ant Colony Optimisation of the Pump Scheduling Problem.** In P. Alberigo, G. Erbacci, F. Garofalo, and S. Monfardini, editors, *Science and Sumpercomputing in Europe*, pages 371–375. CINECA, 2007. ISBN 978-88-86037-21-1.
- [2003] M. López-Ibáñez. *Operational Optimisation of Water Distribution Networks*. PhD thesis, School of Engineering and the Built Environment, Edinburgh Napier University, UK, 2009. URL <https://researchrepository.napier.ac.uk/id/eprint/3044>.
- [2004] M. López-Ibáñez. **Approximating the Hypervolume Indicator using Fast Low-Discrepancy Sequences.** In T. Hu and L. Trujillo, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026*. ACM Press, New York, NY, 2026. doi:10.1145/3795095.3805198.
Annotation: Proposed Rphi-FWE+.
- [2005] M. López-Ibáñez and C. Blum. **Beam-ACO Based on Stochastic Sampling: A Case Study on the TSP with Time Windows.** Technical Report LSI-08-28, Department LSI, Universitat Politècnica de Catalunya, 2008. Extended version published in *Comput. Oper. Res.* [2007].
- [2006] M. López-Ibáñez and C. Blum. **Beam-ACO Based on Stochastic Sampling: A Case Study on the TSP with Time Windows.** In T. Stützle, editor, *LION 3*, volume 5851 of *LNCS*, pages 59–73. Springer, Heidelberg, Germany, 2009. doi:10.1007/978-3-642-11169-3_5.
- [2007] M. López-Ibáñez and C. Blum. **Beam-ACO for the travelling salesman problem with time windows.** *Comput. Oper. Res.*, 37(9):1570–1583, 2010. doi:10.1016/j.cor.2009.11.015.
Keywords: Ant colony optimization, Travelling salesman problem with time windows, Hybridization.

- [2008] M. López-Ibáñez and J. D. Knowles. **Machine Decision Makers as a Laboratory for Interactive EMO**. In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part II*, volume 9019 of *LNCS*, pages 295–309. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-15892-1_20.
- [2009] M. López-Ibáñez and T. Stützle. **An Analysis of Algorithmic Components for Multiobjective Ant Colony Optimization: A Case Study on the Biobjective TSP**. Technical Report TR/IRIDIA/2009-019, IRIDIA, Université Libre de Bruxelles, Belgium, June 2009. Published in the proceedings of Evolution Artificielle, 2009 [2010].
- [2010] M. López-Ibáñez and T. Stützle. **An Analysis of Algorithmic Components for Multiobjective Ant Colony Optimization: A Case Study on the Biobjective TSP**. In P. Collet, N. Monmarché, P. Legrand, M. Schoenauer, and E. Lutton, editors, *Artificial Evolution, EA 2009*, volume 5975 of *LNCS*, pages 134–145. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-14156-0_12.
- [2011] M. López-Ibáñez and T. Stützle. **Automatic Configuration of Multi-Objective ACO Algorithms**. In M. Dorigo et al., editors, *ANTS 2010*, volume 6234 of *LNCS*, pages 95–106. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-15461-4_9.
- [2012] M. López-Ibáñez and T. Stützle. **The impact of design choices of multi-objective ant colony optimization algorithms on performance: An experimental study on the biobjective TSP**. In M. Pelikan and J. Branke, editors, *GECCO*, pages 71–78. ACM Press, New York, NY, 2010. doi:10.1145/1830483.1830494.
- [2013] M. López-Ibáñez and T. Stützle. **The impact of design choices of multi-objective ant colony optimization algorithms on performance: An experimental study on the biobjective TSP**. <https://iridia.ulb.ac.be/supp/IridiaSupp2010-003/>, 2010. Supplementary material of [2012].
- [2014] M. López-Ibáñez and T. Stützle. **The Automatic Design of Multi-Objective Ant Colony Optimization Algorithms**. Technical Report TR/IRIDIA/2011-003, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-003.pdf>. Published in IEEE Transactions on Evolutionary Computation [2020].
- [2015] M. López-Ibáñez and T. Stützle. **The Automatic Design of Multi-Objective Ant Colony Optimization Algorithms: Supplementary material**, 2011. URL <https://iridia.ulb.ac.be/supp/IridiaSupp2011-007/Iridia-2011-007.pdf>.
- [2016] M. López-Ibáñez and T. Stützle. **Automatically Improving the Anytime Behaviour of Optimisation Algorithms**. Technical Report TR/IRIDIA/2012-012, IRIDIA, Université Libre de Bruxelles, Belgium, May 2012. Published in Eur. J. Oper. Res. [2021].
- [2017] M. López-Ibáñez and T. Stützle. **Automatically Improving the Anytime Behaviour of Optimisation Algorithms: Supplementary material**. <https://iridia.ulb.ac.be/supp/IridiaSupp2012-011/>, 2012.
- [2018] M. López-Ibáñez and T. Stützle. **An experimental analysis of design choices of multi-objective ant colony optimization algorithms: Supplementary material**. <https://iridia.ulb.ac.be/supp/IridiaSupp2012-006/>, 2012.
- [2019] M. López-Ibáñez and T. Stützle. **An experimental analysis of design choices of multi-objective ant colony optimization algorithms**. *Swarm Intelligence*, 6(3):207–232, 2012. doi:10.1007/s11721-012-0070-7. Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2012-006/>.

- [2020] M. López-Ibáñez and T. Stützle. **The Automatic Design of Multi-Objective Ant Colony Optimisation Algorithms**. *IEEE Trans. Evol. Comput.*, 16(6):861–875, 2012. doi:[10.1109/TEVC.2011.2182651](https://doi.org/10.1109/TEVC.2011.2182651).
- [2021] M. López-Ibáñez and T. Stützle. **Automatically Improving the Anytime Behaviour of Optimisation Algorithms**. *Eur. J. Oper. Res.*, 235(3):569–582, 2014. doi:[10.1016/j.ejor.2013.10.043](https://doi.org/10.1016/j.ejor.2013.10.043). Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2012-011/>.
- [2022] M. López-Ibáñez, L. Paquete, and T. Stützle. **On the Design of ACO for the Biobjective Quadratic Assignment Problem**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 214–225. Springer, Heidelberg, Germany, 2004. doi:[10.1007/978-3-540-28646-2_19](https://doi.org/10.1007/978-3-540-28646-2_19).
- [2023] M. López-Ibáñez, L. Paquete, and T. Stützle. **Hybrid Population-based Algorithms for the Bi-objective Quadratic Assignment Problem**. Technical Report AIDA-04-11, FG Intellektik, FB Informatik, TU Darmstadt, Dec. 2004. Published in Journal of Mathematical Modelling and Algorithms [2026].
Annotation: First use of EAF differences.
- [2024] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Multi-objective Optimisation of the Pump Scheduling Problem using SPEA2**. In *Proceedings of the 2005 Congress on Evolutionary Computation (CEC 2005)*, volume 1, pages 435–442, Piscataway, NJ, Sept. 2005. IEEE Press. doi:[10.1109/CEC.2005.1554716](https://doi.org/10.1109/CEC.2005.1554716).
- [2025] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Optimal Pump Scheduling: Representation and Multiple Objectives**. In D. A. Savic, G. A. Walters, R. King, and S. Thiam-Khu, editors, *Proceedings of the Eighth International Conference on Computing and Control for the Water Industry (CCWI 2005)*, volume 1, pages 117–122, University of Exeter, UK, Sept. 2005.
- [2026] M. López-Ibáñez, L. Paquete, and T. Stützle. **Hybrid Population-based Algorithms for the Bi-objective Quadratic Assignment Problem**. *Journal of Mathematical Modelling and Algorithms*, 5(1):111–137, 2006. doi:[10.1007/s10852-005-9034-x](https://doi.org/10.1007/s10852-005-9034-x).
- [2027] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Solving Optimal Pump Control Problem using MAX-MIN Ant System**. In D. Thierens et al., editors, *GECCO*, volume 1, page 176. ACM Press, New York, NY, 2007. doi:[10.1145/1276958.1276990](https://doi.org/10.1145/1276958.1276990).
- [2028] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Parallel Optimisation Of Pump Schedules With A Thread-Safe Variant Of EPANET Toolkit**. In J. E. van Zyl, A. A. Ilemobade, and H. E. Jacobs, editors, *Proceedings of the 10th Annual Water Distribution Systems Analysis Conference (WDSA 2008)*. ASCE, Aug. 2008. doi:[10.1061/41024\(340\)40](https://doi.org/10.1061/41024(340)40).
- [2029] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Ant Colony Optimisation for the Optimal Control of Pumps in Water Distribution Networks**. *Journal of Water Resources Planning and Management, ASCE*, 134(4):337–346, 2008. doi:[10.1061/\(ASCE\)0733-9496\(2008\)134:4\(337\)](https://doi.org/10.1061/(ASCE)0733-9496(2008)134:4(337)).
- [2030] M. López-Ibáñez, C. Blum, D. Thiruvady, A. T. Ernst, and B. Meyer. **Beam-ACO based on stochastic sampling for makespan optimization concerning the TSP with time windows**. In C. Cotta and P. Cowling, editors, *EvoCOP*, volume 5482 of *LNCS*, pages 97–108. Springer, Heidelberg, Germany, 2009. doi:[10.1007/978-3-642-01009-5_9](https://doi.org/10.1007/978-3-642-01009-5_9).
- [2031] M. López-Ibáñez, L. Paquete, and T. Stützle. **Exploratory Analysis of Stochastic Local Search Algorithms in Biobjective Optimization**. Technical Report TR/IRIDIA/2009-015, IRIDIA, Université Libre de Bruxelles, Belgium, May 2009. Published as a book chapter [2032].

- [2032] M. López-Ibáñez, L. Paquete, and T. Stützle. **Exploratory Analysis of Stochastic Local Search Algorithms in Biobjective Optimization.** In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 209–222. Springer, Berlin/Heidelberg, 2010. doi:10.1007/978-3-642-02538-9_9.
- [2033] M. López-Ibáñez, L. Paquete, and T. Stützle. **EAF Graphical Tools.** <http://lopez-ibanez.eu/eaftools>, 2010. These tools are described in the book chapter “*Exploratory analysis of stochastic local search algorithms in biobjective optimization*” [2032].
Annotation: Please cite the book chapter, not this.
- [2034] M. López-Ibáñez, J. Dubois-Lacoste, T. Stützle, and M. Birattari. **The irace package, Iterated Race for Automatic Algorithm Configuration.** Technical Report TR/IRIDIA/2011-004, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-004.pdf>. Published in Oper. Res. Perspect. [2044].
- [2035] M. López-Ibáñez, J. D. Knowles, and M. Laumanns. **On Sequential Online Archiving of Objective Vectors.** Technical Report TR/IRIDIA/2011-001, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-001.pdf>. This is a revised version of the paper published in EMO 2011 [2036].
- [2036] M. López-Ibáñez, J. D. Knowles, and M. Laumanns. **On Sequential Online Archiving of Objective Vectors.** In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 46–60. Springer, Berlin/Heidelberg, 2011. doi:10.1007/978-3-642-19893-9_4.
Annotation: Revised version available at <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2011-001.pdf>.
- [2037] M. López-Ibáñez, T. D. Prasad, and B. Paechter. **Representations and Evolutionary Operators for the Scheduling of Pump Operations in Water Distribution Networks.** *Evol. Comput.*, 19(3):429–467, 2011. doi:10.1162/EVCO_a_00035.
- [2038] M. López-Ibáñez, T. Liao, and T. Stützle. **On the anytime behavior of IPOP-CMA-ES.** In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part I*, volume 7491 of *LNCS*, pages 357–366. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32937-1_36.
- [2039] M. López-Ibáñez, T. Liao, and T. Stützle. **On the anytime behavior of IPOP-CMA-ES: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2012-010/IridiaSupp2012-010.pdf>, 2012.
- [2040] M. López-Ibáñez, C. Blum, J. W. Ohlmann, and B. W. Thomas. **The Travelling Salesman Problem with Time Windows: Adapting Algorithms from Travel-time to Makespan Optimization.** *Applied Soft Computing*, 13(9):3806–3815, 2013. doi:10.1016/j.asoc.2013.05.009.
- [2041] M. López-Ibáñez, F. Mascia, M.-E. Marmion, and T. Stützle. **Automatic Design of a Hybrid Iterated Local Search for the Multi-Mode Resource-Constrained Multi-Project Scheduling Problem.** In G. Kendall, G. Vanden Berghe, and B. McCollum, editors, *Multidisciplinary International Conference on Scheduling: Theory and Applications (MISTA 2013)*, pages 1–6, Gent, Belgium, 2013.
- [2042] M. López-Ibáñez, A. Liefooghe, and S. Verel. **Local Optimal Sets and Bounded Archiving on Multi-objective NK-Landscapes with Correlated Objectives.** Technical Report TR/IRIDIA/2014-009, IRIDIA, Université Libre de Bruxelles, Belgium, 2014.

- [2043] M. López-Ibáñez, A. Liefooghe, and S. Verel. **Local Optimal Sets and Bounded Archiving on Multi-objective NK-Landscapes with Correlated Objectives**. In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCs*, pages 621–630. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-10762-2_61.
- [2044] M. López-Ibáñez, J. Dubois-Lacoste, L. Pérez Cáceres, T. Stützle, and M. Birattari. **The irace Package: Iterated Racing for Automatic Algorithm Configuration**. *Oper. Res. Perspect.*, 3:43–58, 2016. doi:10.1016/j.orp.2016.09.002. Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2016-003/>.
- [2045] M. López-Ibáñez, J. Dubois-Lacoste, L. Pérez Cáceres, T. Stützle, and M. Birattari. **The irace Package: Iterated Racing for Automatic Algorithm Configuration (Supplementary Material)**. <https://iridia.ulb.ac.be/supp/IridiaSupp2016-003>, 2016.
- [2046] M. López-Ibáñez, L. Pérez Cáceres, J. Dubois-Lacoste, T. Stützle, and M. Birattari. **The irace package: User Guide**. Technical Report TR/IRIDIA/2016-004, IRIDIA, Université Libre de Bruxelles, Belgium, 2016. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2016-004.pdf>.
- [2047] M. López-Ibáñez, M.-E. Kessaci, and T. Stützle. **Automatic Design of Hybrid Metaheuristics from Algorithmic Components**. Technical Report TR/IRIDIA/2017-012, IRIDIA, Université Libre de Bruxelles, Belgium, Dec. 2017. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2017-012.pdf>.
- [2048] M. López-Ibáñez, T. Stützle, and M. Dorigo. **Ant Colony Optimization: A Component-Wise Overview**. In R. Martí, P. M. Pardalos, and M. G. C. Resende, editors, *Handbook of Heuristics*, pages 371–407. Springer International Publishing, 2018. ISBN 978-3-319-07125-1. doi:10.1007/978-3-319-07124-4_21. Supplementary material: <https://iridia.ulb.ac.be/aco-tsp-qap/>.
Annotation: Proposed ACOTSPQAP software.
- [2049] M. López-Ibáñez, A. Auger, and T. Stützle, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2019, Prague, Czech Republic, July 13-17, 2019*. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.
- [2050] M. López-Ibáñez, A. Auger, and T. Stützle, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO 2019, Prague, Czech Republic, July 13-17, 2019*. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.
- [2051] M. López-Ibáñez, L. Pérez Cáceres, and T. Stützle. **irace: A Tool for the Automatic Configuration of Algorithms**. *International Federation of Operational Research Societies (IFORS) News*, 14(2):30–32, June 2020. URL <https://www.ifors.org/newsletter/ifors-news-june2020.pdf>.
- [2052] M. López-Ibáñez, J. Branke, and L. Paquete. **Reproducibility in Evolutionary Computation**. *Arxiv preprint arXiv:20102.03380 [cs.AI]*, 2021. URL <https://arxiv.org/abs/2102.03380>.
Keywords: Evolutionary Computation, Reproducibility, Empirical study, Benchmarking.
- [2053] M. López-Ibáñez, J. Branke, and L. Paquete. **Reproducibility in Evolutionary Computation**. *ACM Trans. Evol. Learn. Optim.*, 1(4):1–21, 2021. doi:10.1145/3466624.
Keywords: Evolutionary Computation, Reproducibility, Empirical study, Benchmarking.
- [2054] M. López-Ibáñez, N. Álvarez-Gil, S. Fernández Alzueta, and P. Valledor Pellicer. **A Real-world Case Study of Automatic Selection of GRASP Configurations with Single-selectors vs Multi-Selectors**. In *EURO 2022: Conference handbook and abstracts: 32nd European*

Conference on Operational Research (EURO XXXII), Helsinki, Finland, July 3–6, 2022, 2022.
ISBN 978-951-95254-1-9.

- [2055] M. López-Ibáñez, F. Chicano, and R. Gil-Merino. **The Asteroid Routing Problem: A Benchmark for Expensive Black-Box Permutation Optimization**. In J. L. Jiménez Laredo et al., editors, *EvoApplications 2022*, volume 13224 of *LNCS*, pages 124–140. Springer Nature, Switzerland, 2022. doi:10.1007/978-3-031-02462-7_9. Supplementary material: <https://doi.org/10.5281/zenodo.5725837>.
Keywords: Spacecraft Trajectory Optimization, Unbalanced Mallows Model, Combinatorial Efficient Global Optimization, Estimation of Distribution Algorithms, Bayesian Optimization.
- [2056] M. López-Ibáñez, D. Vermetten, J. Dreo, and C. Doerr. **Using the Empirical Attainment Function for Analyzing Single-objective Black-box Optimization Algorithms**. *IEEE Trans. Evol. Comput.*, 29(5):1774–1782, 2025. doi:10.1109/TEVC.2024.3462758.
Keywords: EAF-based ECDF.
Annotation: Pre-print: <https://doi.org/10.48550/arXiv.2404.02031>.
- [2057] A. López Jaimes, C. A. Coello Coello, and D. Chakraborty. **Objective reduction using a feature selection technique**. In C. Ryan, editor, *GECCO*, pages 673–680. ACM Press, New York, NY, 2008.
- [2058] A. López Jaimes, C. A. Coello Coello, and J. E. Urías Barrientos. **Online Objective Reduction to Deal with Many-Objective Problems**. In M. Ehrgott, C. M. Fonseca, X. Gandibleux, J.-K. Hao, and M. Sevaux, editors, *EMO*, volume 5467 of *LNCS*, pages 423–437. Springer, Heidelberg, Germany, 2009.
- [2059] I. Loshchilov and T. Glasmachers. **Black Box Optimization Competition**, 2017. URL <https://bbcomp.ini.rub.de/>.
- [2060] I. Loshchilov and F. Hutter. **SGDR: Stochastic Gradient Descent with Warm Restarts**. In *Proceedings of ICLR 2017*, 2017. doi:10.48550/arXiv.1608.03983.
- [2061] I. Loshchilov, M. Schoenauer, and M. Sebag. **Alternative Restart Strategies for CMA-ES**. In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part I*, volume 7491 of *LNCS*, pages 296–305. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32937-1_30.
- [2062] A. V. Lotov and K. Miettinen. **Visualizing the Pareto Frontier**. In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*, pages 213–243. Springer, Heidelberg, Germany, 2008.
- [2063] S. Loudni and P. Boizumault. **Combining VNS with constraint programming for solving anytime optimization problems**. *Eur. J. Oper. Res.*, 191:705–735, 2008. doi:10.1016/j.ejor.2006.12.062.
- [2064] H. R. Lourenço. **Job-Shop Scheduling: Computational Study of Local Search and Large-Step Optimization Methods**. *Eur. J. Oper. Res.*, 83(2):347–364, 1995.
- [2065] H. R. Lourenço, O. Martin, and T. Stützle. **Iterated Local Search**. In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 321–353. Kluwer Academic Publishers, Norwell, MA, 2002. doi:10.1007/0-306-48056-5_11.
- [2066] H. R. Lourenço, O. Martin, and T. Stützle. **Iterated Local Search: Framework and Applications**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, chapter 9, pages 363–397. Springer, New York, NY, 2nd edition, 2010. doi:10.1007/978-1-4419-1665-5_12.

- [2067] H. R. Lourenço, O. Martin, and T. Stützle. **Iterated Local Search: Framework and Applications**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 272 of *International Series in Operations Research & Management Science*, chapter 5, pages 129–168. Springer, 2019. doi:[10.1007/978-3-319-91086-4_5](https://doi.org/10.1007/978-3-319-91086-4_5).
- [2068] A. Lova and P. Tormos. **Analysis of Scheduling Schemes and Heuristic Rules Performance in Resource-Constrained Multiproject Scheduling**. *Annals of Operations Research*, 102(1-4):263–286, Feb. 2001. doi:[10.1023/A:1010966401888](https://doi.org/10.1023/A:1010966401888).
Keywords: Combinatorics, heuristic based on priority rules, Multiproject scheduling, Operation Research/Decision Theory, Project management, project management software, Resource allocation, Theory of Computation.
- [2069] A. Lova, P. Tormos, M. Cervantes, and F. Barber. **An efficient hybrid genetic algorithm for scheduling projects with resource constraints and multiple execution modes**. *International Journal of Production Economics*, 117(2):302–316, 2009. doi:[10.1016/j.ijpe.2008.11.002](https://doi.org/10.1016/j.ijpe.2008.11.002).
Keywords: genetic algorithm, multi-mode resource-constrained project scheduling.
- [2070] N. Lovell and L. Mainardi, editors. *37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2015, Proceedings*, Annual International Conference of the IEEE Engineering in Medicine and Biology, 2015. IEEE Press.
- [2071] M. Lozano, D. Molina, and C. García-Martínez. **Iterated Greedy for the Maximum Diversity Problem**. *Eur. J. Oper. Res.*, 214(1):31–38, 2011.
- [2072] M. Lozano, F. Glover, C. García-Martínez, F. J. Rodríguez, and R. Martí. **Tabu Search with Strategic Oscillation for the Quadratic Minimum Spanning Tree**. *IIE Transactions*, 46(4):414–428, 2014.
- [2073] Z. Lü, F. Glover, and J.-K. Hao. **A hybrid metaheuristic approach to solving the UBQP problem**. *Eur. J. Oper. Res.*, 207(3):1254–1262, 2010. doi:[10.1016/j.ejor.2010.06.039](https://doi.org/10.1016/j.ejor.2010.06.039).
- [2074] A. Lucas. **Ising formulations of many NP problems**. *Front. Phys.*, 2:5, 2014. doi:[10.3389/fphy.2014.00005](https://doi.org/10.3389/fphy.2014.00005).
- [2075] S. M. Lundberg and S. Lee. **A unified approach to interpreting model predictions**. In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NIPS 30)*, pages 4765–4774, 2017.
Keywords: SHAP, interpretable AI.
- [2076] M. Lundy and A. Mees. **Convergence of an Annealing Algorithm**. *Mathematical Programming*, 34(1):111–124, 1986.
- [2077] H. N. Luong and P. A. N. Bosman. **Elitist Archiving for Multi-Objective Evolutionary Algorithms: To Adapt or Not to Adapt**. In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part II*, volume 7492 of *LNCS*, pages 72–81. Springer, Heidelberg, Germany, 2012. doi:[10.1007/978-3-642-32964-7_8](https://doi.org/10.1007/978-3-642-32964-7_8).
- [2078] T. Lust and A. Jaskiewicz. **Speed-up techniques for solving large-scale biobjective TSP**. *Comput. Oper. Res.*, 37(3):521–533, 2010. doi:[10.1016/j.cor.2009.01.005](https://doi.org/10.1016/j.cor.2009.01.005).
Keywords: Multiobjective combinatorial optimization, Hybrid metaheuristics, TSP, Local search, Speed-up techniques.
- [2079] T. Lust and J. Teghem. **Two-phase Pareto local search for the biobjective traveling salesman problem**. *J. Heuristics*, 16(3):475–510, 2010. doi:[10.1007/s10732-009-9103-9](https://doi.org/10.1007/s10732-009-9103-9).

- [2080] T. Lust and J. Teghem. **The multiobjective traveling salesman problem: A survey and a new approach.** In C. A. Coello Coello, C. Dhaenens, and L. Jourdan, editors, *Advances in Multi-Objective Nature Inspired Computing*, volume 272 of *Studies in Computational Intelligence*, pages 119–141. Springer, 2010.
- [2081] T. Lust and J. Teghem. **The multiobjective multidimensional knapsack problem: a survey and a new approach.** *Arxiv preprint arXiv:1007.4063*, 2010. Published as [2082].
- [2082] T. Lust and J. Teghem. **The multiobjective multidimensional knapsack problem: a survey and a new approach.** *International Transactions in Operational Research*, 19(4): 495–520, 2012. doi:10.1111/j.1475-3995.2011.00840.x.
- [2083] E. Lutton, P. Legrand, P. Parrend, N. Monmarché, and M. Schoenauer, editors. *Artificial Evolution: 13th International Conference, Évolution Artificielle, EA 2017, Paris, France, October 25-27, 2017, Revised Selected*, volume 10764 of *LNCS*. Springer, Heidelberg, Germany, 2017.
- [2084] K. Lwin, R. Qu, and J. Zheng. **Multi-objective Scatter Search with External Archive for Portfolio Optimization.** In *Proceedings of the 5th International Joint Conference on Computational Intelligence - ECTA (IJCCI 2013)*, pages 111–119. SciTePress, 2013. doi:10.5220/0004552501110119.
Annotation: Crowding archive.
- [2085] R. J. Lygoe. *Complexity reduction in high-dimensional multi-objective optimisation*. PhD thesis, University of Sheffield Sheffield, UK, 2010.
- [2086] E. López-Camacho, H. Terashima-Marin, P. Ross, and G. Ochoa. **A unified hyper-heuristic framework for solving bin packing problems.** *Expert Systems with Applications*, 41(15): 6876–6889, 2014. doi:10.1016/j.eswa.2014.04.043.
- [2087] S. S. R. M., R. Mallipeddi, and K. N. Das. **A twin-archive guided decomposition based multi/many-objective evolutionary algorithm.** *Swarm and Evolutionary Computation*, 71: 101082, 2022. doi:10.1016/j.swevo.2022.101082.
- [2088] M. C. Machado, M. G. Bellemare, E. Talvitie, J. Veness, M. Hausknecht, and M. Bowling. **Revisiting the Arcade Learning Environment: Evaluation Protocols and Open Problems for General Agents.** *J. Artif. Intell. Res.*, 61(1):523–562, Jan. 2018. ISSN 1076-9757.
- [2089] G. Mäcke, D. A. Savic, and G. A. Walters. **Application of Genetic Algorithms to Pump Scheduling for Water Supply.** In *Genetic Algorithms in Engineering Systems: Innovations and Applications, GALEZIA’95*, volume 414, pages 400–405, Sheffield, UK, Sept. 1995. IEE Conference Publication. doi:10.1049/cp:19951082.
- [2090] N. K. Madavan. **Multiobjective optimization using a Pareto differential evolution approach.** In D. B. Fogel et al., editors, *WCCI*, pages 1145–1150, Piscataway, NJ, 2002. IEEE Press.
- [2091] S. Madden. **From Databases to Big Data.** *IEEE Internet Computing*, 16(3), 2012.
- [2092] M. Mahdavi, M. Fesanghary, and E. Damangir. **An improved harmony search algorithm for solving optimization problems.** *Applied Mathematics and Computation*, 188(2):1567–1579, 2007. doi:10.1016/j.amc.2006.11.033.
Keywords: Global optimization, Heuristics, Harmony search algorithm, Mathematical programming.
- [2093] M. Maher and J.-F. Puget, editors. *Principles and Practice of Constraint Programming, CP98*, volume 1520 of *LNCS*. Springer, Heidelberg, Germany, 1998.

- [2094] H. R. Maier, A. R. Simpson, A. C. Zecchin, W. K. Foong, K. Y. Phang, H. Y. Seah, and C. L. Tan. **Ant Colony Optimization for Design of Water Distribution Systems**. *Journal of Water Resources Planning and Management, ASCE*, 129(3):200–209, May / June 2003.
- [2095] G. B. Mainieri and D. P. Ronconi. **New heuristics for total tardiness minimization in a flexible flowshop**. *Optimization Letters*, pages 1–20, 2012.
- [2096] C. Maksimović, D. Butler, and F. A. Memon, editors. *Advances in Water Supply Management: Proceedings of the CCWI '03 Conference, London, 15-17 September 2003*. CRC Press, 2003.
- [2097] K. M. Malan and A. Engelbrecht. **A survey of techniques for characterising fitness landscapes and some possible ways forward**. *Inf. Sci.*, 241:148–163, 2013. ISSN 0020-0255. doi:10.1016/j.ins.2013.04.015.
- [2098] R. M. Males, R. M. Clark, P. J. Wehrman, and W. E. Gateset. **Algorithm for mixing problems in water systems**. *Journal of Hydraulic Engineering, ASCE*, 111(2):206–219, 1985.
- [2099] Y. Malitsky and M. Sellmann. **Instance-specific algorithm configuration as a method for non-model-based portfolio generation**. In N. Beldiceanu, N. Jussien, and E. Pinson, editors, *CPAIOR 2012*, volume 7298 of *LNCS*, pages 244–259. Springer, Heidelberg, Germany, 2012. ISBN 978-3-642-29827-1. doi:10.1007/978-3-642-29828-8_16.
- [2100] Y. Malitsky, D. Mehta, B. O’Sullivan, and H. Simonis. **Tuning parameters of large neighborhood search for the machine reassignment problem**. In C. Gomes and M. Sellmann, editors, *CPAIOR 2013*, volume 7874 of *LNCS*, pages 176–192. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-38171-3_12.
- [2101] Y. Malitsky, MehtaDeepak, and B. O’Sullivan. **Evolving Instance Specific Algorithm Configuration**. In M. Helmert and G. Röger, editors, *Proceedings of the Sixth Annual Symposium on Combinatorial Search, SOCS 2013*, pages 133–140. AAAI Press, 2013. doi:10.1609/SOCS.V4I1.18296.
- [2102] V. Maniezzo. **Exact and Approximate Nondeterministic Tree-Search Procedures for the Quadratic Assignment Problem**. *INFORMS Journal on Computing*, 11(4):358–369, 1999.
- [2103] V. Maniezzo and A. Carbonaro. **An ANTS Heuristic for the Frequency Assignment Problem**. *Future Generation Computer Systems*, 16(8):927–935, 2000.
- [2104] V. Maniezzo and A. Colorni. **The Ant System Applied to the Quadratic Assignment Problem**. *IEEE Transactions on Knowledge and Data Engineering*, 11(5):769–778, 1999.
- [2105] V. Maniezzo and M. Milandri. **An Ant-Based Framework for Very Strongly Constrained Problems**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 222–227. Springer, Heidelberg, Germany, 2002.
- [2106] V. Maniezzo, M. Boschetti, and M. Jelasity. **An Ant Approach to Membership Overlay Design**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 37–48. Springer, Heidelberg, Germany, 2004.
- [2107] V. Maniezzo, R. Battiti, and J.-P. Watson, editors. *Learning and Intelligent Optimization, Second International Conference, LION 2007, Trento, Italy, December 8-12, 2007. Selected Papers*, volume 5313 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [2108] V. Maniezzo, T. Stützle, and S. Voß, editors. *Matheuristics—Hybridizing Metaheuristics and Mathematical Programming*, volume 10 of *Annals of Information Systems*. Springer, New York, NY, 2009.

- [2109] R. Männer and B. Manderick, editors. *Parallel Problem Solving from Nature 2, PPSN-II, Brussels, Belgium, September 28-30, 1992*. Elsevier, 1992.
- [2110] C. D. Manning, M. Surdeanu, J. Bauer, J. R. Finkel, S. J. Bethard, and D. McClosky. **The Stanford CoreNLP Natural Language Processing Toolkit**. In *Association for Computational Linguistics (ACL) System Demonstrations*, pages 55–60, 2014.
Annotation: <http://www.aclweb.org/anthology/P/P14/P14-5010>.
- [2111] F. Marcelloni, K. Madani, N. van Stein, and J. Filipe, editors. *Proceedings of the 16th International Joint Conference on Computational Intelligence, IJCCI 2024, Porto, Portugal, November 20-22, 2024*, 2024. SciTePress.
- [2112] H. Marchand, A. Martin, R. Weismantel, and L. Wolsey. **Cutting planes in integer and mixed integer programming**. *Discrete Applied Mathematics*, 123(1-3):397–446, 2002.
- [2113] E. Marchiori and A. G. Steenbeek. **An Iterated Heuristic Algorithm for the Set Covering Problem**. In K. Mehlhorn, editor, *Algorithm Engineering, 2nd International Workshop, WAE'92*, pages 155–166. Max-Planck-Institut für Informatik, Saarbrücken, Germany, 1998.
- [2114] E. Marchiori and A. G. Steenbeek. **An Evolutionary Algorithm for Large Scale Set Covering Problems with Application to Airline Crew Scheduling**. In S. Cagnoni et al., editors, *Real-World Applications of Evolutionary Computing, EvoWorkshops 2000*, volume 1803 of *LNCS*, pages 367–381. Springer, Heidelberg, Germany, 2000.
- [2115] C. E. Mariano and E. Morales. **MOAQ: An Ant-Q Algorithm for Multiple Objective Optimization Problems**. In W. Banzhaf, J. M. Daida, A. E. Eiben, M. H. Garzon, V. Honavar, M. J. Jakiela, and R. E. Smith, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 1999*, pages 894–901. Morgan Kaufmann Publishers, San Francisco, CA, 1999.
- [2116] R. T. Marler and J. S. Arora. **Survey of multi-objective optimization methods for engineering**. *Struct. Multidisc. Optim.*, 26(6):369–395, Apr. 2004. doi:10.1007/s00158-003-0368-6.
Annotation: Discusses a priori (scalarized) methods.
- [2117] M.-E. Marmion, C. Dhaenens, L. Jourdan, A. Liefvooghe, and S. Verel. **NILS: A Neutrality-Based Iterated Local Search and Its Application to Flowshop Scheduling**. In P. Merz and J.-K. Hao, editors, *EvoCOP*, volume 6622 of *LNCS*, pages 191–202. Springer, Heidelberg, Germany, 2011.
- [2118] M.-E. Marmion, F. Mascia, M. López-Ibáñez, and T. Stützle. **Automatic Design of Hybrid Stochastic Local Search Algorithms**. In M. J. Blesa, C. Blum, P. Festa, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 7919 of *LNCS*, pages 144–158. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-38515-5. doi:10.1007/978-3-642-38516-2_12.
- [2119] O. Maron. **Hoeffding Races: Model selection for MRI classification**. Master’s thesis, Massachusetts Institute of Technology, 1994.
- [2120] O. Maron and A. W. Moore. **Hoeffding races: Accelerating model selection search for classification and function approximation**. In J. D. Cowan, G. Tesauro, and J. Alspector, editors, *Advances in Neural Information Processing Systems*, volume 6, pages 59–66. Morgan Kaufmann Publishers, San Francisco, CA, 1994.
- [2121] O. Maron and A. W. Moore. **The Racing Algorithm: Model Selection for Lazy Learners**. *Artificial Intelligence Research*, 11(1-5):193–225, 1997. doi:10.1023/A:1006556606079.
- [2122] K. Marriott and P. Stuckey. *Programming With Constraints*. MIT Press, Cambridge, MA, 1998.

- [2123] S. Martello and P. Toth. **Lower bounds and reduction procedures for the bin packing problem.** *Discrete Applied Mathematics*, 28(1):59–70, 1990. doi:10.1016/0166-218X(90)90094-S.
- [2124] S. Martello and P. Toth. *Knapsack Problems: Algorithms and Computer Implementations*. John Wiley & Sons, Chichester, UK, 1990.
Keywords: bin packing.
- [2125] S. Martello and D. Vigo. **Exact solution of the two-dimensional finite bin packing problem.** *Management Science*, 44(3):388–399, 1998. doi:10.1287/mnsc.44.3.388.
- [2126] D. Martens, M. D. Backer, R. Haesen, J. Vanthienen, M. Snoeck, and B. Baesens. **Classification With Ant Colony Optimization.** *IEEE Trans. Evol. Comput.*, 11(5):651–665, 2007.
- [2127] R. Martí, G. Reinelt, and A. Duarte. **A Benchmark Library and a Comparison of Heuristic Methods for the Linear Ordering Problem.** *Computational Optimization and Applications*, 51(3):1297–1317, 2012.
- [2128] R. Martí, P. M. Pardalos, and M. G. C. Resende, editors. *Handbook of Heuristics*. Springer International Publishing, 2018. ISBN 978-3-319-07125-1.
- [2129] O. Martin and S. W. Otto. **Partitioning of Unstructured Meshes for Load Balancing.** *Concurrency: Practice and Experience*, 7(4):303–314, 1995.
- [2130] O. Martin and S. W. Otto. **Combining Simulated Annealing with Local Search Heuristics.** *Annals of Operations Research*, 63:57–75, 1996.
- [2131] O. Martin, S. W. Otto, and E. W. Felten. **Large-Step Markov Chains for the Traveling Salesman Problem.** *Complex Systems*, 5(3):299–326, 1991.
- [2132] O. Martin, S. W. Otto, and E. W. Felten. **Large-step Markov Chains for the TSP Incorporating Local Search Heuristics.** *Operations Research Letters*, 11(4):219–224, 1992.
- [2133] R. Martín-Santamaría, J. Sánchez-Oro, S. Pérez-Peló, and A. Duarte. **Strategic oscillation for the balanced minimum sum-of-squares clustering problem.** *Inf. Sci.*, 585:529–542, 2022. doi:10.1016/j.ins.2021.11.048.
- [2134] R. Martín-Santamaría, S. Cavero, A. Herrán, A. Duarte, and J. M. Colmenar. **A Practical Methodology for Reproducible Experimentation: An Application to the Double-Row Facility Layout Problem.** *Evol. Comput.*, pages 1–35, Nov. 2023. ISSN 1063-6560. doi:10.1162/evco_a_00317.
Keywords: irace.
- [2135] R. Martín-Santamaría, M. López-Ibáñez, T. Stützle, and J. M. Colmenar. **On the automatic generation of metaheuristic algorithms for combinatorial optimization problems.** *Eur. J. Oper. Res.*, 318(3):740–751, 2024. doi:10.1016/j.ejor.2024.06.001.
Keywords: irace.
- [2136] R. Martín-Santamaría, M. López-Ibáñez, and J. M. Colmenar. **Empirical Complexity Analysis of Optimization Algorithms.** In G. Ochoa and B. Filipič, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2025*, pages 135–138. ACM Press, New York, NY, 2025. doi:10.1145/3712255.3726767.
Annotation: ISBN: 979-8-4007-1464-1.
- [2137] C. Martín-Vide, R. Neruda, and M. A. Vega-Rodríguez, editors. *Theory and Practice of Natural Computing - 6th International Conference, TPNC 2017*, volume 10687 of *LNCS*. Springer International Publishing, Cham, Switzerland, 2017.

- [2138] F. Martínez, V. Bou, V. Hernández, F. Alvarruiz, and J. M. Alonso. **ANN Architectures for Simulating Water Distribution Networks**. In D. A. Savic, G. A. Walters, R. King, and S. Thiam-Khu, editors, *Proceedings of the Eighth International Conference on Computing and Control for the Water Industry (CCWI 2005)*, volume 1, pages 251–256, University of Exeter, UK, Sept. 2005.
- [2139] E. Q. V. Martins. **On a multicriteria shortest path problem**. *Eur. J. Oper. Res.*, 16:236–245, 1984.
- [2140] F. Mascia, M. Birattari, and T. Stützle. **Tuning Algorithms for Tackling Large Instances: An Experimental Protocol**. In P. M. Pardalos and G. Nicosia, editors, *LION 7*, volume 7997 of *LNCS*, pages 410–422. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-44973-4_44.
- [2141] F. Mascia, M. López-Ibáñez, J. Dubois-Lacoste, and T. Stützle. **Grammar-based generation of stochastic local search heuristics through automatic algorithm configuration tools**. Technical Report TR/IRIDIA/2013-015, IRIDIA, Université Libre de Bruxelles, Belgium, 2013.
- [2142] F. Mascia, M. López-Ibáñez, J. Dubois-Lacoste, and T. Stützle. **Grammar-based generation of stochastic local search heuristics through automatic algorithm configuration tools: Supplementary material**. <https://iridia.ulb.ac.be/supp/IridiaSupp2013-009/>, 2013.
- [2143] F. Mascia, M. López-Ibáñez, J. Dubois-Lacoste, and T. Stützle. **From Grammars to Parameters: Automatic Iterated Greedy Design for the Permutation Flow-shop Problem with Weighted Tardiness**. In P. M. Pardalos and G. Nicosia, editors, *LION 7*, volume 7997 of *LNCS*, pages 321–334. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-44973-4_36.
- [2144] F. Mascia, M. López-Ibáñez, J. Dubois-Lacoste, M.-E. Marmion, and T. Stützle. **Algorithm Comparison by Automatically Configurable Stochastic Local Search Frameworks: A Case Study Using Flow-Shop Scheduling Problems**. In M. J. Blesa, C. Blum, and S. Voß, editors, *Hybrid Metaheuristics*, volume 8457 of *LNCS*, pages 30–44. Springer, Heidelberg, Germany, 2014. ISBN 978-3-319-07643-0. doi:10.1007/978-3-319-07644-7_3.
- [2145] F. Mascia, M. López-Ibáñez, J. Dubois-Lacoste, and T. Stützle. **Grammar-Based Generation of Stochastic Local Search Heuristics through Automatic Algorithm Configuration Tools**. *Comput. Oper. Res.*, 51:190–199, 2014. doi:10.1016/j.cor.2014.05.020.
- [2146] F. Mascia, P. Pellegrini, T. Stützle, and M. Birattari. **An Analysis of Parameter Adaptation in Reactive Tabu Search**. *International Transactions in Operational Research*, 21(1):127–152, 2014.
- [2147] F. Massen, Y. Deville, and P. van Hentenryck. **Pheromone-Based Heuristic Column Generation for Vehicle Routing Problems with Black Box Feasibility**. In N. Beldiceanu, N. Jussien, and E. Pinson, editors, *CPAIOR 2012*, volume 7298 of *LNCS*, pages 260–274. Springer, Heidelberg, Germany, 2012. ISBN 978-3-642-29827-1. doi:10.1007/978-3-642-29828-8_17.
- [2148] F. Massen, M. López-Ibáñez, T. Stützle, and Y. Deville. **Experimental Analysis of Pheromone-Based Heuristic Column Generation Using irace**. In M. J. Blesa, C. Blum, P. Festa, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 7919 of *LNCS*, pages 92–106. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-38515-5. doi:10.1007/978-3-642-38516-2_8.
- [2149] R. Massobrio, S. Nesmachnow, and B. Dorronsoro. **Virtual Savant: learning for optimization**. In M. Vlastelica, J. Song, A. Ferber, B. Amos, G. Martius, B. Dilkina, and Y. Yue, editors, *Learning Meets Combinatorial Algorithms Workshop at NeurIPS 2020*, pages 1–5, 2020.

- [2150] R. Masson, T. Vidal, J. Michallet, P. H. Vaz Penna, V. Petrucci, A. Subramanian, and H. Dubedout. **An Iterated Local Search Heuristic for Multi-capacity Bin Packing and Machine Reassignment Problems.** *Expert Systems with Applications*, 40(13):5266–5275, 2013.
- [2151] A. S. Masud and C. L. Hwang. **Interactive Sequential Goal Programming.** *J. Oper. Res. Soc.*, 32(5):391–400, May 1981. ISSN 1476-9360. doi:[10.1057/jors.1981.76](https://doi.org/10.1057/jors.1981.76).
- [2152] Y. Mati, S. Dauzère-Pères, and C. Lahlou. **A General Approach for Optimizing Regular Criteria in the Job-shop Scheduling Problem.** *Eur. J. Oper. Res.*, 212(1):33–42, 2011.
- [2153] N. F. Matsatsinis, Y. Marinakis, and P. M. Pardalos, editors. *13th International Conference, LION 13, Chania, Crete, Greece, May 27-31, 2019, Revised Selected Papers*, volume 11968 of *LNCS*. Springer, Cham, Switzerland, 2019.
- [2154] S. Matsubara, M. Takatsu, T. Miyazawa, T. Shibasaki, Y. Watanabe, K. Takemoto, and H. Tamura. **Digital Annealer for High-Speed Solving of Combinatorial optimization Problems and Its Applications.** In *2020 25th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pages 667–672. IEEE, 2020. doi:[10.1109/ASP-DAC47756.2020.9045100](https://doi.org/10.1109/ASP-DAC47756.2020.9045100).
- [2155] W. J. Matthews. **What might judgment and decision making research be like if we took a Bayesian approach to hypothesis testing?** *Judgment and Decision Making*, 6(8):843–856, 2011. doi:[10.1017/S1930297500004265](https://doi.org/10.1017/S1930297500004265).
- [2156] S. Matwin, S. Yu, and F. Farooq, editors. *KDD’17: The 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Halifax, NS, Canada, August 13-17, 2017*. ACM Press, 2017.
- [2157] M. Maur, M. López-Ibáñez, and T. Stützle. **Pre-scheduled and adaptive parameter variation in MAX–MLN Ant System.** In H. Ishibuchi et al., editors, *Proceedings of the 2010 Congress on Evolutionary Computation (CEC 2010)*, pages 3823–3830, Piscataway, NJ, 2010. IEEE Press. doi:[10.1109/CEC.2010.5586332](https://doi.org/10.1109/CEC.2010.5586332).
- [2158] A. Mazumdar, T. Chugh, K. Miettinen, and M. López-Ibáñez. **On Dealing with Uncertainties from Kriging Models in Offline Data-Driven Evolutionary Multiobjective Optimization.** In K. Deb, E. D. Goodman, C. A. Coello Coello, K. Klamroth, K. Miettinen, S. Mostaghim, and P. M. Reed, editors, *Evolutionary Multi-criterion Optimization, EMO 2019*, volume 11411 of *LNCS*, pages 463–474. Springer International Publishing, Cham, Switzerland, 2019. ISBN 978-3-030-12597-4. doi:[10.1007/978-3-030-12598-1_37](https://doi.org/10.1007/978-3-030-12598-1_37).
- [2159] A. Mazumdar, M. López-Ibáñez, T. Chugh, and K. Miettinen. **Treed Gaussian Process Regression for Solving Offline Data-Driven Continuous Multiobjective Optimization Problems.** *Evol. Comput.*, 31(4):375–399, 2023. doi:[10.1162/evco_a_00329](https://doi.org/10.1162/evco_a_00329).
Keywords: Gaussian processes, Kriging, Regression trees, Metamodelling, Surrogate, Pareto optimality.
- [2160] B. McCollum, A. Schaerf, B. Paechter, P. McMullan, R. M. R. Lewis, A. J. Parkes, L. Di Gaspero, R. Qu, and E. K. Burke. **Setting the Research Agenda in Automated Timetabling: The Second International Timetabling Competition.** *INFORMS*, 22(1):120–130, Feb. 2010. doi:[10.1287/ijoc.1090.0320](https://doi.org/10.1287/ijoc.1090.0320).
- [2161] R. M. McConnell, K. Mehlhorn, S. Näher, and P. Schweitzer. **Certifying algorithms.** *Computer Science Review*, 5(2):119–161, 2011. ISSN 1574-0137. doi:[10.1016/j.cosrev.2010.09.009](https://doi.org/10.1016/j.cosrev.2010.09.009).
Keywords: Algorithms, Software reliability, Certification.

- [2162] G. McCormick and R. S. Powell. **Optimal Pump Scheduling in Water Supply Systems with Maximum Demand Charges**. *Journal of Water Resources Planning and Management, ASCE*, 129(5):372–379, 2003. doi:10.1061/(ASCE)0733-9496(2003)129:5(372).
Keywords: water supply; pumps; Markov processes; cost optimal control.
- [2163] G. McCormick and R. S. Powell. **A progressive mixed integer-programming method for pump scheduling**. In C. Maksimović, D. Butler, and F. A. Memon, editors, *Advances in Water Supply Management*, pages 307–313. CRC Press, 2003.
- [2164] G. McCormick and R. S. Powell. **Derivation of near-optimal pump schedules for water distribution by simulated annealing**. *J. Oper. Res. Soc.*, 55(7):728–736, July 2004. doi:10.1057/palgrave.jors.2601718.
- [2165] J. McDermott. **When and Why Metaheuristics Researchers can Ignore "No Free Lunch" Theorems**. *SN Computer Science*, 1(60):1–18, 2020. doi:10.1007/s42979-020-0063-3.
- [2166] J. McDermott, M. Castelli, L. Sekanina, E. Haasdijk, and P. García-Sánchez, editors. *Genetic Programming, 20th European Conference, EuroGP 2017, Amsterdam, The Netherlands, April 19-21, 2017, Proceedings*, volume 10196 of *LNCS*. Springer, Heidelberg, Germany, 2017. ISBN 978-3-319-55695-6. doi:10.1007/978-3-319-55696-3.
- [2167] C. C. McGeoch. **Analyzing Algorithms by Simulation: Variance Reduction Techniques and Simulation Speedups**. *ACM Comput. Surv.*, 24(2):195–212, 1992. doi:10.1145/130844.130853.
Keywords: experimental analysis of algorithms, move-to-front rule, self-organizing sequential search, statistical analysis of algorithms, transpose rule, variance reduction techniques.
- [2168] C. C. McGeoch. **Toward an Experimental Method for Algorithm Simulation**. *INFORMS Journal on Computing*, 8(1):1–15, 1996. doi:10.1287/ijoc.8.1.1.
- [2169] C. C. McGeoch. *A Guide to Experimental Algorithmics*. Cambridge University Press, 2012.
- [2170] C. C. McGeoch and P. Farré. **The D-Wave Advantage System: An Overview**. Technical report, D-Wave Systems Inc., Burnaby, BC, Canada, 2020. URL https://www.dwavesys.com/media/s3qbjp3s/14-1049a-a_the_d-wave_advantage_system_an_overview.pdf.
- [2171] R. McGill, J. W. Tukey, and W. A. Larsen. **Variations of Box Plots**. *The American Statistician*, 32(1):12–16, 1978. doi:10.1080/00031305.1978.10479236.
Annotation: Proposed notched boxplots.
- [2172] S. A. McIlraith and K. Q. Weinberger, editors. *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, February 2-7, 2018, New Orleans, Louisiana, USA*. AAAI Press, Feb. 2018.
- [2173] M. D. McKay, R. J. Beckman, and W. J. Conover. **A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code**. *Technometrics*, 21(2):239–245, 1979. doi:10.2307/1268522.
Annotation: Republished as [2174].
- [2174] M. D. McKay, R. J. Beckman, and W. J. Conover. **A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code**. *Technometrics*, 42(1):55–61, 2000. doi:10.1080/00401706.2000.10485979.
Annotation: This is a republication of [2173].
- [2175] R. I. McKay, N. X. Hoai, P. A. Whigham, Y. Shan, and M. O'Neill. **Grammar-based Genetic Programming: A Survey**. *Genetic Programming and Evolvable Machines*, 11(3-4):365–396, Sept. 2010. doi:10.1007/s10710-010-9109-y.

- [2176] R. McKenna, V. Bertsch, K. Mainzer, and W. Fichtner. **Combining local preferences with multi-criteria decision analysis and linear optimization to develop feasible energy concepts in small communities.** *Eur. J. Oper. Res.*, 268(3):1092–1110, 2018.
- [2177] H. G. d. Medeiros, E. F. G. Goldberg, and M. C. Goldberg. **Analyzing Limited Size Archivers of Multi-objective Optimizers.** In *2014 Brazilian Conference on Intelligent Systems*, pages 85–90, 2014. doi:10.1109/BRACIS.2014.26.
Keywords: archiving.
- [2178] E. Medvet, G. Pappa, and B. Xue, editors. *Genetic Programming, 25th European Conference, EuroGP 2022, Held as Part of EvoStar 2022, Madrid, Spain, April 20-22, 2022, Proceedings.* LNCS. Springer Nature, Cham, Switzerland, 2022.
- [2179] K. Meer. **Simulated annealing versus Metropolis for a TSP instance.** *Inf. Process. Lett.*, 104(6):216–219, 2007.
- [2180] K. Mehlhorn, editor. *Algorithm Engineering, 2nd International Workshop, WAE’92*, 1998. Max-Planck-Institut für Informatik, Saarbrücken, Germany.
- [2181] J. Mehnen, M. Köppen, A. Saad, and A. Tiwari, editors. *Applications of Soft Computing*, volume 58 of *Advances in Intelligent and Soft Computing*. Springer, Berlin/Heidelberg, 2009.
- [2182] J. F. Meier and U. Clausen. **A versatile heuristic approach for generalized hub location problems.** Preprint, Provided upon personal request, 2014. URL <https://optimization-online.org/wp-content/uploads/2014/12/4690.pdf>.
Keywords: irace.
- [2183] G. Melis, C. Dyer, and P. Blunsom. **On the State of the Art of Evaluation in Neural Language Models.** *Arxiv preprint arXiv:1807.02811*, 2017. URL <http://arxiv.org/abs/1707.05589>.
- [2184] C. S. Mellish, editor. *Proceedings of the 14th International Joint Conference on Artificial Intelligence, IJCAI 95, Montréal Québec, Canada, August 20-25, 1995, 2 Volumes.* Morgan Kaufmann Publishers, 1995.
- [2185] L. Melo, F. Pereira, and E. Costa. **MC-ANT: a Multi-colony Ant Algorithm.** In P. Collet, N. Monmarché, P. Legrand, M. Schoenauer, and E. Lutton, editors, *Artificial Evolution, EA 2009*, volume 5975 of LNCS. Springer, Heidelberg, Germany, 2010.
- [2186] M. T. Melo, S. Nickel, and F. Saldanha-da Gama. **Facility location and supply chain management: A review.** *Eur. J. Oper. Res.*, 196(2):401–412, 2009. doi:10.1016/j.ejor.2008.05.007.
- [2187] M. Melo Silva, A. Subramanian, and L. S. Ochi. **An Iterated Local Search Heuristic for the Split Delivery Vehicle Routing Problem.** *Comput. Oper. Res.*, 53:234–249, 2015.
- [2188] G. Melvin, T. J. Dodd, and R. Groß. **Why ‘GSA: a gravitational search algorithm’ is not genuinely based on the law of gravity.** *Natural Computing*, 11(4):719–720, 2012.
- [2189] A. Menchaca-Mendez and C. A. Coello Coello. **GD-MOEA: A New Multi-Objective Evolutionary Algorithm Based on the Generational Distance Indicator.** In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of LNCS, pages 156–170. Springer, Heidelberg, Germany, 2015.
- [2190] A. Menchaca-Mendez and C. A. Coello Coello. **GDE-MOEA: A New MOEA based on the generational distance indicator and ϵ -dominance.** In *Proceedings of the 2015 Congress on Evolutionary Computation (CEC 2015)*, pages 947–955, Piscataway, NJ, 2015. IEEE Press.

- [2191] H. Mendoza, A. Klein, M. Feurer, J. T. Springenberg, and F. Hutter. **Towards automatically-tuned neural networks**. In *Workshop on Automatic Machine Learning*, pages 58–65, 2016.
- [2192] O. J. Mengshoel. **Understanding the role of noise in stochastic local search: Analysis and experiments**. *Artif. Intell.*, 172(8):955–990, 2008.
- [2193] J.-J. Merelo and C. Cotta. **Building bridges: the role of subfields in metaheuristics**. *SIGEVolution*, 1(4):9–15, 2006. doi:[10.1145/1229735.1229737](https://doi.org/10.1145/1229735.1229737).
- [2194] J.-J. Merelo et al., editors. *Parallel Problem Solving from Nature – PPSN VII*, volume 2439 of *LNCS*. Springer, Heidelberg, Germany, 2002.
Annotation: IC.34.
- [2195] D. Merkle and M. Middendorf. **Prospects for Dynamic Algorithm Control: Lessons from the Phase Structure of Ant Scheduling Algorithms**. In R. B. Heckendorn, editor, *Proceedings of the 2001 Genetic and Evolutionary Computation Conference – Workshop Program. Workshop “The Next Ten Years of Scheduling Research”*, pages 121–126. Morgan Kaufmann Publishers, San Francisco, CA, 2001.
- [2196] D. Merkle and M. Middendorf. **Modeling the Dynamics of Ant Colony Optimization**. *Evol. Comput.*, 10(3):235–262, 2002.
- [2197] D. Merkle and M. Middendorf. **Ant Colony Optimization with Global Pheromone Evaluation for Scheduling a Single Machine**. *Applied Intelligence*, 18(1):105–111, 2003.
- [2198] D. Merkle, M. Middendorf, and H. Schneck. **Ant Colony Optimization for Resource-Constrained Project Scheduling**. In D. Whitley et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2000*, pages 893–900. Morgan Kaufmann Publishers, San Francisco, CA, 2000.
- [2199] D. Merkle, M. Middendorf, and H. Schneck. **Ant Colony Optimization for Resource-Constrained Project Scheduling**. *IEEE Trans. Evol. Comput.*, 6(4):333–346, 2002.
- [2200] O. Mersmann. *mco: Multiple Criteria Optimization Algorithms and Related Functions*, 2014. URL <http://CRAN.R-project.org/package=mco>. R package version 1.0-15.1.
- [2201] O. Mersmann, H. Trautmann, B. Naujoks, and C. Weihs. **Benchmarking Evolutionary Multiobjective Optimization Algorithms**. In H. Ishibuchi et al., editors, *Proceedings of the 2010 Congress on Evolutionary Computation (CEC 2010)*, pages 1–8, Piscataway, NJ, 2010. IEEE Press.
Annotation: TR: <http://hdl.handle.net/2003/26671>.
- [2202] O. Mersmann, B. Bischl, H. Trautmann, M. Preuss, C. Weihs, and G. Rudolph. **Exploratory Landscape Analysis**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 829–836. ACM Press, New York, NY, 2011. doi:[10.1145/2001576.2001690](https://doi.org/10.1145/2001576.2001690).
Keywords: continuous optimization, landscape analysis, instance features.
- [2203] P. Merz and B. Freisleben. **Fitness Landscape Analysis and Memetic Algorithms for the Quadratic Assignment Problem**. *IEEE Trans. Evol. Comput.*, 4(4):337–352, 2000.
- [2204] P. Merz and B. Freisleben. **Memetic Algorithms for the Traveling Salesman Problem**. *Complex Systems*, 13(4):297–345, 2001.
- [2205] P. Merz and B. Freisleben. **Greedy and Local Search Heuristics for Unconstrained Binary Quadratic Programming**. *J. Heuristics*, 8(2):197–213, 2002. doi:[10.1023/A:1017912624016](https://doi.org/10.1023/A:1017912624016).

- [2206] P. Merz and J.-K. Hao, editors. *EvoCOP*, volume 6622 of *LNCS*. Springer, Heidelberg, Germany, 2011.
- [2207] P. Merz and J. Huhse. **An Iterated Local Search Approach for Finding Provably Good Solutions for Very Large TSP Instances.** In G. Rudolph et al., editors, *Parallel Problem Solving from Nature – PPSN X*, volume 5199 of *LNCS*, pages 929–939. Springer, Heidelberg, Germany, 2008.
- [2208] P. Merz and K. Katayama. **Memetic algorithms for the unconstrained binary quadratic programming problem.** *BioSystems*, 78(1):99–118, 2004. doi:[10.1016/j.biosystems.2004.08.002](https://doi.org/10.1016/j.biosystems.2004.08.002).
- [2209] R. G. Mesquita, R. M. A. Silva, C. A. B. Mello, and P. B. C. Miranda. **Parameter tuning for document image binarization using a racing algorithm.** *Expert Systems with Applications*, 42(5):2593–2603, 2015. doi:[10.1016/j.eswa.2014.10.039](https://doi.org/10.1016/j.eswa.2014.10.039).
Keywords: irace.
- [2210] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. Teller, and E. Teller. **Equation of State Calculations by Fast Computing Machines.** *Journal of Chemical Physics*, 21:1087–1092, 1953.
- [2211] N. Meuleau and M. Dorigo. **Ant Colony Optimization and Stochastic Gradient Descent.** *Artificial Life*, 8(2):103–121, 2002.
- [2212] L. Meunier, H. Rakotoarison, P. Wong, B. Rozière, J. Rapin, O. Teytaud, A. Moreau, and C. Doerr. **Black-Box Optimization Revisited: Improving Algorithm Selection Wizards through Massive Benchmarking.** *Arxiv preprint arXiv:2010.04542*, 2020. doi:[10.48550/arXiv.2010.04542](https://doi.org/10.48550/arXiv.2010.04542).
Keywords: Nevergrad, NGOpt.
- [2213] L. Meunier, H. Rakotoarison, P. Wong, B. Rozière, J. Rapin, O. Teytaud, A. Moreau, and C. Doerr. **Black-Box Optimization Revisited: Improving Algorithm Selection Wizards Through Massive Benchmarking.** *IEEE Trans. Evol. Comput.*, 26(3):490–500, 2022. doi:[10.1109/TEVC.2021.3108185](https://doi.org/10.1109/TEVC.2021.3108185).
Keywords: nevergrad, NGOpt.
- [2214] B. Meyer. **Convergence control in ACO.** In *Genetic and Evolutionary Computation Conference (GECCO)*, Seattle, WA, 2004. Late-breaking paper available on CD.
- [2215] B. Meyer and A. T. Ernst. **Integrating ACO and Constraint Propagation.** In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 166–177. Springer, Heidelberg, Germany, 2004.
- [2216] E. Mezura-Montes, J. Velázquez-Reyes, and C. A. Coello Coello. **A comparative study of differential evolution variants for global optimization.** In M. Cattolico et al., editors, *GECCO*, pages 485–492. ACM Press, New York, NY, 2006. doi:[10.1145/1143997.1144086](https://doi.org/10.1145/1143997.1144086).
- [2217] E. Mezura-Montes, M. Reyes-Sierra, and C. A. Coello Coello. **Multi-objective optimization using differential evolution: a survey of the state-of-the-art.** In U. K. Chakraborty, editor, *Advances in differential evolution*, pages 173–196. Springer, Heidelberg, Germany, 2008. doi:[10.1007/978-3-540-68830-3_7](https://doi.org/10.1007/978-3-540-68830-3_7).
- [2218] R. M’Hallah. **An iterated local search variable neighborhood descent hybrid heuristic for the total earliness tardiness permutation flow shop.** *Int. J. Prod. Res.*, 52(13):3802–3819, 2014.
- [2219] MIC. *Proceedings of MIC 2013, the 10th Metaheuristics International Conference*, 2013.

- [2220] Z. Michalewicz, editor. *Proceedings of the First IEEE International Conference on Evolutionary Computation (ICEC'94)*. IEEE Press, Piscataway, NJ, 1994.
- [2221] Z. Michalewicz. *Genetic Algorithms + Data Structures = Evolution Programs*. Springer, Berlin, Germany, 3rd edition, 1996.
- [2222] Z. Michalewicz and D. B. Fogel. *How to Solve It: Modern Heuristics*. Springer, 2nd edition, 2004.
- [2223] Z. Michalewicz, D. Dasgupta, R. G. L. Riche, and M. Schoenauer. **Evolutionary algorithms for constrained engineering problems**. *Computers & Industrial Engineering*, 30(4):851–870, 1996. doi:10.1016/0360-8352(96)00037-X.
- [2224] J. Michallet, C. Prins, F. Yalaoui, and G. Vitry. **Multi-start Iterated Local Search for the Periodic Vehicle Routing Problem with Time Windows and Time Spread Constraints on Services**. *Comput. Oper. Res.*, 41:196–207, 2014.
- [2225] L. D. Michel and P. van Hentenryck. **Iterative Relaxations for Iterative Flattening in Cumulative Scheduling**. In S. Zilberstein, J. Koehler, and S. Koenig, editors, *Proceedings of the Fourteenth International Conference on Automated Planning and Scheduling (ICAPS 2004)*, pages 200–208. AAAI Press/MIT Press, Menlo Park, CA, 2004.
- [2226] R. Michel and M. Middendorf. **An Island Model based Ant System with Lookahead for the Shortest Supersequence Problem**. In A. E. Eiben, T. Bäck, M. Schoenauer, and H.-P. Schwefel, editors, *Parallel Problem Solving from Nature – PPSN V*, volume 1498 of *LNCS*, pages 692–701. Springer, Heidelberg, Germany, 1998.
- [2227] M. Middendorf and C. Blum, editors. *Evolutionary Computation in Combinatorial Optimization – 13th European Conference, EvoCOP 2013, Vienna, Austria, April 3-5, 2013, Proceedings*, volume 7832 of *LNCS*. Springer, Heidelberg, Germany, 2013.
- [2228] K. Miettinen. *Nonlinear Multiobjective Optimization*. Kluwer Academic Publishers, Boston, MA, 1999.
- [2229] K. Miettinen. **IND-NIMBUS for Demanding Interactive Multiobjective Optimization**. In T. Trzaskalik, editor, *Multiple Criteria Decision Making '05*, pages 137–150. Karol Adamiecki University of Economics in Katowice, 2006. ISBN 8372468435.
- [2230] K. Miettinen. **Survey of methods to visualize alternatives in multiple criteria decision making problems**. *OR Spectrum*, 36(1):3–37, 2014. doi:10.1007/s00291-012-0297-0.
- [2231] K. Miettinen, F. Ruiz, and A. P. Wierzbicki. **Introduction to Multiobjective Optimization: Interactive Approaches**. In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of *LNCS*, pages 27–57. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-88908-3_2.
- [2232] K. Miettinen, P. Eskelinen, F. Ruiz, and M. Luque. **NAUTILUS method: An interactive technique in multiobjective optimization based on the nadir point**. *Eur. J. Oper. Res.*, 206(2):426–434, Oct. 2010. ISSN 0377-2217. doi:10.1016/j.ejor.2010.02.041.
Keywords: Reference point methods, Interactive methods, Multiple objective programming, Pareto optimality, Preference information.
- [2233] K. Miettinen, J. Mustajoki, and T. J. Stewart. **Interactive multiobjective optimization with NIMBUS for decision making under uncertainty**. *OR Spectrum*, 36(1):39–56, 2014.
- [2234] R. B. Millar and M. J. Anderson. **Remedies for pseudoreplication**. *Fisheries Research*, 70(2–3):397–407, 2004. doi:10.1016/j.fishres.2004.08.016.

- [2235] G. A. Miller. **The magical number seven, plus or minus two: Some limits on our capacity for processing information.** *Psychological Review*, 63(2):81–97, 1956. doi:10.1037/h0043158.
- [2236] R. E. Miller and W. Thatcher, James, editors. *Complexity of Computer Computations*, The IBM Research Symposia Series, 1972. Springer.
- [2237] G. Minella, R. Ruiz, and M. Ciavotta. **A Review and Evaluation of Multiobjective Algorithms for the Flowshop Scheduling Problem.** *INFORMS Journal on Computing*, 20(3):451–471, 2008.
- [2238] S. Minton. **Automatically configuring constraint satisfaction programs: A case study.** *Constraints*, 1(1):7–43, 1996. doi:10.1007/BF00143877.
- [2239] P. Miranda, R. M. Silva, and R. B. Prudêncio. **Fine-Tuning of Support Vector Machine Parameters Using Racing Algorithms.** In *European Symposium on Artificial Neural Networks, ESSAN*, pages 325–330, 2014.
Keywords: irace.
- [2240] P. Miranda, R. M. Silva, and R. B. Prudêncio. **I/S-Race: An Iterative Multi-objective Racing Algorithm for the SVM Parameter Selection Problem.** In *European Symposium on Artificial Neural Networks, ESSAN*, pages 573–578, 2015.
- [2241] A. Misevičius. **Genetic Algorithm Hybridized with Ruin and Recreate Procedure: Application to the Quadratic Assignment Problem.** *Knowledge-Based Systems*, 16(5–6): 261–268, 2003.
- [2242] A. Misevičius. **Ruin and Recreate Principle Based Approach for the Quadratic Assignment Problem.** In E. Cantú-Paz et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2003, Part I*, volume 2723 of *LNCS*, pages 598–609. Springer, Heidelberg, Germany, 2003.
- [2243] A. Misevičius. **A modified simulated annealing algorithm for the quadratic assignment problem.** *Informatica*, 14(4):497–514, 2003.
- [2244] A. Misevičius and D. Kuznecovaitė. **Investigating some strategies for construction of initial populations in genetic algorithms.** *Computational Science and Techniques*, 5(1): 560–573, 2018.
- [2245] A. Misevičius, D. Kuznecovaitė, and J. Platužienė. **Some Further Experiments with Crossover Operators for Genetic Algorithms.** *Informatica*, 29(3):499–516, 2018.
- [2246] G. Misitano, B. Afsar, G. Larraga, and K. Miettinen. **Towards explainable interactive multiobjective optimization: R-XIMO.** *Autonomous Agents and Multi-Agent Systems*, 36(42), 2022. doi:10.1007/s10458-022-09577-3.
- [2247] D. G. Mitchell, B. Selman, and H. J. Levesque. **Hard and Easy Distributions of SAT Problems.** In W. R. Swartout, editor, *Proceedings of the 10th National Conference on Artificial Intelligence*, pages 459–465. AAAI Press/MIT Press, Menlo Park, CA, 1992.
- [2248] D. Mitra, F. Romeo, and A. Sangiovanni-Vincentelli. **Convergence and Finite-Time Behavior of Simulated Annealing.** In *Decision and Control, 1985 24th IEEE Conference on*, pages 761–767. IEEE, 1985.
- [2249] P. Mitra, C. A. Murthy, and S. K. Pal. **Unsupervised feature selection using feature similarity.** *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(3):301–312, 2002. doi:10.1109/34.990133.

- [2250] ML4AAD Group. **SMAC v3 Project**. <https://github.com/automl/SMAC3>, 2017. Version visited last on August 2017.
- [2251] N. Mladenović and P. Hansen. **Variable Neighborhood Search**. *Comput. Oper. Res.*, 24(11): 1097–1100, 1997.
- [2252] V. Mnih, C. Szepesvári, and J.-Y. Audibert. **Empirical Bernstein stopping**. In W. W. Cohen, A. McCallum, and S. T. Roweis, editors, *Proceedings of the 25th International Conference on Machine Learning, ICML 2008*, pages 672–679, New York, NY, 2008. ACM Press.
- [2253] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, et al. **Human-level control through deep reinforcement learning**. *Nature*, 518(7540):529, 2015.
- [2254] J. Moćkus. **On Bayesian Methods for Seeking the Extremum**. In G. I. Marchuk, editor, *Optimization Techniques IFIP Technical Conference Novosibirsk, July 1–7, 1974*, volume 27 of *LNCS*, pages 400–404. Springer, Berlin/Heidelberg, 1975. doi:10.1007/3-540-07165-2_55. Annotation: Proposed Bayesian optimization.
- [2255] J. Moćkus. *Bayesian Approach to Global Optimization: Theory and Applications*. Kluwer Academic Publishers, 1989.
- [2256] J. Moćkus, V. Tiesis, and A. Zilinskas. **The application of bayesian methods for seeking the extremum**. *Towards global optimization*, pages 117–129, 1978. Annotation: Proposed Bayesian optimization (but later than [2254]).
- [2257] A. Moghaddam, F. Yalaoui, and L. Amodeo. **Lorenz versus Pareto Dominance in a Single Machine Scheduling Problem with Rejection**. In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 520–534. Springer, Berlin/Heidelberg, 2011.
- [2258] I. Molchanov. *Theory of Random Sets*. Springer, 2005. Keywords: Vorob’ev expectation.
- [2259] T. M. Moldovan and P. Abbeel. **Safe Exploration in Markov Decision Processes**. In J. Langford and J. Pineau, editors, *Proceedings of the 29th International Conference on Machine Learning, ICML 2012*, pages 1451–1458. Omnipress, 2012.
- [2260] J. Molina, L. V. Santana, A. G. Hernández-Díaz, C. A. Coello Coello, and R. Caballero. **g-Dominance: Reference point based dominance for Multiobjective Metaheuristics**. *Eur. J. Oper. Res.*, 197(2):685–692, Sept. 2009. doi:10.1016/j.ejor.2008.07.015. Annotation: Proposed g-NSGA-II.
- [2261] A. Mondoro, D. M. Frangopol, and L. Liu. **Multi-criteria robust optimization framework for bridge adaptation under climate change**. *Structural Safety*, 74:14–23, 2018.
- [2262] J.-N. Monette, Y. Deville, and P. van Hentenryck. **Aeon: Synthesizing Scheduling Algorithms from High-Level Models**. In J. W. Chinneck, B. Kristjansson, and M. J. Saltzman, editors, *Operations Research and Cyber-Infrastructure*, volume 47 of *Operations Research/Computer Science Interfaces*, pages 43–59. Springer, New York, NY, 2009.
- [2263] N. Monmarché, G. Venturini, and M. Slimane. **On how *pachycondyla apicalis* ants suggest a new search algorithm**. *Future Generation Computer Systems*, 16(8):937–946, 2000.
- [2264] N. Monmarché, E.-G. Talbi, P. Collet, M. Schoenauer, and E. Lutton, editors. *Artificial Evolution, 8th International Conference, Evolution Artificielle, EA 2007, Tours, France, October 29-31, 2007 Revised Selected Papers*, volume 4926 of *LNCS*. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-79305-2.

- [2265] R. Monroy, G. Arroyo-Figueroa, L. E. Sucar, and H. Sossa, editors. *MICAI 2004: Advances in Artificial Intelligence: Third Mexican International Conference on Artificial Intelligence, Mexico City, Mexico, April 26-30, 2004. Proceedings*, volume 2972 of *LNAI*. Springer, Heidelberg, Germany, 2004.
- [2266] R. Montemanni, L. M. Gambardella, A. E. Rizzoli, and A. V. Donati. **Ant colony system for a dynamic vehicle routing problem**. *Journal of Combinatorial Optimization*, 10:327–343, 2005.
- [2267] E. Montero and M.-C. Riff. **Towards a Method for Automatic Algorithm Configuration: A Design Evaluation Using Tuners**. In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 90–99. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-10762-2_9.
- [2268] E. Montero, M.-C. Riff, and B. Neveu. **An Evaluation of Off-line Calibration Techniques for Evolutionary Algorithms**. In M. Pelikan and J. Branke, editors, *GECCO*, pages 299–300. ACM Press, New York, NY, 2010.
- [2269] E. Montero, L. Pérez Cáceres, M.-C. Riff, and C. A. Coello Coello. **Are State-of-the-Art Fine-Tuning Algorithms Able to Detect a Dummy Parameter?** In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part I*, volume 7491 of *LNCS*, pages 306–315. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32937-1_31.
- [2270] E. Montero, M.-C. Riff, and B. Neveu. **A Beginner’s Buide to Tuning Methods**. *Applied Soft Computing*, 17:39–51, 2014. doi:10.1016/j.asoc.2013.12.017.
- [2271] M. A. Montes de Oca. *Incremental Social Learning in Swarm Intelligence Systems*. PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2011. Annotation: Supervised by Marco Dorigo.
- [2272] M. A. Montes de Oca, T. Stützle, M. Birattari, and M. Dorigo. **Frankenstein’s PSO: A Composite Particle Swarm Optimization Algorithm**. *IEEE Trans. Evol. Comput.*, 13(5): 1120–1132, 2009. doi:10.1109/TEVC.2009.2021465.
- [2273] M. A. Montes de Oca, D. Aydın, and T. Stützle. **An Incremental Particle Swarm for Large-Scale Continuous Optimization Problems: An Example of Tuning-in-the-loop (Re)Design of Optimization Algorithms**. *Soft Computing*, 15(11):2233–2255, 2011. doi:10.1007/s00500-010-0649-0.
- [2274] D. C. Montgomery. *Design and Analysis of Experiments*. John Wiley & Sons, New York, NY, 8th edition, 2012.
- [2275] J. Montgomery. *Solution Biases and Pheromone Representation Selection in Ant Colony Optimisation*. PhD thesis, School of Information Technology, Bond University, Australia, 2005.
- [2276] J. Montgomery, M. Randall, and T. Hendtlass. **Solution bias in ant colony optimisation: Lessons for selecting pheromone models**. *Comput. Oper. Res.*, 35(9):2728–2749, 2008. doi:10.1016/j.cor.2006.12.014.
- [2277] G. Montibeller and H. Yoshizaki. **A Framework for Locating Logistic Facilities with Multi-Criteria Decision Analysis**. In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 505–519. Springer, Berlin/Heidelberg, 2011.
- [2278] A. W. Moore and M. S. Lee. **Efficient Algorithms for Minimizing Cross Validation Error**. In W. W. Cohen and H. Hirsh, editors, *Proceedings of the 11th International Conference on Machine Learning, ICML 1994*, pages 190–198, San Francisco, CA, 1994. Morgan Kaufmann Publishers.

- [2279] J. Moosbauer, J. Herbinger, G. Casalicchio, M. T. Lindauer, and B. Bischl. **Explaining Hyperparameter Optimization via Partial Dependence Plots.** In M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. W. Vaughan, editors, *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pages 2280–2291, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/12ced2db6f0193dda91ba86224ea1cd8-Paper.pdf.
- [2280] A. M. Mora and G. Squillero, editors. *Applications of Evolutionary Computation - 18th European Conference, EvoApplications 2015, Copenhagen, Denmark, April 8 – 10, 2015, Proceedings*, volume 9028 of *LNCS*. Springer, Heidelberg, Germany, 2015.
- [2281] A. M. Mora, J.-J. Merelo, J. L. Jiménez Laredo, C. Millan, and J. Torrecillas. **CHAC, a MOACO algorithm for computation of bi-criteria military unit path in the battlefield: Presentation and first results.** *International Journal of Intelligent Systems*, 24(7):818–843, 2009.
- [2282] M. Morabit, G. Desaulniers, and A. Lodi. **Machine-learning-based column selection for column generation.** *Transportation Science*, 55(4):815–831, 2021.
Keywords: graph neural networks.
- [2283] A. Moraglio and A. Kattan. **Geometric Generalisation of Surrogate Model Based Optimization to Combinatorial Spaces.** In P. Merz and J.-K. Hao, editors, *EvoCOP*, volume 6622 of *LNCS*, pages 142–154. Springer, Heidelberg, Germany, 2011.
- [2284] A. Moraglio, Y. Kim, and Y. Yoon. **Geometric Surrogate-based Optimisation for Permutation-based Problems.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO Companion*, pages 133–134. ACM Press, New York, NY, 2011.
- [2285] A. Moraglio, S. Silva, K. Krawiec, P. Machado, and C. Cotta, editors. *Genetic Programming, 15th European Conference on Genetic Programming, EuroGP 2012, Proceedings*, volume 7244 of *LNCS*. Springer, Heidelberg, Germany, 2012.
- [2286] P. D. Morgan. **Simulation of an adaptive behavior mechanism in an expert decision-maker.** *IEEE Transactions on Systems, Man, and Cybernetics*, 23(1):65–76, 1993.
- [2287] S. Morin, C. Gagné, and M. Gravel. **Ant colony optimization with a specialized pheromone trail for the car-sequencing problem.** *Eur. J. Oper. Res.*, 197(3):1185–1191, 2009. doi:10.1016/j.ejor.2008.03.033.
Keywords: Ant colony optimization, Car-sequencing problem, Pheromone trail, Scheduling.
- [2288] M. D. Morris and T. J. Mitchell. **Exploratory designs for computational experiments.** *Journal of Statistical Planning and Inference*, 43(3):381–402, 1995. doi:10.1016/0378-3758(94)00035-T.
Keywords: Bayesian prediction.
- [2289] P. Morris. **The Breakout Method for Escaping from Local Minima.** In R. Fikes and W. G. Lehnert, editors, *Proceedings of the 11th National Conference on Artificial Intelligence*, pages 40–45. AAAI Press/MIT Press, Menlo Park, CA, 1993.
- [2290] J. N. Morse. **Reducing the size of the nondominated set: Pruning by clustering.** *Comput. Oper. Res.*, 7(1-2):55–66, 1980.
- [2291] P. Moscato. **On Evolution, Search, Optimization, Genetic Algorithms and Martial Arts: Towards Memetic Algorithms.** Caltech Concurrent Computation Program, C3P Report 826, Caltech, 1989.
- [2292] P. Moscato. **Memetic algorithms: a short introduction.** In D. Corne, M. Dorigo, and F. Glover, editors, *New Ideas in Optimization*, pages 219–234. McGraw Hill, London, UK, 1999.

- [2293] P. Moscato and J. F. Fontanari. **Stochastic Versus Deterministic Update in Simulated Annealing.** *Physics Letters A*, 146(4):204–208, 1990.
- [2294] J. D. Moss and C. G. Johnson. **An ant colony algorithm for multiple sequence alignment in bioinformatics.** In D. W. Pearson, N. C. Steele, and R. F. Albrecht, editors, *Artificial Neural Networks and Genetic Algorithms*, pages 182–186. Springer Verlag, 2003.
- [2295] J. Mostow and C. Rich, editors. *Proceedings of the Fifteenth National Conference on Artificial Intelligence and Tenth Innovative Applications of Artificial Intelligence Conference, AAAI 98, IAAI 98, July 26-30, 1998, Madison, Wisconsin, USA*, 1998. AAAI Press/MIT Press, Menlo Park, CA.
- [2296] J. Mote, D. L. Olson, and M. A. Venkataramanan. **A comparative multiobjective programming study.** *Mathematical and Computer Modelling*, 10(10):719–729, 1988. doi:10.1016/0895-7177(88)90085-4.
Keywords: artificial DM, interactive.
Annotation: The purpose of this study was to systematically evaluate a number of multiobjective programming concepts relative to reflection of utility, assurance of nondominated solutions and practicality for larger problems using conventional software. In the problem used, the nonlinear simulated DM utility function applied resulted in a nonextreme point solution. Very often, the preferred solution could end up being an extreme point solution, in which case the techniques relying upon LP concepts would work as well if not better than utilizing constrained objective attainments. The point is that there is no reason to expect linear or near linear utility.
- [2297] J. Mote, I. Murthy, and D. L. Olson. **A parametric approach to solving bicriterion shortest path problems.** *Eur. J. Oper. Res.*, 53(1):81–92, 1991. doi:10.1016/0377-2217(91)90094-C.
- [2298] A. Moura and J. F. Oliveira. **A GRASP Approach to the Container-Loading Problem.** *IEEE Intelligent Systems*, 20(4):50–57, July 2005. doi:10.1109/mis.2005.57.
- [2299] L. Mousin, M.-E. Kessaci, and C. Dhaenens. **Exploiting Promising Sub-Sequences of Jobs to solve the No-Wait Flowshop Scheduling Problem.** *Arxiv preprint arXiv:1903.09035*, 2019. URL <http://arxiv.org/abs/1903.09035>.
- [2300] V. Mousseau. *Elicitation des préférences pour l'aide multicritère à la décision.* PhD thesis, Université Paris-Dauphine, Paris, France, 2003.
- [2301] V. Mousseau and R. Słowiński. **Inferring an ELECTRE TRI model from assignment examples.** *J. Glob. Optim.*, 12(2):157–174, 1998.
- [2302] S. Mouthuy, Y. Deville, and P. van Hentenryck. **Constraint-based Very Large-Scale Neighborhood Search.** *Constraints*, 17(2):87–122, 2012. doi:10.1007/s10601-011-9114-7.
- [2303] J. Moy. **RFC 1583: Open shortest path first protocol**, 1994.
- [2304] M. Mozer, M. I. Jordan, and T. Petsche, editors. *Advances in Neural Information Processing Systems 9, NIPS, Denver, CO, USA, December 2-5, 1996*. MIT Press, 1996.
- [2305] Z. Mu and H. H. Hoos. **On the empirical time complexity of random 3-SAT at the phase transition.** In Q. Yang and M. Wooldridge, editors, *Proc. of IJCAI-15*, pages 367–373. IJCAI/AAAI Press, Menlo Park, CA, 2015.
- [2306] Z. Mu, H. H. Hoos, and T. Stützle. **The Impact of Automated Algorithm Configuration on the Scaling Behaviour of State-of-the-Art Inexact TSP Solvers.** In P. Festa, M. Sellmann, and J. Vanschoren, editors, *LION 10*, volume 10079 of *LNCS*, pages 157–172. Springer, Cham, Switzerland, 2016. doi:10.1007/978-3-319-50349-3_11.

- [2307] Z. Mu, J. Dubois-Lacoste, H. H. Hoos, and T. Stützle. **On the Empirical Scaling of Running Time for Finding Optimal Solutions to the TSP: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2017-010/>, 2017.
- [2308] Z. Mu, J. Dubois-Lacoste, H. H. Hoos, and T. Stützle. **On the empirical scaling of running time for finding optimal solutions to the TSP.** *J. Heuristics*, 24(6):879–898, June 2018. doi:10.1007/s10732-018-9374-0.
Annotation: asymptotic complexity.
- [2309] F. Mueller, editor. *Proceedings of the 2011 IEEE International Parallel & Distributed Processing Symposium*, IPDPS '11, 2011. IEEE Computer Society.
- [2310] H. Mühlenbein. **Evolution in Time and Space—The Parallel Genetic Algorithm.** In G. Rawlins, editor, *Foundations of Genetic Algorithms (FOGA)*, pages 316–337. Morgan Kaufmann Publishers, San Mateo, CA, 1991.
- [2311] H. Mühlenbein and D. Schlierkamp-Voosen. **Predictive models for the breeder genetic algorithm.** *Evol. Comput.*, 1(1):25–49, 1993.
Keywords: crossover, intermediate, line.
- [2312] H. Mühlenbein and J. Zimmermann. **Size of neighborhood more important than temperature for stochastic local search.** In *IEEE CEC*, pages 1017–1024, Piscataway, NJ, July 2000. IEEE Press.
- [2313] M. Mühlenthaler. *Fairness in Academic Course Timetabling*. Springer, 2015. doi:10.1007/978-3-319-12799-6.
Keywords: irace.
- [2314] C. L. Müller and I. F. Sbalzarini. **Energy Landscapes of Atomic Clusters as Black Box Optimization Benchmarks.** *Evol. Comput.*, 20(4):543–573, 2012. doi:10.1162/EVCO_a_00086.
- [2315] M. E. Muller. **A Note on a Method for Generating Points Uniformly on N-Dimensional Spheres.** *Commun. ACM*, 2(4):19–20, 1959. doi:10.1145/377939.377946.
Keywords: hypersphere, sampling.
- [2316] M. A. Muñoz and K. Smith-Miles. **Effects of function translation and dimensionality reduction on landscape analysis.** In *Proceedings of the 2015 Congress on Evolutionary Computation (CEC 2015)*, pages 1336–1342, Piscataway, NJ, 2015. IEEE Press. doi:10.1109/CEC.2015.7257043.
- [2317] M. A. Muñoz and K. Smith-Miles. **Generating New Space-Filling Test Instances for Continuous Black-Box Optimization.** *Evol. Comput.*, 28(3):379–404, Sept. 2020. doi:10.1162/evco_a_00262.
- [2318] M. A. Muñoz, Y. Sun, M. Kirley, and S. K. Halgamuge. **Algorithm selection for black-box continuous optimization problems: a survey on methods and challenges.** *Inf. Sci.*, 317: 224–245, 2015.
- [2319] M. A. Muñoz, L. Villanova, D. Baatar, and K. Smith-Miles. **Instance Spaces for Machine Learning Classification.** *Mach. Learn.*, 107(1):109–147, 2018. doi:10.1007/s10994-017-5629-5.
Keywords: Projecting Instances with Linearly Observable Trends (PILOT).
- [2320] L. J. Murphy, G. C. Dandy, and A. R. Simpson. **Optimum Design and Operation of Pumped Water Distribution Systems.** In *1994 International Conference on Hydraulics and Civil Engineering, Hydraulic working with the Environment*, pages 149–155, Brisbane, Australia, Feb. 1994. The Institution of Engineers.

- [2321] I. Murray, M. Ranzato, and O. Vinyals, editors. *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Workshop Track Proceedings*, 2018. OpenReview.net.
- [2322] T. Mussweiler. **The malleability of anchoring effects**. *Experimental Psychology*, 49(1):67–72, 2002. doi:10.1027//1618-3169.49.1.67.
- [2323] J. Mylopoulos and R. Reiter, editors. *Proceedings of the 12th International Joint Conference on Artificial Intelligence, IJCAI 91, Sydney, Australia, August 24-30, 1991*. Morgan Kaufmann Publishers, 1995.
- [2324] NAFIPS. *Proceedings of the NAFIPS-FLINT International Conference'2002*, Piscataway, New Jersey, June 2002. IEEE Service Center.
- [2325] M. S. Nagano, F. L. Rossi, and N. J. Martarelli. **High-performing heuristics to minimize flowtime in no-idle permutation flowshop**. *Engineering Optimization*, 51(2):185–198, 2019.
- [2326] Y. Nagata and S. Kobayashi. **Edge Assembly Crossover: A High-power Genetic Algorithm for the Traveling Salesman Problem**. In T. Bäck, editor, *ICGA*, pages 450–457. Morgan Kaufmann Publishers, San Francisco, CA, 1997.
- [2327] Y. Nagata and S. Kobayashi. **An analysis of edge assembly crossover for the traveling salesman problem**. In K. Ito, F. Harashima, and K. Tanie, editors, *IEEE SMC'99 Conference Proceedings, 1999 IEEE International Conference on Systems, Man, and Cybernetics*, pages 628–633. IEEE Press, 1999. doi:10.1109/icsmc.1999.823285.
- [2328] Y. Nagata and S. Kobayashi. **A Powerful Genetic Algorithm Using Edge Assembly Crossover for the Traveling Salesman Problem**. *INFORMS Journal on Computing*, 25(2):346–363, 2013. doi:10.1287/ijoc.1120.0506.
Keywords: TSP, EAX, evolutionary algorithms.
- [2329] Y. Nagata and D. Soler. **A New Genetic Algorithm for the Asymmetric TSP**. *Expert Systems with Applications*, 39(10):8947–8953, 2012.
- [2330] R. Nagy, M. Suci, and D. Dumitrescu. **Exploring Lorenz Dominance**. In *Symbolic and Numeric Algorithms for Scientific Computing (SYNASC), 2012 14th International Symposium on*, pages 254–259, 2012.
- [2331] V. Nair and G. E. Hinton. **Rectified linear units improve restricted boltzmann machines**. In J. Fürnkranz and T. Joachims, editors, *Proceedings of the 27th International Conference on Machine Learning, ICML 2010*, pages 807–814, New York, NY, 2010. ACM Press.
- [2332] S. Nallaperuma, M. Wagner, and F. Neumann. **Parameter Prediction Based on Features of Evolved Instances for Ant Colony Optimization and the Traveling Salesperson Problem**. In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 100–109. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-10762-2_10.
- [2333] S. Nallaperuma, P. S. Oliveto, J. P. Heredia, and D. Sudholt. **On the Analysis of Trajectory-Based Search Algorithms: When is it Beneficial to Reject Improvements?** *Algorithmica*, 81(2):858–885, 2019.
- [2334] Y. Nan, K. Shang, H. Ishibuchi, and L. He. **Reverse Strategy for Non-Dominated Archiving**. *IEEE Access*, 8:119458–119469, 2020. doi:10.1109/ACCESS.2020.3005970.
- [2335] V. Nannen and A. E. Eiben. **A Method for Parameter Calibration and Relevance Estimation in Evolutionary Algorithms**. In M. Cattolico et al., editors, *GECCO*, pages 183–190. ACM Press, New York, NY, 2006. doi:10.1145/1143997.1144029.
Keywords: REVAC.

- [2336] V. Nannen and A. E. Eiben. **Relevance Estimation and Value Calibration of Evolutionary Algorithm Parameters**. In M. M. Veloso, editor, *Proc. of IJCAI-07*, pages 975–980. AAAI Press, Menlo Park, CA, 2007.
Keywords: REVAC.
- [2337] K. Naono, K. Teranishi, J. Cavazos, and R. Suda, editors. *Software Automatic Tuning: From Concepts to State-of-the-Art Results*. Springer, 2010.
- [2338] K. Narukawa, Y. Setoguchi, Y. Tanigaki, M. Olhofer, B. Sendhoff, and H. Ishibuchi. **Preference representation using Gaussian functions on a hyperplane in evolutionary multi-objective optimization**. *Soft Computing*, 20(7):2733–2757, July 2016. doi:10.1007/s00500-015-1674-9.
- [2339] J. Nash and R. Varadhan. **Unifying Optimization Algorithms to Aid Software System Users: optimx for R**. *Journal of Statistical Software*, 43(9):1–14, 2011.
- [2340] Y. S. G. Nashed, P. Mesejo, R. Ugolotti, J. Dubois-Lacoste, and S. Cagnoni. **A Comparative Study of Three GPU-Based Metaheuristics**. In C. A. Coello Coello et al., editors, *Parallel Problem Solving from Nature – PPSN XII, Part II*, volume 7492 of *LNCS*, pages 398–407. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-32964-7_40.
- [2341] M. Nawaz, E. Ensore, Jr, and I. Ham. **A Heuristic Algorithm for the m -Machine, n -Job Flow-Shop Sequencing Problem**. *Omega*, 11(1):91–95, 1983.
Keywords: NEH heuristic.
- [2342] B. Nebel, editor. *IJCAI 2001, Proceedings of the 17th International Joint Conference on Artificial Intelligence*, 2001. IEEE Press.
- [2343] A. J. Nebro, F. Luna, E. Alba, B. Dorronsoro, J. J. Durillo, and A. Beham. **AbYSS: Adapting Scatter Search to Multiobjective Optimization**. *IEEE Trans. Evol. Comput.*, 12(4): 439–457, Aug. 2008. doi:10.1109/TEVC.2007.913109.
- [2344] A. J. Nebro, J. J. Durillo, J. García-Nieto, C. A. Coello Coello, F. Luna, and E. Alba. **SMPSO: A new PSO-based metaheuristic for multi-objective optimization**. In *2009 IEEE Symposium on Computational Intelligence in Multi-Criteria Decision-Making (MCDM)*, pages 66–73, 2009. ISBN 9781424427642. doi:10.1109/MCDM.2009.4938830.
- [2345] A. J. Nebro, J. J. Durillo, and C. A. Coello Coello. **Analysis of leader selection strategies in a multi-objective Particle Swarm Optimizer**. In *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, pages 3153–3160, Piscataway, NJ, 2013. IEEE Press. doi:10.1109/CEC.2013.6557955.
- [2346] A. J. Nebro, J. J. Durillo, and M. Vergne. **Redesigning the jMetal Multi-Objective Optimization Framework**. In J. L. Jiménez Laredo, S. Silva, and A. I. Esparcia-Alcázar, editors, *GECCO Companion*, pages 1093–1100. ACM Press, New York, NY, 2015. doi:10.1145/2739482.2768462.
Keywords: jMetal, Multi-objective metaheuristics, Open source, Optimization framework.
- [2347] A. J. Nebro, M. López-Ibáñez, C. Barba-González, and J. García-Nieto. **Automatic Configuration of NSGA-II with jMetal and irace**. In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1374–1381. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.3326832.
- [2348] A. J. Nebro, M. López-Ibáñez, J. García-Nieto, and C. A. Coello Coello. **On the automatic design of multi-objective particle swarm optimizers: experimentation and analysis**. *Swarm Intelligence*, 18:105–139, 2024. doi:10.1007/s11721-023-00227-2.

- [2349] G. L. Nemhauser and L. A. Wolsey. *Integer and Combinatorial Optimization*. John Wiley & Sons, New York, NY, 1988.
- [2350] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. **An Analysis of Approximations for Maximizing Submodular Set Functions—I**. *Mathematical Programming*, 14(1):265–294, 1978. doi:10.1007/BF01588971.
Keywords: Greedy Algorithm,Heuristics,Interchange Algorithm,Linear Programming,Matroid Optimization,Submodular Set Functions.
- [2351] F. Neri, C. Cotta, and P. Moscato, editors. *Handbook of Memetic Algorithms*, volume 379 of *Studies in Computational Intelligence*. Springer, 2011.
- [2352] F. Neri and C. Cotta. **Memetic algorithms and memetic computing optimization: A literature review**. *Swarm and Evolutionary Computation*, 2:1–14, 2012. doi:10.1016/j.swevo.2011.11.003.
- [2353] F. Neumann and C. Witt. **Runtime Analysis of a Simple Ant Colony Optimization Algorithm**. *Electronic Colloquium on Computational Complexity (ECCC)*, 13(084), 2006.
- [2354] F. Neumann, D. Sudholt, and C. Witt. **Analysis of different MMAS ACO algorithms on unimodal functions and plateaus**. *Swarm Intelligence*, 3(1):35–68, 2009.
- [2355] A. Newell and H. A. Simon. **Computer Science as Empirical Inquiry: Symbols and Search**. *Commun. ACM*, 19(3):113–126, Mar. 1976. ISSN 0001-0782. doi:10.1145/360018.360022.
Keywords: cognition, Turing, search, problem solving, symbols, heuristics, list processing, computer science, artificial intelligence, science, empirical.
- [2356] A.-T. Nguyen, S. Reiter, and P. Rigo. **A review on simulation-based optimization methods applied to building performance analysis**. *Applied Energy*, 113:1043–1058, 2014. doi:10.1016/j.apenergy.2013.08.061.
- [2357] S. Nguyen, M. Zhang, M. Johnston, and K. C. Tan. **Genetic Programming for Evolving Due-Date Assignment Models in Job Shop Environments**. *Evol. Comput.*, 22(1):105–138, 2014.
- [2358] S. Nguyen, M. Zhang, M. Johnston, and K. C. Tan. **Automatic Design of Scheduling Policies for Dynamic Multi-objective Job Shop Scheduling via Cooperative Coevolution Genetic Programming**. *IEEE Trans. Evol. Comput.*, 18(2):193–208, 2014.
- [2359] T. T. Nguyen, S. Yang, and J. Branke. **Evolutionary Dynamic Optimization: A Survey of the State of the Art**. *Swarm and Evolutionary Computation*, 6:1–24, 2012.
- [2360] V.-P. Nguyen, C. Prins, and C. Prodhon. **A Multi-start Iterated Local Search with Tabu List and Path Relinking for the Two-echelon Location-routing Problem**. *Engineering Applications of Artificial Intelligence*, 25(1):56–71, 2012.
- [2361] P. Nightingale, O. Akgün, I. P. Gent, C. Jefferson, I. Miguel, and P. Spracklen. **Automatically Improving Constraint Models in Savile Row**. *Artif. Intell.*, 251:35–61, 2017.
- [2362] A. G. Nikolaev and S. H. Jacobson. **Simulated Annealing**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, pages 1–39. Springer, New York, NY, 2nd edition, 2010.
- [2363] M. Nikolić, F. Marić, and P. Janičić. **Instance-based selection of policies for SAT solvers**. In *International Conference on Theory and Applications of Satisfiability Testing*, pages 326–340. Springer, 2009.

- [2364] C. Ning and F. You. **Optimization under uncertainty in the era of big data and deep learning: When machine learning meets mathematical programming.** *Computers & Chemical Engineering*, 125:434–448, 2019. [doi:10.1016/j.compchemeng.2019.03.034](https://doi.org/10.1016/j.compchemeng.2019.03.034).
Keywords: Data-driven optimization, Decision making under uncertainty, Big data, Machine learning, Deep learning.
- [2365] N. Nishimura, K. Tanahashi, K. Suganuma, M. J. Miyama, and M. Ohzeki. **Item listing optimization for e-commerce websites based on diversity.** *Frontiers in Computer Science*, 1:2, 2019.
Keywords: Quantum Annealing.
- [2366] Y. Nishio, A. Oyama, Y. Akimoto, H. E. Aguirre, and K. Tanaka. **Many-objective Optimization of Trajectory Design for DESTINY Mission.** In P. M. Pardalos, M. G. C. Resende, C. Vogiatzis, and J. L. Walteros, editors, *LION 8*, volume 8426 of *LNCS*. Springer, Heidelberg, Germany, 2014.
- [2367] V. Nitivattananon, E. C. Sadowski, and R. G. Quimpo. **Optimization of Water Supply System Operation.** *Journal of Water Resources Planning and Management, ASCE*, 122(5): 374–384, Sept. / Oct. 1996.
- [2368] M. S. Nixon and A. S. Aguado. *Feature extraction & image processing for computer vision*. Academic Press, New York, NY, 2012.
- [2369] J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer, 2nd edition, 2006.
- [2370] B. Nogueira, R. G. S. Pinheiro, and A. Subramanian. **A Hybrid Iterated Local Search Heuristic for the Maximum Weight Independent Set Problem.** *Optimization Letters*, 12(3):567–583, 2018. [doi:10.1007/s11590-017-1128-7](https://doi.org/10.1007/s11590-017-1128-7).
- [2371] B. A. Nosek, G. Alter, G. C. Banks, D. Borsboom, S. D. Bowman, S. J. Breckler, S. Buck, C. D. Chambers, G. Chin, G. Christensen, M. Contestabile, A. Dafoe, E. Eich, J. Freese, R. Glennerster, D. Goroff, D. P. Green, B. Hesse, M. Humphreys, J. Ishiyama, D. Karlan, A. Kraut, A. Lupia, P. Mabry, T. Madon, N. Malhotra, E. Mayo-Wilson, M. McNutt, E. Miguel, E. L. Paluck, U. Simonsohn, C. Soderberg, B. A. Spellman, J. Turitto, G. VandenBos, S. Vazire, E. J. Wagenmakers, R. Wilson, and T. Yarkoni. **Promoting an open research culture.** *Science*, 348(6242):1422–1425, June 2015. [doi:10.1126/science.aab2374](https://doi.org/10.1126/science.aab2374).
- [2372] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. **The Preregistration Revolution.** *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, Mar. 2018. ISSN 0027-8424, 1091-6490. [doi:10.1073/pnas.1708274114](https://doi.org/10.1073/pnas.1708274114).
- [2373] Y. Nourani and B. Andresen. **A Comparison of Simulated Annealing Cooling Strategies.** *Journal of Physics A*, 31(41):8373–8385, 1998.
- [2374] H. E. Nouri, O. B. Driss, and K. Ghédira. **A Classification Schema for the Job Shop Scheduling Problem with Transportation Resources: State-of-the-Art Review.** In R. Silhavy, R. Senkerik, Z. K. Oplatkova, P. Silhavy, and Z. Prokopova, editors, *Artificial Intelligence Perspectives in Intelligent Systems*, volume 464 of *Advances in Intelligent Systems and Computing*, pages 1–11. Springer International Publishing, 2016.
- [2375] K. Nowak, M. Mörtens, and D. Izzo. **Empirical Performance of the Approximation of the Least Hypervolume Contributor.** In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 662–671. Springer, Heidelberg, Germany, 2014.
- [2376] E. Nowicki and C. Smutnicki. **A Fast Taboo Search Algorithm for the Job Shop Problem.** *Management Science*, 42(6):797–813, 1996.

- [2377] E. Nowicki and C. Smutnicki. **A fast tabu search algorithm for the permutation flow-shop problem.** *Eur. J. Oper. Res.*, 91(1):160–175, 1996.
- [2378] S. Obayashi and D. Sasaki. **Visualization and data mining of Pareto solutions using self-organizing map.** In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 796–809. Springer, Heidelberg, Germany, 2003.
Keywords: objective reduction.
- [2379] S. Obayashi et al., editors. *Evolutionary Multi-Criterion Optimization, 4th International Conference, EMO 2007, Matsushima, Japan, March 5-8, 2007, Proceedings*, volume 4403 of *LNCS*. Springer, Heidelberg, Germany, 2007.
- [2380] G. Ochoa and B. Filipič, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO Málaga, Spain, July 14-18, 2025*. ACM Press, New York, NY, 2025.
Annotation: ISBN: 979-8-4007-1464-1.
- [2381] G. Ochoa and N. Veerapen. **Mapping the global structure of TSP fitness landscapes.** *J. Heuristics*, 24(3):265–294, 2018.
- [2382] G. Ochoa, M. Tomassini, S. Verel, and C. Darabos. **A Study of NK Landscapes’ Basins and Local Optima Networks.** In C. Ryan, editor, *GECCO*, pages 555–562. ACM Press, New York, NY, 2008.
- [2383] G. Ochoa, M. Hyde, T. Curtois, J. A. Vazquez-Rodriguez, J. Walker, M. Gendreau, G. Kendall, B. McCollum, A. J. Parkes, S. Petrovic, and E. K. Burke. **Hyflex: A benchmark framework for cross-domain heuristic search.** In J.-K. Hao and M. Middendorf, editors, *EvoCOP*, volume 7245 of *LNCS*, pages 136–147. Springer, Heidelberg, Germany, 2012.
- [2384] A. Oddi, A. Cesta, N. Policella, and S. F. Smith. **Combining Variants of Iterative Flattening Search.** *Engineering Applications of Artificial Intelligence*, 21(5):683–690, 2008.
- [2385] A. Oddi, A. Cesta, N. Policella, and S. F. Smith. **Iterative Flattening Search for Resource Constrained Scheduling.** *Journal of Intelligent Manufacturing*, 21(1):17–30, 2010.
- [2386] A. Oddi, R. Rasconi, A. Cesta, and S. F. Smith. **Iterative Flattening Search for the Flexible Job Shop Scheduling Problem.** In T. Walsh, editor, *Proc. of IJCAI-11*, pages 1991–1996. IJCAI/AAAI Press, Menlo Park, CA, 2011.
- [2387] F. A. Ogbu and D. K. Smith. **The Application of the Simulated Annealing Algorithm to the Solution of the n/m/C Max Flowshop Problem.** *Comput. Oper. Res.*, 17(3):243–253, 1990.
- [2388] J. W. Ohlmann and B. W. Thomas. **A Compressed-Annealing Heuristic for the Traveling Salesman Problem with Time Windows.** *INFORMS Journal on Computing*, 19(1):80–90, 2007. ISSN 1526-5528. doi:10.1287/ijoc.1050.0145.
- [2389] V. Ojalehto, D. Podkopaev, and K. Miettinen. **Towards Automatic Testing of Reference Point Based Interactive Methods.** In J. Handl, E. Hart, P. R. Lewis, M. López-Ibáñez, G. Ochoa, and B. Paechter, editors, *PPSN*, volume 9921 of *LNCS*, pages 483–492. Springer, Heidelberg, Germany, 2016. ISBN 978-3-319-45822-9. doi:10.1007/978-3-319-45823-6_45.
Keywords: artificial DMs.
- [2390] S. M. Oliveira, M. S. Hussin, A. Roli, M. Dorigo, and T. Stützle. **Analysis of the Population-based Ant Colony Optimization Algorithm for the TSP and the QAP.** In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 1734–1741. Piscataway, NJ, 2017. IEEE Press.

- [2391] P. S. Oliveto and C. Witt. **Improved time complexity analysis of the Simple Genetic Algorithm.** *Theor. Comput. Sci.*, 605:21–41, 2015. doi:[10.1016/j.tcs.2015.01.002](https://doi.org/10.1016/j.tcs.2015.01.002).
- [2392] P. S. Oliveto, J. He, and X. Yao. **Time complexity of evolutionary algorithms for combinatorial optimization: A decade of results.** *International Journal of Automation and Computing*, 4(3):281–293, 2007.
- [2393] D. L. Olson. **Review of Empirical Studies in Multiobjective Mathematical Programming: Subject Reflection of Nonlinear Utility and Learning.** *Decision Sciences*, 23(1):1–20, 1992. doi:[10.1111/j.1540-5915.1992.tb00374.x](https://doi.org/10.1111/j.1540-5915.1992.tb00374.x).
Keywords: Decision Analysis, Human Information Processing, Linear Programming.
- [2394] R. S. Olson, N. Bartley, R. J. Urbanowicz, and J. H. Moore. **Evaluation of a Tree-based Pipeline Optimization Tool for Automating Data Science.** In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO*, pages 485–492. ACM Press, New York, NY, 2016. doi:[10.1145/2908812.2908918](https://doi.org/10.1145/2908812.2908918).
Keywords: TPOT.
- [2395] R. S. Olson, R. J. Urbanowicz, P. C. Andrews, N. A. Lavender, L. C. Kidd, and J. H. Moore. **Automating Biomedical Data Science Through Tree-Based Pipeline Optimization.** In G. Squillero and P. Burelli, editors, *Applications of Evolutionary Computation*, volume 9597 of *LNCS*, pages 123–137. Springer, Heidelberg, Germany, 2016. doi:[10.1007/978-3-319-31204-0_9](https://doi.org/10.1007/978-3-319-31204-0_9).
Keywords: TPOT.
- [2396] R. Olsson and A. Løkketangen. **Using Automatic Programming to Generate State-of-the-art Algorithms for Random 3-SAT.** *J. Heuristics*, 19(5):819–844, 2013. doi:[10.1007/s10732-013-9226-x](https://doi.org/10.1007/s10732-013-9226-x).
Annotation: Uses evolution but it is not genetic programming, nor grammatical evolution.
- [2397] M. Oltean. **Evolving Evolutionary Algorithms Using Linear Genetic Programming.** *Evol. Comput.*, 13(3):387–410, 2005. doi:[10.1162/1063656054794815](https://doi.org/10.1162/1063656054794815).
- [2398] E. O’Mahony, E. Hebrard, A. Holland, C. Nugent, and B. O’Sullivan. **Using case-based reasoning in an algorithm portfolio for constraint solving.** In Bridge et al., editors, *Irish Conference on Artificial Intelligence and Cognitive Science*, pages 210–216, 2008.
- [2399] M. O’Neill and C. Ryan. **Grammatical Evolution.** *IEEE Trans. Evol. Comput.*, 5(4):349–358, 2001.
- [2400] Open Science Collaboration. **Estimating the reproducibility of psychological science.** *Science*, 349(6251):aac4716, 2015. doi:[10.1126/science.aac4716](https://doi.org/10.1126/science.aac4716).
- [2401] S. Opricovic and G.-H. Tzeng. **Compromise solution by MCDM methods: A comparative analysis of VIKOR and TOPSIS.** *Eur. J. Oper. Res.*, 156(2):445–455, 2004. doi:[10.1016/S0377-2217\(03\)00020-1](https://doi.org/10.1016/S0377-2217(03)00020-1).
- [2402] L. E. Ormsbee and K. E. Lansey. **Optimal Control of Water Supply Pumping Systems.** *Journal of Water Resources Planning and Management, ASCE*, 120(2):237–252, 1994.
- [2403] L. E. Ormsbee and S. L. Reddy. **Nonlinear Heuristic for Pump Operations.** *Journal of Water Resources Planning and Management, ASCE*, 121(4):302–309, July / Aug. 1995.
- [2404] L. E. Ormsbee, T. M. Walski, D. V. Chase, and W. W. Sharp. **Methodology for improving pump operation efficiency.** *Journal of Water Resources Planning and Management, ASCE*, 115(2):148–164, 1989.

- [2405] J. E. Orosz and S. H. Jacobson. **Analysis of Static Simulated Annealing Algorithms.** *Journal of Optimization Theory and Applications*, 115(1):165–182, 2002.
- [2406] OscaR Team. **OscaR: Scala in OR**, 2012. Available from <https://bitbucket.org/oscarlib/oscar>.
- [2407] I. H. Osman and C. N. Potts. **Simulated Annealing for Permutation Flow-Shop Scheduling.** *Omega*, 17(6):551–557, 1989.
- [2408] A. Ostfeld and E. Salomons. **Optimal Scheduling of Pumping and Chlorine Injections under Unsteady Hydraulics.** In G. Sehlke, D. F. Hayes, and D. K. Stevens, editors, *Critical Transitions In Water And Environmental Resources Management*, pages 1–9, July 2004.
- [2409] F. E. B. Otero, A. A. Freitas, and C. G. Johnson. **cAnt-Miner: An Ant Colony Classification Algorithm to Cope with Continuous Attributes.** In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 6th International Conference, ANTS 2008*, volume 5217 of *LNCS*, pages 48–59. Springer, Heidelberg, Germany, 2008.
- [2410] P. S. Ow and T. E. Morton. **Filtered Beam Search in Scheduling.** *Int. J. Prod. Res.*, 26: 297–307, 1988.
Annotation: Proposed beam search.
- [2411] E. Özcan, E. K. Burke, and B. McCollum, editors. *PATAT 2014: Proceedings of the 10th International Conference of the Practice and Theory of Automated Timetabling*, 2014. PATAT.
- [2412] G. Özerol and E. Karasakal. **Interactive outranking approaches for multicriteria decision-making problems with imprecise information.** *J. Oper. Res. Soc.*, 59:1253–1268, 2007.
- [2413] M. Öztürk, A. Tsoukiàs, and P. Vincke. **Preference Modelling.** In J. R. Figueira, S. Greco, and M. Ehrgott, editors, *Multiple Criteria Decision Analysis, State of the Art Surveys*, chapter 2, pages 27–72. Springer, 2005.
- [2414] PACT. *Proceedings of the 23rd International Conference on Parallel Architectures and Compilation*, New York, NY, 2014. ACM Press.
- [2415] M. Padberg and G. Rinaldi. **A branch-and-cut algorithm for the resolution of large-scale symmetric traveling salesman problems.** *SIAM Review*, 33(1):60–100, 1991.
- [2416] D. Padua, editor. *Encyclopedia of Parallel Computing*. Springer, US, 2011.
- [2417] F. Pagnozzi. *Automatic Design of Hybrid Stochastic Local Search Algorithms*. PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2019.
Annotation: Supervised by Thomas Stützle.
- [2418] F. Pagnozzi and T. Stützle. **Speeding up Local Search for the Insert Neighborhood in the Weighted Tardiness Permutation Flowshop Problem.** *Optimization Letters*, 11: 1283–1292, 2017. doi:10.1007/s11590-016-1086-5.
- [2419] F. Pagnozzi and T. Stützle. **Automatic Design of Hybrid Stochastic Local Search Algorithms for Permutation Flowshop Problems.** Technical Report TR/IRIDIA/2018-005, IRIDIA, Université Libre de Bruxelles, Belgium, Apr. 2018. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2018-005.pdf>.
- [2420] F. Pagnozzi and T. Stützle. **Automatic Design of Hybrid Stochastic Local Search Algorithms for Permutation Flowshop Problems: Supplementary Material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2018-002/>, 2018.

- [2421] F. Pagnozzi and T. Stützle. **Automatic Design of Hybrid Stochastic Local Search Algorithms for Permutation Flowshop Problems.** *Eur. J. Oper. Res.*, 276:409–421, 2019. doi:10.1016/j.ejor.2019.01.018.
Keywords: EMILI.
- [2422] F. Pagnozzi and T. Stützle. **Automatic design of hybrid stochastic local search algorithms for permutation flowshop problems with additional constraints.** <https://iridia.ulb.ac.be/supp/IridiaSupp2018-002/>, 2019.
- [2423] F. Pagnozzi and T. Stützle. **Evaluating the impact of grammar complexity in automatic algorithm design.** *International Transactions in Operational Research*, pages 1–26, 2020. doi:10.1111/itor.12902.
- [2424] F. Pagnozzi and T. Stützle. **Automatic design of hybrid stochastic local search algorithms for permutation flowshop problems with additional constraints.** *Oper. Res. Perspect.*, 8:100180, 2021. doi:10.1016/j.orp.2021.100180.
- [2425] A. Pajares, X. Blasco, J. M. Herrero, and M. A. Martínez. **A Comparison of Archiving Strategies for Characterization of Nearly Optimal Solutions under Multi-Objective Optimization.** *Mathematics*, 9(9):999, 2021. doi:10.3390/math9090999.
Keywords: multi-objective optimization; nearly optimal solutions; non-epsilon dominance; multimodality; decision space diversity; archiving strategy; evolutionary algorithm; non-linear parametric identification.
- [2426] D. Palhazi Cuervo, P. Goos, K. Sörensen, and E. Arráiz. **An Iterated Local Search Algorithm for the Vehicle Routing Problem with Backhauls.** *Eur. J. Oper. Res.*, 237(2):454–464, 2014.
- [2427] I. Palomares, editor. *International Alan Turing Conference on Decision Support and Recommender systems (DSRC-Turing’19)*, London, UK, Nov. 2019. Alan Turing Institute. ISBN 978-1-5262-0820-0.
- [2428] G. Palubeckis. **Iterated tabu search for the unconstrained binary quadratic optimization problem.** *Informatica*, 17(2):279–296, 2006. doi:10.15388/Informatica.2006.138.
- [2429] Q.-K. Pan and R. Ruiz. **Local Search Methods for the Flowshop Scheduling Problem with Flowtime Minimization.** *Eur. J. Oper. Res.*, 222(1):31–43, 2012.
- [2430] Q.-K. Pan and R. Ruiz. **A Comprehensive Review and Evaluation of Permutation Flowshop Heuristics to Minimize Flowtime.** *Comput. Oper. Res.*, 40(1):117–128, 2013.
- [2431] Q.-K. Pan, M. F. Tasgetiren, and Y.-C. Liang. **A Discrete Differential Evolution Algorithm for the Permutation Flowshop Scheduling Problem.** *Computers & Industrial Engineering*, 55(4):795 – 816, 2008.
- [2432] Q.-K. Pan, L. Wang, and B.-H. Zhao. **An improved iterated greedy algorithm for the no-wait flow shop scheduling problem with makespan criterion.** *International Journal of Advanced Manufacturing Technology*, 38(7-8):778–786, 2008.
- [2433] Q.-K. Pan, R. Ruiz, and P. Alfaro-Fernández. **Iterated Search Methods for Earliness and Tardiness Minimization in Hybrid Flowshops with Due Windows.** *Comput. Oper. Res.*, 80:50–60, 2017.
- [2434] S. Pan, D. Vermetten, M. López-Ibáñez, T. Bäck, and H. Wang. **Transfer Learning of Surrogate Models via Domain Affine Transformation.** In J. Handl and X. Li, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2024*. ACM Press, New York, NY, 2024. doi:10.1145/3638529.3654032.

- [2435] S. Pan, D. Vermetten, M. López-Ibáñez, T. Bäck, and H. Wang. **Transfer Learning of Surrogate Models via Domain Affine Transformation: Supplementary Material**. Zenodo, 2024.
- [2436] S. Pan, D. Vermetten, M. López-Ibáñez, T. Bäck, and H. Wang. **Transfer Learning of Surrogate Models: Integrating Domain Warping and Affine Transformations**. In G. Ochoa and B. Filipič, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2025*, pages 2481–2490. ACM Press, New York, NY, 2025. doi:10.1145/3712255.3734291.
Annotation: ISBN: 979-8-4007-1464-1.
- [2437] S. J. Pan and Q. Yang. **A survey on transfer learning**. *IEEE Transactions on Knowledge and Data Engineering*, 22(10):1345–1359, 2009.
- [2438] L. M. Pang, H. Ishibuchi, and K. Shang. **Algorithm configurations of MOEA/D with an unbounded external archive**. In *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 1087–1094. IEEE, 2020.
- [2439] B. K. Panigrahi, P. N. Suganthan, S. Das, and S. S. Dash, editors. *International Conference on Swarm, Evolutionary, and Memetic Computing*, volume 8298 of *Theoretical Computer Science and General Issues*, 2013. Springer International Publishing.
- [2440] D. Papadias, Y. Tao, G. Fu, and B. Seeger. **Progressive skyline computation in database systems**. *ACM Trans. Database Syst.*, 30(1):41–82, 2005. doi:10.1145/1061318.1061320.
- [2441] C. H. Papadimitriou and K. Steiglitz. *Combinatorial Optimization – Algorithms and Complexity*. Prentice Hall, Englewood Cliffs, NJ, 1982.
- [2442] C. H. Papadimitriou and M. Yannakakis. **On the Approximability of Trade-offs and Optimal Access of Web Sources**. In A. Blum, editor, *41st Annual Symposium on Foundations of Computer Science*, pages 86–92. IEEE Computer Society Press, 2000. doi:10.1109/SFCS.2000.892068.
- [2443] G. A. Papadopoulos, editor. *26th IEEE International Conference on Tools with Artificial Intelligence, ICTAI 2014, Limassol, Cyprus, November 10-12, 2014*, 2014. IEEE Press.
- [2444] L. Paquete. **Algoritmos Evolutivos Multiobjetivo para Afectação de Recursos e sua Aplicação à Geração de Horários em Universidades (Multiobjective Evolutionary Algorithms for Resource Allocation and their Application to University Timetabling)**. Master’s thesis, University of Algarve, 2001. In Portuguese.
- [2445] L. Paquete. *Stochastic Local Search Algorithms for Multiobjective Combinatorial Optimization: Methods and Analysis*. PhD thesis, FB Informatik, TU Darmstadt, Germany, 2005.
- [2446] L. Paquete and T. Stützle. **An Experimental Investigation of Iterated Local Search for Coloring Graphs**. In S. Cagnoni et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2002*, volume 2279 of *LNCS*, pages 122–131. Springer, Heidelberg, Germany, 2002.
- [2447] L. Paquete and T. Stützle. **A Two-Phase Local Search for the Biobjective Traveling Salesman Problem**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 479–493. Springer, Heidelberg, Germany, 2003. doi:10.1007/3-540-36970-8_34.
- [2448] L. Paquete and T. Stützle. **A study of stochastic local search algorithms for the biobjective QAP with correlated flow matrices**. *Eur. J. Oper. Res.*, 169(3):943–959, 2006.

- [2449] L. Paquete and T. Stützle. **Clusters of non-dominated solutions in multiobjective combinatorial optimization: An experimental analysis.** In V. Barichard, M. Ehrgott, X. Gandibleux, and V. T'Kindt, editors, *Multiobjective Programming and Goal Programming: Theoretical Results and Practical Applications*, volume 618 of *LNEMS*, pages 69–77. Springer, Berlin, Germany, 2009. doi:10.1007/978-3-540-85646-7_7.
- [2450] L. Paquete and T. Stützle. **Design and analysis of stochastic local search for the multiobjective traveling salesman problem.** *Comput. Oper. Res.*, 36(9):2619–2631, 2009. doi:10.1016/j.cor.2008.11.013.
- [2451] L. Paquete and T. Stützle. **Stochastic Local Search Algorithms for Multiobjective Combinatorial Optimization: A Review.** In T. F. Gonzalez, editor, *Handbook of Approximation Algorithms and Metaheuristics*, pages 411–425. Chapman & Hall/CRC, Boca Raton, FL, 2018. doi:10.1201/9781351236423-24.
- [2452] L. Paquete, M. Chiarandini, and T. Stützle. **Pareto Local Optimum Sets in the Biobjective Traveling Salesman Problem: An Experimental Study.** In X. Gandibleux, M. Sevaux, K. Sörensen, and V. T'Kindt, editors, *Metaheuristics for Multiobjective Optimisation*, volume 535 of *LNEMS*, pages 177–199. Springer, Berlin/Heidelberg, 2004. doi:10.1007/978-3-642-17144-4_7.
Keywords: Pareto local search, PLS.
- [2453] L. Paquete, T. Stützle, and M. López-Ibáñez. **On the design and analysis of SLS algorithms for multiobjective combinatorial optimization problems.** Technical Report TR/IRIDIA/2005-029, IRIDIA, Université Libre de Bruxelles, Belgium, 2005. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2005-029.pdf>.
- [2454] L. Paquete, T. Stützle, and M. López-Ibáñez. **Towards the Empirical Analysis of SLS Algorithms for Multiobjective Combinatorial Optimization Problems through Experimental Design.** In K. F. Doerner, M. Gendreau, P. Greistorfer, W. J. Gutjahr, R. F. Hartl, and M. Reimann, editors, *6th Metaheuristics International Conference (MIC 2005)*, pages 739–746, Vienna, Austria, 2005.
- [2455] L. Paquete, M. Chiarandini, and D. Basso, editors. *Empirical Methods for the Analysis of Algorithms, Workshop EMAA 2006, Proceedings*, Reykjavik, Iceland, 2006.
- [2456] L. Paquete, C. M. Fonseca, and M. López-Ibáñez. **An optimal algorithm for a special case of Klee's measure problem in three dimensions.** Technical Report CSI-RT-I-01/2006, CSI, Universidade do Algarve, 2006. Superseded by paper in *IEEE Trans. Evol. Comput.* [303].
Annotation: Proof of Theorem 3.1 is incorrect.
- [2457] L. Paquete, T. Schiavinotto, and T. Stützle. **On Local Optima in Multiobjective Combinatorial Optimization Problems.** *Annals of Operations Research*, 156:83–97, 2007. doi:10.1007/s10479-007-0230-0.
Keywords: Pareto local search, PLS.
- [2458] L. Paquete, T. Stützle, and M. López-Ibáñez. **Using experimental design to analyze stochastic local search algorithms for multiobjective problems.** In K. F. Doerner, M. Gendreau, P. Greistorfer, W. J. Gutjahr, R. F. Hartl, and M. Reimann, editors, *Metaheuristics: Progress in Complex Systems Optimization*, volume 39 of *Operations Research / Computer Science Interfaces*, pages 325–344. Springer, New York, NY, 2007. doi:10.1007/978-0-387-71921-4_17.
Annotation: Post-Conference Proceedings of the 6th Metaheuristics International Conference (MIC 2005).
- [2459] P. M. Pardalos and D.-Z. Du, editors. *Handbook of Combinatorial Optimization*, volume 2. Kluwer Academic Publishers, 1998.

- [2460] P. M. Pardalos and G. Nicosia, editors. *7th International Conference, LION 7, Catania, Italy, January 7-11, 2013. Selected Papers*, volume 7997 of *LNCS*. Springer, Heidelberg, Germany, 2013.
- [2461] P. M. Pardalos, M. G. C. Resende, C. Vogiatzis, and J. L. Walteros, editors. *8th International Conference, LION 8, Gainesville, Florida, USA, February 16-21, 2014. Revised Selected Papers*, volume 8426 of *LNCS*. Springer, Heidelberg, Germany, 2014.
- [2462] M.-W. Park and Y.-D. Kim. **A systematic procedure for setting parameters in simulated annealing algorithms.** *Comput. Oper. Res.*, 25(3):207–217, 1998. doi:10.1016/S0305-0548(97)00054-3.
- [2463] R. S. Parpinelli, H. S. Lopes, and A. A. Freitas. **Data Mining with an Ant Colony Optimization Algorithm.** *IEEE Trans. Evol. Comput.*, 6(4):321–332, 2002.
- [2464] T. J. Parr and R. W. Quong. **ANTLR: A predicated-LL (k) parser generator.** *Software — Practice & Experience*, 25(7):789–810, 1995.
- [2465] S. N. Parragh, K. F. Doerner, R. F. Hartl, and X. Gandibleux. **A heuristic two-phase solution approach for the multi-objective dial-a-ride problem.** *Networks*, 54(4):227–242, 2009.
- [2466] R. O. Parreiras and J. A. Vasconcelos. **A multiplicative version of PROMETHEE II applied to multiobjective optimization problems.** *Eur. J. Oper. Res.*, 183:729–740, 2007.
- [2467] F. Parreño, R. Alvarez-Valdes, J. M. Tamarit, and J. F. Oliveira. **A Maximal-Space Algorithm for the Container Loading Problem.** *INFORMS Journal on Computing*, 20(3):412–422, Aug. 2008. doi:10.1287/ijoc.1070.0254.
- [2468] R. Parsons and M. Johnson. **A Case Study in Experimental Design Applied to Genetic Algorithms with Applications to DNA Sequence Assembly.** *American Journal of Mathematical and Management Sciences*, 17(3-4):369–396, 1997. doi:10.1080/01966324.1997.10737444.
- [2469] G. Paul. **Comparative performance of tabu search and simulated annealing heuristics for the quadratic assignment problem.** *Operations Research Letters*, 38(6):577–581, 2010.
- [2470] J. Paulli. **A computational comparison of simulated annealing and tabu search applied to the quadratic assignment problem.** In R. V. V. Vidal, editor, *Applied Simulated Annealing*, pages 85–102. Springer, 1993.
- [2471] L. M. Pavelski, M. R. Delgado, and M.-E. Kessaci. **Meta-Learning on Flowshop Using Fitness Landscape Analysis.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 925–933. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.
- [2472] G. S. Peace. *Taguchi Methods: A Hands-On Approach*. Addison-Wesley, 1993.
- [2473] J. Pearl. *Heuristics: Intelligent Search Strategies for Computer Problem Solving*. Addison-Wesley, Reading, MA, 1984.
- [2474] J. Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2nd edition, 2009.
- [2475] J. Pearl. **The do-calculus revisited.** In N. de Freitas and K. Murphy, editors, *Proceedings of the Twenty-Eighth Conference on Uncertainty in Artificial Intelligence (UAI’12), Catalina Island, CA August 14-18 2012*, pages 4–11. AUAI Press, 2013.
- [2476] J. Pearl. **The seven tools of causal inference, with reflections on machine learning.** *Commun. ACM*, 62(3):54–60, 2019.

- [2477] J. Pearl and E. Bareinboim. **Transportability of causal and statistical relations: A formal approach.** In W. Burgard and D. Roth, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 247–254. AAAI Press, 2011.
- [2478] J. Pearl and D. Mackenzie. *The book of why: the new science of cause and effect.* Basic books, 2018.
- [2479] D. W. Pearson, N. C. Steele, and R. F. Albrecht, editors. *Artificial Neural Networks and Genetic Algorithms.* Springer Verlag, 2003.
- [2480] M. Pedemonte, S. Nesmachnow, and H. Cancela. **A survey on parallel ant colony optimization.** *Applied Soft Computing*, 11(8):5181–5197, 2011.
- [2481] J. A. Pedraza, C. García-Martínez, A. Cano, and S. Ventura. **Classification Rule Mining with Iterated Greedy.** In M. M. Polycarpou, A. C. P. L. F. de Carvalho, J. Pan, M. Wozniak, H. Quintián, and E. Corchado, editors, *Hybrid Artificial Intelligence Systems - 9th International Conference, HAIS 2014, Salamanca, Spain, June 11-13, 2014. Proceedings*, volume 8480 of *LNCS*, pages 585–596. Springer, Heidelberg, Germany, 2014.
- [2482] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. **Scikit-learn: Machine learning in Python.** *J. Mach. Learn. Res.*, 12:2825–2830, 2011.
- [2483] L. R. Pedro and R. H. C. Takahashi. **Decision-Maker Preference Modeling in Interactive Multiobjective Optimization.** In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 811–824. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4. doi:10.1007/978-3-642-37140-0_60.
Keywords: decision-maker, interactive, neural networks.
- [2484] M. Pelikan and J. Branke, editors. *Genetic and Evolutionary Computation Conference, GECCO 2010, Proceedings, Portland, Oregon, USA, July 7-11, 2010.* ACM Press, New York, NY, 2010.
- [2485] M. Pelikan and J. Branke, editors. *Genetic and Evolutionary Computation Conference, GECCO 2010, Companion Material Proceedings, Portland, Oregon, USA, July 7-11, 2010.* ACM Press, New York, NY, 2010.
- [2486] P. Pellegrini and M. Birattari. **Implementation Effort and Performance.** In T. Stützle, M. Birattari, and H. H. Hoos, editors, *Engineering Stochastic Local Search Algorithms. Designing, Implementing and Analyzing Effective Heuristics. SLS 2007*, volume 4638 of *LNCS*, pages 31–45. Springer, Heidelberg, Germany, 2007.
- [2487] P. Pellegrini, D. Favaretto, and E. Moretti. **On $MA\mathcal{X}$ - MLN Ant System’s Parameters.** In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*, pages 203–214. Springer, Heidelberg, Germany, 2006.
- [2488] P. Pellegrini, D. Favaretto, and E. Moretti. **Exploration in stochastic algorithms: An application on $MA\mathcal{X}$ - MLN Ant System.** In N. Krasnogor, B. Melián-Batista, J. A. Moreno-Pérez, J. M. Moreno-Vega, and D. A. Pelta, editors, *Nature Inspired Cooperative Strategies for Optimization (NICSO 2008)*, volume 236 of *Studies in Computational Intelligence*, pages 1–13. Springer, Berlin, Germany, 2009. doi:10.1007/978-3-642-03211-0.
- [2489] P. Pellegrini, T. Stützle, and M. Birattari. **Off-line vs. On-line Tuning: A Study on $MA\mathcal{X}$ - MLN Ant System for the TSP.** In M. Dorigo et al., editors, *ANTS 2010*, volume 6234 of *LNCS*, pages 239–250. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-15461-4_21.

- [2490] P. Pellegrini, M. Birattari, and T. Stützle. **A Critical Analysis of Parameter Adaptation in Ant Colony Optimization.** *Swarm Intelligence*, 6(1):23–48, 2012. doi:[10.1007/s11721-011-0061-0](https://doi.org/10.1007/s11721-011-0061-0).
- [2491] P. Pellegrini, L. Castelli, and R. Pesenti. **Metaheuristic algorithms for the simultaneous slot allocation problem.** *IET Intelligent Transport Systems*, 6(4):453–462, Dec. 2012. doi:[10.1049/iet-its.2011.0179](https://doi.org/10.1049/iet-its.2011.0179).
- [2492] P. Pellegrini, F. Mascia, T. Stützle, and M. Birattari. **On the Sensitivity of Reactive Tabu Search to its Meta-parameters.** *Soft Computing*, 18(11):2177–2190, 2014. doi:[10.1007/s00500-013-1192-6](https://doi.org/10.1007/s00500-013-1192-6).
- [2493] M. Péres, G. Ruiz, S. Nesmachnow, and A. C. Olivera. **Multiobjective evolutionary optimization of traffic flow and pollution in Montevideo, Uruguay.** *Applied Soft Computing*, 70:472–485, 2018.
Keywords: Multiobjective evolutionary algorithms, Pollution, Simulation, Traffic flow.
- [2494] L. Pérez Cáceres and T. Stützle. **Exploring Variable Neighborhood Search for Automatic Algorithm Configuration.** *Electronic Notes in Discrete Mathematics*, 58:167–174, 2017. doi:[10.1016/j.endm.2017.03.022](https://doi.org/10.1016/j.endm.2017.03.022).
- [2495] L. Pérez Cáceres and T. Stützle. *Automatic Algorithm Configuration: Analysis, Improvements and Applications.* PhD thesis, IRIDIA, École polytechnique, Université Libre de Bruxelles, Belgium, 2017.
Annotation: Supervised by Thomas Stützle and Manuel López-Ibáñez.
- [2496] L. Pérez Cáceres and S. Verel, editors. *Evolutionary Computation in Combinatorial Optimization – 22nd European Conference, EvoCOP 2022, Held as Part of EvoStar 2022, April 20-22, 2022, Proceedings.* LNCS. Springer, Cham, Switzerland, 2022.
- [2497] L. Pérez Cáceres, M. López-Ibáñez, and T. Stützle. **Ant Colony Optimization on a Budget of 1000.** In M. Dorigo et al., editors, *ANTS 2014*, volume 8667 of LNCS, pages 50–61. Springer, Heidelberg, Germany, 2014. doi:[10.1007/978-3-319-09952-1_5](https://doi.org/10.1007/978-3-319-09952-1_5).
- [2498] L. Pérez Cáceres, M. López-Ibáñez, and T. Stützle. **An Analysis of Parameters of irace.** In C. Blum and G. Ochoa, editors, *EvoCOP*, volume 8600 of LNCS, pages 37–48. Springer, Heidelberg, Germany, 2014. doi:[10.1007/978-3-662-44320-0_4](https://doi.org/10.1007/978-3-662-44320-0_4).
- [2499] L. Pérez Cáceres, M. López-Ibáñez, and T. Stützle. **Ant Colony Optimization on a Budget of 1000: Supplementary material**, 2015. URL <https://iridia.ulb.ac.be/supp/IridiaSupp2015-004>.
- [2500] L. Pérez Cáceres, M. López-Ibáñez, and T. Stützle. **Ant colony optimization on a limited budget of evaluations.** *Swarm Intelligence*, 9(2-3):103–124, 2015. doi:[10.1007/s11721-015-0106-x](https://doi.org/10.1007/s11721-015-0106-x). Supplementary material: <http://iridia.ulb.ac.be/supp/IridiaSupp2015-004>.
- [2501] L. Pérez Cáceres, B. Bischl, and T. Stützle. **Evaluating random forest models for irace.** In P. A. N. Bosman, editor, *GECCO Companion*, pages 1146–1153. ACM Press, New York, NY, 2017. doi:[10.1145/3067695.3082057](https://doi.org/10.1145/3067695.3082057).
- [2502] L. Pérez Cáceres, M. López-Ibáñez, H. H. Hoos, and T. Stützle. **An Experimental Study of Adaptive Capping in irace.** In R. Battiti, D. E. Kvasov, and Y. D. Sergeyev, editors, *LION 11*, volume 10556 of LNCS, pages 235–250. Springer, Cham, Switzerland, 2017. doi:[10.1007/978-3-319-69404-7_17](https://doi.org/10.1007/978-3-319-69404-7_17). Supplementary material: <https://iridia.ulb.ac.be/supp/IridiaSupp2016-007/>.

- [2503] L. Pérez Cáceres, M. López-Ibáñez, H. H. Hoos, and T. Stützle. **An experimental study of adaptive capping in irace: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2016-007/>, 2017.
- [2504] L. Pérez Cáceres, F. Pagnozzi, A. Franzin, and T. Stützle. **Automatic Configuration of GCC Using irace.** In E. Lutton, P. Legrand, P. Parrend, N. Monmarché, and M. Schoenauer, editors, *EA 2017: Artificial Evolution*, volume 10764 of *LNCS*, pages 202–216. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-78133-4_15.
- [2505] L. Pérez Cáceres, F. Pagnozzi, A. Franzin, and T. Stützle. **Automatic configuration of GCC using irace: Supplementary material.** <https://iridia.ulb.ac.be/supp/IridiaSupp2017-009/>, 2017.
- [2506] J. M. Perkel. **Challenge to scientists: does your ten-year-old code still run?** *Nature*, 584:556–658, 2020. doi:10.1038/d41586-020-02462-7.
Keywords: reproducibility; software engineering; ReScience C; Ten Years Reproducibility Challenge; code reusability.
- [2507] G. Pesant, M. Gendreau, J.-Y. Potvin, and J.-M. Rousseau. **An Exact Constraint Logic Programming Algorithm for the Traveling Salesman Problem with Time Windows.** *Transportation Science*, 32:12–29, 1998.
- [2508] A. Pessoa, E. Uchoa, M. Aragão, and R. Rodrigues. **Exact Algorithm over an Arc-time-indexed formulation for Parallel Machine Scheduling Problems.** *Mathematical Programming Computation*, 2(3–4):259–290, 2010.
- [2509] J. Peters, D. Janzing, and B. Schölkopf. *Elements of causal inference: foundations and learning algorithms.* MIT Press, 2017.
- [2510] C. W. Petit. **Touched by nature: putting evolution to work on the assembly line.** *U.S. News & World Report*, 125(4):43–45, July 1998. URL http://dynamics.org/~altenber/IN_THE_MEDIA/US_NEWS.7-27-98/index.html.
Annotation: Evolutionary optimization of turbine design of the Boeing 777 GE.
- [2511] J. Petke, S. O. Haraldsson, M. Harman, W. B. Langdon, D. R. White, and J. R. Woodward. **Genetic Improvement of Software: A Comprehensive Survey.** *IEEE Trans. Evol. Comput.*, 22(3):415–432, 2018. doi:10.1109/TEVC.2017.2693219.
- [2512] M. Petrik and S. Zilberstein. **Learning parallel portfolios of algorithms.** *Annals of Mathematics and Artificial Intelligence*, 48(1):85–106, 2006.
Keywords: algorithm selection.
- [2513] A. E. Petropoulos, E. P. Bonfiglio, D. J. Grebow, T. Lam, J. S. Parker, J. Arrieta, D. F. Landau, R. L. Anderson, E. D. Gustafson, G. J. Whiffen, P. A. Finlayson, and J. A. Sims. **GTOC5: Results from Jet Propulsion Lab.** *Acta Futura*, 8:21–27, 2014. doi:10.2420/AF08.2014.21.
- [2514] J. E. Pettinger and R. M. Everson. **Controlling genetic algorithms with reinforcement learning.** In W. B. Langdon et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2002*, pages 692–692. Morgan Kaufmann Publishers, San Francisco, CA, 2002.
- [2515] S. Pezeshk and O. J. Helweg. **Adaptative Search Optimisation in reducing pump operation costs.** *Journal of Water Resources Planning and Management, ASCE*, 122(1):57–63, Jan. / Feb. 1996.
- [2516] S. Phelps and M. Köksalan. **An interactive evolutionary metaheuristic for multiobjective combinatorial optimization.** *Management Science*, 49(12):1726–1738, 2003.

- [2517] F. Phillipson and H. S. Bhatia. **Portfolio Optimisation Using the D-Wave Quantum Annealer**. In M. Paszynski, D. Kranzlmüller, V. V. Krzhizhanovskaya, J. J. Dongarra, and P. M. A. Sloot, editors, *Computational Science – ICCS 2021*, pages 45–59. Springer International Publishing, Cham, Switzerland, 2021.
- [2518] J. Pihera and N. Musliu. **Application of Machine Learning to Algorithm Selection for TSP**. In G. A. Papadopoulos, editor, *26th IEEE International Conference on Tools with Artificial Intelligence, ICTAI 2014, Limassol, Cyprus, November 10–12, 2014*, pages 47–54. IEEE Press, 2014.
- [2519] M. L. Pilat and T. White. **Using Genetic Algorithms to optimize ACS-TSP**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 282–287. Springer, Heidelberg, Germany, 2002.
- [2520] J. Pineau and K. Sinha. **The Machine Learning Reproducibility Checklist (v2.0)**. <https://www.cs.mcgill.ca/~jpineau/ReproducibilityChecklist-v2.0.pdf>, 2020.
Annotation: Used in NeurIPS 2020.
- [2521] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d’Alché Buc, E. Fox, and H. Larochelle. **Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)**. *Arxiv preprint arXiv:2003.12206 [cs.LG]*, 2020. URL <https://arxiv.org/abs/2003.12206>.
- [2522] M. L. Pinedo. *Scheduling: Theory, Algorithms, and Systems*. Springer, New York, NY, 4th edition, 2012.
- [2523] P. Pinto, T. Runkler, and J. Sousa. **Ant Colony Optimization and its Application to Regular and Dynamic MAX-SAT Problems**. In *Advances in Biologically Inspired Information Systems*, volume 69 of *Studies in Computational Intelligence*, pages 285–304. Springer, Berlin, Germany, 2007. doi:10.1007/978-3-540-72693-7_15.
- [2524] D. Pisinger. **Where are the hard knapsack problems?** *Comput. Oper. Res.*, 32(9):2271–2284, 2005.
- [2525] D. Pisinger and S. Ropke. **A General Heuristic for Vehicle Routing Problems**. *Comput. Oper. Res.*, 34(8):2403–2435, 2007.
- [2526] D. Pisinger and S. Ropke. **Large Neighborhood Search**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, pages 399–419. Springer, New York, NY, 2nd edition, 2010.
- [2527] R. Pitakaso, C. Almeder, K. F. Doerner, and R. F. Hartl. **Combining exact and population-based methods for the Constrained Multilevel Lot Sizing Problem**. *Int. J. Prod. Res.*, 44(22):4755–4771, 2006.
- [2528] R. Pitakaso, C. Almeder, K. F. Doerner, and R. F. Hartl. **A MAX–MLN Ant System for unconstrained multi-level lot-sizing problems**. *Comput. Oper. Res.*, 34(9):2533–2552, 2007. doi:10.1016/j.cor.2005.09.022.
Keywords: Ant colony optimization, Material requirements planning, Multi-level lot-sizing, Wagner-Whitin algorithm.
- [2529] E. Pitzer, A. Beham, and M. Affenzeller. **Automatic Algorithm Selection for the Quadratic Assignment Problem Using Fitness Landscape Analysis**. In M. Middendorf and C. Blum, editors, *EvoCOP*, volume 7832 of *LNCS*, pages 109–120. Springer, Heidelberg, Germany, 2013.
- [2530] H. E. Plesser. **Reproducibility vs. Replicability: A Brief History of a Confused Terminology**. *Frontiers in Neuroinformatics*, 11, Jan. 2018. doi:10.3389/fninf.2017.00076.

- [2531] R. Plotnick. *The Genie in the Machine: How Computer-Automated Inventing Is Revolutionizing Law and Business*. Stanford Law Books, 2009.
Annotation: Mentions evolutionary optimization of Oral-B toothbrushes.
- [2532] D. Plotnikov, D. Melnik, M. Vardanyan, R. Buchatskiy, R. Zhuykov, and J.-H. Lee. **Automatic Tuning of Compiler Optimizations and Analysis of their Impact**. In V. Alexandrov, M. Lees, V. Krzhizhanovskaya, J. Dongarra, and P. M. A. Sloot, editors, *2013 International Conference on Computational Science*, volume 18 of *Procedia Computer Science*, pages 1312–1321. Elsevier, 2013. doi:10.1016/j.procs.2013.05.298.
- [2533] M. E. Pollack, editor. *IJCAI 1997, Proceedings of the 15th International Joint Conference on Artificial Intelligence, IJCAI 97, Nagoya, Japan, August 23-29, 1997, 2 Volumes*. Morgan Kaufmann Publishers, 1997.
- [2534] J. Porta, J. Parapar, R. Doallo, V. Barbosa, I. Santé, R. Crecente, and C. Díaz. **A Population-based Iterated Greedy Algorithm for the Delimitation and Zoning of Rural Settlements**. *Computers, Environment and Urban Systems*, 39:12–26, 2013.
- [2535] V. W. Porto, N. Saravanan, D. Waagen, and A. E. Eiben, editors. *7th International Conference, EP98 San Diego, California, USA, March 25–27, 1998 Proceedings*, volume 1447 of *LNCS*, Heidelberg, Germany, 1998. Springer. doi:10.1007/BFb0040753.
- [2536] V. W. Porto, N. Saravanan, D. E. Waagen, and A. E. Eiben, editors. *Evolutionary Programming VII, 7th International Conference, EP98, San Diego, CA, USA, March 25-27, 1998, Proceedings*, volume 1447 of *LNCS*. Springer, 1998.
- [2537] D. Porumbel, G. Goncalves, H. Allaoui, and T. Hsu. **Iterated Local Search and Column Generation to solve Arc-Routing as a Permutation Set-Covering Problem**. *Eur. J. Oper. Res.*, 256(2):349–367, 2017.
- [2538] J.-Y. Potvin and S. Bengio. **The Vehicle Routing Problem with Time Windows Part II: Genetic Search**. *INFORMS Journal on Computing*, 8:165–172, 1996.
- [2539] M. J. D. Powell. **A Direct Search Optimization Method That Models the Objective and Constraint Functions by Linear Interpolation**. In *Advances in Optimization and Numerical Analysis*, pages 51–67. Springer, Dordrecht, The Netherlands, 1994. ISBN 9789401583305. doi:10.1007/978-94-015-8330-5_4.
Annotation: Proposed COBYLA.
- [2540] M. J. D. Powell. **The BOBYQA algorithm for bound constrained optimization without derivatives**. Technical Report Cambridge NA Report NA2009/06, University of Cambridge, UK, 2009.
- [2541] M. Pöyhönen and R. P. Härmäläinen. **On the convergence of multiattribute weighting methods**. *Eur. J. Oper. Res.*, 129(3):569–585, 2001. doi:10.1016/S0377-2217(99)00467-1.
- [2542] K. Praditwong and X. Yao. **A new multi-objective evolutionary optimisation algorithm: the two-archive algorithm**. In *International Conference on Computational Intelligence and Security*, volume 1, pages 286–291. IEEE, 2006.
- [2543] R. P. Prager and H. Trautmann. **Nullifying the Inherent Bias of Non-invariant Exploratory Landscape Analysis Features**. In J. a. Correia, S. Smith, and R. Qaddoura, editors, *EvoApplications 2023: Applications of Evolutionary Computation*, volume 13989 of *LNCS*, pages 411–425. Springer Nature, Switzerland, 2023. doi:10.1007/978-3-031-30229-9_27.

- [2544] R. P. Prager and H. Trautmann. **Pflacco: Feature-Based Landscape Analysis of Continuous and Constrained Optimization Problems in Python**. *Evol. Comput.*, 32(3):211–216, 2024. doi:10.1162/EVCO_A_00341.
- [2545] R. P. Prager and H. Trautmann. **Exploratory Landscape Analysis for Mixed-Variable Problems**. *IEEE Trans. Evol. Comput.*, 29(4):936–947, 2025. doi:10.1109/TEVC.2024.3399560.
- [2546] R. P. Prager, H. Trautmann, H. Wang, T. Bäck, and P. Kerschke. **Per-Instance Configuration of the Modularized CMA-ES by Means of Classifier Chains and Exploratory Landscape Analysis**. In C. A. Coello Coello, editor, *2020 IEEE Symposium Series on Computational Intelligence, SSCI 2020, Canberra, Australia, December 1-4, 2020*, pages 996–1003. IEEE Press, 2020.
- [2547] M. Pranzo and D. Pacciarelli. **An Iterated Greedy Metaheuristic for the Blocking Job Shop Scheduling Problem**. *J. Heuristics*, 22(4):587–611, 2016. doi:10.1007/s10732-014-9279-5.
- [2548] T. D. Prasad. **Design of pumped water distribution networks with storage**. *Journal of Water Resources Planning and Management, ASCE*, 136(4):129–136, 2009.
- [2549] T. D. Prasad and G. A. Walters. **Optimal rerouting to minimise residence times in water distribution networks**. In C. Maksimović, D. Butler, and F. A. Memon, editors, *Advances in Water Supply Management*, pages 299–306. CRC Press, 2003.
- [2550] M. Prates, P. H. C. Avelar, H. Lemos, L. C. Lamb, and M. Y. Vardi. **Learning to Solve NP-Complete Problems: A Graph Neural Network for Decision TSP**. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01):4731–4738, July 2019. ISSN 2159-5399. doi:10.1609/aaai.v33i01.33014731.
- [2551] F. P. Preparata and M. I. Shamos. *Computational Geometry. An Introduction*. Springer, Berlin, Germany, 2nd edition, 1988.
- [2552] K. Price, R. M. Storn, and J. A. Lampinen. *Differential Evolution: A Practical Approach to Global Optimization*. Springer, New York, NY, 2005. doi:10.1007/3-540-31306-0.
- [2553] K. V. Price, A. Kumar, and P. N. Suganthan. **Trial-based dominance for comparing both the speed and accuracy of stochastic optimizers with standard non-parametric tests**. *Swarm and Evolutionary Computation*, 78:101287, 2023. doi:10.1016/j.swevo.2023.101287. *Keywords:* Benchmarking, Two-variable non-parametric tests, Evolutionary algorithms, Dominance, Stochastic optimization, Numerical optimization, Mann-Whitney test.
- [2554] S. Pricopie, R. Allmendinger, M. López-Ibáñez, C. Fare, M. Benatan, and J. D. Knowles. **Expensive Optimization with Production-Graph Resource Constraints: A First Look at a New Problem Class**. In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022*, pages 840–848. ACM Press, New York, NY, 2022. doi:10.1145/3512290.3528741. *Keywords:* molecular discovery, resource constraints, expensive optimization, production costs.
- [2555] S. Pricopie, R. Allmendinger, M. López-Ibáñez, C. Fare, M. Benatan, and J. D. Knowles. **An Adaptive Approach to Bayesian Optimization with Setup Switching Costs**. In M. Affenzeller, S. M. Winkler, A. V. Kononova, H. Trautmann, T. Tušar, P. Machado, and T. Bäck, editors, *Parallel Problem Solving from Nature – PPSN XVIII*, volume 15149 of *LNCS*, pages 340–355. Springer, Cham, Switzerland, 2024. doi:10.1007/978-3-031-70068-2_21.
- [2556] A. Prieditis and S. Russell, editors. *Proceedings of the Twelfth International Conference on Machine Learning (ML-95)*. Morgan Kaufmann Publishers, Palo Alto, CA, 1995.

- [2557] R. C. Prim. **Shortest connection networks and some generalizations.** *Bell System Technical Journal*, 36(6):1389–1401, 1957.
- [2558] P. Probst, B. Bischl, and A.-L. Boulesteix. **Tunability: Importance of Hyperparameters of Machine Learning Algorithms.** *Arxiv preprint arXiv:1802.09596*, 2018. URL <https://arxiv.org/abs/1802.09596>.
Keywords: parameter importance.
- [2559] P. Probst, B. Bischl, and A.-L. Boulesteix. **Tunability: Importance of Hyperparameters of Machine Learning Algorithms.** *J. Mach. Learn. Res.*, 20(53):1–32, 2019.
- [2560] L. Pronzato and W. G. Müller. **Design of computer experiments: space filling and beyond.** *Statistics and Computing*, 22(3):681–701, 2012.
Keywords: Kriging; Entropy; Design of experiments; Space-filling; Sphere packing; Maximin design; Minimax design.
- [2561] A. Pryke, S. Mostaghim, and A. Nazemi. **Heatmap visualization of population based multi objective algorithms.** In S. Obayashi et al., editors, *EMO*, volume 4403 of *LNCS*, pages 361–375. Springer, Heidelberg, Germany, 2007.
- [2562] H. N. Psaraftis. **Dynamic Vehicle Routing: Status and Prospects.** *Annals of Operations Research*, 61:143–164, 1995.
- [2563] J. M. Puerta, J. A. Gámez, B. Dorronsoro, E. Barrenechea, A. Troncoso, B. Baruque, and M. Galar, editors. *Advances in Artificial Intelligence: 16th Conference of the Spanish Association for Artificial Intelligence, CAEPIA 2015 Albacete, Spain, November 9-12, 2015 Proceedings*, volume 9422 of *LNCS*. Springer, Heidelberg, Germany, 2015.
- [2564] T. Pukkala and T. Heinonen. **Optimizing heuristic search in forest planning.** *Nonlinear Analysis: Real World Applications*, 7(5):1284–1297, 2006.
- [2565] L. Pulina and A. Tacchella. **A self-adaptive multi-engine solver for quantified Boolean formulas.** *Constraints*, 14(1):80–116, 2009.
- [2566] R. C. Purshouse and P. J. Fleming. **Evolutionary many-objective optimisation: an exploratory analysis.** In *IEEE CEC*, pages 2066–2073, Piscataway, NJ, Dec. 2003. IEEE Press. doi:10.1109/CEC.2003.1299927.
Annotation: First to mention NSGA-II failure to deal with many-objectives. Mentions exponential number of nondominated solutions with respect to many objectives (but [1051] already did).
- [2567] R. C. Purshouse and P. J. Fleming. **On the Evolutionary Optimization of Many Conflicting Objectives.** *IEEE Trans. Evol. Comput.*, 11(6):770–784, 2007. doi:10.1109/TEVC.2007.910138.
- [2568] R. C. Purshouse, C. Jalbá, and P. J. Fleming. **Preference-Driven Co-Evolutionary Algorithms Show Promise for Many-Objective optimisation.** In R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors, *EMO*, volume 6576 of *LNCS*, pages 136–150. Springer, Berlin/Heidelberg, 2011.
- [2569] R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors. *Evolutionary Multi-Criterion Optimization – 7th International Conference, EMO 2013, Sheffield, UK, March 19-22, 2013. Proceedings*, volume 7811 of *LNCS*. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4.
- [2570] R. C. Purshouse, K. Deb, M. M. Mansor, S. Mostaghim, and R. Wang. **A review of hybrid evolutionary multiple criteria decision making methods.** COIN Report 2014005, Computational Optimization and Innovation (COIN) Laboratory, University of Michigan, USA, Jan. 2014.

- [2571] M. Püschel, F. Franchetti, and Y. Voronenko. **Spiral**. In D. Padua, editor, *Encyclopedia of Parallel Computing*, pages 1920–1933. Springer, US, 2011. doi:10.1007/978-0-387-09766-4_244.
- [2572] Y. Pushak and H. H. Hoos. **Algorithm Configuration Landscapes: More Benign Than Expected?** In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11101 of *LNCS*, pages 271–283. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-319-99259-4_22. Supplementary material: <http://www.cs.ubc.ca/labs/beta/Projects/ACLandscapes/>.
Annotation: Best paper award at PPSN2018.
- [2573] Y. Pushak and H. H. Hoos. **Golden parameter search: exploiting structure to quickly configure parameters in parallel**. In C. A. Coello Coello, editor, *GECCO*, pages 245–253. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.
Keywords: algorithm configuration.
- [2574] Y. Pushak, Z. Mu, and H. H. Hoos. **Empirical scaling analyzer: An automated system for empirical analysis of performance scaling**. *AI Communications*, 33(2):93–111, Sept. 2020. doi:10.3233/aic-200630.
- [2575] Y. Qi, X. Ma, F. Liu, L. Jiao, J. Sun, and J. Wu. **MOEA/D with adaptive weight adjustment**. *Evol. Comput.*, 22(2):231–264, 2014. doi:10.1162/EVCO_a_00109.
Annotation: Uses an external population.
- [2576] R. Qu, N. Pillay, E. Hart, and M. López-Ibáñez. **Guest Editorial: Machine-Learning-Assisted Evolutionary Computation**. *IEEE Trans. Evol. Comput.*, 9(3):571–573, 2025. doi:10.1109/TEVC.2025.3548888.
Keywords: Special issue.
- [2577] G. Quan, G. W. Greenwood, D. Liu, and S. Hu. **Searching for multiobjective preventive maintenance schedules: Combining preferences with evolutionary algorithms**. *Eur. J. Oper. Res.*, 177(3):1969–1984, 2007. doi:10.1016/j.ejor.2005.12.015.
Keywords: Evolutionary computations, Scheduling, Utility theory, Preventive maintenance, Multi-objective optimization, ranking-based, interactive.
- [2578] G. Quan, M. Sun, and M. López-Ibáñez. **Re-evaluating LLM-based Heuristic Search: A Case Study on the 3D Packing Problem**. *Arxiv preprint arXiv:2509.02297 [cs.AI]*, 2025. doi:10.48550/ARXIV.2509.02297.
- [2579] G. Quan, M. Sun, M. López-Ibáñez, N. Álvarez-Gil, and S. Fernández Alzueta. **Accelerating LLM-Based Algorithm Evolution for the 3D Container Loading Problem**. In T. Hu and L. Trujillo, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026*. ACM Press, New York, NY, 2026. doi:10.1145/3795095.3805178.
- [2580] G. Quan, M. Sun, M. López-Ibáñez, N. Álvarez-Gil, and S. Fernández Alzueta. **Accelerating LLM-Based Algorithm Evolution for the 3D Container Loading Problem: Supplementary Material**. Zenodo, Apr. 2026.
- [2581] J. D. Quinn, P. M. Reed, and K. Keller. **Direct policy search for robust multi-objective management of deeply uncertain socio-ecological tipping points**. *Environmental Modelling & Software*, 92:125–141, 2017.
- [2582] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2008. URL <http://www.R-project.org>.
- [2583] L. Rachmawati and D. Srinivasan. **Preference incorporation in multiobjective evolutionary algorithms: A survey**. In *IEEE CEC*, pages 3385–3391, Piscataway, NJ, July 2006. IEEE Press.

- [2584] S. F. Rad, R. Ruiz, and N. Boroojerian. **New High Performing Heuristics for Minimizing Makespan in Permutation Flowshops**. *Omega*, 37(2):331–345, 2009.
- [2585] A. Radulescu, M. López-Ibáñez, and T. Stützle. **Automatically Improving the Anytime Behaviour of Multiobjective Evolutionary Algorithms**. Technical Report TR/IRIDIA/2012-019, IRIDIA, Université Libre de Bruxelles, Belgium, 2012. Published in the proceedings of EMO 2013 [2586].
- [2586] A. Radulescu, M. López-Ibáñez, and T. Stützle. **Automatically Improving the Anytime Behaviour of Multiobjective Evolutionary Algorithms**. In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 825–840. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4. doi:10.1007/978-3-642-37140-0_61.
- [2587] A. A. M. Rahat, R. M. Everson, and J. E. Fieldsend. **Alternative infill strategies for expensive multi-objective optimisation**. In P. A. N. Bosman, editor, *GECCO*, pages 873–880. ACM Press, New York, NY, 2017.
- [2588] G. R. Raidl and J. Gottlieb, editors. *Proceedings of EvoCOP 2003 – 3rd European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 2611 of *LNCS*. Springer, Heidelberg, Germany, 2003.
- [2589] G. R. Raidl and J. Gottlieb, editors. *Proceedings of EvoCOP 2005 – 5th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3448 of *LNCS*. Springer, Heidelberg, Germany, 2005.
- [2590] G. R. Raidl et al., editors. *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2004*, volume 3005 of *LNCS*. Springer, Heidelberg, Germany, 2004.
- [2591] C. Rajendran. **Heuristic algorithm for scheduling in a flowshop to minimize total flowtime**. *International Journal of Production Economics*, 29(1):65–73, 1993.
- [2592] C. Rajendran and H. Ziegler. **An efficient heuristic for scheduling in a flowshop to minimize total weighted flowtime of jobs**. *Eur. J. Oper. Res.*, 103(1):129–138, 1997. ISSN 0377 – 2217. doi:10.1016/S0377-2217(96)00273-1.
- [2593] C. Rajendran and H. Ziegler. **Ant-colony algorithms for permutation flowshop scheduling to minimize makespan/total flowtime of jobs**. *Eur. J. Oper. Res.*, 155(2):426–438, 2004.
- [2594] C. Ram, G. Montibeller, and A. Morton. **Extending the use of scenario planning and MCDA for the evaluation of strategic options**. *J. Oper. Res. Soc.*, 62(5):817–829, 2011.
- [2595] R. Ramakrishnan, S. J. Stolfo, R. J. Bayardo, and I. Parsa, editors. *Proceedings of the sixth ACM SIGKDD international conference on Knowledge discovery and data mining, Boston, MA, USA, August 20-23, 2000*. ACM Press, New York, NY, 2000.
- [2596] R. Ramamonjison, T. T. Yu, R. Li, H. Li, G. Carenini, B. Ghaddar, S. He, M. Mostajabdaveh, A. Banitalebi-Dehkordi, Z. Zhou, and Y. Zhang. **NL4Opt Competition: Formulating Optimization Problems Based on Their Natural Language Descriptions**. *Arxiv preprint arXiv:2303.08233 [cs.CL]*, 2023. doi:10.48550/arXiv.2303.08233.
- [2597] D. G. Ramos and M. Birattari. **Automatic Design of Collective Behaviors for Robots that Can Display and Perceive Colors**. *Applied Sciences*, 10(13):4654, 2020.
- [2598] J.-M. Ramos-Pérez, G. Miranda, E. Segredo, C. León, and C. Rodríguez-León. **Application of Multi-Objective Evolutionary Algorithms for Planning Healthy and Balanced School Lunches**. *Mathematics*, 9(1):80, Dec. 2021. doi:10.3390/math9010080.

- [2599] M. Randall. **Near Parameter Free Ant Colony Optimisation**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 374–381. Springer, Heidelberg, Germany, 2004.
- [2600] M. Randall and J. Montgomery. **Candidate Set Strategies for Ant Colony Optimisation**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 243–249. Springer, Heidelberg, Germany, 2002.
- [2601] M. Randall, H. A. Abbass, and J. Wiles, editors. *Progress in Artificial Life, Third Australian Conference, ACAL 2007*, volume 4828 of *LNCS*. Springer, Heidelberg, Germany, 2007.
- [2602] M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. W. Vaughan, editors. *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- [2603] C. R. Rao. **Hypercubes of Strength “d” Leading to Confounded Designs in Factorial Experiments**. *Bulletin of the Calcutta Mathematical Society*, 38:67–78, 1946.
Keywords: orthogonal arrays.
- [2604] Z. Rao and E. Salomons. **Development of a real-time, near-optimal control process for water-distribution networks**. *Journal of Hydroinformatics*, 9(1):25–37, 2007. doi:10.2166/hydro.2006.015.
- [2605] Z. Rao, J. Wicks, and S. West. **ENCOMS - An Energy Cost Minimisation System for Real-Time, Operational Control of Water Distribution Networks**. In D. A. Savic, G. A. Walters, R. King, and S. Thiam-Khu, editors, *Proceedings of the Eighth International Conference on Computing and Control for the Water Industry (CCWI 2005)*, volume 1, pages 85–90, University of Exeter, UK, Sept. 2005.
- [2606] J. Rapin and O. Teytaud. **Nevergrad: A gradient-free optimization platform**. <https://github.com/FacebookResearch/Nevergrad>, 2018.
- [2607] E. Raponi, H. Wang, M. Bujny, S. Boria, and C. Doerr. **High Dimensional Bayesian Optimization Assisted by Principal Component Analysis**. In T. Bäck, M. Preuss, A. Deutz, H. Wang, C. Doerr, M. T. M. Emmerich, and H. Trautmann, editors, *Parallel Problem Solving from Nature – PPSN XVI*, volume 12269 of *LNCS*, pages 169–183. Springer, Cham, Switzerland, 2020. doi:10.1007/978-3-030-58112-1_12.
- [2608] M. Rappa, P. Jones, J. Freire, and S. Chakrabarti, editors. *World Wide Web Conference, WWW 2010, Proceedings, Raleigh, North Carolina, USA, April 26-30, 2010*. ACM Press, New York, NY, 2010.
- [2609] R. L. Rardin and R. Uzsoy. **Experimental Evaluation of Heuristic Optimization Algorithms: A Tutorial**. *J. Heuristics*, 7(3):261–304, 2001.
- [2610] J. Rasku, N. Musliu, and T. Kärkkäinen. **Automating the Parameter Selection in VRP: An Off-line Parameter Tuning Tool Comparison**. In W. Fitzgibbon, Y. A. Kuznetsov, P. Neittaanmäki, and O. Pironneau, editors, *Modeling, Simulation and Optimization for Science and Technology*, volume 34 of *Computational Methods in Applied Sciences*, pages 191–209. Springer, 2014. doi:10.1007/978-94-017-9054-3_11.
Keywords: irace.
- [2611] J. Rasku, N. Musliu, and T. Kärkkäinen. **On automatic algorithm configuration of vehicle routing problem solvers**. *Journal on Vehicle Routing Algorithms*, 2(1-4):1–22, Feb. 2019. doi:10.1007/s41604-019-00010-9.
Keywords: irace, SMAC, GGA, REVAC, VRP.
- [2612] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA, 2006. ISBN 026218253X.
Keywords: Gaussian processes, data processing.
Annotation: Kinematics of a Robot Arm Benchmark.

- [2613] A. Rasulo, K. Smith-Miles, M. A. Muñoz, J. Handl, and M. López-Ibáñez. **Extending Instance Space Analysis to Algorithm Configuration Spaces**. In J. Handl and X. Li, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2024*, pages 147–150. ACM Press, New York, NY, 2024. doi:[10.1145/3638530.3654264](https://doi.org/10.1145/3638530.3654264).
Keywords: algorithm configuration, instance space analysis, explainability.
Annotation: ISBN: 979-8-4007-0495-6.
- [2614] A. Rasulo, M. López-Ibáñez, J. Handl, M. A. Muñoz, and K. Smith-Miles. **An Efficient Hybrid Racing Method for Portfolio Configuration**. In T. Hu and L. Trujillo, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026*. ACM Press, New York, NY, 2026. doi:[10.1145/3795095.3805134](https://doi.org/10.1145/3795095.3805134).
- [2615] G. Rawlins, editor. *Foundations of Genetic Algorithms*. Morgan Kaufmann Publishers, San Mateo, CA, 1991.
- [2616] N. Rayner. **Maverick Research: Judgment Day, or Why We Should Let Machines Automate Decision Making**. Gartner research note, Gartner, Inc, Oct. 2011.
- [2617] I. Rechenberg. *Evolutionsstrategie: Optimierung technischer Systeme nach Prinzipien der biologischen Evolution*. PhD thesis, Department of Process Engineering, Technical University of Berlin, 1971.
Annotation: Reprinted as [2618].
- [2618] I. Rechenberg. *Evolutionsstrategie: Optimierung technischer Systeme nach Prinzipien der biologischen Evolution*. Frommann-Holzboog, Stuttgart, Germany, 1973.
- [2619] I. Rechenberg. **Case studies in evolutionary experimentation and computation**. *Computer Methods in Applied Mechanics and Engineering*, 186(2-4):125–140, 2000. doi:[10.1016/S0045-7825\(99\)00381-3](https://doi.org/10.1016/S0045-7825(99)00381-3).
- [2620] P. M. Reed. **Many-Objective Visual Analytics: Rethinking the Design of Complex Engineered Systems**. In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 1–1. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4.
- [2621] P. M. Reed, D. Hadka, J. D. Herman, J. R. Kasprzyk, and J. B. Kollat. **Evolutionary multiobjective optimization in water resources: The past, present, and future**. *Advances in Water Resources*, 51:438–456, 2013.
- [2622] C. R. Reeves. **Genetic algorithms**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, chapter 5, pages 109–140. Springer, New York, NY, 2nd edition, 2010.
- [2623] C. R. Reeves and A. V. Eremeev. **Statistical analysis of local search landscapes**. *J. Oper. Res. Soc.*, 55(7):687–693, 2004.
- [2624] G. R. Reeves and J. J. Gonzalez. **A comparison of two interactive MCDM procedures**. *Eur. J. Oper. Res.*, 41(2):203–209, 1989. doi:[10.1016/0377-2217\(89\)90385-8](https://doi.org/10.1016/0377-2217(89)90385-8).
Keywords: artificial DM, interactive.
- [2625] F. Rehbach, M. Zaefferer, A. Fischbach, G. Rudolph, and T. Bartz-Beielstein. **Benchmark-Driven Configuration of a Parallel Model-Based Optimization Algorithm**. *IEEE Trans. Evol. Comput.*, 26(6):1365–1379, 2022. doi:[10.1109/TEVC.2022.3163843](https://doi.org/10.1109/TEVC.2022.3163843).
- [2626] M. Reimann. **Guiding ACO by Problem Relaxation: A Case Study on the Symmetric TSP**. In T. Bartz-Beielstein, M. J. Blesa, C. Blum, B. Naujoks, A. Roli, G. Rudolph, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 4771 of *LNCS*, pages 45–56. Springer, Heidelberg, Germany, 2007.

- [2627] M. Reimann and M. Laumanns. **Savings based ant colony optimization for the capacitated minimum spanning tree problem.** *Comput. Oper. Res.*, 33(6):1794–1822, 2006. doi:[10.1016/j.cor.2004.11.019](https://doi.org/10.1016/j.cor.2004.11.019).
Keywords: Ant colony Optimization, Capacitated minimum spanning tree problem.
- [2628] M. Reimann, K. F. Doerner, and R. F. Hartl. **D-ants: Savings based ants divide and conquer the vehicle routing problems.** *Comput. Oper. Res.*, 31(4):563–591, 2004.
- [2629] G. Reinelt. **TSPLIB — A Traveling Salesman Problem Library.** *ORSA Journal on Computing*, 3(4):376–384, 1991.
- [2630] G. Reinelt. *The Traveling Salesman: Computational Solutions for TSP Applications*, volume 840 of *LNCS*. Springer, Heidelberg, Germany, 1994.
- [2631] G. Reinelt. **TSPLIB.** <http://comopt.ifl.uni-heidelberg.de/software/TSPLIB95/>, 1995. Version visited last on 24 February 2023.
- [2632] W. Remus and J. Kottemann. **Anchor-and-adjustment behaviour in a dynamic decision environment.** *Decision Support Systems*, 15(1):63–74, 1995. doi:[10.1016/0167-9236\(94\)00051-S](https://doi.org/10.1016/0167-9236(94)00051-S).
- [2633] Z.-G. Ren, Z.-R. Feng, L.-J. Ke, and Z.-J. Zhang. **New Ideas for Applying Ant Colony Optimization to the Set Covering Problem.** *Computers & Industrial Engineering*, 58(4):774–784, 2010.
- [2634] M. G. C. Resende and J. Pinho de Souza, editors. *Proceedings of MIC 1997, the 2nd Metaheuristics International Conference, Sophia-Antipolis, France, July 21-24, 1997*, 1997.
- [2635] M. G. C. Resende and C. C. Ribeiro. **Greedy Randomized Adaptive Search Procedures.** In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 219–249. Kluwer Academic Publishers, Norwell, MA, 2002.
- [2636] M. G. C. Resende and C. C. Ribeiro. **Greedy Randomized Adaptive Search Procedures: Advances, Hybridizations, and Applications.** In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 146 of *International Series in Operations Research & Management Science*, pages 283–319. Springer, New York, NY, 2nd edition, 2010.
- [2637] M. Reyes Sierra and C. A. Coello Coello. **Improving PSO-Based Multi-objective Optimization Using Crowding, Mutation and ϵ -Dominance.** In C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors, *EMO*, volume 3410 of *LNCS*, pages 505–519. Springer, Berlin/Heidelberg, 2005.
Keywords: OMOPSO.
- [2638] M. Reyes-Sierra and C. A. Coello Coello. **Multi-objective particle swarm optimizers: A survey of the state-of-the-art.** *International Journal of Computational Intelligence Research*, 2(3):287–308, 2006.
- [2639] C. W. Reynolds. **Flocks, Herds, and Schools: A Distributed Behavioral Model.** *ACM Computer Graphics*, 21(4):25–34, 1987.
- [2640] J. Rezaei. **Anchoring bias in eliciting attribute weights and values in multi-attribute decision-making.** *Journal of Decision Systems*, 30(1):72–96, 2020. doi:[10.1080/12460125.2020.1840705](https://doi.org/10.1080/12460125.2020.1840705).
- [2641] J. Rezaei, A. Arab, and M. Mehregan. **Analyzing anchoring bias in attribute weight elicitation of SMART, Swing, and best-worst method.** *International Transactions in Operational Research*, 2022. doi:[10.1111/itor.13171](https://doi.org/10.1111/itor.13171).
Keywords: anchoring bias, best-worst method, cognitive bias, MADM, multi-attribute weighting, SMART, Swing.

- [2642] M. Riabacke, M. Danielson, L. Ekenberg, and A. Larsson. **A Prescriptive Approach for Eliciting Imprecise Weight Statements in an MCDA Process.** In F. Rossi and A. Tsoukiàs, editors, *Algorithmic Decision Theory, First International Conference, ADT 2009*, volume 5783 of *LNCS*, pages 168–179. Springer, Heidelberg, Germany, 2009.
- [2643] I. Ribas, R. Companys, and X. Tort-Martorell. **An iterated greedy algorithm for the flowshop scheduling problem with blocking.** *Omega*, 39(3):293 – 301, 2011.
- [2644] I. Ribas, R. Companys, and X. Tort-Martorell. **An Efficient Iterated Local Search Algorithm for the Total Tardiness Blocking Flow Shop Problem.** *Int. J. Prod. Res.*, 51(17):5238–5252, 2013.
- [2645] C. C. Ribeiro and S. Urrutia. **Heuristics for the Mirrored Traveling Tournament Problem.** *Eur. J. Oper. Res.*, 179(3):775–787, 2007.
- [2646] J. R. Rice. **The Algorithm Selection Problem.** *Advances in Computers*, 15:65–118, 1976. doi:10.1016/S0065-2458(08)60520-3.
- [2647] A. J. Richmond and J. E. Beasley. **An Iterative Construction Heuristic for the Ore Selection Problem.** *J. Heuristics*, 10(2):153–167, 2004.
- [2648] E. Ridge and D. Kudenko. **Tuning the Performance of the MMAS Heuristic.** In T. Stützle, M. Birattari, and H. H. Hoos, editors, *Engineering Stochastic Local Search Algorithms. Designing, Implementing and Analyzing Effective Heuristics. SLS 2007*, volume 4638 of *LNCS*, pages 46–60. Springer, Heidelberg, Germany, 2007.
- [2649] E. Ridge and D. Kudenko. **Tuning an Algorithm Using Design of Experiments.** In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 265–286. Springer, Berlin/Heidelberg, 2010.
- [2650] M.-C. Riff and E. Montero. **A new algorithm for reducing metaheuristic design effort.** In *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, pages 3283–3290, Piscataway, NJ, 2013. IEEE Press. doi:10.1109/CEC.2013.6557972.
- [2651] L. Rivadeneira, J.-B. Yang, and M. López-Ibáñez. **Predicting tweet impact using a novel evidential reasoning prediction method.** *Expert Systems with Applications*, 169:114400, May 2021. doi:10.1016/j.eswa.2020.114400.
Keywords: Evidential reasoning rule,Belief rule-based inference,Maximum likelihood data analysis,Twitter,Retweet,Prediction.
- [2652] J. C. Rivera, H. M. Afsar, and C. Prins. **A Multistart Iterated Local Search for the Multitrip Cumulative Capacitated Vehicle Routing Problem.** *Computational Optimization and Applications*, 61(1):159–187, 2015.
- [2653] A. Rivolli, L. P. F. Garcia, C. Soares, J. Vanschoren, and A. C. P. L. F. Carvalho. **Meta-Features for Meta-Learning.** *Knowledge-Based Systems*, 240:108101, 2022. doi:10.1016/j.knosys.2021.108101.
Keywords: Characterization measures,Classification problems,Meta-features,Meta-learning.
- [2654] C. P. Robert. **Simulation of truncated normal variables.** *Statistics and Computing*, 5(2): 121–125, June 1995.
- [2655] M. Roberts. **The Unreasonable Effectiveness of Quasirandom Sequences.** <https://extremelearning.com.au/unreasonable-effectiveness-of-quasirandom-sequences/>, Apr. 2018. Last visited: 13/04/2025.
Annotation: Proposed the Rphi low-discrepancy sequence.

- [2656] T. Robič and B. Filipič. **DEMO: Differential Evolution for Multiobjective Optimization**. In C. A. Coello Coello, A. Hernández Aguirre, and E. Zitzler, editors, *EMO*, volume 3410 of *LNCS*, pages 520–533. Springer, Berlin/Heidelberg, 2005.
- [2657] F. J. Rodríguez, C. Blum, M. Lozano, and C. García-Martínez. **Iterated Greedy Algorithms for the Maximal Covering Location Problem**. In J.-K. Hao and M. Middendorf, editors, *EvoCOP*, volume 7245 of *LNCS*, pages 172–181. Springer, Heidelberg, Germany, 2012.
- [2658] C. A. Rodríguez Villalobos and C. A. Coello Coello. **A new multi-objective evolutionary algorithm based on a performance assessment indicator**. In T. Soule and J. H. Moore, editors, *GECCO*, pages 505–512. ACM Press, New York, NY, 2012.
- [2659] F. Romeo and A. Sangiovanni-Vincentelli. **A Theoretical Framework for Simulated Annealing**. *Algorithmica*, 6(1-6):302–345, 1991.
- [2660] B. Romera-Paredes, M. Barekatin, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, et al. **Mathematical discoveries from program search with large language models**. *Nature*, 625(7995):468–475, 2024. [doi:10.1038/s41586-023-06924-6](https://doi.org/10.1038/s41586-023-06924-6).
Keywords: FunSearch, 1D bin packing.
- [2661] P. A. Romero, A. Krause, and F. H. Arnold. **Navigating the Protein Fitness Landscape with Gaussian Processes**. *Proceedings of the National Academy of Sciences*, 110(3):E193–E201, Dec. 2012. [doi:10.1073/pnas.1215251110](https://doi.org/10.1073/pnas.1215251110).
Keywords: Combinatorial Black-box Expensive.
- [2662] J. Rook, Q. Renau, H. Trautmann, and E. Hart. **Efficient Online Automated Algorithm Selection in the Face of Data-Drift in Optimisation Problem Instances**. In A. V. Kononova, T. Bäck, J. E. Fieldsend, and F. Chicano, editors, *Proceedings of the 18th ACM/SIGEVO Conference on Foundations of Genetic Algorithms*, pages 262–272. ACM, 2025. [doi:10.1145/3729878.3746615](https://doi.org/10.1145/3729878.3746615).
- [2663] D. S. Roos. **Bioinformatics—trying to swim in a sea of data**. *Science*, 291(5507):1260–1261, 2001.
- [2664] S. Ropke and D. Pisinger. **A Unified Heuristic for a Large Class of Vehicle Routing Problems with Backhauls**. *Eur. J. Oper. Res.*, 171(3):750–775, 2006.
- [2665] S. Ropke and D. Pisinger. **An Adaptive Large Neighborhood Search Heuristic for the Pickup and Delivery Problem with Time Windows**. *Transportation Science*, 40(4):455–472, 2006.
- [2666] J. Rose, W. Klebsch, and J. Wolf. **Temperature measurement and equilibrium dynamics of simulated annealing placements**. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 9(3):253–259, 1990.
- [2667] B. C. Ross. **Mutual Information between Discrete and Continuous Data Sets**. *PLoS One*, 9(2):1–5, Feb. 2014. [doi:10.1371/journal.pone.0087357](https://doi.org/10.1371/journal.pone.0087357).
- [2668] P. Ross. **Hyper-Heuristics**. In E. K. Burke and G. Kendall, editors, *Search Methodologies*, pages 529–556. Springer, Boston, MA, 2005. [doi:10.1007/0-387-28356-0_17](https://doi.org/10.1007/0-387-28356-0_17).
- [2669] F. Rossi and A. Tsoukiàs, editors. *Algorithmic Decision Theory, First International Conference, ADT 2009, Venice, Italy, October 20-23, 2009*, volume 5783 of *LNCS*. Springer, Heidelberg, Germany, 2009.
- [2670] L. A. Rossman. *EPANET User’s Guide*. Risk Reduction Engineering Laboratory, Office of Research and Development, U.S. Environmental Protection Agency, Cincinnati, USA, 1994.

- [2671] L. A. Rossman. **The EPANET Programmer's Toolkit for Analysis of Water Distribution Systems.** In *Proceedings of the Annual Water Resources Planning and Management Conference*, Reston, USA, 1999. ASCE.
Annotation: CD-ROM.
- [2672] L. A. Rossman. *EPANET 2 Users Manual*. U.S. Environmental Protection Agency, Cincinnati, USA, 2000.
- [2673] E. Rothberg. **An evolutionary algorithm for polishing mixed integer programming solutions.** *INFORMS Journal on Computing*, 19(4):534–541, 2007.
- [2674] F. Rothlauf, editor. *Genetic and Evolutionary Computation Conference, GECCO 2009, Proceedings, Montreal, Québec, Canada, July 8-12, 2009*. ACM Press, New York, NY, 2009.
- [2675] F. Rothlauf, editor. *Genetic and Evolutionary Computation Conference, GECCO 2009, Proceedings, Montreal, Québec, Canada, July 8-12, 2009, Companion Material*. ACM Press, New York, NY, 2009.
- [2676] D. H. Rothman. **Nonlinear inversion, statistical mechanics, and residual statics estimation.** *Geophysics*, 50(12):2784–2796, 1985.
- [2677] D. H. Rothman. **Automatic estimation of large residual statics corrections.** *Geophysics*, 51(2):332–346, 1986.
- [2678] B. Roy. **Robustness in operational research and decision aiding: A multi-faceted issue.** *Eur. J. Oper. Res.*, 200(3):629–638, 2010. doi:10.1016/j.ejor.2008.12.036.
- [2679] F. Rubin. **An Iterative Technique for Printed Wire Routing.** In *DAC'74, Proceedings of the 11th Design Automation Workshop*, pages 308–313. IEEE Press, 1974.
- [2680] R. Y. Rubinstein and B. Melamed. *Modern simulation and modeling*. Wiley, New York, NY, 1998.
- [2681] I. Rudich, Q. Cappart, and L.-M. Rousseau. **Peel-And-Bound: Generating Stronger Relaxed Bounds with Multivalued Decision Diagrams.** In C. Solnon, editor, *Principles and Practice of Constraint Programming*, volume 235 of *LIPICs*, pages 35:1–35:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2022. ISBN 978-3-95977-240-2. doi:10.4230/LIPICs.CP.2022.35.
- [2682] I. Rudich, Q. Cappart, M. López-Ibáñez, M. Römer, and L.-M. Rousseau. **An Exact Framework for Solving the Space-Time Dependent TSP.** *Arxiv preprint arXiv:2312.01404 [math.OC]*, 2023. doi:10.48550/arXiv.2312.01404.
- [2683] I. Rudich, Q. Cappart, and L.-M. Rousseau. **Improved Peel-and-Bound: Methods for Generating Dual Bounds with Multivalued Decision Diagrams.** *J. Artif. Intell. Res.*, 77: 1489–1538, Aug. 2023. doi:10.1613/jair.1.14607.
- [2684] I. Rudich, M. López-Ibáñez, M. Römer, Q. Cappart, and L.-M. Rousseau. **An Exact Framework for Solving the Space-Time Dependent TSP.** *INFORMS Journal on Computing*, 2026. doi:10.1287/ijoc.2024.0866.
- [2685] G. Rudolph. **Globale Optimierung mit parallelen Evolutionsstrategien.** Diplomarbeit, Department of Computer Science, University of Dortmund, July 1990.
Annotation: Proposed the generalized Rastrigin function.
- [2686] G. Rudolph. **On Correlated Mutations in Evolution Strategies.** In R. Männer and B. Manderick, editors, *Parallel Problem Solving from Nature – PPSN II*, pages 107–116. Elsevier, 1992.

- [2687] G. Rudolph. **Convergence of non-elitist strategies**. In Z. Michalewicz, editor, *Proceedings of the First IEEE International Conference on Evolutionary Computation (ICEC'94)*, pages 63–66. IEEE Press, Piscataway, NJ, 1994.
- [2688] G. Rudolph. **Convergence analysis of canonical genetic algorithms**. *IEEE Transactions on Neural Networks*, 5(1):96–101, 1994.
- [2689] G. Rudolph. **Evolutionary Search for Minimal Elements in Partially Ordered Finite Sets**. In V. W. Porto, N. Saravanan, D. E. Waagen, and A. E. Eiben, editors, *International Conference on Evolutionary Programming*, volume 1447 of *LNCS*, pages 345–353. Springer, 1998. doi:10.1007/BFb0040787.
- [2690] G. Rudolph and A. Agapie. **Convergence Properties of Some Multi-Objective Evolutionary Algorithms**. In *IEEE CEC*, volume 2, pages 1010–1016, Piscataway, NJ, July 2000. IEEE Press.
- [2691] G. Rudolph, H. Trautmann, S. Sengupta, and O. Schütze. **Evenly spaced Pareto front approximations for tricriteria problems based on triangulation**. In R. C. Purshouse, P. J. Fleming, C. M. Fonseca, S. Greco, and J. Shaw, editors, *EMO*, volume 7811 of *LNCS*, pages 443–458. Springer, Heidelberg, Germany, 2013. ISBN 978-3-642-37139-4. Annotation: unbounded archiver, AA_{Δ_1} .
- [2692] G. Rudolph, O. Schütze, C. Grimme, C. Domínguez-Medina, and H. Trautmann. **Optimal averaged Hausdorff archives for bi-objective problems: theoretical and numerical results**. *Computational Optimization and Applications*, 64(2):589–618, 2016.
- [2693] G. Rudolph, A. V. Kononova, H. E. Aguirre, P. Kerschke, G. Ochoa, and T. Tušar, editors. *Parallel Problem Solving from Nature - PPSN XVII, 17th International Conference, PPSN 2022, Dortmund, Germany, September 10-14, 2022, Proceedings, Part I*, volume 13398 of *LNCS*. Springer, Cham, Switzerland, 2022.
- [2694] G. Rudolph et al., editors. *Proceedings of PPSN-X, Tenth International Conference on Parallel Problem Solving from Nature*, volume 5199 of *LNCS*. Springer, Heidelberg, Germany, 2008.
- [2695] A. B. Ruiz, M. Luque, K. Miettinen, and R. Saborido. **An Interactive Evolutionary Multiobjective Optimization Method: Interactive WASF-GA**. In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part II*, volume 9019 of *LNCS*, pages 249–263. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-15892-1_17.
- [2696] A. B. Ruiz, R. Saborido, and M. Luque. **A preference-based evolutionary algorithm for multiobjective optimization: the weighting achievement scalarizing function genetic algorithm**. *J. Glob. Optim.*, 62(1):101–129, May 2015. doi:10.1007/s10898-014-0214-y. Annotation: Proposed WASF-GA.
- [2697] A. B. Ruiz, K. Sindhya, K. Miettinen, F. Ruiz, and M. Luque. **E-NAUTILUS: A decision support system for complex multiobjective optimization problems based on the NAUTILUS method**. *Eur. J. Oper. Res.*, 246(1):218–231, 2015. doi:10.1016/j.ejor.2015.04.027.
- [2698] A. B. Ruiz, F. Ruiz, K. Miettinen, L. Delgado-Antequera, and V. Ojalehto. **NAUTILUS Navigator: free search interactive multiobjective optimization without trading-off**. *J. Glob. Optim.*, 74:213–231, 2019. doi:10.1007/s10898-019-00765-2.
- [2699] R. Ruiz and C. Maroto. **A Comprehensive Review and Evaluation of Permutation Flowshop Heuristics**. *Eur. J. Oper. Res.*, 165(2):479–494, 2005.
- [2700] R. Ruiz and T. Stützle. **A Simple and Effective Iterated Greedy Algorithm for the Permutation Flowshop Scheduling Problem**. *Eur. J. Oper. Res.*, 177(3):2033–2049, 2007.

- [2701] R. Ruiz and T. Stützle. **An Iterated Greedy heuristic for the sequence dependent setup times flowshop problem with makespan and weighted tardiness objectives.** *Eur. J. Oper. Res.*, 187(3):1143 – 1159, 2008.
- [2702] R. Ruiz, C. Maroto, and J. Alcaraz. **Two new robust genetic algorithms for the flowshop scheduling problem.** *Omega*, 34(5):461–476, 2006. doi:10.1016/j.omega.2004.12.006.
- [2703] R. Ruiz, E. Vallada, and C. Fernández-Martínez. **Scheduling in flowshops with no-idle machines.** In *Computational intelligence in flow shop and job shop scheduling*, pages 21–51. Springer, 2009.
- [2704] W. Ruml. **Incomplete Tree Search using Adaptive Probing.** In B. Nebel, editor, *Proc. of IJCAI-01*, pages 235–241. IEEE Press, 2001.
- [2705] T. P. Runarsson, H.-G. Beyer, E. K. Burke, J.-J. Merelo, D. Whitley, and X. Yao, editors. *Proceedings of PPSN-IX, Ninth International Conference on Parallel Problem Solving from Nature*, volume 4193 of *LNCS*. Springer, Heidelberg, Germany, 2006.
- [2706] R. A. Russell. **Hybrid Heuristics for the Vehicle Routing Problem with Time Windows.** *Transportation Science*, 29(2):156–166, 1995.
- [2707] S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*, volume 2. Prentice Hall, Englewood Cliffs, NJ, 2003.
- [2708] J. Rust. **Structural estimation of Markov decision processes.** In *Handbook of Econometrics*, volume 4, pages 3081–3143. Elsevier, 1994. doi:10.1016/S1573-4412(05)80020-0.
- [2709] B. Y. Ryabko. **A fast on-line code.** *Doklady Akademii Nauk SSSR*, 306(3):548–552, 1989. Translation in *Soviet Mathematics - Doklady*, 39(3): 533–537, 1989. Annotation: Proposed Fenwick trees.
- [2710] C. Ryan, editor. *Genetic and Evolutionary Computation Conference, GECCO 2008, Proceedings, Atlanta, Georgia, USA July 12-16, 2008*. ACM Press, New York, NY, 2008.
- [2711] N. R. Sabar, M. Ayob, G. Kendall, and R. Qu. **Grammatical Evolution Hyper-Heuristic for Combinatorial Optimization Problems.** *IEEE Trans. Evol. Comput.*, 17(6):840–861, 2013.
- [2712] N. R. Sabar, M. Ayob, G. Kendall, and R. Qu. **A Dynamic Multiarmed Bandit-Gene Expression Programming Hyper-Heuristic for Combinatorial Optimization Problems.** *IEEE Trans. Cybern.*, 45(2):217–228, 2015.
- [2713] N. R. Sabar, M. Ayob, G. Kendall, and R. Qu. **Automatic Design of a Hyper-Heuristic Framework With Gene Expression Programming for Combinatorial Optimization Problems.** *IEEE Trans. Evol. Comput.*, 19(3):309–325, 2015.
- [2714] M. Sacher, R. Duvigneau, O. Le Maitre, M. Durand, E. Berrini, F. Hauville, and J.-A. Astolfi. **A classification approach to efficient global optimization in presence of non-computable domains.** *Struct. Multidisc. Optim.*, 58(4):1537–1557, 2018. doi:10.1007/s00158-018-1981-8. Keywords: Safe optimization; CMA-ES, Gaussian processes; Least-Squares Support Vector Machine. Annotation: Proposed EGO-LS-SVM.
- [2715] P. J. Sadalage and M. Fowler. **NoSQL distilled.** AddisonWesley Professional, 2012.
- [2716] B. S. Saini, M. López-Ibáñez, and K. Miettinen. **Automatic Surrogate Modelling Technique Selection based on Features of Optimization Problems.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO Companion*, pages 1765–1772. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6748-6. doi:10.1145/3319619.3326890.

- [2717] A. B. A. Sakarya and L. W. Mays. **Optimal Operation of Water Distribution Pumps Considering Water Quality.** *Journal of Water Resources Planning and Management, ASCE*, 126(4):210–220, July / Aug. 2000.
- [2718] A. B. A. Sakarya, F. E. Goldman, and L. W. Mays. **Models for the optimal scheduling of pumps to meet water quality.** In D. A. Savic and G. A. Walters, editors, *Water Industry Systems: Modelling and Optimization Applications*, volume 2, pages 379–391. Research Studies Press Ltd., Baldock, United Kingdom, 1999.
- [2719] Y. Sakurai, K. Takada, T. Kawabe, and S. Tsuruta. **A method to control parameters of evolutionary algorithms by using reinforcement learning.** In *2010 Sixth International Conference on Signal-Image Technology and Internet Based Systems*, pages 74–79. IEEE, 2010.
- [2720] M. Samà, P. Pellegrini, A. D’Ariano, J. Rodriguez, and D. Pacciarelli. **Ant colony optimization for the real-time train routing selection problem.** *Transportation Research Part B: Methodological*, 85:89–108, 2016. doi:10.1016/j.trb.2016.01.005.
Keywords: irace.
- [2721] F. Sambo, B. Di Camillo, A. Franzin, A. Facchinetti, L. Hakaste, J. Kravic, G. Fico, J. Tuomilehto, L. Groop, R. Gabriel, T. Tuomi, and C. Cobelli. **A Bayesian Network analysis of the probabilistic relations between risk factors in the predisposition to type 2 diabetes.** In N. Lovell and L. Mainardi, editors, *37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2015, Proceedings*, Annual International Conference of the IEEE Engineering in Medicine and Biology, pages 2119–2122. IEEE Press, 2015.
- [2722] M. Sambridge. **Geophysical inversion with a neighbourhood algorithm—I. Searching a parameter space.** *Geophysical Journal International*, 138(2):479–494, 1999.
- [2723] J. Sánchez, M. Galán, and E. Rubio. **Applying a traffic lights evolutionary optimization technique to a real case: “Las Ramblas” area in Santa Cruz de Tenerife.** *IEEE Trans. Evol. Comput.*, 12(1):25–40, 2008.
Keywords: Cellular automata, Combinatorial optimization, Genetic algorithms, Microscopic traffic simulator, Traffic lights optimization.
- [2724] J. J. Sánchez-Medina, M. J. Galán-Moreno, and E. Rubio-Royo. **Traffic Signal Optimization in “La Almozara” District in Saragossa Under Congestion Conditions, Using Genetic Algorithms, Traffic Microsimulation, and Cluster Computing.** *IEEE Transactions on Intelligent Transportation Systems*, 11(1):132–141, Mar. 2010. ISSN 1524-9050. doi:10.1109/TITS.2009.2034383.
Keywords: cellular automata; genetic algorithms; road traffic; traffic light programming; urban traffic congestion.
- [2725] E. Sandgren. **Nonlinear integer and discrete programming in mechanical design optimization.** *Journal of Mechanical Design*, 112(2):223–229, 1990. doi:10.1115/1.2912596.
- [2726] N. Sankary and A. Ostfeld. **Stochastic Scenario Evaluation in Evolutionary Algorithms Used for Robust Scenario-Based Optimization.** *Water Resources Research*, 54(4): 2813–2833, 2018.
- [2727] A. Santiago, B. Dorronsoro, A. J. Nebro, J. J. Durillo, O. Castillo, and H. J. Fraire. **A novel multi-objective evolutionary algorithm with fuzzy logic based adaptive selection of operators: FAME.** *Inf. Sci.*, 471:233–251, 2019. ISSN 0020-0255. doi:10.1016/j.ins.2018.09.005.
Keywords: Multi-objective optimization, density estimation, evolutionary algorithm, adaptive algorithm, fuzzy logic, spatial spread deviation.

- [2728] A. Santini, S. Ropke, and L. M. Hvattum. **A comparison of acceptance criteria for the adaptive large neighbourhood search metaheuristic.** *J. Heuristics*, 24:783–815, 2018. doi:[10.1007/s10732-018-9377-x](https://doi.org/10.1007/s10732-018-9377-x).
- [2729] T. J. Santner, B. J. Williams, and W. I. Notz. *The Design and Analysis of Computer Experiments*. Springer Verlag, New York, NY, 2003. doi:[10.1007/978-1-4757-3799-8](https://doi.org/10.1007/978-1-4757-3799-8).
- [2730] V. Santucci. **An Iterative Optimization Algorithm for Planning Spacecraft Pathways Through Asteroids.** *Applied Sciences*, 14(23):10987, Nov. 2024. doi:[10.3390/app142310987](https://doi.org/10.3390/app142310987).
Keywords: FAT-RLS, Asteroid Routing Problem, randomized local search.
- [2731] V. Santucci. **A Simple yet Effective Algorithm for the Asteroid Routing Problem.** In F. Marcelloni, K. Madani, N. van Stein, and J. Filipe, editors, *Proceedings of the 16th International Joint Conference on Computational Intelligence - ECTA*, pages 50–59. SciTePress, 2024. doi:[10.5220/0012908900003837](https://doi.org/10.5220/0012908900003837).
- [2732] V. Santucci and M. Baiocchi. **A Fast Randomized Local Search for Low Budget Optimization in Black-Box Permutation Problems.** In *Proceedings of the 2022 World Congress on Computational Intelligence (WCCI 2022)*, Piscataway, NJ, 2022. IEEE Press.
Keywords: UMM, CEGO.
- [2733] R. Sass, E. Bergman, A. Biedenkapp, F. Hutter, and M. T. Lindauer. **DeepCAVE: An Interactive Analysis Tool for Automated Machine Learning.** *Arxiv preprint arXiv:2206.03493 [cs.LG]*, 2022. doi:[10.48550/arXiv.2206.03493](https://doi.org/10.48550/arXiv.2206.03493).
- [2734] K. Sato and C. Young. **An in-depth look at Google’s first Tensor Processing Unit (TPU).** <https://cloud.google.com/blog/products/ai-machine-learning/an-in-depth-look-at-googles-first-tensor-processing-unit-tpu>, 2017.
- [2735] M. W. P. Savelsbergh. **Local search in routing problems with time windows.** *Annals of Operations Research*, 4(1):285–305, Dec. 1985. doi:[10.1007/BF02022044](https://doi.org/10.1007/BF02022044).
- [2736] D. A. Savic and G. A. Walters, editors. *Water Industry Systems: Modelling and Optimization Applications*, volume 2. Research Studies Press Ltd., Baldock, United Kingdom, 1999.
- [2737] D. A. Savic, G. A. Walters, and M. Schwab. **Multiobjective Genetic Algorithms for Pump Scheduling in Water Supply.** In D. Corne and J. L. Shapiro, editors, *Evolutionary Computing Workshop, AISB’97*, volume 1305 of *LNCS*, pages 227–236. Berlin, Germany, 1997.
- [2738] D. A. Savic, G. A. Walters, R. King, and S. Thiam-Khu, editors. *Proceedings of the Eighth International Conference on Computing and Control for the Water Industry (CCWI 2005)*, volume 1, University of Exeter, UK, Sept. 2005.
- [2739] Y. Sawaragi, H. Nakayama, and T. Tanino. *Theory of multiobjective optimization*. Elsevier, 1985.
- [2740] D. K. Saxena and K. Deb. **Non-linear Dimensionality Reduction Procedures for Certain Large-Dimensional Multi-objective Optimization Problems: Employing Correntropy and a Novel Maximum Variance Unfolding.** In S. Obayashi et al., editors, *EMO*, volume 4403 of *LNCS*, pages 772–787. Springer, Heidelberg, Germany, 2007. doi:[10.1007/978-3-540-70928-2_58](https://doi.org/10.1007/978-3-540-70928-2_58).
- [2741] D. K. Saxena and K. Deb. **Trading on infeasibility by exploiting constraint’s criticality through multi-objectivization: A system design perspective.** In *Proceedings of the 2007 Congress on Evolutionary Computation (CEC 2007)*, pages 919–926, Piscataway, NJ, 2007. IEEE Press. doi:[10.1109/CEC.2007.4424568](https://doi.org/10.1109/CEC.2007.4424568).
Keywords: multi-objectivization.

- [2742] D. K. Saxena and K. Deb. **Dimensionality reduction of objectives and constraints in multi-objective optimization problems: A system design perspective.** In *Proceedings of the 2008 Congress on Evolutionary Computation (CEC 2008)*, pages 3204–3211, Piscataway, NJ, 2008. IEEE Press. doi:10.1109/CEC.2008.4631232.
- [2743] D. K. Saxena, J. A. Duro, A. Tiwari, K. Deb, and Q. Zhang. **Objective Reduction in Many-Objective Optimization: Linear and Nonlinear Algorithms.** *IEEE Trans. Evol. Comput.*, 17(1):77–99, 2013. doi:10.1109/TEVC.2012.2185847.
- [2744] R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors. *Parallel Problem Solving from Nature – PPSN XI*, volume 6238 of *LNCS*. Springer, Heidelberg, Germany, 2010.
- [2745] A. Schaerf. **Combining Local Search and Look-Ahead for Scheduling and Constraint Satisfaction Problems.** In M. E. Pollack, editor, *Proc. of IJCAI-97*, volume 2, pages 1254–1259. Morgan Kaufmann Publishers, 1997.
- [2746] J. D. Schaffer. **Multiple Objective Optimization with Vector Evaluated Genetic Algorithms.** In J. J. Grefenstette, editor, *Proceedings of the First International Conference on Genetic Algorithms (ICGA’85)*, pages 93–100. Lawrence Erlbaum Associates, 1985. ISBN 0805804269.
Keywords: VEGA.
Annotation: Download a scanned copy from: <http://gpbib.cs.ucl.ac.uk/icga/>.
- [2747] J. D. Schaffer, editor. *Proceedings of the 3rd International Conference on Genetic Algorithms (ICGA), George Mason University, Fairfax, Virginia, USA, June 1989*, 1989. Morgan Kaufmann Publishers, San Mateo, CA.
- [2748] J. R. Schames, R. H. Henschman, J. S. Siegel, C. A. Sotriffer, H. Ni, and J. A. McCammon. **Discovery of a Novel Binding Trench in HIV Integrase.** *Journal of Medicinal Chemistry*, 47(8):1879–1881, 2004. doi:10.1021/jm0341913.
Annotation: Evolutionary optimization of the first clinically approved anti-viral drug for HIV.
- [2749] J. C. Schank and T. J. Koehnle. **Pseudoreplication is a pseudoproblem.** *Journal of Comparative Psychology*, 123(4):421–433, 2009.
- [2750] L. Schäpermeier and P. Kerschke. **R2 v2: The Pareto-compliant R2 Indicator for Better Benchmarking in Bi-objective Optimization.** *Evol. Comput.*, pages 1–17, Oct. 2025. doi:10.1162/evco.a.366.
- [2751] E. Schede and K. Tierney. **A method for the automated configuration of anytime portfolios of algorithms.** *Eur. J. Oper. Res.*, 329(2):577–590, 2026. doi:10.1016/j.ejor.2025.07.024.
- [2752] E. Schede, J. Brandt, A. Tornede, M. Wever, V. Bengs, E. Hüllermeier, and K. Tierney. **A survey of methods for automated algorithm configuration.** *J. Artif. Intell. Res.*, 75: 425–487, 2022. doi:10.1613/jair.1.13676.
- [2753] H. Scheffe. *The Analysis of Variance*. John Wiley & Sons, New York, NY, 1st edition, 1959.
- [2754] G. Scheithauer. **Algorithms for the Container Loading Problem.** In W. Gaul, A. Bachem, W. Habenicht, W. Runge, and W. W. Stahl, editors, *Operations Research Proceedings 1991*, pages 445–452. Springer, Berlin/Heidelberg, 1992. doi:10.1007/978-3-642-46773-8_112.
Annotation: Proof that CLP is NP-hard.
- [2755] T. Schiavinotto and T. Stützle. **The Linear Ordering Problem: Instances, Search Space Analysis and Algorithms.** *Journal of Mathematical Modelling and Algorithms*, 3(4):367–402, 2004.

- [2756] T. Schiavinotto and T. Stützle. **A Review of Metrics on Permutations for Search Space Analysis**. *Comput. Oper. Res.*, 34(10):3143–3153, 2007.
- [2757] M. Schilde, K. F. Doerner, R. F. Hartl, and G. Kiechle. **Metaheuristics for the bi-objective orienteering problem**. *Swarm Intelligence*, 3(3):179–201, 2009. doi:10.1007/s11721-009-0029-5.
- [2758] M. Schillinger, B. Hartmann, P. Skalecki, M. Meister, D. Nguyen-Tuong, and O. Nelles. **Safe active learning and safe Bayesian optimization for tuning a PI-controller**. *IFAC-PapersOnLine*, 50(1):5967–5972, 2017. doi:10.1016/j.ifacol.2017.08.1258.
- [2759] M. Schillinger, B. Ortelt, B. Hartmann, J. Schreiter, M. Meister, D. Nguyen-Tuong, and O. Nelles. **Safe active learning of a high pressure fuel supply system**. In *Proceedings of the 9th EUROSIM Congress on Modelling and Simulation, EUROSIM 2016 and the 57th SIMS Conference on Simulation and Modelling SIMS 2016*, pages 286–292. Linköping University Electronic Press, 2018. doi:10.3384/ecp17142286.
- [2760] M. Schlüter, J. A. Egea, and J. R. Banga. **Extended ant colony optimization for non-convex mixed integer nonlinear programming**. *Comput. Oper. Res.*, 36(7):2217–2229, 2009. doi:10.1016/j.cor.2008.08.015.
- [2761] J. Schmea and G. J. Hahn. **A Simple Method for Regression Analysis with Censored Data**. *Technometrics*, 21(4):417–432, 1979. doi:10.2307/1268280.
- [2762] M. Schmidt and H. Lipson. **Age-Fitness Pareto Optimization**. In *Genetic Programming Theory and Practice VIII. Genetic and Evolutionary Computation*, pages 129–146. Springer, 2011. doi:10.1007/978-1-4419-7747-2_8.
- [2763] M. Schmidt, A. Schöbel, and L. Thom. **Min-ordering and max-ordering scalarization methods for multi-objective robust optimization**. *Eur. J. Oper. Res.*, 275(2):446–459, 2019.
- [2764] J. G. Schneider. **Exploiting model uncertainty estimates for safe dynamic control learning**. In M. Mozer, M. I. Jordan, and T. Petsche, editors, *Advances in Neural Information Processing Systems (NIPS 9)*, pages 1047–1053. MIT Press, 1996.
- [2765] M. Schneider and H. H. Hoos. **Quantifying Homogeneity of Instance Sets for Algorithm Configuration**. In Y. Hamadi and M. Schoenauer, editors, *LION 6*, volume 7219 of *LNCS*, pages 190–204. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-34413-8_14. *Keywords:* Quantifying Homogeneity; Empirical Analysis; Parameter Optimization; Algorithm Configuration.
- [2766] M. Schoenauer et al., editors. *PPSN*, volume 1917 of *LNCS*. Springer, Heidelberg, Germany, 2000. *Annotation:* IC.29.
- [2767] M. Schonlau, W. J. Welch, and D. R. Jones. **Global versus Local Search in Constrained Optimization of Computer Models**. In N. Flournoy, W. F. Rosenberger, and W. K. Wong, editors, *New Developments and Applications in Experimental Design*, volume 34 of *IMS Lecture Notes - Monograph Series*, pages 11–25. Institute of Mathematical Statistics, 1998.
- [2768] G. R. Schreiber and O. Martin. **Cut Size Statistics of Graph Bisection Heuristics**. *SIAM J. Optim.*, 10(1):231–251, 1999.
- [2769] J. Schreiter, D. Nguyen-Tuong, M. Eberts, B. Bischoff, H. Markert, and M. Toussaint. **Safe Exploration for Active Learning with Gaussian Processes**. In *Machine Learning and Knowledge Discovery in Databases, ECML PKDD 2015*, volume 9286 of *LNCS*, pages 133–149. Springer, 2015. doi:10.1007/978-3-319-23461-8_9. *Annotation:* Proposed Safe Active Learning (SAL) algorithm.

- [2770] T. Schrijvers, G. Tack, P. Wuille, H. Samulowitz, and P. J. Stuckey. **Search Combinators**. *Constraints*, 18(2):269–305, 2013.
- [2771] G. Schrimpf, J. Schneider, H. Stamm-Wilbrandt, and G. Dueck. **Record Breaking Optimization Results Using the Ruin and Recreate Principle**. *Journal of Computational Physics*, 159(2):139–171, 2000.
- [2772] F. Schroff, D. Kalenichenko, and J. Philbin. **Facenet: A unified embedding for face recognition and clustering**. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 815–823, 2015.
- [2773] C. Schulte, editor. *Principles and Practice of Constraint Programming – CP 2013, 19th International Conference, CP 2013, Uppsala, Sweden, September 16-20, 2013, Proceedings*, volume 8124 of *LNCS*. Springer, Heidelberg, Germany, 2013. doi:10.1007/978-3-642-40627-0.
- [2774] E. Schulz, M. Speekenbrink, and A. Krause. **A tutorial on Gaussian process regression: Modelling, exploring, and exploiting functions**. *Journal of Mathematical Psychology*, 85: 1–16, Aug. 2018. doi:10.1016/j.jmp.2018.03.001.
- [2775] J. Schulze and T. Fahle. **A parallel algorithm for the vehicle routing problem with time window constraints**. *Annals of Operations Research*, 86:585–607, 1999. doi:10.1023/A:1018948011707.
Keywords: parallel, VRPTW.
- [2776] O. Schütze and C. Hernández. *Archiving Strategies for Evolutionary Multi-objective Optimization Algorithms*. Springer, 2021.
- [2777] O. Schütze, M. Laumanns, C. A. Coello Coello, M. Dellnitz, and E.-G. Talbi. **Convergence of stochastic search algorithms to finite size Pareto set approximations**. *J. Glob. Optim.*, 41(4):559–577, 2008.
- [2778] O. Schütze, X. Esquivel, A. Lara, and C. A. Coello Coello. **Some Comments on GD and IGD and Relations to the Hausdorff Distance**. In M. Pelikan and J. Branke, editors, *GECCO*, pages 1971–1974. ACM Press, New York, NY, 2010.
- [2779] O. Schütze, M. Laumanns, E. Tantar, C. A. Coello Coello, and E.-G. Talbi. **Computing gap free Pareto front approximations with stochastic search algorithms**. *Evol. Comput.*, 18(1):65–96, 2010.
- [2780] O. Schütze, A. Lara, and C. A. Coello Coello. **On the Influence of the Number of Objectives on the Hardness of a Multiobjective Optimization Problem**. *IEEE Trans. Evol. Comput.*, 15(4):444–455, 2011.
- [2781] O. Schütze, M. Vasile, and C. A. Coello Coello. **Computing the Set of Epsilon-Efficient Solutions in Multiobjective Space Mission Design**. *Journal of Aerospace Computing, Information, and Communication*, 8(3):53–70, 2011. doi:10.2514/1.46478.
- [2782] O. Schütze, X. Esquivel, A. Lara, and C. A. Coello Coello. **Using the Averaged Hausdorff Distance as a Performance Measure in Evolutionary Multiobjective Optimization**. *IEEE Trans. Evol. Comput.*, 16(4):504–522, 2012. doi:10.1109/TEVC.2011.2161872.
- [2783] O. Schütze, C. Hernández, E.-G. Talbi, J.-Q. Sun, Y. Naranjani, and F.-R. Xiong. **Archivers for the representation of the set of approximate solutions for MOPs**. *J. Heuristics*, 25: 71–105, 2019. doi:10.1007/s10732-018-9383-z.
Keywords: archiving, nearly optimality, epsilon-dominance, epsilon-approximation, hausdorff convergence.

- [2784] D. Schuurmans and M. P. Wellman, editors. *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence, AAAI 2016, February 12-17, 2016, Phoenix, Arizona, USA*. AAAI Press, 2016.
- [2785] H.-P. Schwefel. *Numerische Optimierung von Computer-Modellen mittels der Evolutionsstrategie*. Birkhäuser, Basel, Switzerland, 1977.
- [2786] H.-P. Schwefel and R. Männer, editors. *Parallel Problem Solving from Nature, 1st Workshop, PPSN I Dortmund, FRG, October 1-3, 1990. Proceedings*. Springer, Berlin/Heidelberg, 1991. doi:[10.1007/BFb0029723](https://doi.org/10.1007/BFb0029723).
- [2787] S. Scott and S. Matwin. **Feature engineering for text classification**. In *ICML*, volume 99, pages 379–388, 1999.
- [2788] D. Sculley, J. Snoek, A. Rahimi, and A. B. Wiltschko. **Winner’s Curse? On Pace, Progress and Empirical Rigor**. In I. Murray, M. Ranzato, and O. Vinyals, editors, *6th International Conference on Learning Representations, ICLR 2018, Workshop Track Proceedings*, pages 1–4. OpenReview.net, 2018. URL <https://openreview.net/pdf?id=rJWFOFyw>.
- [2789] H. Seada and K. Deb. **U-NSGA-III: A Unified Evolutionary Optimization Procedure for Single, Multiple, and Many Objectives: Proof-of-Principle Results**. In A. Gaspar-Cunha, C. H. Antunes, and C. A. Coello Coello, editors, *EMO Part I*, volume 9018 of *LNCS*, pages 34–49. Springer, Heidelberg, Germany, 2015.
- [2790] M. Seiler, J. Pohl, J. Bossek, P. Kerschke, and H. Trautmann. **Deep Learning as a Competitive Feature-Free Approach for Automated Algorithm Selection on the Traveling Salesperson Problem**. In T. Bäck, M. Preuss, A. Deutz, H. Wang, C. Doerr, M. T. M. Emmerich, and H. Trautmann, editors, *Parallel Problem Solving from Nature – PPSN XVI*, volume 12269 of *LNCS*, pages 48–64. Springer, Cham, Switzerland, 2020.
- [2791] J. Seipp, S. Sievers, M. Helmert, and F. Hutter. **Automatic Configuration of Sequential Planning Portfolios**. In B. Bonet and S. Koenig, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 3364–3370. AAAI Press, 2015.
- [2792] P. Serafini. **Some Considerations About Computational Complexity for Multiobjective Combinatorial Problems**. In J. Jahn and W. Krabs, editors, *Recent Advances and Historical Development of Vector Optimization*, volume 294 of *LNEMS*, pages 222–231. Springer, Berlin, Germany, 1986.
- [2793] P. Serafini. **Simulated annealing for multiple objective optimization problems**. In G. H. Tzeng and P. L. Yu, editors, *Proceedings of the 10th International Conference on Multiple Criteria Decision Making (MCDM’91)*, volume 1, pages 87–96. Springer Verlag, 1992.
- [2794] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas. **Taking the human out of the loop: A review of Bayesian optimization**. *Proceedings of the IEEE*, 104(1):148–175, 2016.
- [2795] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas. **Taking the Human Out of the Loop: A Review of Bayesian Optimization**. *Proceedings of the IEEE*, 104(1):148–175, 2016.
- [2796] K. Shang, H. Ishibuchi, M.-L. Zhang, and Y. Liu. **A new R2 indicator for better hypervolume approximation**. In H. E. Aguirre and K. Takadama, editors, *GECCO*, pages 745–752. ACM Press, New York, NY, 2018. doi:[10.1145/3205455.3205543](https://doi.org/10.1145/3205455.3205543).
Annotation: Proposed to approximate the hypervolume as an integral over the hypersphere.

- [2797] K. Shang, H. Ishibuchi, and W. Chen. **Greedy approximated hypervolume subset selection for many-objective optimization.** In F. Chicano and K. Krawiec, editors, *GECCO*, pages 448–456. ACM Press, New York, NY, 2021. doi:10.1145/3449639.3459390.
- [2798] K. Shang, H. Ishibuchi, and Y. Nan. **Distance-based subset selection revisited.** In F. Chicano and K. Krawiec, editors, *GECCO*, pages 439–447. ACM Press, New York, NY, 2021.
- [2799] K. Shang, T. Shu, H. Ishibuchi, Y. Nan, and L. M. Pang. **Benchmarking large-scale subset selection in evolutionary multi-objective optimization.** *Inf. Sci.*, 622:755–770, 2023. doi:10.1016/j.ins.2022.11.155.
- [2800] D. F. Shanno. **Conditioning of Quasi-Newton Methods for Function Minimization.** *Mathematics of Computation*, 24(111):647–656, 1970. ISSN 00255718, 10886842.
Keywords: BFGS.
Annotation: One of the four papers that proposed BFGS.
- [2801] W. Shao, D. Pi, and Z. Shao. **Memetic algorithm with node and edge histogram for no-idle flow shop scheduling problem to minimize the makespan criterion.** *Applied Soft Computing*, 54:164–182, 2017.
- [2802] W. Shao, D. Pi, and Z. Shao. **A hybrid discrete teaching-learning based meta-heuristic for solving no-idle flow shop scheduling problem with total tardiness criterion.** *Comput. Oper. Res.*, 94:89–105, 2018.
- [2803] M. Sharma, M. López-Ibáñez, and D. Kazakov. **Performance Assessment of Recursive Probability Matching for Adaptive Operator Selection in Differential Evolution.** In A. Auger, C. M. Fonseca, N. Lourenço, P. Machado, L. Paquete, and D. Whitley, editors, *Parallel Problem Solving from Nature – PPSN XV*, volume 11102 of *LNCS*, pages 321–333. Springer, Cham, Switzerland, 2018. doi:10.1007/978-3-319-99259-4_26. Supplementary material: <https://github.com/mudita11/AOS-comparisons>.
Keywords: Rec-PM.
- [2804] M. Sharma, M. López-Ibáñez, and D. Kazakov. **Performance Assessment of Recursive Probability Matching for Adaptive Operator Selection in Differential Evolution: Supplementary material.** <https://github.com/mudita11/AOS-comparisons>, 2018.
- [2805] M. Sharma, A. Komninos, M. López-Ibáñez, and D. Kazakov. **Deep Reinforcement Learning-Based Parameter Control in Differential Evolution.** In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 709–717. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321813. Supplementary material: <https://dx.doi.org/10.5281/zenodo.2628228>.
Keywords: DE-DDQN.
- [2806] M. Sharma, M. López-Ibáñez, and D. Kazakov. **Deep Reinforcement Learning Based Parameter Control in Differential Evolution: Supplementary material.** <https://github.com/mudita11/DE-DDQN>, 2019.
- [2807] S. M. Shavarani, M. López-Ibáñez, and J. D. Knowles. **Realistic Utility Functions Prove Difficult for State-of-the-Art Interactive Multiobjective Optimization Algorithms.** In F. Chicano and K. Krawiec, editors, *GECCO*, pages 457–465. ACM Press, New York, NY, 2021. doi:10.1145/3449639.3459373.
- [2808] S. M. Shavarani, M. López-Ibáñez, and R. Allmendinger. **Detecting Hidden and Irrelevant Objectives in Interactive Multi-Objective Optimization.** *IEEE Trans. Evol. Comput.*, 28(2):544–557, 2023. doi:10.1109/TEVC.2023.3269348.

- [2809] S. M. Shavarani, M. López-Ibáñez, R. Allmendinger, and J. D. Knowles. **An Interactive Decision Tree-Based Evolutionary Multi-Objective Algorithm: Supplementary material.** In M. T. M. Emmerich et al., editors, *Evolutionary Multi-criterion Optimization, EMO 2023*, volume 13970 of *LNCS*, pages 620–634. Springer International Publishing, Cham, Switzerland, 2023. doi:[10.1007/978-3-031-27250-9_44](https://doi.org/10.1007/978-3-031-27250-9_44).
- [2810] S. M. Shavarani, M. López-Ibáñez, R. Allmendinger, and J. D. Knowles. **An Interactive Decision Tree-Based Evolutionary Multi-Objective Algorithm: Supplementary material.** Zenodo, 2023.
- [2811] S. M. Shavarani, M. López-Ibáñez, and J. D. Knowles. **On Benchmarking Interactive Evolutionary Multi-Objective Algorithms.** *IEEE Trans. Evol. Comput.*, 28(4):1084–1098, 2023. doi:[10.1109/TEVC.2023.3289872](https://doi.org/10.1109/TEVC.2023.3289872).
- [2812] S. M. Shavarani, M. López-Ibáñez, and J. D. Knowles. **On Benchmarking Interactive Evolutionary Multi-Objective Algorithms: Supplementary material.** Zenodo, 2023.
- [2813] B. Shavazipour. *Multi-Objective Optimisation under Deep Uncertainty.* PhD thesis, UCT Statistical sciences, South Africa, 2018.
- [2814] B. Shavazipour and T. J. Stewart. **Multi-objective optimisation under deep uncertainty.** *Operational Research*, Sept. 2019. doi:[10.1007/s12351-019-00512-1](https://doi.org/10.1007/s12351-019-00512-1).
- [2815] B. Shavazipour, J. Stray, and T. J. Stewart. **Sustainable planning in sugar-bioethanol supply chain under deep uncertainty: A case study of South African sugarcane industry.** *Computers & Chemical Engineering*, 143:107091, 2020. doi:[10.1016/j.compchemeng.2020.107091](https://doi.org/10.1016/j.compchemeng.2020.107091).
Keywords: Supply chain management, Multi-objective optimisation, Deep uncertainty, Scenario planning, Renewable energy,.
- [2816] B. Shavazipour, M. López-Ibáñez, and K. Miettinen. **Visualizations for Decision Support in Scenario-based Multiobjective Optimization.** *Inf. Sci.*, 578:1–21, 2021. doi:[10.1016/j.ins.2021.07.025](https://doi.org/10.1016/j.ins.2021.07.025). Supplementary material: <https://doi.org/10.5281/zenodo.5040421>.
- [2817] K. J. Shaw, C. M. Fonseca, A. L. Nortcliffe, M. Thompson, J. Love, and P. J. Fleming. **Assessing the performance of multiobjective genetic algorithms for optimization of a batch process scheduling problem.** In *Proceedings of the 1999 Congress on Evolutionary Computation (CEC 1999)*, volume 1, pages 34–75, Piscataway, NJ, 1999. IEEE Press.
- [2818] P. Shaw. **Using Constraint Programming and Local Search Methods to Solve Vehicle Routing Problems.** In M. Maher and J.-F. Puget, editors, *Principles and Practice of Constraint Programming, CP98*, volume 1520 of *LNCS*, pages 417–431. Springer, Heidelberg, Germany, 1998.
- [2819] J. Shawe-Taylor, R. S. Zemel, P. L. Bartlett, F. Pereira, and K. Q. Weinberger, editors. *Advances in Neural Information Processing Systems 24: Annual Conference on Neural Information Processing Systems 2011.* Curran Associates, Red Hook, NY, 2011.
- [2820] V. Y. Shen, N. Saito, M. R. Lyu, and M. E. Zurko, editors. *Proceedings of the Tenth International World Wide Web Conference, WWW 10, Hong Kong, China, May 1-5, 2001.* ACM Press, New York, NY, 2001. ISBN 1-58113-348-0.
- [2821] D. J. Sheskin. *Handbook of Parametric and Nonparametric Statistical Procedures.* Chapman & Hall/CRC, 2nd edition, 2000.
- [2822] D. J. Sheskin. *Handbook of Parametric and Nonparametric Statistical Procedures.* Chapman & Hall/CRC, 5th edition, 2011.

- [2823] Y. Shi and R. C. Eberhart. **Parameter selection in particle swarm optimization**. In V. W. Porto, N. Saravanan, D. Waagen, and A. E. Eiben, editors, *Evolutionary Programming VII*, volume 1447 of *LNCS*, pages 591–600, Heidelberg, Germany, 1998. Springer. doi:10.1007/BFb0040753.
- [2824] Y. Shi, G. D. van Albada, J. Dongarra, and P. M. A. Slood, editors. *Computational Science – ICCS 2007, 7th International Conference, Proceedings, Part IV*, volume 4490 of *LNCS*. Springer, Heidelberg, Germany, 2007.
- [2825] M. D. Shields and J. Zhang. **The generalization of Latin hypercube sampling**. *Reliability Engineering & System Safety*, 148:96–108, 2016.
- [2826] D. Shilane, J. Martikainen, S. Dudoit, and S. J. Ovaska. **A general framework for statistical performance comparison of evolutionary computation algorithms**. *Inf. Sci.*, 178(14): 2870–2879, 2008. doi:10.1016/j.ins.2008.03.007.
- [2827] B. Shipley. *Cause and Correlation in Biology: a User’s Guide to Path Analysis, Structural Equations and Causal Inference*. Cambridge University Press, 1st edition, 2000.
- [2828] O. M. Shir and T. Bäck. **Niching with derandomized evolution strategies in artificial and real-world landscapes**. *Natural Computing*, 8(1):171–196, 2009. doi:10.1007/s11047-007-9065-5.
- [2829] A. Shirazi, J. Ceberio, and J. A. Lozano. **Spacecraft trajectory optimization: A review of models, objectives, approaches and solutions**. *Progress in Aerospace Sciences*, 102:76–98, Oct. 2018. doi:10.1016/j.paerosci.2018.07.007.
- [2830] A. Shmygelska and H. H. Hoos. **An Ant Colony Optimisation Algorithm for the 2D and 3D Hydrophobic Polar Protein Folding Problem**. *BMC Bioinformatics*, 6:30, 2005. doi:10.1186/1471-2105-6-30.
- [2831] A. Shmygelska, R. Aguirre-Hernández, and H. H. Hoos. **An Ant Colony Optimization Algorithm for the 2D HP Protein Folding Problem**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 40–52. Springer, Heidelberg, Germany, 2002.
- [2832] H. E. Shrobe, T. M. Mitchell, and R. G. Smith, editors. *Proceedings of the 7th National Conference on Artificial Intelligence, St. Paul, MN, August 21-26, AAAI-88*, 1988. AAAI Press/MIT Press, Menlo Park, CA. URL <http://www.aaai.org/Conferences/AAAI/aaai88.php>.
- [2833] J. N. Siddall. *Optimal Engineering Design: Principles and Applications*. Marcel Dekker Inc., New York, NY, 1982.
- [2834] S. Siegel and N. J. Castellan, Jr. *Non Parametric Statistics for the Behavioral Sciences*. McGraw Hill, New York, NY, 2nd edition, 1988.
- [2835] R. Silhavy, R. Senkerik, Z. K. Oplatkova, P. Silhavy, and Z. Prokopova, editors. *Artificial Intelligence Perspectives in Intelligent Systems*, volume 464 of *Advances in Intelligent Systems and Computing*. Springer International Publishing, 2016.
- [2836] C. A. Silva, T. A. Runkler, J. M. Sousa, and R. Palm. **Ant Colonies as Logistic Processes Optimizers**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 76–87. Springer, Heidelberg, Germany, 2002.
- [2837] S. Silva and A. I. Esparcia-Alcázar, editors. *Genetic and Evolutionary Computation Conference, GECCO 2015, Proceedings, Madrid, Spain, July 11-15, 2015*. ACM Press, New York, NY, 2015.

- [2838] S. Silva and L. Paquete, editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2023, Lisbon, Portugal, July 15-19, 2023*. ACM Press, New York, NY, 2023. doi:10.1145/3583131.
Annotation: ISBN: 9798400701191.
- [2839] S. Silva and L. Paquete, editors. *Genetic and Evolutionary Computation Conference Companion, GECCO 2023, Lisbon, Portugal, July 15-19, 2023*. ACM Press, New York, NY, 2023. doi:10.1145/3583133.
Annotation: ISBN: 979-8-4007-0120-7.
- [2840] M. Silva-Muñoz, G. Calderon, A. Franzin, and H. Bersini. **Determining a consistent experimental setup for benchmarking and optimizing databases**. In F. Chicano and K. Krawiec, editors, *GECCO Companion*, pages 1614–1621. ACM Press, New York, NY, 2021. doi:10.1145/3449726.3463180.
- [2841] M. Silva-Muñoz, A. Franzin, and H. Bersini. **Automatic configuration of the Cassandra database using irace**. *PeerJ Computer Science*, 7:e634, 2021. doi:10.7717/peerj-cs.634.
- [2842] M. Silva-Muñoz, C. Contreras-Bolton, C. Rey, and V. Parada. **Automatic generation of a hybrid algorithm for the maximum independent set problem using genetic programming**. *Applied Soft Computing*, page 110474, 2023. doi:10.1016/j.asoc.2023.110474.
- [2843] P. V. Silvestrin and M. Ritt. **An Iterated Tabu Search for the Multi-compartment Vehicle Routing Problem**. *Comput. Oper. Res.*, 81:192–202, 2017.
- [2844] K. Sim, E. Hart, and B. Paechter. **A Lifelong Learning Hyper-heuristic Method for Bin Packing**. *Evol. Comput.*, 23(1):37–67, 2015. doi:10.1162/EVCO_a_00121.
- [2845] J. P. Simmons, R. A. LeBoeuf, and L. D. Nelson. **The effect of accuracy motivation on anchoring and adjustment: Do people adjust from provided anchors?** *Journal of Personality and Social Psychology*, 99(6):917–932, 2010. doi:10.1037/a0021540.
- [2846] J. P. Simmons, L. D. Nelson, and U. Simonsohn. **False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant**. *Psychological Science*, 2011. URL <https://ssrn.com/abstract=1850704>.
Annotation: Proposed the term p-hacking.
- [2847] L. F. Simões, D. Izzo, E. Haasdijk, and A. E. Eiben. **Multi-rendezvous Spacecraft Trajectory Optimization with Beam P-ACO**. In B. Hu and M. López-Ibáñez, editors, *EvoCOP*, volume 10197 of *LNCS*, pages 141–156. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-55453-2.
- [2848] H. A. Simon. **A Behavioral Model of Rational Choice**. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.
- [2849] H. A. Simon and A. Newell. **Heuristic Problem Solving: The Next Advance in Operations Research**. *Operations Research*, 6(1):1–10, 1958. doi:10.1287/opre.6.1.1.
- [2850] O. Simonin, F. Charpillet, and E. Thierry. **Revisiting wavefront construction with collective agents: an approach to foraging**. *Swarm Intelligence*, 9(2):113–138, 2014. doi:10.1007/s11721-014-0093-3.
Keywords: irace.
- [2851] A. R. Simpson, D. C. Sutton, D. S. Keane, and S. J. Sherriff. **Optimal control of pumping at a water filtration plant using genetic algorithms**. In D. A. Savic and G. A. Walters, editors, *Water Industry Systems: Modelling and Optimization Applications*, volume 2. Research Studies Press Ltd., Baldock, United Kingdom, 1999.

- [2852] M. Singer and M. L. Pinedo. **A Computational Study of Branch and Bound Techniques for Minimizing the Total Weighted Tardiness in Job Shops.** *IIE Transactions*, 30(2): 109–118, 1998.
- [2853] H. Singh, T. Ray, J. D. Knowles, X. Li, J. Branke, B. Wang, and A. Oyama, editors. *Evolutionary Multi-Criterion Optimization – 13th International Conference, EMO 2025, Canberra, ACT, Australia, March 4-7, 2025, Proceedings, Part I*, volume 15512 of *LNCS*. Springer Verlag, Singapore, 2025.
- [2854] H. K. Singh, A. Isaacs, and T. Ray. **A Pareto Corner Search Evolutionary Algorithm and Dimensionality Reduction in Many-Objective Optimization Problems.** *IEEE Trans. Evol. Comput.*, 15(4):539–556, 2011. doi:[10.1109/TEVC.2010.2093579](https://doi.org/10.1109/TEVC.2010.2093579).
- [2855] H. K. Singh, K. S. Bhattacharjee, and T. Ray. **Distance-based subset selection for benchmarking in evolutionary multi/many-objective optimization.** *IEEE Trans. Evol. Comput.*, 23(5):904–912, 2019.
- [2856] S. P. Singh and S. Markovitch, editors. *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence, February 4-9, 2017, San Francisco, California, USA*. AAAI Press, Feb. 2017.
- [2857] A. Sinha, D. K. Saxena, K. Deb, and A. Tiwari. **Using objective reduction and interactive procedure to handle many-objective optimization problems.** *Applied Soft Computing*, 13(1):415–427, 2013. doi:[10.1016/j.asoc.2012.08.030](https://doi.org/10.1016/j.asoc.2012.08.030).
Keywords: Evolutionary algorithms, Evolutionary multi- and many-objective optimization, Multi-criteria decision making, Machine learning, Interactive optimization.
- [2858] A. Sioud and C. Gagné. **Enhanced migrating birds optimization algorithm for the permutation flow shop problem with sequence dependent setup times.** *Eur. J. Oper. Res.*, 264(1):66–73, 2018.
- [2859] N. J. A. Sloane. **Covering Arrays and Intersecting Codes.** *Journal of Combinatorial Design*, 1:51–63, 1993. doi:[10.1002/jcd.3180010106](https://doi.org/10.1002/jcd.3180010106).
- [2860] R. Słowiński. **Inducing preference models from pairwise comparisons: implications for preference-guided EMO.** Evolutionary Multi-Criterion Optimization, EMO 2011, 2011. Keynote talk.
- [2861] B. G. Small, B. W. McColl, R. Allmendinger, J. Pahle, G. López-Castejón, N. J. Rothwell, J. D. Knowles, P. Mendes, D. Brough, and D. B. Kell. **Efficient discovery of anti-inflammatory small-molecule combinations using evolutionary computing.** *Nature Chemical Biology*, 7(12):902–908, 2011.
- [2862] SMC. *IEEE International Conference on Systems, Man, and Cybernetics, SMC 2013, Manchester, United Kingdom, October 13-16, 2013*, 2013. IEEE Press.
- [2863] S. K. Smit and A. E. Eiben. **Comparing Parameter Tuning Methods for Evolutionary Algorithms.** In *IEEE CEC*, pages 399–406, Piscataway, NJ, 2009. IEEE Press.
- [2864] S. K. Smit and A. E. Eiben. **Beating the ‘world champion’ evolutionary algorithm via REVAC tuning.** In H. Ishibuchi et al., editors, *Proceedings of the 2010 Congress on Evolutionary Computation (CEC 2010)*, pages 1–8, Piscataway, NJ, 2010. IEEE Press. doi:[10.1109/CEC.2010.5586026](https://doi.org/10.1109/CEC.2010.5586026).
- [2865] S. K. Smit and A. E. Eiben. **Parameter Tuning of Evolutionary Algorithms: Generalist vs. Specialist.** In C. D. Chio, S. Cagnoni, C. Cotta, M. Ebner, A. Ekárt, A. I. Esparcia-Alcázar, C. K. Goh, J.-J. Merelo, F. Neri, M. Preuss, J. Togelius, and G. N. Yannakakis, editors, *Applications of Evolutionary Computation*, volume 6024 of *LNCS*, pages 542–551. Springer, Heidelberg, Germany, 2010. doi:[10.1007/978-3-642-12239-2_56](https://doi.org/10.1007/978-3-642-12239-2_56).

- [2866] S. K. Smit and A. E. Eiben. **Multi-Problem Parameter Tuning using BONESA**. In J.-K. Hao, P. Legrand, P. Collet, N. Monmarché, E. Lutton, and M. Schoenauer, editors, *Artificial Evolution: 10th International Conference, Evolution Artificielle, EA, 2011*, volume 7401 of *LNCS*, pages 222–233. Springer, Heidelberg, Germany, 2012.
Annotation: For some reason, this was not actually published in the LNCS Proceedings of EA.
- [2867] S. K. Smit, A. E. Eiben, and Z. Szilávik. **An MOEA-based Method to Tune EA Parameters on Multiple Objective Functions**. In J. Filipe and J. Kacprzyk, editors, *Proceedings of the International Joint Conference on Computational Intelligence (IJCCI)*, pages 261–268. SciTePress, 2010.
- [2868] J. Smith, C. Stone, and M. Serpell. **Exploiting Diverse Distance Metrics for Surrogate-Based Optimisation of Ordering Problems**. In T. Friedrich, F. Neumann, and A. M. Sutton, editors, *GECCO*, pages 701–708. ACM Press, New York, NY, 2016.
[doi:10.1145/2908812.2908854](https://doi.org/10.1145/2908812.2908854).
- [2869] T. E. Smith and D. E. Setliff. **Knowledge-based constraint-driven software synthesis**. In *Proceedings of the Seventh Knowledge-Based Software Engineering Conference*, pages 18–27. IEEE, 1992. [doi:10.1109/KBSE.1992.252912](https://doi.org/10.1109/KBSE.1992.252912).
- [2870] K. Smith-Miles. **Cross-disciplinary Perspectives on Meta-learning for Algorithm Selection**. *ACM Comput. Surv.*, 41(1):1–25, 2008.
- [2871] K. Smith-Miles. **Towards insightful algorithm selection for optimisation using meta-learning concepts**. In D. Liu et al., editors, *Proceedings of the International Joint Conference on Neural Networks (IJCNN 2008), Hong Kong, China, June 1-6, 2008*, pages 4118–4124. IEEE Press, 2008.
- [2872] K. Smith-Miles and S. Bowly. **Generating New Test Instances by Evolving in Instance Space**. *Comput. Oper. Res.*, 63:102–113, 2015. [doi:10.1016/j.cor.2015.04.022](https://doi.org/10.1016/j.cor.2015.04.022).
Keywords: Benchmarking; Evolving instances; Graph colouring; Instance space; Test instances.
- [2873] K. Smith-Miles and L. Lopes. **Measuring instance difficulty for combinatorial optimization problems**. *Comput. Oper. Res.*, 39:875–889, 2012.
- [2874] K. Smith-Miles and M. A. Muñoz. **Instance Space Analysis for Algorithm Testing: Methodology and Software Tools**. *ACM Comput. Surv.*, 55(12), Mar. 2023. [doi:10.1145/3572895](https://doi.org/10.1145/3572895).
Keywords: test instance diversity, benchmarking, timetabling, Algorithm footprints, MATLAB, software as a service, meta-heuristics, algorithm selection, meta-learning.
- [2875] K. Smith-Miles, J. I. van Hemert, and X. Y. Lim. **Understanding TSP difficulty by Learning from evolved instances**. In C. Blum and R. Battiti, editors, *LION 4*, volume 6073 of *LNCS*, pages 266–280. Springer, Heidelberg, Germany, 2010. [doi:10.1007/978-3-642-13800-3](https://doi.org/10.1007/978-3-642-13800-3).
- [2876] K. Smith-Miles, D. Baatar, B. Wreford, and R. M. R. Lewis. **Towards Objective Measures of Algorithm Performance across Instance Space**. *Comput. Oper. Res.*, 45:12–24, 2014. [doi:10.1016/j.cor.2013.11.015](https://doi.org/10.1016/j.cor.2013.11.015).
Keywords: Algorithm selection; Instance Space Analysis; Graph coloring; Heuristics; Performance prediction.
- [2877] K. Smith-Miles, M. A. Muñoz, and Neelofar. **Melbourne Algorithm Test Instance Library with Data Analytics (MATILDA)**, 2020. URL <https://matilda.unimelb.edu.au/>.
- [2878] K. Smith-Miles, J. Christiansen, and M. A. Muñoz. **Revisiting Where Are the Hard Knapsack Problems? Via Instance Space Analysis**. *Comput. Oper. Res.*, 128:105184, 2021. [doi:10.1016/j.cor.2020.105184](https://doi.org/10.1016/j.cor.2020.105184).

Keywords: 0-1 Knapsack problem; Algorithm portfolios; Algorithm selection; Instance difficulty; Instance generation; Instance Space Analysis; Performance evaluation.

- [2879] G. W. Snedecor and W. G. Cochran. *Statistical Methods*. Iowa State University Press, Ames, IA, USA, 6th edition, 1967.
- [2880] J. Snoek, H. Larochelle, and R. P. Adams. **Practical Bayesian Optimization of Machine Learning Algorithms**. In P. L. Bartlett, F. C. N. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, editors, *Advances in Neural Information Processing Systems (NIPS 25)*, pages 2951–2959. Curran Associates, Red Hook, NY, 2012. URL https://papers.nips.cc/paper_files/paper/2012/hash/05311655a15b75fab86956663e1819cd-Abstract.html.
- [2881] J. Snoek, K. Swersky, R. Zemel, and R. P. Adams. **Input Warping for Bayesian Optimization of Non-Stationary Functions**. In E. P. Xing and T. Jebara, editors, *Proceedings of the 31st International Conference on Machine Learning, ICML 2014*, volume 32, pages 1674–1682. PMLR, 2014.
- [2882] I. M. Sobol. **On the distribution of points in a cube and the approximate evaluation of integrals**. *USSR Computational Mathematics and Mathematical Physics*, 7(4):86–112, Jan. 1967. doi:10.1016/0041-5553(67)90144-9.
Keywords: quasi-random, low-discrepancy.
- [2883] K. Socha. **ACO for Continuous and Mixed-Variable Optimization**. In M. Dorigo et al., editors, *ANTS 2004*, volume 3172 of *LNCS*, pages 25–36. Springer, Heidelberg, Germany, 2004.
- [2884] K. Socha and C. Blum. **An ant colony optimization algorithm for continuous optimization: An application to feed-forward neural network training**. *Neural Computing & Applications*, 16(3):235–247, 2007.
- [2885] K. Socha and M. Dorigo. **Ant Colony Optimization for Mixed-Variable Optimization Problems**. Technical Report TR/IRIDIA/2007-019, IRIDIA, Université Libre de Bruxelles, Belgium, Oct. 2007.
- [2886] K. Socha and M. Dorigo. **Ant Colony Optimization for Continuous Domains**. *Eur. J. Oper. Res.*, 185(3):1155–1173, 2008. doi:10.1016/j.ejor.2006.06.046.
Keywords: ACOR.
Annotation: Proposed ACOR (ACO_R).
- [2887] K. Socha, J. D. Knowles, and M. Sampels. **A MAX-MIN Ant System for the University Course Timetabling Problem**. In M. Dorigo et al., editors, *Ant Algorithms, Third International Workshop, ANTS 2002*, volume 2463 of *LNCS*, pages 1–13. Springer, Heidelberg, Germany, 2002.
- [2888] K. Socha, M. Sampels, and M. Manfrin. **Ant algorithms for the university course timetabling problem with regard to the state-of-the-art**. In S. Cagnoni et al., editors, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops 2003*, volume 2611 of *LNCS*, pages 334–345. Springer, Heidelberg, Germany, 2003.
- [2889] M. L. Soffa and E. Duesterwald, editors. *Proceedings of the 6th Annual IEEE/ACM International Symposium on Code Generation and Optimization*, CGO '08, New York, NY, 2008. ACM Press.
- [2890] D. Soler, E. Martínez, and J. C. Micó. **A Transformation for the Mixed General Routing Problem with Turn Penalties**. *J. Oper. Res. Soc.*, 59:540–547, 2008.
- [2891] C. Solnon. **Ants Can Solve Constraint Satisfaction Problems**. *IEEE Trans. Evol. Comput.*, 6(4):347–357, 2002.

- [2892] C. Solmon. *Ant Colony Optimization and Constraint Programming*. John Wiley & Sons, 2010. doi:10.1002/9781118557563.
- [2893] C. Solmon, editor. *28th International Conference on Principles and Practice of Constraint Programming, CP 2022, July 31 to August 8, 2022, Haifa, Israel*, volume 235 of *LIPIcs*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany, 2022. ISBN 978-3-95977-240-2.
- [2894] M. M. Solomon. **Algorithms for the Vehicle Routing and Scheduling Problems with Time Windows**. *Operations Research*, 35:254–265, 1987.
- [2895] Z. Song, H. Wang, C. He, and Y. Jin. **A Kriging-assisted two-archive evolutionary algorithm for expensive many-objective optimization**. *IEEE Trans. Evol. Comput.*, 25(6):1013–1027, 2021.
- [2896] K. Sörensen. **Metaheuristics—the metaphor exposed**. *International Transactions in Operational Research*, 22(1):3–18, 2015. doi:10.1111/itor.12001.
- [2897] K. Sörensen, F. Arnold, and D. Palhazi Cuervo. **A critical analysis of the “improved Clarke and Wright savings algorithm”**. *International Transactions in Operational Research*, 26(1):54–63, 2017. doi:10.1111/itor.12443.
Keywords: reproducibility, vehicle routing.
- [2898] K. Sörensen, M. Sevaux, and F. Glover. **A history of metaheuristics**. In R. Martí, P. M. Pardalos, and M. G. C. Resende, editors, *Handbook of Heuristics*, pages 1–27. Springer International Publishing, 2018. ISBN 978-3-319-07125-1.
- [2899] J. A. Soria-Alcaraz, G. Ochoa, M. A. Sotelo-Figeroa, and E. K. Burke. **A Methodology for Determining an Effective Subset of Heuristics in Selection Hyper-heuristics**. *Eur. J. Oper. Res.*, 260:972–983, 2017.
- [2900] A. Sotelo, J. Basulado, P. Doldán, and B. Barán. **Algoritmos Evolutivos Multiobjetivo Combinados para la Optimización de la Programación de Bombeo en Sistemas de Suministro de Agua**. In *Congreso Internacional de Tecnologías y Aplicaciones Informáticas, JIT-CITA 2001, Asunción, Paraguay*, 2001. (In Spanish).
- [2901] A. Sotelo, C. von Lücken, and B. Barán. **Multiobjective Evolutionary Algorithms in Pump Scheduling Optimisation**. In B. H. V. Topping and Z. Bittnar, editors, *Proceedings of the Third International Conference on Engineering Computational Technology*. Civil-Comp Press, Stirling, Scotland, 2002.
- [2902] A. Souilah. **Simulated annealing for manufacturing systems layout design**. *Eur. J. Oper. Res.*, 82(3):592–614, 1995.
- [2903] T. Soule and J. H. Moore, editors. *Genetic and Evolutionary Computation Conference, GECCO 2012, Proceedings, Philadelphia, PA, USA, July 7-11, 2012*. ACM Press, New York, NY, 2012.
- [2904] T. Soule and J. H. Moore, editors. *Genetic and Evolutionary Computation Conference, GECCO 2012, Companion Material Proceedings, Philadelphia, PA, USA, July 7-11, 2012*. ACM Press, New York, NY, 2012.
- [2905] C. Spearman. **The proof and measurement of association between two things**. *The American journal of psychology*, 15(1):72–101, 1904.
- [2906] D. A. Spielman and S.-H. Teng. **Smoothed analysis of algorithms: Why the simplex algorithm usually takes polynomial time**. *J. ACM*, 51(3):385–463, 2004.

- [2907] A. Sprecher, R. Kolisch, and A. Drexel. **Semi-active, active, and non-delay schedules for the resource-constrained project scheduling problem.** *Eur. J. Oper. Res.*, 80(1):94–102, 1995. doi:10.1016/0377-2217(93)E0294-8.
Keywords: active schedules, Branch-and-bound methods, non-delay schedules, Resource-constrained project scheduling, Semi-active schedules.
- [2908] A. Sprecher, S. Hartmann, and A. Drexel. **An exact algorithm for project scheduling with multiple modes.** *OR Spektrum*, 19(3):195–203, 1997. doi:10.1007/BF01545587.
Keywords: branch-and-bound, multi-mode resource-constrained project scheduling, project scheduling.
- [2909] G. Squillero and P. Burelli, editors. *Applications of Evolutionary Computation: 19th European Conference, EvoApplications 2016, Porto, Portugal, March 30 – April 1, 2016, Proceedings, Part I*, volume 9597 of *LNCS*. Springer, Heidelberg, Germany, 2016. doi:10.1007/978-3-319-31204-0.
- [2910] G. Squillero and K. Sim, editors. *Applications of Evolutionary Computation: 20th European Conference, EvoApplications 2017, Amsterdam, The Netherlands, April 19-21, 2017, Proceedings, Part I*, volume 10199 of *LNCS*. Springer, Heidelberg, Germany, 2017. doi:10.1007/978-3-319-55849-3.
- [2911] S. Sra, S. Nowozin, and S. J. Wright. *Optimization for machine learning*. MIT Press, Cambridge, MA, 2012.
- [2912] N. Srinivas and K. Deb. **Multiobjective Optimization Using Nondominated Sorting in Genetic Algorithms.** *Evol. Comput.*, 2(3):221–248, 1994.
Annotation: Proposed NSGA.
- [2913] S. Staab and R. Studer, editors. *Handbook on Ontologies*. International Handbooks on Information Systems. Springer, 2009.
- [2914] P. F. Stadler. **Toward a theory of landscapes.** In R. López-Peña, R. Capovilla, R. García-Pelayo, H. Waelbroeck, and F. Zertruche, editors, *Complex Systems and Binary Networks*, pages 77–163. Springer, 1995.
- [2915] M. K. Starr. *Product design and decision theory*. Prentice-Hall Series in Engineering Design, Fundamentals of Engineering Design. Prentice-Hall, Englewood, Cliffs, NJ, 1963.
- [2916] F. Stefanello, V. Aggarwal, L. S. Buriol, J. F. Gonçalves, and M. G. C. Resende. **A Biased Random-key Genetic Algorithm for Placement of Virtual Machines Across Geo-Separated Data Centers.** In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 919–926. ACM Press, New York, NY, 2015. doi:10.1145/2739480.2754768.
Keywords: irace.
- [2917] B. Steffen and G. Woeginger, editors. *Computing and Software Science: State of the Art and Perspectives*, volume 10000 of *LNCS*. Springer, Cham, Switzerland, 2019.
- [2918] H. Stegherr, M. Heider, and J. Hähner. **Classifying Metaheuristics: Towards a unified multi-level classification system.** *Natural Computing*, 2020. doi:10.1007/s11047-020-09824-0.
- [2919] S. Steiner and T. Radzik. **Computing all efficient solutions of the biobjective minimum spanning tree problem.** *Comput. Oper. Res.*, 35(1):198–211, 2008.
- [2920] C. Steinruecken, E. Smith, D. Janz, J. Lloyd, and Z. Ghahramani. **The Automatic Statistician.** In F. Hutter, L. Kotthoff, and J. Vanschoren, editors, *Automated Machine Learning*, pages 161–173. Springer, 2019. doi:10.1007/978-3-030-05318-5_9.

- [2921] R. E. Steuer. *Multiple Criteria Optimization: Theory, Computation and Application*. Wiley Series in Probability and Mathematical Statistics. John Wiley & Sons, New York, NY, 1986.
Keywords: Maximally dispersed weights.
- [2922] R. E. Steuer and L. Gardiner. **On the Computational Testing of Procedures for Interactive Multiple Objective Linear Programming**. In G. Fandel and H. Gehring, editors, *Operations Research*, pages 121–131. Springer, Berlin/Heidelberg, 1991. ISBN 978-3-642-76537-7. doi:10.1007/978-3-642-76537-7_8.
Annotation: Proposed difference between ad hoc and non-ad hoc interactive multi-objective optimization methods.
- [2923] T. J. Stewart. **Robustness of Additive Value Function Methods in MCDM**. *Journal of Multi-Criteria Decision Analysis*, 5(4):301–309, 1996.
Keywords: machine decision-making.
- [2924] T. J. Stewart. **Evaluation and refinement of aspiration-based methods in MCDM**. *Eur. J. Oper. Res.*, 113(3):643–652, 1999.
Keywords: machine decision-making.
- [2925] T. J. Stewart. **Goal programming and cognitive biases in decision-making**. *J. Oper. Res. Soc.*, 56(10):1166–1175, 2005. doi:10.1057/palgrave.jors.2601948.
Keywords: machine decision making.
- [2926] T. J. Stewart, S. French, and J. Rios. **Integrating multicriteria decision analysis and scenario planning: Review and extension**. *Omega*, 41(4):679–688, 2013. doi:10.1016/j.omega.2012.09.003.
Keywords: Multicriteria decision analysis.
- [2927] V. Stodden. **What scientific idea is ready for retirement? Reproducibility**. *Edge*, 2014. URL <https://www.edge.org/annual-question/2014/response/25340>.
Annotation: Introduces computational reproducibility, empirical reproducibility and statistical reproducibility.
- [2928] V. Stodden, M. McNutt, D. H. Bailey, E. Deelman, Y. Gil, B. Hanson, M. A. Heroux, J. P. A. Ioannidis, and M. Taufer. **Enhancing reproducibility for computational methods**. *Science*, 354(6317):1240–1241, Dec. 2016. doi:10.1126/science.aah6168.
- [2929] V. Stodden, J. Seiler, and Z. Ma. **An empirical analysis of journal policy effectiveness for computational reproducibility**. *Proceedings of the National Academy of Sciences*, 115(11):2584–2589, Mar. 2018. doi:10.1073/pnas.1708290115.
- [2930] D. H. Stolfi and E. Alba. **Red Swarm: Reducing travel times in smart cities by using bio-inspired algorithms**. *Applied Soft Computing*, 24:181–195, 2014. doi:10.1016/j.asoc.2014.07.014.
Keywords: Evolutionary algorithm,Road traffic,Smart city,Smart mobility,Traffic light,WiFi connections.
- [2931] D. H. Stolfi and E. Alba. **An Evolutionary Algorithm to Generate Real Urban Traffic Flows**. In J. M. Puerta, J. A. Gámez, B. Dorronsoro, E. Barrenechea, A. Troncoso, B. Baroque, and M. Galar, editors, *Advances in Artificial Intelligence, CAEPIA 2015*, volume 9422 of *LNCIS*, pages 332–343. Springer, Heidelberg, Germany, 2015. doi:10.1007/978-3-319-24598-0_30.
Keywords: Evolutionary algorithm,SUMO,Smart city,Smart mobility,Traffic simulation.
- [2932] R. Storn and K. Price. **Differential Evolution – A Simple and Efficient Heuristic for Global Optimization over Continuous Spaces**. *J. Glob. Optim.*, 11(4):341–359, 1997. doi:10.1023/A:1008202821328.
Annotation: Proposed differential evolution.

- [2933] D. Stracuzzi et al., editors. *Proceedings of the Twenty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2014, Québec City, Québec, Canada, July 27-31, 2014*, 2014. AAAI Press.
- [2934] P. N. Strenski and S. Kirkpatrick. **Analysis of Finite Length Annealing Schedules.** *Algorithmica*, 6(1-6):346–366, 1991.
- [2935] P. Strycharczuk, M. López-Ibáñez, G. Brown, and A. Leemann. **General Northern English: Exploring regional variation in the North of England with machine learning.** *Front. Artif. Intell.*, 3(48), 2020. doi:10.3389/frai.2020.00048.
Keywords: vowels, accent features, dialect leveling, Random forest (bagging), Feature selection.
- [2936] P. J. Stuckey, editor. *Integration of Constraint Programming, Artificial Intelligence, and Operations Research, 18th International Conference, CPAIOR 2021, Vienna, Austria, July 5-8, 2021, Proceedings*, volume 12735 of LNCS. Springer, Cham, Switzerland, 2021.
- [2937] T. Stützle. **MAX-MIN Ant System for the Quadratic Assignment Problem.** Technical Report AIDA-97-4, FG Intellektik, FB Informatik, TU Darmstadt, Germany, July 1997.
- [2938] T. Stützle. **Applying Iterated Local Search to the Permutation Flow Shop Problem.** Technical Report AIDA-98-04, FG Intellektik, FB Informatik, TU Darmstadt, Germany, Aug. 1998.
- [2939] T. Stützle. **An Ant Approach to the Flow Shop Problem.** In *Proceedings of the 6th European Congress on Intelligent Techniques & Soft Computing (EUFIT'98)*, volume 3, pages 1560–1564. Verlag Mainz, Aachen, Germany, 1998.
- [2940] T. Stützle. *Local Search Algorithms for Combinatorial Problems — Analysis, Improvements, and New Applications.* PhD thesis, FB Informatik, TU Darmstadt, Germany, 1998.
- [2941] T. Stützle. **ACOTSP: A Software Package of Various Ant Colony Optimization Algorithms Applied to the Symmetric Traveling Salesman Problem**, 2002. URL <http://www.aco-metaheuristic.org/aco-code>.
Annotation: <http://www.aco-metaheuristic.org/aco-code>.
- [2942] T. Stützle. **Iterated Local Search for the Quadratic Assignment Problem.** *Eur. J. Oper. Res.*, 174(3):1519–1539, 2006.
- [2943] T. Stützle, editor. *Third International Conference, LION 3, Trento, Italy, January 14-18, 2009. Selected Papers*, volume 5851 of LNCS. Springer, Heidelberg, Germany, 2009.
- [2944] T. Stützle. **Some Thoughts on Engineering Stochastic Local Search Algorithms.** In A. Viana et al., editors, *Proceedings of the EU/MEeting 2009: Debating the future: new areas of application and innovative approaches*, pages 47–52, 2009.
- [2945] T. Stützle and M. Dorigo. **ACO Algorithms for the Quadratic Assignment Problem.** In D. Corne, M. Dorigo, and F. Glover, editors, *New Ideas in Optimization*, pages 33–50. McGraw Hill, London, UK, 1999.
Annotation: Also available as Technical Report IRIDIA/99-2, Université Libre de Bruxelles, Belgium.
- [2946] T. Stützle and M. Dorigo. **A Short Convergence Proof for a Class of ACO Algorithms.** *IEEE Trans. Evol. Comput.*, 6(4):358–365, 2002.
- [2947] T. Stützle and S. Fernandes. **New Benchmark Instances for the QAP and the Experimental Analysis of Algorithms.** In J. Gottlieb and G. R. Raidl, editors, *Proceedings of EvoCOP 2004 – 4th European Conference on Evolutionary Computation in Combinatorial Optimization*, volume 3004 of LNCS, pages 199–209. Springer, Heidelberg, Germany, 2004. doi:10.1007/978-3-540-24652-7_20.

- [2948] T. Stützle and H. H. Hoos. **Improving the Ant System: A Detailed Report on the $MA\mathcal{X}$ - MLN Ant System**. Technical Report AIDA-96-12, FG Intellektik, FB Informatik, TU Darmstadt, Germany, Aug. 1996.
- [2949] T. Stützle and H. H. Hoos. **The $MA\mathcal{X}$ - MLN Ant System and Local Search for the Traveling Salesman Problem**. In T. Bäck, Z. Michalewicz, and X. Yao, editors, *Proceedings of the 1997 IEEE International Conference on Evolutionary Computation (ICEC'97)*, pages 309–314. IEEE Press, Piscataway, NJ, 1997.
- [2950] T. Stützle and H. H. Hoos. **$MA\mathcal{X}$ - MLN Ant System and Local Search for Combinatorial Optimization Problems**. In S. Voß, S. Martello, I. H. Osman, and C. Roucairol, editors, *Meta-Heuristics: Advances and Trends in Local Search Paradigms for Optimization*, pages 137–154. Kluwer Academic Publishers, Dordrecht, The Netherlands, 1999.
- [2951] T. Stützle and H. H. Hoos. **$MA\mathcal{X}$ - MLN Ant System**. *Future Generation Computer Systems*, 16(8):889–914, 2000.
- [2952] T. Stützle and H. H. Hoos. **Analysing the Run-time Behaviour of Iterated Local Search for the Travelling Salesman Problem**. In P. Hansen and C. Ribeiro, editors, *Essays and Surveys on Metaheuristics*, Operations Research/Computer Science Interfaces Series, pages 589–611. Kluwer Academic Publishers, Boston, MA, 2001.
- [2953] T. Stützle and M. López-Ibáñez. **Automatic (Offline) Configuration of Algorithms**. In J. L. Jiménez Laredo, S. Silva, and A. I. Esparcia-Alcázar, editors, *GECCO Companion*, pages 681–702. ACM Press, New York, NY, 2015. doi:10.1145/2739482.2756581.
- [2954] T. Stützle and M. López-Ibáñez. **Automated Offline Design of Algorithms**. In P. A. N. Bosman, editor, *GECCO Companion*, pages 1038–1065. ACM Press, New York, NY, 2017. doi:10.1145/3067695.3067722.
- [2955] T. Stützle and M. López-Ibáñez. **Automated Design of Metaheuristic Algorithms**. In M. Gendreau and J.-Y. Potvin, editors, *Handbook of Metaheuristics*, volume 272 of *International Series in Operations Research & Management Science*, pages 541–579. Springer, 2019. doi:10.1007/978-3-319-91086-4_17.
Keywords: automatic design, automatic configuration.
- [2956] T. Stützle and R. Ruiz. **Iterated Greedy**. In R. Martí, P. M. Pardalos, and M. G. C. Resende, editors, *Handbook of Heuristics*, pages 1–31. Springer International Publishing, 2018. ISBN 978-3-319-07125-1. doi:10.1007/978-3-319-07153-4_10-1.
- [2957] T. Stützle and R. Ruiz. **Iterated Local Search**. In R. Martí, P. M. Pardalos, and M. G. C. Resende, editors, *Handbook of Heuristics*, pages 1–27. Springer International Publishing, 2018. ISBN 978-3-319-07125-1. doi:10.1007/978-3-319-07153-4_8-1.
- [2958] T. Stützle, M. Birattari, and H. H. Hoos, editors. *Engineering Stochastic Local Search Algorithms. Designing, Implementing and Analyzing Effective Heuristics. SLS 2007*, volume 4638 of *LNCS*. Springer, Heidelberg, Germany, 2007.
- [2959] T. Stützle, M. Birattari, and H. H. Hoos, editors. *Engineering Stochastic Local Search Algorithms. Designing, Implementing and Analyzing Effective Heuristics. SLS 2009*, volume 5752 of *LNCS*. Springer, Heidelberg, Germany, 2009.
- [2960] T. Stützle, M. López-Ibáñez, P. Pellegrini, M. Maur, M. A. Montes de Oca, M. Birattari, and M. Dorigo. **Parameter Adaptation in Ant Colony Optimization**. Technical Report TR/IRIDIA/2010-002, IRIDIA, Université Libre de Bruxelles, Belgium, Jan. 2010. Published as a book chapter [2962].

- [2961] T. Stützle, M. López-Ibáñez, and M. Dorigo. **A Concise Overview of Applications of Ant Colony Optimization.** In J. J. Cochran, editor, *Wiley Encyclopedia of Operations Research and Management Science*, volume 2, pages 896–911. John Wiley & Sons, 2011. doi:[10.1002/9780470400531.eorms0001](https://doi.org/10.1002/9780470400531.eorms0001).
- [2962] T. Stützle, M. López-Ibáñez, P. Pellegrini, M. Maur, M. A. Montes de Oca, M. Birattari, and M. Dorigo. **Parameter Adaptation in Ant Colony Optimization.** In Y. Hamadi, E. Monfroy, and F. Saubion, editors, *Autonomous Search*, pages 191–215. Springer, Berlin, Germany, 2012. doi:[10.1007/978-3-642-21434-9_8](https://doi.org/10.1007/978-3-642-21434-9_8).
- [2963] J. Styles and H. H. Hoos. **Ordered racing protocols for automatically configuring algorithms for scaling performance.** In C. Blum and E. Alba, editors, *GECCO*, pages 551–558. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8. doi:[10.1145/2463372.2463438](https://doi.org/10.1145/2463372.2463438).
- [2964] J. Styles, H. H. Hoos, and M. Müller. **Automatically Configuring Algorithms for Scaling Performance.** In Y. Hamadi and M. Schoenauer, editors, *LION 6*, volume 7219 of *LNCS*, pages 205–219. Springer, Heidelberg, Germany, 2012.
- [2965] Z. Su, G. Zhang, F. Yue, D. Zhan, M. Li, B. Li, and X. Yao. **Enhanced Constraint Handling for Reliability-Constrained Multiobjective Testing Resource Allocation.** *IEEE Trans. Evol. Comput.*, 25(3):537–551, 2021.
- [2966] A. Subramanian and M. Battarra. **An Iterated Local Search Algorithm for the Travelling Salesman Problem with Pickups and Deliveries.** *J. Oper. Res. Soc.*, 64(3):402–409, 2013.
- [2967] A. Subramanian, M. Battarra, and C. N. Potts. **An Iterated Local Search Heuristic for the Single Machine Total Weighted Tardiness Scheduling Problem with Sequence-dependent Setup Times.** *Int. J. Prod. Res.*, 52(9):2729–2742, 2014.
- [2968] P. N. Suganthan, N. Hansen, J. J. Liang, K. Deb, Y. P. Chen, A. Auger, and S. Tiwari. **Problem definitions and evaluation criteria for the CEC 2005 special session on real-parameter optimization.** Technical report, Nanyang Technological University, Singapore, 2005.
Keywords: CEC’05 benchmark.
Annotation: Also known as KanGAL Report Number 2005005 (Kanpur Genetic Algorithms Laboratory, IIT Kanpur).
- [2969] Y. Sui, A. Gotovos, J. W. Burdick, and A. Krause. **Safe Exploration for Optimization with Gaussian Processes.** In F. Bach and D. Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning, ICML 2015*, volume 37, pages 997–1005. PMLR, 2015.
Keywords: Safe Optimization, SafeOpt.
- [2970] Y. Sui, V. Zhuang, J. W. Burdick, and Y. Yue. **Stagewise Safe Bayesian Optimization with Gaussian Processes.** *Arxiv preprint arXiv:1806.07555*, 2018. URL <https://arxiv.org/abs/1806.07555>.
Keywords: Safe Optimization, StageOpt, Published as [2971].
- [2971] Y. Sui, V. Zhuang, J. W. Burdick, and Y. Yue. **Stagewise Safe Bayesian Optimization with Gaussian Processes.** In J. G. Dy and A. Krause, editors, *Proceedings of the 35th International Conference on Machine Learning, ICML 2018*, volume 80 of *Proceedings of Machine Learning Research*, pages 4788–4796. PMLR, 2018.
Keywords: StageOpt.
- [2972] Y. Sun, G. G. Yen, and Z. Yi. **IGD Indicator-based Evolutionary Algorithm for Many-objective Optimization Problems.** *IEEE Trans. Evol. Comput.*, 23(2):173–187, 2019. doi:[10.1109/TEVC.2018.2791283](https://doi.org/10.1109/TEVC.2018.2791283).

- [2973] Y. Sun et al., editors. *KDD '25: The 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Toronto, Canada, August 3-7, 2025*. ACM Press, New York, NY, Aug. 2025.
- [2974] Z. Sun and M. Han. **Multi-criteria Decision Making Based on PROMETHEE Method**. In *Proceedings of the 2010 International Conference on Computing, Control and Industrial Engineering*, pages 416–418, Los Alamitos, CA, 2010. IEEE Computer Society Press.
- [2975] A. Suppapitnarm, K. A. Seffen, G. T. Parks, and P. J. Clarkson. **A simulated annealing algorithm for multiobjective optimization**. *Engineering Optimization*, 33(1):59–85, 2000.
- [2976] D. C. Sutton, D. S. Keane, and S. J. Sherriff. **Optimizing the Real Time Operation of a Pumping Station at a Water Filtration Plant using Genetic Algorithms**. Honors thesis, Department of Civil and Environmental Engineering, The University of Adelaide, 1998.
- [2977] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 1998.
- [2978] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2nd edition, 2018.
- [2979] J. A. K. Suykens and J. Vandewalle. **Least Squares Support Vector Machine Classifiers**. *Neural Processing Letters*, 9(3):293–300, 1999. doi:10.1023/A:1018628609742. Keywords: LS-SVM.
- [2980] J. Swan, E. Özcan, and G. Kendall. **Hyperion: A Recursive Hyper-heuristic Framework**. In C. A. Coello Coello, editor, *LION 5*, volume 6683 of *LNCS*, pages 616–630. Springer, Heidelberg, Germany, 2011.
- [2981] J. Swan, J. R. Woodward, E. Özcan, G. Kendall, and E. K. Burke. **Searching the Hyper-heuristic Design Space**. *Cognitive Computation*, 6(1):66–73, Mar. 2014. doi:10.1007/s12559-013-9201-8.
- [2982] J. Swan, S. Adriaensen, A. D. Barwell, K. Hammond, and D. R. White. **Extending the “Open-Closed Principle” to Automated Algorithm Configuration**. *Evol. Comput.*, 27(1):173–193, 2019. doi:10.1162/evco_a_00245.
- [2983] J. Swan, S. Adriaensen, A. E. I. Brownlee, K. Hammond, C. G. Johnson, A. Kheiri, F. Krawiec, J.-J. Merelo, L. L. Minku, E. Özcan, G. Pappa, P. García-Sánchez, K. Sörensen, S. Voß, M. Wagner, and D. R. White. **Metaheuristics “In the Large”**. *Eur. J. Oper. Res.*, 297(2):393–406, Mar. 2022. doi:10.1016/j.ejor.2021.05.042.
- [2984] J. Swan et al. **A Research Agenda for Metaheuristic Standardization**. In E.-G. Talbi, editor, *Proc. of MIC 2015, the 11th Metaheuristics International Conference*, 2015.
- [2985] W. R. Swartout, editor. *Proceedings of the 10th National Conference on Artificial Intelligence*, 1992. AAAI Press/MIT Press, Menlo Park, CA.
- [2986] G. Syswerda. **Uniform Crossover in Genetic Algorithms**. In J. D. Schaffer, editor, *Proceedings of the Third International Conference on Genetic Algorithms (ICGA'89)*, pages 2–9. Morgan Kaufmann Publishers, San Mateo, CA, 1989. Keywords: uniform crossover.
- [2987] S. Szeider. **CP-Agent: Agentic Constraint Programming**. *Arxiv preprint arXiv:2508.07468*, 2025. doi:10.48550/ARXIV.2508.07468. Keywords: LLM, CP-Bench.
- [2988] H. Szu and R. Hartley. **Fast Simulated Annealing**. *Physics Letters A*, 122(3):157–162, 1987.

- [2989] K. Tagawa, H. Shimizu, and H. Nakamura. **Indicator-based Differential Evolution Using Exclusive Hypervolume Approximation and Parallelization for Multi-core Processors**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 657–664. ACM Press, New York, NY, 2011.
- [2990] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf. **Deepface: Closing the gap to human-level performance in face verification**. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1701–1708, 2014.
- [2991] É. D. Taillard. **Some Efficient Heuristic Methods for the Flow Shop Sequencing Problem**. *Eur. J. Oper. Res.*, 47(1):65–74, 1990.
- [2992] É. D. Taillard. **Robust Taboo Search for the Quadratic Assignment Problem**. *Parallel Computing*, 17(4-5):443–455, 1991.
Annotation: faster 2-exchange delta evaluation in QAP.
- [2993] É. D. Taillard. **Benchmarks for Basic Scheduling Problems**. *Eur. J. Oper. Res.*, 64(2): 278–285, 1993.
- [2994] É. D. Taillard. **Comparison of Iterative Searches for the Quadratic Assignment Problem**. *Location Science*, 3(2):87–105, 1995.
- [2995] R. H. C. Takahashi, K. Deb, E. F. Wanner, and S. Greco, editors. *Evolutionary Multi-Criterion Optimization. 6th International Conference, EMO 2011, Ouro Preto, Brazil, April 5-8, 2011, Proceedings*, volume 6576 of *LNCS*. Springer, Berlin/Heidelberg, 2011.
- [2996] E.-G. Talbi. **A Taxonomy of Hybrid Metaheuristics**. *J. Heuristics*, 8(5):541–564, 2002.
- [2997] E.-G. Talbi, editor. *Hybrid Metaheuristics*, volume 434 of *Studies in Computational Intelligence*. Springer Verlag, 2013. URL <http://www.springer.com/engineering/computational+intelligence+and+complexity/book/978-3-642-30670-9>.
- [2998] E.-G. Talbi, editor. *Proceedings of MIC 2015, the 11th Metaheuristics International Conference*, 2015.
- [2999] E.-G. Talbi, P. Liardet, P. Collet, E. Lutton, and M. Schoenauer, editors. *Artificial Evolution: 7th International Conference, Evolution Artificielle, EA 2005, Lille, France*, volume 3871 of *LNCS*. Springer, Heidelberg, Germany, 2005.
- [3000] K. Y. Tam. **A Simulated Annealing Algorithm for Allocating Space to Manufacturing Cells**. *Int. J. Prod. Res.*, 30(1):63–87, 1992.
- [3001] M. Tamiz and D. F. Jones. **Interactive Frameworks for Investigation of Goal Programming Models: Theory and Practice**. *Journal of Multi-Criteria Decision Analysis*, 6(1):52–60, 1997. doi:10.1002/(SICI)1099-1360(199701)6:1<52::AID-MCDA124>3.0.CO;2-3.
- [3002] M. Tamiz, D. F. Jones, and E. El-Darzi. **A review of Goal Programming and its applications**. *Annals of Operations Research*, 58(1):39–53, Jan. 1995. doi:10.1007/BF02032309.
Keywords: Goal Programming, lexicographic, weighted.
- [3003] R. Tanabe and H. Ishibuchi. **An easy-to-use real-world multi-objective optimization problem suite**. *Applied Soft Computing*, 89:106078, 2020.
Annotation: Proposed the RE benchmark suite.
- [3004] R. Tanabe and A. Oyama. **Benchmarking MOEAs for multi-and many-objective optimization using an unbounded external archive**. In P. A. N. Bosman, editor, *GECCO*, pages 633–640. ACM Press, New York, NY, 2017.

- [3005] R. Tanabe, H. Ishibuchi, and A. Oyama. **Benchmarking Multi- and Many-Objective Evolutionary Algorithms Under Two Optimization Scenarios.** *IEEE Access*, 5: 19597–19619, 2017.
Annotation: compared a number of MOEAs using a wide range of numbers of objectives and stopping criteria, with and without archivers; unbounded archive.
- [3006] S. Tanaka and M. Araki. **An Exact Algorithm for the Single-machine Total Weighted Tardiness Problem with Sequence-dependent Setup Times.** *Comput. Oper. Res.*, 40(1): 344–352, 2013.
- [3007] L. Tang and X. Wang. **Iterated local search algorithm based on very large-scale neighborhood for prize-collecting vehicle routing problem.** *International Journal of Advanced Manufacturing Technology*, 29(11):1246–1258, 2006.
- [3008] A. J. Tarquin and J. Dowdy. **Optimal pump operation in water distribution.** *Journal of Hydraulic Engineering, ASCE*, 115(2):158–169 or 496–501, Feb. 1989.
- [3009] M. F. Tasgetiren, Y.-C. Liang, M. Sevkli, and G. Gencyilmaz. **A particle swarm optimization algorithm for makespan and total flowtime minimization in the permutation flowshop sequencing problem.** *Eur. J. Oper. Res.*, 177(3):1930–1947, 2007. doi:10.1016/j.ejor.2005.12.024.
- [3010] M. F. Tasgetiren, O. Buyukdagli, Q.-K. Pan, and P. N. Suganthan. **A general variable neighborhood search algorithm for the no-idle permutation flowshop scheduling problem.** In B. K. Panigrahi, P. N. Suganthan, S. Das, and S. S. Dash, editors, *Swarm, Evolutionary, and Memetic Computing*, volume 8298 of *Theoretical Computer Science and General Issues*, pages 24–34. Springer International Publishing, 2013.
- [3011] M. F. Tasgetiren, Q.-K. Pan, P. N. Suganthan, and O. Buyukdagli. **A variable iterated greedy algorithm with differential evolution for the no-idle permutation flowshop scheduling problem.** *Comput. Oper. Res.*, 40(7):1729–1743, 2013.
- [3012] M. F. Tasgetiren, D. Kizilay, Q.-K. Pan, and P. N. Suganthan. **Iterated Greedy Algorithms for the Blocking Flowshop Scheduling Problem with Makespan Criterion.** *Comput. Oper. Res.*, 77:111–126, 2017.
- [3013] J. Tavares and F. B. Pereira. **Automatic Design of Ant Algorithms with Grammatical Evolution.** In A. Moraglio, S. Silva, K. Krawiec, P. Machado, and C. Cotta, editors, *EuroGP 2012*, volume 7244 of *LNCS*, pages 206–217. Springer, Heidelberg, Germany, 2012.
- [3014] J. C. Tay and N. B. Ho. **Evolving dispatching rules using genetic programming for solving multi-objective flexible job-shop problems.** *Computers & Industrial Engineering*, 54(3):453 – 473, 2008. doi:10.1016/j.cie.2007.08.008.
- [3015] C. Teixeira, J. Covas, T. Stützle, and A. Gaspar-Cunha. **Application of Pareto Local Search and Multi-Objective Ant Colony Algorithms to the Optimization of Co-Rotating Twin Screw Extruders.** In A. Viana et al., editors, *Proceedings of the EU/MEeting 2009: Debating the future: new areas of application and innovative approaches*, pages 115–120, 2009.
- [3016] C. Teixeira, J. Covas, T. Stützle, and A. Gaspar-Cunha. **Engineering an Efficient Two-Phase Local Search for the Co-Rotating Twin-Screw Configuration Problem.** *International Transactions in Operational Research*, 18(2):271–291, 2011.
- [3017] C. Teixeira, J. Covas, T. Stützle, and A. Gaspar-Cunha. **Multi-Objective Ant Colony Optimization for Solving the Twin-Screw Extrusion Configuration Problem.** *Engineering Optimization*, 44(3):351–371, 2012.

- [3018] C. Teixeira, J. Covas, T. Stützle, and A. Gaspar-Cunha. **Hybrid Algorithms for the Twin-Screw Extrusion Configuration Problem.** *Applied Soft Computing*, 23:298–307, 2014.
- [3019] F. Teklu, A. Sumalee, and D. Watling. **A Genetic Algorithm Approach for Optimizing Traffic Control Signals Considering Routing.** *Computer-Aided Civil and Infrastructure Engineering*, 22(1):31–43, Jan. 2007. doi:[10.1111/j.1467-8667.2006.00468.x](https://doi.org/10.1111/j.1467-8667.2006.00468.x).
- [3020] J. B. Tenenbaum, V. D. Silva, and J. C. Langford. **A global geometric framework for nonlinear dimensionality reduction.** *Science*, 290(5500):2319–2323, 2000.
- [3021] J. Teo and H. A. Abbass. **Automatic generation of controllers for embodied legged organisms: A Pareto evolutionary multi-objective approach.** *Evol. Comput.*, 12(3): 355–394, 2004. doi:[10.1162/1063656041774974](https://doi.org/10.1162/1063656041774974).
- [3022] K. T. K. Teo, W. Y. Kow, and Y. K. Chin. **Optimization of traffic flow within an urban traffic light intersection with genetic algorithm.** In *Proceedings - 2nd International Conference on Computational Intelligence, Modelling and Simulation, CIMSIm 2010*, pages 172–177. IEEE, IEEE Press, 2010.
Keywords: Genetic algorithm,T-junction,Traffic control system,Traffic flows.
- [3023] H. Terashima-Marín, P. Ross, and M. Valenzuela-Rendón. **Evolution of Constraint Satisfaction Strategies in Examination Timetabling.** In W. Banzhaf, J. M. Daida, A. E. Eiben, M. H. Garzon, V. Honavar, M. J. Jakiela, and R. E. Smith, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 1999*, pages 635–642. Morgan Kaufmann Publishers, San Francisco, CA, 1999.
- [3024] K. Terayama, M. Sumita, R. Tamura, and K. Tsuda. **Black-Box Optimization for Automated Discovery.** *Accounts of Chemical Research*, 54(6):1334–1346, Mar. 2021. doi:[10.1021/acs.accounts.0c00713](https://doi.org/10.1021/acs.accounts.0c00713).
- [3025] Teredesai et al., editors. *KDD’19: The 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Anchorage, AK, USA, August 4-8, 2019*. ACM Press, New York, NY, July 2019.
- [3026] The Turing Way Community, B. Arnold, L. Bowler, S. Gibson, P. Herterich, R. Higman, A. Krystalli, A. Morley, M. O’Reilly, and K. Whitaker. *The Turing Way: A Handbook for Reproducible Data Science*. Zenodo, Mar. 2019. doi:[10.5281/zenodo.3233986](https://doi.org/10.5281/zenodo.3233986).
Annotation: Available from <https://the-turing-way.netlify.app>. This work was supported by The UKRI Strategic Priorities Fund under the EPSRC Grant EP/T001569/1, particularly the "Tools, Practices and Systems" theme within that grant, and by The Alan Turing Institute under the EPSRC grant EP/N510129/1.
- [3027] P. Thibodeau. **Machine-based decision-making is coming.** *Computer World*, Nov. 2011. URL http://www.computerworld.com/s/article/359630/Machine_Based_Decision_Making_Is_Coming. Last accessed: 15 January 2014.
- [3028] L. Thiele, K. Miettinen, P. Korhonen, and J. Molina. **A Preference-Based Evolutionary Algorithm for Multi-Objective Optimization.** *Evol. Comput.*, 17(3):411–436, 2009. doi:[10.1162/evco.2009.17.3.411](https://doi.org/10.1162/evco.2009.17.3.411).
- [3029] D. Thierens. **Population-based Iterated Local Search: Restricting the Neighborhood Search by Crossover.** In K. Deb et al., editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2004, Part II*, volume 3103 of *LNCS*, pages 234–245. Springer, Heidelberg, Germany, 2004.
- [3030] D. Thierens. **An Adaptive Pursuit Strategy for Allocating Operator Probabilities.** In H.-G. Beyer and U.-M. O’Reilly, editors, *GECCO*, pages 1539–1546. ACM Press, New York, NY, 2005.

- [3031] D. Thierens. **Adaptive strategies for operator allocation**. In F. Lobo, C. F. Lima, and Z. Michalewicz, editors, *Parameter Setting in Evolutionary Algorithms*, pages 77–90. Springer, Berlin, Germany, 2007.
- [3032] D. Thierens. **Adaptive operator selection for iterated local search**. In T. Stützle, M. Birattari, and H. H. Hoos, editors, *SLS 2009*, volume 5752 of *LNCS*, pages 140–144. Springer, Heidelberg, Germany, 2009.
- [3033] D. Thierens et al., editors. *Genetic and Evolutionary Computation Conference, GECCO 2007, Proceedings, London, England, UK, July 7-11, 2007*. ACM Press, New York, NY, 2007.
- [3034] A. Thomaser, M. Vogt, T. Bäck, and A. V. Kononova. **Real-World Optimization Benchmark from Vehicle Dynamics: Specification of Problems in 2D and Methodology for Transferring (Meta-)Optimized Algorithm Parameters**. In N. van Stein, F. Marcelloni, H. K. Lam, M. Cottrell, and J. Filipe, editors, *Proceedings of the 15th International Joint Conference on Computational Intelligence, IJCCI 2023, Rome, Italy, November 13-15, 2023*, pages 31–40. SciTePress, 2023. doi:10.5220/0012158000003595.
- [3035] C. Thornton, F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Auto-WEKA: Combined Selection and Hyperparameter Optimization of Classification Algorithms**. In I. S. Dhillon, Y. Koren, R. Ghani, T. E. Senator, P. Bradley, R. Parekh, J. He, R. L. Grossman, and R. Uthrusamy, editors, *The 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD 2013*, pages 847–855. ACM Press, New York, NY, 2013.
- [3036] S. Thrun and L. Pratt. *Learning to learn*. Springer, 1998.
- [3037] S. Thrun, L. Saul, and B. Schölkopf, editors. *Proceedings of the 16th International Conference on Neural Information Processing Systems, NIPS*. MIT Press, 2003.
- [3038] Y. Tian, R. Cheng, X. Zhang, F. Cheng, and Y. Jin. **An Indicator-Based Multiobjective Evolutionary Algorithm With Reference Point Adaptation for Better Versatility**. *IEEE Trans. Evol. Comput.*, 22(4):609–622, 2018. doi:10.1109/TEVC.2017.2749619. Annotation: IGD-based archiver.
- [3039] T.-O. Ting, M. V. C. Rao, C. K. Loo, and S. S. Ngu. **Solving Unit Commitment Problem Using Hybrid Particle Swarm Optimization**. *J. Heuristics*, 9(6):507–520, 2003. doi:10.1023/B:HEUR.0000012449.84567.1a.
- [3040] R. Tinós, D. Whitley, and G. Ochoa. **Generalized Asymmetric Partition Crossover (GAPX) for the Asymmetric TSP**. In C. Igel and D. V. Arnold, editors, *GECCO*, pages 501–508. ACM Press, New York, NY, 2014.
- [3041] S. Tiwari, P. Koch, G. Fadel, and K. Deb. **AMGA: An archive-based micro genetic algorithm for multi-objective optimization**. In C. Ryan, editor, *GECCO*, pages 729–736. ACM Press, New York, NY, 2008. doi:10.1145/1389095.1389235.
- [3042] S. Tiwari, G. Fadel, and K. Deb. **AMGA2: Improving the performance of the archive-based micro-genetic algorithm for multi-objective optimization**. *Engineering Optimization*, 43(4):377–401, 2011.
- [3043] V. T’Kindt, N. Monmarché, F. Tercinet, and D. Laügt. **An ant colony optimization algorithm to solve a 2-machine bicriteria flowshop scheduling problem**. *Eur. J. Oper. Res.*, 142(2):250–257, 2002. doi:10.1016/S0377-2217(02)00265-5.
- [3044] M. K. Tomczyk and M. Kadziński. **Decomposition-based interactive evolutionary algorithm for multiple objective optimization**. *IEEE Trans. Evol. Comput.*, 24(2):320–334, 2019. doi:10.1109/TEVC.2019.2915767.
Keywords: interactive multi-objective; decision-making.

- [3045] M. K. Tomczyk and M. Kadziński. **EMOSOR: Evolutionary multiple objective optimization guided by interactive stochastic ordinal regression**. *Comput. Oper. Res.*, 108:134–154, 2019. doi:[10.1016/j.cor.2019.04.008](https://doi.org/10.1016/j.cor.2019.04.008).
Keywords: Multiple objective optimization, Interactive evolutionary hybrids, Stochastic ordinal regression, Preference disaggregation, Pairwise comparisons, Active learning.
- [3046] M. K. Tomczyk and M. Kadziński. **Robust Indicator-Based Algorithm for Interactive Evolutionary Multiple Objective Optimization**. In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 629–637. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:[10.1145/3321707.3321742](https://doi.org/10.1145/3321707.3321742).
Keywords: preference learning, indicator-based algorithms, interactive algorithms, multiple objective optimization, pairwise comparisons, evolutionary algorithms.
- [3047] M. K. Tomczyk and M. Kadziński. **Decomposition-based co-evolutionary algorithm for interactive multiple objective optimization**. *Inf. Sci.*, 549:178–199, 2021. doi:[10.1016/j.ins.2020.11.030](https://doi.org/10.1016/j.ins.2020.11.030).
Keywords: Evolutionary multiple objective optimization, Co-evolution, Decomposition, Indirect preference information, Preference learning.
- [3048] H. Topaloglu, editor. *Theory Driven by Influential Applications*. INFORMS, 2013.
- [3049] C. E. Torres, L. F. Rossi, J. Keffer, K. Li, and C.-C. Shen. **Modeling, analysis and simulation of ant-based network routing protocols**. *Swarm Intelligence*, 4(3):221–244, 2010.
- [3050] G. Toscano Pulido and C. A. Coello Coello. **The Micro Genetic Algorithm 2: Towards Online Adaptation in Evolutionary Multiobjective Optimization**. In C. M. Fonseca, P. J. Fleming, E. Zitzler, K. Deb, and L. Thiele, editors, *EMO*, volume 2632 of *LNCS*, pages 252–266. Springer, Heidelberg, Germany, 2003. doi:[10.1007/3-540-36970-8_18](https://doi.org/10.1007/3-540-36970-8_18).
- [3051] P. Toth and D. Vigo. *The vehicle routing problem*. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 2002.
- [3052] F. Toyama, K. Shoji, H. Mori, and J. Miyamichi. **An Iterated Greedy Algorithm for the Binary Quadratic Programming Problem**. In *Joint 6th International Conference on Soft Computing and Intelligent Systems (SCIS) and 13th International Symposium on Advanced Intelligent Systems (ISIS), 2012*, pages 2183–2188. IEEE Press, 2012.
- [3053] R. Trajanov, A. Nikolikj, G. Cenikj, F. Teytaud, M. Videau, O. Teytaud, T. Eftimov, M. López-Ibáñez, and C. Doerr. **Improving Nevergrad’s Algorithm Selection Wizard NGOpt Through Automated Algorithm Configuration**. In G. Rudolph, A. V. Kononova, H. E. Aguirre, P. Kerschke, G. Ochoa, and T. Tušar, editors, *Parallel Problem Solving from Nature – PPSN XVII*, volume 13398 of *LNCS*, pages 18–31. Springer, Cham, Switzerland, 2022. doi:[10.1007/978-3-031-14714-2_2](https://doi.org/10.1007/978-3-031-14714-2_2).
- [3054] H. Trautmann and J. Mehnen. **Preference-based Pareto optimization in certain and noisy environments**. *Engineering Optimization*, 41(1):23–38, Jan. 2009.
- [3055] H. Trautmann, O. Mersmann, and D. Arnu. *cmaes: Covariance Matrix Adapting Evolutionary Strategy*, 2011. URL <http://cran.r-project.org/package=cmaes>. R package.
- [3056] H. Trautmann, G. Rudolph, K. Klamroth, O. Schütze, M. M. Wiecek, Y. Jin, and C. Grimme, editors. *Evolutionary Multi-Criterion Optimization – 9th International Conference, EMO 2017, Münster, Germany, March 19 - 22, 2017. Proceedings*, volume 10173 of *LNCS*. Springer International Publishing, Cham, Switzerland, 2017.

- [3057] C. Treude and M. Wagner. **Predicting Good Configurations for GitHub and Stack Overflow Topic Models**. In *Proceedings of the 16th International Conference on Mining Software Repositories*, MSR '19, pages 84–95, Piscataway, NJ, 2019. IEEE Press. doi:[10.1109/MSR.2019.00022](https://doi.org/10.1109/MSR.2019.00022).
Keywords: algorithm portfolio, corpus features, topic modelling.
- [3058] V. Trianni and M. López-Ibáñez. **Advantages of Multi-Objective Optimisation in Evolutionary Robotics: Survey and Case Studies**. Technical Report TR/IRIDIA/2014-014, IRIDIA, Université Libre de Bruxelles, Belgium, 2014. URL <https://iridia.ulb.ac.be/IridiaTrSeries/link/IridiaTr2014-014.pdf>.
- [3059] V. Trianni and M. López-Ibáñez. **Advantages of Task-Specific Multi-Objective Optimisation in Evolutionary Robotics**. *PLoS One*, 10(8):e0136406, 2015. doi:[10.1371/journal.pone.0136406](https://doi.org/10.1371/journal.pone.0136406).
- [3060] V. Trianni and S. Nolfi. **Engineering the evolution of self-organizing behaviors in swarm robotics: A case study**. *Artificial Life*, 17(3):183–202, 2011.
- [3061] M. A. Trick. **Graph Coloring Instances**. <https://mat.gsia.cmu.edu/COLOR/instances.html>, 2018.
- [3062] A. Trivedi, D. Srinivasan, K. Sanyal, and A. Ghosh. **A survey of multiobjective evolutionary algorithms based on decomposition**. *IEEE Trans. Evol. Comput.*, 21(3):440–462, 2016.
- [3063] L.-Y. Tseng and Y.-T. Lin. **A hybrid genetic local search algorithm for the permutation flowshop scheduling problem**. *Eur. J. Oper. Res.*, 198(1):84–92, 2009.
- [3064] S. Tsutsui. **An Enhanced Aggregation Pheromone System for Real-Parameter Optimization in the ACO Metaphor**. In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*, pages 60–71. Springer, Heidelberg, Germany, 2006.
- [3065] S. Tsutsui. **cAS: Ant Colony Optimization with Cunning Ants**. In T. P. Runarsson, H.-G. Beyer, E. K. Burke, J.-J. Merelo, D. Whitley, and X. Yao, editors, *Parallel Problem Solving from Nature – PPSN IX*, volume 4193 of *LNCS*, pages 162–171. Springer, Heidelberg, Germany, 2006.
- [3066] S. Tsutsui. **Ant Colony Optimization with Cunning Ants**. *Transactions of the Japanese Society for Artificial Intelligence*, 22:29–36, 2007. doi:[10.1527/tjsai.22.29](https://doi.org/10.1527/tjsai.22.29).
Keywords: ant colony optimization, traveling salesman problem, cunning ant, donor ant, local search.
- [3067] E. R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT, 2nd edition, 2001. ISBN 0-9613921-4-2.
Keywords: data visualization, information graphics, cognitive science.
- [3068] A. Tugilimana, A. P. Thrall, and R. Filomeno Coelho. **Conceptual Design of Modular Bridges Including Layout Optimization and Component Reusability**. *Journal of Bridge Engineering*, 22(11):04017094, 2017. doi:[10.1061/\(ASCE\)BE.1943-5592.0001138](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001138).
Keywords: scenario-based.
- [3069] M. Turchetta, F. Berkenkamp, and A. Krause. **Safe Exploration in Finite Markov Decision Processes with Gaussian Processes**. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NIPS 29)*, pages 4312–4320, 2016.
Keywords: SafeMDP.

- [3070] M. Turchetta, F. Berkenkamp, and A. Krause. **Safe Exploration for Interactive Machine Learning**. In H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. B. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NeurIPS 32)*, pages 2887–2897, 2019.
Keywords: Reinforcement Learning; Markov Decision Process; SafeML.
- [3071] R. Turkeš, K. Sörensen, and L. M. Hvattum. **Meta-analysis of metaheuristics: Quantifying the effect of adaptiveness in adaptive large neighborhood search**. *Eur. J. Oper. Res.*, 292(2):423–42, 2021. doi:[10.1016/j.ejor.2020.10.045](https://doi.org/10.1016/j.ejor.2020.10.045).
Keywords: Metaheuristics, Meta-analysis, Adaptive large neighborhood search.
- [3072] T. Tušar. **Design of an Algorithm for Multiobjective Optimization with Differential Evolution**. M.sc. thesis, Faculty of Computer and Information Science, University of Ljubljana, 2007.
- [3073] T. Tušar and B. Filipič. **Differential Evolution versus Genetic Algorithms in Multiobjective Optimization**. In S. Obayashi et al., editors, *EMO*, volume 4403 of *LNCS*, pages 257–271. Springer, Heidelberg, Germany, 2007.
- [3074] T. Tušar and B. Filipič. **Visualizing 4D approximation sets of multiobjective optimizers with projections**. In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 737–744. ACM Press, New York, NY, 2011.
- [3075] T. Tušar and B. Filipič. **Visualizing Exact and Approximated 3D Empirical Attainment Functions**. *Mathematical Problems in Engineering*, 2014:1–18, 2014. doi:[10.1155/2014/569346](https://doi.org/10.1155/2014/569346).
Annotation: Article ID 569346, 18 pages.
- [3076] T. Tušar and B. Filipič. **Visualization of Pareto front approximations in evolutionary multiobjective optimization: A critical review and the projection method**. *IEEE Trans. Evol. Comput.*, 19(2):225–245, 2015. doi:[10.1109/TEVC.2014.2313407](https://doi.org/10.1109/TEVC.2014.2313407).
- [3077] D. Tuytens, J. Teghem, P. Fortemps, and K. V. Nieuwenhuyze. **Performance of the MOSA Method for the Bicriteria Assignment Problem**. *J. Heuristics*, 6:295–310, 2000.
- [3078] A. Tversky. **Choice by elimination**. *Journal of Mathematical Psychology*, 9(4):341–367, 1972.
- [3079] A. Tversky and D. Kahneman. **Judgment under uncertainty: Heuristics and biases**. *Science*, 185(4157):1124–1131, 1974.
- [3080] A. Tversky and D. Kahneman. **Loss aversion in riskless choice: a reference-dependent model**. *The Quarterly Journal of Economics*, 106(4):1039–1061, 1991.
- [3081] C. Twomey, T. Stützle, M. Dorigo, M. Manfrin, and M. Birattari. **An Analysis of Communication Policies for Homogeneous Multi-colony ACO Algorithms**. *Inf. Sci.*, 180(12):2390–2404, 2010. doi:[10.1016/j.ins.2010.02.017](https://doi.org/10.1016/j.ins.2010.02.017).
- [3082] G. H. Tzeng and P. L. Yu, editors. *Proceedings of the 10th International Conference on Multiple Criteria Decision Making (MCDM'91)*. Springer Verlag, 1992.
- [3083] K. Uchida, R. Hamano, M. Nomura, S. Saito, and S. Shirakawa. **CMA-ES for Safe Optimization**. In J. Handl and X. Li, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2024*, pages 722–730. ACM Press, New York, NY, 2024. doi:[10.1145/3638529.3654193](https://doi.org/10.1145/3638529.3654193).
- [3084] D. Uciniski, A. C. Atkinson, and M. Patan, editors. *mODa 10 – Advances in Model-Oriented Design and Analysis, Proceedings of the 10th International Workshop in Model-Oriented Design and Analysis Held in Łagów Lubuski, Poland, June 10-14, 2013*. Springer International Publishing, Heidelberg, Germany, 2013.

- [3085] N. L. J. Ulder, E. H. L. Aarts, H.-J. Bandelt, P. J. M. van Laarhoven, and E. Pesch. **Genetic Local Search Algorithms for the Travelling Salesman Problem**. In H.-P. Schwefel and R. Männer, editors, *Parallel Problem Solving from Nature – PPSN I*, pages 109–116. Springer, Berlin/Heidelberg, 1991. doi:[10.1007/BFb0029723](https://doi.org/10.1007/BFb0029723).
- [3086] T. Ulrich, J. Bader, and L. Thiele. **Defining and Optimizing Indicator-Based Diversity Measures in Multiobjective Search**. In R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 707–717. Springer, Heidelberg, Germany, 2010. doi:[10.1007/978-3-642-15844-5_71](https://doi.org/10.1007/978-3-642-15844-5_71).
Annotation: Two archive; two populations; decision space diversity.
- [3087] E. Ulungu and J. Teghem. **The two phases method: An efficient procedure to solve bi-objective combinatorial optimization problems**. *Foundations of Computing and Decision Sciences*, 20(2):149–165, 1995.
- [3088] E. Ulungu, J. Teghem, P. H. Fortemps, and D. Tuytens. **MOSA method: a tool for solving multiobjective combinatorial optimization problems**. *Journal of Multi-Criteria Decision Analysis*, 8(4):221–236, 1999.
- [3089] S. Urbanek. *multicore: Parallel Processing of R Code on Machines with Multiple Cores or CPUs*, 2010. URL <http://www.rforge.net/multicore/>. R package version 0.1-3.
- [3090] T. Urlings, R. Ruiz, and F. Sivrikaya-Şerifoğlu. **Genetic Algorithms for Complex Hybrid Flexible Flow Line Problems**. *International Journal of Metaheuristics*, 1(1):30–54, 2010.
- [3091] T. Urlings, R. Ruiz, and T. Stützle. **Shifting Representation Search for Hybrid Flexible Flowline Problems**. *Eur. J. Oper. Res.*, 207(2):1086–1095, 2010. doi:[10.1016/j.ejor.2010.05.041](https://doi.org/10.1016/j.ejor.2010.05.041).
- [3092] R. J. M. Vaessens, E. H. L. Aarts, and J. K. Lenstra. **A Local Search Template**. *Comput. Oper. Res.*, 25(11):969–979, 1998. doi:[10.1016/S0305-0548\(97\)00093-2](https://doi.org/10.1016/S0305-0548(97)00093-2).
- [3093] E. Vallada and R. Ruiz. **Genetic algorithms with path relinking for the minimum tardiness permutation flowshop problem**. *Omega*, 38(1-2):57–67, 2010. doi:[10.1016/j.omega.2009.04.002](https://doi.org/10.1016/j.omega.2009.04.002).
- [3094] E. Vallada, R. Ruiz, and G. Minella. **Minimising total tardiness in the m-machine flowshop problem: A review and evaluation of heuristics and metaheuristics**. *Comput. Oper. Res.*, 35(4):1350–1373, 2008.
- [3095] E. Vallada, R. Ruiz, and J. M. Framiñán. **New hard benchmark for flowshop scheduling problems minimising makespan**. *Eur. J. Oper. Res.*, 240(3):666–677, 2015. doi:[10.1016/j.ejor.2014.07.033](https://doi.org/10.1016/j.ejor.2014.07.033).
- [3096] M. Vallati, C. Fawcett, A. E. Gerevini, H. H. Hoos, and A. Saetti. **Generating Fast Domain-Optimized Planners by Automatically Configuring a Generic Parameterised Planner**. In E. Karpas, S. Jiménez Celorrio, and S. Kambhampati, editors, *Proc. of ICAPS-PAL11*, 2011.
- [3097] A. Valsecchi, J. Dubois-Lacoste, T. Stützle, S. Damas, J. Santamaría, and L. Marrakchi-Kacem. **Evolutionary Medical Image Registration using Automatic Parameter Tuning**. In *Proceedings of the 2013 Congress on Evolutionary Computation (CEC 2013)*, pages 1326–1333, Piscataway, NJ, 2013. IEEE Press.
- [3098] L. van der Maaten and G. Hinton. **Visualizing Data using t-SNE**. *J. Mach. Learn. Res.*, 9(86):2579–2605, 2008.

- [3099] H. van Hasselt, A. Guez, and D. Silver. **Deep Reinforcement Learning with Double Q-Learning**. In D. Schuurmans and M. P. Wellman, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press, 2016.
- [3100] J. I. van Hemert. **Evolving Combinatorial Problem Instances That Are Difficult to Solve**. *Evol. Comput.*, 14(4):433–462, 2006. doi:[10.1162/evco.2006.14.4.433](https://doi.org/10.1162/evco.2006.14.4.433).
- [3101] P. van Hentenryck. *The OPL optimization programming language*. MIT Press, Cambridge, MA, 1999.
- [3102] P. van Hentenryck, editor. *Principles and Practice of Constraint Programming, CP 2002*. LNCS. Springer, Heidelberg, Germany, 2002.
- [3103] P. van Hentenryck and L. D. Michel. *Constraint-based Local Search*. MIT Press, Cambridge, MA, 2005.
- [3104] P. van Hentenryck and L. D. Michel. **Synthesis of constraint-based local search algorithms from high-level models**. In R. C. Holte and A. Howe, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 273–278. AAAI Press/MIT Press, Menlo Park, CA, 2007.
- [3105] P. J. M. van Laarhoven and E. H. L. Aarts. *Simulated Annealing: Theory and Applications*, volume 37. Springer, 1987.
- [3106] P. J. M. van Laarhoven, E. H. L. Aarts, and J. K. Lenstra. **Job Shop Scheduling by Simulated Annealing**. *Operations Research*, 40(1):113–125, 1992.
- [3107] J. N. van Rijn and F. Hutter. **Hyperparameter Importance Across Datasets**. In Y. Guo and F. Farooq, editors, *24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2367–2376. ACM Press, New York, NY, July 2018. doi:[10.1145/3219819.3220058](https://doi.org/10.1145/3219819.3220058).
Keywords: hyperparameter optimization, meta-learning, hyperparameter importance.
- [3108] S. van Rijn. **Modular CMA-ES framework from [3109], v0.3.0**. <https://github.com/sjvrijn/ModEA>, 2018. Available also as pypi package at <https://pypi.org/project/ModEA/0.3.0/>.
- [3109] S. van Rijn, H. Wang, M. van Leeuwen, and T. Bäck. **Evolving the structure of Evolution Strategies**. In X. Chen and A. Stafylopatis, editors, *Computational Intelligence (SSCI), 2016 IEEE Symposium Series on*, pages 1–8, 2016. doi:[10.1109/SSCI.2016.7850138](https://doi.org/10.1109/SSCI.2016.7850138).
Keywords: automated design, automatic configuration, CMA-ES, Gaussian distribution.
- [3110] N. van Stein, D. Vermetten, and T. Bäck. **In-the-loop Hyper-Parameter Optimization for LLM-Based Automated Design of Heuristics**. *ACM Trans. Evol. Learn. Optim.*, Apr. 2025. doi:[10.1145/3731567](https://doi.org/10.1145/3731567).
- [3111] D. A. Van Veldhuizen. *Multiobjective evolutionary algorithms: Classifications, analyses, and new innovations*. PhD thesis, Department of Electrical and Computer Engineering, Graduate School of Engineering, Air Force Institute of Technology, Wright-Patterson AFB, Ohio, 1999.
- [3112] D. A. Van Veldhuizen and G. B. Lamont. **Evolutionary Computation and Convergence to a Pareto Front**. In J. R. Koza, editor, *Late Breaking Papers at the Genetic Programming 1998 Conference*, pages 221–228, Stanford University, California, July 1998. Stanford University Bookstore.
Keywords: generational distance.

- [3113] D. A. Van Veldhuizen and G. B. Lamont. **Multiobjective Evolutionary Algorithms: Analyzing the State-of-the-art.** *Evol. Comput.*, 8(2):125–147, 2000. doi:10.1162/106365600568158.
- [3114] E. Van Wolputte, E. Korneva, and H. Blockeel. **MERCS: multi-directional ensembles of regression and classification trees.** In S. A. McIlraith and K. Q. Weinberger, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 4276–4283. AAAI Press, Feb. 2018.
- [3115] J. E. van Zyl. *A Methodology for Improved Operational Optimization of Water Distribution Systems.* PhD thesis, School of Engineering and Computer Science, University of Exeter, UK, 2001.
- [3116] J. E. van Zyl, D. A. Savic, and G. A. Walters. **Operational Optimization of Water Distribution Systems using a Hybrid Genetic Algorithm.** *Journal of Water Resources Planning and Management, ASCE*, 130(2):160–170, Mar. 2004.
- [3117] J. Vanschoren, J. N. van Rijn, B. Bischl, and L. Torgo. **OpenML: Networked Science in Machine Learning.** *ACM SIGKDD Explorations Newsletter*, 15(2):49–60, June 2014. doi:10.1145/2641190.2641198.
- [3118] P. Vansteenwegen and M. Mateo. **An Iterated Local Search Algorithm for the Single-vehicle Cyclic Inventory Routing Problem.** *Eur. J. Oper. Res.*, 237(3):802–813, 2014.
- [3119] P. Vansteenwegen, W. Souffriau, G. Vanden Berghe, and D. V. Oudheusden. **Iterated Local Search for the Team Orienteering Problem with Time Tindows.** *Comput. Oper. Res.*, 36(12):3281–3290, 2009.
- [3120] T. K. Varadharajan and C. Rajendran. **A multi-objective simulated-annealing algorithm for scheduling in flowshops to minimize the makespan and total flowtime of jobs.** *Eur. J. Oper. Res.*, 167(3):772–795, 2005.
- [3121] F. J. Varela and P. Bourguine, editors. *Proceedings of the First European Conference on Artificial Life*, 1992. MIT Press, Cambridge, MA.
- [3122] A. Vargha and H. D. Delaney. **A critique and improvement of the CL common language effect size statistics of McGraw and Wong.** *Journal of Educational and Behavioral Statistics*, 25(2):101–132, 2000.
Keywords: effect size test, A12 test.
- [3123] A. Vasan and S. P. Simonovic. **Optimization of Water Distribution Network Design Using Differential Evolution.** *Journal of Water Resources Planning and Management, ASCE*, 136(2):279–287, 2010.
- [3124] M. Vasile and P. De Pascale. **Preliminary design of multiple gravity-assist trajectories.** *Journal of Spacecraft and Rockets*, 43(4):794–805, 2006. doi:10.2514/1.17413.
- [3125] S. Vassilvitskii and M. Yannakakis. **Efficiently computing succinct trade-off curves.** *Theor. Comput. Sci.*, 348(2-3):334–356, 2005.
- [3126] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. **Attention Is All You Need.** *Arxiv preprint arXiv:1706.03762*, 2017. URL <http://arxiv.org/abs/1706.03762>.
- [3127] D. Vaz, L. Paquete, C. M. Fonseca, K. Klamroth, and M. Stiglmayr. **Representation of the non-dominated set in biobjective discrete optimization.** *Comput. Oper. Res.*, 63:172–186, 2015. ISSN 0305-0548. doi:10.1016/j.cor.2015.05.003.

- [3128] P. H. Vaz Penna, A. Subramanian, and L. S. Ochi. **An Iterated Local Search Heuristic for the Heterogeneous Fleet Vehicle Routing Problem.** *J. Heuristics*, 19(2):201–232, 2013.
- [3129] J. A. Vázquez-Rodríguez and G. Ochoa. **On the Automatic Discovery of Variants of the NEH Procedure for Flow Shop Scheduling Using Genetic Programming.** *J. Oper. Res. Soc.*, 62(2):381–396, 2010.
- [3130] V. Černý. **A Thermodynamical Approach to the Traveling Salesman Problem: An Efficient Simulation Algorithm.** *Journal of Optimization Theory and Applications*, 45(1):41–51, 1985.
- [3131] A. Vedaldi and B. Fulkerson. **VLFeat: An open and portable library of computer vision algorithms.** In *Proceedings of the 18th ACM international conference on Multimedia*, pages 1469–1472. ACM, 2010.
- [3132] P. Vellanki, S. Rana, S. Gupta, D. Rubin, A. Sutti, T. Dorin, M. Height, P. G. Sanders, and S. Venkatesh. **Process-constrained batch Bayesian optimisation.** In I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NIPS 30)*, pages 3414–3423, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/1f71e393b3809197ed66df836fe833e5-Abstract.html>.
- [3133] M. M. Veloso, editor. *IJCAI 2007, Proceedings of the 20th International Joint Conference on Artificial Intelligence, Hyderabad, India, January 6-12, 2007*, 2007. AAAI Press, Menlo Park, CA.
- [3134] S. Verel, A. Liefooghe, and C. Dhaenens. **Set-based Multiobjective Fitness Landscapes: A Preliminary Study.** In N. Krasnogor and P. L. Lanzi, editors, *GECCO*, pages 769–776. ACM Press, New York, NY, 2011. doi:10.1145/2001576.2001681.
- [3135] S. Verel, A. Liefooghe, L. Jourdan, and C. Dhaenens. **On the Structure of Multiobjective Combinatorial Search Space: MNK-landscapes with Correlated Objectives.** *Eur. J. Oper. Res.*, 227(2):331–342, 2013. doi:10.1016/j.ejor.2012.12.019.
- [3136] A. Verma and M. Lewis. **Penalty and partitioning techniques to improve performance of QUBO solvers.** *Discrete Optimization*, page 100594, 2020. ISSN 1572-5286. doi:10.1016/j.disopt.2020.100594.
Keywords: Quadratic Unconstrained Binary Optimization, Nonlinear optimization, Pseudo-Boolean optimization, Equality constraint, Inequality constraint.
- [3137] D. Vermetten. **modde: Modular Differential Evolution.** <https://pypi.org/project/modde/>, 2024.
- [3138] D. Vermetten, H. Wang, C. Doerr, and T. Bäck. **Integrated vs. Sequential Approaches for Selecting and Tuning CMA-ES Variants.** In C. A. Coello Coello, editor, *GECCO*. ACM Press, New York, NY, 2020. ISBN 978-1-4503-7128-5. doi:10.1145/3377930.3389831.
- [3139] D. Vermetten, F. Caraffini, B. van Stein, and A. V. Kononova. **Using Structural Bias to Analyse the Behaviour of Modular CMA-ES.** In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO Companion 2022*, pages 1674–1682. ACM Press, New York, NY, 2022. ISBN 9781450392686. doi:10.1145/3520304.3534035.
- [3140] D. Vermetten, H. Wang, M. López-Ibáñez, C. Doerr, and T. Bäck. **Analyzing the Impact of Undersampling on the Benchmarking and Configuration of Evolutionary Algorithms.** In J. E. Fieldsend and M. Wagner, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2022*, pages 867–875. ACM Press, New York, NY, 2022.

[doi:10.1145/3512290.3528799](https://doi.org/10.1145/3512290.3528799).

Keywords: parameter tuning, evolution strategies, algorithm configuration, performance measures.

- [3141] D. Vermetten, J. Rook, O. L. Preuß, J. de Nobel, C. Doerr, M. López-Ibáñez, H. Trautmann, and T. Bäck. **MO-IOHinspector: Anytime Benchmarking of Multi-objective Algorithms Using IOHprofiler**. In H. Singh, T. Ray, J. D. Knowles, X. Li, J. Branke, B. Wang, and A. Oyama, editors, *Evolutionary Multi-criterion Optimization, EMO 2025*, volume 15512 of *LNCS*, pages 242–256. Springer Verlag, Singapore, 2025. [doi:10.1007/978-981-96-3506-1_17](https://doi.org/10.1007/978-981-96-3506-1_17).
- [3142] D. Vermetten, F. Ye, T. Bäck, and C. Doerr. **MA-BBOB: A Problem Generator for Black-Box Optimization Using Affine Combinations and Shifts**. *ACM Trans. Evol. Learn. Optim.*, 5(1):1–19, 2025. [doi:10.1145/3673908](https://doi.org/10.1145/3673908).
- [3143] A. Viana et al., editors. *Proceedings of the EU/MEeting 2009: Debating the future: new areas of application and innovative approaches*, 2009.
- [3144] P. Viappiani, B. Faltings, and P. Pu. **Preference-based Search using Example-Critiquing with Suggestions**. *J. Artif. Intell. Res.*, 27:465–503, 2006.
- [3145] P. Viappiani, P. Pu, and B. Faltings. **Preference-based Search with Adaptive Recommendations**. *AI Communications*, 21(2):155–175, 2008.
- [3146] R. V. V. Vidal, editor. *Applied Simulated Annealing*. Springer, 1993.
- [3147] T. Vidal, T. G. Crainic, M. Gendreau, and C. Prins. **Heuristics for Multi-attribute Vehicle Routing Problems: A Survey and Synthesis**. *Eur. J. Oper. Res.*, 231(1):1–21, 2013.
- [3148] T. Vidal, T. G. Crainic, M. Gendreau, and C. Prins. **A Unified Solution Framework for Multi-attribute Vehicle Routing Problems**. *Eur. J. Oper. Res.*, 234(3):658–673, 2014.
- [3149] M. Videau, A. Leite, O. Teytaud, and M. Schoenauer. **Multi-Objective Genetic Programming for Explainable Reinforcement Learning**. In E. Medvet, G. Pappa, and B. Xue, editors, *Proceedings of the 25th European Conference on Genetic Programming, EuroGP 2022*, LNCS, pages 256–281. Springer Nature, Cham, Switzerland, 2022.
Keywords: genetic algorithms, genetic programming: Poster.
- [3150] C. Vieira, L. Pérez Cáceres, and L. C. T. Bezerra. **Evaluating Anytime Performance on NAS-Bench-101**. In *Proceedings of the 2021 Congress on Evolutionary Computation (CEC 2021)*, pages 1249–1256, Piscataway, NJ, 2021. IEEE Press. [doi:10.1109/CEC45853.2021.9504902](https://doi.org/10.1109/CEC45853.2021.9504902).
- [3151] M. G. Vilas Boas, H. G. Santos, L. H. d. C. Merschmann, and G. Vanden Berghe. **Optimal decision trees for the algorithm selection problem: integer programming based approaches**. *International Transactions in Operational Research*, 28(5):2759–2781, 2021. [doi:10.1111/itor.12724](https://doi.org/10.1111/itor.12724).
- [3152] A. Violin. *Mathematical Programming Approaches to Pricing Problems*. PhD thesis, Faculté de Sciences, Université Libre de Bruxelles and Dipartimento di Ingegneria e Architettura, Università degli studi di Trieste, 2014.
Annotation: Supervised by Dr. Martine Labbé and Dr. Lorenzo Castelli.
- [3153] P. Virtanen et al. **SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python**. *Nature Methods*, 17:261–272, 2020. [doi:10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2).
- [3154] B. Vitoriano, E. Pinson, and F. Valente, editors. *ICORES 2014 - Proceedings of the 3rd International Conference on Operations Research and Enterprise Systems, Angers, Loire Valley, France*. SciTePress, 2014.

- [3155] M. Vlastelica, J. Song, A. Ferber, B. Amos, G. Martius, B. Dilkina, and Y. Yue, editors. *Learning Meets Combinatorial Algorithms Workshop at NeurIPS 2020, LMCA 2020, Vancouver, Canada, December 12, 2020*, 2020.
- [3156] H.-M. Voigt et al., editors. *The 4th International Conference on Parallel Problem Solving from Nature Berlin, Germany, September 22 - 26, 1996. Proceedings*, volume 1141 of *LNCS*. Springer, Heidelberg, Germany, 1996.
- [3157] V. Volz, B. Naujoks, P. Kerschke, and T. Tušar. **Single- and Multi-Objective Game-Benchmark for Evolutionary Algorithms**. In M. López-Ibáñez, A. Auger, and T. Stützle, editors, *GECCO*, pages 647–655. ACM Press, New York, NY, 2019. ISBN 978-1-4503-6111-8. doi:10.1145/3321707.3321805.
Annotation: MarioGAN, TopTrumps.
- [3158] C. von Lüken, B. Barán, and C. Brizuela. **A survey on multi-objective evolutionary algorithms for many-objective problems**. *Computational Optimization and Applications*, 58(3):707–756, 2014.
- [3159] D. von Winterfeldt and W. Edwards. *Decision Analysis and Behavioral Research*. Cambridge University Press, Aug. 1986. ISBN 978-0-521-25308-6.
- [3160] M. D. Vose. **A linear algorithm for generating random numbers with a given distribution**. *IEEE Trans. Software Eng.*, 17(9):972–975, 1991. doi:10.1109/32.92917.
Annotation: Proposed Walker-Vose alias method for sampling from a discrete distribution.
- [3161] S. Voß and D. L. Woodruff, editors. *Optimization Software Class Libraries*. Kluwer Academic Publishers, Boston, MA, 2002.
- [3162] T. Voß, N. Hansen, and C. Igel. **Improved Step Size Adaptation for the MO-CMA-ES**. In M. Pelikan and J. Branke, editors, *GECCO*, pages 487–494. ACM Press, New York, NY, 2010.
- [3163] C. Voudouris and E. P. K. Tsang. **Guided Local Search and its Application to the Travelling Salesman Problem**. *Eur. J. Oper. Res.*, 113(2):469–499, 1999.
- [3164] C. Voudouris and E. P. K. Tsang. **Guided Local Search**. In F. Glover and G. A. Kochenberger, editors, *Handbook of Metaheuristics*, pages 185–218. Kluwer Academic Publishers, Norwell, MA, 2002.
- [3165] A. Wachi, Y. Sui, Y. Yue, and M. Ono. **Safe Exploration and Optimization of Constrained MDPs Using Gaussian Processes**. In S. A. McIlraith and K. Q. Weinberger, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6548–6556. AAAI Press, Feb. 2018. doi:10.1609/aaai.v32i1.12103.
Keywords: Markov Decision Process, Gaussian Processes.
- [3166] M. Wagner and F. Neumann. **A Fast Approximation-guided Evolutionary Multi-objective Algorithm**. In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 687–694. ACM Press, New York, NY, 2015.
- [3167] M. Wagner, T. Friedrich, and M. T. Lindauer. **Improving local search in a minimum vertex cover solver for classes of networks**. In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 1704–1711, Piscataway, NJ, 2017. IEEE Press. doi:10.1109/CEC.2017.7969507.
Keywords: graph theory;search problems;local search;minimum vertex cover solver;network classes;straightforward alternative approach;benchmark sets;graphs;algorithm portfolio;single integrated approach;Training;Portfolios;Algorithm design and analysis;Prediction algorithms;Machine learning algorithms;Optimization;Benchmark testing;smac,paramils.

- [3168] T. Wagner, N. Beume, and B. Naujoks. **Pareto-, Aggregation-, and Indicator-Based Methods in Many-Objective Optimization.** In S. Obayashi et al., editors, *EMO*, volume 4403 of *LNCS*, pages 742–756. Springer, Heidelberg, Germany, 2007.
- [3169] B. W. Wah and Y. X. Chen. **Optimal Anytime Constrained Simulated Annealing for Constrained Global Optimization.** In R. Dechter, editor, *Principles and Practice of Constraint Programming, CP 2000*, volume 1894 of *LNCS*, pages 425–440. Springer, Heidelberg, Germany, 2000. doi:10.1007/3-540-45349-0_31.
- [3170] D. J. Walker, R. M. Everson, and J. E. Fieldsend. **Visualizing mutually nondominating solution sets in many-objective optimization.** *IEEE Trans. Evol. Comput.*, 17(2):165–184, 2012.
- [3171] H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. B. Fox, and R. Garnett, editors. *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, 8-14 December 2019, Vancouver, BC, Canada*, 2019.
- [3172] J. Wallenius. **Comparative Evaluation of Some Interactive Approaches to Multicriterion Optimization.** *Management Science*, 21(12):1387–1396, 1975.
- [3173] J. Wallenius, J. S. Dyer, P. C. Fishburn, R. E. Steuer, S. Zionts, and K. Deb. **Multiple Criteria Decision Making, Multiattribute Utility Theory: Recent Accomplishments and What Lies Ahead.** *Management Science*, 54(7):1336–1349, 2008. doi:10.1287/mnsc.1070.0838.
- [3174] J. P. Walser. **Solving Linear Pseudo-Boolean Constraint Problems with Local Search.** In B. Kuipers and B. L. Webber, editors, *Proc. of AAAI 1997 – Fourteenth National Conference on Artificial Intelligence*, pages 269–274. AAAI Press/MIT Press, Menlo Park, CA, 1997.
- [3175] J. P. Walser. *Integer Optimization by Local Search: A Domain-Independent Approach*, volume 1637 of *LNCS*. Springer, Heidelberg, Germany, 1999.
- [3176] J. P. Walser, R. Iyer, and N. Venkatasubramanian. **An Integer Local Search Method with Application to Capacitated Production Planning.** In J. Mostow and C. Rich, editors, *Proc. of AAAI 1998 – Fifteenth National Conference on Artificial Intelligence*, pages 373–379. AAAI Press/MIT Press, Menlo Park, CA, 1998.
- [3177] T. Walsh. **Depth-bounded Discrepancy Search.** In M. E. Pollack, editor, *Proc. of IJCAI-97*, pages 1388–1395. Morgan Kaufmann Publishers, 1997.
- [3178] T. Walsh, editor. *IJCAI 2011, Proceedings of the 22nd International Joint Conference on Artificial Intelligence, Barcelona, Spain, July 16-22, 2011*, 2011. IJCAI/AAAI Press, Menlo Park, CA.
- [3179] C. Walshaw and M. Cross. **Mesh Partitioning: A Multilevel Balancing and Refinement Algorithm.** *SIAM Journal on Scientific Computing*, 22(1):63–80, 2000. doi:10.1137/S1064827598337373.
- [3180] T. M. Walski, D. V. Chase, D. A. Savic, W. Grayman, S. Beckwith, and E. Koelle. *Advanced Water Distribution Modeling and Management*. Haestad Methods, Inc., Haestad Press, 1st edition, 2003.
- [3181] C. Wang, C. Chu, and J.-M. Proth. **Heuristic Approaches for n/m/F/ΣCi Scheduling Problems.** *Eur. J. Oper. Res.*, 96(3):636–644, 1997. doi:10.1016/0377-2217(95)00347-9.
- [3182] H. Wang, L. Jiao, and X. Yao. **TwoArch2: An improved two-archive algorithm for many-objective optimization.** *IEEE Trans. Evol. Comput.*, 19(4):524–541, 2015.

- [3183] H. Wang, J. Doherty, and Y. Jin. **Hierarchical surrogate-assisted evolutionary multi-scenario airfoil shape optimization**. In *Proceedings of the 2018 Congress on Evolutionary Computation (CEC 2018)*, pages 1–8, Piscataway, NJ, 2018. IEEE Press.
Keywords: scenario-based.
- [3184] H. Wang, C. Sun, Y. Jin, S. Qin, and H. Yu. **A Multi-indicator based Selection Strategy for Evolutionary Many-objective Optimization**. In *Proceedings of the 2019 Congress on Evolutionary Computation (CEC 2019)*, pages 2042–2049, Piscataway, NJ, June 2019. IEEE Press.
Annotation: unbounded archive.
- [3185] H. Wang, D. Vermetten, F. Ye, C. Doerr, and T. Bäck. **IOHanalyzer: Detailed Performance Analyses for Iterative Optimization Heuristics**. *ACM Trans. Evol. Learn. Optim.*, 2(1): 3:1–3:29, 2022. doi:10.1145/3510426.
- [3186] R. Wang, R. C. Purshouse, and P. J. Fleming. **Preference-Inspired Coevolutionary Algorithms for Many-Objective Optimization**. *IEEE Trans. Evol. Comput.*, 17(4):474–494, 2013.
- [3187] R. Wang, J. Xiong, M.-f. He, L. Gao, and L. Wang. **Multi-objective optimal design of hybrid renewable energy system under multiple scenarios**. *Renewable Energy*, 151:226–237, 2020. doi:10.1016/j.renene.2019.11.015.
- [3188] S. Wang, Y. Mei, and M. Zhang. **Two-stage multi-objective genetic programming with archive for uncertain capacitated arc routing problem**. In F. Chicano and K. Krawiec, editors, *GECCO*, pages 287–295. ACM Press, New York, NY, 2021.
- [3189] X. Wang, Y. Jin, S. Schmitt, M. Olhofer, and R. Allmendinger. **Transfer learning based surrogate assisted evolutionary bi-objective optimization for objectives with different evaluation times**. *Knowledge-Based Systems*, 227:107190, 2021. doi:10.1016/j.knosys.2021.107190.
- [3190] Y. Wang, Z. Lü, F. Glover, and J.-K. Hao. **Path relinking for unconstrained binary quadratic programming**. *Eur. J. Oper. Res.*, 223(3):595–604, 2012. doi:10.1016/j.ejor.2012.07.012.
- [3191] Y. Wang, Z. Lü, F. Glover, and J.-K. Hao. **Probabilistic GRASP-Tabu Search algorithms for the UBQP problem**. *Comput. Oper. Res.*, 40(12):3100–3107, 2013. doi:10.1016/j.cor.2011.12.006.
- [3192] Y. Wang, Z. Lü, F. Glover, and J.-K. Hao. **Backbone Guided Tabu Search for Solving the UBQP Problem**. *J. Heuristics*, 19(4):679–695, 2013. doi:10.1007/s10732-011-9164-4.
- [3193] Y. Wang, X. Dong, P. Chen, and Y. Lin. **Iterated local search algorithms for the sequence-dependent setup times flow shop scheduling problem minimizing makespan**. In *Foundations of Intelligent Systems*, pages 329–338. Springer, 2014.
- [3194] M. O. Ward. **Multivariate data glyphs: Principles and practice**. In C.-h. Chen, W. K. Härdle, and A. Unwin, editors, *Handbook of Data Visualization*, pages 179–198. Springer, 2008.
- [3195] A. A. Watson and J. R. Kasprzyk. **Incorporating deeply uncertain factors into the many objective search process**. *Environmental Modelling & Software*, 89:159–171, 2017.
Keywords: scenario-based.
- [3196] J.-P. Watson, L. Barbulescu, D. Whitley, and A. E. Howe. **Contrasting Structured and Random Permutation Flow-Shop Scheduling Problems: Search Space Topology and Algorithm Performance**. *INFORMS Journal on Computing*, 14(2):98–123, 2002.

- [3197] J.-P. Watson, J. C. Beck, A. E. Howe, and D. Whitley. **Problem Difficulty for Tabu Search in Job-Shop Scheduling**. *Artif. Intell.*, 143(2):189–217, 2003.
- [3198] J.-P. Watson, A. E. Howe, and D. Whitley. **Deconstructing Nowicki and Smutnicki’s i-TSAB tabu search algorithm for the job-shop scheduling problem**. *Comput. Oper. Res.*, 33(9):2623–2644, 2006.
- [3199] T. Wauters. **10 years of Eternity II—from \$2 million puzzle to challenging optimization problem**. In *International Workshop on Cutting, Packing and Related Topics*, Gent, Belgium, 2017. URL <https://lirias.kuleuven.be/1675982?limo=0>.
- [3200] WCCI. *Proceedings of the First IEEE Conference on Evolutionary Computation, IEEE World Congress on Computational Intelligence, Orlando, Florida, USA, June 27-29, 1994*, Piscataway, NJ, June 1994. IEEE Press.
- [3201] WCCI. *Proceedings of the 2022 World Congress on Computational Intelligence (WCCI 2022)*, Piscataway, NJ, 2022. IEEE Press.
- [3202] I. Wegener. **Simulated annealing beats metropolis in combinatorial optimization**. In L. Caires, G. F. Italiano, L. Monteiro, C. Palamidessi, and M. Yung, editors, *Proceedings of the 32nd International Colloquium on Automata, Languages and Programming, ICALP 2005*, volume 3580 of *LNCS*, pages 589–601. Springer, Heidelberg, Germany, 2005.
- [3203] C. Wegley, M. Eusuff, and K. E. Lansey. **Determining Pump Operations Using Particle Swarm Optimization**. In R. H. Hotchkiss and M. Glade, editors, *Building Partnerships, Proceedings of the Joint Conference on Water Resources Engineering and Water Resources Planning and Management*, Minneapolis, USA, 2000.
- [3204] E. J. Wegman. **Hyperdimensional data analysis using parallel coordinates**. *Journal of the American Statistical Association*, 85(411):664–675, 1990.
- [3205] P. Wegner. **Research paradigms in computer science**. In *ICSE’76: Proceedings of the 2nd international conference on Software engineering*, pages 322–330, Oct. 1976.
- [3206] E. D. Weinberger. **Local properties of Kauffman’s N-k model: A tunably rugged energy landscape**. *Physical Review A*, 44(10):6399, 1991.
- [3207] K. Q. Weinberger and L. K. Saul. **An Introduction to Nonlinear Dimensionality Reduction by Maximum Variance Unfolding**. In A. Cohn, editor, *Proceedings of the 21st National Conference on Artificial Intelligence*, volume 6, pages 1683–1686. AAAI Press/MIT Press, Menlo Park, CA, 2006.
- [3208] K. Q. Weinberger, F. Sha, and L. K. Saul. **Learning a kernel matrix for nonlinear dimensionality reduction**. In C. E. Brodley, editor, *Proceedings of the 21st International Conference on Machine Learning, ICML 2004*, New York, NY, 2004. ACM Press. doi:10.1145/1015330.1015345.
- [3209] K. Weiss, T. M. Khoshgoftaar, and D. Wang. **A survey of transfer learning**. *Journal of Big Data*, 3(1):1–40, 2016.
- [3210] B. L. Welch. **The significance of the difference between two means when the population variances are unequal**. *Biometrika*, 29(3/4):350–362, 1938.
- [3211] S. Wessing and M. López-Ibáñez. **Latin Hypercube Designs with Branching and Nested Factors for Initialization of Automatic Algorithm Configuration**. *Evol. Comput.*, 27(1):129–145, 2018. doi:10.1162/evco_a_00241.

- [3212] S. Wessing, N. Beume, G. Rudolph, and B. Naujoks. **Parameter Tuning Boosts Performance of Variation Operators in Multiobjective Optimization.** In R. Schaefer, C. Cotta, J. Kolodziej, and G. Rudolph, editors, *Parallel Problem Solving from Nature, PPSN XI*, volume 6238 of *LNCS*, pages 728–737. Springer, Heidelberg, Germany, 2010. doi:10.1007/978-3-642-15844-5_73.
- [3213] D. Weyland. **A Rigorous Analysis of the Harmony Search Algorithm: How the Research Community can be misled by a “novel” Methodology.** *International Journal of Applied Metaheuristic Computing*, 12(2):50–60, 2010.
- [3214] D. Weyland. **A critical analysis of the harmony search algorithm: How not to solve Sudoku.** *Oper. Res. Perspect.*, 2:97–105, 2015.
- [3215] C. R. Whaley. **ATLAS: Automatically Tuned Linear Algebra Software.** In D. Padua, editor, *Encyclopedia of Parallel Computing*, pages 95–101. Springer, US, 2011.
- [3216] L. While and L. Bradstreet. **Applying the WFG Algorithm to Calculate Incremental Hypervolumes.** In *IEEE CEC*, pages 1–8, Piscataway, NJ, 2012. IEEE Press.
- [3217] L. While, L. Bradstreet, and L. Barone. **A Fast Way of Calculating Exact Hypervolumes.** *IEEE Trans. Evol. Comput.*, 16(1):86–95, 2012.
- [3218] D. R. White, A. Arcuri, and J. A. Clark. **Evolutionary Improvement of Programs.** *IEEE Trans. Evol. Comput.*, 15(4):515–538, 2011.
- [3219] T. White, B. Paturek, and F. Oppacher. **Connection Management Using Adaptive Mobile Agents.** In H. R. Arabnia, editor, *Proceedings of the International Conference on Parallel and Distributed Processing Techniques and Applications (PDPTA’98)*, pages 802–809. CSREA Press, 1998.
- [3220] D. Whitley, editor. *Proceedings of the Second Workshop on Foundations of Genetic Algorithms.* Morgan Kaufmann Publishers, 1993. ISBN 1-55860-263-1.
- [3221] D. Whitley, S. Rana, J. Dzubera, and K. E. Mathias. **Evaluating Evolutionary Algorithms.** *Artif. Intell.*, 85:245–296, 1996.
- [3222] D. Whitley et al., editors. *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2000.* Morgan Kaufmann Publishers, San Francisco, CA, 2000.
- [3223] A. Wiegele. **Biq Mac Library – A collection of Max-Cut and quadratic 0-1 programming instances of medium size.** Technical report, Institut für Mathematik, Alpen-Adria-Universität Klagenfurt, 2007. URL <http://biqmac.aau.at/biqmaclib.pdf>.
- [3224] A. Wiegele. **Biq Mac Library – Binary Quadratic and Max Cut Library.** <http://biqmac.aau.at/biqmaclib.html>, 2007.
- [3225] A. P. Wierzbicki. **The Use of Reference Objectives in Multiobjective Optimisation.** In G. Fandel and T. Gal, editors, *Multiple Criteria Decision Making Theory and Application*, number 177 in *LNEMS*, pages 468–486. Springer, Heidelberg, Germany, 1980. doi:10.1007/978-3-642-48782-8_32.
- [3226] W. Wiesemann and T. Stützle. **Iterated Ants: An Experimental Study for the Quadratic Assignment Problem.** In M. Dorigo et al., editors, *Ant Colony Optimization and Swarm Intelligence, 5th International Workshop, ANTS 2006*, volume 4150 of *LNCS*, pages 179–190. Springer, Heidelberg, Germany, 2006.
- [3227] R. J. Williams. **Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning.** *Mach. Learn.*, 8(3):229–256, 1992.

- [3228] D. P. Williamson and D. B. Shmoys. *The design of approximation algorithms*. Cambridge University Press, 2011.
- [3229] P. Winkler. **Random Orders**. *Order*, 1:317–331, 1985.
Annotation: Showed that fraction of Pareto-optimal increases with number of objectives.
- [3230] C. Witt. **Analysis of an Iterated Local Search Algorithm for Vertex Cover in Sparse Random Graphs**. *Theor. Comput. Sci.*, 425:117–125, 2012.
- [3231] S. Wolf and P. Merz. **Iterated Local Search for Minimum Power Symmetric Connectivity in Wireless Networks**. In C. Cotta and P. Cowling, editors, *EvoCOP*, volume 5482 of *LNCS*, pages 192–203. Springer, Heidelberg, Germany, 2009.
- [3232] R. Wolfler Calvo. **A New Heuristic for the Traveling Salesman Problem with Time Windows**. *Transportation Science*, 34(1):113–124, 2000. doi:10.1287/trsc.34.1.113.12284.
- [3233] D. H. Wolpert and W. G. Macready. **No Free Lunch Theorems for Optimization**. *IEEE Trans. Evol. Comput.*, 1(1):67–82, 1997. doi:10.1109/4235.585893.
- [3234] H. S. Woo and D. S. Yim. **A Heuristic Algorithm for Mean Flowtime Objective in Flowshop Scheduling**. *Comput. Oper. Res.*, 25(3):175–182, 1998.
- [3235] D. L. Woodruff, U. Ritzinger, and J. Oppen. **Research Note: The Point of Diminishing Returns in Heuristic Search**. *International Journal of Metaheuristics*, 1(3):222–231, 2011. doi:10.1504/IJMHeur.2011.041195.
Keywords: anytime.
- [3236] M. J. Woodruff, P. M. Reed, and T. W. Simpson. **Many objective visual analytics: rethinking the design of complex engineered systems**. *Struct. Multidisc. Optim.*, 48(1): 201–219, 2013. doi:10.1007/s00158-013-0891-z.
- [3237] M. N. Wright and A. Ziegler. **ranger: A Fast Implementation of Random Forests for High Dimensional Data in C++ and R**. *Arxiv preprint arXiv:1508.04409 [stat.ML]*, 2015. URL <https://arxiv.org/abs/1508.04409>.
- [3238] M. N. Wright and A. Ziegler. **ranger: A Fast Implementation of Random Forests for High Dimensional Data in C++ and R**. *Journal of Statistical Software*, 77(1):1–17, 2017. doi:10.18637/jss.v077.i01.
- [3239] J. Wu and S. Azam. **Metrics for Quality Assessment of a Multiobjective Design Optimization Solution Set**. *J. Mech. Des.*, 123(1):18–25, 2001. doi:10.1115/1.1329875.
Annotation: Proposed inclusion-exclusion method for computing hypervolume.
- [3240] X. Wu, X. Zhu, G.-Q. Wu, and W. Ding. **Data mining with big data**. *IEEE Transactions on Knowledge and Data Engineering*, 26(1):97–107, 2014.
- [3241] Y. Wu, M. Schuster, Z. Chen, Q. V. Le, M. Norouzi, W. Macherey, M. Krikun, Y. Cao, Q. Gao, K. Macherey, et al. **Google’s neural machine translation system: Bridging the gap between human and machine translation**. *Arxiv preprint arXiv:1609.08144 [cs.CL]*, 2016. URL <https://arxiv.org/abs/1609.08144>.
- [3242] A. E. Xavier and V. L. Xavier. **Solving the minimum sum-of-squares clustering problem by hyperbolic smoothing and partition into boundary and gravitational regions**. *Pattern Recognition*, 44(1):70–77, 2011. ISSN 0031-3203. doi:10.1016/j.patcog.2010.07.004.
Keywords: Cluster analysis, Min-sum-min problems, Nondifferentiable programming, Smoothing.

- [3243] B. Xin, L. Chen, J. Chen, H. Ishibuchi, K. Hirota, and B. Liu. **Interactive Multiobjective Optimization: A Review of the State-of-the-Art**. *IEEE Access*, 6:41256–41279, 2018. doi:[10.1109/ACCESS.2018.2856832](https://doi.org/10.1109/ACCESS.2018.2856832).
Keywords: Decision making, Evolutionary computation, Pareto optimization, Evolutionary multiobjective optimization, interactive multiobjective optimization, multiple criteria decision making, preference information, preference models.
- [3244] E. P. Xing and T. Jebara, editors. *Proceedings of the 31st International Conference on Machine Learning, ICML 2014, Beijing, China, 21-26 June 2014*, volume 32, 2014. PMLR.
- [3245] D.-L. Xu and J.-B. Yang. **Intelligent Decision System for Self-Assessment**. *Journal of Multi-Criteria Decision Analysis*, 12(1):43–60, 2003. doi:[10.1002/mcda.343](https://doi.org/10.1002/mcda.343).
Keywords: decision support system, business excellence, MCDA, quality award, self-assessment, the evidential reasoning approach.
- [3246] H. Xu, Z. Lü, and T. C. E. Cheng. **Iterated Local Search for Single-machine Scheduling with Sequence-dependent Setup Times to Minimize Total Weighted Tardiness**. *Journal of Scheduling*, 17(3):271–287, 2014.
- [3247] J. Xu, S. Y. Chiu, and F. Glover. **Fine-tuning a tabu search algorithm with statistical tests**. *International Transactions in Operational Research*, 5(3):233–244, 1998. doi:[10.1111/j.1475-3995.1998.tb00117.x](https://doi.org/10.1111/j.1475-3995.1998.tb00117.x).
- [3248] J. Xu, P. P. Pokharel, A. R. C. Paiva, and J. C. Príncipe. **Nonlinear Component Analysis Based on Correntropy**. In *Proceedings of the International Joint Conference on Neural Networks, IJCNN 2006*, pages 1851–1855. IEEE, 2006. doi:[10.1109/IJCNN.2006.246905](https://doi.org/10.1109/IJCNN.2006.246905).
- [3249] L. Xu, F. Hutter, H. H. Hoos, and K. Leyton-Brown. **SATzilla: Portfolio-based Algorithm Selection for SAT**. *J. Artif. Intell. Res.*, 32:565–606, June 2008. doi:[10.1613/jair.2490](https://doi.org/10.1613/jair.2490).
- [3250] L. Xu, H. H. Hoos, and K. Leyton-Brown. **Hydra: Automatically Configuring Algorithms for Portfolio-Based Selection**. In M. Fox and D. Poole, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 210–216. AAAI Press, 2010. doi:[10.1609/aaai.v24i1.7565](https://doi.org/10.1609/aaai.v24i1.7565).
Keywords: automated algorithm design; portfolio-based algorithm selection; automated algorithm configuration; SAT; stochastic local search.
- [3251] L. Xu, F. Hutter, H. H. Hoos, and K. Leyton-Brown. **Hydra-MIP: Automated Algorithm Configuration and Selection for Mixed Integer Programming**. Technical Report TR-2011-01, Department of Computer Science, University of British Columbia, Canada, 2011. URL <https://www.cs.ubc.ca/tr/2011/tr-2011-01>.
- [3252] L. Xu, A. R. KhudaBukhsh, H. H. Hoos, and K. Leyton-Brown. **Quantifying the similarity of algorithm configurations**. In P. Festa, M. Sellmann, and J. Vanschoren, editors, *LION 10*, volume 10079 of *LNCS*, pages 203–217. Springer, Cham, Switzerland, 2016.
- [3253] M. Yagiura, M. Kishida, and T. Ibaraki. **A 3-Flip Neighborhood Local Search for the Set Covering Problem**. *Eur. J. Oper. Res.*, 172(2):472–499, 2006.
- [3254] Y. Yamada. **How to Crack Pre-registration: Toward Transparent and Open Science**. *Frontiers in Psychology*, 9, Sept. 2018. doi:[10.3389/fpsyg.2018.01831](https://doi.org/10.3389/fpsyg.2018.01831).
Keywords: HARKing; PARKing.
- [3255] A. Yaman, A. Hallawa, M. Coler, and G. Iacca. **Presenting the ECO: evolutionary computation ontology**. In G. Squillero and K. Sim, editors, *Applications of Evolutionary Computation*, volume 10199 of *LNCS*, pages 603–619. Springer, Heidelberg, Germany, 2017. doi:[10.1007/978-3-319-55849-3](https://doi.org/10.1007/978-3-319-55849-3).

- [3256] C. Yang, X. Wang, Y. Lu, H. Liu, Q. V. Le, D. Zhou, and X. Chen. **Large Language Models as Optimizers**. In *Proc. of ICLR 2024*, 2024. doi:10.48550/arXiv.2309.03409.
- [3257] K. Yang, M. T. M. Emmerich, A. H. Deutz, and T. Bäck. **Multi-Objective Bayesian Global Optimization using Expected Hypervolume Improvement Gradient**. *Swarm and Evolutionary Computation*, 44:945–956, Feb. 2019. doi:10.1016/j.swevo.2018.10.007.
Keywords: Bayesian Optimisation with preferences.
- [3258] Q. Yang and M. Wooldridge, editors. *IJCAI 2015, Proceedings of the 24th International Joint Conference on Artificial Intelligence, Buenos Aires, Argentina, July 25-31, 2015*, 2015. IJCAI/AAAI Press, Menlo Park, CA.
- [3259] S. Yang, M. Li, X. Liu, and J. Zheng. **A Grid-Based Evolutionary Algorithm for Many-Objective Optimization**. *IEEE Trans. Evol. Comput.*, 17(5):721–736, 2013. doi:10.1109/TEVC.2012.2227145.
Annotation: epsilon-grid.
- [3260] Y. Yang, S. Kreipl, and M. L. Pinedo. **Heuristics for Minimizing Total Weighted Tardiness in Flexible Flow Shops**. *Journal of Scheduling*, 3(2):89–108, 2000.
- [3261] X. Yao. *Evolutionary Computation: Theory and Applications*. World Scientific Singapore, River Edge, NJ, 1999. ISBN 9810223064.
Keywords: Evolutionary programming (Computer science); Neural networks (Computer science); Evolutionary computation.
- [3262] X. Yao et al., editors. *Proceedings of PPSN-VIII, Eighth International Conference on Parallel Problem Solving from Nature, Birmingham, UK*, volume 3242 of LNCS. Springer, Heidelberg, Germany, 2004.
- [3263] A. Yarimcam, S. Asta, E. Özcan, and A. J. Parkes. **Heuristic Generation via Parameter Tuning for Online Bin Packing**. In P. Angelov et al., editors, *Evolving and Autonomous Learning Systems (EALS), 2014 IEEE Symposium on*, pages 102–108. IEEE, 2014. doi:10.1109/EALS.2014.7009510.
Keywords: irace.
- [3264] C. Yasojima, T. Araújo, B. Meiguins, N. Neto, and J. Morais. **A Comparison of Genetic Algorithms and Particle Swarm Optimization to Estimate Cluster-Based Kriging Parameters**. In P. Moura Oliveira, P. Novais, and L. P. Reis, editors, *Progress in Artificial Intelligence*, pages 750–761. Springer International Publishing, Cham, Switzerland, 2019. ISBN 978-3-030-30241-2.
- [3265] G. Yavuz, D. Aydın, and T. Stützle. **Self-adaptive Search Equation-based Artificial Bee Colony Algorithm on the CEC 2014 Benchmark Functions**. In *Proceedings of the 2016 Congress on Evolutionary Computation (CEC 2016)*, pages 1173–1180, Piscataway, NJ, 2016. IEEE Press. ISBN 978-1-5090-0623-6.
- [3266] D. Yazdani, R. Cheng, D. Yazdani, J. Branke, Y. Jin, and X. Yao. **A Survey of Evolutionary Continuous Dynamic Optimization Over Two Decades - Part B**. *IEEE Trans. Evol. Comput.*, 25(4):630–650, 2021. doi:10.1109/TEVC.2021.3060012.
- [3267] D. Yazdani, M. N. Omidvar, R. Cheng, J. Branke, T. T. Nguyen, and X. Yao. **Benchmarking Continuous Dynamic Optimization: Survey and Generalized Test Suite**. *IEEE Trans. Cybern.*, 52(5):3380–3393, 2022. doi:10.1109/TCYB.2020.3011828.
Annotation: Proposed Generalized Moving Peaks Benchmark (GMPB).
- [3268] F. Ye, C. Doerr, H. Wang, and T. Bäck. **Automated Configuration of Genetic Algorithms by Tuning for Anytime Performance**. *IEEE Trans. Evol. Comput.*, 26(6):1526–1538, 2022. doi:10.1109/TEVC.2022.3159087.

- [3269] C. Young, D. S. Johnson, D. R. Karger, and M. D. Smith. **Near-optimal Intraprocedural Branch Alignment**. In M. C. Chen, R. K. Cytron, and A. M. Berman, editors, *Proceedings of the ACM SIGPLAN'97 Conference on Programming Language Design and Implementation (PLDI), Las Vegas, Nevada*, pages 183–193. ACM Press, 1997.
- [3270] G. Yu, R. S. Powell, and M. J. H. Sterling. **Optimized Pump Scheduling in Water Distribution Systems**. *Journal of Optimization Theory and Applications*, 83(3):463–488, 1994.
- [3271] H. Yu. *Rmpi: Interface (Wrapper) to MPI (Message-Passing Interface)*, 2010. URL <http://cran.r-project.org/package=Rmpi>. R package version 0.5-8.
- [3272] P. L. H. Yu, W. M. Wan, and P. H. Lee. **Decision Tree Modeling for Ranking Data**. In J. Fürnkranz and E. Hüllermeier, editors, *Preference Learning*, pages 83–106. Springer, Heidelberg, Germany, 2011. ISBN 978-3-642-14125-6. doi:10.1007/978-3-642-14125-6_5.
- [3273] V. F. Yu and S.-W. Lin. **Iterated Greedy Heuristic for the Time-dependent Prize-collecting Arc Routing Problem**. *Computers & Industrial Engineering*, 90:54–66, 2015.
- [3274] B. Yuan and M. Gallagher. **Statistical Racing Techniques for Improved Empirical Evaluation of Evolutionary Algorithms**. In X. Yao et al., editors, *PPSN*, volume 3242 of *LNCS*, pages 172–181. Springer, Heidelberg, Germany, 2004.
- [3275] B. Yuan and M. Gallagher. **Combining Meta-EAs and racing for difficult EA parameter tuning tasks**. In F. Lobo, C. F. Lima, and Z. Michalewicz, editors, *Parameter Setting in Evolutionary Algorithms*, pages 121–142. Springer, Berlin, Germany, 2007.
- [3276] Z. Yuan, A. Fügenschuh, H. Homfeld, P. Balaprakash, T. Stützle, and M. Schoch. **Iterated Greedy Algorithms for a Real-World Cyclic Train Scheduling Problem**. In M. J. Blesa, C. Blum, C. Cotta, A. J. Fernández, J. E. Gallardo, A. Roli, and M. Sampels, editors, *Hybrid Metaheuristics*, volume 5296 of *LNCS*, pages 102–116. Springer, Heidelberg, Germany, 2008.
- [3277] Z. Yuan, M. A. Montes de Oca, T. Stützle, and M. Birattari. **Continuous Optimization Algorithms for Tuning Real and Integer Algorithm Parameters of Swarm Intelligence Algorithms**. *Swarm Intelligence*, 6(1):49–75, 2012.
- [3278] Z. Yuan, M. A. Montes de Oca, T. Stützle, H. C. Lau, and M. Birattari. **An Analysis of Post-selection in Automatic Configuration**. In C. Blum and E. Alba, editors, *GECCO*, pages 1557–1564. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8.
- [3279] L. Yuefeng, W. Du, and T. Stützle. **Three L-SHADE Based Algorithms on Mixed-variables Optimization Problems**. In *Proceedings of the 2017 Congress on Evolutionary Computation (CEC 2017)*, pages 2274–2281, Piscataway, NJ, 2017. IEEE Press.
- [3280] J. Yuen, S. Gao, M. Wagner, and F. Neumann. **An adaptive data structure for evolutionary multi-objective algorithms with unbounded archives**. In *IEEE CEC*, pages 1–8, Piscataway, NJ, 2012. IEEE Press.
- [3281] X. Yun and S. L. Epstein. **Learning Algorithm Portfolios for Parallel Execution**. In Y. Hamadi and M. Schoenauer, editors, *LION 6*, volume 7219 of *LNCS*, pages 323–338. Springer, Heidelberg, Germany, 2012. doi:10.1007/978-3-642-34413-8_23.
- [3282] M. Zaefferer, J. Stork, and T. Bartz-Beielstein. **Distance Measures for Permutations in Combinatorial Efficient Global Optimization**. In T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, editors, *Parallel Problem Solving from Nature – PPSN XIII*, volume 8672 of *LNCS*, pages 373–383. Springer, Heidelberg, Germany, 2014. doi:10.1007/978-3-319-10762-2_37. *Keywords:* CEGO, Bayesian optimization.

- [3283] M. Zaefferer, J. Stork, M. Friese, A. Fischbach, B. Naujoks, and T. Bartz-Beielstein. **Efficient Global Optimization for Combinatorial Problems**. In C. Igel and D. V. Arnold, editors, *GECCO*, pages 871–878. ACM Press, New York, NY, 2014. doi:10.1145/2576768.2598282.
Keywords: CEGO, Bayesian optimization.
Annotation: Proposed CEGO algorithm.
- [3284] C. Zarges and S. Verel, editors. *Evolutionary Computation in Combinatorial Optimization – 21st European Conference, EvoCOP 2021, Held as Part of EvoStar 2021, Virtual Event, April 7-9, 2021, Proceedings*, volume 12692 of *LNCS*. Springer, Cham, Switzerland, 2021.
- [3285] E. Zarpas. **Benchmarking SAT solvers for bounded model checking**. In F. Bacchus and T. Walsh, editors, *International Conference on Theory and Applications of Satisfiability Testing*, volume 3569, pages 340–354, 2005.
- [3286] Q. Zeng and Z. Yang. **Integrating Simulation and Optimization to Schedule Loading Operations in Container Terminals**. *Comput. Oper. Res.*, 36(6):1935–1944, 2009. doi:10.1016/j.cor.2008.06.010.
- [3287] J. Zhang and A. C. Sanderson. **JADE: Adaptive differential evolution with optional external archive**. *IEEE Trans. Evol. Comput.*, 13(5):945–958, 2009. doi:10.1109/TEVC.2009.2014613.
- [3288] Q. Zhang. **MOEA/D homepage**. <https://sites.google.com/view/moead/>, 2007.
Annotation: Previous URL was at <https://dces.essex.ac.uk/staff/zhang/webofmoead.htm>.
- [3289] Q. Zhang and H. Li. **MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition**. *IEEE Trans. Evol. Comput.*, 11(6):712–731, 2007. doi:10.1109/TEVC.2007.892759.
Annotation: Introduces penalty-based boundary intersection (PBI) function.
- [3290] Q. Zhang and P. N. Suganthan. **Special Session on Performance Assessment of Multiobjective Optimization Algorithms/CEC’09 MOEA Competition**. https://www3.ntu.edu.sg/home/epnsugan/index_files/CEC09-MOEA/CEC09-MOEA.htm, 2009.
Annotation: Previously available at <http://dces.essex.ac.uk/staff/qzhang/moeacompetition09.htm>.
- [3291] Q. Zhang, W. Liu, and H. Li. **The Performance of a New Version of MOEA/D on CEC09 Unconstrained MOP Test Instances**. In *IEEE CEC*, pages 203–208, Piscataway, NJ, 2009. IEEE Press.
- [3292] Q. Zhang, A. Zhou, S. Zhao, P. N. Suganthan, W. Liu, and S. Tiwari. **Multiobjective Optimization Test Instances for the CEC 2009 Special Session and Competition**. Working Report CES-487, School of Computer Science and Electronic Engineering, University of Essex, Apr. 2009.
Annotation: Proposed UF benchmark.
- [3293] T. Zhang, M. Georgiopoulos, and G. C. Anagnostopoulos. **S-Race: A Multi-Objective Racing Algorithm**. In C. Blum and E. Alba, editors, *GECCO*, pages 1565–1572. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8.
- [3294] T. Zhang, M. Georgiopoulos, and G. C. Anagnostopoulos. **SPRINT: Multi-Objective Model Racing**. In S. Silva and A. I. Esparcia-Alcázar, editors, *GECCO*, pages 1383–1390. ACM Press, New York, NY, 2015. doi:10.1145/2739480.2754791.
Keywords: model selection, multi-objective optimization, racing algorithm, sequential probability ratio test.
Annotation: Extended version published as [3295].
- [3295] T. Zhang, M. Georgiopoulos, and G. C. Anagnostopoulos. **Multi-Objective Model Selection via Racing**. *IEEE Trans. Cybern.*, 46(8):1863–1876, 2016.

- [3296] H. Zhao and S. Ram. **Constrained cascade generalization of decision trees.** *IEEE Transactions on Knowledge and Data Engineering*, 16(6):727–739, 2004. doi:10.1109/TKDE.2004.3.
- [3297] L. Zhen and D.-F. Chang. **A bi-objective model for robust berth allocation scheduling.** *Computers & Industrial Engineering*, 63(1):262–273, 2012.
- [3298] Y. Zhong, X. Wang, Y. Sun, and Y. Gong. **SDDObench: A Benchmark for Streaming Data-Driven Optimization with Concept Drift.** In J. Handl and X. Li, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2024*, pages 59–67. ACM Press, New York, NY, 2024. doi:10.1145/3638529.3654063.
- [3299] A. Zhou, Q. Zhang, and Y. Jin. **Approximating the set of Pareto-optimal solutions in both the decision and objective spaces by an estimation of distribution algorithm.** *IEEE Trans. Evol. Comput.*, 13(5):1167–1189, 2009. doi:10.1109/TEVC.2009.2021467.
Keywords: multi-modal, IGDx.
- [3300] H. Zhu, M. Ji, and W. Guo. **Integer Linear Programming Models for the Containership Stowage Problem.** *Mathematical Problems in Engineering*, 2020:1–14, Oct. 2020. doi:10.1155/2020/4382745.
Keywords: Container loading problem.
Annotation: Article ID 4382745.
- [3301] W. Zhu, W.-C. Oon, A. Lim, and Y. Weng. **The six elements to block-building approaches for the single container loading problem.** *Applied Intelligence*, 37(3):431–445, Jan. 2012. doi:10.1007/s10489-012-0337-0.
Annotation: Proposed G2LA.
- [3302] S. Zilberstein. **Using Anytime Algorithms in Intelligent Systems.** *AI Magazine*, 17(3):73–83, 1996. doi:10.1609/aimag.v17i3.1232.
Keywords: performance profiles.
- [3303] S. Zilberstein, J. Koehler, and S. Koenig, editors. *Proceedings of the Fourteenth International Conference on Automated Planning and Scheduling (ICAPS 2004)*. AAAI Press/MIT Press, Menlo Park, CA, 2004.
- [3304] H. Zille, H. Ishibuchi, S. Mostaghim, and Y. Nojima. **Mutation operators based on variable grouping for multi-objective large-scale optimization.** In X. Chen and A. Stafylopatis, editors, *Computational Intelligence (SSCI), 2016 IEEE Symposium Series on*, pages 1–8, 2016. doi:10.1109/SSCI.2016.7850214.
Annotation: linked polynomial mutation.
- [3305] S. Zionts and J. Wallenius. **An interactive multiple objective linear programming method for a class of underlying nonlinear utility functions.** *Management Science*, 29(5):519–529, 1983.
- [3306] E. Zitzler. *Evolutionary Algorithms for Multiobjective Optimization: Methods and Applications*. PhD thesis, ETH Zürich, Switzerland, 1999.
- [3307] E. Zitzler and S. Künzli. **Indicator-based Selection in Multiobjective Search.** In X. Yao et al., editors, *PPSN*, volume 3242 of *LNCS*, pages 832–842. Springer, Heidelberg, Germany, 2004.
Keywords: IBEA.
- [3308] E. Zitzler and L. Thiele. **Multiobjective Optimization Using Evolutionary Algorithms - A Comparative Case Study.** In A. E. Eiben, T. Bäck, M. Schoenauer, and H.-P. Schwefel, editors, *Parallel Problem Solving from Nature – PPSN V*, volume 1498 of *LNCS*, pages 292–301.

- Springer, Heidelberg, Germany, 1998. doi:10.1007/BFb0056872.
Annotation: Proposed hypervolume measure.
- [3309] E. Zitzler and L. Thiele. **Multiobjective Evolutionary Algorithms: A Comparative Case Study and the Strength Pareto Approach.** *IEEE Trans. Evol. Comput.*, 3(4):257–271, 1999. doi:10.1109/4235.797969.
Annotation: Proposed SPEA, <http://www.tik.ee.ethz.ch/sop/publicationListFiles/zt1999a.pdf>.
- [3310] E. Zitzler, L. Thiele, and K. Deb. **Comparison of Multiobjective Evolutionary Algorithms: Empirical Results.** *Evol. Comput.*, 8(2):173–195, 2000. doi:10.1162/106365600568202.
Keywords: ZDT benchmark.
- [3311] E. Zitzler, K. Deb, L. Thiele, C. A. Coello Coello, and D. Corne, editors. *Evolutionary Multi-Criterion Optimization, First International Conference, EMO 2001, Zurich, Switzerland, March 7-9, 2001, Proceedings*, volume 1993 of LNCS. Springer, Berlin/Heidelberg, 2001.
- [3312] E. Zitzler, M. Laumanns, and L. Thiele. **SPEA2: Improving the Strength Pareto Evolutionary Algorithm.** Technical Report 103, Computer Engineering and Networks Laboratory (TIK), Swiss Federal Institute of Technology (ETH), Zürich, Switzerland, 2001.
Annotation: Published as [3313].
- [3313] E. Zitzler, M. Laumanns, and L. Thiele. **SPEA2: Improving the Strength Pareto Evolutionary Algorithm for Multiobjective Optimization.** In K. C. Giannakoglou, D. T. Tsahalis, J. Periaux, K. D. Papaliliou, and T. Fogarty, editors, *EUROGEN*, pages 95–100. CIMNE, Barcelona, Spain, 2002. ISBN 84-89925-97-6.
Annotation: Proposed SPEA2.
- [3314] E. Zitzler, L. Thiele, M. Laumanns, C. M. Fonseca, and V. Grunert da Fonseca. **Performance Assessment of Multiobjective Optimizers: an Analysis and Review.** *IEEE Trans. Evol. Comput.*, 7(2):117–132, 2003. doi:10.1109/TEVC.2003.810758.
Annotation: Proposed the combination of quality indicators; proposed epsilon-indicator.
- [3315] E. Zitzler, M. Laumanns, and S. Bleuler. **A tutorial on evolutionary multiobjective optimization.** In X. Gandibleux, M. Sevaux, K. Sörensen, and V. T’Kindt, editors, *Metaheuristics for Multiobjective Optimisation*, volume 535 of LNEMS, pages 3–37. Springer, Berlin/Heidelberg, 2004.
- [3316] E. Zitzler, D. Brockhoff, and L. Thiele. **The Hypervolume Indicator Revisited: On the Design of Pareto-compliant Indicators Via Weighted Integration.** In S. Obayashi et al., editors, *EMO*, volume 4403 of LNCS, pages 862–876. Springer, Heidelberg, Germany, 2007. doi:10.1007/978-3-540-70928-2_64. Supplementary material: <https://sop.tik.ee.ethz.ch/download/supplementary/weightedHypervolume/>.
- [3317] E. Zitzler, J. D. Knowles, and L. Thiele. **Quality Assessment of Pareto Set Approximations.** In J. Branke, K. Deb, K. Miettinen, and R. Słowiński, editors, *Multiobjective Optimization: Interactive and Evolutionary Approaches*, volume 5252 of LNCS, pages 373–404. Springer, Heidelberg, Germany, 2008. doi:10.1007/978-3-540-88908-3_14.
- [3318] E. Zitzler, L. Thiele, and J. Bader. **SPAM: Set Preference Algorithm for Multiobjective Optimization.** In G. Rudolph et al., editors, *Parallel Problem Solving from Nature – PPSN X*, volume 5199 of LNCS, pages 847–858. Springer, Heidelberg, Germany, 2008.
- [3319] E. Zitzler, L. Thiele, and J. Bader. **On Set-Based Multiobjective Optimization.** *IEEE Trans. Evol. Comput.*, 14(1):58–79, 2010. doi:10.1109/TEVC.2009.2016569.
Keywords: Performance assessment; Preference articulation; refinement; Set Partitioning; Set-preference.
Annotation: Proposed SPAM and explores combination of quality indicators.

- [3320] M. Zlochin, M. Birattari, N. Meuleau, and M. Dorigo. **Model-Based Search for Combinatorial Optimization: A Critical Survey**. *Annals of Operations Research*, 131 (1–4):373–395, 2004.
- [3321] A. Zujevs and J. Eiduks. **New decision maker model for multiobjective optimization interactive methods**. In *17th International Conference on Information and Software Technologies, Kaunas, Lithuania*, pages 51–58, 2011.
Keywords: Machine Decision Maker.